MATH5703M Statistical Theory

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Introduction

These are the lecture notes for the Statistical Theory (MATH5703M) module. The question we will consider in this module, in many different forms, is the following: What can we learn about the parameters of a model if we are given data from the model?

The first half of the module is concerned with estimation. We will start by considering statistical models and properties of an estimator, using the bias and mean squared error as measures of quality. We will then consider two different approaches to constructing estimators: maximum likelihood estimators and the method of moments. The unifying framework for this part of the module is the exponential family of distributions. Sufficiency, completeness and the Rao–Blackwell and Lehmann–Scheffé theorems will be used to study how to improve an estimator and to identify cases where an estimator cannot be improved. Finally, we will consider Fisher information and the Cramér–Rao bound to study limitations on the precision of unbiased estimators. We will conclude the study of estimation by considering large-sample theory for the maximum likelihood estimator and the EM algorithm.

The second half of the module is concerned with testing and interval estimation. We will develop the Neyman–Pearson theory of hypothesis tests and the likelihood-ratio test, and will then consider confidence intervals. A section on Bayesian inference will cover the choice of prior, credible intervals and Bayesian testing. We will conclude the study of testing by considering standard tests used in practice.

Alongside the theory, we will use simulation in R to illustrate the ideas of the module. Appendix B collects the R commands we use.

How to Read the Proofs

Many of the benefits of this module can be gained by studying the proofs, but a different approach is needed to read and understand a proof than is needed to understand ordinary text. The technique we will use here, called *self-explanation*, is the only technique for reading proofs which is supported by strong experimental evidence. In three experiments with undergraduate mathematics students, Hodds et al. (2014) showed that a short self-study training in self-explanation, consisting of around twenty minutes of slides and one worked example and one practice proof, significantly improved scores in proof-comprehension tests. The improvement in the laboratory experiment was close to one standard deviation. A fifteen-minute booklet version of the training, completed during a scheduled lecture, also showed a smaller but significant improvement, and the effect of this was still visible three weeks later. We will practise this technique together in

the first week of the module, and my hope is that you will use the technique throughout the module.

The idea of self-explanation is that you should pause after every line of the proof, and try to explain the line to yourself, before you read the next line. A good self-explanation answers the following three questions.

- What does this line state?
- Why is this statement true? Be specific about how you know: is it from an earlier line of the proof, or from a definition, or from an assumption of the theorem, or from a previous result?
- How does this line contribute to the proof? How does it move the argument towards the statement?

There are two bad habits which look similar to self-explanation, but are not: paraphrasing, *i.e.* restating a line in different words without understanding the reason, and 'monitoring', *i.e.* saying 'yes, this makes sense' and then carrying on. Both of these techniques help your understanding less than self-explanation does. It is not enough to just have a feeling that you understand a proof; you also need to be able to reconstruct the argument line by line. If you cannot name the reason for a statement, you have a gap in your understanding and you need to find it: write down exactly what is missing, look it up or work it out, and only then continue.

In practice this means reading with a pencil. Before you read the proof, read the statement carefully, and make sure you understand what is assumed and what is claimed. Then, as you read the proof, write down the justification for each step in the margin. If you cannot find a justification for a step, try to fill the gap yourself before you read the next line. This approach will make reading a proof slower, but it will also make sure that you understand the proof. The same technique can be used to understand the worked solutions in appendix C: try to solve the exercise yourself first, and only then look at the solution line by line to check your work.

What About AI?

Large language models are very good at the kind of material we cover in this module. Using one of these models, it is possible to get a solution to any of the exercises in these notes, to get answers for the examination questions, and even to get R code for most of the tasks required for the practical report.

The problem with this is that the aim of the module is for you to learn the material in these notes, not to solve the exercises (solutions for all of the exercises are in the appendix). While AI can explain every line of a proof to you, this is not the same as self-explanation. Similarly, the practical report requires you to show your judgement and to explain your work, but a report written by the model does not give you practice in either of these skills.

There is evidence of this problem in scientific studies: Strömberg et al. (2026) studied the effect of such tools on 26 811 secondary-school students over a period of 30 months: homework scores increased by 18%, homework time decreased by 30% and after six months scores in the monthly, closed-book examinations had

decreased by 20%. The losses were concentrated amongst the approximately 80% of users who completed their homework faster than any non-user and scored high marks for the homework, and the authors of the study interpreted these results as indicating that these users had "outsourced" their homework to the tool. Users who spent as much time on their homework as non-users did performed about as well in the examinations, so the loss in this group was not due to the tools themselves, but due to the way the tools were used.

A randomised trial in mathematics (Bastani et al., 2025) showed the same effect. Around 1000 secondary-school students either used a language model to learn the material or used only the textbook and notes. The students were then tested in a closed-book exam. The students who used the unrestricted language model scored 17% worse than the others, mainly because they used the model to find answers to the questions. When a version of the model was used which only gave hints instead of answers, the loss in performance disappeared. The students were not aware of the difference: students who used the unrestricted model did not believe that they had learned less than the other students.

I want to also mention that the exam for our module is closed book, pen and paper based. In the exam, you will not have access to AI support, and will need to rely on what you have personally learned.

There are ways to use these models to help understanding, but these require a slightly different approach: you ask the model for something other than the answer to the question at hand.

- Ask for more practice material. The model can easily generate more exercises for you. But remember that the only way to learn from these exercises is to solve them yourself!

- Ask for a second explanation. If a definition, a result or a step in a proof is not clear to you, ask the model to explain it in different words; sometimes a second explanation helps you to understand it. Afterwards, close the window and write it down from memory: a definition or result in full, a step in a proof by giving the reason why it holds in your own words. Reading the model's explanation is only monitoring, while writing it down from memory shows that you have understood or at least memorised it.

- Ask which result a step relies on. If you are stuck on a step of a proof which you cannot yet justify, the model can sometimes name the result the step uses, for example Jensen's inequality or the tower property of conditional expectation. Ask for the name only, look up the result in the index and work out the step yourself.

- Ask why, not what. A question like "Why do we divide by $n - 1$ in the sample variance?" is a question which you can first try to answer yourself, and then check your answer with the model's answer. In contrast, a question like "solve this exercise" leaves nothing for you to do.

- Ask about R. What does an error message mean? What does a given argument of a command do? These are questions which can be answered by the help pages (see appendix B). It is not expected that you will know these details by heart from this module, so asking the model is fine.

One way to test whether you have understood an argument is to try to re-create the argument on paper, from memory, a week later. If you can do this, you have learned something! If you cannot do this, you only have a solution, not an understanding.

Reading

These notes are self-contained, but the following books can be used as additional reading. The links lead to the University of Leeds Library catalogue.

- G. Casella and R. L. Berger, *Statistical Inference* (2nd edition; available online). This is our main text. It covers most of the material of the module, from point estimation to sufficiency and the information inequality, and from hypothesis testing to confidence intervals.
- G. A. Young and R. L. Smith, *Essentials of Statistical Inference* (print only). A concise and rigorous graduate text at the level of this module. The book is particularly good on likelihood, decision theory and the large-sample behaviour of the maximum likelihood estimator.
- G. H. Givens and J. A. Hoeting, *Computational Statistics* (2nd edition; available online). This book is useful for learning about the EM algorithm and about numerical optimisation of the likelihood. It also introduces the computational ideas which we use in the R sessions.
- P. M. Lee, *Bayesian Statistics: An Introduction* (4th edition; available online). A self-contained introduction to the Bayesian material we cover, including the choice of prior, Jeffreys priors and Bayesian testing.
- W. J. Braun and D. J. Murdoch, *A First Course in Statistical Programming with R* (3rd edition; available online). This book is designed to support the R sessions and the practical report. It teaches programming and simulation in R from the ground up.

The following list has additional resources which either are more specialised and go deeper, or offer a gentler alternative to the more standard texts listed above.

- D. R. Cox and D. V. Hinkley, *Theoretical Statistics* (available online). The classic deeper treatment of the theory behind the module.
- E. L. Lehmann and G. Casella, *Theory of Point Estimation* (2nd edition; print only). The advanced reference for the estimation half: sufficiency, completeness and efficiency in full detail.
- A. C. Davison, *Statistical Models* (print only). Broad and likelihood-centred, with the EM algorithm and many worked applications.
- A. Gelman et al., *Bayesian Data Analysis* (3rd edition; available online). The standard modern reference for Bayesian statistics, well beyond what we need but the natural next step.

- M. H. DeGroot and M. J. Schervish, *Probability and Statistics* (4th edition; available online). A gentle and thorough account, useful for revising the prerequisite material from MATH2701.
- J. A. Rice, *Mathematical Statistics and Data Analysis* (3rd edition; print only). An accessible companion pitched a little below the core, good for a second explanation.

Lecture 1

Statistical Models and the Estimation Problem

The subjects of probability and statistics are in some sense the opposite of each other. In a probability course we are given a distribution $\mathbb{P}$ and we ask questions about the data generated by this distribution: how often should we expect ten tosses of a fair coin to result in seven heads? In contrast, in a statistics course we are given data and we ask questions about the distribution: what can we learn about the coin by observing seven heads?

$$\text{probability:} \quad \mathbb{P} \longrightarrow X_1, \ldots, X_n, \qquad \text{statistics:} \quad X_1, \ldots, X_n \longrightarrow \mathbb{P}.$$

The topic of this module is the second arrow in the above discussion. We will consider two questions: in *estimation* we consider how (and how well) we can determine an unknown quantity from data. In *hypothesis testing* we consider whether the data are compatible with a given claim. We will answer these questions first for the classical approach, and then for the Bayesian approach. In this first lecture we will lay the foundations for these questions by introducing the model and the estimators computed from the data.

1.1 Statistical Models

The data are given as $x_1, \ldots, x_n$, a list of n numbers. Once we have computed an estimate from these numbers, we want to know how far it is from the truth, and this only makes sense if there is a true value to compare the estimate to. The data themselves do not contain a 'true value'. Instead, this ground truth is provided by a model, a mathematical description of how the data were generated.

Our description of how the data were generated is called a model. We use a model to describe the data as the outcome of a random experiment. The data we have observed, $x_1, \ldots, x_n$, are seen as the observed values of random variables $X_1, \ldots, X_n$, where the distribution of the random variables depends on the unknown true value. Here we use the convention that upper case letters like X_i stand for random variables, and that lower case letters like x_i stand for the observed values for our dataset.

In the simplest and most common situation the observations are a *random sample*: the X_i are independent and identically distributed (i.i.d.), but we do not know the distribution. Instead, we assume that the distribution belongs to a known family of distributions, labelled by a parameter.

Definition 1.1. A *parametric model* for data $X_1, \dots, X_n$ consists of a family

$$(\mathbb{P}_\theta \,|\, \theta \in \Theta)$$

of distributions, and the assumption that the $X_1, \dots, X_n$ are i.i.d. with distribution $\mathbb{P}_\theta$ for an unknown value of the *parameter* θ. The set Θ of possible parameter values is called the *parameter space*.

Our assumption here is that the actually observed data, $x_1, \dots, x_n$, are a sample from $\mathbb{P}_\theta$ for one (unknown to us) θ. This value is called the *true parameter value* and is denoted by θ_0. This is the value which our estimates are compared to: an estimate is good, if it is close to θ_0, and bad if it is far away from θ_0.

Since the distribution is determined by the parameter, learning about the distribution also means learning about the parameter. Thus, the diagram from the beginning of this lecture can be updated to

$$\begin{aligned} \text{probability:} \quad & \theta_0 \longrightarrow \mathbb{P}_{\theta_0} \longrightarrow X_1, \dots, X_n, \\ \text{statistics:} \quad & X_1, \dots, X_n \longrightarrow \theta_0. \end{aligned}$$

The true parameter value can never be observed directly, and all information about the true parameter value comes from $X_1, \dots, X_n$.

The parameter θ does not necessarily have to be a single number. For example, if we assume that the data are normally distributed with unknown mean μ and unknown variance σ^2, then $\mathbb{P}_\theta = N(\mu, \sigma^2)$ and the parameter is the pair $\theta = (\mu, \sigma^2)$. In this case, $\Theta = \mathbb{R} \times (0, \infty)$ is a subset of the plane. While we allow for more general parameters from the start, most of the examples in this text will involve a scalar parameter θ.

The distribution $\mathbb{P}_\theta$ describes, for every set A of possible values, the probability $\mathbb{P}_\theta(A)$ that an observation falls into A. In practice, a distribution is almost always given by a function $f(x; \theta)$ of the possible values x, and this can be done in two different ways. For discrete distributions, where the observations can take finitely or countably many values, *e.g.* in $\{0, 1\}$ or in the set $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ of non-negative integers (we write $\mathbb{N} = \{1, 2, 3, \dots\}$ for the positive integers), we have $f(x; \theta) = \mathbb{P}_\theta(\{x\})$, giving the probability of observing the value x. These numbers are called the *probability weights* of the distribution (sometimes also called the *probability mass function*). These weights are numbers in the interval $[0, 1]$, summing to one, and satisfying

$$\mathbb{P}_\theta(A) = \sum_{x \in A} f(x; \theta).$$

For continuous distributions, $f(x; \theta)$ is a *density*: a non-negative function, which is integrable and has integral one, such that

$$\mathbb{P}_\theta(A) = \int_A f(x; \theta) \, \mathrm{d}x.$$

For continuous distributions, each value x has probability $\mathbb{P}_\theta(\{x\}) = 0$, and a density can take values larger than one. The function values $f(x; \theta)$ of a density are not probabilities, but integrals of the density over sets give probabilities.

In both cases, the family of functions $(f(x; \theta) \,|\, \theta \in \Theta)$ determines the model, and we will often write the model in this form. Much of the theory in this

module is common to both cases. Where a statement is true for both cases, we sometimes only state it for densities, and then the discrete case can be deduced by interpreting $f(x;\theta)$ as the probability weights and all integrals over x as sums.

We use the notation $\mathbb{P}_\theta$ also for the probabilities of events involving the data, and we write $\mathbb{E}_\theta$ for expectations, to indicate that the result depends on the value of the parameter. For example, $\mathbb{P}_\theta(X_1 \leq t) = \mathbb{P}_\theta((-\infty, t])$ is the probability that a single observation is at most t, when the data have distribution $\mathbb{P}_\theta$.

We will consider a small number of families of models throughout the module. The discrete families are listed in table 1.1 and the continuous ones in table 1.2. The corresponding means and variances, which we will need frequently in calculations, are shown in tables A.1 and A.2 in appendix A.

Family	θ	Weights $f(x;\theta)$	Support	Θ
Bernoulli	p	$p^x(1-p)^{1-x}$	$\{0,1\}$	$(0,1)$
Binomial	(m,p)	$\binom{m}{x} p^x(1-p)^{m-x}$	$\{0,1,\ldots,m\}$	$\mathbb{N} \times (0,1)$
Poisson	λ	$\frac{\lambda^x}{x!}\, e^{-\lambda}$	$\mathbb{N}_0$	$(0,\infty)$

Table 1.1: Discrete families of models considered in the module.

Family	θ	Density $f(x;\theta)$	Support	Θ
Exponential	λ	$\lambda\, e^{-\lambda x}$	$[0,\infty)$	$(0,\infty)$
Normal	(μ,σ^2)	$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(-\frac{(x-\mu)^2}{2\sigma^2}\Big)$	$\mathbb{R}$	$\mathbb{R} \times (0,\infty)$
Uniform	θ	$1/\theta$	$[0,\theta]$	$(0,\infty)$

Table 1.2: Continuous families of models considered in the module.

For families where θ is a vector, usually one component is assumed known and fixed, and only the remaining component is estimated. For example, for the binomial distribution the number m of trials is often assumed to be known, and for the normal distribution the variance σ^2 is often assumed to be known.

Example 1.2. Consider the model $X_1, \ldots, X_n \sim \text{Uniform}(0,\theta)$ with $\theta > 0$, *i.e.* each observation is equally likely to take a value anywhere in the interval $[0,\theta]$ and $f(x;\theta) = 1/\theta$ for $0 \leq x \leq \theta$, and 0 otherwise. Since no observation can exceed θ, if θ increases, larger data values are possible. The range of the data can be used to draw inference about the parameter. This is in contrast to the normal distribution or the exponential distribution, where all values are possible for all θ. We will see later that the moving support of the uniform distribution leads to different behaviour than for the other models, and this example will return as a test case.

1.2 Statistics and Estimators

Once a model is fixed, we want to extract information about the unknown θ from the data. Since we do not know the value of θ, we need to consider quantities which can be computed from the data.

Definition 1.3. A *statistic* is a function $T = T(X_1, \ldots, X_n)$ of the data, which does not depend on the parameter θ.

Under this restriction the sample mean $\bar{X} = (X_1 + \cdots + X_n)/n$ is a statistic, but $X_1 - \theta$ is not, since we need to know the parameter value to compute it.

Definition 1.4. An *estimator* for θ is a statistic $\hat{\theta} = \hat{\theta}(X_1, \ldots, X_n)$, which can be used to approximate θ. The number $\hat{\theta}(x_1, \ldots, x_n)$, computed from a given dataset $x_1, \ldots, x_n$, is called an *estimate*.

Throughout the notes we will use hats to denote estimators: $\hat{\theta}$ is an estimator for θ, $\hat{\sigma}^2$ is an estimator for σ^2, and so on. If two estimators for the same quantity are compared, the second one is denoted by a tilde: $\tilde{\theta}$ is a competitor of $\hat{\theta}$.

Most estimators are defined by a formula which works for all sample sizes, *e.g.* $\bar{X} = (X_1 + \cdots + X_n)/n$. Technically, this formula defines a *family of estimators* $(\hat{\theta}_n)_{n \in \mathbb{N}}$, one for each sample size n. When we need to consider the dependence on n, and in particular when we study the behaviour of an estimator as the amount of data increases, we write $\hat{\theta}_n$ instead of $\hat{\theta}$.

An important note is that the definition does not imply anything about quality. Any statistic is an estimator, for example the constant function $\hat{\theta} = 0$ is a (bad) estimator for a Bernoulli success probability. We will discuss how to find and recognise good estimators in the following lectures.

The difference between an estimator and an estimate is a subtle but important one. An *estimate* is a single number, computed from the data, for example $\hat{\theta} = 0.7$. An *estimator* is a rule for computing such a number, typically a function of the random variables $X_1, \ldots, X_n$, such that different outcomes of the experiment give rise to different values of the estimate.

We can only make statements about how good an estimator is, because $\hat{\theta}(X_1, \ldots, X_n)$ is a random variable. The *sampling distribution* of an estimator is the distribution of the resulting estimates. We can ask questions like the following: If the data are generated by the true value θ_0, how are the estimates distributed around θ_0? Is the distribution centred around the truth? If so, how large is the spread? A good estimator will have a sampling distribution which is concentrated around the true parameter value, for all possible values of the true parameter. In the next lecture we will learn how to quantify these ideas using the concepts of bias and mean squared error.

Summary.

- Statistics is the inverse of probability: we use data to learn about the model, rather than using the model to learn about the data.
- A parametric model assumes that the data are i.i.d. from a family $(\mathbb{P}_\theta \mid \theta \in \Theta)$ of distributions, for an unknown true parameter value θ_0. The model provides the true value which an estimate is compared to.
- Discrete distributions are given by probability weights, which are probabilities, while continuous distributions are given by a density, which is not a probability. Probabilities are calculated as sums of weights or as integrals of the density.
- A statistic is a function of the data. An estimator is a statistic which is used to approximate θ, and an estimate is the value of the estimator for a given dataset.
- An estimator is a random variable, and we study its sampling distribution. A good estimator will have a sampling distribution concentrated around the true value θ_0, for all possible values of θ_0. In the next lecture we will learn how to quantify this idea.

Exercise 1.1. A coin with unknown probability θ of heads is tossed n times. The results of the individual tosses are given by $X_i = 1$, if the i-th toss is heads, and $X_i = 0$ otherwise.

1. Write down a parametric model for $X_1, \ldots, X_n$, including the probability weights and the parameter space.
2. Determine an estimator $\hat{\theta}$ for θ.
3. In one sentence each, explain why $\hat{\theta}(X_1, \ldots, X_n)$ is a random variable while $\hat{\theta}(x_1, \ldots, x_n)$ is a number.

Exercise 1.2. Let $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ be i.i.d., where both the mean μ and the variance σ^2 are unknown, so that the parameter is $\theta = (\mu, \sigma^2)$. Which of the following quantities are statistics?

(a) $\max_{1 \le i \le n} X_i - \min_{1 \le i \le n} X_i$, (b) $\frac{1}{n} \sum_{i=1}^{n} (X_i - \mu)^2$, (c) $\frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2$.

Exercise 1.3. Let $X_1, \ldots, X_n \sim \text{Uniform}(0, \theta)$ be i.i.d., as in example 1.2.

1. Write down the joint density $f(x_1, \ldots, x_n; \theta)$, and state the values of the data for which it is non-zero.
2. Show that the estimator $\hat{\theta} = \max_i X_i$ always satisfies $\hat{\theta} \le \theta$.
3. Show that in fact $\hat{\theta} < \theta$ with probability one, and deduce that $\mathbb{E}_\theta(\hat{\theta}) < \theta$, so that on average the estimator underestimates θ.

Lecture 2

Bias, Mean Squared Error and Consistency

We finished the previous lecture with the idea that a good estimator is one which has a sampling distribution concentrated around the true value θ_0, for all possible true values. In this lecture we quantify this idea: the bias measures whether the estimator is on average right, the mean squared error measures how far away the estimator typically is from the truth, and consistency is the property that the estimator converges to the truth as the amount of data increases. The main idea to take away is the decomposition of the mean squared error into a variance and a squared bias.

2.1 Bias and Unbiasedness

The most basic question we can ask about an estimator is whether it gives the correct answer on average.

Definition 2.1. The *bias* of an estimator $\hat{\theta}$ for θ is given by

$$\text{bias}(\hat{\theta}) = \mathbb{E}_\theta(\hat{\theta}) - \theta.$$

The estimator is *unbiased*, if $\hat{\theta}$ has finite mean and $\text{bias}(\hat{\theta}) = 0$ for all $\theta \in \Theta$, *i.e.* if $\mathbb{E}_\theta(\hat{\theta}) = \theta$ for all θ.

The bias in general depends on the true value θ, as shown by the expectation with θ as a subscript.

The following lemma shows that the sample mean is an unbiased estimator for the mean. For future reference, we also compute the variance of the sample mean here.

Lemma 2.2. *Let $X_1, \ldots, X_n$ be i.i.d. with $\mathbb{E}(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$. Define $\bar{X} = (X_1 + \cdots + X_n)/n$. Then*

$$\mathbb{E}(\bar{X}) = \mu \qquad \textit{and} \qquad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

Proof. Using the linearity of expectation we find

$$\mathbb{E}(\bar{X}) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}(X_i) = \frac{1}{n} \cdot n\mu = \mu.$$

Since the X_i are independent, the variance of the sum can be found as the sum of the variances:

$$\mathrm{Var}(\bar{X}) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{1}{n^2}\cdot n\sigma^2 = \frac{\sigma^2}{n}.$$

This completes the proof. □

Whenever the parameter of interest is the mean of the distribution $\mathbb{P}_\theta$, *i.e.* $\theta = \mu$, then lemma 2.2 shows that $\mathbb{E}_\theta(\bar{X}) = \theta$ and thus $\mathrm{bias}(\bar{X}) = \mu - \mu = 0$.

A more delicate example, which we will consider again in section 2.3, is the problem of estimating a variance. If a sample is observed with unknown mean, then one can estimate the variance σ^2 using the *sample variance*

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

The unexpected divisor $n-1$ instead of the seemingly more obvious n is required to make this estimator unbiased.

Lemma 2.3. *Let $n \geq 2$ and let $X_1, \ldots, X_n$ be i.i.d. with $\mathrm{Var}(X_i) = \sigma^2$. Then $\mathbb{E}(S^2) = \sigma^2$ and thus the sample variance is an unbiased estimator for σ^2.*

Proof. Let $\mu = \mathbb{E}(X_i)$. We prove the algebraic identity

$$\sum_{i=1}^{n}(X_i - \bar{X})^2 = \sum_{i=1}^{n}(X_i - \mu)^2 - n(\bar{X} - \mu)^2, \tag{2.1}$$

which holds for all numbers. Writing $X_i - \bar{X} = (X_i - \mu) - (\bar{X} - \mu)$, we can expand the square and use the relation $\sum_i(X_i - \mu) = n(\bar{X} - \mu)$ to simplify the expression of the cross term. Taking expectations on (2.1) and using the relations $\mathbb{E}\big((X_i - \mu)^2\big) = \sigma^2$ as well as $\mathbb{E}\big((\bar{X} - \mu)^2\big) = \mathrm{Var}(\bar{X}) = \sigma^2/n$ from lemma 2.2 we find

$$\mathbb{E}\Big(\sum_{i=1}^{n}(X_i - \bar{X})^2\Big) = n\sigma^2 - n\cdot\frac{\sigma^2}{n} = (n-1)\sigma^2.$$

Dividing by $n-1$ gives the result $\mathbb{E}(S^2) = \sigma^2$, as claimed. □

The calculation also shows what is wrong with the divisor n: the estimator $(1/n)\sum_i(X_i - \bar{X})^2$ has expectation $(n-1)\sigma^2/n$ and thus systematically underestimates the variance, since the observations are on average slightly closer to the sample mean $\bar{X}$ than the true mean μ would be. We will return to this comparison, and to the surprising fact that the biased estimator is in some sense better, in section 2.3.

The bias of this estimator is small if the sample is large. A property which is weaker than unbiasedness, but which holds for large sample sizes, is that the bias goes to zero in the limit.

Definition 2.4. Let $(\hat{\theta}_n)_{n\in\mathbb{N}}$ be a family of estimators for θ, depending on the sample size n. The estimator $\hat{\theta}_n$ is *asymptotically unbiased*, if

$$\lim_{n\to\infty}\mathrm{bias}(\hat{\theta}_n) = 0$$

for all $\theta \in \Theta$.

Example 2.5. Let $X_1, \dots, X_n$ be i.i.d. with $\mathrm{Var}(X_i) = \sigma^2$ and consider the family of estimators

$$\hat{\sigma}_n^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

for $n \geq 2$. Since $\hat{\sigma}_n^2 = (n-1)S^2/n$, we can use lemma 2.3 to conclude that $\mathbb{E}(\hat{\sigma}_n^2) = (n-1)\sigma^2/n$ and thus

$$\mathrm{bias}(\hat{\sigma}_n^2) = \frac{n-1}{n}\sigma^2 - \sigma^2 = -\frac{\sigma^2}{n}.$$

While the bias for every n is non-zero, it converges to 0 as $n \to \infty$. Thus, $\hat{\sigma}_n^2$ is a biased but asymptotically unbiased estimator.

2.2 Mean Squared Error

If an estimator is unbiased, this does not mean that it is accurate. The mean squared error is a numerical measure of the typical size of the error $\hat{\theta} - \theta$.

Definition 2.6. The *mean squared error* (MSE) of an estimator $\hat{\theta}$ for θ is given by

$$\mathrm{MSE}(\hat{\theta}) = \mathbb{E}_\theta\Big((\hat{\theta} - \theta)^2\Big).$$

We want the mean squared error to be small. It combines two different kinds of defects, a systematic offset and random scatter, and the following decomposition separates the two.

Lemma 2.7. *For any estimator $\hat{\theta}$ for θ with finite variance,*

$$\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + \mathrm{bias}(\hat{\theta})^2.$$

Proof. Let $m = \mathbb{E}_\theta(\hat{\theta})$. Then we can write $\mathrm{bias}(\hat{\theta}) = m - \theta$. Adding and subtracting m inside the square we get

$$\begin{aligned}\mathrm{MSE}(\hat{\theta}) &= \mathbb{E}_\theta\Big(\big((\hat{\theta} - m) + (m - \theta)\big)^2\Big) \\ &= \mathbb{E}_\theta\big((\hat{\theta} - m)^2\big) + 2(m-\theta)\,\mathbb{E}_\theta(\hat{\theta} - m) + (m-\theta)^2.\end{aligned}$$

Since $\mathbb{E}_\theta(\hat{\theta} - m) = m - m = 0$, the middle term disappears. The first term is $\mathrm{Var}(\hat{\theta})$ and the last term is $\mathrm{bias}(\hat{\theta})^2$. This is the required identity and the proof is complete. □

The decomposition allows us to understand the rôle of unbiasedness: for an unbiased estimator, the bias term disappears and we have $\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta})$, and thus, when comparing two unbiased estimators, we can only consider the variances. This is the basis of all optimality theory: among the unbiased estimators, the best one is the one with the smallest variance. In lecture 13 we will see that there is a lower bound for the variance of an unbiased estimator.

Example 2.8. Using lemma 2.7 and lemma 2.2, we find that the mean squared error of the sample mean is given by $\mathrm{MSE}(\bar{X}) = \sigma^2/n$.

2.3 When Unbiasedness Is Not Enough

Unbiasedness is an appealing property, but it is a constraint, not a goal, and insisting on it can cost us accuracy. We will show that for the variance of a normal sample the unbiased sample variance is beaten, in terms of mean squared error, by a biased competitor.

Let $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ be i.i.d. and consider the family of estimators

$$\hat{\sigma}_c^2 = c \sum_{i=1}^{n} (X_i - \bar{X})^2,$$

for every constant $c > 0$. For $c = 1/(n-1)$ this is the unbiased sample variance S^2 and for $c = 1/n$ it is the underestimating version from the previous section. To compare these estimators, we need to know the distribution of the sum of squares. From proposition A.4 in appendix A we know that the scaled sum of squares satisfies

$$\frac{1}{\sigma^2} \sum_{i=1}^{n} (X_i - \bar{X})^2 \sim \chi^2_{n-1}.$$

We first use this result to find the mean squared error of $\hat{\sigma}_c^2$ for a general constant c. We write $W = \sum_{i=1}^{n} (X_i - \bar{X})^2$, so that $\hat{\sigma}_c^2 = cW$. Since the χ^2_{n-1} distribution has mean $n-1$ and variance $2(n-1)$ by lemma A.3, we have

$$\mathbb{E}(W) = (n-1)\sigma^2 \qquad \text{and} \qquad \operatorname{Var}(W) = 2(n-1)\sigma^4.$$

Multiplying by c, we find that the estimator has expectation $\mathbb{E}(\hat{\sigma}_c^2) = c(n-1)\sigma^2$ and variance $\operatorname{Var}(\hat{\sigma}_c^2) = c^2 \operatorname{Var}(W) = 2(n-1)c^2\sigma^4$. Thus its bias is given by

$$\operatorname{bias}(\hat{\sigma}_c^2) = \mathbb{E}(\hat{\sigma}_c^2) - \sigma^2 = \big(c(n-1) - 1\big)\sigma^2,$$

and lemma 2.7 gives

$$\operatorname{MSE}(\hat{\sigma}_c^2) = \operatorname{Var}(\hat{\sigma}_c^2) + \operatorname{bias}(\hat{\sigma}_c^2)^2 = \Big(2(n-1)c^2 + \big(c(n-1) - 1\big)^2\Big)\sigma^4. \tag{2.2}$$

This formula holds for every $c > 0$, and we can now use it to compare the two estimators.

If we choose $c = 1/(n-1)$, the bias term disappears and we get

$$\operatorname{MSE}(S^2) = \frac{2}{n-1}\sigma^4.$$

If we choose $c = 1/n$, then $c(n-1) - 1 = -1/n$ and equation (2.2) gives

$$\operatorname{MSE}(\hat{\sigma}_{1/n}^2) = \Big(\frac{2(n-1)}{n^2} + \frac{1}{n^2}\Big)\sigma^4 = \frac{2n-1}{n^2}\,\sigma^4.$$

An easy calculation shows that $(2n-1)/n^2 < 2/(n-1)$ for all n. Thus, for every sample size and every value of σ^2, the biased estimator $\hat{\sigma}_{1/n}^2$ has smaller MSE than the unbiased estimator S^2.

Remark. The value $c = 1/n$ is not optimal. The mean squared error (2.2) is a quadratic function of c and minimising this function over all $c > 0$ gives $c = 1/(n+1)$. For this value we get

$$\operatorname{MSE}(\hat{\sigma}_{1/(n+1)}^2) = \frac{2\sigma^4}{n+1}.$$

Thus, neither the unbiased estimator nor the maximum-likelihood estimator $c = 1/n$ (which we will see in lecture 5) are optimal.

By allowing a small, controlled bias, we have been able to reduce the variance and thus have strictly improved the mean squared error. The lesson to be learned here is not that bias is good, but that an estimator should be evaluated using its mean squared error instead of just the bias.

2.4 Consistency

The mean squared errors we have computed all shrink as the sample grows. Consistency is the property which describes this limiting behaviour: as the number of observations increases, a consistent estimator converges to the true value.

Definition 2.9. Consider a family $(\hat{\theta}_n)_{n\in\mathbb{N}}$ of estimators for θ, one for each sample size n. The estimator $\hat{\theta}_n$ is said to be *consistent* for θ, if $\hat{\theta}_n \xrightarrow{\mathrm{p}} \theta$ as $n \to \infty$, *i.e.* if for every $\varepsilon > 0$ we have

$$\lim_{n\to\infty} \mathbb{P}_\theta(|\hat{\theta}_n - \theta| \geq \varepsilon) = 0,$$

for all $\theta \in \Theta$.

Here we write $\xrightarrow{\mathrm{p}}$ for convergence in probability, as recalled in definition A.12 in appendix A: the probability of the estimate being off the truth by any fixed amount eventually becomes negligible.

To show that an estimator is consistent, we need an upper bound for the probability in definition 2.9. The basic tool for this is Markov's inequality, which bounds the probability of a non-negative random variable being large in terms of its mean.

Lemma 2.10 (Markov's inequality)**.** *Let Y be a non-negative random variable and $c > 0$. Then*

$$\mathbb{P}(Y \geq c) \leq \frac{\mathbb{E}(Y)}{c}.$$

Proof. We define a new random variable Y_c by

$$Y_c = \begin{cases} c & \text{if } Y \geq c, \\ 0 & \text{if } Y < c. \end{cases}$$

If $Y \geq c$ we have $Y_c = c \leq Y$, and if $Y < c$ we have $Y_c = 0 \leq Y$, since Y is non-negative. Thus we have $Y_c \leq Y$, as shown in figure 2.1. Since Y_c only takes the values 0 and c, taking expectations gives

$$\mathbb{E}(Y) \geq \mathbb{E}(Y_c) = 0 \cdot \mathbb{P}(Y_c = 0) + c \cdot \mathbb{P}(Y_c = c) = c\,\mathbb{P}(Y \geq c).$$

Dividing by the positive number c completes the proof. □

Applying Markov's inequality to the squared distance between the estimator and the parameter gives a criterion for consistency in terms of the mean squared error.

Lemma 2.11. *If $\mathrm{MSE}(\hat{\theta}_n) \to 0$ as $n \to \infty$, then $\hat{\theta}_n$ is consistent.*

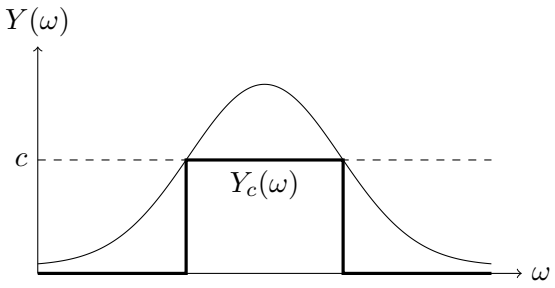


Figure 2.1: The random variable Y and the variable Y_c from the proof of Markov's inequality, as functions of the outcome ω. Since Y_c is never above Y, its mean $c\,\mathbb{P}(Y \geq c)$ is at most $\mathbb{E}(Y)$.

Proof. Using Markov's inequality, lemma 2.10, for the non-negative random variable $(\hat{\theta}_n - \theta)^2$, for any $\varepsilon > 0$ we get

$$\mathbb{P}_\theta(|\hat{\theta}_n - \theta| \geq \varepsilon) = \mathbb{P}_\theta((\hat{\theta}_n - \theta)^2 \geq \varepsilon^2) \leq \frac{\mathbb{E}_\theta((\hat{\theta}_n - \theta)^2)}{\varepsilon^2} = \frac{\mathrm{MSE}(\hat{\theta}_n)}{\varepsilon^2} \longrightarrow 0$$

as $n \to \infty$. Thus $\hat{\theta}_n \xrightarrow{\mathrm{p}} \theta$ as required. □

The most basic example of a consistent estimator is the sample mean, where consistency is a manifestation of the law of large numbers.

Example 2.12. Let $X_1, \ldots, X_n$ be i.i.d. with $\mathbb{E}_\theta(X_i) = \theta$ and finite variance σ^2. From example 2.8 we know that $\mathrm{MSE}(\bar{X}) = \sigma^2/n$. Since this converges to zero as $n \to \infty$, we can use lemma 2.11 to show that $\bar{X}$ is a consistent estimator for the mean.

Combining this criterion with the bias-variance decomposition gives a sufficient condition in terms of the two quantities we know how to compute.

Corollary 2.13. *If $\hat{\theta}_n$ is asymptotically unbiased and $\mathrm{Var}(\hat{\theta}_n) \to 0$ as $n \to \infty$, then $\hat{\theta}_n$ is consistent.*

Proof. From lemma 2.7 we know that $\mathrm{MSE}(\hat{\theta}_n) = \mathrm{Var}(\hat{\theta}_n) + \mathrm{bias}(\hat{\theta}_n)^2$ and since both terms go to zero by assumption, we can conclude from lemma 2.11 that the result holds. □

All estimators in this lecture where we have computed a mean squared error have $\mathrm{MSE}(\hat{\theta}_n)$ of order $1/n$ and thus are consistent. Consistency is a coarse property: two consistent estimators can have very different rates of convergence and the mean squared error is a more detailed measure.

Summary.

- The bias $\text{bias}(\hat{\theta}) = \mathbb{E}_\theta(\hat{\theta}) - \theta$ measures whether an estimator is right on average; an unbiased estimator has zero bias for all θ.
- The mean squared error $\text{MSE}(\hat{\theta}) = \mathbb{E}_\theta((\hat{\theta} - \theta)^2)$ is a measure for the typical squared error, given by $\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{bias}(\hat{\theta})^2$.
- Unbiasedness is desirable, but it is not the only aim: a biased estimator can have smaller mean squared error, as shown by the normal variance.
- An estimator is consistent, if $\hat{\theta}_n$ converges in probability to θ. A sufficient condition for this is $\text{MSE}(\hat{\theta}_n) \to 0$, *i.e.* being asymptotically unbiased with vanishing variance.

Exercise 2.1. Let $X_1, \ldots, X_n$ be an i.i.d. sample from the Poisson distribution with parameter θ, *i.e.* $\mathbb{E}_\theta(X_i) = \text{Var}_\theta(X_i) = \theta$ and let $\hat{\theta} = \bar{X}$.

1. Show that $\hat{\theta}$ is an unbiased estimator for θ.
2. Determine the mean squared error of $\hat{\theta}$ and show that $\hat{\theta}$ is a consistent estimator.

Exercise 2.2. Let $X_1, \ldots, X_n \sim \text{Uniform}(0, \theta)$ be i.i.d., as in example 1.2, and let $\hat{\theta} = \max_i X_i$. You may use that $\hat{\theta}$ has density $f(t) = n\,t^{n-1}/\theta^n$ for $0 \le t \le \theta$.

1. Show that $\mathbb{E}_\theta(\hat{\theta}) = n\theta/(n+1)$, and hence that the bias is $\text{bias}(\hat{\theta}) = -\theta/(n+1)$. Is $\hat{\theta}$ asymptotically unbiased?
2. Show that $\text{MSE}(\hat{\theta}) = 2\theta^2/((n+1)(n+2))$, and deduce that $\hat{\theta}$ is consistent.
3. Compare the rate at which this mean squared error shrinks with the $1/n$ rate found for the sample mean, and comment.

Exercise 2.3. Let $X_1, \ldots, X_n$ be an i.i.d. sample from the exponential distribution with parameter θ, *i.e.* $f(x;\theta) = \theta e^{-\theta x}$ for $x \ge 0$ and $\mathbb{E}_\theta(X_i) = 1/\theta$. If we define an estimator by $\hat{\theta} = 1/\bar{X}$, and if we use the fact that $T = X_1 + \cdots + X_n$ has the gamma density $f_T(x) = \theta^n x^{n-1} e^{-\theta x}/(n-1)!$ for $x \ge 0$, show that we have

$$\mathbb{E}_\theta(\hat{\theta}) = \frac{n\theta}{n-1}$$

for all $n \ge 2$. From this result, find the bias of $\hat{\theta}$ and show that $\hat{\theta}$ is biased but asymptotically unbiased.

Lecture 3

R Basics and a First Simulation

This is the first of the R sessions, which illustrate the theory on the computer and build up the R skills needed for the practical report. The lecture has two aims: to get R up and running on the computer, and to learn how to simulate a random sample. The second part of the lecture uses simulation to illustrate the ideas from lectures 1 and 2: an estimator is a random quantity, and the bias and spread of an estimator can be seen by simulation. Appendix B covers the same ground at a slower pace and in more detail. This is the written back-stop for the R sessions, and is the place to look up R commands later. The commands we use in the lecture are shown in a framed box, and the R output is shown below, with each line of the output marked by `##`.

3.1 First Steps in R

Appendix B explains how to install R or how to use R on the university computers. Either way, we will end up at the *console* where we type in commands and R prints the results.

The commands we need to use today are as follows: To construct a list of numbers, called a *vector*, we can use `c`. To store a list of numbers under a name, we use the assignment arrow `<-`. Finally, we can use the built-in functions `mean`, `median` and `var` to summarise a vector.

```
x <- c(4, 7, 1, 5, 3)
x
mean(x)
median(x)
var(x)
```

```
## [1] 4 7 1 5 3
## [1] 4
## [1] 4
## [1] 5
```

The `[1]` at the start of each output line is a position marker, not part of the answer: it gives the index of the first value printed on that line. The assignment itself prints nothing; typing `x` on its own shows the stored values.

The function `var` does not use a general convention, but computes the sample variance S^2 from lecture 2, using the divisor $n - 1$. To verify this, we can compute the formula manually, using the fact that arithmetic in R operates on the individual elements of a vector, and that `length` can be used to determine the number of elements.

```
sum((x - mean(x))^2) / (length(x) - 1)
```

```
## [1] 5
```

The answer coincides with the result of `var(x)` above, and we now know that R uses the unbiased estimator from lemma 2.3 for variances. Further details about vectors, indexing and arithmetic can be found in appendix B. The appendix also explains how to understand the help page of an unfamiliar command; we will not repeat this information here.

3.2 Simulating a Normal Sample

The theory from the last two lectures considers data generated from a model with known true parameter. In this section we will see that we can simulate such data on a computer. The command `rnorm` can be used to generate a random sample from a normal distribution. The arguments of the command are the sample size and the values for the `mean` and `sd`.

Warning. R uses the standard deviation as a parameter of the normal distribution, not the variance. Thus, if we write `sd = 2`, we mean $\sigma = 2$ and thus $\sigma^2 = 4$. A common mistake is to forget this and to get simulations which are mysteriously too spread out.

Since random draws are different every time, it is often a nuisance to check our work and in a report it would be a disaster if the random numbers changed every time the report was printed. The solution to this problem is the `set.seed` command: if we call this command with a fixed number, the random-number generator will be set to a fixed state and the same seed will always return the same random numbers.

```
set.seed(201234567)
rnorm(3)
set.seed(201234567)
rnorm(3)
```

```
## [1] -1.0333701 -0.7724437  0.1556372
## [1] -1.0333701 -0.7724437  0.1556372
```

Any fixed number can be used as the seed. In the practical report, it is a good idea to use a personal number, like your student ID, to seed the random number generator. This way, your report is exactly reproducible, while your dataset is still different from the datasets of your classmates.

Finally, we can produce a proper sample: $n = 100$ observations from the normal distribution with mean $\mu = 10$ and variance $\sigma^2 = 4$. The command `summary` can be used to print the standard descriptive statistics for the sample, and `hist` can be used to produce a histogram. Finally, `abline` can be used to

add a straight line to an existing plot (in this case, a dashed vertical line at the sample mean).

```
set.seed(201234567)
sample <- rnorm(100, mean = 10, sd = 2)
summary(sample)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   4.710   9.045  10.166  10.006  11.350  13.877
```

The summary already tells a plausible story: the sample mean 10.006 is close to the true mean 10, and the quartiles are roughly symmetric around the mean. To get a better picture of the sample, we consider the histogram in figure 3.1. We adjust the horizontal range using the `xlim` argument so that the axis can be compared to the axis in a second figure, below.

```
hist(sample, col = "grey85", main = "", xlab = "x", xlim = c(4, 16))
abline(v = mean(sample), lty = 2)
```

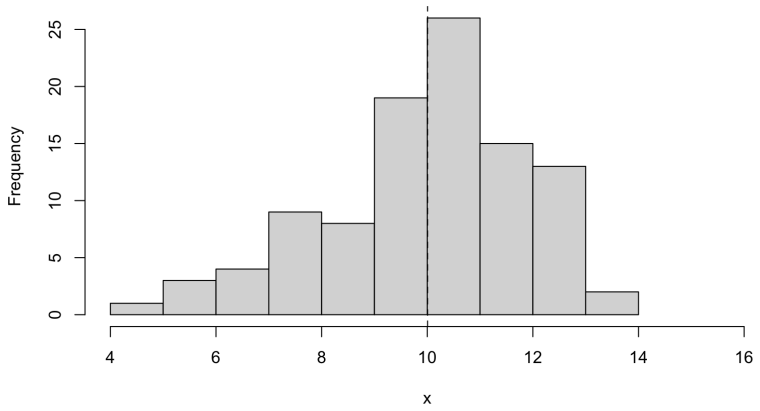


Figure 3.1: A simulated sample of $n = 100$ observations from a normal distribution with mean $\mu = 10$ and variance $\sigma^2 = 4$. The dashed line gives the sample mean.

3.3 The Sample Mean as a Random Quantity

In lecture 1 we have discussed the difference between an estimate, a single number computed from data, and an estimator, which is a random variable. We can use simulation to visualise this difference: every call to `rnorm` below generates a fresh sample of size $n = 25$ from the same model as before, and the sample mean of each sample is a fresh realisation of the estimator $\bar{X}$. Before you execute the following three lines of R, make a prediction: will the three printed numbers be equal? If not, how much do you expect the three numbers to be away from the true value 10?

```
mu <- 10
n  <- 25
set.seed(2026)
mean(rnorm(n, mean = mu, sd = 2))
mean(rnorm(n, mean = mu, sd = 2))
mean(rnorm(n, mean = mu, sd = 2))
```

```
## [1] 9.340722
## [1] 10.5942
## [1] 9.335904
```

Three samples, three different estimates, scattered around the true value 10. The scatter we see here is the sampling distribution, introduced in lecture 1. By repeating the experiment many times, we can map out this distribution. The `replicate` command can be used to repeat a command a given number of times, and to store the results in a vector. Loops and functions can be used to achieve the same effect (see appendix B). We repeat the experiment 10000 times, and compare the results to the predictions from lecture 2: From lemma 2.2 we know that the estimator $\bar{X}$ is unbiased, with standard deviation $\sigma/\sqrt{n} = 2/\sqrt{25} = 0.4$. From lemma 2.7 we know that the mean squared error of the estimator equals the variance $\sigma^2/n = 4/25 = 0.16$.

```
estimates <- replicate(10000, mean(rnorm(n, mean = mu, sd = 2)))
mean(estimates) - mu           # empirical bias
sd(estimates)                  # empirical spread
mean((estimates - mu)^2)       # empirical mean squared error
```

```
## [1] 0.004472127
## [1] 0.4006136
## [1] 0.1604952
```

The agreement is striking: the empirical bias is close to zero, the empirical spread 0.4006 matches the theoretical value 0.4, and the empirical mean squared error 0.1605 matches the theoretical value 0.16. This shows that the theory from lecture 2 is not an abstraction, but accurately predicts the output of the machine with two decimal digits of accuracy.

Finally we draw the histogram of all 10000 estimates, on the same horizontal axis as figure 3.1 so that the two plots can be compared directly. Before looking at the result, sketch the histogram you expect: where should it be centred, and how wide should it be compared with the data histogram in figure 3.1?

```
hist(estimates, breaks = 40, col = "grey85", main = "",
     xlab = "estimate", xlim = c(4, 16))
abline(v = mu, lty = 2)
```

While the data in figure 3.1 are spread out over most of the axis, the simulated estimates form a narrow spike around the truth. This is a consequence of the fact that, by averaging, the estimator has exchanged the spread $\sigma = 2$ of an individual observation for the much smaller spread 0.4 of the estimator. The picture of a histogram of an estimator which is close to the truth is the one we had in mind in lecture 2, when we discussed small bias and small mean squared

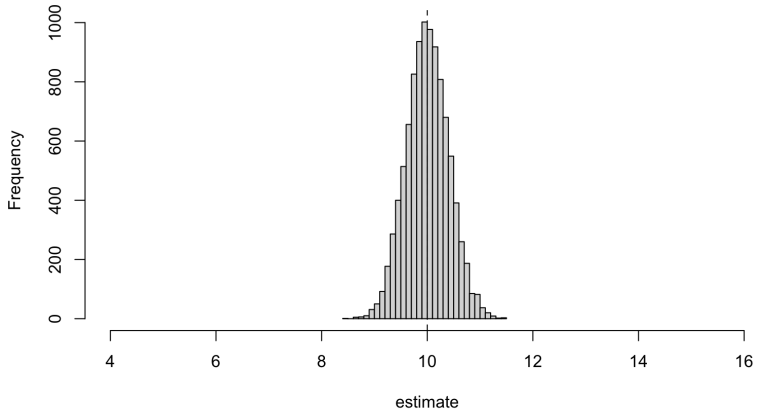


Figure 3.2: The sampling distribution of the sample mean $\bar{X}$ for $n = 25$: a histogram of 10000 simulated estimates, where the true mean $\mu = 10$ is marked by the dashed line. The axis is the same as in figure 3.1.

error. In the next R session, lecture 6, we will use the same simulation setup to compare different estimators. In lecture 9 we will watch the spike get narrower as n increases. This is an example of consistency.

Before the next session, please type in the commands of this session and try to understand the results.

Summary.

- Information about most R commands used in this module can be found in appendix B.
- The command `rnorm` can be used to simulate samples from a normal distribution. The standard deviation (!) of the distribution can be controlled using the `sd` argument of the command. The commands `mean`, `median`, `var`, `summary` and `hist` can be used to summarise and display the simulated samples.
- If `set.seed` is called with a fixed number, *e.g.* the student ID, then the output of all subsequent simulation commands is exactly reproducible.
- If an estimate is re-computed using many simulated samples, a histogram of the resulting sampling distribution can be plotted. The bias, spread and mean squared error of the sampling distribution can be computed and shown to agree with the theoretically predicted values from lecture 2.

Lecture 4

Likelihood and the Method of Moments

In the previous two lectures we have introduced the problem of estimation, but all the estimators we have considered so far were just guesses. In this lecture we introduce the *likelihood function*, the most important object in this module: the concepts of sufficiency (in lecture 8), Fisher information (in lecture 11), maximum likelihood (in lecture 5) and the Bayesian approach (in lecture 28) are all based on this function. We also discuss the *method of moments*, a simple method for constructing an estimator from a model. We will see the weaknesses of this approach which will lead to the introduction of the maximum likelihood estimator in the next lecture.

4.1 The Likelihood Function

Before we can introduce the likelihood, we need the distribution of the whole sample, not only of a single observation. Recall from lecture 1 that under a parametric model the observations $X_1, \dots, X_n$ are i.i.d. with density $f(x;\theta)$. Since the observations are independent, the density of the vector $(X_1, \dots, X_n)$ is the product of the densities of the individual observations, and since the observations are identically distributed, every factor uses the same function f. Thus, for each $\theta \in \Theta$, the *joint density* of the sample is given by

$$f(x_1, \dots, x_n; \theta) = \prod_{i=1}^{n} f(x_i; \theta).$$

In the discrete case the same formula holds for the probability weights, and $f(x_1, \dots, x_n; \theta)$ is then the probability $\mathbb{P}_\theta(X_1 = x_1, \dots, X_n = x_n)$ of observing the values $x_1, \dots, x_n$.

The joint density describes, for a fixed value of θ, which data are likely to be observed. The idea of this section is to consider the same expression as a function of the parameter θ instead of as a function of the data x_i.

Definition 4.1. Let $x_1, \dots, x_n$ be data from the model $\big(f(x;\theta) \,\big|\, \theta \in \Theta\big)$. Then the *likelihood function* is the function $L\colon \Theta \to [0, \infty)$, given by

$$L(\theta) = \prod_{i=1}^{n} f(x_i; \theta),$$

and the *log-likelihood* is $\ell(\theta) = \log L(\theta)$, given by

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log f(x_i; \theta).$$

While the formula for L is the same as the joint density of the sample, the interpretation of this function has changed: the joint density is a function of the data with the parameter fixed, but the likelihood is a function of the parameter with the data fixed. To emphasise that the data are used to compute the likelihood, we sometimes write $L(\theta; x_1, \ldots, x_n)$. For discrete data, $L(\theta)$ equals $\mathbb{P}_\theta(X_1 = x_1, \ldots, X_n = x_n)$, *i.e.* the probability that the model with parameter θ generates the observed data.

Example 4.2. Assume we have observed 7 heads in $n = 10$ tosses of a coin with unknown heads probability p. Using the Bernoulli weights from table 1.1, we find the likelihood to be

$$L(p) = \prod_{i=1}^{10} p^{x_i}(1-p)^{1-x_i} = p^7(1-p)^3, \qquad p \in (0, 1).$$

For a fair coin with $p = 1/2$ we have $L(0.5) = 0.5^{10} \approx 0.00098$ and for $p = 0.7$ we get $L(0.7) = 0.7^7 \cdot 0.3^3 \approx 0.00222$. Both values are very small, since any sequence of ten tosses is unlikely, but the ratio of these values is more informative: the observed data are about 2.3 times as probable under $p = 0.7$ as under $p = 0.5$.

Warning. The likelihood is not a probability density over θ: L does not integrate to one and also the parameter is unknown but not random. Thus, statements like "the probability that $\theta = 0.7$ is small" make no sense. We will learn more about this in lecture 28, where we will see how the Bayesian approach can be used to turn the parameter into a random variable.

In practice we almost always use the log-likelihood: the likelihood is a product of n factors, typically each less than one, so for realistic sample sizes it is astronomically small. In contrast, ℓ is a sum of moderate size and it is much easier to take derivatives of a sum than of a product. Since the logarithm is strictly increasing, the functions L and ℓ order the parameter values in the same way, so no information is lost when we use the logarithm.

Example 4.3. For an i.i.d. sample $x_1, \ldots, x_n$ from the exponential model with density $f(x; \theta) = \theta e^{-\theta x}$ for $x \geq 0$, the likelihood is

$$L(\theta) = \prod_{i=1}^{n} \theta e^{-\theta x_i} = \theta^n \exp\Big(-\theta \sum_{i=1}^{n} x_i\Big).$$

Thus, the log-likelihood is

$$\ell(\theta) = n \log \theta - \theta \sum_{i=1}^{n} x_i, \qquad \theta > 0.$$

The product in this expression has been replaced by a smooth function of θ, which uses only two summaries of the data, n and $\sum_i x_i$. We will learn more about this approach to summarising data in lecture 8.

4.2 The Score Function

If the log-likelihood function is differentiable, then the derivative is sometimes called the *score* function. In this section we only use the score function to find the maximum of the likelihood. (We will learn the deeper meaning of the score function, as the carrier of information in the sample, in lecture 11.)

Definition 4.4. Let $\ell(\theta)$ be the log-likelihood function and assume that it is differentiable at θ. Then the *score function* is given by

$$\ell'(\theta) = \frac{\mathrm{d}}{\mathrm{d}\theta}\,\ell(\theta) = \sum_{i=1}^{n} \frac{\partial}{\partial\theta} \log f(x_i;\theta).$$

The score is the sum of n terms, one for each observation, and if we assume that the data are random, these terms are i.i.d. This additive structure allows us to apply the law of large numbers and the central limit theorem to the likelihood function later in the module. Since the score is the derivative of the log-likelihood, a zero of the score is a good starting point to find the parameter value which gives the best explanation for the data.

Example 4.5. Continuing from example 4.3, the score for the exponential sample is given by

$$\ell'(\theta) = \frac{n}{\theta} - \sum_{i=1}^{n} x_i, \qquad \theta > 0.$$

The score is positive for $\theta < 1/\bar{x}$ and negative for $\theta > 1/\bar{x}$, where $\bar{x} = (1/n)\sum_i x_i$. Thus the log-likelihood increases up to $1/\bar{x}$ and then decreases, so $\theta = 1/\bar{x}$ is the best explanation for the data. The idea of using the maximiser of the likelihood to estimate parameters is the basis of maximum likelihood estimation, as discussed in lecture 5.

4.3 The Method of Moments

Before we consider how to maximise the likelihood, we will discuss a much simpler method, the oldest general purpose method for constructing estimators. This method is based on the fact that the model often gives the moments of the observations as functions of the model parameters θ, and that the data can be used to compute the empirical moments. The idea of the method is then to find an estimator such that the two sets of moments coincide.

Let

$$\mu_j(\theta) = \mathbb{E}_\theta(X_1^j), \qquad m_j = \frac{1}{n}\sum_{i=1}^{n} X_i^j$$

for $j = 1, 2, \ldots$. Then $\mu_j(\theta)$ is the j-th *population moment*, a deterministic function of θ, and m_j is the corresponding *sample moment*, a statistic. By the law of large numbers (proposition A.14 in appendix A), we have $m_j \xrightarrow{\mathrm{p}} \mu_j(\theta_0)$: for large sample size the sample moments are close to the population moments corresponding to the true parameter value.

Definition 4.6. Let $\theta = (\theta_1, \ldots, \theta_k)$ be a parameter with k components. Then the *method of moments estimator* $\hat{\theta}$ is found by solving the system of k equations

$$\mu_j(\hat{\theta}) = m_j, \qquad j = 1, \ldots, k,$$

for $\hat{\theta} \in \Theta$, if this system of equations has a unique solution.

The recipe uses as many moment equations as there are unknown parameter components, and the required moments can usually be found in tables like tables A.1 and A.2. We illustrate the method for three of our running examples.

Example 4.7. For the exponential model with rate θ, we have $\mu_1(\theta) = 1/\theta$ and the only moment equation is $1/\hat{\theta} = \bar{X}$. Thus, the method of moments estimator is given by

$$\hat{\theta} = \frac{1}{\bar{X}}.$$

Coincidentally, this is the same as the zero of the score from example 4.5.

Example 4.8. For the normal model with $\theta = (\mu, \sigma^2)$ the parameter has two components. Thus we need to match the first two moments, $\mu_1(\theta) = \mu$ and $\mu_2(\theta) = \sigma^2 + \mu^2$. We can solve these two moment equations (see exercise 4.3) to find

$$\hat{\mu} = \bar{X} \qquad \text{and} \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Note the divisor n: since we are matching moments without considering the bias, the method gives the biased plug-in variance estimator from lecture 2, instead of the unbiased sample variance S^2. In this example, from the mean squared error comparison in lecture 2, this is not a problem.

Example 4.9. Let $X_1, \ldots, X_n \sim \text{Gamma}(\alpha, \beta)$ be i.i.d. where the shape α and the rate β are unknown. Then $\theta = (\alpha, \beta)$. From table A.2 we know $\mathbb{E}_\theta(X_1) = \alpha/\beta$ and $\text{Var}_\theta(X_1) = \alpha/\beta^2$. Instead of matching μ_1 and μ_2 we can equivalently match the mean and the variance: if we write $V = m_2 - m_1^2 = (1/n)\sum_i (X_i - \bar{X})^2$ for the plug-in variance, we get the equations

$$\frac{\hat{\alpha}}{\hat{\beta}} = \bar{X} \qquad \text{and} \qquad \frac{\hat{\alpha}}{\hat{\beta}^2} = V.$$

Dividing the first equation by the second we find $\hat{\beta} = \bar{X}/V$ and thus

$$\hat{\alpha} = \frac{\bar{X}^2}{V} \qquad \text{and} \qquad \hat{\beta} = \frac{\bar{X}}{V}.$$

This example shows the method in a good light: for the gamma distribution the maximum likelihood equations from lecture 5 have no closed-form solution, whereas the method of moments finds explicit formulas in two lines.

We now weigh the merits of the method. On the credit side, it is quick, often gives explicit formulas where the likelihood does not, and is consistent under weak conditions: the sample moments converge by the law of large numbers, and if the solution of the moment equations depends continuously on the moments, the continuous mapping theorem (proposition A.17) carries the convergence over to the estimator.

The method has three main weaknesses. First, the estimates can be contradicted by the data: for the uniform model $X_1, \ldots, X_n \sim \text{Uniform}(0, \theta)$ we have $\mu_1(\theta) = \theta/2$ and thus $\hat{\theta} = 2\bar{X}$, but for the dataset $x_1 = 0.1$, $x_2 = 0.2$, $x_3 = 0.9$ this gives $\hat{\theta} = 2 \cdot 0.4 = 0.8$, an estimate where the observation $x_3 = 0.9$ could

never have occurred. This problem is immediately detected by the likelihood, since $L(0.8) = 0$, but the moment equations do not detect this problem. Secondly, the choice of moments is arbitrary: if we had used the second moment instead of the first, we would in general have obtained a different estimator. Finally, by taking only a few moments, much of the information in the data is lost. A method which makes use of all the information in the data should be more successful. This is the aim of the next few lectures.

Remark. In some cases, both approaches coincide: for the exponential model the moment estimator $1/\bar{X}$ is also the maximiser of the likelihood (from example 4.5), the same is true for the Poisson distribution (exercise 4.1) and for the normal distribution the two methods coincide (lecture 5). In contrast, for the uniform model considered above, the maximum likelihood estimator is $\max_i X_i$ instead of $2\bar{X}$ and has a very different mean squared error (as shown in exercises 2.2 and 4.4).

4.4 Identifiability

Both recipes of this lecture, and the whole estimation programme, assume that the parameter can be recovered from the distribution of the data: if two parameter values lead to the same distribution, no data can distinguish between them and no good estimator for θ can exist. This property of a model deserves a name.

Definition 4.10. A parametric model $(\mathbb{P}_\theta \mid \theta \in \Theta)$ is *identifiable*, if different parameter values lead to different distributions, *i.e.* if we have $\mathbb{P}_{\theta_1} \neq \mathbb{P}_{\theta_2}$ for all $\theta_1, \theta_2 \in \Theta$ with $\theta_1 \neq \theta_2$.

All the families of models listed in tables 1.1 and 1.2 are identifiable. Non-identifiability usually occurs in models which are over-parametrised, as the following example shows.

Example 4.11. Assume that we want to model a measurement which is subject to two additive effects. We can assume that $X_1, \ldots, X_n \sim N(\alpha + \beta, 1)$ are i.i.d. with parameter $\theta = (\alpha, \beta) \in \mathbb{R}^2$. Then the distribution of the data depends on θ only via the sum $\alpha + \beta$. Thus, the parameters $(1, 2)$ and $(0, 3)$ give rise to the same distribution and the model is not identifiable. Both recipes from this lecture fail in this case: the likelihood is constant on the lines $\alpha + \beta = \text{const}$ and the moment equations only contain the unknowns in the combination $\alpha + \beta$. To solve the problem we can either reparametrise the model using $\mu = \alpha + \beta$ or we can restrict the parameter space, for example by setting $\beta = 0$.

Identifiability is a question about the parametrisation, not about the data or the sample size. A more subtle version of the problem is the case of the mixture models from lecture 19, where the models are only identifiable up to relabelling the mixture components. For the rest of the module, we will assume that the models are identifiable, unless we explicitly state otherwise.

Summary.

- The likelihood $L(\theta) = \prod_i f(x_i;\theta)$ is the joint density of the observed data, as a function of θ with the data fixed, and is not a probability distribution over θ.
- The score $\ell'(\theta)$, the derivative of the log-likelihood function $\ell(\theta) = \sum_i \log f(x_i;\theta)$, finds the parameter value which best explains the data.
- The method of moments solves the equations $\mu_j(\hat{\theta}) = m_j$ for $j = 1,\dots,k$. The method is fast, explicit and consistent, but it can return impossible estimates and does not make good use of the data. The maximum likelihood method from lecture 5 is better.
- A model is identifiable, if different parameter values lead to different distributions. If the model is not identifiable, no estimator can work and we need to change the parametrisation.

Exercise 4.1. Let $X_1,\dots,X_n$ be an i.i.d. sample from a Poisson distribution with parameter $\theta > 0$ and let $x_1,\dots,x_n$ be the observed data.

1. Write down the likelihood $L(\theta)$ and show that the log-likelihood is $\ell(\theta) = \log\theta \sum_i x_i - n\theta - \sum_i \log(x_i!)$.
2. Compute the score $\ell'(\theta)$ and find the value of θ where the score equals zero.
3. Compute the method of moments estimator for θ and compare your result to your answer in the previous part.

Exercise 4.2. Let $X_1,\dots,X_n$ be an i.i.d. sample from the geometric distribution with parameter $p \in (0,1)$, *i.e.* from the distribution with $f(x;p) = (1-p)^{x-1}p$ for $x \in \mathbb{N}$ and $\mathbb{E}_p(X_1) = 1/p$. (There is some disagreement among authors whether the geometric distribution counts the number of trials until the first success, inclusive of this trial, starting at 1 as in the present exercise, or the number of failures before the first success, starting at 0. In this module we always count trials.)

1. Using the method of moments, determine an estimator $\hat{p}$ for p.
2. Show that the value of $\hat{p}$ always lies in the interval $(0,1]$, *i.e.* that the estimate can never be contradicted by the data (this is in contrast to the example of the uniform distribution in the text).

Exercise 4.3. Let $X_1,\dots,X_n \sim N(\mu,\sigma^2)$ i.i.d. with parameter $\theta = (\mu,\sigma^2)$, as in example 4.8.

1. Using the relation $\mathrm{Var}(X_1) = \mathbb{E}(X_1^2) - \mathbb{E}(X_1)^2$, show that the first two population moments are $\mu_1(\theta) = \mu$ and $\mu_2(\theta) = \sigma^2 + \mu^2$.
2. Write down the two moment equations and solve them to verify the estimators $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = (1/n)\sum_i (X_i - \bar{X})^2$ from example 4.8.

Exercise 4.4. Let $X_1,\dots,X_n \sim \mathrm{Uniform}(0,\theta)$ i.i.d. and let $\hat{\theta} = 2\bar{X}$ be the method of moments estimator as given in the text.

1. Show that $\hat{\theta}$ is unbiased and that $\mathrm{MSE}(\hat{\theta}) = \theta^2/(3n)$.
2. In exercise 2.2 we have seen that $\mathrm{MSE}(\max_i X_i) = 2\theta^2/((n+1)(n+2))$ for the biased estimator $\max_i X_i$. Show that $\max_i X_i$ has strictly smaller mean squared error for all $n \geq 3$ and comment on your result.

Exercise 4.5. Consider the model $X_1, \ldots, X_n \sim N(0, \theta^2)$ i.i.d. for the parameter $\theta \in \Theta = \mathbb{R} \setminus \{0\}$.

1. Show that the model is not identifiable.
2. Find a smaller parameter space $\Theta' \subseteq \Theta$, where the model is identifiable.
3. On your space Θ', determine the method of moments estimator for θ, using the second moment.

Lecture 5

Maximum Likelihood Estimation

In the previous lecture we have seen that for the exponential model, the parameter value which maximises the likelihood is the one which 'best' explains the data. In this lecture we will make this idea into the main estimation method of the module: we define the maximum likelihood estimator, practise computing it for the families from tables 1.1 and 1.2, and will see problems where more thought is needed. We will see that the estimator is not unbiased in general, but is invariant under reparametrisation; the deeper justification of the method will be given in lectures 13, 16 and 17.

5.1 The Maximum Likelihood Estimator

Example 4.5 already contains the idea: for the exponential sample the log-likelihood increases up to $\theta = 1/\bar{x}$ and then decreases, so this value fits the data best. We now formalise this principle.

Definition 5.1. A *maximum likelihood estimator* (MLE) for θ is an estimator $\hat{\theta} = \hat{\theta}(X_1, \ldots, X_n)$, which maximises the likelihood function over the parameter space, *i.e.* which satisfies

$$L(\hat{\theta}) \geq L(\theta) \qquad \text{for all } \theta \in \Theta.$$

Since the logarithm is strictly increasing, $\hat{\theta}$ maximises L if and only if it maximises the log-likelihood ℓ and for the reasons given in lecture 4 we mostly work with ℓ. The definition is careful to avoid the problem of a maximum not existing or not being unique. For example, the likelihood could increase as the edge of the parameter space is approached, so that no maximum exists. Similarly, the maximum may not be unique. In such cases, any maximiser can be used. For the standard families of models and typical data, neither of these problems will occur and it will be safe to speak of "the" MLE.

If the set Θ is an interval and the log-likelihood function is differentiable, then calculus can be used to find the MLE: at a maximum inside the interval Θ, the derivative of ℓ equals zero and thus the MLE satisfies the *likelihood equation*

$$\ell'(\theta) = 0,$$

i.e. the MLE is a zero of the score function from definition 4.4. The general procedure is to find the log-likelihood function, to solve the likelihood equation and to verify that the solution is a maximum, for example by checking that $\ell''(\theta) < 0$ or by checking that the score changes from positive to negative.

Warning. The solution of the likelihood equation only gives *interior* critical points. These could be minima or inflection points and the checking step is required to avoid wrong answers. Also, the above procedure assumes that the maximum is interior and that ℓ is differentiable. We will see that these assumptions can be violated.

5.2 Worked Examples

We start our analysis by considering the MLE for commonly used families of probability distributions. For each of them we find the MLE using the recipe from the previous section, and then we check whether the resulting estimator is unbiased.

The Exponential Distribution

The exponential distribution is the simplest of the three cases considered here: the likelihood equation has a single root and the moments of the MLE can be computed exactly.

Example 5.2. From examples 4.3 and 4.5 we know that the score function of an exponential sample $x_1, \dots, x_n$ is given by

$$\ell'(\theta) = \frac{n}{\theta} - \sum_{i=1}^{n} x_i$$

for all $\theta > 0$. Setting $\ell'(\theta) = 0$ gives $n/\theta = \sum_i x_i$, and thus the likelihood equation has the unique solution

$$\theta = \frac{n}{\sum_{i=1}^{n} x_i} = \frac{1}{\bar{x}}.$$

To check that this root is a maximum, we differentiate once more and find

$$\ell''(\theta) = -\frac{n}{\theta^2} < 0$$

for all $\theta > 0$. Thus, the function ℓ is strictly concave and the root $1/\bar{x}$ is the only global maximum. We conclude that the MLE for the rate of the exponential distribution is

$$\hat{\theta} = \frac{1}{\bar{X}}.$$

This coincides with the method of moments estimator from example 4.7.

To find the bias of this estimator, let $T = X_1 + \cdots + X_n$, so that $\hat{\theta} = n/T$. In exercise 2.3 we found, using the gamma density of T, that $\mathbb{E}_\theta(\hat{\theta}) = n\theta/(n-1)$ for $n \geq 2$, while for $n = 1$ the expectation is infinite. Thus, for $n \geq 2$, the MLE is biased with bias

$$\operatorname{bias}(\hat{\theta}) = \mathbb{E}_\theta(\hat{\theta}) - \theta = \frac{n\theta}{n-1} - \theta = \frac{\theta}{n-1}.$$

Thus, the MLE for the rate is biased but asymptotically unbiased in the sense of definition 2.4. Multiplying $\hat{\theta}$ by $(n-1)/n$ removes the bias: the estimator $\tilde{\theta} = (n-1)\hat{\theta}/n = (n-1)/T$ satisfies $\mathbb{E}_\theta(\tilde{\theta}) = \theta$. The same computation gives every inverse moment of T, since integrating t^{-k} against the gamma density of T leaves another gamma density, and we find

$$\mathbb{E}_\theta(T^{-k}) = \theta^k \frac{\Gamma(n-k)}{\Gamma(n)} = \frac{\theta^k}{(n-1)(n-2)\cdots(n-k)}$$

for $k < n$. Exercise 5.3 uses the case $k = 2$ to find the variance of $\tilde{\theta}$.

The Bernoulli Distribution

For the Bernoulli distribution the recipe works whenever the data contain both successes and failures. The remaining two cases need separate treatment.

Example 5.3. Let $X_1, \ldots, X_n$ be an i.i.d. Bernoulli sample with success probability θ, and let $s = \sum_i x_i$ be the observed number of successes. As in example 4.2 we have the likelihood $L(\theta) = \theta^s(1-\theta)^{n-s}$ and thus

$$\ell(\theta) = s \log\theta + (n-s)\log(1-\theta)$$

for all $\theta \in (0,1)$. Assume $0 < s < n$. Then the likelihood equation is

$$0 = \ell'(\theta) = \frac{s}{\theta} - \frac{n-s}{1-\theta}.$$

This equation implies $s(1-\theta) = (n-s)\theta$ and the only solution to this equation is $\theta = s/n$. Since $\ell''(\theta) = -s/\theta^2 - (n-s)/(1-\theta)^2 < 0$ everywhere, this root is the global maximum and the MLE for the proportion of successes is

$$\hat{\theta} = \frac{s}{n} = \bar{X}.$$

Since $\mathbb{E}_\theta(\bar{X}) = \theta$, the MLE for the Bernoulli parameter is unbiased.

Remark. The two cases $s = 0$ and $s = n$ are special. For $s = 0$ the function $\ell(\theta) = n\log(1-\theta)$ is strictly decreasing, and for $s = n$ the function $\ell(\theta) = n\log\theta$ is strictly increasing. In both cases the supremum is approached at the respective boundary point. We cannot find these points by setting $\ell'(\theta) = 0$. Additionally, in this module we use the open interval $\Theta = (0,1)$ as the parameter space, so for us no $\theta \in \Theta$ maximises the log-likelihood and if we are pedantic the MLE does not exist.

The Normal Distribution

The normal distribution with both parameters unknown has a two-dimensional parameter vector, and thus the likelihood equation becomes a system of two equations.

Example 5.4. Consider $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, where both parameters are unknown. Then we have $\theta = (\mu, \sigma^2)$. Taking logarithms we get

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2.$$

To find the maximum, we set both partial derivatives equal to zero. The partial derivatives are

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^{n} (x_i - \mu)$$

and

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^{n} (x_i - \mu)^2.$$

The first derivative is zero only for $\mu = \bar{x}$. Substituting this value into the second derivative and setting it to zero gives $\sigma^2 = (1/n) \sum_i (x_i - \bar{x})^2$.

To check that this solution is the maximum, we note that for every fixed σ^2 the sum of squares $\sum_i (x_i - \mu)^2$ is smallest for $\mu = \bar{x}$, and thus $\ell(\mu, \sigma^2) \leq \ell(\bar{x}, \sigma^2)$. At $\mu = \bar{x}$ the derivative with respect to σ^2 can be written as $\big(\sum_i (x_i - \bar{x})^2 - n\sigma^2\big)/(2\sigma^4)$, which is positive to the left of the solution and negative to the right. Thus, the MLE is the pair

$$\hat{\mu} = \bar{X} \qquad \text{and} \qquad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (X_i - \bar{X})^2.$$

The MLE $\hat{\mu} = \bar{X}$ for the mean is unbiased. The MLE $\hat{\sigma}^2$ for the variance is the estimator from example 2.5, where we found the bias $-\sigma^2/n$. Thus, $\hat{\sigma}^2$ is biased but asymptotically unbiased.

In all three examples considered in this section, the MLE is either unbiased or asymptotically unbiased. Note that there are other examples, where this property does not hold. The behaviour of the MLE for large samples in general is the subject of lectures 16 and 17.

5.3 When the Recipe Fails

The recipe for finding the MLE can fail in three different ways: the maximum may sit at a point where the log-likelihood is not differentiable, the likelihood equation may have several roots, and the likelihood equation may have no closed-form solution. We consider these three cases in turn.

A Maximum at the Boundary

Both assumptions in the recipe, *i.e.* an interior maximum and differentiable log-likelihood, break down for the uniform model. In this case we have to return to definition 5.1 and maximise the expression directly.

Example 5.5. Let $X_1, \ldots, X_n \sim \text{Uniform}(0, \theta)$ be i.i.d., as in example 1.2. From the joint density found in exercise 1.3, the likelihood is

$$L(\theta) = \begin{cases} \theta^{-n} & \text{if } \theta \geq \max_i x_i, \\ 0 & \text{otherwise.} \end{cases}$$

Calculus is useless here: on the region where L is positive its derivative $-n\theta^{-n-1}$ is negative everywhere, so the likelihood equation has no solution, and at $\theta = \max_i x_i$ the function is not even continuous. But maximising directly is easy:

the likelihood is zero for $\theta < \max_i x_i$ and strictly decreasing for $\theta \geq \max_i x_i$, and thus largest at the left edge of the positive region. The MLE is the sample maximum,

$$\hat{\theta} = \max_i X_i.$$

This identifies the estimator we have been tracking since exercise 1.3, which by exercise 4.4 beats the method of moments estimator $2\bar{X}$ in mean squared error for every $n \geq 3$. The likelihood also rules out impossible parameter values automatically: for the dataset $x_1 = 0.1$, $x_2 = 0.2$, $x_3 = 0.9$ of lecture 4, any $\theta < 0.9$ has likelihood zero, so the contradiction the method of moments overlooked cannot arise.

Several Roots

The uniform model fails the recipe in an obvious way. A failure which is harder to spot is a likelihood equation with several roots. Consider the t location model with $\nu \geq 1$ degrees of freedom, where ν is fixed and known and the density is given by

$$f(x;\theta) = c_\nu \Big(1 + \frac{(x-\theta)^2}{\nu}\Big)^{-(\nu+1)/2}, \qquad x \in \mathbb{R},$$

with location $\theta \in \mathbb{R}$ and a normalising constant c_ν free of θ. The log-likelihood is differentiable everywhere, so the recipe seems to apply, but after clearing denominators the likelihood equation

$$\ell'(\theta) = (\nu+1) \sum_{i=1}^{n} \frac{x_i - \theta}{\nu + (x_i - \theta)^2} = 0$$

turns into a polynomial equation of degree $2n-1$ in θ and for $n \geq 2$ this equation can have several roots. Exercise 5.4 shows that for $n = 2$ there are three roots when the observations are more than $2\sqrt{\nu}$ apart: the mid-point of the interval between the observations, which is then a local *minimum* of the likelihood, and two global maxima of equal likelihood, one close to each of the observations. Thus, the model prefers to fit one data point and to consider the other as an outlier, and in this case the data cannot choose between the two maxima and the MLE will not be unique. The exercise also shows that a root of the likelihood equation can be a minimum and that the checking step in the recipe is not optional.

No Closed-Form Solution

For larger samples the likelihood equation of the t location model cannot be solved in closed form, and similarly the likelihood equations for many other models, *e.g.* the two-parameter gamma distribution from example 4.9, cannot be solved in closed form. One solution to this problem is to maximise ℓ numerically. We will discuss this approach, together with its pitfalls, in lecture 18.

5.4 Invariance Under Reparametrisation

Often the quantity we want to estimate is not the parameter itself, but a function of the parameter, *e.g.* a probability, a mean or a median. The following propo-

sition shows that under maximum likelihood such quantities transform together with the parameter.

Proposition 5.6 (Invariance of the MLE)**.** *Let $g\colon \Theta \to \Psi$ be a bijection and let $\hat{\theta}$ be a maximum likelihood estimator for θ. Then $g(\hat{\theta})$ is a maximum likelihood estimator for the transformed parameter $\psi = g(\theta)$.*

The idea of the proof is that relabelling the parameter does not change the family of distributions: in terms of ψ the likelihood is $\tilde{L}(\psi) = L(g^{-1}(\psi))$, so maximising $\tilde{L}$ over Ψ is the same maximisation as before. Writing this down is exercise 5.5.

Remark. If g is not a bijection, then the likelihood of ψ is defined to be the maximal likelihood of the values of θ which map to this value. While the statement $\hat{\psi} = g(\hat{\theta})$ is still true, we will not require the general case in this module.

As an application, we find the MLE for the median waiting time in the exponential model without any further maximisation.

Example 5.7. Let $X_1, \ldots, X_n$ be i.i.d. exponential with rate $\theta > 0$. The *median* waiting time is the value m such that $\mathbb{P}_\theta(X_1 \leq m) = 1/2$. Since $\mathbb{P}_\theta(X_1 \leq m) = 1 - e^{-\theta m}$, this condition is equivalent to $e^{-\theta m} = 1/2$, and taking logarithms gives

$$m = g(\theta) = \frac{\log 2}{\theta}.$$

The map g is strictly decreasing on $(0, \infty)$, with $g(\theta) \to \infty$ as $\theta \to 0$ and $g(\theta) \to 0$ as $\theta \to \infty$, and thus it maps $(0, \infty)$ bijectively onto itself. Since the MLE for the rate is $\hat{\theta} = 1/\bar{X}$ by example 5.2, proposition 5.6 shows that the MLE for the median is

$$\hat{m} = g(\hat{\theta}) = \frac{\log 2}{\hat{\theta}} = (\log 2)\, \bar{X}.$$

Exercise 5.8 shows that this example contains a surprise: $\hat{m}$ is unbiased, despite the MLE for the rate being biased. In general, unbiasedness is lost under nonlinear reparametrisation, and thus there is no analogue of proposition 5.6 for unbiasedness.

Since we have seen in this lecture that the MLE can be biased, non-unique and even non-existent, we need to do more work to understand the method. The arguments for the validity of the MLE can be found in lectures 13, 16 and 17.

Summary.

- The maximum likelihood estimator $\hat{\theta}$ is the parameter value which maximises the likelihood (or log-likelihood) of the data.
- For smooth problems, where the maximum is an interior point of the parameter space, the MLE can be found by solving the likelihood equation $\ell'(\theta) = 0$ and verifying that the root is a maximum.
- If this recipe fails, the definition can still be used: for the uniform distribution the maximum sits at the boundary, at $\hat{\theta} = \max_i X_i$, and for the t location model with two observations far apart the MLE is not unique.
- The MLE can be biased.
- The MLE is invariant under reparametrisation: the MLE for $g(\theta)$ is $g(\hat{\theta})$. Unbiasedness is not invariant.

Exercise 5.1. Let $X_1, \ldots, X_n$ be an i.i.d. sample from the geometric distribution with parameter $p \in (0, 1)$, *i.e.* $f(x; p) = (1-p)^{x-1}p$ for $x \in \mathbb{N}$ as in exercise 4.2.

1. Show that the log-likelihood function is given by $\ell(p) = n \log p + \left(\sum_i x_i - n\right) \log(1-p)$.
2. Solve the likelihood equation and show that your solution is a maximum. Deduce that the MLE is $\hat{p} = 1/\bar{X}$.
3. Discuss your results in the light of the method of moments estimator from exercise 4.2.

Exercise 5.2. Let $X_1, \ldots, X_n$ be i.i.d. with density $f(x; \theta) = \theta^{-2} x\, e^{-x/\theta}$ for $x \geq 0$, where $\theta > 0$. This is the gamma distribution with known shape 2 and scale θ, *i.e.* Gamma$(2, 1/\theta)$ in the rate parametrisation from table A.2.

1. Find the MLE $\hat{\theta}$ for θ.
2. Show that your estimator $\hat{\theta}$ is unbiased.
3. Determine MSE$(\hat{\theta})$ and use this to show that $\hat{\theta}$ is consistent.

Exercise 5.3. Let $X_1, \ldots, X_n$ be i.i.d. exponential with rate θ, let $\hat{\theta} = 1/\bar{X}$ be the MLE and assume $n \geq 3$.

1. Using the gamma density of $T = X_1 + \cdots + X_n$ as in exercise 2.3, show that $\mathbb{E}_\theta(\hat{\theta}^2) = n^2\theta^2/((n-1)(n-2))$.
2. Deduce that
$$\mathrm{Var}_\theta(\hat{\theta}) = \frac{n^2\theta^2}{(n-1)^2(n-2)}$$
and that the unbiased estimator $\tilde{\theta} = (n-1)\hat{\theta}/n$ from the text has variance $\mathrm{Var}_\theta(\tilde{\theta}) = \theta^2/(n-2)$.

Exercise 5.4. Let $x_1 > x_2$ be any two observations from the t location model given in section 5.3. Let $m = (x_1 + x_2)/2$ be the mid-point and $d = (x_1 - x_2)/2$ be half the distance between the two observations.

1. Let $a = x_1 - \theta$ and $b = x_2 - \theta$. Show that the likelihood equation $\ell'(\theta) = 0$ is equivalent to $(a + b)(\nu + ab) = 0$.

2. Show that for $\theta = m + t$ the log-likelihood function is $\ell(m + t) = C - \frac{\nu+1}{2} \log P(t)$, where C does not depend on t and
$$P(t) = (\nu + d^2)^2 + 2(\nu - d^2)\, t^2 + t^4.$$

3. Show that for $d^2 < \nu$ the mid-point m is the unique MLE, but for $d^2 > \nu$ the likelihood equation has three roots m, $m \pm \sqrt{d^2 - \nu}$, where m is a local minimum of ℓ and the other two are maximum likelihood estimates with equal likelihood.

Exercise 5.5. Prove the statement of proposition 5.6: Let $g\colon \Theta \to \Psi$ be a bijection and let the likelihood in the new parametrisation be given by $\tilde{L}(\psi) = L\big(g^{-1}(\psi)\big)$. Show that $g(\hat{\theta})$ is the maximum of $\tilde{L}$ over Ψ.

Exercise 5.6. Let X be an observation from the exponential distribution with rate $\theta > 0$, *i.e.* $\mathbb{P}_\theta(X > x) = e^{-\theta x}$ for $x \geq 0$. The following properties of the exponential distribution will be needed later in the module.

1. Show that the exponential distribution has no memory, *i.e.* that for all $s, t \geq 0$ we have
$$\mathbb{P}_\theta(X > s + t \,|\, X > s) = \mathbb{P}_\theta(X > t).$$

2. Show that for all $c \geq 0$ we have
$$\mathbb{E}_\theta(X \,|\, X > c) = c + \frac{1}{\theta}.$$
Comment upon your result in the light of the first part of the question.

Exercise 5.7. Let $X_1, \ldots, X_n$ be an i.i.d. sample from the Poisson distribution with parameter $\theta > 0$, and assume that $\sum_i x_i \geq 1$.

1. In exercise 4.1 we have seen that the score equals zero at $\bar{x}$. Show that $\bar{X}$ is indeed the MLE for θ.

2. Determine the MLE for the probability $p_0 = \mathbb{P}_\theta(X_1 = 0)$ of observing zero.

3. Using the relation $\mathbb{E}_\theta\big(z^{X_1}\big) = e^{\theta(z-1)}$ for $z > 0$, show that your estimator from the previous part is biased, but asymptotically unbiased. Comment on your result in the light of the fact that $\bar{X}$ is unbiased.

Exercise 5.8. Let $\hat{m} = (\log 2)\, \bar{X}$ be the MLE for the median waiting time $m = (\log 2)/\theta$ of the exponential distribution from example 5.7. Show that $\hat{m}$ is an unbiased estimator for m, and comment on why this does not contradict the bias of the MLE $\hat{\theta} = 1/\bar{X}$ for the rate.

Lecture 6

Estimators by Simulation

In the previous two lectures we introduced two estimators for the uniform model: the method of moments estimator $2\bar{X}$ in lecture 4 and the maximum likelihood estimator $\max_i X_i$ in lecture 5. In exercise 4.4 you used algebra to determine which of these two estimators is better. Today we will use the Monte Carlo methods from lecture 3 to answer the same question, and we will learn how to write our own function in R and how to repeat this function in a loop. (Appendix B shows both approaches, but at a slower pace.) The approach we follow here is a good one to get into: before you do any simulation, try to predict what the result will be. Only then can the output of your simulation either confirm or contradict your expectation.

6.1 Functions and Loops

The model we consider today is $X_1, \ldots, X_n \sim \text{Uniform}(0, \theta)$, and to study the estimators for this model using simulation, we will need to repeat the same experiment thousands of times: we have to sample from the model, compute the estimate, and store the result. While we could just type in the commands to do this over and over again in the console, as we did in lecture 3, this approach is not scalable. Instead, we will encapsulate the experiment into a *function* and then use R to repeat this function many times.

To generate samples from the uniform distribution, we can use the function `runif` and to specify the boundaries of the distribution we use `min` and `max`, as explained in appendix B. To create a function, we use the keyword `function`, followed by a list of arguments and then the body of the function enclosed in braces. The return value of a function is the value of the last line of the body. The following function performs a complete experiment for the method of moments estimator: it simulates a fresh sample of size `n` from the model with true parameter `theta`, and then returns the computed value $2\bar{x}$ of the estimate.

```
est_mom <- function(n, theta) {
  sample <- runif(n, min = 0, max = theta)
  2 * mean(sample)
}
set.seed(2026)
est_mom(10, 1)
est_mom(10, 1)
```

```
## [1] 0.8844232
## [1] 0.6339234
```

Defining the function prints nothing; only the two calls do. Each call draws its own sample, so two calls give two different estimates, both scattered around the true value $\theta = 1$: once more the estimator is a random quantity, as seen in lecture 3 for the sample mean.

To repeat the experiment systematically, we can use a `for`-loop: we set aside a vector to store the results using `numeric(5)`, then use a counter `i` which takes the values 1 to 5 (generated by `seq_len(5)`), and then store the estimate for each iteration of the loop in the results vector. (Appendix B explains why we use `seq_len` instead of the shorter notation `1:5`.)

```
set.seed(2026)
estimates <- numeric(5)
for (i in seq_len(5)) {
  estimates[i] <- est_mom(10, 1)
}
estimates
```

```
## [1] 0.8844232 0.6339234 0.5469486 0.9469991 0.8432259
```

Since we reset the seed, the first two values are exactly the two estimates computed above: the loop repeats the same call and keeps the answers. For this everyday pattern of repeating an expression and collecting the results, R provides the shortcut `replicate` which can be used to achieve the same effect in one line.

```
set.seed(2026)
replicate(5, est_mom(10, 1))
```

```
## [1] 0.8844232 0.6339234 0.5469486 0.9469991 0.8432259
```

The five values agree with the loop, as they must. We use `replicate` for the rest of this lecture; the explicit loop returns in lecture 9, where the sample size changes from one run to the next.

6.2 Empirical Bias and Mean Squared Error

With `est_mom` in place we can now use the Monte Carlo approach from lecture 3: we generate many estimates and then compute summary statistics. If $\hat{\theta}^{(k)}$ is the estimate using the k-th of N simulated samples, then by the law of large numbers (proposition A.14), the average of the deviations $\hat{\theta}^{(k)} - \theta$ is approximately equal to the bias $\mathbb{E}_\theta(\hat{\theta}) - \theta$ from lecture 2, and the average of the squared deviations is approximately equal to the mean squared error $\mathbb{E}_\theta((\hat{\theta} - \theta)^2)$. In R, both summary statistics can be computed in a single line: the empirical bias is given by `mean(estimates) - theta` and the empirical mean squared error is given by `mean((estimates - theta)^2)`.

We now have all the ingredients for a complete Monte Carlo study: a function to perform the experiment, `replicate` to repeat it and two one-liners for the summaries. The box below shows such a study in full, but with the lines of the

R script intentionally shuffled. Before you continue, arrange the lines of this box in the correct order to form an R script which performs a Monte Carlo study with $\theta = 1$, $n = 10$ and 10000 repetitions.

```
estimates <- replicate(10000, est_mom(n, theta))
mean((estimates - theta)^2)    # empirical MSE
theta <- 1
n     <- 10
est_mom <- function(n, theta) {
  sample <- runif(n, min = 0, max = theta)
  2 * mean(sample)
}
mean(estimates) - theta        # empirical bias
set.seed(2026)
```

The rule is that all names must be defined before they are used: the function and the values for `theta` and `n` must come before the `replicate` line, the seed must be set before any random draws are made, and the two summaries need the finished vector `estimates` to be computed, so they come last. There is more than one order which satisfies these conditions, so any of these orders is correct. The reconstructed study is shown below; for comparison the theoretical mean squared error is also shown. Before we run the program, we make a final prediction: from exercise 4.4 we know that $2\bar{X}$ is unbiased and has $\text{MSE}(2\bar{X}) = \theta^2/(3n)$, where $1/30 \approx 0.0333$, so should the two printed summaries be exactly 0 and 0.0333, or just close to these values? How close?

```
est_mom <- function(n, theta) {
  sample <- runif(n, min = 0, max = theta)
  2 * mean(sample)
}
theta <- 1
n     <- 10
set.seed(2026)
estimates <- replicate(10000, est_mom(n, theta))

mean(estimates) - theta        # empirical bias
mean((estimates - theta)^2)    # empirical MSE
theta^2 / (3 * n)              # the theoretical value
```

```
## [1] 0.00226647
## [1] 0.03303851
## [1] 0.03333333
```

The numbers are close to the theory, but not identical, and this is expected: the empirical bias and mean squared error are averages over 10000 random repetitions, so are estimates of the true values, but not identical to the true values. Changing the seed of the random number generator would result in slightly different results. Similarly, if the number of repetitions was increased, the results would get closer to the theoretical values. So, agreement to two decimal places is a good result, and an empirical bias of exactly zero would be a cause for concern (not for celebration).

6.3 Comparing Two Estimators

The real use of the Monte Carlo method is to compare estimators in models where the algebra runs out. Before trusting it there, we should watch it succeed on a contest whose outcome we know: by exercises 2.2 and 4.4, the maximum likelihood estimator $\max_i X_i$ of example 5.5 is biased but has smaller mean squared error than the unbiased estimator $2\bar{X}$ for all $n \geq 3$. If the simulation reproduces these known answers, we may trust the same recipe in the practical report and in lecture 18, where no formula is available.

We can write the maximum likelihood estimator as a function, in analogy to `est_mom`. To allow for a fair comparison, we use the same seed `2026` as we used above, to ensure that both estimators use the same 10000 samples. Before you try the code, what should the two printed numbers be close to? From exercise 2.2 we know that the bias of the maximum likelihood estimator is $-\theta/(n+1) = -1/11$ and the mean squared error is $2\theta^2/((n+1)(n+2)) = 2/132 \approx 0.0152$.

```
est_mle <- function(n, theta) {
  sample <- runif(n, min = 0, max = theta)
  max(sample)
}
set.seed(2026)
mle <- replicate(10000, est_mle(n, theta))

mean(mle) - theta              # empirical bias
mean((mle - theta)^2)          # empirical MSE
```

```
## [1] -0.08960997
## [1] 0.01479181
```

Both predictions are correct: the empirical bias -0.0896 is close to the theoretical value $-1/11 \approx -0.0909$ and the empirical mean squared error 0.0148 is close to the theoretical value 0.0152, which is less than half of the 0.0330 for $2\bar{X}$ shown above. Thus, in this example, the biased estimator performs better than the unbiased estimator, as shown in exercise 4.4.

The histograms make the comparison visible, and this is the time for the main prediction of the day. Before you run the code, sketch the two histograms you expect, using the theory above for the centres and the widths (the standard deviation of $2\bar{X}$ is $\sqrt{1/30} \approx 0.18$). Which of the two histograms will be more spread out? Where will the two histograms be centred? Will the second histogram be bell-shaped like the first one? The command `par(mfrow = c(2, 1))` can be used to arrange the following two plots as a column, and by using `xlim` for both plots we keep the axes directly comparable, as in lecture 3.

```
par(mfrow = c(2, 1))
hist(estimates, breaks = 40, col = "grey85", main = "method of moments",
     xlab = "estimate", xlim = c(0.4, 1.6))
abline(v = theta, lty = 2, lwd = 3)
hist(mle, breaks = 40, col = "grey85", main = "maximum likelihood",
     xlab = "estimate", xlim = c(0.4, 1.6))
abline(v = theta, lty = 2, lwd = 3)
```

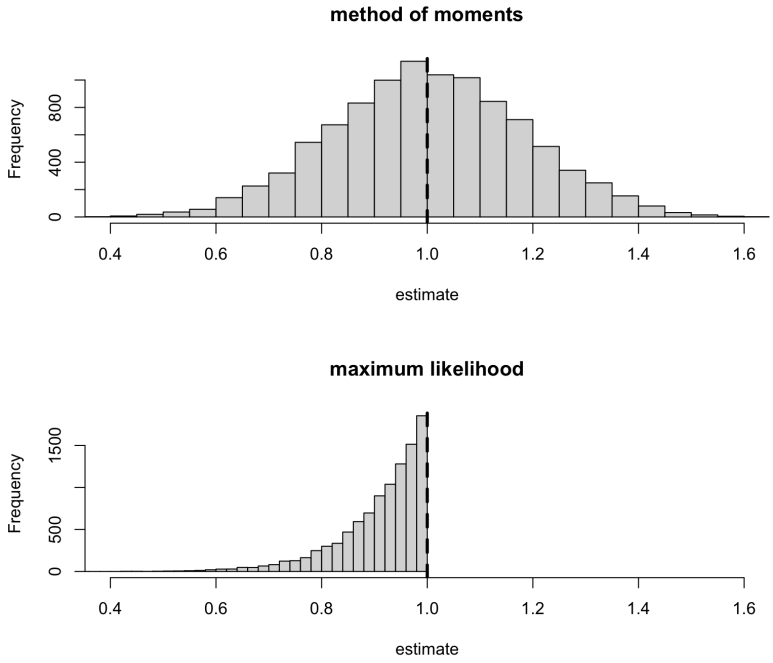


Figure 6.1: Simulated sampling distributions of the two estimators for the uniform parameter, using the same axis: the method of moments estimator $2\bar{X}$ (top) and the maximum likelihood estimator $\max_i X_i$ (bottom). The true value $\theta = 1$ is given by the dashed line.

Figure 6.1 shows that the first two guesses were right, and that the third guess is wrong for most people. The top histogram is a symmetric, bell-shaped histogram, centred around the dashed line: the estimator $2\bar{X}$ is unbiased. The bell-shape is expected, since it is an average and by the central limit theorem (theorem A.15), averages are asymptotically normal. (We will learn more about this result in lecture 12.) The histogram for the maximum likelihood estimator is much narrower, and is shifted to the left of the dashed line. This indicates a bias $-1/11$. The histogram is not bell-shaped at all: no simulated estimates are above 1, since the maximum of a sample from Uniform$(0, \theta)$ can never be above θ and the estimates are converging to this boundary. This shows that not every estimator has a bell-shaped sampling distribution: the central limit theorem only applies to averages and does not apply to the maximum. We will see this histogram again in lecture 13, where the uniform model will be used as a counterexample to the general theory.

In lecture 9 we will vary the sample size and will see the histograms getting tighter around the truth.

Just type the commands you see in this solution in your own R session, to make sure you understand what they do.

Summary.

- Appendix B is the written reference for functions, loops and `replicate`.
- A function created with `function(n, theta) {...}` packages one simulation experiment; a `for`-loop or, more conveniently, `replicate` repeats it and collects the estimates.
- The one-liner `mean(estimates) - theta` gives the empirical bias, and `mean((estimates - theta)^2)` gives the empirical mean squared error; both approximate the true values up to a simulation error that shrinks as the number of repetitions grows.
- For the uniform model with $n = 10$, the simulation reproduced the analytic results of exercise 4.4: the biased estimator $\max_i X_i$ beats the unbiased $2\bar{X}$ in mean squared error, and its sampling distribution is a skewed pile below θ, not a bell.
- Predict before you run: a simulation only teaches when its output can confirm or contradict a stated expectation.

Lecture 7

Exponential Families

In lectures 4 and 5 we repeatedly collapsed products of n densities to expressions which summarised the data using only one or two sums. This lecture explains why: almost all families of models considered here have the same algebraic shape, and models of this shape are called *exponential families*. Many of the results in the following weeks are almost automatic for this class.

7.1 Recognising the Shape

We start by considering a familiar distribution and then rewrite its probability weights so that the data and the parameter are as far apart as possible. For the Bernoulli distribution with $x \in \{0, 1\}$ we can write

$$\begin{aligned} f(x;\theta) = \theta^x (1-\theta)^{1-x} &= \exp\big(x \log\theta + (1-x)\log(1-\theta)\big) \\ &= \exp\Big(x \log\frac{\theta}{1-\theta} + \log(1-\theta)\Big). \end{aligned}$$

Here we have written the weights as the exponential of the product of one function of θ and one function of x, plus a function of only θ. The Poisson weights and the exponential density can be written in a similar way: for $x \in \mathbb{N}_0$ and $x \geq 0$ respectively we have

$$\frac{\theta^x}{x!} e^{-\theta} = \frac{1}{x!} \exp\big(x \log\theta - \theta\big) \qquad \text{and} \qquad \theta e^{-\theta x} = \exp\big(-\theta x + \log\theta\big).$$

For the Poisson weights an additional factor $1/x!$ appears, which depends on x alone. We will give a name to this structure.

Definition 7.1. A parametric model $\big(f(x;\theta) \,\big|\, \theta \in \Theta\big)$ with $\Theta \subseteq \mathbb{R}$ is a *one-parameter exponential family*, if the density or the probability weights can be written as

$$f(x;\theta) = h(x) \exp\big(\eta(\theta)\, T(x) - B(\theta)\big)$$

for all x and all $\theta \in \Theta$, where $h \geq 0$ and T are functions of x alone and η and B are functions of θ alone. The statistic $T(X)$ is called the *natural statistic* and $\eta(\theta)$ is called the *natural parameter* of the family.

Since the exponential is always positive, $f(x;\theta)$ is positive if and only if $h(x)$ is. Thus the support $\{x \mid f(x;\theta) > 0\}$ of the distribution is $\{x \mid h(x) > 0\}$ and does not depend on θ. The representation is not unique, since we can replace η by

$c\eta$ and T by T/c for any constant $c \neq 0$. Finally, since the density must integrate (or sum) to one, the function B is determined by the other three functions: we have

$$B(\theta) = \log \int h(x) \exp\bigl(\eta(\theta)\, T(x)\bigr)\, \mathrm{d}x. \tag{7.1}$$

These rewritings give the following examples.

Example 7.2. The Bernoulli distribution with $\theta \in (0, 1)$ is a one-parameter exponential family with $h(x) = 1$ for $x \in \{0, 1\}$ and $h(x) = 0$ otherwise, natural statistic $T(x) = x$, natural parameter $\eta(\theta) = \log(\theta/(1-\theta))$, the *log-odds* of success, and $B(\theta) = -\log(1-\theta)$.

Example 7.3. The Poisson distribution with $\theta > 0$ is a one-parameter exponential family with $h(x) = 1/x!$ for $x \in \mathbb{N}_0$ and $h(x) = 0$ otherwise, natural statistic $T(x) = x$, natural parameter $\eta(\theta) = \log\theta$ and $B(\theta) = \theta$.

Example 7.4. The exponential distribution with rate $\theta > 0$ is a one-parameter exponential family with $h(x) = 1$ for $x \geq 0$ and $h(x) = 0$ otherwise, natural statistic $T(x) = x$, natural parameter $\eta(\theta) = -\theta$ and $B(\theta) = -\log\theta$.

The normal distribution is also an exponential family, but since both parameters are unknown, we need to use the two-parameter version of the definition from section 7.2.

Remark. The uniform distribution on $[0, \theta]$ is not an exponential family, since the support of the distribution changes with the parameter. In contrast, the support of an exponential family is always fixed. This is the same reason why the likelihood equation was useless in example 5.5. Similarly, the t location model from section 5.3 is not an exponential family, despite the fact that its support is all of $\mathbb{R}$; we do not prove this.

The shape survives the passage from one observation to a sample.

Lemma 7.5. *Let $X_1, \ldots, X_n$ be i.i.d. from the one-parameter exponential family in definition 7.1. Then the joint density of the sample is*

$$f(x_1, \ldots, x_n; \theta) = \prod_{i=1}^{n} h(x_i) \cdot \exp\Bigl(\eta(\theta) \sum_{i=1}^{n} T(x_i) - nB(\theta)\Bigr),$$

which is of the form of definition 7.1 with the same natural parameter $\eta(\theta)$, natural statistic $T_n = \sum_{i=1}^{n} T(X_i)$, the function $\prod_i h(x_i)$ instead of h and $nB(\theta)$ instead of $B(\theta)$.

Proof. Since the observations are independent, the joint density is the product $\prod_{i=1}^{n} h(x_i) \exp\bigl(\eta(\theta)\, T(x_i) - B(\theta)\bigr)$ of the marginal densities. Combining the n exponents using the rule $e^a e^b = e^{a+b}$ gives the required form. This completes the proof. □

Read as a function of θ, the joint density is the likelihood, and thus the log-likelihood of an exponential family sample is

$$\ell(\theta) = \sum_{i=1}^{n} \log h(x_i) + \eta(\theta)\, T_n - nB(\theta).$$

The first term does not involve θ. Thus, up to a factor which does not depend on θ, the likelihood depends on the data only through the single number $T_n = \sum_i T(x_i)$. This is an extension of example 4.3 to the whole class; in lecture 8 we will see in which sense the value T_n contains all the information about θ.

7.2 Several Parameters

The normal distribution where both μ and σ^2 are unknown does not fit definition 7.1, since the exponent contains the two products $x\mu/\sigma^2$ and $-x^2/(2\sigma^2)$, which cannot be combined into a single product $\eta(\theta)\,T(x)$. The extension is to allow a sum of k such products in the exponent.

Definition 7.6. A parametric model $\big(f(x;\theta)\,\big|\,\theta \in \Theta\big)$ is a *k-parameter exponential family*, if the density or the probability weights can be written as

$$f(x;\theta) = h(x)\exp\Big(\sum_{j=1}^{k} \eta_j(\theta)\,T_j(x) - B(\theta)\Big)$$

for all x and all $\theta \in \Theta$, where $h \geq 0$ and $T_1, \ldots, T_k$ are functions of x alone and $\eta_1, \ldots, \eta_k$ and B are functions of θ alone. The vector $\big(T_1(X), \ldots, T_k(X)\big)$ is the *natural statistic* and $\big(\eta_1(\theta), \ldots, \eta_k(\theta)\big)$ is the *natural parameter* of the family.

The number k counts the products in the exponent, not the components of θ. In most of our examples the two agree, but they do not have to: the normal distribution $N(\theta, \theta^2)$ considered below has the single parameter $\theta > 0$, but its exponent needs two products. An exponential family in which k is larger than the dimension of Θ is called a *curved exponential family*.

The support of the distribution is again $\{x \mid h(x) > 0\}$ and the proof of lemma 7.5 applies without change: an i.i.d. sample is again a k-parameter exponential family, where the natural statistic is the vector of the k sums $\sum_i T_j(X_i)$.

Example 7.7. By expanding the square in the exponent of the density of the normal distribution $N(\mu, \sigma^2)$ from table 1.2, we can write

$$f(x;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\Big(-\frac{x^2}{2\sigma^2} + \frac{\mu}{\sigma^2}\,x - \frac{\mu^2}{2\sigma^2}\Big).$$

If only μ is unknown, the factor $(2\pi\sigma^2)^{-1/2}$ and the term $-x^2/(2\sigma^2)$ contain no unknown quantities and we can define the function $h(x) = (2\pi\sigma^2)^{-1/2}e^{-x^2/(2\sigma^2)}$: we have found a one-parameter exponential family with natural statistic $T(x) = x$, natural parameter $\eta(\mu) = \mu/\sigma^2$ and $B(\mu) = \mu^2/(2\sigma^2)$. If both parameters are unknown, we have to interpret the term $-x^2/(2\sigma^2)$ as a second product and we get a two-parameter exponential family with $h(x) = 1$, natural statistic (x, x^2), natural parameter $\big(\mu/\sigma^2,\, -1/(2\sigma^2)\big)$ and $B(\theta) = \mu^2/(2\sigma^2) + \frac{1}{2}\log(2\pi\sigma^2)$. For a sample, the natural statistic is $\big(\sum_i x_i, \sum_i x_i^2\big)$. Since $\sum_i (x_i - \bar{x})^2 = \sum_i x_i^2 - n\bar{x}^2$, for fixed n this pair and the pair $\big(\bar{x}, \sum_i (x_i - \bar{x})^2\big)$ used to derive the maximum likelihood estimator in example 5.4 are one-to-one functions of each other.

The gamma distribution with both parameters unknown is a second example of a two-parameter exponential family. If we write $x^{\alpha-1} = e^{(\alpha-1)\log x}$, then the natural statistic is $(\log x,\, x)$ and the details are worked out in exercise 7.1. For a sample of size n, the natural statistic is the pair $\big(\sum_i \log x_i, \sum_i x_i\big)$, which is different from the pair of moments used in example 4.9.

For the completeness statements in lecture 10 it is important that the natural parameter varies in as many directions as there are natural statistics. Mathematically, this can be expressed as the following condition.

Definition 7.8. A one-parameter exponential family is of *full rank*, if the set $\eta(\Theta) = \{\eta(\theta) \mid \theta \in \Theta\}$ of natural parameter values contains a non-degenerate open interval. A k-parameter exponential family is of *full rank*, if the set $\{(\eta_1(\theta), \ldots, \eta_k(\theta)) \mid \theta \in \Theta\}$ contains a non-empty open subset of $\mathbb{R}^k$.

In the one-parameter case, the condition is almost always satisfied, since the image of an interval Θ under a continuous, non-constant η is an interval with more than one point; all the one-parameter examples above are of full rank. For several parameters, the condition can be more interesting: the natural parameter values of the normal distribution fill the open half-plane $\{(\eta_1, \eta_2) \mid \eta_2 < 0\}$ (example 7.7) and the natural parameter values of the gamma distribution fill an open quadrant (exercise 7.1), but the natural parameter values of $N(\theta, \theta^2)$ with $\theta > 0$, where the standard deviation equals the mean, form a curve in $\mathbb{R}^2$ which does not contain an open set, and thus this curved exponential family is not of full rank (exercise 7.2).

7.3 The Natural Parametrisation and the Cumulant Function

We return to one-parameter exponential families. Since the parameter enters the density only via $\eta(\theta)$, it is often convenient to use η as the parameter. In this case, the normalising function (7.1) deserves a name, since its first two derivatives are the mean and the variance of the natural statistic.

Definition 7.9. Let h and T be as in definition 7.1. Then the *natural parametrisation* of the family is given by

$$f(x;\eta) = h(x)\exp(\eta\, T(x) - K(\eta)), \qquad \eta \in \mathcal{N}.$$

Here the *cumulant function* (also called the *log-partition function*) is given by

$$K(\eta) = \log \int h(x)\, e^{\eta T(x)}\, \mathrm{d}x,$$

with a sum in place of the integral for discrete models. The *natural parameter space* is the set $\mathcal{N} = \{\eta \in \mathbb{R} \mid K(\eta) < \infty\}$ of all η for which the integral is finite.

Since K makes $f(x;\eta)$ integrate to one, the formula defines a probability distribution for every $\eta \in \mathcal{N}$, whether or not η is of the form $\eta(\theta)$; comparing with (7.1) shows that $\eta(\theta) \in \mathcal{N}$ and $B(\theta) = K(\eta(\theta))$, so that the original family sits inside this larger one. For the three examples of section 7.1 we find $K(\eta) = \log(1 + e^\eta)$ with $\mathcal{N} = \mathbb{R}$ for the Bernoulli distribution, $K(\eta) = e^\eta$ with $\mathcal{N} = \mathbb{R}$ for the Poisson distribution and $K(\eta) = -\log(-\eta)$ with $\mathcal{N} = (-\infty, 0)$ for the exponential distribution. In all three cases $\mathcal{N}$ is an open interval and $\theta \mapsto \eta(\theta)$ is a bijection onto $\mathcal{N}$, making the natural parametrisation a relabelling in the sense of proposition 5.6.

The first two derivatives of K give the mean and the variance of $T(X)$, and the name of K comes from a related object. The *moment generating function* of a random variable Y is the function $s \mapsto \mathbb{E}(e^{sY})$, defined for all s for which the expectation is finite. Its logarithm $s \mapsto \log \mathbb{E}(e^{sY})$ is called the *cumulant generating function* of Y, and the derivatives of this function at $s = 0$ are called

the *cumulants* of Y. The first two cumulants are the mean and the variance of Y. For a random variable Y with values in $\mathbb{N}_0$ the closely related *probability generating function* is given by

$$G_Y(z) = \mathbb{E}(z^Y) = \sum_{y=0}^{\infty} \mathbb{P}(Y = y)\, z^y, \tag{7.2}$$

defined for all $z \geq 0$ for which the sum is finite. Substituting $z = e^s$ shows that $s \mapsto G_Y(e^s)$ is the moment generating function of Y. In exercise 7.6 we show that the function

$$s \mapsto K(\eta + s) - K(\eta)$$

is the cumulant generating function of $T(X)$.

Lemma 7.10. *Let η be an interior point of the natural parameter space $\mathcal{N}$. Then, in the natural parametrisation,*

$$\mathbb{E}_\eta(T(X)) = K'(\eta) \qquad \text{and} \qquad \mathrm{Var}_\eta(T(X)) = K''(\eta).$$

Proof. Set $Z(\eta) = \int h(x) e^{\eta T(x)}\, \mathrm{d}x$, so that $K = \log Z$. Since the density $f(x;\theta) = h(x) \exp\big(\eta(\theta)\, T(x) - B(\theta)\big)$ integrates to one, the function $h \geq 0$ is not identically zero. Together with $e^{\eta T(x)} > 0$ this gives $Z(\eta) > 0$, and thus we may divide by $Z(\eta)$ below. Since η is in the interior of $\mathcal{N}$, we can take derivatives inside the integral as many times as required; this is a result from analysis which we use without proof. Taking derivatives once, we get

$$Z'(\eta) = \int T(x)\, h(x)\, e^{\eta T(x)}\, \mathrm{d}x.$$

Dividing by $Z(\eta) = e^{K(\eta)}$ changes the integrand to $T(x)$ times the density $f(x;\eta)$ and thus we get

$$K'(\eta) = \frac{Z'(\eta)}{Z(\eta)} = \int T(x)\, h(x)\, e^{\eta T(x) - K(\eta)}\, \mathrm{d}x = \mathbb{E}_\eta(T(X)).$$

Taking derivatives again, we find

$$K''(\eta) = \frac{Z''(\eta)}{Z(\eta)} - \Big(\frac{Z'(\eta)}{Z(\eta)}\Big)^2,$$

and using the same argument as above, with $T(x)^2$ in the integrand, we find $Z''(\eta)/Z(\eta) = \mathbb{E}_\eta(T(X)^2)$. Thus we have $K''(\eta) = \mathbb{E}_\eta(T(X)^2) - \mathbb{E}_\eta(T(X))^2 = \mathrm{Var}_\eta(T(X))$ and the proof is complete. □

The lemma allows us to compute moments by differentiation: for the Poisson distribution $K(\eta) = e^\eta$ we find $K'(\eta) = K''(\eta) = e^\eta = \theta$, the mean and variance from table A.1, and in exercise 7.5 we consider the exponential and normal distributions. For T_n, being a sum of n i.i.d. copies of $T(X)$, we can use the lemma to find $\mathbb{E}_\eta(T_n) = nK'(\eta)$ and $\mathrm{Var}_\eta(T_n) = nK''(\eta)$.

7.4 What the Structure Buys

We conclude this lecture by collecting what the exponential family shape tells us about the questions of lectures 4 and 5, and by indicating where the shape will be used again.

The first consequence concerns the likelihood equation. By lemma 7.5, written in the natural parametrisation, the log-likelihood of a sample is

$$\ell(\eta) = \sum_{i=1}^{n} \log h(x_i) + \eta\, T_n - nK(\eta),$$

and thus we have $\ell'(\eta) = T_n - nK'(\eta)$ and, by lemma 7.10, $\ell''(\eta) = -nK''(\eta) = -n\operatorname{Var}_\eta(T(X))$. If $T(X)$ is not constant, this variance is positive and ℓ is strictly concave on the interior of $\mathcal{N}$. Since ℓ' is then strictly decreasing, the likelihood equation $\ell'(\eta) = 0$ has at most one root, and if there is a root, it is the maximum likelihood estimator for η. When $\theta \mapsto \eta(\theta)$ is a bijection, proposition 5.6 transfers this estimator back to θ. The argument shows uniqueness, but not existence: for a Bernoulli sample with no successes we have $T_n = 0$, and the equation $K'(\eta) = e^\eta/(1+e^\eta) = 0$ has no solution. This is the case $s = 0$ from the remark after example 5.3.

The second consequence concerns smoothness. Many of the results later in the module are proved by differentiating with respect to the parameter, and they require the support of the distribution not to depend on θ, the density to be smooth in θ and derivatives to be taken inside integrals. For an exponential family the support is fixed by definition, the density is smooth in θ whenever η and B are, and the argument in the proof of lemma 7.10 allows differentiation under the integral sign whenever $\eta(\theta)$ lies in the interior of $\mathcal{N}$. Similarly, if η is one-to-one and $T(X)$ is not constant, then different values of θ give different distributions, and the parameter is identifiable in the sense of definition 4.10. Under these mild conditions on η, the exponential families are the standard examples for the theory in the rest of the module.

Finally, the observation that the likelihood depends on the data only through T_n and n suggests that, once T_n is known, the individual observations carry no further information about θ. In lecture 8 we make this idea precise, and lemma 7.5 will do most of the work there. From then on, the natural statistic T_n will be the starting point whenever we look for good estimators, and again when we update our knowledge about θ using the Bayesian approach in the second half of the module.

Summary.

- A one-parameter exponential family has density of the form $f(x;\theta) = h(x)\exp(\eta(\theta)T(x) - B(\theta))$, where the natural statistic is T, the natural parameter is η and the support of the distribution does not depend on θ. Examples of exponential families include the Bernoulli, Poisson, exponential, normal and gamma distributions. The uniform distribution is not an exponential family.
- An i.i.d. sample from an exponential family is again an exponential family. In this case, the natural statistic is $T_n = \sum_i T(X_i)$ and the likelihood depends on the data only through T_n.
- A k-parameter exponential family is of full rank, if the set of natural parameter values contains a non-empty open subset of $\mathbb{R}^k$. This condition is required for completeness in lecture 10. The curved exponential family $N(\theta, \theta^2)$ is not of full rank.
- In the natural parametrisation $f(x;\eta) = h(x)\exp(\eta T(x) - K(\eta))$ the cumulant function K can be used to find the mean and variance of the natural statistic: we have $\mathbb{E}_\eta(T(X)) = K'(\eta)$ and $\mathrm{Var}_\eta(T(X)) = K''(\eta)$.
- If $T(X)$ is not constant, the log-likelihood is strictly concave in the natural parameter, and thus the likelihood equation has at most one root. A root need not exist.
- Under mild conditions on η, exponential families satisfy the smoothness conditions of the later theory.

Exercise 7.1. Let X be gamma distributed, $X \sim \mathrm{Gamma}(\alpha, \beta)$ with shape $\alpha > 0$ and rate $\beta > 0$, both unknown, *i.e.* with density $f(x;\alpha,\beta) = \beta^\alpha x^{\alpha-1} e^{-\beta x}/\Gamma(\alpha)$ for $x > 0$, as listed in table A.2.

1. Show that this model is a two-parameter exponential family, as given in definition 7.6, with $h(x) = 1$ for $x > 0$ and $h(x) = 0$ otherwise, natural statistic $(\log x,\ x)$, natural parameter $(\alpha - 1,\ -\beta)$, and function $B(\alpha,\beta) = \log\Gamma(\alpha) - \alpha\log\beta$.

2. Show that the set of natural parameter values is the open quadrant

$$\{(\eta_1,\eta_2) \mid \eta_1 > -1,\ \eta_2 < 0\}$$

   and thus the family has full rank as given in definition 7.8.

3. Assume that the shape α is known, as in exercise 5.2. Show that then the factor $x^{\alpha-1}$ can be incorporated into the function h to yield a one-parameter exponential family with natural statistic $T(x) = x$ and natural parameter $-\beta$.

Exercise 7.2. Consider the normal distribution $N(\theta, \theta^2)$ with mean and standard deviation both equal to $\theta > 0$.

1. Using example 7.7, show that this model is a curved two-parameter exponential family with natural statistic and natural parameter given by

$$T(x) = (x, x^2) \qquad \text{and} \qquad \eta(\theta) = (1/\theta,\ -1/(2\theta^2)).$$

2. Show that as θ varies, the natural parameter values follow the curve $\eta_2 = -\eta_1^2/2$ with $\eta_1 > 0$. Conclude that this family is not of full rank in the sense of definition 7.8.

Exercise 7.3. Let X be geometrically distributed with parameter $p \in (0, 1)$, *i.e.* $f(x; p) = (1-p)^{x-1}p$ for $x \in \mathbb{N}$, as in exercises 4.2 and 5.1.

1. Show that this distribution is a one-parameter exponential family and identify the functions h, T, η and B. What is the natural statistic of an i.i.d. sample $X_1, \ldots, X_n$?

2. Determine the natural parameter space $\mathcal{N}$ and the cumulant function K. Check that $B(p) = K\big(\eta(p)\big)$. Is this family of full rank?

3. Using lemma 7.10, determine the mean and variance of X and compare the result with table A.1.

Exercise 7.4. Let X be binomially distributed with a known number m of trials and unknown success probability $\theta \in (0, 1)$, *i.e.* $f(x; \theta) = \binom{m}{x}\theta^x(1-\theta)^{m-x}$ for $x \in \{0, 1, \ldots, m\}$.

1. Show that this distribution is a one-parameter exponential family, with the same natural parameter as the Bernoulli distribution from example 7.2, and identify the functions h, T and B.

2. Determine the natural parameter space and the cumulant function.

3. Using lemma 7.10, determine the mean and variance of X, and the mean and variance of the natural statistic T_n for an i.i.d. sample of size n.

Exercise 7.5.

1. For the exponential distribution with rate θ, use the cumulant function $K(\eta) = -\log(-\eta)$ from the text to find the mean and variance of an observation.

2. For the normal distribution with known variance σ^2 and unknown mean μ, where h, T and η are as in example 7.7, use the method of completing the square to show that $K(\eta) = \sigma^2\eta^2/2$ with $\mathcal{N} = \mathbb{R}$ and use this to derive the mean and variance of an observation. Check that your result satisfies $B(\mu) = K\big(\eta(\mu)\big)$.

Exercise 7.6. Let X be a random variable with density

$$f(x; \eta) = h(x)\exp\big(\eta T(x) - K(\eta)\big)$$

given by definition 7.9, and let s be such that $\eta + s \in \mathcal{N}$.

1. Show that $\mathbb{E}_\eta\big(e^{sT(X)}\big) = \exp\big(K(\eta+s) - K(\eta)\big)$, *i.e.* that the increment $C(s) = K(\eta+s) - K(\eta)$ is the cumulant generating function of $T(X)$.

2. Let η be an interior point of $\mathcal{N}$. Then, since C is differentiable at $s = 0$, we can find $C'(0)$ and $C''(0)$ from the first part of this exercise. Use these results to give an alternative proof of lemma 7.10.

3. Using the first part of this exercise, show that for the Poisson distribution with parameter θ the probability generating function (7.2) is given by $\mathbb{E}_\theta(z^X) = e^{\theta(z-1)}$ for all $z > 0$. This is the same identity as in exercise 5.7.

Exercise 7.7. Let $X_1, \ldots, X_n$ be an i.i.d. sample from a one-parameter exponential family in its natural parametrisation, with $\mathcal{N}$ an open interval and $\mathrm{Var}_\eta(T(X_1)) > 0$ for all $\eta \in \mathcal{N}$.

1. Show that the likelihood equation $\ell'(\eta) = 0$ from lecture 5 can be written as
$$\frac{1}{n}\sum_{i=1}^{n} T(x_i) = \mathbb{E}_\eta(T(X_1)),$$
i.e. the maximum likelihood estimator is the parameter value under which the population mean of the natural statistic equals its sample mean.

2. Show that every root of the likelihood equation is the unique maximum likelihood estimator for η.

3. Apply the equation from the first part to the Poisson distribution and to the Bernoulli distribution, and use proposition 5.6 to recover the estimators found in exercise 5.7 and example 5.3.

4. Explain how the first part of this exercise accounts for the coincidences between the method of moments and maximum likelihood which we noted in lecture 4, and why no such coincidence is to be expected for the gamma distribution of exercise 7.1.

Lecture 8

Sufficiency

In example 4.2 we have observed ten coin tosses and got seven heads. Writing θ for the heads probability, the likelihood in this example is $L(\theta) = \theta^7(1-\theta)^3$. This function depends on the data only via the number 7 of heads, but not on the order of heads and tails, *i.e.* for learning about θ we can replace the full record of observations by this single number. This lecture makes the idea precise: a *sufficient statistic* contains all information about the parameter. The factorisation theorem helps to find sufficient statistics, and the concept of sufficiency forms the basis of improvements of estimators in lecture 10.

In this lecture we write $X = (X_1, \ldots, X_n)$ for the data, $x = (x_1, \ldots, x_n)$ for an observed value, and

$$f(x;\theta) = \prod_{i=1}^{n} f(x_i;\theta)$$

for the joint density or joint probability weights of the sample; the vector argument distinguishes it from the density of a single observation. As a function of θ for fixed data, this is the likelihood $L(\theta)$ from lecture 4.

8.1 Sufficient Statistics

A statistic $T = T(X_1, \ldots, X_n)$ in the sense of definition 1.3 reduces the data to a smaller summary, and usually some information is lost in the process. The following definition expresses that no information about θ is lost.

Definition 8.1. A statistic $T = T(X_1, \ldots, X_n)$ is *sufficient* for θ, if the conditional distribution of $(X_1, \ldots, X_n)$ given the value $T = t$ does not depend on θ, for all values t of T.

If T is sufficient, anyone who knows the value t of T can generate a new data set with the same distribution as the original data set, by sampling from the conditional distribution of X given $T = t$. Since this distribution does not depend on θ, all information about θ that we could obtain by studying the data set can now be obtained by studying t alone.

Example 8.2. Let $X_1, \ldots, X_n$ be i.i.d. Bernoulli with success probability p and let $T = \sum_{i=1}^{n} X_i$ be the number of successes. We know that $T \sim \text{Binomial}(n, p)$, as a sum of n Bernoulli variables. Let $t \in \{0, 1, \ldots, n\}$ and let $x \in \{0, 1\}^n$ be a

data vector with $\sum_i x_i = t$. Since the event $\{X = x\}$ is contained in the event $\{T = t\}$, we have

$$\mathbb{P}_p(X = x|T = t) = \frac{\mathbb{P}_p(X = x)}{\mathbb{P}_p(T = t)} = \frac{p^t(1-p)^{n-t}}{\binom{n}{t}p^t(1-p)^{n-t}} = \frac{1}{\binom{n}{t}}.$$

If $\sum_i x_i \neq t$, the events $\{X = x\}$ and $\{T = t\}$ are disjoint and the conditional probability is zero. In both cases the value does not depend on p, and thus T is sufficient for p. Given the number of successes, each of the $\binom{n}{t}$ possible arrangements of successes and failures is equally likely, whatever the value of p: the positions of the successes do not contain any information about the coin.

Two simple observations follow from the definition. First, the whole sample $T = (X_1, \ldots, X_n)$ is always sufficient, since given the whole sample there is no randomness left. Secondly, if φ is a one-to-one function, the events $\{\varphi(T) = \varphi(t)\}$ and $\{T = t\}$ coincide, and thus $\varphi(T)$ is sufficient whenever T is: a one-to-one function of a sufficient statistic is sufficient again. For the coin tosses the sample mean $\bar{X} = T/n$ is thus sufficient too.

8.2 The Factorisation Theorem

Directly verifying the definition of sufficiency requires knowledge of the distribution of T and involves a conditional probability which, for most models, is tedious to compute. The following proposition replaces the computation with a condition on the density or weights.

Proposition 8.3 (Fisher–Neyman factorisation theorem)**.** *The statistic T is sufficient for θ, if and only if there are functions h and g such that*

$$f(x;\theta) = h(x)\, g\big(T(x);\theta\big) \tag{8.1}$$

for all data vectors x and all $\theta \in \Theta$.

The function h may not depend on θ at all, and the function g may depend on the data only through $T(x)$: the parameter and the data interact only through T. Nothing in the statement requires T or θ to be one-dimensional. We give a proof for discrete data only.

Proof for discrete data. Since we have discrete observations, the joint probability weight $f(x;\theta) = \mathbb{P}_\theta(X = x)$ is the probability of observing the data vector x.

Assume first that T is sufficient and let x be a data vector with $t = T(x)$. Since the event $\{X = x\}$ is contained in the event $\{T = t\}$, we have

$$\mathbb{P}_\theta(X = x) = \mathbb{P}_\theta(X = x,\, T = t) = \mathbb{P}_\theta(X = x|T = t)\, \mathbb{P}_\theta(T = t)$$

whenever $\mathbb{P}_\theta(T = t) > 0$. Using the sufficiency of the statistic, the conditional probability does not depend on θ and thus is a function $h(x)$ of the data. The remaining factor is a function $g(t;\theta)$ of t and θ. If $\mathbb{P}_\theta(T = t) = 0$ for the given θ, both sides in (8.1) equal zero since $g(t;\theta) = 0$. For the data vectors x with $\mathbb{P}_\theta(T = T(x)) = 0$ for every θ, which are never observed, we define $h(x) = 0$. Thus (8.1) holds for all x and θ.

Conversely, assume that (8.1) holds and let t be a value of T and θ a parameter value with $\mathbb{P}_\theta(T = t) > 0$. Summing the factorisations for all data vectors y with $T(y) = t$, we find

$$\mathbb{P}_\theta(T = t) = \sum_{y\colon T(y)=t} f(y;\theta) = \sum_{y\colon T(y)=t} h(y)\, g(t;\theta) = H(t)\, g(t;\theta),$$

where $H(t)$ is the sum of the $h(y)$, which does not depend on θ. Since the left-hand side is positive, we have $g(t;\theta) \neq 0$. For data x with $T(x) = t$ we get

$$\mathbb{P}_\theta(X = x | T = t) = \frac{\mathbb{P}_\theta(X = x)}{\mathbb{P}_\theta(T = t)} = \frac{h(x)\, g(t;\theta)}{H(t)\, g(t;\theta)} = \frac{h(x)}{H(t)},$$

and for x with $T(x) \neq t$ the conditional probability is zero. Thus, in both cases the value does not depend on θ and T is sufficient. This completes the proof. □

The proof for the continuous case is more technical, and we omit it here. As a first step, one needs to define what it means to condition the distribution of X on $T = t$, since we typically have $\mathbb{P}(T = t) = 0$ in the continuous case. To avoid these complications, we will use (8.1) as the definition of sufficiency for the continuous case from now on. Once conditioning on $T = t$ has been given a meaning, it can be shown that the factorisation again implies that the conditional distribution of the sample given T does not depend on θ, as in the discrete case. We will use this property without proof in lecture 10; exercise 10.3 verifies it for an exponential sample.

When the support depends on θ, the density has to be written down for all x, including the values where it is zero, since this is where the parameter and the data may interact. To write such densities compactly we use the *indicator function* of a condition, $\mathbf{1}_{\{A\}}$, which equals 1 if the condition A holds and 0 otherwise. The uniform distribution is the standard example.

Example 8.4. Let $X_1, \ldots, X_n \sim \text{Uniform}(0, \theta)$ be i.i.d. with $\theta > 0$, as in example 1.2. The density of a single observation, written for all $x_i \in \mathbb{R}$, is $f(x_i;\theta) = \theta^{-1}\mathbf{1}_{\{0 \leq x_i \leq \theta\}}$. A product of indicators is the indicator of the intersection of the conditions, and all n conditions $0 \leq x_i \leq \theta$ hold if and only if the smallest observation is at least 0 and the largest is at most θ. Thus the joint density is given by

$$f(x;\theta) = \theta^{-n} \prod_{i=1}^{n} \mathbf{1}_{\{0 \leq x_i \leq \theta\}} = \mathbf{1}_{\{\min_i x_i \geq 0\}}\, \theta^{-n}\, \mathbf{1}_{\{\max_i x_i \leq \theta\}}.$$

If we set $h(x) = \mathbf{1}_{\{\min_i x_i \geq 0\}}$ and $g(t;\theta) = \theta^{-n}\mathbf{1}_{\{t \leq \theta\}}$ we have

$$f(x;\theta) = h(x) g\big(\max_i x_i;\, \theta\big)$$

and by proposition 8.3 the sample maximum $X_{(n)} = \max_i X_i$ is sufficient for θ. It is a common mistake to try to write the joint density simply as θ^{-n} and to conclude that no statistic is needed: the indicator is where the data and the parameter interact.

The factorisation has a consequence for maximum likelihood estimators. For fixed data the likelihood is $L(\theta) = h(x)\, g(T(x);\theta)$, and the factor $h(x)$ does not

affect where the maximum over θ is attained: the maximum likelihood estimator, whenever it is unique, is a function of any sufficient statistic. The examples of lecture 5 confirm this: the Bernoulli MLE is T/n with the sufficient statistic T of example 8.2, and the uniform MLE is the sample maximum, which example 8.4 shows to be sufficient.

8.3 Sufficiency in Exponential Families

For the exponential families from lecture 7, no work is needed to find a sufficient statistic, since the densities are already given in factorised form.

Corollary 8.5. *Let $X_1, \dots, X_n$ be an i.i.d. sample from a one-parameter exponential family as given in definition 7.1 and let T be the natural statistic. Then $T_n = \sum_{i=1}^n T(X_i)$ is a sufficient statistic for θ.*

For a k-parameter exponential family as given in definition 7.6, the vector $(\sum_i T_1(X_i), \dots, \sum_i T_k(X_i))$ of summed natural statistics is sufficient for the parameter vector.

Proof. From lemma 7.5 we know that the joint density of the sample is

$$f(x;\theta) = \Big(\prod_{i=1}^n h(x_i)\Big) \exp\Big(\eta(\theta) \sum_{i=1}^n T(x_i) - nB(\theta)\Big).$$

Thus, we can write the joint density in the form (8.1) with the product $\prod_i h(x_i)$ in the role of $h(x)$ and with $g(t;\theta) = \exp(\eta(\theta)t - nB(\theta))$. Consequently, T_n is sufficient by proposition 8.3. A very similar argument gives the proof for k-parameter exponential families. This completes the proof. □

Example 8.6. Let $X_1, \dots, X_n$ be i.i.d. Poisson with parameter $\theta > 0$. The joint probability weights are given by

$$f(x;\theta) = \prod_{i=1}^n \frac{\theta^{x_i}}{x_i!}\, e^{-\theta} = \theta^{\sum_i x_i}\, e^{-n\theta} \cdot \frac{1}{\prod_i x_i!},$$

and thus the sum $T = \sum_i X_i$ is sufficient for θ, with $g(t;\theta) = \theta^t e^{-n\theta}$ and $h(x) = 1/\prod_i x_i!$.

Example 8.7. Let $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ be i.i.d. with both parameters unknown. Example 7.7 identifies the natural statistics as x and x^2, and thus by corollary 8.5 the pair $T = (\sum_i X_i, \sum_i X_i^2)$ is sufficient for (μ, σ^2). Since $\sum_i (x_i - \bar{x})^2 = \sum_i x_i^2 - n\bar{x}^2$, the pair $(\bar{X}, S^2)$ maps to T one-to-one and thus is also sufficient. This is the pair of statistics on which the maximum likelihood estimator from example 5.4 is based; the factorisation is worked out in exercise 8.4.

8.4 The Sample Maximum and Minimum

Example 8.4 shows that a sufficient statistic need not be a sum. To find the bias or the mean squared error of the sample maximum we need its distribution, which the following result gives for the maximum and minimum of an i.i.d. sample.

Proposition 8.8. *Let $X_1, \ldots, X_n$ be i.i.d. with distribution function F and density f. Furthermore, let $X_{(n)} = \max_i X_i$ and $X_{(1)} = \min_i X_i$. Then the random variables $X_{(n)}$ and $X_{(1)}$ have densities*

$$f_{(n)}(y) = n\,F(y)^{n-1} f(y) \qquad \text{and} \qquad f_{(1)}(y) = n\,\big(1 - F(y)\big)^{n-1} f(y).$$

Proof. The maximum is at most y, if and only if all observations are at most y. Using independence we get

$$\mathbb{P}(X_{(n)} \leq y) = \mathbb{P}(X_1 \leq y, \ldots, X_n \leq y) = \prod_{i=1}^{n} \mathbb{P}(X_i \leq y) = F(y)^n.$$

Using the chain rule for differentiation with respect to y, we find that the maximum has density $nF(y)^{n-1}f(y)$. Similarly, the minimum is strictly greater than y, if and only if all observations are strictly greater than y. Thus we get

$$\mathbb{P}(X_{(1)} \leq y) = 1 - \mathbb{P}(X_1 > y, \ldots, X_n > y) = 1 - \big(1 - F(y)\big)^n.$$

Using the chain rule for differentiation we find $n\big(1 - F(y)\big)^{n-1} f(y)$. This completes the proof. □

Example 8.9. Let $X_1, \ldots, X_n \sim \text{Uniform}(0, \theta)$ be i.i.d., as in example 8.4. The distribution function is $F(y) = y/\theta$ for $0 \leq y \leq \theta$, and the density is $f(y) = 1/\theta$ on $[0, \theta]$. By proposition 8.8 the sample maximum has density

$$f_{(n)}(y) = n\Big(\frac{y}{\theta}\Big)^{n-1} \frac{1}{\theta} = \frac{n\,y^{n-1}}{\theta^n}, \qquad 0 \leq y \leq \theta,$$

and $f_{(n)}(y) = 0$ outside this interval. This is the density supplied without proof in exercise 2.2, from which we computed $\mathbb{E}_\theta(X_{(n)}) = n\theta/(n+1)$ there. The support of the maximum moves with θ, as the support of a single observation does.

8.5 Minimal Sufficiency and Completeness

As we have seen in section 8.1, the whole sample is always sufficient. For an i.i.d. sample, the vector of ordered observations is also sufficient, since the joint density does not change when the x_i are interchanged. The aim of this section is to characterise the statistic which allows us to reduce the data as much as possible, while still retaining all the information.

Definition 8.10. A sufficient statistic T is *minimal sufficient* for θ, if for every sufficient statistic S there is a function φ such that $T = \varphi(S)$.

A minimal sufficient statistic can thus be computed from every other sufficient statistic. The definition is awkward to check directly, because it refers to all sufficient statistics at once; the following criterion instead compares the likelihood functions of two data sets. In the criterion we write

$$\mathcal{X} = \{\, x \mid f(x;\theta) > 0 \text{ for some } \theta \in \Theta \,\}$$

for the set of data vectors which can be observed under some parameter value, for the Poisson distribution the vectors of non-negative integers.

Proposition 8.11. *Let T be a statistic such that for all $x, y \in \mathcal{X}$ the following two statements are equivalent:*

1. *there is a constant $c(x, y) > 0$ such that $f(x; \theta) = c(x, y)\, f(y; \theta)$ for all $\theta \in \Theta$;*
2. *$T(x) = T(y)$.*

Then T is minimal sufficient for θ.

When both $f(x; \theta)$ and $f(y; \theta)$ are positive for all θ, the first statement says that the ratio $f(x; \theta)/f(y; \theta)$ does not depend on θ. We only sketch the proof. For sufficiency, we choose for every value t of T a data vector $x_t \in \mathcal{X}$ with $T(x_t) = t$; the first statement then gives the factorisation (8.1) with $g(t; \theta) = f(x_t; \theta)$ and $h(x) = c(x, x_t)$. For minimality, let S be sufficient and $x, y \in \mathcal{X}$ with $S(x) = S(y)$. The factorisation for S shows that $f(x; \theta)$ and $f(y; \theta)$ are proportional, and the criterion gives $T(x) = T(y)$: the value of T is determined by the value of S.

The criterion states that $T(x)$ determines the likelihood function of the data up to a constant factor, and that two data sets with different values of T have likelihood functions which are not proportional. We apply the criterion once, to the Poisson sum. The two-parameter normal case is worked out in exercise 8.4.

Example 8.12. Let $X_1, \ldots, X_n$ be i.i.d. Poisson with parameter $\theta > 0$, and let $T = \sum_i X_i$. For two vectors x and y of non-negative integers, both probability weights are positive and the joint probability weights of example 8.6 give

$$\frac{f(x; \theta)}{f(y; \theta)} = \frac{\theta^{\sum_i x_i} e^{-n\theta} / \prod_i x_i!}{\theta^{\sum_i y_i} e^{-n\theta} / \prod_i y_i!} = \theta^{\sum_i x_i - \sum_i y_i} \frac{\prod_i y_i!}{\prod_i x_i!}.$$

If $\sum_i x_i = \sum_i y_i$, the power of θ disappears and the ratio is a positive constant. If the sums differ by $k \neq 0$, the ratio contains the factor θ^k, which is not constant as θ ranges over $(0, \infty)$. Thus the ratio is free of θ if and only if $\sum_i x_i = \sum_i y_i$, and T is minimal sufficient by proposition 8.11.

Summary.

- A statistic T is sufficient for θ, if the conditional distribution of the data given T does not depend on θ: the data can be reduced to T without losing information about θ.
- The factorisation theorem states that T is sufficient if and only if $f(x;\theta) = h(x)\, g(T(x);\theta)$. When the support depends on θ, its indicator is part of the density and belongs to g. When the MLE is unique, it is a function of any sufficient statistic.
- For exponential families, the summed natural statistic $\sum_i T(X_i)$ is sufficient. This includes the Bernoulli total, the Poisson sum and the normal pair $(\sum_i X_i, \sum_i X_i^2)$.
- The densities of the sample maximum and minimum are $nF^{n-1}f$ and $n(1-F)^{n-1}f$, obtained by differentiating F^n and $1-(1-F)^n$, respectively.
- A minimal sufficient statistic is a function of any sufficient statistic. A statistic T is minimal sufficient, if $T(x) = T(y)$ holds exactly when the likelihood functions of x and y are proportional.

Exercise 8.1. Let $X_1, \dots, X_n$ be i.i.d. Bernoulli with success probability $\theta \in (0,1)$, where $n \geq 2$, and let $T = \sum_{i=1}^n X_i$ be the number of successes.

1. Let $x \in \{0,1\}^n$ be a data vector. Compute the conditional probability $\mathbb{P}_\theta(X = x | X_1 = x_1)$ and show that the first observation X_1 on its own is not sufficient for θ.

2. Using the factorisation theorem (proposition 8.3), show that T is sufficient for θ, confirming example 8.2.

3. Is the statistic (X_1, T) sufficient for θ? Justify your answer.

Exercise 8.2. Let $X_1, \dots, X_n \sim \text{Uniform}(-\theta, \theta)$ be i.i.d. with $\theta > 0$, so that a single observation has density $f(x_i;\theta) = (2\theta)^{-1}\mathbf{1}_{\{-\theta \leq x_i \leq \theta\}}$.

1. Show that the joint density can be written as

$$f(x;\theta) = (2\theta)^{-n}\, \mathbf{1}_{\{\max_i |x_i| \leq \theta\}}.$$

2. Deduce from proposition 8.3 that $M = \max_i |X_i|$ is sufficient for θ. Identify the functions g and h.

3. Show that the pair $(X_{(1)}, X_{(n)})$ of the sample minimum and maximum is also sufficient for θ, and express M as a function of this pair.

Exercise 8.3. Let $X_1, \dots, X_n$ be i.i.d. exponential with rate $\theta > 0$, *i.e.* $f(x;\theta) = \theta e^{-\theta x}$ for $x \geq 0$.

1. Show that $T = \sum_i X_i$ is sufficient for θ.

2. Using proposition 8.11, show that T is minimal sufficient.

3. In example 5.2 we have found the MLE $\hat{\theta} = 1/\bar{X}$. Write $\hat{\theta}$ as a function of T and explain why the factorisation in the first part of the question guarantees that such a function exists.

Exercise 8.4. Let $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ be i.i.d.

1. Assume that μ is known and σ^2 is unknown. Show that $\sum_i (X_i - \mu)^2$ is sufficient for σ^2.

2. Assume that both parameters are unknown. By expanding the squares in the exponent of the joint density, show that the joint density depends on the data only via the pair $T = (\sum_i X_i, \sum_i X_i^2)$. Using proposition 8.3, show that T is sufficient for (μ, σ^2).

3. Using proposition 8.11, show that the pair T from the previous part is minimal sufficient for (μ, σ^2).

4. Is the sample mean $\bar{X}$ on its own sufficient for (μ, σ^2) when both parameters are unknown? Justify your answer using the previous part.

Exercise 8.5. Let $X_1, \ldots, X_n$ be i.i.d. exponential with rate $\theta > 0$, and let $X_{(1)} = \min_i X_i$.

1. Show that $X_{(1)}$ is exponentially distributed with rate $n\theta$, first by proposition 8.8 and then directly from $\mathbb{P}_\theta(X_{(1)} > y)$.

2. Show that $nX_{(1)}$ is an unbiased estimator for the mean $1/\theta$, and compute its variance.

3. Compare the variance of $nX_{(1)}$ with that of the sample mean $\bar{X}$, which is also unbiased for $1/\theta$, and comment on the behaviour of the two estimators as n grows. Is $nX_{(1)}$ a function of the sufficient statistic $\sum_i X_i$?

Lecture 9

Consistency by Simulation

In this lecture we use simulation to watch an estimator get closer to the truth as the sample size grows, *i.e.* the consistency from lecture 2. We see this in three ways: through a single estimate as observations are added, through the sampling distribution at three sample sizes and through the mean squared error as a function of n. On the way we extend the Monte Carlo skills from lecture 6 by a loop which varies the sample size. We conclude with a simulation which shows what lecture 8 means by a sufficient statistic. As before, appendix B contains a written reference for this topic.

9.1 A Running Estimate

In this section we simulate one long sample and watch the estimate change as the observations arrive one by one.

We will consider the exponential distribution with rate $\theta = 2$, where the maximum likelihood estimator for the rate is $\hat{\theta}_n = 1/\bar{X}_n$, as found in example 5.2. This estimator is consistent for θ by example 2.12 and proposition A.17. The most direct way to see the estimator at work is to simulate one long sample and to re-compute the estimate after each observation. The command `cumsum` can be used to compute the running totals X_1, $X_1 + X_2$, $X_1 + X_2 + X_3, \ldots$ in one step, and dividing the count n by these totals gives all 2000 running estimates at once. Before you look at the plot, sketch the path you expect. Where is it most volatile?

```
theta   <- 2
set.seed(2026)
x       <- rexp(2000, rate = theta)
n       <- seq_len(2000)
running <- n / cumsum(x)
plot(n, running, type = "l", log = "x", ylim = c(1, 3.5),
     xlab = "n", ylab = "running estimate")
abline(h = theta, lty = 2)
abline(h = theta + c(-0.1, 0.1), lty = 3)
```

Figure 9.1 shows the expected behaviour: the path fluctuates strongly for small n, crosses the edge of the band $\theta \pm 0.1$ several times and ends inside it. A single path, however, cannot tell us how likely it is to be outside the band at a given n, and this is what consistency is about.

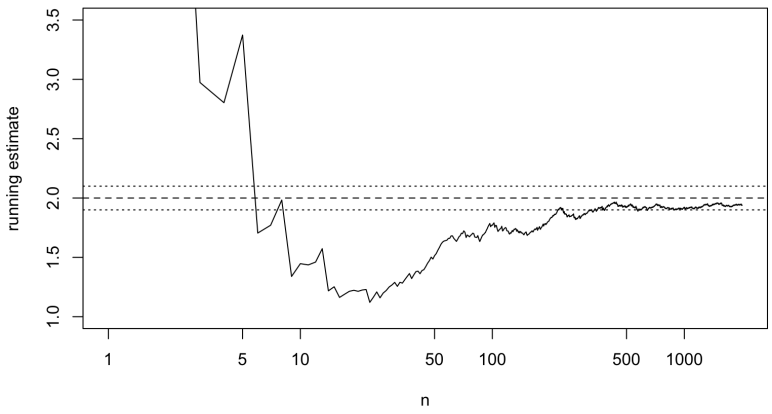


Figure 9.1: The running estimate $1/\bar{X}_n$ for the exponential rate $\theta = 2$, plotted against n on a logarithmic axis. The truth is given by the dashed line, the band $\theta \pm 0.1$ is given by the dotted lines.

9.2 The Sampling Distribution as n Grows

In this section we compare the sampling distribution at three sample sizes and which fraction of estimates misses the truth by more than a fixed amount.

By definition 2.9, consistency means that for every band around the truth, here $\theta \pm 0.1$, the probability $\mathbb{P}_\theta(|\hat{\theta}_n - \theta| \geq 0.1)$ of missing the band goes to zero as n increases. A probability is a property of many repetitions, and thus, as in lecture 6, we use `replicate` to compute 10000 independent estimates, at each of the sample sizes $n = 10$, $n = 100$ and $n = 1000$. The fraction of estimates outside the band estimates the probability; as in appendix B, the mean of a condition is the fraction of cases where it holds. Before you run the code, write down three guesses for the three fractions. Will the fraction for $n = 1000$ be close to zero?

```
set.seed(2026)
est10   <- replicate(10000, 1 / mean(rexp(10, rate = theta)))
est100  <- replicate(10000, 1 / mean(rexp(100, rate = theta)))
est1000 <- replicate(10000, 1 / mean(rexp(1000, rate = theta)))
c(mean(abs(est10 - theta) >= 0.1),
  mean(abs(est100 - theta) >= 0.1),
  mean(abs(est1000 - theta) >= 0.1))
```

```
## [1] 0.8756 0.6211 0.1160
```

The probability of missing the band decreases from 0.88 at $n = 10$ to 0.12 at $n = 1000$, as consistency requires, but at $n = 1000$ it is still far from zero: the band is narrow, and consistency only says that the probability goes to zero eventually. To see the whole sampling distribution we draw the three histograms with a common horizontal axis, as in lecture 6. New is the argument `freq = FALSE`, which draws the histogram on the density scale, so that the total area of the bars is one and a narrower histogram must be a taller one. Values outside the

horizontal range are not shown; for $n = 10$ these are about 8% of the estimates, almost all of them in the long right tail. The dotted lines mark the band.

```
par(mfrow = c(3, 1))
hist(est10, breaks = 60, freq = FALSE, col = "grey85", main = "n = 10",
     xlab = "estimate", xlim = c(1, 3.5))
abline(v = theta + c(-0.1, 0.1), lty = 3)
hist(est100, breaks = 40, freq = FALSE, col = "grey85", main = "n =
    100",
     xlab = "estimate", xlim = c(1, 3.5))
abline(v = theta + c(-0.1, 0.1), lty = 3)
hist(est1000, breaks = 40, freq = FALSE, col = "grey85", main = "n =
    1000",
     xlab = "estimate", xlim = c(1, 3.5))
abline(v = theta + c(-0.1, 0.1), lty = 3)
```

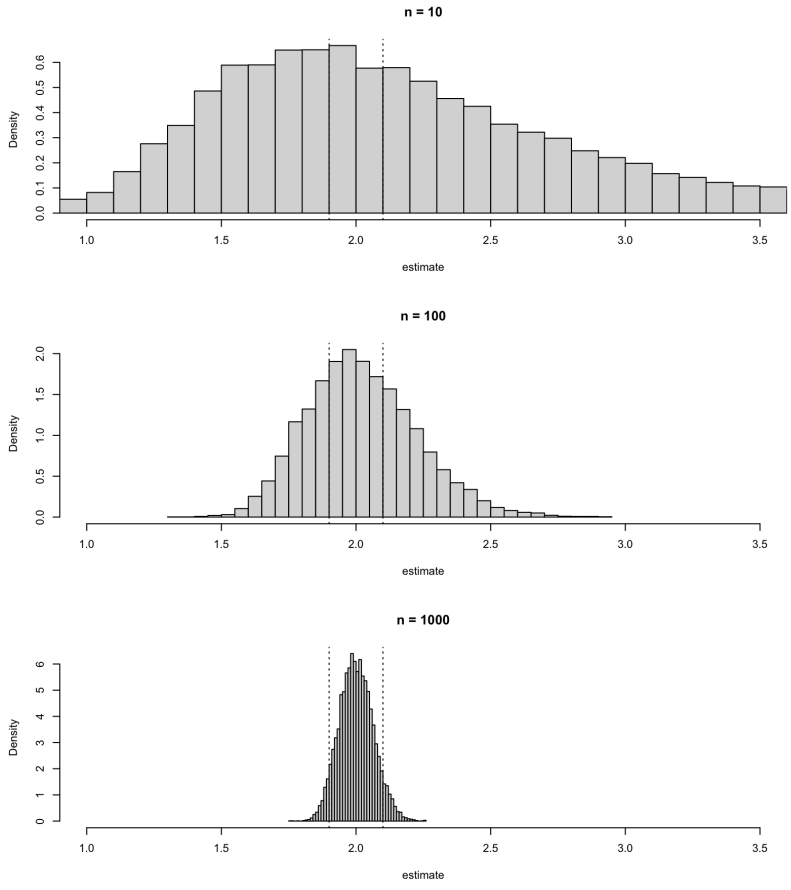


Figure 9.2: Histograms of 10000 simulated values of $1/\bar{X}_n$ for the exponential rate $\theta = 2$, at $n = 10$, $n = 100$ and $n = 1000$, on the density scale and with a common horizontal axis. The dotted lines mark the band $\theta \pm 0.1$.

Figure 9.2 is the picture promised in lectures 3 and 6: the histogram gets tighter around the truth as n grows. For $n = 10$ it is also visibly skewed to the right, since $1/\bar{X}_n$ is large whenever the sample mean is small, while for $n = 1000$ it is close to symmetric.

9.3 The Mean Squared Error as a Function of the Sample Size

In this section we estimate the mean squared error of two estimators for a range of sample sizes, collect the results in a table and plot them.

For a report a single number per sample size is often more useful than a histogram. In lecture 6 we estimated the mean squared error of the two uniform estimators $2\bar{X}$ and $\max_i X_i$ for $n = 10$. We now repeat this study for several sample sizes, which is the usual shape of a simulation study: an outer loop over a setting, an inner Monte Carlo run and a table of results. We first put both estimates into one function, so that both are computed from the same sample and returned as a vector of length two. The command `replicate` then collects the 10000 repetitions into a *matrix*, a rectangular table of numbers with two rows, one per estimator, and 10000 columns, one per repetition, and the command `rowMeans` can be used to average each row.

```
est_both <- function(n, theta) {
  sample <- runif(n, min = 0, max = theta)
  c(2 * mean(sample), max(sample))
}
```

The outer loop runs over a vector `ns` of sample sizes, using `seq_along(ns)` to count through its positions. The command `matrix(0, nrow = length(ns), ncol = 2)` can be used to set aside a matrix for the results, `mse[j, ]` refers to the j-th row of this matrix, and row and column names, set using `rownames` and `colnames`, can be used to make the matrix a readable table. As in lecture 6, the box below shows the complete script, with the lines of the loop kept together for clarity, but with the lines in a shuffled order.

Exercise. Before you read the script, you need to work out the correct order of lines to make the script work.

```
signif(mse, 3)
for (j in seq_along(ns)) {
  ests    <- replicate(10000, est_both(ns[j], theta))
  mse[j, ] <- rowMeans((ests - theta)^2)
}
mse   <- matrix(0, nrow = length(ns), ncol = 2)
set.seed(2026)
rownames(mse) <- ns
theta <- 1
colnames(mse) <- c("2 * mean", "max")
ns    <- c(5, 10, 20, 50, 100, 200, 500, 1000)
```

The rule for writing R scripts is still in force: all names must be defined before they are used, so `ns` must come before the matrix is created and `theta` before the loop, the seed must be set before the loop starts, and the display (using `signif` to round the numbers) must come last. As in lecture 6, several orders satisfy these rules.

For $n = 10$ the mean squared error of the maximum was approximately half of the one for $2\bar{X}$. Before you run the reconstructed script, make a prediction. Will this ratio stay close to two for $n = 1000$?

```
theta <- 1
ns    <- c(5, 10, 20, 50, 100, 200, 500, 1000)
mse   <- matrix(0, nrow = length(ns), ncol = 2)
set.seed(2026)
for (j in seq_along(ns)) {
  ests    <- replicate(10000, est_both(ns[j], theta))
  mse[j, ] <- rowMeans((ests - theta)^2)
}
rownames(mse) <- ns
colnames(mse) <- c("2 * mean", "max")
signif(mse, 3)
```

```
##      2 * mean      max
## 5    0.065900 4.58e-02
## 10   0.033300 1.52e-02
## 20   0.016600 4.48e-03
## 50   0.006610 7.51e-04
## 100  0.003310 1.91e-04
## 200  0.001660 5.07e-05
## 500  0.000668 7.86e-06
## 1000 0.000331 1.99e-06
```

The values in each column decrease towards zero as n increases, and thus both estimators are consistent by lemma 2.11. The ratio between the two columns is not close to two anymore: for $n = 1000$ the mean squared error for the maximum is smaller by a factor of approximately 170. This is expected, since the exact mean squared error $\theta^2/(3n)$ of exercise 4.4 decreases like $1/n$, while the exact value $2\theta^2/((n+1)(n+2))$ of exercise 2.2 decreases like $1/n^2$. On log-log axes, which the argument `log = "xy"` selects, a quantity proportional to n^{-k} is a straight line with slope $-k$. The command `points` can be used to add a second set of points to an existing plot, with the plotting symbol chosen by the argument `pch`, and the command `legend` can be used to add a key. Will the two sets of points be parallel?

```
plot(ns, mse[, 1], log = "xy", pch = 16, xlab = "n", ylab = "MSE",
     ylim = range(mse))
points(ns, mse[, 2], pch = 1)
lines(ns, theta^2 / (3 * ns), lty = 2)
lines(ns, 2 * theta^2 / ((ns + 1) * (ns + 2)), lty = 3)
legend("bottomleft", legend = c("2 * mean", "max"),
       pch = c(16, 1), lty = c(2, 3))
```

Both sets of points lie on straight lines: the slope is -1 for $2\bar{X}$ and the slope is -2 for the maximum. The rate $1/n$ is the usual rate of decrease for a mean squared error; it is the rate of the sample mean in example 2.8. The faster rate for the maximum is a consequence of the support of the uniform distribution moving with θ, the same feature which, by the remark after example 7.4 in lecture 7, keeps the uniform distribution out of the exponential families. We will consider this example again in lecture 13.

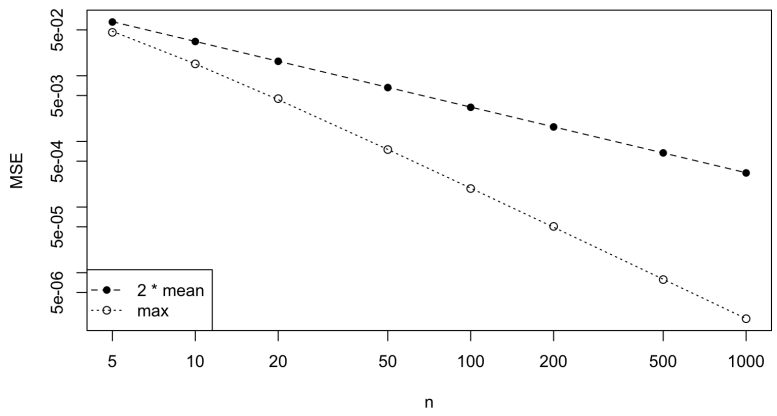


Figure 9.3: Simulated mean squared errors for $2\bar{X}$ (filled circles) and $\max_i X_i$ (open circles), for the uniform parameter $\theta = 1$, on logarithmic axes. The exact values $\theta^2/(3n)$ (dashed line) and $2\theta^2/((n+1)(n+2))$ (dotted line) are also shown.

9.4 A Sufficient Statistic in R

In this section we check by simulation that the conditional distribution of a Poisson sample given its total does not depend on θ.

To conclude this lecture, we consider an example from lecture 8: for Poisson samples, the total $\sum_i X_i$ is sufficient by example 8.6. By definition of sufficiency, given the total, the conditional distribution of the sample does not depend on θ. We can see this by simulation, without any algebra. We generate many Poisson samples of size $n = 5$, keep only those whose total equals 10, and look at the first observation in the kept samples; we do this for two different values of θ.

The command `replicate` returns a matrix with 5 rows and one column per sample, `colSums` computes the column totals, and `samples[1, totals == 10]` selects from the first row the columns where the condition holds. The command `table` can be used to count how often each value occurs, and dividing by the number of kept samples turns the counts into proportions. Before you run the code, make a prediction. Will the two tables of proportions be different, since the two values of θ are different? What will the mean of the first observation be in each case?

```
first_given_total <- function(theta, N) {
  samples <- replicate(N, rpois(5, lambda = theta))
  totals  <- colSums(samples)
  samples[1, totals == 10]
}
set.seed(2026)
first1 <- first_given_total(1.5, 100000)
first3 <- first_given_total(3, 100000)
c(length(first1), length(first3))
round(table(first1) / length(first1), 3)
round(table(first3) / length(first3), 3)
c(mean(first1), mean(first3))
```

```
## [1] 8749 4870
## first1
##     0     1     2     3     4     5     6     7
## 0.110 0.269 0.296 0.203 0.087 0.028 0.006 0.001
## first3
##     0     1     2     3     4     5     6     7
## 0.105 0.266 0.300 0.210 0.087 0.025 0.007 0.001
## [1] 2.002857 2.014168
```

The total equals 10 for about 9% of the samples when $\theta = 1.5$ and for about 5% when $\theta = 3$, so the two values of θ clearly matter for how often the total is 10. Once the total is fixed, however, the two tables agree up to simulation error, and both means are close to $10/5 = 2$: given the total, the first observation has the same distribution whether $\theta = 1.5$ or $\theta = 3$. This is definition 8.1 in action. All that the data can tell us about θ is carried by the total, and the distribution of the individual counts around it is the same for every θ. (The proportions in the table are themselves estimates; with about 9000 and 5000 kept samples the simulation error is between 0.005 and 0.007 for the larger entries.)

9.5 A Task to Take Home

The following task is a review of the topics of this lecture, in the style of a practical report. Following the structure of section 9.3, for the exponential rate with $\theta = 2$ and sample sizes given in `ns`, estimate the bias and mean squared error of the maximum likelihood estimator $1/\bar{X}$ using 10000 repetitions each, in a table with named rows and columns. Create a plot of the mean squared error against n, using logarithmic axes. From the plot, determine the slope. Compare the empirical bias to the exact value from exercise 2.3. Write three sentences about what the table and plot show about the consistency of the estimator $1/\bar{X}$.

Before the next session, type in the commands from this session and check that the output matches what is shown in the text.

Summary.

- Appendix B is the written reference for `cumsum`, the matrices returned by `replicate`, `colSums`, `rpois`, `table` and `legend`.
- The command `cumsum` can be used to compute all running estimates of a growing sample at once. One path shows an estimate settling down, but consistency is a statement about the probability of missing a band at a fixed n, which only many repetitions can show.
- When the Monte Carlo study is repeated at several sample sizes, the fraction of estimates outside the band decreases and the histograms get tighter around the truth. With `freq = FALSE` a histogram is drawn on the density scale.
- A loop over a vector of sample sizes, with the results of each Monte Carlo run stored in one row of a matrix, gives the mean squared error as a function of n. On log-log axes a mean squared error proportional to n^{-k} is a straight line with slope $-k$: for the uniform parameter, $2\bar{X}$ shows slope -1, the usual rate, and $\max_i X_i$ slope -2.
- Keeping only the Poisson samples with a given total, the distribution of the first observation is the same for two different values of θ: the total is sufficient, in the sense of definition 8.1.

Lecture 10

Improving Estimators: Rao–Blackwell and Lehmann–Scheffé

In lecture 8 we have seen that a sufficient statistic T can be used to summarise all the information in the data about the parameter. In this lecture we will use this idea to construct improved versions of estimators. The Rao–Blackwell theorem states that conditioning an unbiased estimator on the value of T never increases its variance. The Lehmann–Scheffé theorem states that, if T is a complete sufficient statistic, this procedure gives the best unbiased estimator.

10.1 The Rao–Blackwell Theorem

Assume that we want to estimate a function $g(\theta)$ of the parameter, *e.g.* we could be interested in the probability $\mathbb{P}_\theta(X_1 = 0)$ of zero counts. As in lecture 2, an estimator U with finite mean is *unbiased* for $g(\theta)$, if $\mathbb{E}_\theta(U) = g(\theta)$ for all $\theta \in \Theta$. Unbiased estimators are often easy to find, but a crude one, such as the indicator of the event $\{X_1 = 0\}$, uses only a small part of the available sample. The Rao–Blackwell theorem uses a sufficient statistic T to improve an existing estimator by replacing U with the conditional expectation $V = \mathbb{E}_\theta(U|T)$, *i.e.* the average of U for all samples which have the same value of T as the observed sample.

Proposition 10.1 (Rao–Blackwell)**.** *Let T be a sufficient statistic for θ and let U be an unbiased estimator for $g(\theta)$ with $\mathrm{Var}_\theta(U) < \infty$ for all θ. Then $V = \mathbb{E}_\theta(U|T)$ has the following properties.*

1. *The random variable V does not depend on θ, but is a function of T alone and thus a statistic.*
2. *The statistic V is an unbiased estimator for $g(\theta)$.*
3. *For all $\theta \in \Theta$ we have $\mathrm{Var}_\theta(V) \leq \mathrm{Var}_\theta(U)$.*

Proof. Since T is a sufficient statistic, the conditional distribution of the sample $(X_1, \ldots, X_n)$ given T does not depend on θ; for discrete data this is definition 8.1, and for continuous data it is the property stated after proposition 8.3. Since U is

a function of the sample, the conditional distribution of U given T, and thus the conditional expectation V, do not depend on θ either. Like every conditional expectation given T, the random variable V is a function of T, as explained in appendix A, and since this function does not involve the parameter θ, the random variable V is a statistic in the sense of definition 1.3.

For the second statement, using the tower property from proposition A.9, we find

$$\mathbb{E}_\theta(V) = \mathbb{E}_\theta\big(\mathbb{E}_\theta(U\,|\,T)\big) = \mathbb{E}_\theta(U) = g(\theta)$$

for all θ. This shows that V is unbiased.

Finally, for the third statement, we can use the law of total variance from proposition A.11, with U instead of X and T instead of Y. Since $\mathbb{E}_\theta(U\,|\,T) = V$ we get

$$\mathrm{Var}_\theta(U) = \mathbb{E}_\theta\big(\mathrm{Var}_\theta(U\,|\,T)\big) + \mathrm{Var}_\theta(V).$$

Since the conditional variance is non-negative, as explained in appendix A, we get $\mathrm{Var}_\theta(V) \leq \mathrm{Var}_\theta(U)$. This completes the proof. □

Since both estimators are unbiased, their mean squared errors equal their variances by lemma 2.7, and thus V is at least as good as U. The procedure of passing from U to V is called *Rao–Blackwellisation*. Sufficiency is needed here: conditioning on a statistic which is not sufficient will in general give a quantity which depends on θ and is thus not an estimator.

Example 10.2. Let $X_1, \ldots, X_n$ be i.i.d. Poisson with parameter θ and assume that we want to estimate $g(\theta) = \mathbb{P}_\theta(X_1 = 0) = e^{-\theta}$. The indicator $U = \mathbf{1}_{\{X_1=0\}}$ is unbiased for $g(\theta)$, since $\mathbb{E}_\theta(U) = \mathbb{P}_\theta(X_1 = 0)$. By corollary 8.5 the sum $T = \sum_i X_i$ is sufficient. Given $T = t$, the first observation is distributed as $\mathrm{Binomial}(t, 1/n)$; this is shown in exercise 10.1. Thus we have

$$V = \mathbb{E}_\theta(U\,|\,T) = \mathbb{P}_\theta(X_1 = 0\,|\,T) = \Big(1 - \frac{1}{n}\Big)^T.$$

The Rao–Blackwellised estimator now uses all n observations. Exercise 10.3 applies the same procedure to a continuous model.

10.2 Completeness and the Lehmann–Scheffé Theorem

The Rao–Blackwell theorem improves an existing estimator, but two different unbiased estimators could in principle be improved to two different functions of T. To rule this out we need one more property of a statistic, concerning functions of the statistic which are unbiased estimators for zero.

Definition 10.3. A statistic T is *complete*, if for every function w such that $w(T)$ has finite mean under $\mathbb{P}_\theta$ for all $\theta \in \Theta$, the condition $\mathbb{E}_\theta\big(w(T)\big) = 0$ for all $\theta \in \Theta$ implies $\mathbb{P}_\theta\big(w(T) = 0\big) = 1$ for all $\theta \in \Theta$.

This means that a function of a complete statistic which has expectation zero for all parameter values is itself zero with probability one. The condition is violated for statistics which do not reduce the data enough: in exercise 10.2 we will see that the whole Bernoulli sample is sufficient, but not complete, whereas

the total $\sum_i X_i$ is complete by proposition 10.6 below. Completeness is a separate property from minimal sufficiency: a minimal sufficient statistic need not be complete. If T is complete, two unbiased estimators for the same quantity which are both functions of T agree with probability one, and thus every Rao–Blackwellisation leads to the same estimator. Before we prove this, we name the target.

Definition 10.4. An unbiased estimator V for $g(\theta)$ with $\operatorname{Var}_\theta(V) < \infty$ for all $\theta \in \Theta$ is a *uniformly minimum variance unbiased estimator* (UMVUE) for $g(\theta)$, if $\operatorname{Var}_\theta(V) \leq \operatorname{Var}_\theta(W)$ for all $\theta \in \Theta$ and for every unbiased estimator W for $g(\theta)$ with finite variance.

The word "uniformly" refers to the parameter: the inequality has to hold for every θ at once, and there is no reason a priori why such an estimator should exist. The following theorem shows that, whenever a complete sufficient statistic exists, a UMVUE exists as well and is easy to identify.

Theorem 10.5 (Lehmann–Scheffé)**.** *Let T be a complete sufficient statistic for θ and let $V = \varphi(T)$ be a function of T which is unbiased for $g(\theta)$ and has $\operatorname{Var}_\theta(V) < \infty$ for all θ. Then V is a UMVUE for $g(\theta)$.*

Proof. We first show that V is the only unbiased estimator for $g(\theta)$ among the functions of T. Let $\psi(T)$ be another unbiased estimator for $g(\theta)$ which is a function of T. Then we have $\mathbb{E}_\theta(\psi(T)) = g(\theta) = \mathbb{E}_\theta(\varphi(T))$ for all θ. The function $w = \psi - \varphi$ satisfies $\mathbb{E}_\theta(w(T)) = 0$ for all θ and, since T is complete, by definition 10.3, we have $\mathbb{P}_\theta(w(T) = 0) = 1$ for all θ, *i.e.* $\psi(T) = \varphi(T)$ with probability one.

Now let W be any unbiased estimator for $g(\theta)$ with finite variance. Then, by the Rao–Blackwell theorem, proposition 10.1, the random variable $\mathbb{E}_\theta(W|T)$ is a function of T which is unbiased for $g(\theta)$. Thus, by the first step of the proof, it equals V with probability one. The Rao–Blackwell theorem then gives

$$\operatorname{Var}_\theta(V) = \operatorname{Var}_\theta\big(\mathbb{E}_\theta(W|T)\big) \leq \operatorname{Var}_\theta(W)$$

for all θ, *i.e.* V is a UMVUE. This completes the proof. □

The theorem turns the search for a best unbiased estimator into a two-step recipe: find a complete sufficient statistic T, then find any function of T which is unbiased for $g(\theta)$, by adjusting a constant or by Rao–Blackwellising a crude unbiased estimator. For the models of this module the following two facts settle the question of completeness.

Proposition 10.6.

1. *Let $X_1, \ldots, X_n$ be an i.i.d. sample from a one-parameter exponential family of full rank, as given in definitions 7.1 and 7.8. Then the natural statistic $T_n = \sum_{i=1}^n T(X_i)$ of the sample is complete. For a k-parameter exponential family of full rank, as given in definition 7.6, the vector*

$$\Big(\sum_{i=1}^n T_1(X_i), \ldots, \sum_{i=1}^n T_k(X_i)\Big)$$

of summed natural statistics is complete.

2. *Let $X_1, \dots, X_n$ be i.i.d. and uniformly distributed on the interval $[0, \theta]$ where $\theta > 0$. Then the sample maximum $\max_i X_i$ is complete.*

We state this proposition without proof, and both statements may be used without proof in what follows. The first statement rests on the uniqueness theorem for Laplace transforms. The second statement rests on taking derivatives, with respect to θ, of the relation $\int_0^\theta w(t)\, t^{n-1}\, \mathrm{d}t = 0$, which is the condition $\mathbb{E}_\theta(w(\max_i X_i)) = 0$ up to a constant factor. Together with corollary 8.5 and example 8.4 they provide a complete sufficient statistic for every family in tables 1.1 and 1.2 (with m known for the binomial). A one-to-one function $\varphi(T)$ of a complete sufficient statistic T is again complete and sufficient: sufficiency was shown in lecture 8, and if $\mathbb{E}_\theta(w(\varphi(T))) = 0$ for all θ, then $w \circ \varphi$ is a function of T with mean zero and thus, since T is complete, $w(\varphi(T)) = 0$ with probability one. In particular, we can use the sample mean $\bar{X}$ instead of $\sum_i X_i$ if required.

Example 10.7. For the Poisson sample from example 10.2, the natural parameter $\eta = \log \theta$ takes values on all of $\mathbb{R}$, and thus the family is of full rank in the sense of definition 7.8. By corollary 8.5 and proposition 10.6 the sum $T = \sum_i X_i$ is a complete and sufficient statistic. Since $\bar{X} = T/n$ is a function of T, is unbiased for θ and has finite variance θ/n, it is the UMVUE for θ by theorem 10.5. Similarly, the Rao–Blackwellised indicator $V = (1 - 1/n)^T$ takes values in $[0, 1]$ and thus has finite variance, and it is the UMVUE for $e^{-\theta}$.

Example 10.8. Let $X_1, \dots, X_n$ be i.i.d. $N(\mu, \sigma^2)$ with both parameters unknown. This is a two-parameter exponential family with natural statistic $T(x) = (x, x^2)$, and the pair $(\bar{X}, S^2)$ is sufficient by example 8.7. The family has full rank, and thus the pair is also complete by proposition 10.6; exercise 10.5 verifies the rank condition. Since $\bar{X}$ and S^2 are unbiased for μ and σ^2 by lemmas 2.2 and 2.3, they are the UMVUEs for these parameters.

The second example puts section 2.3 into perspective: there the biased estimator $\frac{1}{n+1}\sum_i (X_i - \bar{X})^2$ had smaller mean squared error than S^2. This is no contradiction, since the Lehmann–Scheffé theorem compares S^2 to unbiased estimators only.

In lecture 13 we will see a second way to derive optimality, using the Cramér–Rao bound, which does not require conditioning.

Summary.

- Rao–Blackwell: If an unbiased estimator is conditioned on a sufficient statistic T, the conditioned estimator is unbiased and has no larger variance.
- A UMVUE is an unbiased estimator which has minimal variance among all unbiased estimators, for all θ.
- A statistic T is complete, if every function of T with expectation zero for all θ is zero with probability one.
- Lehmann–Scheffé: If T is complete and sufficient, then the only function of T which is unbiased for $g(\theta)$ is the UMVUE for $g(\theta)$.
- The natural statistic for a full-rank exponential family and the maximum of a uniform sample are complete (stated without proof).

Exercise 10.1. Let $n \geq 2$, let $X_1, \ldots, X_n$ be i.i.d. Poisson with parameter $\theta > 0$, let $T = \sum_{i=1}^n X_i$ and let $U = \mathbf{1}_{\{X_1=0\}}$, as in example 10.2. You may use the fact that a sum of independent Poisson random variables is again Poisson, with the parameters added.

1. Show that U is an unbiased estimator for $e^{-\theta}$ and determine the variance of this estimator.

2. Show that for $t \in \mathbb{N}_0$ and $x \in \{0, 1, \ldots, t\}$ we have

$$\mathbb{P}_\theta(X_1 = x | T = t) = \binom{t}{x} \Big(\frac{1}{n}\Big)^x \Big(1 - \frac{1}{n}\Big)^{t-x},$$

   i.e. that the conditional distribution of X_1 given $T = t$ is Binomial$(t, 1/n)$ and that this distribution does not depend on θ.

3. Deduce that $V = \mathbb{E}_\theta(U|T) = (1 - 1/n)^T$.

4. Show directly that V is an unbiased estimator for $e^{-\theta}$, using the probability generating function (7.2) of the Poisson distribution with parameter $n\theta$, which is given by $\mathbb{E}_\theta(z^T) = \exp\big(n\theta(z-1)\big)$ (exercise 7.6).

5. Explain why V is the UMVUE for $e^{-\theta}$ and discuss the differences between this UMVUE and the MLE $e^{-\bar{X}}$, introduced in exercise 5.7.

Exercise 10.2. Let $X_1, \ldots, X_n$ be i.i.d. Bernoulli with success probability $\theta \in (0, 1)$, where $n \geq 2$, and let $T = (X_1, \ldots, X_n)$ be the whole sample.

1. Show that $w(T) = X_1 - X_2$ satisfies $\mathbb{E}_\theta\big(w(T)\big) = 0$ for all θ and $\mathbb{P}_\theta\big(w(T) = 0\big) < 1$.

2. Using definition 10.3, show that T is not complete, despite T being sufficient.

Exercise 10.3. Let $n \geq 2$ and let $X_1, \ldots, X_n$ be i.i.d. exponential with rate $\theta > 0$, *i.e.* $f(x;\theta) = \theta e^{-\theta x}$ for $x \geq 0$ and let $T = \sum_{i=1}^n X_i$. You may use the fact that the sum of k independent random variables of this type has density $f_k(t) = \theta^k t^{k-1} e^{-\theta t}/(k-1)!$ for $t \geq 0$.

1. Explain why X_1 is an unbiased estimator for the mean $1/\theta$ and why T is sufficient for θ.

2. Let $R = \sum_{i=2}^n X_i$. Use the independence of X_1 and R to find the joint density of (X_1, T). Deduce the conditional density of X_1 given $T = s$. The result is

$$f(x|s) = \frac{(n-1)(s-x)^{n-2}}{s^{n-1}} \qquad \text{for } 0 < x < s,$$

   and zero elsewhere. Check that this density does not depend on θ.

3. Show that $\mathbb{E}_\theta(X_1|T = s) = s/n$ and conclude that Rao–Blackwellising X_1 leads to the sample mean $\bar{X}$, again unbiased for $1/\theta$.

4. Find a second argument for $\mathbb{E}_\theta(X_1|T) = T/n$, based on the symmetry of the sample, without any integration.

Exercise 10.4. Let $X_1, \ldots, X_n$ be i.i.d. uniformly distributed on the interval $[0, \theta]$ where $\theta > 0$, and let $M = \max_i X_i$.

1. Using the mean of M from exercise 2.2, show that $\hat{\theta} = (n+1)M/n$ is an unbiased estimator for θ.
2. Show that $\hat{\theta}$ is the UMVUE for θ.
3. The estimator $\tilde{\theta} = 2\bar{X}$ is also unbiased for θ. Determine the variances of $\hat{\theta}$ and of $\tilde{\theta}$ and verify that they are ordered as in theorem 10.5.
4. Without computing any conditional expectations, find $\mathbb{E}_\theta(\tilde{\theta}|M)$.

Exercise 10.5. Let $X_1, \ldots, X_n$ be i.i.d. $N(\mu, \sigma^2)$, where both parameters are unknown. This is the situation of example 10.8.

1. Write the density of an observation in the form from definition 7.6, confirm that the natural statistic is $T(x) = (x, x^2)$ and determine the natural parameter $\eta(\mu, \sigma^2)$ and the function $B(\mu, \sigma^2)$.
2. Show that the family of densities has full rank, in the sense of definition 7.8, by determining the range of η.
3. Deduce that $(\sum_i X_i, \sum_i X_i^2)$ is complete and sufficient and argue that the same holds for the pair $(\bar{X}, S^2)$.

Lecture 11

Fisher Information

Since the data are random, the score function from lecture 4 is a random variable, and the Fisher information, given by the variance of the score, measures how much information the sample contains about the parameter. In lecture 13 we will use the Fisher information to obtain a lower bound for the variance of any unbiased estimator, and in lecture 17 we will see that the reciprocal of the Fisher information is the asymptotic variance of the maximum likelihood estimator. In this lecture we will define the Fisher information, prove two identities which allow us to compute the Fisher information, and we will compute the Fisher information for the standard families of models.

11.1 Setting and Assumptions

Throughout this lecture, $X_1, \dots, X_n$ are i.i.d. with density or probability weights $f(x;\theta)$, where the parameter space $\Theta \subseteq \mathbb{R}$ is an open interval. As in lecture 4, the log-likelihood is the sum $\ell(\theta) = \sum_{i=1}^n \ell_i(\theta)$ of the terms $\ell_i(\theta) = \log f(X_i;\theta)$ and the score of definition 4.4 is its derivative $\ell'(\theta) = \sum_i \ell_i'(\theta)$. In this section we study the distribution of the score under $\mathbb{P}_\theta$, for a fixed value of θ.

The results of this lecture are obtained by differentiating the identity

$$\int f(x;\theta)\,\mathrm{d}x = 1$$

with respect to θ. Throughout the section we make the following assumptions:

1. The support $\{x \mid f(x;\theta) > 0\}$ of the distribution does not depend on θ.
2. For every x the function $\theta \mapsto f(x;\theta)$ is twice continuously differentiable on Θ.
3. The integral of $f(x;\theta)$ over the support may be differentiated under the integral sign twice, so that

$$\frac{\mathrm{d}^k}{\mathrm{d}\theta^k} \int f(x;\theta)\,\mathrm{d}x = \int \frac{\partial^k}{\partial\theta^k} f(x;\theta)\,\mathrm{d}x \qquad \text{for } k = 1, 2, \tag{11.1}$$

   where the integrals on the right-hand side exist, with sums in place of integrals for a discrete model.
4. The score of a single observation has finite variance, $\mathrm{Var}_\theta(\ell_1'(\theta)) < \infty$ for all $\theta \in \Theta$, so that the quantities defined below are finite.

These four assumptions are referred to as the conditions of this section. The one-parameter models from tables 1.1 and 1.2, and the two-parameter models with one parameter held fixed (with m known for the binomial distribution), satisfy these conditions, with the exception of the uniform distribution (see section 11.6). So do the exponential families of lecture 7, as explained after lemma 7.10 there.

11.2 The Score Has Mean Zero

The first identity is that, if θ is the correct parameter value, the data do not systematically deviate from this value: the score at the true parameter value is zero on average.

Lemma 11.1. *Let the situation be as described in section 11.1. Then*

$$\mathbb{E}_\theta\big(\ell_i'(\theta)\big) = 0$$

for $i = 1, \ldots, n$, and thus

$$\mathbb{E}_\theta\big(\ell'(\theta)\big) = 0$$

for all $\theta \in \Theta$.

Proof. Using the chain rule, we find the derivative of $\ell_i(\theta) = \log f(X_i; \theta)$ as

$$\ell_i'(\theta) = \frac{\frac{\partial}{\partial\theta} f(X_i;\theta)}{f(X_i;\theta)}. \tag{11.2}$$

Since X_i lies in the support with probability one, the denominator is positive and the ratio is well defined. Cancelling the density against the denominator we find

$$\begin{aligned}\mathbb{E}_\theta\big(\ell_i'(\theta)\big) &= \int \frac{\frac{\partial}{\partial\theta} f(x;\theta)}{f(x;\theta)}\, f(x;\theta)\,\mathrm{d}x = \int \frac{\partial}{\partial\theta} f(x;\theta)\,\mathrm{d}x \\ &= \frac{\mathrm{d}}{\mathrm{d}\theta}\int f(x;\theta)\,\mathrm{d}x = \frac{\mathrm{d}}{\mathrm{d}\theta}\,1 = 0,\end{aligned}$$

where the integral is over the support of the distribution, which does not depend on θ (with a sum in place of the integral for a discrete model), and the third equality comes from the case $k = 1$ of (11.1). Summing over i gives the statement for the whole sample. This completes the proof. □

If the parameter value is wrong, the score is not mean zero: for the exponential model from example 4.5, the score $\ell'(\theta) = n/\theta - \sum_i X_i$ has expectation $\mathbb{E}_{\theta_0}\big(\ell'(\theta)\big) = n/\theta - n/\theta_0$ if the true parameter value is θ_0, positive for $\theta < \theta_0$ and negative for $\theta > \theta_0$. The score on average points towards the correct parameter value from both sides. In lecture 16 we will see the corresponding property of the log-likelihood, which on average is larger at the true parameter value than at any other value, and use it to prove that the maximum likelihood estimator is consistent.

11.3 Fisher Information

The next question we could ask is about the fluctuations of the score around zero. If the score is typically large in magnitude, this indicates that the data allow us to distinguish well between nearby parameter values. In contrast, if the score is typically close to zero, this indicates that the log-likelihood is nearly flat at the truth. The variance of the score is a measure for how informative the sample is.

Definition 11.2. Under the conditions of section 11.1, the *Fisher information* of the sample $X_1, \dots, X_n$ about θ is given by

$$\mathcal{I}_n(\theta) = \mathrm{Var}_\theta(\ell'(\theta)),$$

where the variance of the score is taken at θ. The Fisher information of a single observation is denoted by $\mathcal{I}(\theta) = \mathcal{I}_1(\theta)$ and is given by the variance of $\ell_1'(\theta)$.

The subscript in $\mathcal{I}_n$ is the sample size. Where we compare two different parametrisations of the same model, we write $\mathcal{I}_\theta$, $\mathcal{I}_\eta$ or $\mathcal{I}_\psi$ for the information of a single observation about the parameter named in the subscript.

The variance of the score is often awkward to compute directly, and the second expression of the following lemma is nearly always the quickest route to the information.

Lemma 11.3. *Suppose we are in the situation of section 11.1. Then*

$$\mathcal{I}_n(\theta) = \mathbb{E}_\theta(\ell'(\theta)^2) = -\mathbb{E}_\theta(\ell''(\theta))$$

for all $\theta \in \Theta$.

Proof. The first equality follows from the fact that $\ell'(\theta)$ has mean zero by lemma 11.1, and thus the variance equals the second moment. The second equality is obtained by taking the derivative of the ratio (11.2) using the quotient rule:

$$\ell_i''(\theta) = \frac{\frac{\partial^2}{\partial\theta^2} f(X_i;\theta)}{f(X_i;\theta)} - \Big(\frac{\frac{\partial}{\partial\theta} f(X_i;\theta)}{f(X_i;\theta)}\Big)^2 = \frac{\frac{\partial^2}{\partial\theta^2} f(X_i;\theta)}{f(X_i;\theta)} - \ell_i'(\theta)^2.$$

Similar to the proof of lemma 11.1, the expectation of the first term vanishes, using the case $k = 2$ of (11.1):

$$\mathbb{E}_\theta\Big(\frac{\frac{\partial^2}{\partial\theta^2} f(X_i;\theta)}{f(X_i;\theta)}\Big) = \int \frac{\partial^2}{\partial\theta^2} f(x;\theta)\,\mathrm{d}x = \frac{\mathrm{d}^2}{\mathrm{d}\theta^2}\int f(x;\theta)\,\mathrm{d}x = 0.$$

Since the second moment of $\ell_i'(\theta)$ is finite by the conditions of section 11.1, we can take expectations term by term and get $\mathbb{E}_\theta(\ell_i''(\theta)) = -\mathbb{E}_\theta(\ell_i'(\theta)^2)$ for all i.

To go from ℓ_i to ℓ we also need to consider the cross terms: $\ell_1'(\theta), \dots, \ell_n'(\theta)$ are independent of each other, since the X_i are independent. Thus the variance of the sum is the sum of the variances. Since each term also has mean zero we find

$$\begin{aligned}\mathrm{Var}_\theta(\ell'(\theta)) &= \sum_{i=1}^n \mathrm{Var}_\theta(\ell_i'(\theta)) = \sum_{i=1}^n \mathbb{E}_\theta(\ell_i'(\theta)^2) \\ &= -\sum_{i=1}^n \mathbb{E}_\theta(\ell_i''(\theta)) = -\mathbb{E}_\theta(\ell''(\theta)),\end{aligned}$$

where we used $\ell''(\theta) = \sum_i \ell_i''(\theta)$ in the last step. This completes the proof. □

The second expression motivates the name Fisher *Information*: the quantity is the expected curvature of the log-likelihood at the true parameter value. Large information means that the log-likelihood has a sharp maximum, well-defined by the data. Small information means the log-likelihood is flat. Note that the identity holds in expectation only: for a given sample the curvature $-\ell''(\theta)$ is random and in general differs from $\mathcal{I}_n(\theta)$.

The proof also shows how the information of the sample relates to the information of one observation: each observation contributes the same amount, and n observations carry n times the information of one.

Proposition 11.4. *With the same assumptions as in section 11.1 we have*

$$\mathcal{I}_n(\theta) = n\,\mathcal{I}(\theta)$$

for all $\theta \in \Theta$.

Proof. The last display of the proof of lemma 11.3 writes $\mathcal{I}_n(\theta)$ as the sum of the n variances $\operatorname{Var}_\theta(\ell_i'(\theta))$, which are all equal to $\mathcal{I}(\theta)$ since the observations are identically distributed. This completes the proof. □

11.4 Examples

We now compute the information for four of the standard families of tables 1.1 and 1.2; the exercises cover the binomial, gamma and geometric distributions. By proposition 11.4 it suffices to work with a single observation $X = X_1$.

Example 11.5. For the Bernoulli distribution with success probability $\theta \in (0,1)$ we have $\ell_1(\theta) = X \log\theta + (1-X)\log(1-\theta)$ and thus

$$\ell_1'(\theta) = \frac{X}{\theta} - \frac{1-X}{1-\theta}, \qquad \ell_1''(\theta) = -\frac{X}{\theta^2} - \frac{1-X}{(1-\theta)^2}.$$

Using $\mathbb{E}_\theta(X) = \theta$ we find

$$\mathcal{I}(\theta) = -\mathbb{E}_\theta(\ell_1''(\theta)) = \frac{\theta}{\theta^2} + \frac{1-\theta}{(1-\theta)^2} = \frac{1}{\theta(1-\theta)}.$$

Thus, the information of the sample is $\mathcal{I}_n(\theta) = n/(\theta(1-\theta))$. Exercise 11.1 checks these identities for this model directly.

Example 11.6. For the Poisson distribution with mean $\theta > 0$ we have $\ell_1(\theta) = X\log\theta - \theta - \log X!$ and thus

$$\ell_1'(\theta) = \frac{X}{\theta} - 1, \qquad \ell_1''(\theta) = -\frac{X}{\theta^2}, \qquad \mathcal{I}(\theta) = \frac{\mathbb{E}_\theta(X)}{\theta^2} = \frac{1}{\theta}.$$

Example 11.7. In the case of the normal distribution with unknown mean μ and known variance σ^2, the parameter is μ and $\ell_1(\mu) = -\frac{1}{2}\log(2\pi\sigma^2) - (X-\mu)^2/(2\sigma^2)$. Thus we have

$$\ell_1'(\mu) = \frac{X-\mu}{\sigma^2}, \qquad \ell_1''(\mu) = -\frac{1}{\sigma^2}, \qquad \mathcal{I}(\mu) = \frac{1}{\sigma^2}.$$

Since the second derivative does not depend on the data, we do not need to consider the expectation. Note that $\operatorname{Var}_\mu(\bar{X}) = \sigma^2/n = 1/\mathcal{I}_n(\mu)$ by lemma 2.2, and in lecture 13 we will see that no unbiased estimator can have a smaller variance.

Example 11.8. For the exponential distribution with rate $\theta > 0$, we have $\ell_1(\theta) = \log\theta - \theta X$ and thus

$$\ell_1'(\theta) = \frac{1}{\theta} - X, \qquad \ell_1''(\theta) = -\frac{1}{\theta^2}, \qquad \mathcal{I}(\theta) = \frac{1}{\theta^2}.$$

Again, the second derivative is non-random. Section 11.5 shows how the information changes if we describe the model by the mean $1/\theta$ instead of the rate.

All four models from section 11.4 are exponential families, and exercise 11.2 shows that for such models the information is a variance which can be read off tables A.1 and A.2.

11.5 Reparametrisation

The information depends on which parameter we use to describe a model. The exponential distribution, for example, can be described by its rate or by its mean, and the following proposition gives the general rule for how the information changes.

Proposition 11.9. *Let $\psi = g(\theta)$ where $g\colon \Theta \to g(\Theta)$ is a twice continuously differentiable bijection with $g'(\theta) \neq 0$ for all $\theta \in \Theta$. Let $\mathcal{I}_\theta$ denote the Fisher information for a single observation when the model is parametrised by θ and let $\mathcal{I}_\psi$ be the information when the model is parametrised by ψ. Then,*

$$\mathcal{I}_\psi(\psi) = \frac{\mathcal{I}_\theta(\theta)}{g'(\theta)^2} \qquad \text{for } \psi = g(\theta).$$

Proof. With the new parametrisation, the density is $\tilde{f}(x;\psi) = f\big(x; g^{-1}(\psi)\big)$ and the log-likelihood for a single observation is $\tilde{\ell}_1(\psi) = \ell_1\big(g^{-1}(\psi)\big)$. Since g' is continuous and never vanishes on the interval Θ, it has the same sign throughout, so that g is strictly monotonic and g^{-1} is twice continuously differentiable. Using the chain rule, the conditions of section 11.1 for the parametrisation by θ carry over to the parametrisation by ψ; we omit the verification. The derivative of the inverse function is $(g^{-1})'(\psi) = 1/g'(\theta)$ for $\theta = g^{-1}(\psi)$. This gives

$$\tilde{\ell}_1'(\psi) = \ell_1'(\theta)\,(g^{-1})'(\psi) = \frac{\ell_1'(\theta)}{g'(\theta)}.$$

Since θ is fixed for fixed ψ, $1/g'(\theta)$ is a constant and taking variances under $\mathbb{P}_\theta$ gives $\mathcal{I}_\psi(\psi) = \mathcal{I}_\theta(\theta)/g'(\theta)^2$, as claimed. □

For the exponential distribution, reparametrising from the rate θ to the mean $\mu = 1/\theta$ changes the information $1/\theta^2$ from example 11.8 to $\mathcal{I}_\mu(\mu) = 1/\mu^2$; see exercise 11.6. Thus, the information is not a property of the model in isolation, but depends on the parametrisation. In lecture 28 we will use the transformation rule of the proposition to construct a prior distribution, the Jeffreys prior, which does not depend on the parametrisation chosen.

11.6 When the Identities Fail

The uniform distribution from example 1.2 shows what can go wrong if the conditions from section 11.1 are violated.

Example 11.10. Let X be uniformly distributed on $[0, \theta]$, *i.e.* $f(x;\theta) = 1/\theta$ for $0 \leq x \leq \theta$. The support of the distribution depends on θ, but we can still compute the derivatives on the support to see whether the identities hold. For $0 < X < \theta$ we have $\ell_1(\theta) = -\log\theta$ and thus $\ell_1'(\theta) = -1/\theta$ and $\ell_1''(\theta) = 1/\theta^2$. The score is a negative constant and thus we have $\mathbb{E}_\theta\big(\ell_1'(\theta)\big) = -1/\theta \neq 0$, in contradiction to lemma 11.1. The variance is zero, whereas $\mathbb{E}_\theta\big(\ell_1'(\theta)^2\big) = 1/\theta^2$ and $-\mathbb{E}_\theta\big(\ell_1''(\theta)\big) = -1/\theta^2$: the three expressions for lemma 11.3 take three different values, one of which is negative. The proof breaks down at the interchange between differentiation and integration: for this model, the two sides of (11.1) with $k = 1$ differ by the term $f(\theta;\theta) = 1/\theta$ from the moving upper limit, which is missed by the differentiation under the integral sign. Exercise 11.7 can be used to work out the two computations.

For this model the Fisher information is not defined, and the two results based on it, in lectures 13 and 17, do not apply.

Summary.

- Under the conditions of section 11.1 (a support free of θ, twice continuously differentiable density, differentiation under the integral sign and a score with finite variance), the score at the true parameter value has mean zero.
- The Fisher information $\mathcal{I}_n(\theta)$ is the variance of the score, and it equals $\mathbb{E}_\theta\big(\ell'(\theta)^2\big)$ and $-\mathbb{E}_\theta\big(\ell''(\theta)\big)$, the expected curvature of the log-likelihood at the truth.
- Information from independent observations can be combined, $\mathcal{I}_n(\theta) = n\mathcal{I}(\theta)$. The information depends on the parametrisation, and under $\psi = g(\theta)$ we get $\mathcal{I}_\psi(\psi) = \mathcal{I}_\theta(\theta)/g'(\theta)^2$.
- We have learned the following standard values: $1/\big(\theta(1-\theta)\big)$ for the Bernoulli distribution, $1/\theta$ for the Poisson distribution, $1/\sigma^2$ for the mean of the normal distribution and $1/\theta^2$ for the rate parameter in the exponential distribution. For the uniform distribution these relations do not hold, since the support of the distribution moves with the parameter.
- The information is the currency of the two results presented later: it gives a bound on the variance of unbiased estimators in lecture 13 and it determines the asymptotic variance of the MLE in lecture 17.

Exercise 11.1. Let $X_1, \ldots, X_n$ be i.i.d. Bernoulli with success probability $\theta \in (0, 1)$, and let $\ell_1'(\theta)$ be the score of a single observation from example 11.5.

1. Verify lemma 11.1 for this model by computing $\mathbb{E}_\theta(\ell_1'(\theta))$ directly from the two possible values of X_1.
2. Compute $\mathrm{Var}_\theta(\ell_1'(\theta))$ directly, and check that it agrees with the value of $\mathcal{I}(\theta)$ found in example 11.5.
3. Compare $1/\mathcal{I}_n(\theta)$ with the variance of the estimator $\bar{X}$ for θ.
4. Now let Y be a single observation from the binomial distribution with known number of trials m and success probability θ. Show that the information of Y about θ is $m/(\theta(1-\theta))$, and explain the relation to the Bernoulli case using proposition 11.4.

Exercise 11.2. Let $f(x;\eta) = h(x)\exp\big(\eta T(x) - K(\eta)\big)$ be an exponential family in its natural parametrisation from definition 7.9, where η ranges over an open interval inside the natural parameter space $\mathcal{N}$, and let X be a single observation.

1. Show that the Fisher information about η is given by $\mathcal{I}_\eta(\eta) = K''(\eta) = \mathrm{Var}_\eta(T(X))$.
2. Now let the family be written as $f(x;\theta) = h(x)\exp\big(\eta(\theta)T(x) - B(\theta)\big)$ as in definition 7.1, where η is twice continuously differentiable and takes values in the interior of $\mathcal{N}$. Show that $B(\theta) = K\big(\eta(\theta)\big)$ and that the information about θ is given by $\mathcal{I}_\theta(\theta) = \eta'(\theta)^2\,\mathrm{Var}_\theta(T(X))$.
3. Use the previous part to recover the information
$$\mathcal{I}(\theta) = \frac{1}{\theta(1-\theta)}$$
of the Bernoulli distribution from example 7.2.

Exercise 11.3. Let $X_1, \dots, X_n$ be i.i.d. with density $f(x;\theta) = \theta^{-2} x\, e^{-x/\theta}$ for $x \geq 0$, where $\theta > 0$, the gamma distribution with known shape 2 and scale θ from exercise 5.2, *i.e.* $\mathrm{Gamma}(2, 1/\theta)$ in the rate parametrisation from table A.2.

1. Compute the score $\ell_1'(\theta)$ of a single observation and verify that it has mean zero.
2. Show that $\mathcal{I}(\theta) = 2/\theta^2$, once via $-\mathbb{E}_\theta(\ell_1''(\theta))$ and once via the variance of the score, using $\mathrm{Var}_\theta(X_1) = 2\theta^2$.
3. Write the density in the form of definition 7.1 with $T(x) = x$, identify $\eta(\theta)$, and recover $\mathcal{I}(\theta)$ from exercise 11.2.
4. In exercise 5.2 we found that the MLE $\hat{\theta} = \bar{X}/2$ is unbiased with $\mathrm{MSE}(\hat{\theta}) = \theta^2/(2n)$. Compare this with $1/\mathcal{I}_n(\theta)$.

Exercise 11.4. Let $X_1, \dots, X_n$ be i.i.d. from the geometric distribution with parameter $p \in (0,1)$, *i.e.* $f(x;p) = (1-p)^{x-1}p$ for $x \in \mathbb{N}$, where $\mathbb{E}_p(X_1) = 1/p$ and $\mathrm{Var}_p(X_1) = (1-p)/p^2$, as in exercises 4.2 and 5.1.

1. Compute the score $\ell_1'(p)$ of a single observation and verify that it has mean zero.
2. Show that $\mathcal{I}(p) = 1/\big(p^2(1-p)\big)$ and write down $\mathcal{I}_n(p)$.

3. Verify the value from the previous part by taking the variance of the score.

Exercise 11.5. Let $X_1, \ldots, X_n$ be i.i.d. Poisson with mean $\theta > 0$.

1. From lecture 7 we know that the natural parameter is $\eta = \log \theta$ and the cumulant function is $K(\eta) = e^\eta$. Using exercise 11.2, find the information $\mathcal{I}_\eta(\eta)$ about η in terms of θ.

2. Recover $\mathcal{I}_\theta(\theta) = 1/\theta$ from the previous part using proposition 11.9.

3. In exercise 5.7 we have estimated the probability $p_0 = \mathbb{P}_\theta(X_1 = 0) = e^{-\theta}$. Determine the information $\mathcal{I}_{p_0}(p_0)$ about p_0 and describe its behaviour as $\theta \to \infty$. Give a one-sentence answer as to why this behaviour is reasonable.

Exercise 11.6. Let $X_1, \ldots, X_n$ be i.i.d. exponential with rate $\theta > 0$ and let $\mu = 1/\theta$ be the mean.

1. Using proposition 11.9 and example 11.8, verify the value $\mathcal{I}_\mu(\mu) = 1/\mu^2$ from section 11.5.

2. Show that the information of the sample about μ is n/μ^2 and compare the reciprocal of this quantity to the variance of the estimator $\bar{X}$ for μ.

Exercise 11.7. Let X be uniformly distributed on the interval $[0, \theta]$ as described in example 11.10. Compute both sides of (11.1) for this model, using $k = 1$, and verify that the difference between the two quantities corresponds to the contribution $f(\theta; \theta) = 1/\theta$ of the (moving) upper limit of the integral.

Lecture 12

Sampling Distributions and Standard Errors

This is the R session for week 4. We will watch the sampling distribution of a maximum likelihood estimator converge to a normal distribution, we will use the Fisher information from this week's lectures to compute a standard error and a confidence interval, and we will extend the simulation idea of lecture 9 from estimating a probability to estimating the coverage of an interval. (The latter skill will be needed for the coverage checks in the practical report, and for the size and power calculations in lecture 24.)

12.1 The Sampling Distribution of an Estimator

The model of the day is $X_1, \ldots, X_n$ i.i.d. from the exponential distribution with rate θ, where the maximum likelihood estimator is $\hat{\theta} = 1/\bar{X}$ by example 5.2. The command `rexp` can be used to simulate such a sample, and as in lecture 6 we package one run as a function. Before you run the code below, predict the two histograms: for $n = 10$, will the estimates form a symmetric bell around the true value $\theta = 1$? Will the peak sit above, below or at 1? How will the picture change for $n = 100$?

```
est_rate <- function(n, theta) {
  sample <- rexp(n, rate = theta)
  1 / mean(sample)
}
theta <- 1
set.seed(2026)
small <- replicate(10000, est_rate(10, theta))
large <- replicate(10000, est_rate(100, theta))
```

As in lecture 9, the histograms are drawn with `freq = FALSE`, so that a density can be drawn on top of them. The command `curve` can be used to plot any function of `x` on top of the existing plot, if the argument `add = TRUE` is used; we use it to overlay the density. The density we overlay is the normal density with mean θ and standard deviation $\theta/\sqrt{n}$, the normal approximation to the distribution of $\hat{\theta}$ for large n; section 12.2 explains where this spread comes from. The command `qqnorm` can be used to plot the sorted estimates against the quantiles of a standard normal distribution, and `qqline` can be used to add

the straight line which the points would follow if the estimates were exactly normally distributed. Points above the line at the right-hand end indicate a right tail which is heavier than that of the normal distribution, and points above the line at the left-hand end indicate a left tail which is lighter.

```
par(mfrow = c(2, 2))
hist(small, breaks = 40, freq = FALSE, col = "grey85", main = "n = 10",
     xlab = "estimate")
curve(dnorm(x, mean = theta, sd = theta / sqrt(10)),
      add = TRUE, lwd = 2)
qqnorm(small, main = "n = 10")
qqline(small)
hist(large, breaks = 40, freq = FALSE, col = "grey85", main = "n = 100",
     xlab = "estimate")
curve(dnorm(x, mean = theta, sd = theta / sqrt(100)),
      add = TRUE, lwd = 2)
qqnorm(large, main = "n = 100")
qqline(large)
```

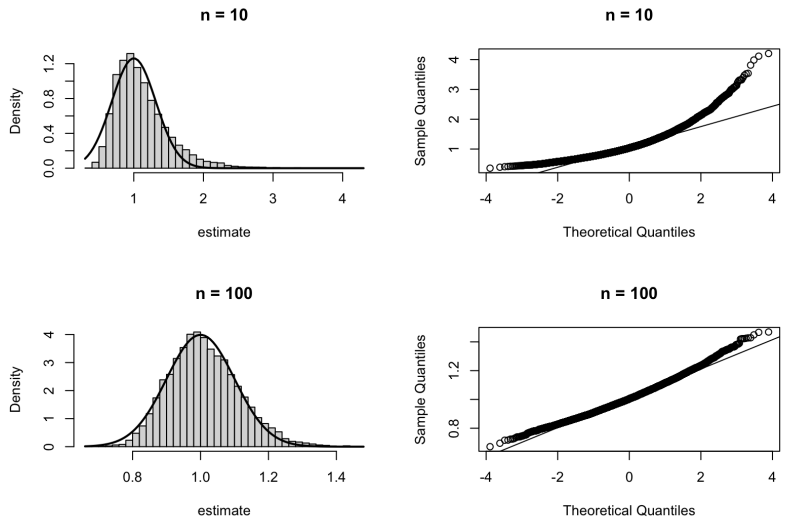


Figure 12.1: The sampling distribution of $\hat{\theta} = 1/\bar{X}$ for the exponential model with $\theta = 1$: histograms with the normal approximation overlaid (left column) and normal quantile plots (right column), for 10000 samples of size $n = 10$ (top row) and $n = 100$ (bottom row).

Figure 12.1 shows that for $n = 10$ the estimates are not normally distributed: the histogram has a long right tail, the peak is to the left of 1 and the quantile plot shows quantiles diverging from the line. This is no contradiction: the normal approximation is only claimed for large n. For $n = 100$ the distortion is gone.

12.2 Standard Errors and a Wald Interval

The standard deviation $\theta/\sqrt{n}$ of the normal curves in figure 12.1 can be read off the Fisher information from lecture 11: by example 11.8 and proposition 11.4 the sample contains information $\mathcal{I}_n(\theta) = n/\theta^2$, and thus $1/\sqrt{\mathcal{I}_n(\theta)} = \theta/\sqrt{n}$.

This is not a coincidence. Lecture 17 will show that every maximum likelihood estimator is, under conditions, approximately normal with spread $1/\sqrt{\mathcal{I}_n(\theta)}$ for large n, and that θ may be replaced by the estimate inside the information. We use these two facts on trust in this session. Since the value of θ is normally unknown, the *standard error* of the estimate is given by

$$\operatorname{se}(\hat{\theta}) = \frac{1}{\sqrt{\mathcal{I}_n(\hat{\theta})}} = \frac{\hat{\theta}}{\sqrt{n}}.$$

This is a measure of the likely size of the estimation error, computed from the data. Since the estimator is approximately normally distributed with this spread, the *Wald interval* $\hat{\theta} \pm 1.96 \operatorname{se}(\hat{\theta})$ covers the true value with probability approximately 95%. This is a confidence interval of the kind we have seen in MATH2701, and is used informally here and more formally in lecture 26. Before you try the code, make a prediction: will the interval cover the true value 1?

```
n <- 30
set.seed(2026)
x <- rexp(n, rate = theta)
theta_hat <- 1 / mean(x)
se <- theta_hat / sqrt(n)
theta_hat
se
theta_hat + c(-1, 1) * 1.96 * se
```

```
## [1] 0.6337667
## [1] 0.1157094
## [1] 0.4069762 0.8605572
```

The interval $[0.407, 0.861]$ does not contain the true value 1. This is not a programming error: this seed happens to give one of the misses, and in the long run a 95% interval misses the truth in about one sample out of twenty. From a single sample it is not possible to know whether this is one of the misses; how often the recipe is wrong in the long run can be answered by simulation.

12.3 Estimating a Probability by Simulation

Recall from lecture 9 that a probability is the expected value of an indicator, and thus by the law of large numbers (proposition A.14) it can be estimated by the `mean` of a logical vector over many independent repetitions. The same principle applies to all probabilities we estimate by simulation: for the coverage probability of an interval, the indicator is that the interval contains the truth, and for the rejection rate of a test (see lecture 24) it is that the test rejects.

We apply this to the coverage of the Wald interval: the box below contains a complete script, with its lines shuffled, except that the lines of the function are kept together. Before you continue, arrange the lines on paper into a working script which estimates, from 10000 samples of size $n = 10$, the probability that the Wald interval contains θ. Several orders satisfy the rules below.

```
mean(ci[1, ] <= theta & theta <= ci[2, ])    # empirical coverage
theta <- 1
```

```
wald <- function(n, theta) {
  x <- rexp(n, rate = theta)
  theta_hat <- 1 / mean(x)
  se <- theta_hat / sqrt(n)
  theta_hat + c(-1, 1) * 1.96 * se
}
ci <- replicate(10000, wald(10, theta))
set.seed(2026)
```

Names must be defined before they are used and the seed set before the first random draw (lecture 6): the function, `theta` and the seed come first, in any order, then `replicate`, then the summary. One detail is new: the function returns the two endpoints, which `replicate` collects as the columns of a two-row matrix, so that `ci[1, ]` and `ci[2, ]` hold the lower and upper endpoints, and `&` combines two logical vectors elementwise (see appendix B). Below, the reconstructed script is extended by a loop over $n = 10$, 30 and 100. Before running it, make a prediction about the misses at $n = 10$: the standard error $\hat{\theta}/\sqrt{n}$ is proportional to the estimate, so will an interval which misses the truth more often lie entirely below or entirely above θ?

```
wald <- function(n, theta) {
  x <- rexp(n, rate = theta)
  theta_hat <- 1 / mean(x)
  se <- theta_hat / sqrt(n)
  theta_hat + c(-1, 1) * 1.96 * se
}
theta <- 1
ns <- c(10, 30, 100)
coverage <- numeric(3)
set.seed(2026)
for (i in seq_along(ns)) {
  ci <- replicate(10000, wald(ns[i], theta))
  coverage[i] <- mean(ci[1, ] <= theta & theta <= ci[2, ])
}
data.frame(n = ns, coverage = coverage)
```

```
##     n coverage
## 1  10   0.9535
## 2  30   0.9539
## 3 100   0.9504
```

The coverage is close to the nominal 0.95 for all three sample sizes, even for $n = 10$ where the sampling distribution was far from normal. Since 10000 repetitions estimate a probability near 0.95 with a standard error of about 0.002, the differences between the three values are within simulation error. The total hides an imbalance: small estimates get short intervals, and at $n = 10$ most misses are intervals whose upper endpoint lies below the truth. Check this yourself, using the logical vectors `ci[2, ] < theta` and `ci[1, ] > theta` in place of the coverage indicator.

12.4 A Task to Take Home

The following task is in the style of a short practical report. Choose a rate θ, let $n = 50$, set the random number generator seed to your student ID, and simulate one sample from an exponential distribution. Determine the estimate $\hat{\theta} = 1/\bar{x}$, the standard error and the Wald interval for this sample. Write down your results in one sentence each. Then, using 10000 samples, estimate the coverage of the interval at $n = 50$. Also determine how often the upper endpoint of the interval lies below θ and how often the lower endpoint lies above θ, and report the two proportions separately. In two sentences, discuss whether you would trust the interval for this sample size.

Before the next session, type the commands used in this session into R, and check that your answers match the text.

Summary.

- The written reference for the commands used in this session is appendix B.
- The normal approximation to the sampling distribution of $\hat{\theta} = 1/\bar{X}$ is poor for $n = 10$ and good for $n = 100$. The standard error $\text{se}(\hat{\theta}) = 1/\sqrt{\mathcal{I}_n(\hat{\theta})}$ and the Wald interval $\hat{\theta} \pm 1.96\ \text{se}(\hat{\theta})$ can be computed from data.
- A probability can be estimated by simulation as the mean of a logical vector, *i.e.* by taking the average of an indicator function over replicates. This method can be used to compute both coverage probabilities and rejection rates. For the Wald interval of the exponential rate, the coverage is close to 95% already for $n = 10$, but most misses are intervals which lie below the truth, since small estimates come with short intervals.

Lecture 13

The Cramér–Rao Inequality

In lecture 10 we have found the best unbiased estimators by conditioning on a complete sufficient statistic. In contrast, in this lecture we will discuss a question which could be asked about any unbiased estimator: What are the limitations of unbiased estimators? In the situation of lecture 11, we will see that the variance of an unbiased estimator cannot be smaller than $1/\mathcal{I}_n(\theta)$. This is the Cramér–Rao inequality, which gives a benchmark against which every unbiased estimator can be measured and which we will use in lecture 17 to characterise the maximum likelihood estimator. In this lecture we will prove the inequality, in a form which covers estimators for functions of the parameter, and then see which estimators achieve the bound. We will conclude by considering the uniform distribution, where the conditions for the bound to hold are violated and where it is possible to beat the bound.

13.1 The Inequality

Throughout this lecture, $X_1, \dots, X_n$ are i.i.d. with density or probability weights $f(x;\theta)$, the parameter space $\Theta \subseteq \mathbb{R}$ is an open interval, and the conditions of section 11.1 hold. The score $\ell'(\theta)$ and the Fisher information $\mathcal{I}_n(\theta) = \operatorname{Var}_\theta(\ell'(\theta))$ are as in lecture 11; we assume in addition that $0 < \mathcal{I}_n(\theta) < \infty$ for all $\theta \in \Theta$, and we write $f(x;\theta) = \prod_{i=1}^n f(x_i;\theta)$ for the joint density of a value $x = (x_1, \dots, x_n)$ of the sample.

The idea of the proof can be stated in advance: the covariance between an unbiased estimator W for $g(\theta)$ and the score equals $g'(\theta)$, for any model, and then the Cauchy–Schwarz inequality shows that a small Fisher information forces a large variance. We state the result for the covariance as a separate lemma.

Lemma 13.1. *Let $W = W(X_1, \dots, X_n)$ be a statistic such that $\operatorname{Var}_\theta(W) < \infty$ for all $\theta \in \Theta$, and assume that the expectation of W can be differentiated under the integral sign, i.e. that*

$$\frac{\mathrm{d}}{\mathrm{d}\theta} \int W(x)\, f(x;\theta)\, \mathrm{d}x = \int W(x)\, \frac{\partial}{\partial\theta} f(x;\theta)\, \mathrm{d}x, \tag{13.1}$$

with a sum in place of the integral for discrete models. Then we have

$$\frac{\mathrm{d}}{\mathrm{d}\theta} \mathbb{E}_\theta(W) = \operatorname{Cov}_\theta(W, \ell'(\theta))$$

for all $\theta \in \Theta$.

Proof. Since the score has mean zero (lemma 11.1), the covariance is given by the expectation of the product $\operatorname{Cov}_\theta(W, \ell'(\theta)) = \mathbb{E}_\theta(W\ell'(\theta))$. As in equation (11.2), the score of the sample is the ratio $\ell'(\theta) = \frac{\partial}{\partial\theta} f(X;\theta)/f(X;\theta)$ of the joint density and thus the density cancels when we take the expectation:

$$\begin{aligned}
\mathbb{E}_\theta(W\ell'(\theta)) &= \int W(x)\, \frac{\frac{\partial}{\partial\theta} f(x;\theta)}{f(x;\theta)}\, f(x;\theta)\, \mathrm{d}x \\
&= \int W(x)\, \frac{\partial}{\partial\theta} f(x;\theta)\, \mathrm{d}x \\
&= \frac{\mathrm{d}}{\mathrm{d}\theta} \int W(x)\, f(x;\theta)\, \mathrm{d}x \\
&= \frac{\mathrm{d}}{\mathrm{d}\theta} \mathbb{E}_\theta(W),
\end{aligned}$$

where the integrals are over the support of the distribution, which does not depend on θ. □

The assumption (13.1) is an additional exchange of differentiation and integration. For exponential families, this relation holds for all statistics with finite variance, by a similar argument from analysis as given in the proof of lemma 7.10, but we will not verify this relation for individual examples.

Often the quantity of interest is a function $g(\theta)$ of the parameter rather than θ itself, for example the probability $e^{-\theta}$ of zero counts in example 10.2. We therefore state the inequality for an unbiased estimator for $g(\theta)$.

Theorem 13.2 (Cramér–Rao inequality)**.** *Assume the conditions of section 11.1 and $0 < \mathcal{I}_n(\theta) < \infty$, and let $g\colon \Theta \to \mathbb{R}$ be differentiable. Let W be an unbiased estimator for $g(\theta)$ with $\operatorname{Var}_\theta(W) < \infty$ which satisfies (13.1). Then we have*

$$\operatorname{Var}_\theta(W) \geq \frac{g'(\theta)^2}{\mathcal{I}_n(\theta)}$$

for all $\theta \in \Theta$.

Proof. Since W is unbiased, we have $\mathbb{E}_\theta(W) = g(\theta)$, and lemma 13.1 gives $\operatorname{Cov}_\theta(W, \ell'(\theta)) = g'(\theta)$. Using the Cauchy–Schwarz inequality, lemma A.8, for W and $\ell'(\theta)$, we find

$$g'(\theta)^2 = \operatorname{Cov}_\theta(W, \ell'(\theta))^2 \leq \operatorname{Var}_\theta(W)\, \operatorname{Var}_\theta(\ell'(\theta)) = \operatorname{Var}_\theta(W)\, \mathcal{I}_n(\theta),$$

where we used definition 11.2 for the last step. Dividing by the positive number $\mathcal{I}_n(\theta)$ gives the claim. This completes the proof. □

The most important special case is the estimation of θ itself.

Corollary 13.3. *Assume the conditions of section 11.1 and $0 < \mathcal{I}_n(\theta) < \infty$. Let W be an unbiased estimator for θ with $\operatorname{Var}_\theta(W) < \infty$ which satisfies (13.1). Then we have*

$$\operatorname{Var}_\theta(W) \geq \frac{1}{\mathcal{I}_n(\theta)}$$

for all $\theta \in \Theta$.

Proof. This is the case $g(\theta) = \theta$ of theorem 13.2, where $g'(\theta) = 1$. □

Two comments help to understand the corollary. First, by proposition 11.4 the bound is $1/(n\mathcal{I}(\theta))$: no unbiased estimator can have a standard deviation which decreases faster than the $1/\sqrt{n}$ of the sample mean (lemma 2.2). Second, the bound is independent of the estimator used, and only depends on the model: the bound measures how much the data can reveal about θ, regardless of the unbiased method used to estimate it. For the normal mean with known variance the bound is $\sigma^2/n = \operatorname{Var}_\mu(\bar{X})$, as shown in example 11.7, and thus $\bar{X}$ achieves this bound.

It is the general form of theorem 13.2 which is worth memorising: identify the function g the estimator is unbiased for, differentiate, square and divide by the information. Both numerator and denominator depend on how the model is parametrised, but by proposition 11.9 the bound itself does not: it is a property of the estimator and the model, not of the name we give to the parameter. Exercise 13.6 verifies this.

13.2 Efficiency and Attainment

We give a name to the estimators which attain the bound.

Definition 13.4. Let W be an unbiased estimator for $g(\theta)$ that satisfies the assumptions of theorem 13.2. Then the *efficiency* of W is given by

$$\operatorname{eff}_\theta(W) = \frac{g'(\theta)^2/\mathcal{I}_n(\theta)}{\operatorname{Var}_\theta(W)}.$$

The estimator W is called *efficient*, if $\operatorname{eff}_\theta(W) = 1$ for all $\theta \in \Theta$, *i.e.* if its variance equals the Cramér–Rao bound for every θ.

From theorem 13.2 we know that the efficiency lies in the interval $[0, 1]$. An efficient estimator is a UMVUE in the sense of definition 10.4 amongst the unbiased estimators satisfying (13.1), since none of these can beat the bound. The converse is not true, because the bound is not always attained: in exercise 13.4, the UMVUE for the rate of the exponential distribution has efficiency $(n-2)/n$.

The proof of theorem 13.2 suggests when the bound is attained: equality in the Cauchy–Schwarz step requires $W - g(\theta)$ to be a multiple of the score. To make this precise, we compare $W - g(\theta)$ with the multiple $c(\theta)\,\ell'(\theta)$, where $c(\theta) = g'(\theta)/\mathcal{I}_n(\theta)$ is the regression coefficient $\operatorname{Cov}_\theta(W, \ell'(\theta))/\operatorname{Var}_\theta(\ell'(\theta))$ of W on the score, and we show that the variance of W exceeds the bound by the mean squared difference between the two.

Proposition 13.5. *Under the assumptions of theorem 13.2, the estimator W is efficient if and only if there is a function $c(\theta)$ such that, for every $\theta \in \Theta$, we have*

$$W(X) - g(\theta) = c(\theta)\,\frac{\partial}{\partial\theta}\log f(X;\theta) \tag{13.2}$$

with probability one, where $X = (X_1, \ldots, X_n)$ and $f(x;\theta)$ is the joint density of the sample. In this case we have $c(\theta) = g'(\theta)/\mathcal{I}_n(\theta)$.

Proof. The derivative in (13.2) is the score $\ell'(\theta)$ of the sample. We first derive an identity which holds for every unbiased estimator W, efficient or not, and then use it to prove the two directions of the claim. Consider the random variable

$$R = W - g(\theta) - \frac{g'(\theta)}{\mathcal{I}_n(\theta)}\,\ell'(\theta).$$

Since W is unbiased and the score has mean zero (lemma 11.1), we have $\mathbb{E}_\theta(R) = 0$ and thus $\mathbb{E}_\theta(R^2) = \operatorname{Var}_\theta(R)$. By lemma 13.1 we have $\operatorname{Cov}_\theta(W, \ell'(\theta)) = g'(\theta)$, and by definition 11.2 we have $\operatorname{Var}_\theta(\ell'(\theta)) = \mathcal{I}_n(\theta)$. Expanding the variance of R thus gives

$$\begin{aligned}\mathbb{E}_\theta(R^2) &= \operatorname{Var}_\theta(W) - 2\,\frac{g'(\theta)}{\mathcal{I}_n(\theta)}\operatorname{Cov}_\theta(W, \ell'(\theta)) + \frac{g'(\theta)^2}{\mathcal{I}_n(\theta)^2}\operatorname{Var}_\theta(\ell'(\theta)) \\ &= \operatorname{Var}_\theta(W) - 2\,\frac{g'(\theta)^2}{\mathcal{I}_n(\theta)} + \frac{g'(\theta)^2}{\mathcal{I}_n(\theta)} \\ &= \operatorname{Var}_\theta(W) - \frac{g'(\theta)^2}{\mathcal{I}_n(\theta)}.\end{aligned}$$

Now assume that W is efficient. Then the right-hand side is zero for every θ. Since the non-negative random variable R^2 then has expectation zero, we have $R = 0$ with probability one, which is (13.2) with $c(\theta) = g'(\theta)/\mathcal{I}_n(\theta)$. Conversely, assume that (13.2) holds for some function $c(\theta)$. Taking the covariance of both sides with $\ell'(\theta)$ gives $g'(\theta) = c(\theta)\mathcal{I}_n(\theta)$ by lemma 13.1, and thus $c(\theta) = g'(\theta)/\mathcal{I}_n(\theta)$. Then $R = 0$ with probability one, the variance of W equals the bound and W is efficient. This completes the proof. □

Since the left-hand side $\mathbb{E}_\theta(R^2)$ of the identity in the proof is non-negative, the identity also gives a second proof of theorem 13.2. The bound is the part of the variance of W which is explained by the score, and an efficient estimator is one for which nothing is left over.

In an exponential family, the attainment condition identifies an efficient estimator directly.

Corollary 13.6. *Let the model be a one-parameter exponential family as in definition 7.1, with density*

$$f(x;\theta) = h(x)\exp\bigl(\eta(\theta)T(x) - B(\theta)\bigr),$$

where η is twice continuously differentiable with $\eta'(\theta) \neq 0$ for all θ, and let $T_n = \sum_{i=1}^n T(X_i)$ be the natural statistic. Then T_n/n is an efficient estimator for the mean $\mu(\theta) = \mathbb{E}_\theta\bigl(T(X_1)\bigr)$.

Proof. The score of a single observation is $\ell_i'(\theta) = \eta'(\theta)T(X_i) - B'(\theta)$. Since the score has mean zero (lemma 11.1), taking expectations gives $B'(\theta) = \eta'(\theta)\mu(\theta)$, and thus $\ell_i'(\theta) = \eta'(\theta)\bigl(T(X_i) - \mu(\theta)\bigr)$. Summing over i and dividing by $n\eta'(\theta) \neq 0$, we find

$$\frac{T_n}{n} - \mu(\theta) = \frac{1}{n\,\eta'(\theta)}\,\ell'(\theta),$$

which is (13.2) for $W = T_n/n$, $g = \mu$ and $c(\theta) = 1/\bigl(n\eta'(\theta)\bigr)$. It remains to check the assumptions of theorem 13.2. The conditions of section 11.1 hold for exponential families, as noted there. The estimator T_n/n is unbiased for $\mu(\theta)$, and it has finite variance, since we have $T(X_i) = \mu(\theta) + \ell_i'(\theta)/\eta'(\theta)$ and the score has finite variance by the conditions of section 11.1. Finally, T_n/n satisfies (13.1) by the remark after lemma 13.1. Thus, by proposition 13.5, the estimator T_n/n is efficient. This completes the proof. □

The condition (13.2) is restrictive. With the score from the proof of corollary 13.6, an efficient estimator W for $g(\theta)$ in an exponential family satisfies

$$W = g(\theta) + c(\theta)\, n\, \eta'(\theta)\Big(\frac{T_n}{n} - \mu(\theta)\Big).$$

Since W is a statistic, it cannot depend on θ, and thus $W = a + bT_n/n$ for constants a and b. Then $g(\theta) = \mathbb{E}_\theta(W) = a + b\,\mu(\theta)$: the only functions of θ which have an efficient estimator are the affine functions of the mean of the natural statistic. For the Poisson distribution, an exponential family with $T(x) = x$ and $\mu(\theta) = \theta$, the sample mean is thus an efficient estimator for θ; exercise 13.3 considers $g(\theta) = e^{-\theta}$ instead.

13.3 When the Conditions Fail

As we have seen, the proof of the inequality in this lecture involves two instances of differentiating under the integral sign, once in lemma 11.1 and once in the assumption (13.1). If the support of the distribution moves as θ does, both of these exchanges can fail, since differentiating an integral whose limits depend on θ produces an additional term from the boundary, which the exchange omits.

Example 13.7. Let $X_1, \dots, X_n$ be i.i.d. with a uniform distribution on $(0, \theta)$ and $\theta > 0$. The support of the distribution depends on θ and the conditions from section 11.1 are violated. Thus, theorem 13.2 does not apply. We can still see what the formula would be for this model: in example 11.10 the three expressions of lemma 11.3 had three different values, and taking the only positive one of these three values, $\mathbb{E}_\theta(\ell_1'(\theta)^2) = 1/\theta^2$, and multiplying by n as in proposition 11.4, we find the formal expression n/θ^2 and the formal bound θ^2/n.

Now consider the estimators for θ based on the maximum $M = \max_i X_i$. The MLE M is biased by exercise 2.2, and thus its small mean squared error does not violate the inequality, which is a statement about unbiased estimators only. The bias-corrected estimator $W = (n+1)M/n$ is unbiased and by exercise 10.4 has variance

$$\operatorname{Var}_\theta(W) = \frac{\theta^2}{n(n+2)} < \frac{\theta^2}{n} \qquad \text{for every } n \geq 1,$$

which is of order $1/n^2$ instead of $1/n$. An unbiased estimator with variance smaller than the formal bound in this way is said to be *super-efficient*.

There is no contradiction: the inequality has hypotheses, and the uniform model violates these hypotheses. The simulation in lecture 9 has already shown the $1/n^2$ decay for the mean squared error of M. Exercise 16.3 proves the consistency of M by a direct argument.

Summary.

- Under the conditions of section 11.1, all unbiased estimators W for $g(\theta)$ with (13.1) satisfy $\operatorname{Var}_\theta(W) \geq g'(\theta)^2/\mathcal{I}_n(\theta)$. For $g(\theta) = \theta$ we get the bound $1/\mathcal{I}_n(\theta)$. The proof uses $\operatorname{Cov}_\theta(W, \ell'(\theta)) = g'(\theta)$ and the Cauchy–Schwarz inequality.
- The efficiency of an unbiased estimator is the ratio of the bound to the variance, with values in $[0, 1]$. An efficient estimator is a UMVUE amongst the estimators satisfying (13.1), but a UMVUE need not be efficient.
- The bound is attained if and only if $W - g(\theta) = c(\theta)\ell'(\theta)$ with probability one. For a one-parameter exponential family, the average T_n/n of the natural statistic is efficient for its mean, and the only functions of θ which can be used to construct an efficient estimator are affine functions of the mean.
- The bound depends on regularity conditions. For example, for the uniform distribution on $(0, \theta)$, the support of the distribution depends on the parameter. The unbiased estimator $(n + 1)M/n$ has variance $\theta^2/(n(n + 2))$, which is smaller than the formal value θ^2/n. Thus, the estimator is super-efficient.

Exercise 13.1. Let $X_1, \ldots, X_n$ be i.i.d. exponentially distributed with rate $\theta > 0$. Define $f(x; \theta) = \theta e^{-\theta x}$ for $x \geq 0$ and $T = \sum_{i=1}^n X_i$. Use the fact that the chi-squared distribution χ^2_k from definition A.2 coincides with the Gamma$(k/2, 1/2)$ distribution from table A.2 and that T is Gamma(n, θ) distributed.

1. Show that $Y_1 = 2\theta X_1$ has density $\frac{1}{2}e^{-y/2}$ for $y \geq 0$ and consequently $Y_1 \sim \chi^2_2$.
2. Using definition A.2, show that $2\theta T \sim \chi^2_{2n}$.
3. Using the gamma density of T and the change of variables formula from proposition A.19, show that you get the same result as in the previous part.
4. Using the mean and variance of the chi-squared distribution, find $\mathbb{E}_\theta(T)$ and $\operatorname{Var}_\theta(T)$. Check your values by computing the moments of a single observation.

Exercise 13.2. Let $X_1, \ldots, X_n$ be i.i.d. Bernoulli with success probability $\theta \in (0, 1)$, and let $T_n = \sum_{i=1}^n X_i$.

1. Write down the Cramér–Rao bound for θ, using example 11.5.
2. Show that $\bar{X}$ is an efficient estimator for θ.
3. Verify condition (13.2) for $W = \bar{X}$ directly, by writing the score $\ell'(\theta)$ in terms of $\bar{X} - \theta$, and check that the resulting $c(\theta)$ equals $g'(\theta)/\mathcal{I}_n(\theta)$.
4. Write down the Cramér–Rao bound for the odds $g(\theta) = \theta/(1 - \theta)$, and explain why no efficient estimator for the odds exists.

5. Show that in fact no unbiased estimator for the odds exists at all.

Exercise 13.3. Let $X_1, \ldots, X_n$ be i.i.d. Poisson with mean $\theta > 0$ and let $T_n = \sum_{i=1}^n X_i$.

1. Write down the Cramér–Rao bound for θ, using example 11.6, and show that $\bar{X}$ achieves this bound.

2. Verify condition (13.2) for $W = \bar{X}$ directly, by writing the score $\ell'(\theta)$ in terms of $\bar{X} - \theta$, and by checking that the resulting $c(\theta)$ equals $g'(\theta)/\mathcal{I}_n(\theta)$.

3. Explain why no efficient estimator for $g(\theta) = e^{-\theta}$ exists, and why this does not contradict the fact that $(1 - 1/n)^{T_n}$ is the UMVUE for $e^{-\theta}$, using example 10.7.

Exercise 13.4. Let $X_1, \ldots, X_n$, where $n \geq 3$, be i.i.d. exponential with rate $\theta > 0$, let $T = \sum_{i=1}^n X_i$ and let $\tilde{\theta} = (n-1)/T$ be the unbiased version of the MLE from lecture 5.

1. Write down the Cramér–Rao bound for θ, using example 11.8.

2. In exercise 5.3 we found $\mathrm{Var}_\theta(\tilde{\theta}) = \theta^2/(n-2)$. Determine the efficiency of $\tilde{\theta}$ and describe how it behaves as $n \to \infty$.

3. Show that no unbiased estimator for θ can be efficient.

4. Show that $\tilde{\theta}$ is the UMVUE for θ.

5. Show that $\bar{X}$ is an efficient estimator for the mean $\mu = 1/\theta$, and explain how this is compatible with the previous results.

Exercise 13.5. Let $X_1, \ldots, X_n$ be i.i.d. $N(\mu, \sigma^2)$. Assume first that μ is known and $\sigma^2 > 0$ is the parameter.

1. Show that $\mathcal{I}_n(\sigma^2) = n/(2\sigma^4)$ and find the Cramér–Rao bound for σ^2.

2. Show that $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \mu)^2$ is an efficient estimator for σ^2, using definition A.2 and lemma A.3.

3. Now assume that μ is unknown. From lemma 2.3 we know that the sample variance S^2 is an unbiased estimator for σ^2. Using proposition A.4, find $\mathrm{Var}(S^2)$ and compare the result with the bound from the first part of the question. Lecture 14 shows that this bound still holds when μ is unknown. What can you say about S^2, in light of example 10.8?

Exercise 13.6. Let $\psi = g(\theta)$ where $g\colon \Theta \to g(\Theta)$ is a twice continuously differentiable bijection with $g'(\theta) \neq 0$ for all $\theta \in \Theta$. Furthermore, let $\mathcal{I}_{n,\psi}$ and $\mathcal{I}_{n,\theta}$ be the Fisher information of the sample with model parametrised by ψ and by θ, respectively.

1. Using propositions 11.4 and 11.9, show that $\mathcal{I}_{n,\psi}(\psi) = \mathcal{I}_{n,\theta}(\theta)/g'(\theta)^2$.

2. Let W be an unbiased estimator for $g(\theta)$ which satisfies the assumptions of theorem 13.2. Show that the bound from corollary 13.3, for W as an unbiased estimator for the parameter ψ, coincides with the bound from theorem 13.2 for $g(\theta)$.

Lecture 14

Multiparameter Fisher Information

In lectures 11 and 13 the parameter was a single number, but most models we actually fit, *e.g.* the normal distribution with unknown mean and variance, have several parameters. In this lecture we will extend the concepts of Fisher information and the Cramér–Rao bound to a vector parameter $\theta = (\theta_1, \dots, \theta_k)$: the score will be a vector, the information will be a $k \times k$ matrix, and the bound will be an inequality of covariance matrices. The new feature here is that we will need to invert the matrix, and the inverse will give the cost of estimating one parameter in the presence of the other parameters.

14.1 The Score Vector and the Information Matrix

Setting the stage as in lecture 11, we now assume that the parameter space is an open set $\Theta \subseteq \mathbb{R}^k$. The data $X_1, \dots, X_n$ are again i.i.d. with density or probability weights $f(x;\theta)$ and the log-likelihood function is $\ell(\theta) = \sum_{i=1}^n \ell_i(\theta)$, where $\ell_i(\theta) = \log f(X_i;\theta)$, but θ is now a vector. Derivatives with respect to θ are now partial derivatives, denoted by ∂_j for the derivative with respect to θ_j. We assume the conditions of section 11.1 with partial derivatives in place of $\mathrm{d}/\mathrm{d}\theta$, including the mixed derivatives $\partial_j \partial_l$, and we assume that each coordinate of the score has finite variance under $\mathbb{P}_\theta$.

Definition 14.1. The *score vector* of the sample is given by the gradient of the log-likelihood function

$$\nabla \ell(\theta) = (\partial_1 \ell(\theta), \dots, \partial_k \ell(\theta)),$$

as a column vector in $\mathbb{R}^k$.

For $k = 1$ the score vector is the score $\ell'(\theta)$ from definition 4.4. As in lecture 11, we consider the score vector as a random vector under $\mathbb{P}_\theta$. The covariance matrix $\mathrm{Cov}(Y)$ of a random vector $Y = (Y_1, \dots, Y_k)$ is the $k \times k$ matrix with entries $\mathrm{Cov}(Y_j, Y_l)$, and its diagonal contains the variances.

Definition 14.2. Given the conditions of section 11.1, the *Fisher information matrix* of the sample about θ is given by

$$\mathcal{I}_n(\theta) = \mathrm{Cov}_\theta\big(\nabla \ell(\theta)\big), \qquad \textit{i.e.} \qquad \mathcal{I}_n(\theta)_{jl} = \mathrm{Cov}_\theta\big(\partial_j \ell(\theta), \partial_l \ell(\theta)\big).$$

The information matrix for a single observation is denoted by $\mathcal{I}(\theta) = \mathcal{I}_1(\theta)$.

The diagonal element $\mathcal{I}_n(\theta)_{jj}$ is the Fisher information about θ_j in the sense of definition 11.2 with all other coordinates fixed to their true values; the off-diagonal elements are new and measure the strength of correlations between the components of the score. Both identities from lecture 11 carry over to this situation, separately for each coordinate.

Proposition 14.3. *With the assumptions of section 11.1 in place, the score has mean* $\mathbb{E}_\theta(\nabla \ell(\theta)) = 0$*, and we have*

$$\mathcal{I}_n(\theta)_{jl} = \mathbb{E}_\theta\big(\partial_j \ell(\theta)\, \partial_l \ell(\theta)\big) = -\mathbb{E}_\theta\big(\partial_j \partial_l \ell(\theta)\big) \qquad \text{for } j, l = 1, \dots, k.$$

Thus, $\mathcal{I}_n(\theta)$ *is the negative of the expected Hessian matrix of the log-likelihood.*

The proof of this result is similar to the proof of lemma 11.1 and lemma 11.3, but with partial derivatives instead of $\mathrm{d}/\mathrm{d}\theta$. In outline, the ratio $\partial_j f / f$ has mean zero since the density in the numerator and denominator cancels. Using the quotient rule we find $\partial_j \partial_l \ell_i = \partial_j \partial_l f(X_i;\theta)/f(X_i;\theta) - \partial_j \ell_i\, \partial_l \ell_i$ where the first term has mean zero. Finally, the independence of the observations allows us to carry over the resulting identity from one observation to the sample. As for the scalar case, the second expression is often the more practical one.

14.2 Properties of the Information Matrix

Two properties of the scalar information from lecture 11 have matrix counterparts. Recall that a symmetric matrix A is *positive semi-definite*, if $a^\top A a \geq 0$ for all $a \in \mathbb{R}^k$, and *positive definite*, if $a^\top A a > 0$ for all $a \neq 0$; a positive definite matrix has a positive definite inverse.

Proposition 14.4. *Suppose the conditions from section 11.1 hold. Then the following statements hold.*

1. *The matrix* $\mathcal{I}_n(\theta)$ *is symmetric and we have* $a^\top \mathcal{I}_n(\theta) a = \mathrm{Var}_\theta(a^\top \nabla \ell(\theta)) \geq 0$ *for all* $a \in \mathbb{R}^k$*. In particular,* $\mathcal{I}_n(\theta)$ *is positive semi-definite.*
2. *Information is additive:* $\mathcal{I}_n(\theta) = n\,\mathcal{I}(\theta)$*.*

Proof. For the first statement, the symmetry of the matrix follows from the fact that $\mathrm{Cov}(Y_j, Y_l) = \mathrm{Cov}(Y_l, Y_j)$ for all j and l. For $a \in \mathbb{R}^k$, the variance of the linear combination $a^\top \nabla \ell(\theta)$ is given by

$$\mathrm{Var}_\theta\Big(\sum_{j=1}^k a_j\, \partial_j \ell(\theta)\Big) = \sum_{j=1}^k \sum_{l=1}^k a_j a_l\, \mathcal{I}_n(\theta)_{jl} = a^\top \mathcal{I}_n(\theta) a,$$

and thus is non-negative. The second statement is proved by using the fact that the score vector of the sample is the sum of n i.i.d. score vectors of the individual observations and that the covariance matrix of the sum of independent random vectors equals the sum of the covariance matrices. This completes the proof. □

Strict positive definiteness is violated in cases where the log-likelihood does not respond to changes of θ in some direction, for example for the model $N(\theta_1 + \theta_2, 1)$. From now on we assume that $\mathcal{I}_n(\theta)$ is positive definite so that the inverse exists.

14.3 The Cramér–Rao Bound for Vector Parameters

Theorem 13.2 bounds the variance of an unbiased estimator for a scalar parameter from below. For a vector parameter the natural object to bound is the covariance matrix, and the natural reading of "at least" between symmetric matrices is that the difference is positive semi-definite.

For real numbers, $a > b$ is the same as $a - b > 0$. For symmetric matrices A and B we use the second form as the definition: we think of a matrix A as larger than B, if $A - B$ is positive definite, and as at least as large as B, if $A - B$ is positive semi-definite. Thus A is at least as large as B exactly when $a^\top A a \geq a^\top B a$ for every $a \in \mathbb{R}^k$; unlike for numbers, two matrices may not be comparable at all.

Proposition 14.5. *Suppose we have the conditions of section 11.1 and that the matrix $\mathcal{I}_n(\theta)$ is positive definite. Let $T = (T_1, \dots, T_k)$ be an unbiased estimator for θ, so that $\mathbb{E}_\theta(T_j) = \theta_j$ for all j and all θ, where the variances are finite and each T_j satisfies (13.1) in every component of θ. Then the matrix*

$$\operatorname{Cov}_\theta(T) - \mathcal{I}_n(\theta)^{-1}$$

is positive semi-definite for all $\theta \in \Theta$. In particular we have

$$\operatorname{Var}_\theta(T_j) \geq \left(\mathcal{I}_n(\theta)^{-1}\right)_{jj} \qquad \text{and} \qquad \operatorname{Var}_\theta(a^\top T) \geq a^\top \mathcal{I}_n(\theta)^{-1} a$$

for $j = 1, \dots, k$ and for all $a \in \mathbb{R}^k$.

Proof. The argument is the proof of theorem 13.2, applied to linear combinations. If we hold all coordinates of θ other than θ_l fixed, lemma 13.1 applies to the one-parameter model with score $\partial_l \ell(\theta)$ and gives $\operatorname{Cov}_\theta(T_j, \partial_l \ell(\theta)) = \partial_l \theta_j$, which is 1 for $l = j$ and 0 otherwise. Thus, by bilinearity, we have $\operatorname{Cov}_\theta(a^\top T, b^\top \nabla \ell(\theta)) = a^\top b$ for all $a, b \in \mathbb{R}^k$. Using the Cauchy–Schwarz inequality, lemma A.8, and the first statement of proposition 14.4, we find

$$(a^\top b)^2 \leq (a^\top \operatorname{Cov}_\theta(T)\, a)(b^\top \mathcal{I}_n(\theta)\, b).$$

Choosing $b = \mathcal{I}_n(\theta)^{-1} a$, both $a^\top b$ and $b^\top \mathcal{I}_n(\theta) b$ equal $a^\top \mathcal{I}_n(\theta)^{-1} a$, which is positive for $a \neq 0$, and dividing by this value gives $a^\top \operatorname{Cov}_\theta(T)\, a \geq a^\top \mathcal{I}_n(\theta)^{-1} a$. This is the claim, since $a^\top \operatorname{Cov}_\theta(T)\, a = \operatorname{Var}_\theta(a^\top T)$, and for the diagonal case we have $a = e_j$, with the jth unit vector. This completes the proof. □

The same argument gives a bound for a function $g(\theta)$ of the parameter. Let W be an unbiased estimator for $g(\theta)$ with finite variance which satisfies (13.1) in every component of θ. Then lemma 13.1 gives $\operatorname{Cov}_\theta(W, b^\top \nabla \ell(\theta)) = b^\top \nabla g(\theta)$, and the proof above with $a^\top T$ replaced by W and a replaced by $\nabla g(\theta)$ shows

$$\operatorname{Var}_\theta(W) \geq \nabla g(\theta)^\top \mathcal{I}_n(\theta)^{-1} \nabla g(\theta).$$

For $k = 1$ this is theorem 13.2. The case $g(\theta) = a^\top \theta$ is used in exercise 14.3.

As an example, we apply the bound to the most important case, the normal distribution with both parameters unknown.

Example 14.6. Let $X_1, \dots, X_n$ be i.i.d. $N(\mu, \sigma^2)$ where $\theta = (\mu, \sigma^2)$, both unknown. From example 5.4 we know that the log-likelihood function is

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2,$$

and the first partial derivatives are given by

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \mu), \qquad \frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(X_i - \mu)^2.$$

Taking second derivatives, where we treat σ^2 as a single variable, we find the three second derivatives as

$$\frac{\partial^2 \ell}{\partial \mu^2} = -\frac{n}{\sigma^2}, \qquad \frac{\partial^2 \ell}{\partial \mu\, \partial \sigma^2} = -\frac{1}{\sigma^4}\sum_{i=1}^{n}(X_i - \mu),$$
$$\frac{\partial^2 \ell}{\partial (\sigma^2)^2} = \frac{n}{2\sigma^4} - \frac{1}{\sigma^6}\sum_{i=1}^{n}(X_i - \mu)^2.$$

To compute the expectations, we use the fact that $\mathbb{E}_\theta(X_i - \mu) = 0$ and $\mathbb{E}_\theta((X_i - \mu)^2) = \sigma^2$. The mixed derivative has expectation zero. The last term has expectation $n/(2\sigma^4) - n\sigma^2/\sigma^6 = -n/(2\sigma^4)$. Thus, by proposition 14.3, we find the information matrix and its inverse to be

$$\mathcal{I}_n(\mu, \sigma^2) = \begin{pmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{pmatrix} \qquad \text{and} \qquad \mathcal{I}_n(\mu, \sigma^2)^{-1} = \begin{pmatrix} \frac{\sigma^2}{n} & 0 \\ 0 & \frac{2\sigma^4}{n} \end{pmatrix}. \tag{14.1}$$

The upper left entry is the information n/σ^2 about the mean, where the variance is known from example 11.7. The sample mean, with variance σ^2/n, achieves the bound for μ. For the variance the bound is $2\sigma^4/n$. The UMVUE S^2 from example 10.8 has variance $2\sigma^4/(n-1)$, as shown in the computation of section 2.3, and thus does not achieve the bound. Since S^2 is the UMVUE, no unbiased estimator for σ^2 achieves the bound when μ is unknown.

The information matrix for the normal distribution is diagonal. The following section explains the significance of this fact and what happens when this is not the case.

14.4 Nuisance Parameters

Assume that we are interested in θ_1 only, and that the remaining components $\theta_2, \dots, \theta_k$ are *nuisance parameters*, *i.e.* unknown but not of interest. If the values of the nuisance parameters were known, θ_1 would be the only parameter and the Cramér–Rao bound for θ_1 would be $1/\mathcal{I}_n(\theta)_{11}$. Since the values are unknown, the bound is $\left(\mathcal{I}_n(\theta)^{-1}\right)_{11}$ by proposition 14.5. The following lemma compares the two bounds.

Lemma 14.7. *For every $\theta \in \Theta$ we have $\left(\mathcal{I}_n(\theta)^{-1}\right)_{jj} \geq 1/\mathcal{I}_n(\theta)_{jj}$ for $j = 1, \dots, k$. For $k = 2$, write*

$$\mathcal{I}_n(\theta) = \begin{pmatrix} a & b \\ b & c \end{pmatrix}.$$

Then we have

$$\bigl(\mathcal{I}_n(\theta)^{-1}\bigr)_{11} = \frac{1}{a} \cdot \frac{1}{1-\rho^2}, \qquad \text{where } \rho = \frac{b}{\sqrt{ac}}$$

is the correlation between the two components of the score vector, and $0 \le \rho^2 < 1$.

Proof. For the first statement, we apply the Cauchy–Schwarz inequality from lemma A.8 to the two random variables $e_j^\top \mathcal{I}_n(\theta)^{-1} \nabla \ell(\theta)$ and $\partial_j \ell(\theta)$. By the first statement of proposition 14.4, these have variances $\bigl(\mathcal{I}_n(\theta)^{-1}\bigr)_{jj}$ and $\mathcal{I}_n(\theta)_{jj}$ and covariance $e_j^\top \mathcal{I}_n(\theta)^{-1} \mathcal{I}_n(\theta) e_j = 1$, and thus we find $1 \le \bigl(\mathcal{I}_n(\theta)^{-1}\bigr)_{jj} \mathcal{I}_n(\theta)_{jj}$. For $k = 2$ the inverse matrix is

$$\mathcal{I}_n(\theta)^{-1} = \frac{1}{ac - b^2} \begin{pmatrix} c & -b \\ -b & a \end{pmatrix},$$

where $ac - b^2 > 0$ since the matrix is positive definite. Dividing numerator and denominator of $c/(ac-b^2)$ by ac gives the claimed formula. Since a and c are the variances of the two components of the score vector and b is their covariance, ρ is their correlation, and $ac - b^2 > 0$ gives $\rho^2 < 1$. This completes the proof. □

Thus, not knowing the other parameters can never decrease the bound. For $k = 2$ the bound for θ_1 with θ_2 unknown is $1/(1-\rho^2)$ times the bound with θ_2 known, and the two bounds agree if and only if $b = 0$. This case has a name.

Definition 14.8. The two components of a parameter $\theta = (\theta_1, \theta_2)$ are called *orthogonal*, if $\mathcal{I}(\theta)_{12} = 0$ for all $\theta \in \Theta$.

By the additivity of information, the condition does not depend on the sample size. For orthogonal parameters, not knowing θ_2 does not increase the bound for θ_1, and vice versa. We consider one example of each case.

Example 14.9. For the normal distribution with $\theta = (\mu, \sigma^2)$, the information matrix (14.1) is diagonal, and thus μ and σ^2 are orthogonal. The bound for μ is σ^2/n whether or not σ^2 is known, and the bound for σ^2 is $2\sigma^4/n$ whether or not μ is known. Equal bounds do not mean that not knowing the nuisance parameter is harmless, though: exercise 13.5 shows that the bound for σ^2 is attained when μ is known, while example 14.6 shows that it is not attained when μ is unknown.

Orthogonality is a special feature of the normal distribution, not the rule, as the following example shows.

Example 14.10. Consider the gamma distribution with shape $\alpha > 0$ and rate $\beta > 0$ both unknown, as in exercise 7.1. The log-likelihood $\ell_1(\alpha, \beta)$ of one observation has second derivatives $-\psi'(\alpha)$, $1/\beta$ and $-\alpha/\beta^2$ with respect to α twice, to α and β, and to β twice, where ψ' denotes the *trigamma function*, the second derivative of $\log \Gamma$. Since these are non-random, the information matrix of one observation is

$$\mathcal{I}(\alpha, \beta) = \begin{pmatrix} \psi'(\alpha) & -\dfrac{1}{\beta} \\[2ex] -\dfrac{1}{\beta} & \dfrac{\alpha}{\beta^2} \end{pmatrix};$$

exercise 14.2 checks this computation. The off-diagonal entry is non-zero, so the parameters are not orthogonal. One can show that this matrix is positive definite. By lemma 14.7 we have

$$\rho^2 = \frac{1}{\alpha\psi'(\alpha)}.$$

For $\alpha = 2$ we use, without proof, the series $\psi'(\alpha) = \sum_{m=0}^{\infty} 1/(\alpha+m)^2$. With $\psi'(2) = \sum_{m=2}^{\infty} 1/m^2 = \pi^2/6 - 1 \approx 0.645$, this gives $1/(1-\rho^2) \approx 4.45$: the bound for α with β unknown is about four and a half times the bound with β known.

The inverse information matrix will appear again in lecture 17, where we will see that for large samples the maximum likelihood estimator is approximately normal with covariance matrix $\mathcal{I}_n(\theta_0)^{-1}$.

Summary.

- For a vector parameter the score is the gradient $\nabla\ell(\theta)$, and the Fisher information matrix $\mathcal{I}_n(\theta)$ is its covariance matrix, equal to minus the expected Hessian of the log-likelihood.
- The information matrix is symmetric and positive semi-definite, and adds over independent observations, $\mathcal{I}_n = n\mathcal{I}$.
- By the Cramér–Rao inequality, for an unbiased estimator T for θ the matrix $\operatorname{Cov}_\theta(T) - \mathcal{I}_n(\theta)^{-1}$ is positive semi-definite; in particular $\operatorname{Var}_\theta(T_j) \geq \big(\mathcal{I}_n(\theta)^{-1}\big)_{jj}$.
- For $N(\mu, \sigma^2)$ with both parameters unknown, $\mathcal{I}_n = \operatorname{diag}(n/\sigma^2,\, n/(2\sigma^4))$ with inverse $\operatorname{diag}(\sigma^2/n,\, 2\sigma^4/n)$.
- The diagonal of the inverse dominates the reciprocal of the diagonal, with equality for orthogonal parameters: nuisance parameters inflate the bound by $1/(1-\rho^2)$ in the two-parameter case, which is one for the normal distribution and about 4.45 for the gamma distribution with shape 2.

Exercise 14.1. Let $X_1, \ldots, X_n$ be i.i.d. $N(\mu, \sigma^2)$ and consider the parametrisation $\varphi = (\mu, \sigma)$, where the standard deviation is used.

1. Write down the log-likelihood $\ell(\mu, \sigma)$ and compute its first partial derivatives.

2. By taking derivatives a second time and taking expectations, show that the information matrix for (μ, σ) is $\operatorname{diag}(n/\sigma^2,\, 2n/\sigma^2)$.

3. Write down the Cramér–Rao bound for an unbiased estimator for σ. Using Jensen's inequality, lemma A.7, show that the sample standard deviation $S = \sqrt{S^2}$ is not unbiased for σ and thus cannot be compared to $\operatorname{Var}(S)$ directly.

Exercise 14.2. Consider the gamma distribution with shape $\alpha > 0$ and rate $\beta > 0$. The log-likelihood of observing one value is $\ell_1(\alpha, \beta) = \alpha \log \beta - \log \Gamma(\alpha) + (\alpha - 1) \log X - \beta X$ from exercise 7.1, and the information matrix $\mathcal{I}(\alpha, \beta)$ is given in section 14.4.

1. Compute the first and second partial derivatives of $\ell_1(\alpha, \beta)$ and check the matrix.

2. Assume that the shape parameter is known to be $\alpha = 2$. In this case, β is the only parameter. From the matrix derive the Fisher information about β. Why is this a diagonal element of $\mathcal{I}(\alpha, \beta)$ and not of the inverse matrix?

3. Exercise 11.3 considers parametrisation of the same model in terms of the scale parameter $\theta = 1/\beta$. Use proposition 11.9 to derive the value $\mathcal{I}(\theta) = 2/\theta^2$ found in that exercise.

4. Now assume that both parameters are unknown. Show that the Cramér–Rao bound for an unbiased estimator for β from a sample of size n is given by
$$\frac{\beta^2 \psi'(\alpha)}{n(\alpha\psi'(\alpha) - 1)},$$
and determine the ratio of this bound to the bound for known shape parameter $\alpha = 2$.

Exercise 14.3. Let $X_1, \ldots, X_m$ be i.i.d. with distribution $N(\mu_1, \sigma^2)$ and, independently of these, let $Y_1, \ldots, Y_n$ be i.i.d. with distribution $N(\mu_2, \sigma^2)$, where the common variance σ^2 is known and $\theta = (\mu_1, \mu_2)$ is unknown. The score of all $m + n$ observations is the sum of the scores of the two samples, and the results of this lecture carry over with $\mathcal{I}_n(\theta)$ replaced by the covariance matrix $\mathcal{I}_{m+n}(\theta)$ of this score.

1. Write down the log-likelihood of all $m + n$ observations and show that the information matrix is $\mathcal{I}_{m+n}(\mu_1, \mu_2) = \operatorname{diag}(m/\sigma^2,\, n/\sigma^2)$.

2. Using proposition 14.5, find the Cramér–Rao bound for an unbiased estimator for the difference $\mu_1 - \mu_2$. Show that $\bar{X} - \bar{Y}$ achieves this bound.

3. Assume now that σ^2 is also unknown, *i.e.* when we have $\theta = (\mu_1, \mu_2, \sigma^2)$. Show that the 3×3 information matrix is still diagonal and discuss the implications for the estimation of $\mu_1 - \mu_2$ when σ^2 is unknown.

Lecture 15

Cramér–Rao and Efficiency in R

In lecture 13 we proved the Cramér–Rao inequality, theorem 13.2. In this lecture we compare two unbiased estimators with the bound by simulation. The sample mean of a Poisson sample attains the bound. The bias-corrected maximum likelihood estimator for the exponential rate does not, and we use proposition 13.5 to see why. As in the previous R sessions, appendix B covers the R commands we use, and we predict the outcome of each simulation before we run it.

15.1 An Efficient Estimator Against the Bound

In this section we consider $X_1, \ldots, X_n$ i.i.d. Poisson with mean $\theta = 3$. From example 11.6 we know that the Cramér–Rao bound for an unbiased estimator for θ is θ/n. The sample mean, $\bar{X}$, is an unbiased estimator for θ with $\mathrm{Var}_\theta(\bar{X}) = \theta/n$ (see lemma 2.2) and thus, by definition 13.4, is an efficient estimator. We check this by simulation, with the loop over `ns` from lecture 9. The command `var` can be used to compute the sample variance of a vector of estimates, and dividing the bound by this value allows us to estimate the efficiency of the estimator. The full script for this section, with lines intentionally shuffled (except for the function body and the loop which are left as one piece), is given in the following box. Before you can run the script, you will need to arrange the lines yourself.

```
bound <- theta / ns
data.frame(n = ns, variance = signif(v, 3), bound = signif(bound, 3),
           efficiency = round(bound / v, 3))
est_pois <- function(n, theta) {
  mean(rpois(n, lambda = theta))
}
set.seed(2026)
for (j in seq_along(ns)) {
  v[j] <- var(replicate(10000, est_pois(ns[j], theta)))
}
theta <- 3
v     <- numeric(length(ns))
ns    <- c(5, 10, 20, 50, 100, 200)
```

As in lectures 6 and 9, all names must be defined before they are used: the function, `theta` and `ns` come first, `v` needs `length(ns)`, the seed must be set before the loop, and the table must be created last. Before you try the reconstructed script, make a prediction for the efficiency column. Will all entries be exactly 1? Can an entry be larger than 1, despite the fact that theorem 13.2 states that no unbiased estimator can have a variance smaller than the bound?

```
est_pois <- function(n, theta) {
  mean(rpois(n, lambda = theta))
}
theta <- 3
ns    <- c(5, 10, 20, 50, 100, 200)
v     <- numeric(length(ns))
set.seed(2026)
for (j in seq_along(ns)) {
  v[j] <- var(replicate(10000, est_pois(ns[j], theta)))
}
bound <- theta / ns
data.frame(n = ns, variance = signif(v, 3), bound = signif(bound, 3),
           efficiency = round(bound / v, 3))
```

```
##     n variance bound efficiency
## 1   5   0.5970 0.600      1.005
## 2  10   0.3000 0.300      1.001
## 3  20   0.1510 0.150      0.995
## 4  50   0.0590 0.060      1.017
## 5 100   0.0297 0.030      1.010
## 6 200   0.0150 0.015      0.999
```

The empirical variances in the table follow the bound for all sample sizes, and the estimated efficiencies are clustered around 1 with four of the six entries above this value. This is no contradiction: the theorem only bounds the true variance $\mathrm{Var}_\theta(\bar{X})$, whereas the table shows the sample variance of $N = 10000$ simulated estimates, which is itself random. For a normal sample of size N, example 14.6 gives $\mathrm{Var}(S^2) = 2\sigma^4/(N-1)$, so that the relative standard error of a sample variance is about $\sqrt{2/N}$, *i.e.* 1.4% for $N = 10000$. Estimates with a heavier tail than the normal distribution give a larger error. An estimated efficiency of 1.017 thus means that, within the limits of error of the simulation, the estimator is efficient.

15.2 An Estimator Not Attaining the Bound

In this section we consider $X_1, \ldots, X_n$ i.i.d. exponential with rate $\theta > 0$, and an unbiased estimator for θ which does not attain the bound. From exercise 13.4 we know that the bias-corrected maximum likelihood estimator $\tilde{\theta} = (n-1)/\sum_i X_i$ is unbiased and has variance $\theta^2/(n-2)$. On the other hand, the bound from example 11.8 is θ^2/n and thus the efficiency of this estimator is $(n-2)/n$. By the same exercise, $\tilde{\theta}$ is the UMVUE, and thus no unbiased estimator for the rate has efficiency larger than $(n-2)/n$.

We repeat the analysis of the previous section for $\tilde{\theta}$ with $\theta = 2$. To do this, we replace `est_pois` by the function `est_exp` below, both in its definition and

in the loop. We then set `theta <- 2` and `ns <- c(5, 10, 20, 50, 100)`, recreate `v`, reset the seed as before and replace the bound by `theta^2 / ns`. The resulting script is `R/S15-efficiency.R`. Before you run the script, write down $(n-2)/n$ for $n = 5$, 10, 20, 50 and 100. Considering the simulation error we have observed, for which sample sizes could the simulation be expected to distinguish $\tilde{\theta}$ from an efficient estimator?

```
est_exp <- function(n, theta) {
  (n - 1) / sum(rexp(n, rate = theta))
}
data.frame(n = ns, efficiency = round(bound / v, 3), exact = (ns - 2) /
    ns)
```

```
##     n efficiency exact
## 1   5      0.607  0.60
## 2  10      0.803  0.80
## 3  20      0.903  0.90
## 4  50      0.929  0.96
## 5 100      1.008  0.98
```

For $n = 5$, 10 and 20 the simulated efficiencies are close to the exact values and clearly below 1. The estimator $\tilde{\theta}$ has a heavy right tail, and repeating the simulation with other seeds shows that the estimated efficiency fluctuates by about 0.03 for $n = 5$ and by about 0.015 for $n = 50$ and 100. For these two sample sizes, the exact loss of 4% and 2% is comparable to this error: the simulated 0.929 suggests a larger loss than the true one, and the simulated 1.008 suggests no loss at all. We would need considerably more than 10000 repetitions to resolve these values. The loss of efficiency, $2/n$, goes to zero as n increases: $\tilde{\theta}$ is asymptotically efficient in the sense of lecture 17, but not efficient.

To explain the difference between the two estimators, we return to the proof of theorem 13.2 in lecture 13, which uses the Cauchy–Schwarz inequality between the estimator T and the score $\ell'(\theta)$. Dividing both sides of this inequality by $\mathrm{Var}_\theta(T)\,\mathcal{I}_n(\theta)$ shows that the efficiency of T is given by the squared correlation between T and the score. This equals 1, if and only if $T - \theta$ is a multiple of the score. For the Poisson sample $\bar{X} - \theta$ equals θ/n times the score, and thus the correlation is 1 without any calculation. For the exponential sample $\tilde{\theta} - \theta$ is a function of the score, but not a linear one. The following function computes the estimation error and the score for a given sample of size $n = 5$. The command `cor` can be used to compute the correlation of two vectors. Predict the value of the squared correlation before you run the code.

```
n <- 5
err_score <- function(n, theta) {
  x <- rexp(n, rate = theta)
  c((n - 1) / sum(x) - theta, n / theta - sum(x))
}
set.seed(2026)
expo <- replicate(2000, err_score(n, 2))
cor(expo[1, ], expo[2, ])^2
```

```
## [1] 0.6147813
```

The squared correlation 0.615 coincides with the efficiency 3/5 from the table, up to simulation error. To see where the remaining variance comes from, we plot the estimation error against the score and add the line $c(\theta)\,\ell'(\theta)$ with $c(\theta) = \theta^2/n$ from proposition 13.5. Before you look at the plot, predict where the points will lie for an efficient estimator.

```
plot(expo[2, ], expo[1, ], pch = 20, xlab = "score",
     ylab = "estimation error")
abline(a = 0, b = theta^2 / n, lty = 2)
```

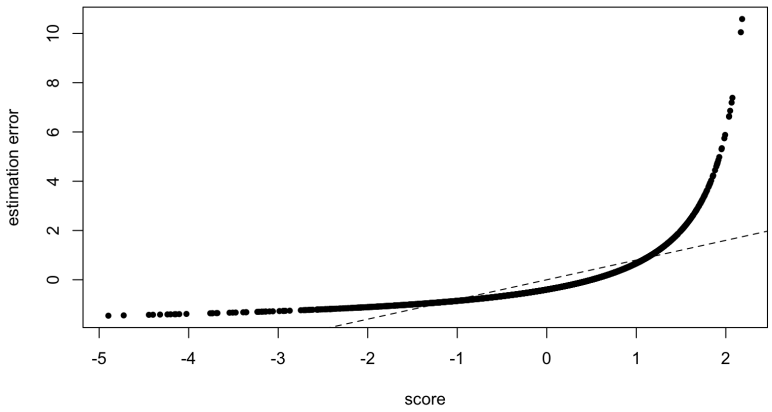


Figure 15.1: Estimation error $\tilde{\theta} - \theta$ against the score $\ell'(\theta)$ for 2000 exponential samples of size $n = 5$ with $\theta = 2$. The dashed line is $c(\theta)\,\ell'(\theta)$ with $c(\theta) = \theta^2/n$.

By proposition 13.5, every point lies on the dashed line for an efficient estimator. In figure 15.1 the points lie on a curve instead. The vertical distances between the points and the line are the part of the error which the score does not explain, and this part is the reason for the loss of efficiency.

15.3 A Task to Take Home

The following task is in a style similar to the practical report. Consider the sample variance S^2, computed using `var`, as an estimator for σ^2 for a normal sample with $\mu = 0$ and $\sigma^2 = 1$. The bound $2\sigma^4/n$ for this estimator is given in example 14.6, together with the variance $2\sigma^4/(n-1)$ of S^2. Repeat the study of section 15.1 for S^2, and create a table which shows the estimated and the exact efficiencies for the sample sizes in `ns`. Finally, in three sentences, discuss whether S^2 is efficient, how large its loss of efficiency is, and, using the relative error $\sqrt{2/N}$ from section 15.1, how many repetitions N would be required to see the loss of efficiency of S^2 at $n = 100$.

Before the next session, you should type the commands used in this session, and should check that you get the same results as the text.

Summary.

- The estimated efficiency, *i.e.* the ratio of the Cramér–Rao bound to the sample variance of the simulated estimates, fluctuates around the true efficiency by about $\sqrt{2/N}$ for N repetitions, *i.e.* 1.4% for $N = 10000$, and more for estimators with heavy tails. It can be greater than 1. A simulation can only detect a loss of efficiency larger than this error.
- The Poisson mean is efficient for all n. The bias-corrected exponential-rate estimator has efficiency $(n-2)/n$. The efficiency is the square of the correlation between the estimator and the score, and equals 1, if and only if the estimator is a linear function of the score.

Lecture 16

Consistency of the MLE

In lecture 5 we have seen that the maximum likelihood estimator can be biased, that it may not be unique and that it may not even exist. The following two lectures present the results which, despite these problems, allow us to justify the use of the MLE: for large sample size the maximum likelihood estimator converges to the true parameter value, and does so with minimal spread. In this lecture we will prove the first of these two results, the consistency of the MLE, from a set of assumptions which we will explicitly state. The proof of the asymptotic normality in lecture 17 builds on these same assumptions and on the consistency result. The numerical aspects, concerning estimates and standard errors, will be discussed in lecture 18.

16.1 The Regularity Conditions

Throughout this and the following lectures, $X_1, \ldots, X_n$ are i.i.d. with density or probability weights $f(x;\theta)$, the parameter θ is one-dimensional and the parameter space $\Theta \subseteq \mathbb{R}$ is an open interval. As in lecture 11, the log-likelihood is the sum $\ell(\theta) = \sum_{i=1}^n \ell_i(\theta)$ of the terms $\ell_i(\theta) = \log f(X_i;\theta)$, and for random data the $\ell_i(\theta)$ are i.i.d. random variables. We write θ_0 for the true parameter value, and $\mathbb{P}_{\theta_0}$ and $\mathbb{E}_{\theta_0}$ for probabilities and expectations under this value.

The large-sample results are true for most models of interest, but not for all of them. The following assumptions are what the theorems require.

Definition 16.1. The model $\big(f(x;\theta)\,\big|\,\theta \in \Theta\big)$ satisfies the *regularity conditions*, if the following four statements hold.

(R1) The parameter is identifiable in the sense of definition 4.10, *i.e.* different values of θ lead to different distributions.

(R2) The support $\{x \mid f(x;\theta) > 0\}$ does not depend on θ.

(R3) The conditions of section 11.1 hold, for every x the function $\theta \mapsto f(x;\theta)$ is three times continuously differentiable on Θ, and the Fisher information $\mathcal{I}(\theta)$ of a single observation is a continuous function of θ with $\mathcal{I}(\theta) > 0$ for all $\theta \in \Theta$.

(R4) For every $\theta_0 \in \Theta$ there are a $\delta > 0$ and a function $M(x)$ such that

$\mathbb{E}_{\theta_0}(M(X_1)) < \infty$ and

$$\Big|\frac{\partial^3}{\partial\theta^3}\log f(x;\theta)\Big| \le M(x) \qquad \text{for all } x \text{ and all } \theta \text{ with } |\theta-\theta_0|<\delta.$$

The four conditions play different roles. Condition (R1) is needed for the question to make sense: if two parameter values give the same distribution, no estimator can converge to both. Condition (R2) excludes models like the uniform distribution from example 5.5, where the parameter moves the edge of the support and the likelihood jumps to zero. The arguments of this lecture fail for such models, as we will see in section 16.4. Condition (R3) is the smoothness needed to differentiate the log-likelihood. Since it includes the conditions of section 11.1, it also makes the two identities of lecture 11 available (lemma 11.1 and lemma 11.3),

$$\mathbb{E}_\theta(\ell_i'(\theta)) = 0 \qquad \text{and} \qquad \mathcal{I}(\theta) = \operatorname{Var}_\theta(\ell_i'(\theta)) = -\mathbb{E}_\theta(\ell_i''(\theta)). \tag{16.1}$$

Condition (R4), and the parts of (R3) beyond the differentiability of the likelihood, are not needed in this lecture. Condition (R4) controls the remainder term in the Taylor expansion in the proof of asymptotic normality in lecture 17, and it is listed here so that both lectures work from the same definition. Although the condition looks technical, it is usually easy to verify, because the third derivative of the log-likelihood is an explicit function of x and θ; exercise 16.1 does this for the exponential distribution.

The one-parameter models from tables 1.1 and 1.2, and the two-parameter models with one parameter held fixed (with m known for the binomial distribution), satisfy the regularity conditions, with the exception of the uniform distribution. The same is true for the t location model from section 5.3, despite the fact that the likelihood equation may have more than one solution. We will return to this problem in section 16.4.

16.2 Comparing the True Value with a Wrong Value

If θ_0 is the true value, it seems natural that for a large sample the likelihood $L(\theta_0)$ should be larger than $L(\theta)$ for each fixed wrong value θ. This is the main step in the consistency proof, and thus we prove it first.

Lemma 16.2. *Suppose that (R1) and (R2) hold. Let $\theta \neq \theta_0$ and assume that $\log(f(X_1;\theta)/f(X_1;\theta_0))$ has finite expectation under $\mathbb{P}_{\theta_0}$. Then*

$$\lim_{n\to\infty} \mathbb{P}_{\theta_0}(L(\theta_0) > L(\theta)) = 1.$$

Proof. Let $Z_i = f(X_i;\theta)/f(X_i;\theta_0)$ for $i = 1,\dots,n$, where, by (R2), the ratios are well defined and positive for all observations. Using logarithms and dividing by n, we can rewrite the event $L(\theta_0) > L(\theta)$ as

$$\frac{1}{n}\sum_{i=1}^{n} \log Z_i < 0.$$

The left-hand side is the sample mean of the i.i.d. random variables $\log Z_i$ and by the law of large numbers, proposition A.14, converges in probability to $\mu =$

$\mathbb{E}_{\theta_0}(\log Z_1)$. Therefore, we only need to show that μ is strictly negative: then on the event that the sample mean is non-negative, it differs from μ by at least $|\mu|$, and by definition A.12 the probability of this event tends to zero.

To study the expectation μ, let $Z = Z_1$, so that $\mu = \mathbb{E}_{\theta_0}(\log Z)$. Writing the expectation of Z as an integral over the common support, we can cancel the density $f(x;\theta_0)$ and get

$$\mathbb{E}_{\theta_0}(Z) = \int \frac{f(x;\theta)}{f(x;\theta_0)}\, f(x;\theta_0)\,\mathrm{d}x = \int f(x;\theta)\,\mathrm{d}x = 1,$$

with a sum instead of an integral for a discrete model. Since Z has finite mean and the function $g(z) = -\log z$ is strictly convex on $(0,\infty)$, we can use Jensen's inequality, lemma A.7, to get

$$-\mu = \mathbb{E}_{\theta_0}(-\log Z) \geq -\log \mathbb{E}_{\theta_0}(Z) = -\log 1 = 0,$$

and the inequality is strict unless Z is constant with probability one. If Z were constant, we would have $f(x;\theta) = c\, f(x;\theta_0)$ on the common support and, since both functions have integral one, this would imply $c = 1$, *i.e.* the two distributions would be the same, which (R1) excludes for $\theta \neq \theta_0$. Thus the inequality is strict and we have $\mu < 0$. This completes the proof. □

The quantity $-\mathbb{E}_{\theta_0}(\log Z) = \mathbb{E}_{\theta_0}(\log(f(X_1;\theta_0)/f(X_1;\theta)))$ from the proof is called the *Kullback–Leibler divergence*. The proof shows that, on average, each observation increases the log-likelihood of the truth by this positive amount, compared to the log-likelihood of the competitor θ; the amount depends on θ. Note that the lemma compares θ_0 with one fixed value θ at a time. It does not state that $L(\theta_0)$ eventually exceeds $L(\theta)$ for all $\theta \neq \theta_0$ at once, and it says nothing directly about the maximiser of the likelihood. In the next section we apply the lemma to the two fixed values $\theta_0 \pm \varepsilon$ and turn the result into a statement about the MLE.

16.3 Consistency

From definition 2.9 we know that an estimator $\hat{\theta}_n$ is consistent for θ, if $\hat{\theta}_n \xrightarrow{\mathrm{p}} \theta$ for all possible true values. In lecture 2 we have already seen how to prove consistency using the mean squared error. For the maximum likelihood estimator this approach is not feasible in general, because there is no general formula for the mean squared error. Instead, we will use the likelihood equation from lecture 5 to show that with probability tending to one there is a root of the equation which is close to θ_0. If we assume that the root is unique and is the MLE, it follows that the MLE is close to θ_0.

Theorem 16.3. *Suppose that the regularity conditions (R1), (R2) and (R3) hold and that, in addition, the following two statements hold.*

1. *For all $\theta, \theta' \in \Theta$ the log-likelihood ratio $\log(f(X_1;\theta)/f(X_1;\theta'))$ has finite expectation under $\mathbb{P}_{\theta'}$.*

2. *For every sample size $n \geq 1$ and every sample whose observations lie in the common support from (R2), the likelihood equation $\ell'(\theta) = 0$ has a unique root, and this root is the maximum likelihood estimator $\hat{\theta}_n$.*

Then $\hat{\theta}_n$ is a consistent estimator for θ.

Proof. Let the true value be θ_0 and let $\varepsilon > 0$ be small enough that the closed interval $[\theta_0 - \varepsilon, \theta_0 + \varepsilon]$ is contained in the open set Θ. Let $\theta_1 = \theta_0 - \varepsilon$ and $\theta_2 = \theta_0 + \varepsilon$ and consider the event

$$S_\varepsilon = \{L(\theta_1) < L(\theta_0) \text{ and } L(\theta_2) < L(\theta_0)\}.$$

By the assumptions of the theorem, lemma 16.2 applies to both θ_1 and θ_2, and thus the two events $L(\theta_1) \geq L(\theta_0)$ and $L(\theta_2) \geq L(\theta_0)$ have probability tending to zero. Since the complement of S_ε is the union of these two events, we find

$$\mathbb{P}_{\theta_0}(S_\varepsilon) \geq 1 - \mathbb{P}_{\theta_0}(L(\theta_1) \geq L(\theta_0)) - \mathbb{P}_{\theta_0}(L(\theta_2) \geq L(\theta_0)) \longrightarrow 1$$

as $n \to \infty$.

We now show that on the event S_ε the MLE is within ε of θ_0. Assume that the sample lies in S_ε. By (R3) the likelihood L is a continuous function of θ, and a continuous function on a closed, bounded interval attains its maximum. Thus there is a point $\theta^* \in [\theta_1, \theta_2]$ with $L(\theta^*) \geq L(\theta)$ for all $\theta \in [\theta_1, \theta_2]$. Since $L(\theta_0)$ is larger than both $L(\theta_1)$ and $L(\theta_2)$, the maximum is not attained at either end of the interval, *i.e.* we have $\theta^* \in (\theta_1, \theta_2)$. The function L is differentiable and has a maximum at the interior point θ^*, and thus $L'(\theta^*) = 0$. Since the observations lie in the common support, we have $L > 0$, and thus θ^* is also a root of the likelihood equation $\ell'(\theta) = L'(\theta)/L(\theta) = 0$. By assumption, this root is unique and equals the MLE, and thus $\hat{\theta}_n = \theta^*$. Since θ^* is in (θ_1, θ_2), we have shown that

$$S_\varepsilon \subseteq \{|\hat{\theta}_n - \theta_0| < \varepsilon\}.$$

Taking probabilities we find $\mathbb{P}_{\theta_0}(|\hat{\theta}_n - \theta_0| < \varepsilon) \geq \mathbb{P}_{\theta_0}(S_\varepsilon) \to 1$, *i.e.* $\mathbb{P}_{\theta_0}(|\hat{\theta}_n - \theta_0| \geq \varepsilon) \to 0$ as $n \to \infty$. At the start of the proof we restricted ε to small values, but this is no loss, since the probability $\mathbb{P}_{\theta_0}(|\hat{\theta}_n - \theta_0| \geq \varepsilon)$ is non-increasing in ε. Since θ_0 was an arbitrary point of Θ, this is the statement $\hat{\theta}_n \xrightarrow{\mathrm{p}} \theta$ for every true value. This completes the proof. □

In the standard examples, the uniqueness of the root is typically proved by showing that the log-likelihood is strictly concave, as for the exponential distribution in example 5.2; exercise 16.1 carries out the argument for this model.

Remark. The assumption that the likelihood equation has a unique root for *every* sample is slightly stronger than required: the proof is unchanged if the condition only holds on an event with probability tending to one, since we can intersect S_ε with such an event. This covers the Bernoulli distribution of example 5.3, where the likelihood equation has no root when all observations are equal, an event of probability $\theta_0^n + (1-\theta_0)^n \to 0$. It also covers the Poisson distribution, where the same happens when all observations are zero, an event of probability $e^{-n\theta_0} \to 0$.

16.4 When the Assumptions Fail

We now look at what happens when the main hypotheses of theorem 16.3 fail. These are the cases where more care is needed in practice.

If the support of the model depends on the parameter, *i.e.* if (R2) fails, the likelihood is not a smooth function of θ and the MLE is typically not a root

of the likelihood equation. For example, for the uniform distribution on $(0, \theta)$, the MLE is the sample maximum, as shown in example 5.5, and the theorem does not apply in this case. Nevertheless, the estimator is consistent: the mean squared error of the estimator goes to zero as the sample size increases, as shown in exercise 2.2, and thus lemma 2.11 applies; exercise 16.3 gives a direct proof of the same fact. We have seen in lecture 13 that for this model the Cramér–Rao bound can be violated, and in lecture 17 we will see that the limiting distribution of the MLE is not normal.

If the likelihood equation has several roots, the regularity conditions may all hold and the theorem still does not apply, because we cannot identify the root found in the proof with the MLE; the t location model of section 5.3 is of this type. What survives is this: under the other assumptions of the theorem, for each fixed ε, with probability tending to one there is a root within ε of θ_0. What is lost is the guarantee that the root we find, or the global maximiser, is this one. The practical consequence of this is that numerical maximisation of the likelihood should be started with a simple consistent estimator, so that the iteration converges to the nearby root. This is a lesson we will take up in lecture 18.

Summary.

- The regularity conditions (R1) to (R4) from definition 16.1 state that the parameter must be identifiable, the support of the model must not depend on θ, the log-likelihood must be three times differentiable with finite and positive Fisher information, and the third derivative of the log-likelihood must be bounded near each θ_0 by a function with finite expectation.
- For a fixed wrong value θ, the likelihood of the true value θ_0 exceeds the likelihood of θ with probability tending to one. The proof uses the law of large numbers and Jensen's inequality.
- Under conditions (R1) to (R3), if the log-likelihood ratios have finite expectations and the likelihood equation has a unique root which is the MLE, then the MLE is consistent. The proof applies the lemma at $\theta_0 \pm \varepsilon$ and finds a maximum of the likelihood, and thus a root of the likelihood equation, between these two points.
- The theorem does not apply if the support of the model depends on the parameter (since then the MLE is typically not a root of the likelihood equation, as for the uniform distribution, even though it may still be consistent), or if the likelihood equation has multiple roots (since then we only know that, with probability tending to one, there is a root near θ_0).

Exercise 16.1. Let $X_1, \ldots, X_n$ be i.i.d. exponentially distributed with rate $\theta > 0$, *i.e.* $f(x; \theta) = \theta e^{-\theta x}$ for all $x \geq 0$.

1. Show that the regularity conditions (R1) to (R4) hold for this model. Hint: For (R3) you can use the value $\mathcal{I}(\theta) = 1/\theta^2$ from example 11.8 and the fact from lecture 11 that exponential families satisfy the conditions of section 11.1. For (R4) you should take derivatives of $\log f(x;\theta)$ with respect to θ and then find a bound $M(x)$.

2. Show that the likelihood equation has only one root for each sample with $\sum_i x_i > 0$ and that this root is the MLE. Use theorem 16.3 and the remark after its proof to conclude that $\hat{\theta}_n = 1/\bar{X}$ is consistent for θ.

3. Find an alternative proof of the consistency of $1/\bar{X}$, using the law of large numbers (proposition A.14), the continuous mapping theorem (proposition A.17), and any other arguments you need.

Exercise 16.2. Let X_1 be the only observation of a Bernoulli random variable with success probability $\theta_0 \in (0,1)$, and let $\theta \in (0,1)$ be another parameter value.

1. Compute the expectation $\mathbb{E}_{\theta_0}\big(\log\big(f(X_1;\theta)/f(X_1;\theta_0)\big)\big)$ from the proof of lemma 16.2, as an explicit function of θ and θ_0.

2. Using the inequality $\log y \leq y - 1$ for $y > 0$, with equality only for $y = 1$, show that this expectation is negative for all $\theta \neq \theta_0$.

Exercise 16.3. Let $X_1, \ldots, X_n$ be i.i.d. uniformly distributed on the interval $(0,\theta)$ and let $\hat{\theta}_n = \max_i X_i$ be the MLE from example 5.5.

1. Show that for $0 < \varepsilon < \theta$ we have $\mathbb{P}_\theta\big(|\hat{\theta}_n - \theta| \geq \varepsilon\big) = (1 - \varepsilon/\theta)^n$ and thus $\hat{\theta}_n$ is consistent for θ.

2. Which of the assumptions of theorem 16.3 are violated for this model? Why can the theorem not be applied in this case, despite the fact that the conclusion is still true?

Lecture 17

Asymptotic Normality and Efficiency of the MLE

In lecture 16 we have seen that under regularity conditions the maximum likelihood estimator converges to the true parameter value. In this lecture we show that for large samples the error $\hat{\theta}_n - \theta_0$ is approximately normal with mean zero and variance given by the Cramér–Rao bound from lecture 13. The proof is challenging, but uses only tools we have already learned. The resulting theorem justifies standard errors and approximate confidence statements, which we compute in lecture 18 and make precise in lecture 26.

17.1 The Theorem

As in lecture 11, we write $\mathcal{I}(\theta)$ and $\mathcal{I}_n(\theta) = n\mathcal{I}(\theta)$ for the Fisher information of one observation and of the sample, and as in lecture 16 we write θ_0 for the true parameter value. Since the error $\hat{\theta}_n - \theta_0$ converges to zero, we need to magnify it to see the shape of its distribution. We use the factor $\sqrt{n}$, which for the sample mean gives the central limit theorem, and thus we study the scaled error $\sqrt{n}\,(\hat{\theta}_n - \theta_0)$.

Theorem 17.1. *Suppose that the regularity conditions (R1) to (R4) from definition 16.1 are satisfied, that* $\log\bigl(f(X_1;\theta)/f(X_1;\theta')\bigr)$ *has finite expectation under* $\mathbb{P}_{\theta'}$ *for all* $\theta, \theta' \in \Theta$ *and that for every* $\theta_0 \in \Theta$ *the likelihood equation* $\ell'(\theta) = 0$ *has, with* $\mathbb{P}_{\theta_0}$*-probability tending to one as* $n \to \infty$*, a unique root* $\hat{\theta}_n$*, which is the maximum likelihood estimator. Then for every* $\theta_0 \in \Theta$ *we have, under* $\mathbb{P}_{\theta_0}$*,*

$$\sqrt{n}\,(\hat{\theta}_n - \theta_0) \xrightarrow{\mathrm{d}} N\Bigl(0, \frac{1}{\mathcal{I}(\theta_0)}\Bigr) \qquad (n \to \infty).$$

The assumptions are those of theorem 16.3, in the weaker form of the remark after its proof, plus condition (R4). Informally, the theorem states that, as n gets large,

$$\hat{\theta}_n \approx N\Bigl(\theta_0, \frac{1}{n\mathcal{I}(\theta_0)}\Bigr) = N\Bigl(\theta_0, \frac{1}{\mathcal{I}_n(\theta_0)}\Bigr).$$

This statement contains three different aspects: (1) the approximating distribution is centred at the true value, (2) the spread of the distribution decreases as $1/\sqrt{n}$, as for the sample mean, and (3) the variance of the approximating distribution equals the Cramér–Rao lower bound $1/\mathcal{I}_n(\theta_0)$ from lecture 13. The

bound concerns unbiased estimators, while the MLE is often biased, but example 17.4 shows that its bias disappears in the limit. Property (3) is called asymptotic efficiency, and the definition applies to any estimator.

Definition 17.2. An estimator $\tilde{\theta}_n$ for θ is *asymptotically efficient*, if $\sqrt{n}\,(\tilde{\theta}_n - \theta) \xrightarrow{\mathrm{d}} N\big(0, 1/\mathcal{I}(\theta)\big)$ under $\mathbb{P}_\theta$, for all $\theta \in \Theta$.

In this language theorem 17.1 states that under its assumptions the MLE is asymptotically efficient.

17.2 Proof of the Theorem

The idea of the proof is easiest to see in a first-order approximation. Since the MLE is a root of the score, Taylor expansion of ℓ' around the true value gives

$$0 = \ell'(\hat{\theta}_n) \approx \ell'(\theta_0) + \ell''(\theta_0)\,(\hat{\theta}_n - \theta_0),$$

and solving for the scaled error we find

$$\sqrt{n}\,(\hat{\theta}_n - \theta_0) \approx \frac{\ell'(\theta_0)/\sqrt{n}}{-\ell''(\theta_0)/n}.$$

The numerator is a scaled mean of the i.i.d. terms $\ell_i'(\theta_0)$, which by (16.1) and (R3) have mean zero and finite, positive variance $\mathcal{I}(\theta_0)$. Thus the central limit theorem, theorem A.15, gives

$$A_n = \frac{\ell'(\theta_0)}{\sqrt{n}} = \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} \ell_i'(\theta_0) - 0\right) \xrightarrow{\mathrm{d}} N\big(0, \mathcal{I}(\theta_0)\big). \tag{17.1}$$

The denominator is the mean of the i.i.d. terms $-\ell_i''(\theta_0)$, which by (16.1) have finite mean $\mathcal{I}(\theta_0)$, and thus the law of large numbers, proposition A.14, gives

$$B_n = -\frac{\ell''(\theta_0)}{n} = \frac{1}{n}\sum_{i=1}^{n}\big(-\ell_i''(\theta_0)\big) \xrightarrow{\mathrm{p}} \mathcal{I}(\theta_0). \tag{17.2}$$

The rigorous proof follows this outline, and most of the work goes into showing that the remainder of the expansion is negligible.

Proof of theorem 17.1. Let the true value be θ_0, let $\delta > 0$ be as in (R4), and let E_n be the event that the likelihood equation has a unique root, that this root is the MLE $\hat{\theta}_n$ and that $|\hat{\theta}_n - \theta_0| < \delta$. By the assumptions of the theorem, and since $\hat{\theta}_n$ is consistent by theorem 16.3 and the remark after its proof, we have $\mathbb{P}_{\theta_0}(E_n) \to 1$.

On the event E_n we expand the score around θ_0. By (R3), the function ℓ' is twice continuously differentiable. Thus, by Taylor's theorem with the Lagrange form of the remainder, we get

$$0 = \ell'(\hat{\theta}_n) = \ell'(\theta_0) + \ell''(\theta_0)\,(\hat{\theta}_n - \theta_0) + \tfrac{1}{2}\,\ell'''(\xi_n)\,(\hat{\theta}_n - \theta_0)^2,$$

where ξ_n is a point between θ_0 and $\hat{\theta}_n$. Outside the event E_n we set $\xi_n = \theta_0$. We define

$$C_n = \frac{1}{2n}\,\ell'''(\xi_n)\,(\hat{\theta}_n - \theta_0),$$

so that, like the terms A_n and B_n from (17.1) and (17.2), the term C_n is defined for every sample. Dividing the expansion by $\sqrt{n}$ and rearranging, we find that on E_n we have

$$A_n = \sqrt{n}\,(\hat{\theta}_n - \theta_0)\,(B_n - C_n).$$

To avoid division by zero, let $D_n = B_n - C_n$ if $B_n \neq C_n$ and $D_n = \mathcal{I}(\theta_0)$ otherwise. Then, on the event F_n that E_n occurs and $B_n \neq C_n$, we can write the scaled error as a ratio:

$$\sqrt{n}\,(\hat{\theta}_n - \theta_0) = \frac{A_n}{D_n}. \tag{17.3}$$

We have already seen the limits of A_n and B_n in (17.1) and (17.2), and it remains to control the remainder term C_n, for which we use consistency and condition (R4). Let M be the function from (R4). On the event E_n, the point ξ_n lies between θ_0 and $\hat{\theta}_n$, and thus within δ of θ_0. Writing $\ell'''(\xi_n) = \sum_i \ell_i'''(\xi_n)$ and using (R4) for each term, we find

$$\Big|\frac{1}{n}\,\ell'''(\xi_n)\Big| \leq \frac{1}{n}\sum_{i=1}^{n}|\ell_i'''(\xi_n)| \leq \frac{1}{n}\sum_{i=1}^{n} M(X_i) = \bar{M}_n$$

on this event, where $\bar{M}_n$ denotes the mean of $M(X_1), \ldots, M(X_n)$. Let $m = \mathbb{E}_{\theta_0}\big(M(X_1)\big) < \infty$ and $K = m + 1$. If $|\ell'''(\xi_n)|/n > K$, then either E_n does not occur, or $\bar{M}_n > m + 1$ and thus $|\bar{M}_n - m| \geq 1$. Since $\mathbb{P}_{\theta_0}(E_n) \to 1$ and since $\bar{M}_n \xrightarrow{\mathrm{p}} m$ by the law of large numbers, we find

$$\mathbb{P}_{\theta_0}\Big(\frac{|\ell'''(\xi_n)|}{n} > K\Big) \leq \mathbb{P}_{\theta_0}(E_n^{\mathrm{c}}) + \mathbb{P}_{\theta_0}\big(|\bar{M}_n - m| \geq 1\big) \longrightarrow 0.$$

The other factor of C_n is $\hat{\theta}_n - \theta_0$, which converges in probability to zero, as we have seen at the start of the proof. Let $\eta > 0$. Since $|C_n| = |\ell'''(\xi_n)|\,|\hat{\theta}_n - \theta_0|/(2n)$, the event $|C_n| \geq \eta$ is contained in the union of the events $|\ell'''(\xi_n)|/n > K$ and $|\hat{\theta}_n - \theta_0| \geq 2\eta/K$. Both of these events have probability tending to zero, and thus

$$C_n \xrightarrow{\mathrm{p}} 0. \tag{17.4}$$

It remains to combine the three limits (17.1), (17.2) and (17.4) using Slutsky's theorem, proposition A.16. By (17.2) and (17.4), the denominator satisfies $B_n - C_n \xrightarrow{\mathrm{p}} \mathcal{I}(\theta_0) > 0$. Since $|B_n - C_n - \mathcal{I}(\theta_0)| = \mathcal{I}(\theta_0)$ on the event $B_n = C_n$, this event has probability tending to zero, and since also $\mathbb{P}_{\theta_0}(E_n) \to 1$, we find $\mathbb{P}_{\theta_0}(F_n) \to 1$. Since $|D_n - \mathcal{I}(\theta_0)| \leq |B_n - C_n - \mathcal{I}(\theta_0)|$ for every sample, we have $D_n \xrightarrow{\mathrm{p}} \mathcal{I}(\theta_0)$. Let $Z \sim N\big(0, \mathcal{I}(\theta_0)\big)$. By the quotient part of Slutsky's theorem we have

$$\frac{A_n}{D_n} \xrightarrow{\mathrm{d}} \frac{Z}{\mathcal{I}(\theta_0)} \sim N\Big(0, \frac{1}{\mathcal{I}(\theta_0)}\Big),$$

since $Z/\mathcal{I}(\theta_0)$ has variance $\mathcal{I}(\theta_0)/\mathcal{I}(\theta_0)^2$. Finally, by (17.3) the scaled error coincides with this ratio on the event F_n. Thus, for every $\varepsilon > 0$, the event $\big|\sqrt{n}\,(\hat{\theta}_n - \theta_0) - A_n/D_n\big| \geq \varepsilon$ is contained in F_n^{c}, and since $\mathbb{P}_{\theta_0}(F_n) \to 1$ the difference converges in probability to zero. Slutsky's theorem now shows that the scaled error has the same limit as A_n/D_n. This completes the proof. □

Exercise 17.1 carries out these steps for the exponential distribution.

17.3 Standard Errors

The variance $1/\mathcal{I}(\theta_0)$ in theorem 17.1 depends on the unknown θ_0. The next result shows that we may replace θ_0 by its estimate.

Corollary 17.3. *Under the assumptions of theorem 17.1 we have*

$$\sqrt{n\mathcal{I}(\hat{\theta}_n)}\,(\hat{\theta}_n - \theta_0) \xrightarrow{\mathrm{d}} N(0,1) \qquad (n \to \infty).$$

Proof. By (R3) the function $\mathcal{I}$ is continuous and positive on Θ, and thus the function $\theta \mapsto \sqrt{\mathcal{I}(\theta)/\mathcal{I}(\theta_0)}$ is continuous at θ_0. Since $\hat{\theta}_n \xrightarrow{\mathrm{p}} \theta_0$ by theorem 16.3 and the remark after its proof, the continuous mapping theorem, proposition A.17, implies that $\sqrt{\mathcal{I}(\hat{\theta}_n)/\mathcal{I}(\theta_0)} \xrightarrow{\mathrm{p}} 1$. Multiplying the result of theorem 17.1 by the constant $\sqrt{\mathcal{I}(\theta_0)}$, which is allowed by the continuous mapping theorem, gives $\sqrt{n\mathcal{I}(\theta_0)}\,(\hat{\theta}_n - \theta_0) \xrightarrow{\mathrm{d}} Z$ where $Z \sim N(0,1)$. Using the product part of Slutsky's theorem, proposition A.16, we find

$$\sqrt{n\mathcal{I}(\hat{\theta}_n)}\,(\hat{\theta}_n - \theta_0) = \sqrt{\frac{\mathcal{I}(\hat{\theta}_n)}{\mathcal{I}(\theta_0)}} \cdot \sqrt{n\mathcal{I}(\theta_0)}\,(\hat{\theta}_n - \theta_0) \xrightarrow{\mathrm{d}} 1 \cdot Z = Z.$$

This completes the proof. □

The standard error $\mathrm{se}(\hat{\theta}_n) = 1/\sqrt{\mathcal{I}_n(\hat{\theta}_n)}$ of lecture 12 is now justified by the corollary: for large n, the distribution of $\hat{\theta}_n$ is approximately normal with mean θ_0 and standard deviation $\mathrm{se}(\hat{\theta}_n)$. This is a statement about distributions, not about moments: the MLE for the log-odds in exercise 17.2 does not exist when all observations are equal, an event of positive probability for every n, and thus it has no variance.

The corollary also justifies the Wald interval $\hat{\theta}_n \pm 1.96\ \mathrm{se}(\hat{\theta}_n)$ from lecture 12, whose coverage probability tends to 0.95, as we will show in lecture 26.

If we do not have a closed formula for $\mathcal{I}$, the denominator $\mathcal{I}_n(\hat{\theta}_n)$ can be replaced by the *observed information* $-\ell''(\hat{\theta}_n)$, the curvature of the log-likelihood at the maximum. Expanding ℓ'' around θ_0 gives the term B_n from the proof plus a remainder which we can handle like C_n, and we find

$$-\frac{1}{n}\,\ell''(\hat{\theta}_n) \xrightarrow{\mathrm{p}} \mathcal{I}(\theta_0).$$

By Slutsky's theorem the corollary thus remains true with this replacement. This is the form we will use with a numerical optimiser in lecture 18.

17.4 Examples

The Bernoulli and Poisson MLEs from lecture 5 are sample means, and for them theorem 17.1 only restates the central limit theorem (exercises 17.2 and 17.3). The first new case is an MLE which is not a sample mean.

Example 17.4. Let $X_1, \ldots, X_n$ be i.i.d. with exponential rate θ. Then the MLE for θ is $\hat{\theta}_n = 1/\bar{X}$ from example 5.2, and by exercise 16.1 all assumptions

of theorem 17.1 hold for this model. From example 11.8 we know that the Fisher information is given by $\mathcal{I}(\theta) = 1/\theta^2$. Thus, by theorem 17.1,

$$\sqrt{n}\left(\frac{1}{\bar{X}} - \theta\right) \xrightarrow{\mathrm{d}} N(0, \theta^2),$$

with standard error $\operatorname{se}(\hat{\theta}_n) = \hat{\theta}_n/\sqrt{n}$, and the approximate 95% Wald interval for θ is $\hat{\theta}_n \pm 1.96\,\hat{\theta}_n/\sqrt{n}$.

To compare with exact results, recall that in lecture 5 we found $\mathbb{E}_\theta(\hat{\theta}_n) = n\theta/(n-1)$, which is a bias of $\theta/(n-1)$. In exercise 5.3 we found the variance $\operatorname{Var}_\theta(\hat{\theta}_n) = n^2\theta^2/((n-1)^2(n-2))$. Multiplied by n, this variance converges to θ^2, and multiplied by $\sqrt{n}$, the bias converges to zero: the bias is of smaller order than the spread, and thus disappears in the limit.

We can also treat functions of the parameter. The mean of the exponential distribution is $\mu = g(\theta) = 1/\theta$, where g is a bijection of $(0, \infty)$, and by proposition 5.6 the MLE for this parameter is $\hat{\mu}_n = 1/\hat{\theta}_n = \bar{X}$. Using the delta method, proposition A.18 with $g'(\theta) = -1/\theta^2$, we get

$$\sqrt{n}\,(\bar{X} - \mu) \xrightarrow{\mathrm{d}} N\big(0,\ g'(\theta)^2\,\theta^2\big) = N\Big(0, \frac{1}{\theta^2}\Big) = N(0, \mu^2).$$

This is the central limit theorem for $\bar{X}$, and it is also what theorem 17.1 gives in the parametrisation by μ, where the information is $\mathcal{I}_\mu(\mu) = 1/\mu^2$ by proposition 11.9. Replacing θ by $\hat{\theta}_n$, we find that the standard error of $\bar{X}$ is

$$|g'(\hat{\theta}_n)|\,\operatorname{se}(\hat{\theta}_n) = \frac{1}{\hat{\theta}_n^2} \cdot \frac{\hat{\theta}_n}{\sqrt{n}} = \frac{\bar{X}}{\sqrt{n}}.$$

The same works for every twice continuously differentiable bijection g with $g'(\theta) \neq 0$: the MLE $g(\hat{\theta}_n)$ has standard error $|g'(\hat{\theta}_n)|\,\operatorname{se}(\hat{\theta}_n)$, and since its asymptotic variance $g'(\theta)^2/\mathcal{I}(\theta)$ is the reciprocal of the information for $g(\theta)$, it is again asymptotically efficient. Thus asymptotic efficiency holds much more widely than the exact efficiency of lecture 13: by exercise 13.3 there is no efficient estimator for the probability $e^{-\theta}$ of observing zero in a Poisson sample, but by exercise 17.3 its MLE $e^{-\bar{X}}$ is asymptotically efficient. The conditions on g matter: the Bernoulli variance $\theta(1-\theta)$ has derivative zero at $\theta = 1/2$, where the delta method does not apply.

17.5 Quality of the Approximation

Theorem 17.1 is a limit statement and does not say how large n must be. For small samples the estimator can be skewed: for the exponential rate, $1/\bar{X}$ is right-skewed, and its bias $\theta/(n-1)$ is visible for n around 10. Near the boundary of the parameter space the approximation is poor: for the Poisson distribution with θ_0 close to zero, all observations are zero with appreciable probability, and then $\bar{X} = 0$ lies outside $\Theta = (0, \infty)$ and the Wald interval has zero width. Finally, without the regularity conditions the limit need not be normal: for the uniform distribution, where (R2) fails, the error is of order $1/n$, and exercise 17.5 shows that $n(\theta - \hat{\theta}_n)$ converges in distribution to an exponential distribution.

17.6 Vector Parameters

We conclude with the extension of theorem 17.1 to a vector parameter $\theta = (\theta_1, \dots, \theta_k)$, which we state without proof. Under appropriate regularity conditions, the MLE is approximately normal with mean θ_0 and covariance matrix $\mathcal{I}_n(\theta_0)^{-1}$, the inverse of the information matrix from lecture 14. For the normal distribution with unknown mean and variance, the inverse matrix is $\operatorname{diag}(\sigma^2/n,\, 2\sigma^4/n)$, and thus $\bar{X}$ and the MLE $\hat{\sigma}^2$ are approximately normal with variances σ^2/n and $2\sigma^4/n$, and they are asymptotically independent.

Summary.

- Under regularity conditions and assuming the uniqueness of the likelihood root, we have $\sqrt{n}\,(\hat{\theta}_n - \theta_0) \xrightarrow{\mathrm{d}} N\big(0, 1/\mathcal{I}(\theta_0)\big)$: the limit is centred at θ_0, the error is of order $1/\sqrt{n}$ and the asymptotic variance equals the Cramér–Rao bound, *i.e.* the MLE is asymptotically efficient.
- The proof uses Taylor expansion of the score around θ_0, the central limit theorem for the score, the law of large numbers for the second derivative, consistency and (R4) for the remainder, and Slutsky's theorem to combine these results. Events of vanishing probability do not affect the limit.
- Replacing θ_0 by $\hat{\theta}_n$ in the expression for the variance does not change the limit, and thus we get the standard error $\operatorname{se}(\hat{\theta}_n) = 1/\sqrt{\mathcal{I}_n(\hat{\theta}_n)}$ and the approximate 95% Wald interval $\hat{\theta}_n \pm 1.96 \operatorname{se}(\hat{\theta}_n)$ for θ_0. The observed information $-\ell''(\hat{\theta}_n)$ may be used in place of $\mathcal{I}_n(\hat{\theta}_n)$.
- Using the delta method, we can apply the result to a twice continuously differentiable bijection g of the parameter with $g'(\theta) \neq 0$. The standard error for such an application is $|g'(\hat{\theta}_n)| \operatorname{se}(\hat{\theta}_n)$.
- The approximation can be poor for small n, especially when the estimator is skewed or θ_0 is near the boundary of the parameter space, and it breaks down completely when the regularity conditions are violated, *e.g.* for the uniform distribution.
- For a vector parameter, the MLE is approximately normal with covariance matrix $\mathcal{I}_n(\theta_0)^{-1}$.

Exercise 17.1. Let $X_1, \dots, X_n$ be i.i.d. exponentially distributed with rate $\theta_0 > 0$, and let $\hat{\theta}_n = 1/\bar{X}$ be the MLE. In this exercise we follow the proof of theorem 17.1 for this model.

1. Find $\ell'(\theta)$, $\ell''(\theta)$ and $\ell'''(\theta)$, and express the terms A_n, B_n and C_n from the proof in terms of $\bar{X}$, θ_0, $\hat{\theta}_n$ and the intermediate point ξ_n.

2. Use the central limit theorem to show directly that $A_n \xrightarrow{\mathrm{d}} N(0, 1/\theta_0^2)$, and note that B_n is not random.

3. Without using the bound M from condition (R4), show that $C_n \xrightarrow{\text{p}} 0$. Hint: on the event $|\hat{\theta}_n - \theta_0| < \theta_0/2$ we have $\xi_n > \theta_0/2$.

Exercise 17.2. Let $X_1, \ldots, X_n$ be i.i.d. Bernoulli with success probability $\theta \in (0,1)$, where the MLE $\hat{\theta}_n = \bar{X}$ and the Fisher information $\mathcal{I}(\theta) = 1/(\theta(1-\theta))$ are given in examples 5.3 and 11.5.

1. State the result of theorem 17.1 for this model and verify that it coincides with the central limit theorem for $\bar{X}$.

2. The *log-odds* are given by $\psi = g(\theta) = \log(\theta/(1-\theta))$. Find the MLE for ψ on the event $0 < \bar{X} < 1$, and use the delta method to show that $\sqrt{n}\,(\hat{\psi}_n - \psi) \xrightarrow{\text{d}} N\big(0, 1/(\theta(1-\theta))\big)$. Hint: the event $0 < \bar{X} < 1$ has probability tending to one, and thus, as at the end of the proof of theorem 17.1, the value of $\hat{\psi}_n$ outside this event does not affect the limit.

3. Determine the standard error of $\hat{\psi}_n$ and the approximate 95% Wald interval for ψ, in terms of $\hat{\theta}_n$ and n.

Exercise 17.3. Let $X_1, \ldots, X_n$ be i.i.d. Poisson with parameter $\theta > 0$, where the MLE $\hat{\theta}_n = \bar{X}$ and the Fisher information $\mathcal{I}(\theta) = 1/\theta$ are given in exercise 5.7 and example 11.6.

1. State the result of theorem 17.1 for this model and verify that the result coincides with the central limit theorem.

2. In exercise 5.7 we have found the MLE $\hat{p}_0 = e^{-\bar{X}}$ for the probability $p_0 = \mathbb{P}_\theta(X_1 = 0) = e^{-\theta}$ and we have seen that this MLE is biased. Using the delta method, determine the asymptotic distribution of $\sqrt{n}\,(\hat{p}_0 - p_0)$. Why does the bias found in exercise 5.7 not contradict the result from this part?

3. Show that the asymptotic variance from the previous part equals n times the Cramér–Rao bound for unbiased estimators for $e^{-\theta}$. Compare this with exercise 13.3, where we have seen that no efficient estimator for $e^{-\theta}$ exists.

Exercise 17.4. Let $X_1, \ldots, X_n$ be i.i.d. from the geometric distribution with parameter $p \in (0,1)$, where the MLE $\hat{p}_n = 1/\bar{X}$ and the Fisher information $\mathcal{I}(p) = 1/(p^2(1-p))$ are given in exercises 5.1 and 11.4. You may use that the assumptions of theorem 17.1 hold for this model.

1. State the result of theorem 17.1 for this model.

2. Derive the same result from the central limit theorem for $\bar{X}$ and the delta method, using $\mathbb{E}_p(X_1) = 1/p$ and $\mathrm{Var}_p(X_1) = (1-p)/p^2$.

3. A sample of size $n = 40$ has $\bar{x} = 2.5$. Compute the estimate $\hat{p}_n$, its standard error and the approximate 95% Wald interval for p.

Exercise 17.5. Let $X_1, \ldots, X_n$ be i.i.d. uniformly distributed on the interval $(0, \theta)$. The MLE for this model is $\hat{\theta}_n = \max_i X_i$.

1. For every $x \geq 0$, show that $\mathbb{P}_\theta\big(n(\theta - \hat{\theta}_n) > x\big) \to e^{-x/\theta}$ as $n \to \infty$. Deduce that $n(\theta - \hat{\theta}_n)$ converges in distribution to the exponential distribution with rate $1/\theta$, *i.e.* with mean θ.

2. Show that $\sqrt{n}\,(\hat{\theta}_n - \theta) \xrightarrow{\text{p}} 0$. Why is the MLE not asymptotically normal in the sense of theorem 17.1, and which of the regularity conditions fails? How does this relate to the super-efficiency of the uniform model from lecture 13?

Lecture 18

Maximum Likelihood in R

This R session turns the large-sample theory of lectures 16 and 17 into numbers. We find the maximum likelihood estimate numerically for a model where the likelihood equations cannot be solved explicitly, read off the standard errors from the output of the optimiser, and use simulation to check how often the interval $\hat{\theta} \pm 1.96 \, \mathrm{se}(\hat{\theta})$ from corollary 17.3 contains the truth.

18.1 The Log-Likelihood as an R Function

The model of the day is $X_1, \ldots, X_n$ i.i.d. from the gamma distribution with shape α and rate β, both unknown. Example 4.9 gave the method of moments estimates $\tilde{\alpha} = \bar{X}^2/V$ and $\tilde{\beta} = \bar{X}/V$, with V the plug-in variance, and section 5.3 noted that the likelihood equations have no closed-form solution. We simulate one sample of size $n = 40$ with $\alpha = 2.5$ and $\beta = 0.5$. The command `dgamma` with `log = TRUE` can be used to evaluate the logarithm of the density at every observation, and the sum of these values is $\ell(\alpha, \beta)$. Since the R optimisers *minimise*, the function returns the negative log-likelihood, and since they hand over all parameters as one vector, it takes the vector `par`, shape first and rate second.

```
alpha <- 2.5
beta  <- 0.5
n     <- 40
set.seed(2026)
x <- rgamma(n, shape = alpha, rate = beta)
nll <- function(par, data) {
  -sum(dgamma(data, shape = par[1], rate = par[2], log = TRUE))
}
nll(c(2.5, 0.5), x)
nll(c(1, 1), x)
```

```
## [1] 102.0691
## [1] 217.3799
```

The true parameter values lead to a much smaller negative log-likelihood than the pair $(1, 1)$. The command `optim` can be used to minimise a function of a vector, starting from a given value. From section 16.4 we know that a good starting value is a simple, consistent estimator, such as the method of moments

estimate. Before you try the code, predict whether the maximum likelihood estimates will agree with the method of moments estimates to one decimal place.

```
V     <- mean((x - mean(x))^2)
start <- c(mean(x)^2 / V, mean(x) / V)
start
fit <- optim(start, nll, data = x)
fit$par
fit$convergence
fit$counts
```

```
## [1] 2.2942540 0.4221649
## [1] 2.258769 0.415672
## [1] 0
## function gradient
##       43       NA
```

The extra argument `data = x` is passed on to our function. The result `fit` is a list: `fit$par` holds the estimates, $\hat{\alpha} = 2.26$ and $\hat{\beta} = 0.42$, which agree with the start to one decimal place; `fit$convergence` is 0 if the search finished properly, and any other value means the estimates cannot be trusted; `fit$counts` records that 43 evaluations were needed. By default `optim` uses the Nelder–Mead method, which needs no derivatives; appendix B shows the alternative `"BFGS"`.

18.2 Standard Errors from the Observed Information

From theorem 17.1 and the vector version of this result from lecture 17 we know that the MLE is approximately normally distributed with covariance matrix $\mathcal{I}_n(\theta_0)^{-1}$. As in section 17.3, we can replace $\mathcal{I}_n(\hat{\theta})$ by the observed information, *i.e.* the matrix of second derivatives of the negative log-likelihood at the maximum. The command `optim` can be used to compute this matrix, if we ask for it explicitly with `hessian = TRUE`. The matrix can be inverted using the `solve` command, the `diag` command gives the diagonal of this matrix, and the square roots of the diagonal elements are the standard errors. Before you look at the output, predict which of the two standard errors will be larger and by how much.

```
fit <- optim(start, nll, data = x, hessian = TRUE)
fit$hessian
covmat <- solve(fit$hessian)
se <- sqrt(diag(covmat))
result <- cbind(estimate = fit$par, se = se,
                lower = fit$par - 1.96 * se, upper = fit$par + 1.96 * se
    )
rownames(result) <- c("shape", "rate")
round(result, 3)
cov2cor(covmat)[1, 2]
```

```
##           [,1]     [,2]
## [1,]  22.18712 -96.2299
## [2,] -96.22990 522.9200
##       estimate    se lower upper
## shape    2.259 0.473 1.333 3.185
```

```
## rate     0.416 0.097 0.225 0.606
## [1] 0.8933916
```

The table prints the results of a maximum likelihood estimate in an easy to read format. Each row corresponds to one parameter, and shows the parameter estimate, the standard error and the interval $\hat{\theta} \pm 1.96$ se$(\hat{\theta})$ from corollary 17.3. Both intervals capture the true parameter values. The standard error for the shape parameter is approximately five times larger than the standard error for the rate parameter, but, since the estimates differ by a similar factor, the two standard errors are comparable relative to the estimates. The final row of the table, computed using the `cov2cor` function, shows that the two estimates have correlation 0.89. It is easy to show that, if we increase the shape and the rate simultaneously, while keeping the mean α/β fixed, the log-likelihood changes only slightly. Thus the ratio α/β is well determined by the data, but each parameter individually is determined only poorly. This is the problem of a nuisance parameter as discussed in lecture 14.

18.3 Starting Values

The gamma log-likelihood function has only one peak, so the choice of starting value matters little. Before you try the code, make a prediction: what happens if we start with very different values, *e.g.* with shape 30 and rate 30? Will `optim` still find the maximum? Will it take more than 43 evaluations to find the maximum, when we used the "good" start?

```
fit_far <- optim(c(30, 30), nll, data = x)
fit_far$par
fit_far$counts
```

```
## [1] 2.2571696 0.4154238
## function gradient
##       97       NA
```

The search reaches the same maximum, up to two decimal places, but it needs more than twice as many evaluations. When the likelihood has several local maxima, as in section 5.3, a poor start can end at the wrong peak, since an optimiser only finds a local maximum. Thus we always start from a consistent estimator, and we re-run `optim` from a different start to see whether the solution changes.

18.4 Coverage of the Wald Interval

While corollary 17.3 shows that the interval $\hat{\theta} \pm 1.96$ se$(\hat{\theta})$ covers the truth with probability approaching 0.95, it says nothing about moderate n. To check the coverage for the gamma shape and rate, we use simulation in this section, following the approach to coverage introduced in lecture 12. The complete script for our simulation is given in the following box, with the lines shuffled (except for the bodies of the function and of the loop). Before you read the rest of this section, try to do the following: (1) write down the lines in a way which forms a working script, and (2) predict the coverages for $n = 20$: will they be above or below 0.95? Which of the two, shape or rate, will be worse?

```
data.frame(n = ns, shape = coverage[, 1], rate = coverage[, 2])
fit_gamma <- function(n, alpha, beta) {
  x     <- rgamma(n, shape = alpha, rate = beta)
  V     <- mean((x - mean(x))^2)
  start <- c(mean(x)^2 / V, mean(x) / V)
  fit   <- optim(start, nll, data = x, hessian = TRUE)
  c(fit$par, sqrt(diag(solve(fit$hessian))))
}
set.seed(2026)
for (i in seq_along(ns)) {
  res   <- replicate(5000, fit_gamma(ns[i], alpha, beta))
  lower <- res[1:2, ] - 1.96 * res[3:4, ]
  upper <- res[1:2, ] + 1.96 * res[3:4, ]
  coverage[i, 1] <- mean(lower[1, ] <= alpha & alpha <= upper[1, ])
  coverage[i, 2] <- mean(lower[2, ] <= beta & beta <= upper[2, ])
}
alpha <- 2.5
coverage <- matrix(0, nrow = length(ns), ncol = 2)
ns    <- c(20, 100)
beta  <- 0.5
```

Every name must exist before it is used: the function and the four assignments come first, with `coverage` after `ns`; then the seed, then the loop, and the table last. The one new detail is the return value, the two estimates followed by the two standard errors, which `replicate` stacks as the columns of a matrix; thus `res[1:2, ]` holds the estimates and `res[3:4, ]` the standard errors. The script in working order is `R/S18-coverage.R`; it fits the model 5000 times for each sample size, takes a few seconds, and prints the following table.

```
##     n  shape   rate
## 1  20 0.9624 0.9604
## 2 100 0.9508 0.9526
```

At $n = 100$ both coverages are within simulation error of the nominal 0.95, as expected from corollary 17.3, and at $n = 20$ both are slightly above it: the interval is a little too wide rather than too narrow.

18.5 A Task to Take Home

The following task is in the style of a practical report. Use your student ID to seed the random number generator, simulate a gamma sample of size 30 with a shape and rate of your choice, and fit the model by maximum likelihood. For each parameter, report the estimate and its standard error.

Before the next session, you should type the commands used in this session, and should check that you get the same results as the text.

Summary.

- The written documentation for `optim`, `solve` and the Hessian matrix can be found in appendix B.
- The negative log-likelihood can be represented as an R function for one parameter vector, using the density with `log = TRUE`, and `optim` can be used to find its minimum, starting from a given point. It is important to check `fit$convergence` to see whether the optimisation was successful.
- If `hessian = TRUE`, the optimiser returns the observed information matrix. The inverse of this matrix can be used to approximately determine the covariance matrix of the MLE, and the square roots of the diagonal elements can be used to determine the standard errors reported in section 17.3. The results of a fit can be printed in a table which shows the estimates, standard errors and intervals for each parameter.
- A good starting point for the optimisation procedure is a consistent estimator, *e.g.* based on the method of moments. Since an optimiser finds a local maximum, it is important to re-run the optimisation procedure with a different starting point.
- For the gamma distribution, with shape and rate parameters, the coverage of the Wald interval is slightly above 95% for $n = 20$ and close to 95% for $n = 100$.

Lecture 19

The EM Algorithm

In lecture 5 we found the MLE by solving the likelihood equation, in closed form where this is possible and numerically otherwise. In this lecture we consider models with latent variables or missing data, where the likelihood would be easy to maximise if the missing values were known. The expectation-maximisation (EM) algorithm, introduced in this lecture, makes use of this structure: it alternates between filling in the missing information and re-estimating the parameter values. We illustrate the EM algorithm with the help of a worked example, considering a mixture of two normal distributions. This example also gives the non-identifiable model from lecture 4, to which we return in lecture 20. There we will also prove that each round of the EM algorithm cannot decrease the likelihood, and in lecture 21 we will discuss how to implement the EM algorithm in R.

19.1 Incomplete Data

Assume that we are given data consisting of i.i.d. pairs (X_i, Z_i), $i = 1, \ldots, n$, with joint density or joint probability weights $f_c(x, z; \theta)$, but we have observed only $X = (X_1, \ldots, X_n)$. The pair (X, Z) with $Z = (Z_1, \ldots, Z_n)$ is called the *complete data*, the values X are called the *observed data* and the unobserved values Z_i are called *latent variables*. The *complete-data log-likelihood* is given by

$$\ell_c(\theta) = \sum_{i=1}^{n} \log f_c(X_i, Z_i; \theta),$$

and the *observed-data log-likelihood* is given by

$$\ell(\theta) = \sum_{i=1}^{n} \log f(X_i; \theta),$$

where f is the marginal density of a single observation, given by

$$f(x; \theta) = \sum_{z} f_c(x, z; \theta),$$

with the integral used instead of the sum when Z_i is continuous. The MLE for θ is the maximiser of ℓ, but the algebra from lecture 5 is not applicable, since the sum inside the logarithm prevents any simplifications. The function ℓ_c, being a sum of standard log-likelihoods, would be easy to maximise, but it cannot even be evaluated as it depends on Z. The EM algorithm is based on this observation.

We will use the following example throughout this lecture and the next.

Example 19.1. Suppose that a population consists of two groups, with proportion π for group 1 and $1-\pi$ for group 2. The measurement has distribution $N(\mu_1, \sigma^2)$ in group 1 and $N(\mu_2, \sigma^2)$ in group 2, where we assume that we know σ^2, while π, μ_1 and μ_2 are unknown. We are given observations of the measurement X_i, but we do not know which group $Z_i \in \{1, 2\}$ the observation belongs to. If we write φ_σ for the density of $N(0, \sigma^2)$, then the observations are i.i.d. with density

$$f(x; \theta) = \pi\, \varphi_\sigma(x - \mu_1) + (1 - \pi)\, \varphi_\sigma(x - \mu_2),$$

where the parameter is

$$\theta = (\pi, \mu_1, \mu_2) \in (0, 1) \times \mathbb{R} \times \mathbb{R}.$$

This is an example of a *two-component normal mixture*. The observed-data log-likelihood is

$$\ell(\theta) = \sum_{i=1}^{n} \log\big(\pi\, \varphi_\sigma(X_i - \mu_1) + (1 - \pi)\, \varphi_\sigma(X_i - \mu_2)\big),$$

with a sum of two terms inside each logarithm. If we knew the groups, the MLEs for this example would be easy to find: we would get the proportion of group 1 and the two group means by using examples 5.3 and 5.4, but without the groups we cannot find the MLEs in closed form.

19.2 The Algorithm

The EM algorithm starts from a guess θ_0 and then alternates between two steps: it replaces ℓ_c by its conditional expectation given the observed data, computed under the current parameter value, and it maximises the result over θ.

Definition 19.2. Let $\theta_0 \in \Theta$ be given. Then the *EM algorithm* for finding the parameter values $\theta_1, \theta_2, \ldots$ consists of the following iterative steps for $k = 0, 1, 2, \ldots$:

(E) Expectation step: compute

$$Q(\theta | \theta_k) = \mathbb{E}_{\theta_k}\big(\ell_c(\theta) \,\big|\, X\big),$$

the conditional expectation of the complete-data log-likelihood given the observed data, where the conditional distribution of Z is computed using the parameter value θ_k.

(M) Maximisation step: set

$$\theta_{k+1} = \arg\max_{\theta \in \Theta} Q(\theta | \theta_k).$$

The second argument θ_k of Q enters only through the conditional distribution of Z given X. By Bayes' rule, proposition A.20 in appendix A, this conditional distribution has density or probability weights

$$g(z | x; \theta) = \frac{f_c(x, z; \theta)}{f(x; \theta)}.$$

Since the pairs (X_i, Z_i) are independent, once the data are observed $Q(\theta|\theta_k)$ is a sum over the observations, given by

$$Q(\theta|\theta_k) = \sum_{i=1}^{n} \sum_{z} g(z|x_i;\theta_k) \log f_c(x_i, z;\theta).$$

For a discrete latent variable the complete-data log-likelihood is linear in the indicators $\mathbf{1}_{\{Z_i=z\}}$, since

$$\log f_c(x_i, Z_i;\theta) = \sum_{z} \mathbf{1}_{\{Z_i=z\}} \log f_c(x_i, z;\theta).$$

Comparing this with the formula for $Q(\theta|\theta_k)$, we see that the E step replaces each indicator $\mathbf{1}_{\{Z_i=z\}}$ by its conditional probability $g(z|x_i;\theta_k)$ given the data.

We now apply the algorithm to the two-component normal mixture from example 19.1. We will see that the M step gives the complete-data MLEs from lecture 5, with each group indicator replaced by the conditional probability that the observation belongs to the group.

Example 19.1 (continued)**.** In this model, the latent variables $Z_1, \dots, Z_n$ are i.i.d. with $\mathbb{P}(Z_i = 1) = \pi$ and $\mathbb{P}(Z_i = 2) = 1 - \pi$, and given $Z_i = j$ we have $X_i \sim N(\mu_j, \sigma^2)$. The complete-data density is $f_c(x, 1;\theta) = \pi\varphi_\sigma(x - \mu_1)$ and $f_c(x, 2;\theta) = (1 - \pi)\varphi_\sigma(x - \mu_2)$, and thus the complete-data log-likelihood is

$$\ell_c(\theta) = \sum_{i=1}^{n} \Big(\mathbf{1}_{\{Z_i=1\}} \big(\log\pi + \log\varphi_\sigma(x_i-\mu_1)\big) + \mathbf{1}_{\{Z_i=2\}} \big(\log(1-\pi) + \log\varphi_\sigma(x_i-\mu_2)\big) \Big),$$

which is linear in the indicators. For the E step, using Bayes' rule with the current value $\theta_k = (\pi_k, \mu_{1,k}, \mu_{2,k})$, we find

$$\gamma_{i,k} = \mathbb{P}_{\theta_k}(Z_i = 1 | X_i = x_i) = \frac{\pi_k\, \varphi_\sigma(x_i - \mu_{1,k})}{\pi_k\, \varphi_\sigma(x_i - \mu_{1,k}) + (1 - \pi_k)\, \varphi_\sigma(x_i - \mu_{2,k})}.$$

This is called the *responsibility* of group 1 for the observation i. By this linearity, taking the conditional expectation replaces $\mathbf{1}_{\{Z_i=1\}}$ by $\gamma_{i,k}$ and $\mathbf{1}_{\{Z_i=2\}}$ by $1-\gamma_{i,k}$, and thus

$$Q(\theta|\theta_k) = \sum_{i=1}^{n} \Big(\gamma_{i,k} \big(\log\pi + \log\varphi_\sigma(x_i-\mu_1)\big) + (1-\gamma_{i,k}) \big(\log(1-\pi) + \log\varphi_\sigma(x_i-\mu_2)\big) \Big).$$

For the M step we maximise over $\theta = (\pi, \mu_1, \mu_2)$ with the $\gamma_{i,k}$ fixed. Since $\log\varphi_\sigma(x - \mu) = -(x - \mu)^2/(2\sigma^2) + \text{const}$, the partial derivatives of Q are

$$\frac{\partial Q}{\partial \pi} = \frac{\sum_i \gamma_{i,k}}{\pi} - \frac{\sum_i (1 - \gamma_{i,k})}{1 - \pi},$$
$$\frac{\partial Q}{\partial \mu_1} = \frac{1}{\sigma^2} \sum_{i=1}^{n} \gamma_{i,k}(x_i - \mu_1), \qquad \frac{\partial Q}{\partial \mu_2} = \frac{1}{\sigma^2} \sum_{i=1}^{n} (1 - \gamma_{i,k})(x_i - \mu_2),$$

and setting these to zero gives the updates

$$\pi_{k+1} = \frac{1}{n} \sum_{i=1}^{n} \gamma_{i,k}, \qquad \mu_{1,k+1} = \frac{\sum_i \gamma_{i,k} x_i}{\sum_i \gamma_{i,k}}, \qquad \mu_{2,k+1} = \frac{\sum_i (1 - \gamma_{i,k}) x_i}{\sum_i (1 - \gamma_{i,k})}. \quad (19.1)$$

The second derivatives are negative and thus these are maxima. They are the complete-data MLEs of examples 5.3 and 5.4 with indicators replaced by responsibilities.

Summary.

- In an incomplete-data model, the complete data (X, Z) has an easy log-likelihood ℓ_c, but only X is observed and the observed-data log-likelihood ℓ has a sum inside the logarithm.
- EM iteratively computes the expectation $Q(\theta|\theta_k) = \mathbb{E}_{\theta_k}(\ell_c(\theta) \,|\, X)$ and then maximises the expectation Q over θ.
- For a two-component normal mixture, the E step computes the responsibilities $\gamma_{i,k} = \mathbb{P}_{\theta_k}(Z_i = 1|X_i = x_i)$ and the M step computes the weight as the average responsibility and the means as responsibility-weighted means of the data.

Exercise 19.1. Let X be geometrically distributed with parameter $p \in (0, 1)$, *i.e.* $f(x; p) = (1-p)^{x-1}p$ for $x \in \mathbb{N}$, as in exercise 5.1.

1. Show that $\mathbb{P}_p(X \geq m) = (1-p)^{m-1}$ for $m \in \mathbb{N}$ and use this result to show that $\mathbb{P}_p(X \geq m + k|X \geq m) = \mathbb{P}_p(X \geq k + 1)$ for all $k \geq 0$. Comment upon your result in the light of the memorylessness of the exponential distribution from exercise 5.6.

2. Show that

$$\mathbb{E}_p(X\,|\,X \geq m) = m - 1 + \mathbb{E}_p(X) = m - 1 + \frac{1}{p} \qquad \text{for all } m \in \mathbb{N}.$$

Exercise 19.2. Consider the two-component normal mixture from example 19.1, but with known weight $\pi \in (0, 1)$ and known variance σ^2. The only unknown quantities are the means $\theta = (\mu_1, \mu_2)$.

1. Show that the EM updates for this case are given by the two mean updates from (19.1), where the responsibilities are computed using the known π.

2. Compute the derivative of the observed-data log-likelihood and show that

$$\frac{\partial \ell}{\partial \mu_1}(\theta) = \frac{1}{\sigma^2} \sum_{i=1}^{n} \gamma_i(\theta)\,(x_i - \mu_1),$$

   where $\gamma_i(\theta)$ is the responsibility at θ. Deduce that the likelihood equations coincide with the fixed-point equations of the EM iteration.

3. What are the updates in the limit $\pi \to 1$? Comment upon your result in the context of example 5.4.

Lecture 20

Properties of the EM Algorithm

In lecture 19 we introduced the EM algorithm for models with latent variables: starting from a guess θ_0, the algorithm alternates between computing the function $Q(\theta|\theta_k)$ in the E step and maximising it over θ in the M step, as in definition 19.2. In this lecture we prove that no step of the algorithm can decrease the observed-data log-likelihood, and that the fixed points of the iteration solve the likelihood equation. We then discuss the issues which arise when the algorithm is used in practice. We use the notation of lecture 19 throughout, and we continue with the two-component normal mixture of example 19.1.

20.1 The Ascent Property

The EM algorithm only ever maximises Q, and it is not obvious that this does anything for the observed-data log-likelihood ℓ. The central result of this lecture is that it does: no step of the algorithm can decrease ℓ. The proof rests on writing ℓ as the difference of Q and a second function H, and we prove this decomposition first. We make three assumptions, which hold for the rest of this lecture:

- The set of values z with $f_c(x, z; \theta) > 0$ does not depend on θ. This is the analogue for the latent variables of condition (R2) in definition 16.1.
- We have $f(x; \theta) > 0$ for every data point x and every $\theta \in \Theta$, so that $g(z|x; \theta)$ is defined.
- The conditional expectation $\mathbb{E}_{\theta'}\big(\log f_c(x, Z; \theta) \,\big|\, X = x\big)$ is finite for all $\theta, \theta' \in \Theta$ and every data point x, so that $Q(\theta|\theta')$ is finite.

The following lemma gives the decomposition. The function H in it is defined like Q, with the conditional density g in place of the complete-data density f_c.

Lemma 20.1. *For all $\theta, \theta' \in \Theta$ we have*

$$\ell(\theta) = Q(\theta|\theta') - H(\theta|\theta'), \tag{20.1}$$

where H is given by

$$H(\theta|\theta') = \sum_{i=1}^{n} \mathbb{E}_{\theta'}\big(\log g(Z_i|X_i; \theta) \,\big|\, X_i\big).$$

Proof. We start from the formula $g(z|x;\theta) = f_c(x,z;\theta)/f(x;\theta)$ for the conditional distribution of the latent variable. Solving it for $f(x;\theta)$ and taking logarithms gives

$$\log f(x;\theta) = \log f_c(x,z;\theta) - \log g(z|x;\theta)$$

for every z with $f_c(x,z;\theta) > 0$. By the first assumption, these are the values which Z can take, whatever the parameter. We now replace z by the random variable Z and take conditional expectations given $X = x$ under θ'. Since the left-hand side does not depend on z, it stays the same, and we find

$$\log f(x;\theta) = \mathbb{E}_{\theta'}\big(\log f_c(x,Z;\theta)\,\big|\,X = x\big) - \mathbb{E}_{\theta'}\big(\log g(Z|x;\theta)\,\big|\,X = x\big).$$

The first term on the right is finite by the third assumption, and thus so is the second. Summing this identity over the observations $X_1,\dots,X_n$ gives (20.1). This completes the proof. □

We now use the lemma to show that the EM algorithm can only increase the observed-data log-likelihood.

Proposition 20.2. *Let $\theta_0,\theta_1,\theta_2,\dots$ be generated by the EM algorithm from definition 19.2. Then*

$$\ell(\theta_{k+1}) \geq \ell(\theta_k) \qquad \text{for all } k = 0,1,2,\dots.$$

Proof. We first show that $H(\theta|\theta_k) \leq H(\theta_k|\theta_k)$ for all θ. Since H is a sum over the observations, we only need to consider a single observation with observed value x. Let $Y = g(Z|x;\theta)/g(Z|x;\theta_k)$. This value is well defined and positive by our assumption on the supports. Since $-\log$ is convex, we can use Jensen's inequality, lemma A.7, for the conditional distribution of Z given $X = x$ under θ_k to get

$$\begin{aligned}
&\mathbb{E}_{\theta_k}\big(\log g(Z|x;\theta)\,\big|\,X = x\big) - \mathbb{E}_{\theta_k}\big(\log g(Z|x;\theta_k)\,\big|\,X = x\big)\\
&\quad= \mathbb{E}_{\theta_k}\big(\log Y\,\big|\,X = x\big) \leq \log \mathbb{E}_{\theta_k}\big(Y\,\big|\,X = x\big)\\
&\quad= \log \sum_z \frac{g(z|x;\theta)}{g(z|x;\theta_k)}\, g(z|x;\theta_k) = \log \sum_z g(z|x;\theta) = \log 1 = 0,
\end{aligned}$$

since $z \mapsto g(z|x;\theta)$ is a probability distribution over z (with sums replaced by integrals for a continuous latent variable). This is the same argument as in the proof of lemma 16.2.

Now we can use lemma 20.1 with $\theta' = \theta_k$ at $\theta = \theta_{k+1}$ and at $\theta = \theta_k$, and taking the difference we get

$$\ell(\theta_{k+1}) - \ell(\theta_k) = \big(Q(\theta_{k+1}|\theta_k) - Q(\theta_k|\theta_k)\big) - \big(H(\theta_{k+1}|\theta_k) - H(\theta_k|\theta_k)\big).$$

The first bracket is non-negative, since θ_{k+1} is the maximiser of $\theta \mapsto Q(\theta|\theta_k)$ in the M step, and the second bracket is non-positive by the first step of the proof. Thus the difference is non-negative. This completes the proof. □

The statement of the proposition does not imply that EM finds the MLE: the sequence $\ell(\theta_k)$ converges whenever ℓ is bounded above, but the iteration may stop at a local maximum or, in principle, at a saddle point. In general we can only say that the fixed points of the iteration are stationary points of ℓ.

Proposition 20.3. *Let Θ be open and ℓ as well as $\theta \mapsto Q(\theta|\theta')$ differentiable for all $\theta' \in \Theta$. Then*

$$\ell'(\theta') = \frac{\partial}{\partial\theta} Q(\theta|\theta')\Big|_{\theta=\theta'} \qquad \text{for all } \theta' \in \Theta.$$

In particular, if $\theta_{k+1} = \theta_k$ for some k, the point θ_k is a root of the likelihood equation $\ell'(\theta) = 0$.

Proof. From lemma 20.1 we know that the function $\theta \mapsto H(\theta|\theta') = Q(\theta|\theta') - \ell(\theta)$ is differentiable. Furthermore, from the first step of the proof of proposition 20.2 we know that this function has its maximum at an interior point θ'. Thus, the derivative at this point must be zero. Taking derivatives of the decomposition at $\theta = \theta'$ gives the identity. If $\theta_{k+1} = \theta_k$, then θ_k is an interior maximiser of $\theta \mapsto Q(\theta|\theta_k)$ and the right-hand side equals zero at $\theta' = \theta_k$. Thus we have $\ell'(\theta_k) = 0$. This completes the proof. □

For a vector parameter the same argument applies, but now using gradients instead of derivatives.

20.2 Practical Matters

Three issues arise whenever EM is used, and the mixture shows all three. The first is the promised failure of identifiability: the parameter values (π, μ_1, μ_2) and $(1-\pi, \mu_2, \mu_1)$ give the same density $f(x;\theta)$, and thus the model is not identifiable in the sense of definition 4.10, the difficulty foreshadowed after example 4.11. This is called *label switching*: the likelihood has two global maxima which are mirror images of each other. For the data of exercise 20.2, EM started from $(1/2, 2, 4)$ converges to $(0.4, 0.5, 6.0)$, but started from $(1/2, 4, 2)$ it converges to $(0.6, 6.0, 0.5)$. Either answer is fine on its own, but averaging different runs is not, and the remedy of lecture 4 is to restrict the parameter space by a convention such as $\mu_1 < \mu_2$. The second issue is local maxima: proposition 20.2 guarantees ascent and nothing more, and a poor starting value can lead the iteration to a local maximum of ℓ without warning. The standard precaution is to run EM from several starting values and to keep the run with the largest observed-data log-likelihood, which we can always evaluate. The third issue is when to stop: since $\ell(\theta_k)$ is non-decreasing, we stop when its increase falls below a small tolerance; convergence is typically geometric near the fixed point, but can be slow. Finally, EM produces an estimate but no standard error; that is obtained afterwards from the observed information at the fixed point, as in section 17.3. Lecture 21 shows label switching, a starting value for which the iteration converges to a saddle point and the stopping rule in R.

Summary.

- We have shown that $\ell(\theta_{k+1}) \geq \ell(\theta_k)$ for all k. The proof uses $\ell = Q - H$ and Jensen's inequality to show that the maximum of $\theta \mapsto H(\theta|\theta_k)$ is attained at θ_k.
- A fixed point of the EM iteration is a root of the likelihood equation.
- Mixture parameters are only identifiable up to relabelling the components (label switching). Local maxima of the likelihood function can be avoided by running EM with different starting values and by comparing runs of the algorithm using the observed-data log-likelihood.

Exercise 20.1. Suppose f_1 and f_2 are two known, distinct densities, and that $X_1, \ldots, X_n$ are i.i.d. with density

$$f(x;\pi) = \pi f_1(x) + (1-\pi) f_2(x),$$

where $\pi \in (0,1)$ is the only unknown parameter. Let $Z_i \in \{1,2\}$ be the component observation i belongs to, as in example 19.1.

1. Write down the observed-data log-likelihood $\ell(\pi)$ and the likelihood equation $\ell'(\pi) = 0$.

2. Write down the complete-data log-likelihood and show that the E step determines the responsibilities $\gamma_i(\pi_k) = \pi_k f_1(x_i)/(\pi_k f_1(x_i) + (1-\pi_k) f_2(x_i))$ and the M step determines the update $\pi_{k+1} = \frac{1}{n}\sum_i \gamma_i(\pi_k)$.

3. Show that $\gamma_i(\pi) - \pi = \pi(1-\pi)\,(f_1(x_i) - f_2(x_i))/f(x_i;\pi)$ and thus that $\frac{1}{n}\sum_i \gamma_i(\pi) - \pi = \pi(1-\pi)\,\ell'(\pi)/n$. Deduce that $\pi^* \in (0,1)$ is a fixed point of the EM iteration if and only if it solves the likelihood equation. This shows that for this model the converse of proposition 20.3 holds as well.

Exercise 20.2. Consider the normal mixture of example 19.1 with $\sigma = 1$, the observations $x = (0, 1, 5, 6, 7)$ and the initial value $\theta_0 = (1/2, 2, 4)$.

1. Show that the responsibilities are $\gamma_{i,0} = 1/(1 + e^{2x_i - 6})$, compute them to four decimal digits and use the updates (19.1) to find the first iterate θ_1.

2. Compute the observed-data log-likelihood $\ell(\theta) = \sum_{i=1}^{5} \log(\pi\varphi_1(x_i - \mu_1) + (1-\pi)\varphi_1(x_i - \mu_2))$ at θ_0 and at θ_1, and show that the likelihood has increased. Which of the five observations contributes most to the increase in likelihood? Why?

Lecture 21

EM in R

In lecture 19 the EM algorithm was introduced, and in lecture 20 the ascent property was proved. In example 19.1 the E and M steps were derived for a two-component normal mixture. In this lecture we will type the two steps into R functions, check the functions against the hand computation of exercise 20.2, and then run the functions on a simulated sample of size 200. We will conclude by considering the problem of label switching and the effect of a bad start. The written reference for this session is appendix B.

21.1 The E and M Steps as R Functions

Throughout, the parameter is the vector $\theta = (\pi, \mu_1, \mu_2)$ and the common standard deviation σ is known. The E step of example 19.1 computes the responsibilities $\gamma_{i,k}$ and the M step computes the updates (19.1). Both steps need only vector arithmetic and the normal density, which the command `dnorm` can be used to compute. To check the iteration, we also compute the observed-data log-likelihood ℓ, which by proposition 20.2 cannot decrease from one iteration to the next.

```
resp <- function(x, theta, sigma) {
  a <- theta[1] * dnorm(x, mean = theta[2], sd = sigma)
  b <- (1 - theta[1]) * dnorm(x, mean = theta[3], sd = sigma)
  a / (a + b)
}
mstep <- function(x, gamma) {
  c(mean(gamma),
    sum(gamma * x) / sum(gamma),
    sum((1 - gamma) * x) / sum(1 - gamma))
}
loglik <- function(x, theta, sigma) {
  sum(log(theta[1] * dnorm(x, mean = theta[2], sd = sigma) +
          (1 - theta[1]) * dnorm(x, mean = theta[3], sd = sigma)))
}
```

We first test our functions on the data from exercise 20.2. For this we use the five observations $x = (0, 1, 5, 6, 7)$ where $\sigma = 1$ and the initial value is $\theta_0 = (1/2, 2, 4)$. The results of one iteration of the algorithm are then compared to the responsibilities and the iterate θ_1, and to the two values of the log-likelihood computed by hand.

```
x     <- c(0, 1, 5, 6, 7)
theta <- c(0.5, 2, 4)
gamma <- resp(x, theta, 1)
round(gamma, 4)
theta1 <- mstep(x, gamma)
theta1
c(loglik(x, theta, 1), loglik(x, theta1, 1))
```

```
## [1] 0.9975 0.9820 0.0180 0.0025 0.0003
## [1] 0.4000671 0.5444727 5.9709581
## [1] -17.518842  -9.212934
```

Everything agrees with the hand computation ($\mu_{2,1}$ differs in the last digit because the hand computation rounded the intermediate sums).

21.2 A Larger Sample

The two groups in the five observations of exercise 20.2 are far apart, and the iteration converges to $(0.4, 0.5, 6.0)$, as stated in section 20.2. Here we simulate a sample where the groups overlap. We can simulate such a sample by first randomly choosing a group, using `rbinom` to get $Z_i = 1$ with probability $\pi = 0.3$. For the other group R gives the value 0 instead of 2, and thus `z == 0` marks group 2. We then simulate the measurements from the normal distribution corresponding to the group, *i.e.* with means 0 and 3 and $\sigma = 1$. The command `ifelse` can be used to choose the correct mean for each observation. The box below contains the complete script for simulation and EM iteration, with lines shuffled (except for the body of the loop). Before you can run the script, you have to arrange these lines into a working script. Following the stopping rule of section 20.2, the command `break` ends the loop as soon as the log-likelihood increases by less than 10^{-8}. The vector `ll` stores the log-likelihood after each iteration, with the initial value at position 1, and grows by one entry per iteration.

```
theta <- c(0.5, q[1], q[2])
for (k in 1:1000) {
  theta <- mstep(x, resp(x, theta, 1))
  ll[k + 1] <- loglik(x, theta, 1)
  if (ll[k + 1] - ll[k] < 1e-8) break
}
z <- rbinom(n, size = 1, prob = 0.3)
n <- 200
q     <- quantile(x, c(0.25, 0.75), names = FALSE)
set.seed(2026)
ll    <- loglik(x, theta, 1)
x <- rnorm(n, mean = ifelse(z == 1, 0, 3), sd = 1)
```

The rule is the one from lecture 9: each line must come after all lines on which it depends. As in that lecture, several orders satisfy the rule. One of them is seed, `n`, `z` and `x`, quartiles `q`, start value, first entry of `ll`, and then the loop; the seed and `n` can also be swapped. The start values put the weight at 1/2 and the two means at the two quartiles, which is a cheap way to split the

sample into two groups. After the loop, the variable `k` keeps the number of the last iteration. Before you look at the output, answer two questions. Will the loop need fewer than 10 iterations, between 10 and 50, or more than 50? Will the log-likelihood increase by the same amount in each iteration?

```
table(z)
k
round(theta, 4)
round(ll[k + 1], 3)
d <- diff(ll)
signif(d[1:7], 2)
signif(d[21:27], 2)
```

```
## z
##   0   1
## 152  48
## [1] 28
## [1] 0.2397 0.2202 3.0761
## [1] -354.505
## [1] 5.00 4.40 3.80 3.00 2.10 1.20 0.66
## [1] 3.3e-06 1.3e-06 5.4e-07 2.2e-07 8.8e-08 3.6e-08 1.4e-08
```

The fixed point $\hat{\theta} = (0.240, 0.220, 3.076)$ is close to the proportion $48/200 = 0.24$ of group-1 observations which were actually drawn, rather than to $\pi = 0.3$, and it recovers the two means to within about two tenths. By proposition 20.3 the fixed point is a root of the likelihood equation. The command `diff` can be used to compute the differences of consecutive entries of a vector, and thus `d` holds the increases of the log-likelihood in the 28 iterations. The last two lines of the output show the first seven and seven of the last increases. They are positive, as proposition 20.2 guarantees, but they are not equal. Towards the end, each increase is about 0.4 times the previous one, for example $3.3 \cdot 10^{-6}$, then $1.3 \cdot 10^{-6}$ and $5.4 \cdot 10^{-7}$. This is the geometric convergence mentioned in section 20.2. Since $0.66 \cdot 0.4^{20}$ is approximately $7 \cdot 10^{-9}$, at this rate about 20 further iterations are needed to reduce the increase from 0.66 in the seventh iteration to below the tolerance 10^{-8}, which explains the total of 28 iterations. The factor 0.4 depends on the data, and the take-home task below shows what happens when the groups overlap more.

At the fixed point the responsibility of group 1 is a decreasing function of x which equals $1/2$ where $\pi\varphi_\sigma(x - \mu_1) = (1 - \pi)\varphi_\sigma(x - \mu_2)$. Before you look at figure 21.1, predict where the crossing lies: at the midpoint 1.648 of the two fitted means, or above or below this point?

The crossing happens at 1.244, well below the midpoint 1.648, since the weights are unequal: an observation which is half-way between the means has three times the probability of being from the larger group. The filled and open circles give the true group of each observation, which is not known to EM. The responsibilities agree with the truth when the observations are far away from the crossing point, but between 0.7 and 1.6 both types of observations can be found and the group cannot be decided. The figure is drawn by the script `R/S21-mixture.R`.

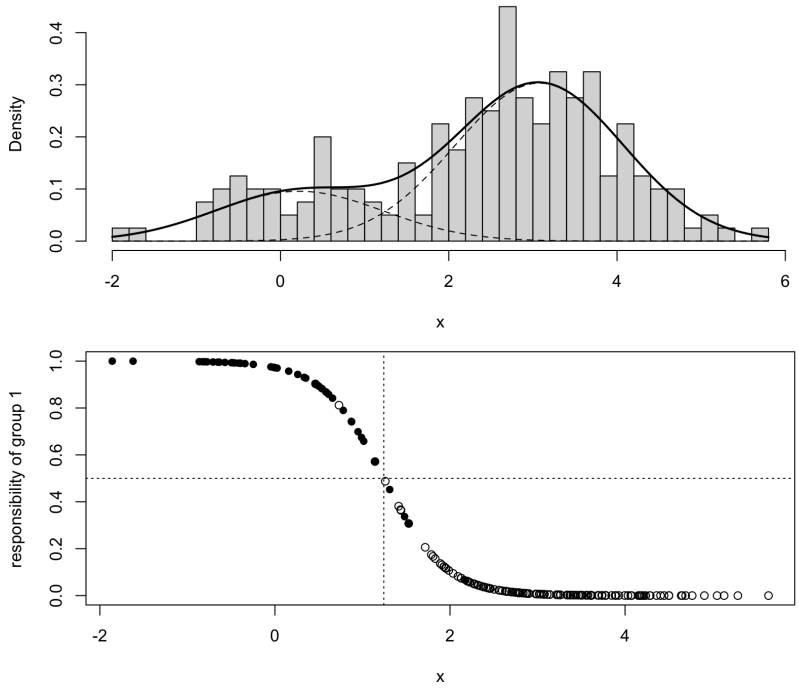


Figure 21.1: The fitted mixture density with its two weighted components (top panel) and the responsibilities of group 1 at the fixed point as a function of the observations (bottom panel). The filled circles mark the observations which were simulated from group 1. The dotted lines give the responsibility 1/2 and the crossing point 1.244.

21.3 Label Switching and a Bad Start

Section 20.2 discusses label switching and the danger of a poor start. In the script `R/S21-starts.R` the loop is packaged as a function `em`, which is then run on our sample from three starts: the quartile start, the same start with the two means swapped, and the start $(1/2, 3, 3)$ with both means equal. Before you look at table 21.1, predict where the start with the swapped means ends up, and what happens to the responsibilities, and thus to the next update, when $\mu_1 = \mu_2$.

start	π	μ_1	μ_2	ℓ	iterations
quartiles	0.2397	0.2202	3.0761	−354.5046	28
swapped	0.7603	3.0761	0.2202	−354.5046	28
equal	0.5000	2.3915	2.3915	−421.8067	2

Table 21.1: The fixed points, their log-likelihoods and the numbers of iterations of the EM algorithm for three starts, computed by the script `R/S21-starts.R`.

The start with swapped means ends in the mirror image $(0.760, 3.076, 0.220)$ after the same 28 iterations and with the same log-likelihood. Both parameter values give the same density, so the model is not identifiable in the sense of definition 4.10, and the labelling in the start determines which of the two maxima we reach. When $\mu_1 = \mu_2$, the two normal densities in the responsibility cancel, all γ_i are equal to π, and the M step sets both means equal to the sample

mean while leaving the weight unchanged. The next step changes nothing, and the loop stops after two iterations at $(1/2, 2.3915, 2.3915)$, effectively a single normal distribution, with a log-likelihood 67 below the maximum. This resting point is a root of the likelihood equation by proposition 20.3. Since the density does not depend on π when $\mu_1 = \mu_2$, every point $(\pi, \bar{x}, \bar{x})$ is such a root, and the weight $1/2$ is simply the weight of the start.

The resting point is not a local maximum. The log-likelihood at the nearby start $(1/2, 2.39, 2.41)$ is already equal to -421.800. This is slightly above the value -421.807 at the resting point, and figure 21.2 shows that EM started there climbs all the way to the maximum. Neither is it a local minimum, since among the points with $\mu_1 = \mu_2$ the log-likelihood is largest at the sample mean. The resting point is thus a saddle point, and EM cannot tell a saddle point apart from a maximum. The figure also shows that near the saddle point the log-likelihood increases only slowly at first: the first iteration from the nearby start gains only 0.07. With a coarse tolerance such as 0.1, the loop would have stopped there, far away from the maximum.

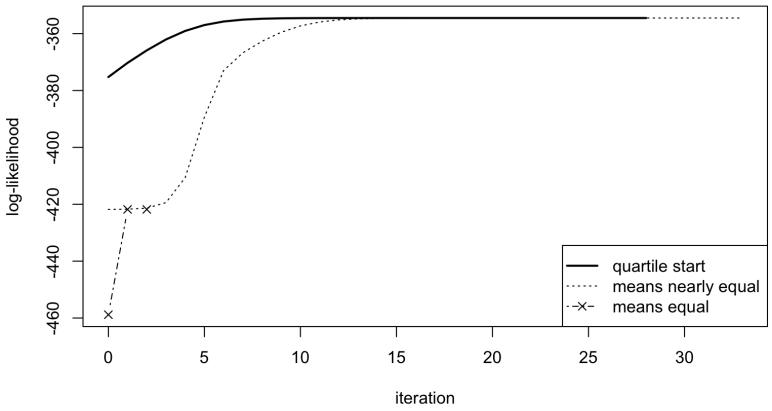


Figure 21.2: Log-likelihood of the given data as a function of the iteration number for the EM algorithm, for the start with quartile-spaced means (solid line), for the start $(1/2, 2.39, 2.41)$ with nearly equal means (dotted line) and for the start with equal means (dash-dotted line with crosses). The trace of the start with interchanged means coincides with the solid line.

21.4 A Task to Take Home

The published scripts of this lecture can be used as a starting point for the following task. Choose a weight π between 0.2 and 0.5 and two means which are two to three units apart. Using your student ID as the seed, simulate a sample of size $n = 300$ from the mixture with $\sigma = 1$ and run the EM algorithm from the quartile start. The EM algorithm does not provide standard errors. To obtain them, use `optim` with `hessian = TRUE` to minimise the negative of `loglik`, starting from the fixed point, as in lecture 18; the script `R/S21-mixture.R` contains the necessary command. Report each estimate with its standard error. Then repeat

the task with the two means only one unit apart. How many iterations does the loop need now, and what happens to the standard errors? If the loop runs through all 1000 iterations, the stopping rule was never met and the result is not a fixed point, so you should say so in your answer.

Typing the commands yourself is an important step towards understanding how the R scripts work. Before the next session, please type in the commands from this session, and check that you get the same results as shown in the text.

Summary.

- The written reference for the R code in this session is appendix B.
- The E step and M step from example 19.1 involve only a few lines of vector arithmetic. For five observations, these lines reproduce the manual computation.
- For a sample of size 200 with overlapping groups, it takes 28 iterations for EM to converge. The log-likelihood increases in every step, as required by proposition 20.2. Near the fixed point each increase is about 0.4 times the previous one, and this geometric convergence sets the number of iterations.
- The responsibility of group 1 is a decreasing S-shaped function of x. The function crosses 1/2 below the midpoint of the means when group 1 is the smaller group. Close to the crossing point it is no longer possible to decide which group an observation belongs to.
- If the labels are swapped before starting the EM algorithm, the algorithm converges to the mirror-image maximum with the same log-likelihood. If the two means are equal at the start, the EM algorithm stops at a saddle point immediately, *i.e.* at a root of the likelihood equation which is not a maximum. To avoid this, the EM algorithm should be started with different values and the best of the resulting runs should be chosen, based on the log-likelihood of the observed data.

Lecture 22

Hypothesis Testing

In this lecture we turn from estimation to the second question from lecture 1: are the data compatible with a given claim about θ? We first set up the language of hypothesis testing, which lectures 23, 25, 26 and 31 build on, and then construct two tests for the mean of a normal sample.

22.1 Hypotheses

Throughout, $X_1, \ldots, X_n$ follow a parametric model in the sense of definition 1.1. Questions such as whether a coin is fair or whether a treatment has an effect ask whether the true value of θ belongs to a subset of the parameter space Θ.

Definition 22.1. A *hypothesis* is a statement of the form $\theta \in \Theta'$ for a subset $\Theta' \subseteq \Theta$. In a testing problem, the *null hypothesis* $H_0 \colon \theta \in \Theta_0$ and the *alternative hypothesis* $H_1 \colon \theta \in \Theta_1$ are given, where Θ_0 and Θ_1 are disjoint subsets of Θ. A hypothesis is called *simple*, if the corresponding subset consists of a single point, and *composite* otherwise.

From now on we write θ_0 for the value of θ named by a null hypothesis. In lectures 16 and 17 the same symbol denoted the true value of the parameter; the context makes clear which meaning is intended.

Often we have $\Theta_1 = \Theta \setminus \Theta_0$, and then exactly one of the two hypotheses is true. For example, for the mean θ of a normal sample, the hypothesis $H_0 \colon \theta = \theta_0$ is simple, whereas the one-sided alternative $H_1 \colon \theta > \theta_0$, the two-sided alternative $H_1 \colon \theta \neq \theta_0$ and the one-sided null hypothesis $H_0 \colon \theta \leq \theta_0$ are all composite.

The two hypotheses have different roles: the null hypothesis is the default position, and the alternative is the claim which requires evidence before we can act on it. Section 22.3 makes this distinction precise.

22.2 Tests, Critical Regions and Errors

A test decides between the two hypotheses, and can be described by the set of data values where it rejects H_0.

Definition 22.2. A *test* of H_0 against H_1 is a rule which for every possible data vector $x = (x_1, \ldots, x_n)$ decides whether to "reject H_0" or "do not reject H_0". The set

$$C = \{\, x \mid \text{the test rejects } H_0 \text{ for the data } x \,\}$$

is called the *critical region* (or rejection region) of the test. Alternatively, a test can be described by its *test function* $\varphi(x) = 1$ for $x \in C$ and $\varphi(x) = 0$ otherwise.

In practice, the critical region usually has the form $C = \{x \mid T(x) > c\}$ or $C = \{x \mid T(x) \geq c\}$ for a statistic T given by definition 1.3, called the *test statistic*, and a constant c called the *critical value*. The two forms give tests with different error probabilities only if $T(X) = c$ has positive probability, as can happen for discrete distributions (exercise 22.3).

A test can decide wrongly in two different ways.

Definition 22.3. A *type I error* occurs, if the test wrongly rejects H_0 when H_0 is true. A *type II error* occurs, if the test wrongly does not reject H_0 when H_1 is true.

Table 22.1 summarises the four possible outcomes.

	do not reject H_0	reject H_0
H_0 true	correct decision	type I error
H_1 true	type II error	correct decision

Table 22.1: The four possible outcomes of a test.

As an example, in a criminal trial the null hypothesis is the innocence of the defendant, and the alternative is guilt. Convicting an innocent person is a type I error, and acquitting a guilty person is a type II error. Under the standard of proof "beyond reasonable doubt", many acquittals of the guilty are accepted to avoid wrongful convictions, because a type I error is more serious. Since an acquittal does not prove innocence, the table says "do not reject H_0" instead of "accept H_0".

22.3 The Power Function, Size and Level

Both error probabilities depend on the true parameter value, and can be read off a single function of θ.

Definition 22.4. The *power function* of a test with critical region C is given by

$$\beta(\theta) = \mathbb{P}_\theta(X \in C) = \mathbb{E}_\theta(\varphi(X)), \qquad \theta \in \Theta,$$

i.e. the probability of rejecting H_0 when the true parameter value is θ. For $\theta \in \Theta_1$ the value $\beta(\theta)$ is called the *power* of the test at the alternative θ.

For $\theta \in \Theta_0$ the value $\beta(\theta)$ is the probability of a type I error, and for $\theta \in \Theta_1$ the value $1-\beta(\theta)$ is the probability of a type II error (some books write β for the latter). Enlarging the critical region can only increase $\beta(\theta)$, which reduces type II errors at the cost of type I errors, and thus we have to choose which probability to control. Since a type I error abandons the default position without good reason, we restrict its probability first and only then consider the power of the test.

Definition 22.5. The *size* of a test with power function β is given by

$$\sup_{\theta \in \Theta_0} \beta(\theta),$$

For $\alpha \in (0,1)$, the test is a *level α test*, if the size of the test is less than or equal to α.

For a simple null hypothesis $H_0\colon \theta = \theta_0$, the size of the test equals $\beta(\theta_0)$. The number α is called the *significance level* and the values 0.05 and 0.01 are only conventions. Among all level α tests, we prefer the test with the largest power at the alternatives. Lecture 23 shows how to find the most powerful test when both hypotheses are simple.

22.4 Testing the Mean of a Normal Sample

We now construct tests for $X_1, \dots, X_n$ i.i.d. $N(\theta, \sigma^2)$ with known variance σ^2. We write Φ for the standard normal distribution function and z_q for the q-quantile of the standard normal distribution, *i.e.* we have $\Phi(z_q) = q$. The most commonly used quantiles are $z_{0.95} = 1.645$ and $z_{0.975} = 1.96$.

Example 22.6. We want to test $H_0\colon \theta = \theta_0$ against $H_1\colon \theta > \theta_0$ at significance level α. Since a value of $\bar{X}$ much larger than θ_0 is evidence against H_0, we use $\bar{X}$ as the test statistic and reject if $\bar{X} \geq c$. By proposition A.4, under H_0 the standardised mean $\sqrt{n}\,(\bar{X} - \theta_0)/\sigma$ is standard normally distributed, and thus the probability of a type I error is

$$\beta(\theta_0) = \mathbb{P}_{\theta_0}(\bar{X} \geq c) = \mathbb{P}_{\theta_0}\Big(\frac{\sqrt{n}\,(\bar{X} - \theta_0)}{\sigma} \geq \frac{\sqrt{n}\,(c - \theta_0)}{\sigma}\Big) = 1 - \Phi\Big(\frac{\sqrt{n}\,(c - \theta_0)}{\sigma}\Big).$$

Setting this probability equal to α, we find the critical value as $\sqrt{n}\,(c - \theta_0)/\sigma = z_{1-\alpha}$ and thus the resulting test rejects H_0 if

$$\bar{X} \geq \theta_0 + z_{1-\alpha}\,\frac{\sigma}{\sqrt{n}}, \qquad \text{equivalently if} \qquad Z = \frac{\sqrt{n}\,(\bar{X} - \theta_0)}{\sigma} \geq z_{1-\alpha}.$$

This is the one-sided z-test. To determine its power, we let $\delta = \sqrt{n}\,(\theta - \theta_0)/\sigma$. Since $\sqrt{n}\,(\bar{X} - \theta)/\sigma$ is standard normally distributed under $\mathbb{P}_\theta$, and since $1 - \Phi(x) = \Phi(-x)$ by the symmetry of the normal distribution, the power function is

$$\beta(\theta) = \mathbb{P}_\theta\Big(\frac{\sqrt{n}\,(\bar{X} - \theta)}{\sigma} \geq z_{1-\alpha} - \delta\Big) = \Phi(\delta - z_{1-\alpha}). \tag{22.1}$$

The left-hand panel of figure 22.1 shows this function. As n increases the curve gets steeper, and exercise 22.1 uses this to determine the required sample size.

For $n = 25$, $\sigma = 2$, $\theta_0 = 10$ and $\alpha = 0.05$ the test rejects if $\bar{X} \geq 10 + 1.645 \cdot 2/5 = 10.658$. At the alternative $\theta = 11$ we have $\delta = 2.5$ and the power is $\Phi(2.5 - 1.645) = \Phi(0.855) = 0.804$, at $\theta = 10.5$ it is only $\Phi(1.25 - 1.645) = \Phi(-0.395) = 0.346$: a shift of a quarter of a standard deviation is detected in barely a third of all samples.

The values $\theta < \theta_0$ belong to neither hypothesis of this example. Since the power function (22.1) is increasing, the test also has size $\sup_{\theta \leq \theta_0} \beta(\theta) = \beta(\theta_0) = \alpha$ for the composite null hypothesis $H_0\colon \theta \leq \theta_0$, which covers these values. This is why definition 22.5 takes a supremum.

Example 22.7. To detect departures in both directions, we test $H_0\colon \theta = \theta_0$ against $H_1\colon \theta \neq \theta_0$ and we reject for $|\bar{X} - \theta_0| \geq c$. Since $Z = \sqrt{n}\,(\bar{X} - \theta_0)/\sigma$

is standard normally distributed under H_0 and the normal distribution is symmetric, we have

$$\mathbb{P}_{\theta_0}(|\bar{X} - \theta_0| \geq c) = \mathbb{P}_{\theta_0}\Big(|Z| \geq \frac{\sqrt{n}\,c}{\sigma}\Big) = 2\Big(1 - \Phi\Big(\frac{\sqrt{n}\,c}{\sigma}\Big)\Big).$$

Setting this equal to α, we find $\sqrt{n}\,c/\sigma = z_{1-\alpha/2}$ and thus the two-sided z-test rejects H_0 if

$$|\bar{X} - \theta_0| \geq z_{1-\alpha/2}\,\frac{\sigma}{\sqrt{n}}, \qquad \text{equivalently if} \qquad |Z| \geq z_{1-\alpha/2}.$$

With δ as in example 22.6, the same standardisation gives the power function

$$\beta(\theta) = \Phi(-z_{1-\alpha/2} - \delta) + \Phi(\delta - z_{1-\alpha/2}),$$

shown in the right-hand panel of figure 22.1. This function is symmetric about θ_0 and has its minimum value α at this point (exercise 22.2). For $n = 25$, $\sigma = 2$, $\theta_0 = 10$ and $\alpha = 0.05$ the test rejects if $|\bar{X} - 10| \geq 1.96 \cdot 2/5 = 0.784$ and at $\theta = 11$ we have $\delta = 2.5$ and the power $\Phi(-4.46) + \Phi(0.54) = 0.705$.

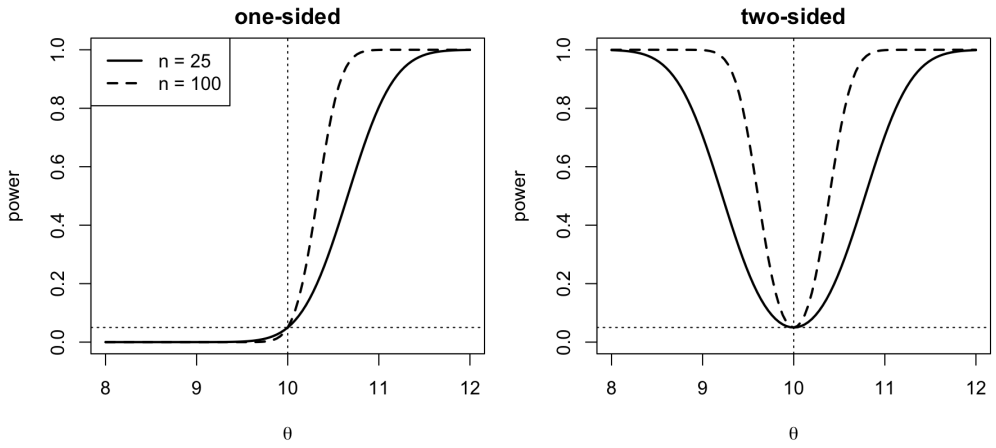


Figure 22.1: Power functions of the one-sided z-test (left) and the two-sided z-test (right) of $H_0\colon \theta = 10$ at level 0.05, for $\sigma = 2$ and the sample sizes $n = 25$ (solid) and $n = 100$ (dashed).

At $\theta = 11$ the one-sided test has power 0.804 against 0.705, since it places all of its type I error probability in the upper tail, but it is useless for alternatives below θ_0. Lecture 23 shows that the one-sided test is most powerful against every alternative above θ_0, and that no single test is best for both directions. Thus, before we consider the data, we must decide which departures from H_0 are of interest.

22.5 The p-Value

A test at a fixed level only states whether H_0 is rejected. The p-value also records how extreme the observed value of the test statistic is under H_0.

Definition 22.8. For a test which rejects H_0 for large values of a test statistic T, the *p-value* of the data x is given by

$$p(x) = \sup_{\theta \in \Theta_0} \mathbb{P}_\theta\big(T(X) \geq T(x)\big). \tag{22.2}$$

For a test which rejects for small values of T the inequality is reversed, and the two-sided z-test uses $T = |Z|$.

For a simple null hypothesis the supremum reduces to $\mathbb{P}_{\theta_0}$. Since $p(x)$ is the size of the test with critical region $\{T \geq T(x)\}$, it is the smallest level at which a test of the form $\{T \geq c\}$ rejects x, and thus H_0 is rejected at level α if and only if $p(x) \leq \alpha$. In this case we call the data *significant* at level α. For the one-sided z-test we find $p(x) = 1 - \Phi(z)$, where $z = \sqrt{n}\,(\bar{x} - \theta_0)/\sigma$ is the observed value of Z, and for the two-sided test we find $p(x) = 2\big(1 - \Phi(|z|)\big)$. In the situation of example 22.6, the observed mean $\bar{x} = 10.9$ gives $z = 2.25$ and thus $p = 1 - \Phi(2.25) = 0.0122$ for the one-sided test and $p = 0.0244$ for the two-sided test.

For a simple null hypothesis and a continuous test statistic, the distribution of the p-value under H_0 does not depend on the model.

Lemma 22.9. *Let $H_0 \colon \theta = \theta_0$ be simple and let $T(X)$ have a continuous distribution function F under $\mathbb{P}_{\theta_0}$. Then the p-value of a test which rejects for large values of T satisfies $\mathbb{P}_{\theta_0}\big(p(X) \leq u\big) = u$ for all $u \in (0,1)$, i.e. it is uniformly distributed on $(0,1)$ under H_0.*

Proof. Fix $u \in (0,1)$. Since F is continuous, we have $p(x) = 1 - F\big(T(x)\big)$. Let t be the smallest number with $F(t) = 1 - u$, which exists since F is continuous with limits 0 and 1. Since F is non-decreasing, $p(X) \leq u$ holds if and only if $T(X) \geq t$, and thus $\mathbb{P}_{\theta_0}\big(p(X) \leq u\big) = 1 - F(t) = u$. This completes the proof. □

For a discrete test statistic, *e.g.* the number of successes in a Bernoulli sample, the p-value takes only countably many values and is not uniformly distributed. For an integer-valued T, the test which rejects if $p(x) \leq \alpha$ is the test $\{T \geq c\}$ with the smallest integer c of level α, and thus it still has size at most α. A test whose size is smaller than its level in this way is called *conservative*.

Since the p-value is a probability, it is tempting to misread it. It is not the probability that H_0 is true: the parameter θ is a fixed, unknown number, and the probability in (22.2) refers to the data. Nor does a large p-value establish H_0, since data compatible with H_0 may be compatible with many alternatives as well: the test from example 22.6 has power only 0.346 at $\theta = 10.5$.

Summary.

- A testing problem contrasts a null hypothesis $H_0 \colon \theta \in \Theta_0$ against an alternative $H_1 \colon \theta \in \Theta_1$. A hypothesis is simple, if the corresponding subset of Θ is a single point, and composite otherwise.
- A test is given by its critical region C, typically of the form $T(x) \geq c$ for a test statistic T and critical value c.
- Type I errors are cases where H_0 is true but is wrongly rejected by the test, type II errors are cases where H_1 is true but H_0 is not rejected. The power function $\beta(\theta) = \mathbb{P}_\theta(X \in C)$ gives the probability of type I errors for Θ_0 and the power of the test for Θ_1.
- The size of a test is $\sup_{\Theta_0} \beta$. A level α test has size at most α. The level is chosen first, and then the power of the test can be considered.
- For the mean of a normal sample with known variance, the one-sided z-test has an S-shaped power function and the two-sided z-test a U-shaped one. Before looking at the data, we decide which departures from H_0 are of interest.
- The p-value is the probability under H_0 of a test statistic at least as extreme as the observed one, and H_0 is rejected at level α if and only if $p(x) \leq \alpha$. It is not the probability that H_0 is true, and a large p-value does not establish H_0. For a simple null hypothesis and a continuous test statistic, the p-value is uniformly distributed on $(0, 1)$ under H_0.

Exercise 22.1. Consider the one-sided z-test from example 22.6, at level α, and let $\theta_1 > \theta_0$ be the alternative and $1 - \gamma \in (\alpha, 1)$ be the target power.

1. Show that the test has power at least $1 - \gamma$ at θ_1 if and only if
$$n \geq \Big(\frac{(z_{1-\alpha} + z_{1-\gamma})\, \sigma}{\theta_1 - \theta_0} \Big)^2 .$$
2. For $\sigma = 2$, $\theta_1 - \theta_0 = 0.5$, $\alpha = 0.05$ and target power 0.9 (so that $z_{0.9} = 1.282$), determine the smallest sample size which achieves the target power.
3. How does the required sample size change, when the difference $\theta_1 - \theta_0$ is halved? What happens when σ is doubled?

Exercise 22.2. Consider the two-sided z-test from example 22.7, with power function $\beta(\theta) = g(\delta)$ where $g(\delta) = \Phi(-z - \delta) + \Phi(\delta - z)$, $z = z_{1-\alpha/2}$ and $\delta = \sqrt{n}\,(\theta - \theta_0)/\sigma$.

1. Show that $g(-\delta) = g(\delta)$ for all $\delta \in \mathbb{R}$, and conclude that β is symmetric about θ_0.
2. Show that g is strictly increasing on $[0, \infty)$. Conclude that β has its minimum value α at θ_0.

Exercise 22.3. Let $X_1, \ldots, X_{10}$ be i.i.d. Bernoulli with success probability θ and consider testing $H_0 \colon \theta = 1/2$ against $H_1 \colon \theta > 1/2$ using the test which rejects if $T = \sum_i X_i \geq c$ for an integer c.

1. Write down the size of the test as a function of c, using the binomial distribution of T under H_0.

2. Determine the smallest c such that the test has level 0.05. Compute the size of the test for this case. Justify your answer by showing that no choice of c can lead to a size of exactly 0.05.

3. Determine the power of the test from the previous part when $\theta = 0.7$, $\theta = 0.8$ and $\theta = 0.9$. Comment on the capabilities and limitations of the test with ten observations.

4. Nine successes are observed. Compute the p-value and state the decision at level 0.05.

Exercise 22.4. Let $X_1, \ldots, X_n \sim \text{Uniform}(0, \theta)$ be i.i.d. as in example 1.2 and let $M = \max_i X_i$, where the density was found in example 8.9. We test $H_0 \colon \theta = \theta_0$ against $H_1 \colon \theta > \theta_0$.

1. Show that $\mathbb{P}_\theta(M \leq m) = (m/\theta)^n$ for $0 \leq m \leq \theta$.

2. Consider the test which rejects if $M > c$, for a constant $c \leq \theta_0$. Determine the size of this test and the value of c which makes the size equal to α.

3. Determine the power function of the test from the previous part, and evaluate this function at $n = 5$, $\theta_0 = 1$, $\alpha = 0.05$ and $\theta = 1.2$.

4. Consider the second test, which rejects if $M > \theta_0$. Show that this test has size 0. Determine the power function of this test and compare the power functions of the two tests. Which of the two tests would you choose and why?

Exercise 22.5. A sample of size $n = 16$ from $N(\theta, \sigma^2)$ with known $\sigma = 4$ has mean $\bar{x} = 52.3$. We want to test $\theta_0 = 50$.

1. Determine the p-value of the one-sided z-test with $H_0 \colon \theta = 50$ against $H_1 \colon \theta > 50$ and state the decisions for 0.05 and 0.01.

2. Determine the p-value of the two-sided test against $H_1 \colon \theta \neq 50$ and state the same two decisions.

3. Your colleague summarises the first part of the question as "there is a 1.1% chance that the mean is 50". What is wrong with this sentence? Write one correct sentence to summarise the first part of the question.

4. Verify lemma 22.9 directly for the one-sided z-test: show that under H_0 the p-value $p(X) = 1 - \Phi(Z)$ with $Z = \sqrt{n}\,(\bar{X} - \theta_0)/\sigma$ satisfies $\mathbb{P}_{\theta_0}(p(X) \leq u) = u$ for all $u \in (0, 1)$.

Lecture 23

The Neyman–Pearson Lemma

In lecture 22 it seemed clear that for the mean of a normal sample we could not do better than to reject for large values of $\bar{X}$. This lecture proves that this is correct: for the most basic testing problem, where one distribution is tested against another, the Neyman–Pearson lemma identifies the most powerful test, in analogy to the role of the Cramér–Rao bound from lecture 13 for estimators. We then see how far this optimality extends to composite alternatives.

23.1 Simple Hypotheses and Most Powerful Tests

We consider the problem of testing a simple null hypothesis $H_0 \colon \theta = \theta_0$ against a simple alternative $H_1 \colon \theta = \theta_1$, where $\theta_0 \neq \theta_1$ are two fixed points of the parameter space, for data $X = (X_1, \ldots, X_n)$ with joint density or probability weights $f(x; \theta)$ as in lecture 8. Under either hypothesis the distribution of the data is completely specified; section 23.4 shows that the solution of this problem also settles more realistic ones.

We will compare different tests, so we write $\beta_C(\theta) = \mathbb{P}_\theta(X \in C)$ for the power function of definition 22.4 of the test with critical region C. For simple hypotheses, the size from definition 22.5 and the power are the single numbers $\beta_C(\theta_0)$ and $\beta_C(\theta_1)$. Among all tests with level α, *i.e.* with $\beta_C(\theta_0) \leq \alpha$, we want to find the test which makes type II errors least often.

Definition 23.1. Let $0 < \alpha < 1$. A test with critical region C is said to be *most powerful at level* α for testing $H_0 \colon \theta = \theta_0$ against $H_1 \colon \theta = \theta_1$, if $\beta_C(\theta_0) \leq \alpha$ and

$$\beta_C(\theta_1) \geq \beta_{C'}(\theta_1)$$

for all critical regions C' with $\beta_{C'}(\theta_0) \leq \alpha$.

As with the uniformly minimum variance unbiased estimator from definition 10.4, the definition does not say how to find such a test, or whether one exists; the next section answers both questions.

23.2 The Lemma

If $f(x; \theta_1)$ is much larger than $f(x; \theta_0)$, the data favour the alternative, and we should reject. The natural test statistic is the *likelihood ratio*

$$R(x) = \frac{f(x; \theta_1)}{f(x; \theta_0)} = \frac{L(\theta_1)}{L(\theta_0)},$$

and the natural test rejects if $R(X)$ exceeds some threshold k. The lemma states that no test with the same level can have more power. We write the inequality $f(x;\theta_1) > k\, f(x;\theta_0)$ instead of $R(x) > k$ so that points with $f(x;\theta_0) = 0$ do not require separate treatment and so that the boundary of the critical region can include any points where equality holds, as required for discrete models and in exercise 23.3.

Theorem 23.2 (Neyman–Pearson lemma)**.** *Let $k \geq 0$ and let C be a critical region with*

$$\{x \mid f(x;\theta_1) > k\, f(x;\theta_0)\} \subseteq C \subseteq \{x \mid f(x;\theta_1) \geq k\, f(x;\theta_0)\},$$

where the size $\alpha = \beta_C(\theta_0)$ satisfies $0 < \alpha < 1$. Then the following statements hold.

1. *The test with critical region C is the most powerful test at level α for the hypothesis $H_0\colon \theta = \theta_0$ against $H_1\colon \theta = \theta_1$.*
2. *If C' is another critical region, which is most powerful at level α for the same problem, then C and C' differ at most on the boundary set $\{x \mid f(x;\theta_1) = k\, f(x;\theta_0)\}$, except for a set of observations with probability zero for every $\theta \in \Theta$.*

Proof. Let C' be a critical region with $\beta_{C'}(\theta_0) \leq \alpha$ and let $\mathbf{1}_C$ and $\mathbf{1}_{C'}$ be the indicator functions for the two regions. Then the proof is based on the observation that

$$h(x) = \big(\mathbf{1}_C(x) - \mathbf{1}_{C'}(x)\big)\big(f(x;\theta_1) - k\, f(x;\theta_0)\big) \geq 0 \qquad \text{for all } x. \tag{23.1}$$

If $x \in C$, the first factor is either 0 or 1 and the second factor is non-negative by the choice of C. If $x \notin C$, the first factor is either 0 or -1 and the second factor is zero or negative by the choice of C.

Integrating (23.1) over all x, with a sum in place of the integral for a discrete model, and using $\int \mathbf{1}_A(x)\, f(x;\theta)\, \mathrm{d}x = \mathbb{P}_\theta(X \in A) = \beta_A(\theta)$ for each of the four terms, we find

$$0 \leq \int h(x)\, \mathrm{d}x = \big(\beta_C(\theta_1) - \beta_{C'}(\theta_1)\big) - k\big(\beta_C(\theta_0) - \beta_{C'}(\theta_0)\big). \tag{23.2}$$

Since $\beta_C(\theta_0) = \alpha \geq \beta_{C'}(\theta_0)$ and $k \geq 0$, the subtracted term is non-negative and thus we have

$$\beta_C(\theta_1) - \beta_{C'}(\theta_1) \geq k\big(\alpha - \beta_{C'}(\theta_0)\big) \geq 0,$$

and this is the first statement.

For the second statement assume that C' is also most powerful at level α. Then the two powers at θ_1 are equal, and thus the right-hand side of (23.2) equals $-k\big(\alpha - \beta_{C'}(\theta_0)\big) \leq 0$. Together with the inequality on the left we find $\int h(x)\, \mathrm{d}x = 0$. By the first paragraph, $h(x)$ is strictly positive exactly at those x outside the boundary set which lie in one of the regions C and C' but not in the other. For a discrete model a sum of non-negative terms is zero only if every term is zero, and for a model with densities a non-negative function with integral zero vanishes outside a set of measure zero, which has probability zero under every density $f(x;\theta)$. This completes the proof. □

Two comments help in using the lemma. First, the constant k is rarely computed: whenever R is a strictly monotone function of a simpler statistic T, we rewrite $R(x) > k$ as $T(x) > c$ or $T(x) < c$ and choose c from the distribution of $T(X)$ under θ_0 to give size α. Second, the lemma shows that the test is most powerful at the level equal to its own size. For models with densities, $R(X)$ usually has a continuous distribution, and thus any size can be achieved and the boundary set in the second statement has probability zero. In this case the most powerful test is essentially unique. For discrete models only certain sizes are attainable, and a test of size $\alpha' < \alpha$ is guaranteed to be most powerful only at level α'; at level α another critical region can have more power.

23.3 The Normal Mean

We now apply the lemma to the normal-mean example from lecture 22: the test constructed there using common sense is the most powerful one.

Example 23.3. Let $X_1, \dots, X_n$ be i.i.d. $N(\theta, \sigma^2)$ where σ^2 is known, and consider the hypothesis $H_0\colon \theta = \theta_0$ against the alternative $H_1\colon \theta = \theta_1$, where $\theta_1 > \theta_0$. The joint density of the observations is $f(x;\theta) = (2\pi\sigma^2)^{-n/2} \exp\bigl(-\sum_i (x_i - \theta)^2/(2\sigma^2)\bigr)$. In the ratio of two densities, the constant factors cancel. After expanding the squares in the exponent, we find

$$\sum_{i=1}^n (x_i - \theta_1)^2 - \sum_{i=1}^n (x_i - \theta_0)^2 = -2(\theta_1 - \theta_0) \sum_{i=1}^n x_i + n(\theta_1^2 - \theta_0^2),$$

and thus

$$\log R(x) = \frac{n(\theta_1 - \theta_0)}{\sigma^2}\, \bar{x} - \frac{n(\theta_1^2 - \theta_0^2)}{2\sigma^2}.$$

Since $\theta_1 - \theta_0 > 0$, the right-hand side is a strictly increasing function of $\bar{x}$ and the inequality $R(x) > k$ is equivalent to $\bar{x} > c$ for a constant c which depends on k. We choose c by considering the size of the test: under H_0 we have $\bar{X} \sim N(\theta_0, \sigma^2/n)$ by proposition A.4 and thus

$$\mathbb{P}_{\theta_0}(\bar{X} > c) = 1 - \Phi\Bigl(\frac{\sqrt{n}\,(c - \theta_0)}{\sigma}\Bigr) = \alpha$$

if and only if

$$c = c_\alpha = \theta_0 + z_{1-\alpha} \frac{\sigma}{\sqrt{n}},$$

where $z_{1-\alpha}$ is the $(1-\alpha)$-quantile of the standard normal distribution. By theorem 23.2 the test which rejects for $\bar{X} > c_\alpha$ is most powerful at level α and, since the boundary set $\{\bar{x} = c_\alpha\}$ has probability zero, this is the only test up to sets of probability zero. This is the test from example 22.6. The power of this test for the alternative is

$$\beta(\theta_1) = \mathbb{P}_{\theta_1}(\bar{X} > c_\alpha) = \Phi\Bigl(\frac{\sqrt{n}\,(\theta_1 - \theta_0)}{\sigma} - z_{1-\alpha}\Bigr).$$

For $n = 25$, $\sigma = 2$, $\theta_0 = 10$ and $\alpha = 0.05$ we have $z_{0.95} = 1.645$ and thus $c_\alpha = 10 + 1.645 \cdot 2/5 = 10.658$, and the power against $\theta_1 = 11$ is $\Phi(2.5 - 1.645) = \Phi(0.855) = 0.804$, the value we found in lecture 22. The lemma allows us to turn this number into a statement about all possible procedures: no test of level 0.05, based on the given 25 observations, can be better.

23.4 One-Sided Alternatives

The threshold $c_\alpha = \theta_0 + z_{1-\alpha}\sigma/\sqrt{n}$ in example 23.3 does not involve θ_1: the alternative entered the computation only through the sign of $\theta_1 - \theta_0$, and thus every $\theta_1 > \theta_0$ leads to the same critical region. This has a consequence for the composite problem of testing $H_0\colon \theta = \theta_0$ against $H_1\colon \theta > \theta_0$, the problem of interest in lecture 22.

Proposition 23.4. *Consider testing $H_0\colon \theta = \theta_0$ against $H_1\colon \theta > \theta_0$. Assume that C is a critical region with $\beta_C(\theta_0) = \alpha$ which, for every $\theta_1 > \theta_0$, satisfies the condition from theorem 23.2 for testing θ_0 against θ_1, for some constant $k = k(\theta_1)$ which may depend on θ_1. Then, for every $\theta_1 > \theta_0$ and every critical region C' with $\beta_{C'}(\theta_0) \le \alpha$, we have $\beta_C(\theta_1) \ge \beta_{C'}(\theta_1)$.*

Proof. Let $\theta_1 > \theta_0$ and let C' be a critical region with $\beta_{C'}(\theta_0) \le \alpha$. Then, since the null hypothesis is simple, C' is also a test of level α for the simple hypothesis θ_0 against θ_1 and by assumption C satisfies the condition from theorem 23.2 for this problem. Thus, by the first statement of the theorem, we have $\beta_C(\theta_1) \ge \beta_{C'}(\theta_1)$ and since θ_1 was arbitrary this completes the proof. □

A test which has this property is most powerful against every alternative in H_1 simultaneously and is said to be *uniformly most powerful* at level α. We only use this name as a description and do not consider the general theory of such tests.

For the normal mean, example 23.3 shows that the critical region $\{\bar{x} > \theta_0 + z_{1-\alpha}\sigma/\sqrt{n}\}$ is the Neyman–Pearson region for every $\theta_1 > \theta_0$, and thus the one-sided test of example 22.6 is uniformly most powerful against $H_1\colon \theta > \theta_0$, with power function

$$\beta_+(\theta) = \Phi\Big(\frac{\sqrt{n}\,(\theta - \theta_0)}{\sigma} - z_{1-\alpha}\Big).$$

For alternatives $\theta_1 < \theta_0$ the same computation with the inequality reversed gives the region $\{\bar{x} < \theta_0 - z_{1-\alpha}\sigma/\sqrt{n}\}$, uniformly most powerful against $H_1\colon \theta < \theta_0$, with power function

$$\beta_-(\theta) = \Phi\Big(-\frac{\sqrt{n}\,(\theta - \theta_0)}{\sigma} - z_{1-\alpha}\Big).$$

23.5 Two-Sided Alternatives

We now consider the alternative $H_1\colon \theta \ne \theta_0$. The two one-sided tests disagree: one rejects for large values of $\bar{X}$, the other for small values, and each one is poor on the other side of θ_0, since for $\theta < \theta_0$ we have $\beta_+(\theta) < \alpha$. Thus, no single test can agree with both tests simultaneously, and the uniqueness statement in the lemma turns this observation into the following proposition.

Proposition 23.5. *Let $X_1, \dots, X_n$ be i.i.d. $N(\theta, \sigma^2)$ with known σ^2 and let $0 < \alpha < 1$. Then there is no critical region C^* with $\beta_{C^*}(\theta_0) \le \alpha$ such that $\beta_{C^*}(\theta) \ge \beta_{C'}(\theta)$ for all $\theta \ne \theta_0$ and any critical region C' with $\beta_{C'}(\theta_0) \le \alpha$.*

Proof. Suppose that C^* were such a region, and fix some $\theta_1 > \theta_0$. Then C^* is most powerful at level α for testing θ_0 against θ_1, and so is the region $C_+ =$

$\{\bar{x} > c_\alpha\}$ of example 23.3. The boundary set $\{\bar{x} = c_\alpha\}$ has probability zero, and thus by the second statement of theorem 23.2 the regions C^* and C_+ differ only on a set which has probability zero for every θ. Consequently the two tests have the same power function, $\beta_{C^*}(\theta) = \beta_+(\theta)$ for all θ. Now take any $\theta_2 < \theta_0$. From the formulas of section 23.4 we get

$$\beta_{C^*}(\theta_2) = \beta_+(\theta_2) < \Phi(-z_{1-\alpha}) = \alpha < \beta_-(\theta_2),$$

since Φ is strictly increasing and $\sqrt{n}\,(\theta_2 - \theta_0)/\sigma < 0$. The left-sided test $C_- = \{\bar{x} < \theta_0 - z_{1-\alpha}\sigma/\sqrt{n}\}$ has size α and thus is an admissible competitor C', and it has strictly larger power at θ_2 than C^*. This contradicts the assumed property of C^*, which completes the proof. □

The argument is not particular to the normal distribution: whenever the one-sided most powerful regions disagree, no uniformly most powerful two-sided test exists, and we have to settle for a compromise such as the two-sided test of example 22.7. In lecture 25 we will see that this test arises from applying the likelihood-ratio idea to the composite alternative, which gives a general-purpose tool for such problems.

Summary.

- A test is most powerful at level α for a simple null hypothesis against a simple alternative, if it has level α and no other test of level α has larger power.
- The Neyman–Pearson lemma states that the test which rejects when $f(x;\theta_1) > k\,f(x;\theta_0)$, where k is chosen so that the size is α, is most powerful at level α. Any other most powerful test coincides with this test outside the boundary set, up to a set of probability zero. The proof involves integration of $(\mathbf{1}_C - \mathbf{1}_{C'})(f(x;\theta_1) - k\,f(x;\theta_0)) \geq 0$.
- In practice, $R(x) > k$ can be rewritten as $T(x) > c$ for a simple statistic T and c can be found by considering the distribution of T under H_0; for the normal mean this gives $\bar{X} > \theta_0 + z_{1-\alpha}\sigma/\sqrt{n}$.
- If the critical region does not depend on the alternative θ_1, the same test will be most powerful against all $\theta_1 > \theta_0$: the test is then uniformly most powerful for the one-sided problem.
- For the two-sided alternative, no uniformly most powerful test can exist, since the two one-sided tests disagree; instead, the likelihood-ratio tests from lecture 25 can be used.

Exercise 23.1. Let $X_1, \ldots, X_n$ be i.i.d. Poisson with mean $\theta > 0$ and let $S = \sum_{i=1}^n X_i$, Poisson with mean $n\theta$. We test $H_0\colon \theta = \theta_0$ against $H_1\colon \theta = \theta_1$ where $\theta_1 > \theta_0$.

1. Show that the likelihood ratio $R(x)$ is a strictly increasing function of $s = \sum_i x_i$ and that the most powerful test rejects when $S > c$. Why does c not depend on θ_1? What are the consequences for testing against the composite alternative $H_1 \colon \theta > \theta_0$?

2. Let $n = 5$, $\theta_0 = 1$ and $\alpha = 0.05$. Determine the smallest integer c such that the test has size at most 0.05. Compute the actual size of the test for this value. You may use $\mathbb{P}(S \le 8) = 0.9319$ and $\mathbb{P}(S \le 9) = 0.9682$ for a Poisson random variable S with mean 5.

3. Determine the power of the test at $\theta_1 = 2$, using $\mathbb{P}(S \le 9) = 0.4579$ for a Poisson random variable with mean 10.

4. Why cannot a test of the form $\{S > c\}$ have size exactly 0.05? At which level is the test from part (b) most powerful?

Exercise 23.2. Let $X_1, \dots, X_n$ be i.i.d. Bernoulli with success probability $p \in (0,1)$, and let $S = \sum_{i=1}^n X_i$. In exercise 22.3 we tested $H_0 \colon p = 1/2$ against $H_1 \colon p > 1/2$ by rejecting for large values of S.

1. For a fixed alternative $p_1 > 1/2$, show that the likelihood ratio is given by

$$R(x) = \Big(\frac{p_1}{1-p_1}\Big)^s \big(2(1-p_1)\big)^n, \qquad s = \sum_{i=1}^n x_i,$$

   and that it is strictly increasing in s. Deduce that the most powerful test rejects when $S > c$, and that the test of exercise 22.3 is uniformly most powerful against $H_1 \colon p > 1/2$.

2. Let $n = 20$. Using $\mathbb{P}_{1/2}(S \ge 14) = 0.0577$ and $\mathbb{P}_{1/2}(S \ge 15) = 0.0207$, find the critical region of the largest size not exceeding 0.05, and compute its power at $p_1 = 0.7$ using $\mathbb{P}_{0.7}(S \le 14) = 0.5836$. Compare the size and the power with those found for $n = 10$ in exercise 22.3.

3. For an alternative $p_1 < 1/2$, find the most powerful critical region, and explain why the two one-sided problems lead to different tests.

Exercise 23.3. Let $X_1, \dots, X_n$ be i.i.d. uniformly distributed on the interval $(0, \theta)$, with joint density $f(x;\theta) = \theta^{-n} \mathbf{1}_{\{m \le \theta\}}$ for $x_1, \dots, x_n \ge 0$, where $m = \max_i x_i$, and let $M = \max_i X_i$. From example 8.9 we know that $\mathbb{P}_\theta(M \le y) = (y/\theta)^n$ for $0 \le y \le \theta$. We want to test $H_0 \colon \theta = \theta_0$ against $H_1 \colon \theta = \theta_1$, where $\theta_1 > \theta_0$, at significance level $\alpha \in (0,1)$. In exercise 22.4 we have considered tests based on M for this model, and in the following we will see that these tests are optimal.

1. Let $k = (\theta_0/\theta_1)^n$. Determine the sets $\{x \mid f(x;\theta_1) > k\, f(x;\theta_0)\}$ and $\{x \mid f(x;\theta_1) = k\, f(x;\theta_0)\}$ in terms of m.

2. Show that the critical region $C = \{m > c\}$ for $c = \theta_0(1-\alpha)^{1/n}$ has size α and satisfies the condition from theorem 23.2 for this k. Conclude that this is the most powerful test at level α. Compute the power of the test and explain why the test is uniformly most powerful against $H_1 \colon \theta > \theta_0$.

3. Show that the region $C'' = \{m > \theta_0\} \cup \{m \leq \theta_0 \alpha^{1/n}\}$ also has size α and the same power as C. Explain, using the second statement of theorem 23.2, why this does not contradict the theorem, even though the model has densities.

Exercise 23.4. Let T be a sufficient statistic for the model $f(x;\theta)$ and let $\theta_0 \neq \theta_1$ be two parameter values. Using the factorisation theorem (proposition 8.3), show that the likelihood ratio $R(x) = f(x;\theta_1)/f(x;\theta_0)$ depends on the data only through $T(x)$, for all x where the denominator is positive.

Lecture 24

Power and Size by Simulation

This is the R session for week 8. The lectures this week have been about the language of tests and about the Neyman–Pearson lemma. In this session we will see how tests perform in practice. The R technique we will use here is the indicator average from lecture 12, where the indicator now marks the samples for which the test rejects. The written reference for this is in appendix B.

24.1 The Size of a Test by Simulation

The test we consider today is the one-sided z-test from example 22.6, in the numerical setting of the example: we have $n = 25$ observations from $N(\theta, \sigma^2)$ with known $\sigma = 2$ and we want to test $H_0 \colon \theta = 10$ against $H_1 \colon \theta > 10$ at level $\alpha = 0.05$, *i.e.* we reject if and only if $\bar{X} > 10.658$. By definition 22.4, the power function $\beta(\theta) = \mathbb{P}_\theta(\bar{X} > 10.658)$ is a probability, and by proposition A.14 we can approximate this probability by the proportion of samples where the test rejects. As in lecture 12, we write a function which, for one run of the test, returns `TRUE` if the test rejects and `FALSE` otherwise. The command `mean` can then be used to get the proportion of rejections. Before you run the code, predict the value it prints. Will it be exactly 0.05? How far away from 0.05 would it have to be before you would suspect a mistake in the code?

```
n      <- 25
sigma  <- 2
theta0 <- 10
crit   <- theta0 + qnorm(0.95) * sigma / sqrt(n)
z_reject <- function(theta) {
  x <- rnorm(n, mean = theta, sd = sigma)
  mean(x) > crit
}
set.seed(2026)
mean(replicate(10000, z_reject(theta0)))
```

```
## [1] 0.0523
```

The rejection rate under H_0 is 0.0523. This is the estimated size of the test in the sense of definition 22.5. The value is not exactly 0.05, and it should not be: as in lecture 12, the estimate has standard deviation $\sqrt{0.05 \cdot 0.95/10000} = 0.0022$. Thus, a value between 0.046 and 0.054 is what we expect for a correct test, while

a value of 0.07 would point to an error in the code. Calling the same function with a value of θ above θ_0 gives the power at this point; for $\theta = 10.5$ and $\theta = 11$ the exact values are 0.346 and 0.804, from example 22.6.

24.2 The Power Curve

A single number is a poor summary of a test, so we now consider the power function over a grid of alternatives. Figure 24.1 shows the rejection rates on the grid of values from 9 to 12 in steps of 0.1 for three level 0.05 tests: the one-sided z-test from above, the two-sided z-test from example 22.7 and the one-sided z-test applied to the first 12 observations only. The last test is wasteful, but it is a legal level 0.05 test. The script `R/S24-power.R` can be used to reproduce the figure. Before you look at the figure, predict how the two-sided test compares with the one-sided test. Is its rejection rate above or below that of the one-sided test for $\theta < 10$? Is it above or below for $\theta > 10$?

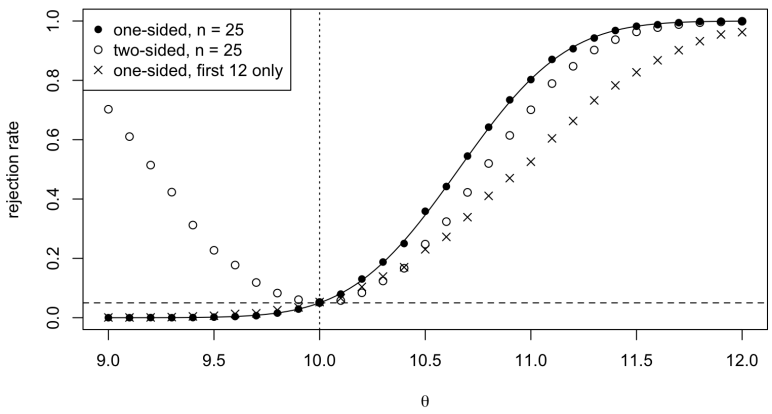


Figure 24.1: Estimated power functions for three level 0.05 tests for $H_0\colon \theta = 10$, using 5000 samples of size $n = 25$ for each grid point: one-sided z-test (filled circles), two-sided z-test (open circles) and one-sided z-test on the first 12 observations only (crosses). For comparison, the line gives the exact power function of the one-sided test from lecture 22.

The line in the figure is the exact power function (22.1) of the one-sided test, and the filled circles lie on it. All three curves pass through 0.05 at $\theta_0 = 10$, as required for level 0.05 tests, and the tests differ only in power. The two-sided test has lower power than the one-sided test for $\theta > 10$ and much higher power for $\theta < 10$. Thus, neither curve lies above the other everywhere, and this illustrates proposition 23.5. From example 23.3 and proposition 23.4 we know that no level 0.05 test based on these 25 observations can reject at $\theta = 11$ more than 80% of the time. The crosses show the cost of having discarded thirteen of the observations: the power at $\theta = 11$ decreases from 0.80 to 0.53.

24.3 Looking at the Data More Than Once

The recipe of the first section gives the size of every procedure which ends in a decision, and we now apply it to a procedure whose size is not what it seems to be. Assume that the observations arrive one at a time, up to 100 in total, and that we perform the two-sided z-test from example 22.7 after every ten observations. We stop and reject H_0 as soon as one of the p-values is below 0.05: this is the case of the researcher who "collects a few more data points and looks again". The following function simulates one run of this procedure under H_0, using the p-value $2(1 - \Phi(|z|)) = 2\,\Phi(-|z|)$ from lecture 22, and returns `TRUE` if the procedure rejects. Before you run the code, predict the rejection rate. Each of the ten tests has level 0.05. Will the rejection rate be 0.05? Could it be as large as 0.5?

```
peek <- function() {
  x <- rnorm(100, mean = theta0, sd = sigma)
  for (k in seq(10, 100, by = 10)) {
    z <- sqrt(k) * (mean(x[1:k]) - theta0) / sigma
    if (2 * pnorm(-abs(z)) < 0.05) return(TRUE)
  }
  FALSE
}
set.seed(2026)
mean(replicate(10000, peek()))
```

```
## [1] 0.1993
```

The procedure rejects a true null hypothesis in 20% of all runs, four times as often as the level of the single tests suggests. By lemma 22.9, each of the ten p-values is uniformly distributed under H_0, and thus each single look rejects with probability 0.05. The procedure, however, rejects if at least one of the ten looks rejects. Since the probability of a union is at most the sum of the probabilities, the rejection rate cannot exceed $10 \cdot 0.05 = 0.5$, and ten independent tests would give $1 - 0.95^{10} = 0.40$. Here the looks share observations: a sample which is far from rejection at one look is likely to be far from rejection at the next look as well, and for this reason the rate 0.20 is smaller than both values. The level is a property of the whole procedure, including the rule which determines when to stop, and for this reason the sample size must be fixed before the data are seen.

24.4 A Task to Take Home

Set the seed of the random number generator to your student ID and consider the setting from section 24.2, but with sample size $n = 10$ instead of $n = 25$. Estimate the size of the one-sided z-test for 10000 samples and state whether the estimated size is consistent with 0.05, within simulation error. Next, estimate the power of the test on the grid `seq(9, 13, by = 0.1)` and create a plot of the estimated power curve, together with the exact power function from lecture 22; the script `R/S24-power.R` shows how the command `sapply` can be used to apply the rejection-rate recipe to every point of a grid. From the plot, determine the smallest θ at which the power is greater than 0.9. Finally, modify the function

`peek` from section 24.3 to test after every five observations, estimate the rejection rate under H_0 from 10000 runs and compare it with 0.199.

Before the next session, please type the commands of this session into R, and check that what you get is what the text says you should get.

Summary.

- The written reference for the commands used in this session is appendix B.
- We can estimate the size and power of a test by simulation, by taking the average of the indicator function for the rejection. An estimated rate for 10000 samples has standard deviation of approximately 0.002 near 0.05.
- The power curve of every level 0.05 test passes through the point 0.05 at θ_0, and the tests differ in power only. The two-sided test is weaker than the one-sided test on one side and stronger on the other, and a test which discards observations is weaker for all $\theta > \theta_0$.
- A procedure which tests after every ten observations and stops at the first p-value below 0.05 rejects a true null hypothesis with probability 0.20. The level is a property of the whole procedure, including the stopping rule.

Lecture 25

Likelihood-Ratio Tests

Lecture 23 ended with a gap: the Neyman–Pearson lemma does not consider composite null hypotheses or two-sided alternatives, but these are the most common situations in statistical tests. The purpose of this lecture is to describe a generalisation of the likelihood-ratio test, which can be used for such cases. The generalised likelihood-ratio test is based on the same principles as the Neyman–Pearson ratio, but is modified to account for the more general case. In the second part of this lecture, we will see that the large-sample behaviour of the generalised likelihood-ratio test, described by Wilks' theorem, is based on the asymptotic theory from lecture 17. The generalised likelihood-ratio test is the test computed by most statistical software packages when a likelihood-ratio test is requested.

25.1 The Generalised Likelihood Ratio

As in lecture 23, we write $L(\theta)$ for the likelihood of the data $X = (X_1, \dots, X_n)$ and $\ell(\theta) = \log L(\theta)$ for the log-likelihood, where the parameter space Θ is now allowed to be a subset of $\mathbb{R}^k$. We consider the following testing problem:

$$H_0 \colon \theta \in \Theta_0 \qquad \text{against} \qquad H_1 \colon \theta \in \Theta \setminus \Theta_0$$

for a subset $\Theta_0 \subseteq \Theta$, where both hypotheses can be composite. Since the alternative is the complement of the null hypothesis within Θ, the null hypothesis is a restriction of the full model and we have *nested* hypotheses.

If a hypothesis is composite, we represent it in the Neyman–Pearson comparison of the two likelihoods by the parameter value which gives the best fit to the data, *i.e.* by the maximum of the likelihood over the hypothesis.

Definition 25.1. The *generalised likelihood ratio* for testing $H_0 \colon \theta \in \Theta_0$ against $H_1 \colon \theta \in \Theta \setminus \Theta_0$ is given by

$$\Lambda(x) = \frac{\sup_{\theta \in \Theta_0} L(\theta)}{\sup_{\theta \in \Theta} L(\theta)}.$$

The *generalised likelihood-ratio test* at level α rejects H_0, if $\Lambda(X) \leq c$, where the constant c is chosen as large as possible subject to $\sup_{\theta \in \Theta_0} \mathbb{P}_\theta\big(\Lambda(X) \leq c\big) \leq \alpha$. The test can equivalently be written as rejecting H_0 for large values of the statistic

$$W = -2 \log \Lambda(X).$$

The denominator of Λ is the likelihood at the maximum likelihood estimator $\hat{\theta}$ from lecture 5, the numerator is the likelihood at the *restricted* maximum likelihood estimator $\hat{\theta}_0$, which is the best value within H_0; if H_0 is simple, we have $\Theta_0 = \{\theta_0\}$ and the numerator is just $L(\theta_0)$. Since $\Theta_0 \subseteq \Theta$ we always have $0 \leq \Lambda \leq 1$: values close to one mean that the restriction to Θ_0 costs little in terms of fit, values close to zero mean that H_0 is a much worse fit for the data than the full model is, and this is the evidence on which we base the rejection. The factor -2 in W is the scaling which, as we will see in section 25.3, gives the limiting distribution of W its standard form. In terms of the log-likelihood we have

$$W = 2\big(\ell(\hat{\theta}) - \ell(\hat{\theta}_0)\big), \tag{25.1}$$

which is twice the amount the maximised log-likelihood decreases when the null hypothesis is imposed.

The test is an extension of the Neyman–Pearson test, not a replacement. For example, for a two-point parameter space $\Theta = \{\theta_0, \theta_1\}$ where $\Theta_0 = \{\theta_0\}$, and using the notation $R = L(\theta_1)/L(\theta_0)$ for the likelihood ratio from lecture 23, we have

$$\Lambda = \frac{L(\theta_0)}{\max\big(L(\theta_0), L(\theta_1)\big)} = \min\Big(1, \frac{1}{R}\Big),$$

and for $c < 1$ the event $\Lambda \leq c$ is the same as the event $R \geq 1/c$; thus, for this case, the test is the most powerful test from theorem 23.2 with threshold $1/c$ for R. For composite hypotheses no optimality is guaranteed and in fact proposition 23.5 shows that for the two-sided normal problem no such test can exist.

One more property follows directly from the definition. If $\psi = \tau(\theta)$ is a one-to-one reparametrisation, the suprema in the definition of Λ do not change and thus Λ and W do not depend on the choice of parametrisation of the model. This is the same invariance as in proposition 5.6, and exercise 25.3 verifies it in an example.

25.2 The Two-Sided Test for the Mean of a Normal Distribution

We will now consider the question which was left open at the end of lecture 23, *i.e.* we will consider the two-sided alternative for the mean of a normal sample with known variance. This example is convenient, since for this case the distribution of W under H_0 can be found exactly.

Example 25.2. Let $X_1, \ldots, X_n$ be i.i.d. $N(\theta, \sigma^2)$ with known σ^2 and consider $H_0 \colon \theta = \theta_0$ against the alternative $H_1 \colon \theta \neq \theta_0$. Thus we have $\Theta = \mathbb{R}$ and $\Theta_0 = \{\theta_0\}$. From example 5.4 we know that the log-likelihood

$$\ell(\theta) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \theta)^2$$

attains its maximum over $\mathbb{R}$ at $\hat{\theta} = \bar{X}$. Since the null hypothesis is simple, the restricted maximiser is θ_0 and using formula (25.1) we find

$$W = 2\big(\ell(\bar{X}) - \ell(\theta_0)\big) = \frac{1}{\sigma^2}\Big(\sum_{i=1}^{n}(X_i - \theta_0)^2 - \sum_{i=1}^{n}(X_i - \bar{X})^2\Big).$$

Using the identity (2.1) from lecture 2, where θ_0 is substituted for μ, we find that the difference of the two sums of squares is $n(\bar{X} - \theta_0)^2$. Thus we have

$$W = \frac{n(\bar{X} - \theta_0)^2}{\sigma^2} = Z^2, \qquad \text{where } Z = \frac{\sqrt{n}\,(\bar{X} - \theta_0)}{\sigma}. \tag{25.2}$$

The test rejects for large values of W, *i.e.* for large values of $|Z|$. Thus we have recovered the two-sided z-test, which example 22.7 derived by common sense.

Under $\mathbb{P}_{\theta_0}$, the statistic Z is standard normally distributed by proposition A.4 and thus, by definition A.2, its square $W = Z^2$ satisfies $W \sim \chi^2_1$ for every sample size n. Let $\chi^2_{r,p}$ be the p-quantile of the χ^2_r distribution. Then the level α test rejects if $W \geq \chi^2_{1,1-\alpha}$. Since $W = Z^2$, the events $W \geq \chi^2_{1,1-\alpha}$ and $|Z| \geq z_{1-\alpha/2}$ coincide and thus we have $\chi^2_{1,1-\alpha} = z^2_{1-\alpha/2}$; from the tables we find $\chi^2_{1,0.95} = 3.841 = 1.96^2$. For $n = 25$, $\sigma = 2$, $\theta_0 = 10$ and observed mean $\bar{x} = 10.9$, for the data from lecture 22, we find $w = 25 \cdot 0.81/4 = 5.06$, which is larger than 3.841 and thus H_0 is rejected at level 0.05. The p-value is $\mathbb{P}(\chi^2_1 \geq 5.06) = 2\big(1 - \Phi(\sqrt{5.06})\big) = 2\big(1 - \Phi(2.25)\big) = 0.0244$, which is the same value as obtained for the two-sided test in section 22.5.

A familiar test has been recovered from the general recipe, and the next section shows that the chi-squared distribution of W is the general case for large samples, even in models where normality is not exactly satisfied.

25.3 Wilks' Theorem

In example 25.2 the distribution of W under H_0 was known, because $\bar{X}$ is exactly normal; in most models this will not be the case and Wilks' theorem provides the large-sample approximation which allows the test to be used practically.

The theorem is stated for hypotheses which fix some coordinates of the parameter. Let $\Theta \subseteq \mathbb{R}^k$ be open, write $\theta = (\theta_1, \ldots, \theta_k)$ for the coordinates of the parameter and let the null hypothesis fix the first r coordinates at given values,

$$\Theta_0 = \{\, \theta \in \Theta \mid \theta_j = \theta_j^0 \text{ for all } j \leq r \,\}, \tag{25.3}$$

where $1 \leq r \leq k$. For $r = k$ the null hypothesis is simple, and for $r < k$ the free coordinates $\theta_{r+1}, \ldots, \theta_k$ are *nuisance parameters*, unknown parameters about which the hypothesis says nothing. The number r of constraints is the difference between the dimensions of Θ and Θ_0.

Theorem 25.3 (Wilks)**.** *Let $X_1, \ldots, X_n$ be i.i.d. from a model with parameter $\theta \in \Theta \subseteq \mathbb{R}^k$ which is regular enough for the vector form of theorem 17.1, stated without its conditions in section 17.6, to hold. Let Θ_0 be of the form (25.3), suppose that with probability tending to one the restricted maximum likelihood estimator is the unique root of the likelihood equations over Θ_0, and let $W_n = -2 \log \Lambda_n$ be the generalised likelihood-ratio statistic for a sample of size n. Then for every $\theta \in \Theta_0$ we have*

$$W_n \xrightarrow{\text{d}} \chi^2_r \qquad (n \to \infty)$$

under $\mathbb{P}_\theta$.

Whatever the model, the only feature of the problem which enters the limit is the number r of constraints. In example 25.2 we have $k = r = 1$ and there the statement holds exactly for every n. We do not prove the theorem: together with the vector case of theorem 17.1, on which its general form rests, it is one of the few results of this module which we state without proof. The following argument shows where the chi-squared distribution comes from.

Consider the case $k = r = 1$ of a one-dimensional parameter and a simple null hypothesis $H_0 \colon \theta = \theta_0$, so that $W_n = 2\big(\ell(\hat\theta_n) - \ell(\theta_0)\big)$. We will expand the log-likelihood around the maximum likelihood estimator (MLE) instead of around θ_0 as in lecture 17: using Taylor's theorem with the Lagrange form of the remainder we find

$$\ell(\theta_0) = \ell(\hat\theta_n) + \ell'(\hat\theta_n)(\theta_0 - \hat\theta_n) + \tfrac{1}{2}\,\ell''(\theta_n^*)(\theta_0 - \hat\theta_n)^2$$

for some θ_n^* between θ_0 and $\hat\theta_n$. Since $\hat\theta_n$ solves the likelihood equation, the first derivative at this point is zero and rearranging this gives

$$W_n = -\ell''(\theta_n^*)\,(\hat\theta_n - \theta_0)^2 = \Big(-\frac{1}{n}\,\ell''(\theta_n^*)\Big)\Big(\sqrt{n}\,(\hat\theta_n - \theta_0)\Big)^2. \tag{25.4}$$

Now assume that H_0 is true. Then, by theorem 17.1, the second term is the square of a quantity which converges in distribution to $N\big(0, 1/\mathcal{I}(\theta_0)\big)$, and by the continuous mapping theorem, proposition A.17, this shows that the second term converges in distribution to $Y^2/\mathcal{I}(\theta_0)$ where Y is standard normally distributed. The first term is the term B_n from the proof of theorem 17.1 with θ_n^* instead of θ_0 and since θ_n^* is between θ_0 and the consistent estimator $\hat\theta_n$, the bound (R4) on the third derivative shows that this term still converges in probability to $\mathcal{I}(\theta_0)$. Finally, by Slutsky's theorem, proposition A.16, we find

$$W_n \xrightarrow{\mathrm{d}} \mathcal{I}(\theta_0) \cdot \frac{Y^2}{\mathcal{I}(\theta_0)} = Y^2 \sim \chi^2_1,$$

by definition A.2. Since the Fisher information cancels, the limit is independent of the details of the model.

For a simple null hypothesis in one dimension, this argument is close to a complete proof. The proof for the general case is more challenging: the expansion will be in k variables and will use the vector version of theorem 17.1, and the quadratic form which replaces Y^2 will consist of r independent, squared standard normal variables, one for each constraint. This is the origin of the number r of degrees of freedom in the argument.

25.4 Using the Theorem

Wilks' theorem turns the generalised likelihood-ratio test into a recipe: compute the statistic $W = 2\big(\ell(\hat\theta) - \ell(\hat\theta_0)\big)$ from the two maximised log-likelihoods, found numerically if need be as in lecture 18; count the degrees of freedom $r = \dim\Theta - \dim\Theta_0$; reject H_0 at level α, if $W \geq \chi^2_{r,1-\alpha}$; and report the approximate p-value $\mathbb{P}(\chi^2_r \geq w)$ for the observed value w, which for $r = 1$ equals $2\big(1 - \Phi(\sqrt{w})\big)$. Only the counting of degrees of freedom requires some care. For example, in the case of the mean of a normal sample with unknown variance, and $H_0 \colon \mu = \mu_0$, we have $k = 2$ and $r = 1$, since the variance is unrestricted under both hypotheses.

For a multinomial distribution with m categories, the parameter $(p_1, \dots, p_{m-1})$ ranges over an open subset of $\mathbb{R}^{m-1}$, with $p_m = 1 - p_1 - \cdots - p_{m-1}$, and a fully specified null hypothesis fixes all $m - 1$ coordinates, so that $r = m - 1$.

The theorem is a limit statement and there are three cases where problems can occur when applying the theorem to a fixed n. First, the approximation may be poor for small sample size: for the mean of a normal sample with unknown variance, exercise 25.1 shows that the level 0.05 test based on $\chi^2_{1,0.95}$ rejects too often at $n = 10$. Lecture 27 uses simulation to show how the approximation improves as n increases. Whenever the exact distribution of W or of an equivalent test statistic is known, as in the situation of the exercise, this distribution should be used. Lecture 31 takes this approach, using the exact distribution, for the standard tests. Secondly, if the regularity conditions are violated, for example if the support of the distribution depends on the parameter or if the null value coincides with the boundary of the parameter space, the expansion (25.4) will break down and the limit will not be χ^2_r. Finally, the theorem only describes W under H_0. The theorem gives the size of the test, but the power of the test must be determined separately.

Summary.

- For nested hypotheses $H_0 \colon \theta \in \Theta_0$ against $H_1 \colon \theta \in \Theta \setminus \Theta_0$, the generalised likelihood ratio is $\Lambda = \sup_{\Theta_0} L / \sup_{\Theta} L \in [0, 1]$, and the generalised likelihood-ratio test rejects for small Λ, *i.e.* for large $W = -2 \log \Lambda = 2(\ell(\hat{\theta}) - \ell(\hat{\theta}_0))$.
- For two simple hypotheses the test is the Neyman–Pearson test; in general it is unchanged by reparametrisation.
- For a normal mean with known variance and $H_0 \colon \theta = \theta_0$, the statistic is $W = n(\bar{X} - \theta_0)^2/\sigma^2 = Z^2$, which is exactly χ^2_1 under H_0 and the test is the two-sided z-test.
- Wilks' theorem states that, under regularity conditions, if H_0 fixes r of the k parameters, then $W \xrightarrow{\mathrm{d}} \chi^2_r$ under H_0. The theorem is given without proof, but from the arguments for $k = r = 1$, using Taylor expansion of ℓ around $\hat{\theta}_n$ and theorem 17.1, it is clear why the limit is χ^2_1.
- To use the theorem, compute W from the two maximised log-likelihoods, count $r = \dim \Theta - \dim \Theta_0$, reject if $W \geq \chi^2_{r,1-\alpha}$. The approximation can be poor for small n, the test fails if the regularity conditions are violated, and no information about power is available.

Exercise 25.1. Let $X_1, \dots, X_n$ be i.i.d. $N(\mu, \sigma^2)$, where both parameters are unknown. Consider the hypothesis $H_0 \colon \mu = \mu_0$ against the alternative $H_1 \colon \mu \neq \mu_0$, where σ^2 is a nuisance parameter. Let $Q = \sum_i (X_i - \bar{X})^2$, $S^2 = Q/(n-1)$ and $t = \sqrt{n}\,(\bar{X} - \mu_0)/S$.

1. Following example 5.4, show that the maximised log-likelihood function over Θ is given by
$$-\frac{n}{2}\log(2\pi Q/n) - \frac{n}{2}.$$

2. Show that, restricting ourselves to the region $\mu = \mu_0$, the maximum of the likelihood function is attained for $\hat{\sigma}_0^2 = \frac{1}{n}\sum_i (X_i - \mu_0)^2$ and find expressions for $\hat{\sigma}_0^2$ in terms of Q and $\bar{X} - \mu_0$ using the identity (2.1).

3. Deduce that
$$W = n\log\Big(1 + \frac{t^2}{n-1}\Big),$$
and that the generalised likelihood-ratio test rejects for large values of $|t|$.

4. State the approximate distribution of W under H_0 given by theorem 25.3 and justify your answer for the degrees of freedom. Recall that, under H_0, the statistic t follows the t_{n-1} distribution by definition A.5 and proposition A.4. For $n = 10$, determine the value of $|t|$ above which the approximate test with critical value $\chi^2_{1,0.95} = 3.841$ rejects and compare your result with the 0.975-quantile 2.262 of the t_9 distribution. Comment upon your result.

Exercise 25.2. An experiment with m possible outcomes is repeated n times independently. The number of times each outcome occurs is given by N_j for outcome j. Thus, $(N_1, \dots, N_m)$ is multinomially distributed with probability weights
$$\mathbb{P}_p(N_1 = n_1, \dots, N_m = n_m) = \frac{n!}{n_1! \cdots n_m!}\, p_1^{n_1} \cdots p_m^{n_m}, \qquad n_1 + \cdots + n_m = n,$$
where $p = (p_1, \dots, p_m)$ satisfies $p_j > 0$ and $\sum_j p_j = 1$. We test $H_0 \colon p = p^0$ for a given vector $p^0 = (p_1^0, \dots, p_m^0)$ against the alternative $p \neq p^0$.

1. Show that the log-likelihood function is $\sum_j n_j \log p_j$ up to an additive constant and use the Lagrange multiplier method, with constraint $\sum_j p_j = 1$, to show that the maximum of the likelihood function is at $\hat{p}_j = n_j/n$.

2. Show that
$$W = 2\sum_{j=1}^{m} n_j \log \frac{n_j}{n\, p_j^0},$$
where $0 \log 0 = 0$.

3. Why does theorem 25.3 state that $W \xrightarrow{\mathrm{d}} \chi^2_{m-1}$ under H_0?

4. A die is thrown 60 times and the six faces appear 13, 8, 11, 6, 12 and 10 times, respectively. Test the hypothesis that the die is fair at level 0.05, using $\chi^2_{5,0.95} = 11.07$.

5. Let $n_j = np_j^0 + d_j$ so that $\sum_j d_j = 0$. Show that, by expanding the logarithm to second order in $d_j/(np_j^0)$, we find
$$W \approx \sum_{j=1}^{m} \frac{(n_j - np_j^0)^2}{np_j^0}.$$
This is Pearson's chi-squared statistic. Compute the value for the die in the previous part of the question.

Exercise 25.3. Let $X_1, \dots, X_n$ be i.i.d. with Bernoulli distribution, *i.e.* with success probability $p \in (0,1)$. Furthermore, let $S = \sum_i X_i$ and $\hat{p} = S/n$ and consider the hypothesis $H_0 \colon p = p_0$ against the alternative $H_1 \colon p \neq p_0$.

1. Show that
$$W = 2\Big(S \log \frac{\hat{p}}{p_0} + (n-S) \log \frac{1-\hat{p}}{1-p_0}\Big),$$
where we use the convention $0 \log 0 = 0$.

2. Approximately determine the distribution of W under H_0 for large n and use the resulting distribution to derive a test at level α.

3. In $n = 50$ trials, we observed 32 successes. Test the hypothesis $H_0 \colon p = 1/2$ at level 0.05 and determine the approximate p-value.

4. In exercise 17.2, the model was reparametrised using the log-odds $\psi = \log\big(p/(1-p)\big)$. Show that the generalised likelihood-ratio statistic for testing $H_0 \colon \psi = \psi_0$ coincides with the one in part (a) for $p_0 = e^{\psi_0}/(1+e^{\psi_0})$, the value of the null proportion.

Lecture 26

Confidence Intervals

So far we have considered estimators which output a single number. In this lecture we will consider intervals of parameter values, constructed such that the true parameter value lies inside the interval with a given probability. We will see two different ways to construct such an interval, one using the concept of a pivotal quantity and the other one based on the inversion of a family of hypothesis tests. This latter approach also allows us to relate the testing theory from lectures 22 to 25 to the topic of estimation. The Bayesian equivalents of confidence intervals are called credible intervals and we will learn about these in lecture 29.

26.1 Definition and Interpretation

Here, $X = (X_1, \ldots, X_n)$ is a sample from a parametric model with parameter $\theta \in \Theta \subseteq \mathbb{R}$, and $\mathbb{P}_\theta$ is the distribution of the sample under θ. For a fixed small $\alpha \in (0, 1)$, typically $\alpha = 0.05$, the aim is to find an interval which misses the truth with probability at most α.

Definition 26.1. Let $L(X)$ and $U(X)$ be statistics such that $L(X) \leq U(X)$. The random interval $[L(X), U(X)]$ is a *confidence interval* for θ with *confidence level* $1 - \alpha$, if

$$\mathbb{P}_\theta\big(L(X) \leq \theta \leq U(X)\big) \geq 1 - \alpha \qquad \text{for all } \theta \in \Theta.$$

The probability on the left-hand side is called the *coverage probability* of the interval at θ.

If the model also has a nuisance parameter ν, here and below $\mathbb{P}_\theta$ is replaced by $\mathbb{P}_{\theta,\nu}$ and the requirements must hold for all values of (θ, ν). The endpoints of the interval are statistics, *i.e.* functions of the data, while the parameter θ is an unknown constant which does not change between samples. For continuous data we can usually achieve equality in the definition, but for discrete data the coverage probability of the interval can jump as θ changes. For this reason, the definition only requires the left-hand side to satisfy the inequality. More generally, any random subset $C(X) \subseteq \Theta$ of the parameter space such that $\mathbb{P}_\theta\big(\theta \in C(X)\big) \geq 1 - \alpha$ for all θ is called a *confidence set*.

Some care is needed when interpreting a confidence interval. For example, if the data lead to the 95% interval $[9.32, 11.28]$, both endpoints are now fixed

numbers, and so is the unknown θ; the statement $9.32 \leq \theta \leq 11.28$ is either true or false. The 95% describes the procedure: 95% of the computed intervals would cover the truth in repeated samples, but we do not know whether the interval in front of us is one of these. The statement "θ is in $[9.32, 11.28]$ with probability 0.95" is wrong, since it assumes that the parameter is random (this only holds in the Bayesian model from lecture 28). The statement "95% of the observations are in the interval" is wrong, since this would be a confusion of parameter with data. The correct interpretation of such an interval can be illustrated by simulating many intervals and seeing how many of these cover the truth. This is done in lecture 27.

26.2 Pivotal Quantities

The most direct approach to constructing a confidence interval is to construct an interval based on a quantity whose distribution is known exactly, regardless of the parameter value. Such a quantity is called a *pivotal quantity*.

Definition 26.2. A function $Q(X, \theta)$ of the sample and the parameter is a *pivotal quantity* or *pivot*, if the distribution of $Q(X, \theta)$ under $\mathbb{P}_\theta$ does not depend on $\theta \in \Theta$.

A pivot is not a statistic, because it contains the unknown θ, but we know the quantiles of the pivot distribution. To get a confidence interval, we can choose constants $a < b$ such that $\mathbb{P}_\theta(a \leq Q(X, \theta) \leq b) = 1 - \alpha$ and then solve the inequalities $a \leq Q(X, \theta) \leq b$ for θ. If Q is monotone in θ, the solution of these inequalities forms an interval which, with probability exactly $1 - \alpha$, contains the unknown parameter value. Using the quantile notation z_p from lecture 22, we have $z_{0.975} = 1.96$ and, using symmetry, $z_{1-p} = -z_p$. Similarly, we write $t_{k,p}$ and $\chi^2_{k,p}$ for the p-quantiles of the t_k and the χ^2_k distribution, respectively. These quantiles are all tabulated.

Example 26.3. Let $X_1, \ldots, X_n$ be i.i.d. $N(\theta, \sigma^2)$ where σ^2 is known. Then, by proposition A.4, $\bar{X} \sim N(\theta, \sigma^2/n)$ and thus

$$Q(X, \theta) = \frac{\sqrt{n}\,(\bar{X} - \theta)}{\sigma} \sim N(0, 1)$$

for all θ. Thus, Q is a pivot. We can find a confidence interval for θ by bracketing Q between $-z_{1-\alpha/2}$ and $z_{1-\alpha/2}$ and then solving the resulting inequalities for θ. The resulting confidence interval is

$$\Big[\, \bar{X} - z_{1-\alpha/2}\,\frac{\sigma}{\sqrt{n}},\ \bar{X} + z_{1-\alpha/2}\,\frac{\sigma}{\sqrt{n}} \,\Big]$$

with level $1 - \alpha$. For $n = 16$ observations, with $\sigma = 2$ and $\bar{x} = 10.3$, the 95% interval is $10.3 \pm 1.96 \cdot 0.5 = [9.32, 11.28]$. This is the same interval as we discussed above.

If σ^2 is unknown, this pivot is not available. If we replace σ by the sample standard deviation S, we get a new pivot and we have to consider the t distribution from appendix A.

Example 26.4. Let $X_1, \dots, X_n$ be i.i.d. $N(\mu, \sigma^2)$, where both parameters are unknown, and let S^2 be the sample variance. By proposition A.4, the two values

$$Z = \frac{\sqrt{n}\,(\bar{X} - \mu)}{\sigma} \sim N(0,1) \qquad \text{and} \qquad V = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$$

are independent. By definition A.5, the ratio

$$T = \frac{Z}{\sqrt{V/(n-1)}} = \frac{\sqrt{n}\,(\bar{X} - \mu)}{S}$$

is distributed as t_{n-1} and, since the unknown σ cancels, T is a pivot for μ. The *t-interval*

$$\Big[\, \bar{X} - t_{n-1,1-\alpha/2}\,\frac{S}{\sqrt{n}},\ \bar{X} + t_{n-1,1-\alpha/2}\,\frac{S}{\sqrt{n}} \,\Big],$$

with level $1-\alpha$, can be obtained by bracketing the pivot as above. For the data from example 26.3, with sample standard deviation $s = 2.1$, we can use the table to find $t_{15,0.975} = 2.131$ and the 95% interval $10.3 \pm 2.131 \cdot 2.1/4 = [9.18, 11.42]$. The wider interval is the price we pay for not knowing σ. As n increases, this difference decreases, since t_{n-1} converges to the standard normal distribution.

Example 26.5. In the same setting, $V = (n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$ is a pivot for σ^2. If we bracket this value between $\chi^2_{n-1,\alpha/2}$ and $\chi^2_{n-1,1-\alpha/2}$ and then solve for σ^2, the order of the endpoints is reversed, and we obtain the confidence interval

$$\Big[\, \frac{(n-1)S^2}{\chi^2_{n-1,1-\alpha/2}},\ \frac{(n-1)S^2}{\chi^2_{n-1,\alpha/2}} \,\Big]$$

for σ^2 with level $1-\alpha$. For $n = 16$ and $s^2 = 4.41$ the table gives $\chi^2_{15,0.025} = 6.262$ and $\chi^2_{15,0.975} = 27.488$, and thus the 95% interval is given by

$$[66.15/27.488,\ 66.15/6.262] = [2.41, 10.56].$$

This interval is far from symmetric around the estimate 4.41.

If we take the square root of both endpoints, the interval $[1.55, 3.25]$ for σ has the same coverage probability: applying a strictly increasing function to both endpoints does not change the coverage probability, whereas applying a decreasing function swaps the endpoints.

26.3 Inverting a Test

A level-α test of $H_0\colon \theta = \theta_0$ decides whether a single value θ_0 is compatible with the data; a confidence interval collects all the values which are. The following lemma, using the notions of critical region and level from definitions 22.2 and 22.5, makes this correspondence exact, in both directions.

Lemma 26.6. *For every $\theta_0 \in \Theta$, let $A(\theta_0)$ be the acceptance region of a level-α test of $H_0\colon \theta = \theta_0$, that is the set of samples for which H_0 is not rejected. Then*

$$C(X) = \{\theta_0 \in \Theta \mid X \in A(\theta_0)\},$$

is a confidence set for θ with level $1-\alpha$. Conversely, if $C(X)$ is a confidence set with level $1-\alpha$, then for each $\theta_0 \in \Theta$ the test which rejects $H_0\colon \theta = \theta_0$ if and only if $\theta_0 \notin C(X)$ has level α.

Proof. By the definition of $C(X)$, for every θ_0 and every sample we have $\theta_0 \in C(X)$ if and only if $X \in A(\theta_0)$. Let θ be the true value. Then, for $\theta_0 = \theta$, under $\mathbb{P}_\theta$ the null hypothesis $H_0 \colon \theta_0 = \theta$ is true and the test with acceptance region $A(\theta)$ rejects with probability at most α. We have

$$\mathbb{P}_\theta\big(\theta \in C(X)\big) = \mathbb{P}_\theta\big(X \in A(\theta)\big) = 1 - \mathbb{P}_\theta\big(X \notin A(\theta)\big) \geq 1 - \alpha.$$

Since θ was arbitrary, $C(X)$ is a confidence set with level $1-\alpha$. For the converse statement, the test for $H_0 \colon \theta = \theta_0$ rejects if and only if $\theta_0 \notin C(X)$. Thus, under $\mathbb{P}_{\theta_0}$, this event has probability $1 - \mathbb{P}_{\theta_0}\big(\theta_0 \in C(X)\big) \leq \alpha$ and the test has level α. This completes the proof. □

The lemma is a change of viewpoint, rather than a new result: the event "θ_0 is not rejected by X" can be read both as a test (with θ_0 fixed and X random) and as a confidence set (with X fixed and θ_0 varying). As an illustration, the size-α two-sided test from example 22.7 for the mean of an i.i.d. $N(\theta, \sigma^2)$ sample with known σ^2 rejects $H_0 \colon \theta = \theta_0$ if $|\bar{X} - \theta_0| > z_{1-\alpha/2}\,\sigma/\sqrt{n}$ and the values not rejected by the test form the interval from example 26.3. Similarly, by inverting the one-sided test from example 22.6, we find a one-sided confidence interval. We will consider this type of interval again in section 26.5. Finally, in exercise 26.4, the same approach is repeated with unknown σ. In practice, the duality is most often used in the opposite direction: a value θ_0 is rejected at level α if it does not belong to the $1-\alpha$ interval.

26.4 Approximate Intervals from the MLE

Exact pivots only exist for special models. For every regular model, however, corollary 17.3 provides an approximate pivot: the standardised error $(\hat{\theta}_n - \theta)/\operatorname{se}(\hat{\theta}_n)$ of the maximum likelihood estimator, with $\operatorname{se}(\hat{\theta}_n) = 1/\sqrt{\mathcal{I}_n(\hat{\theta}_n)}$, converges in distribution to $N(0,1)$ whatever the value of θ. Solving the event $|(\hat{\theta}_n - \theta)/\operatorname{se}(\hat{\theta}_n)| \leq z_{1-\alpha/2}$ for θ gives the following interval.

Definition 26.7. Let $\hat{\theta}_n$ be the maximum likelihood estimator and $\operatorname{se}(\hat{\theta}_n) = 1/\sqrt{\mathcal{I}_n(\hat{\theta}_n)}$ its standard error. The *Wald interval* with level $1-\alpha$ is given by

$$\big[\,\hat{\theta}_n - z_{1-\alpha/2}\,\operatorname{se}(\hat{\theta}_n),\ \hat{\theta}_n + z_{1-\alpha/2}\,\operatorname{se}(\hat{\theta}_n)\,\big].$$

The level is justified only asymptotically, as the following result shows.

Proposition 26.8. *With the same assumptions as in corollary 17.3, the coverage probability of the Wald interval converges to $1-\alpha$ as $n \to \infty$, for all $\theta \in \Theta$.*

Proof. Let $Z_n = (\hat{\theta}_n - \theta)/\operatorname{se}(\hat{\theta}_n)$ and $z = z_{1-\alpha/2}$. The event that the interval contains θ is the event $|\hat{\theta}_n - \theta| \leq z\,\operatorname{se}(\hat{\theta}_n)$. This event is the same as $-z \leq Z_n \leq z$. From corollary 17.3 we know that $Z_n \xrightarrow{\mathrm{d}} Z$ with $Z \sim N(0,1)$. Writing F_n for the distribution function of Z_n, for every $\varepsilon > 0$ we have

$$\begin{aligned}
\mathbb{P}_\theta(-z \leq Z_n \leq z) &\geq \mathbb{P}_\theta(-z < Z_n \leq z) = F_n(z) - F_n(-z),\\
\mathbb{P}_\theta(-z \leq Z_n \leq z) &\leq \mathbb{P}_\theta(-z - \varepsilon < Z_n \leq z) = F_n(z) - F_n(-z-\varepsilon).
\end{aligned}$$

Since Φ is continuous everywhere, definition A.13 gives $F_n(t) \to \Phi(t)$ for every t, and thus the lower bound converges to $\Phi(z) - \Phi(-z) = 1 - \alpha$ and the upper bound converges to $\Phi(z) - \Phi(-z - \varepsilon)$ as $n \to \infty$. Letting $\varepsilon \downarrow 0$ and using the continuity of Φ again, we find that $\mathbb{P}_\theta(-z \le Z_n \le z) \to 1 - \alpha$ as $n \to \infty$ and thus the coverage probability converges to $1 - \alpha$. This completes the proof. □

The Wald interval, for 95%, is the interval $\hat{\theta}_n \pm 1.96 \operatorname{se}(\hat{\theta}_n)$. This is the interval used in lectures 12 and 18. Its coverage is only guaranteed in the limit, and section 17.5 lists the cases to be aware of.

Example 26.9. Let $X_1, \ldots, X_n$ be i.i.d. Bernoulli with success probability $\theta \in (0, 1)$. The MLE is $\hat{\theta}_n = \bar{X}$ by example 5.3 and the Fisher information is $\mathcal{I}(\theta) = 1/(\theta(1-\theta))$ by example 11.5. Thus the Wald interval is $\bar{X} \pm z_{1-\alpha/2}\sqrt{\bar{X}(1-\bar{X})/n}$. For 37 successes in $n = 100$ trials we have $\bar{x} = 0.37$, the standard error is $\sqrt{0.37 \cdot 0.63/100} = 0.048$, and the 95% interval is $0.37 \pm 1.96 \cdot 0.048 = [0.275, 0.465]$. If θ is close to 0 or 1, or if n is small, the coverage of this interval may be less than $1 - \alpha$, since the normal approximation to $\bar{X}$ is poor in these cases and the formula gives the standard error zero when $\bar{X}$ is 0 or 1.

26.5 One-Sided Intervals and the Choice of Interval

Sometimes it is only important to consider one direction of a confidence interval, for example when we want to show that a failure rate is less than or equal to some value. A *lower confidence bound* for θ with level $1 - \alpha$ is a statistic $L(X)$ such that $\mathbb{P}_\theta(L(X) \le \theta) \ge 1 - \alpha$ for all θ. The corresponding *one-sided confidence interval* is then $[L(X), \infty)$. An upper confidence bound is defined symmetrically. In the pivot method, a one-sided interval can be obtained by considering the single inequality $Q(X, \theta) \le b$, where b is the $(1 - \alpha)$-quantile of the pivot. For example, for the normal distribution with known variance, this gives the lower bound $\bar{X} - z_{1-\alpha}\,\sigma/\sqrt{n}$ as promised in section 26.3, where $z_{1-\alpha}$ is used instead of $z_{1-\alpha/2}$ since all of the error probability is allocated to one side. In the context of the pivot method, any pair of values $a < b$ with $\mathbb{P}(a \le Q \le b) = 1 - \alpha$ defines a valid two-sided interval. Here we use the *equal-tailed* choice, *i.e.* a and b are the $(\alpha/2)$- and $(1 - \alpha/2)$-quantiles, respectively, which the tables give. For the z and t pivots this gives the shortest interval, but for the chi-squared pivot of example 26.5 it does not. Exercise 26.2 shows an example where the shortest interval puts all of the error probability in one tail of the pivot. In lecture 29 we will see the same choice of interval again, when we consider credible intervals.

Summary.

- A confidence interval covers the true parameter with probability at least $1-\alpha$ for all θ. The interval is random, the parameter is fixed, and the level describes the procedure, not the individual interval.
- An exact interval can be obtained by bracketing the pivot between two quantiles and solving for θ. For normal samples this can be used to derive the z-, t- and chi-squared intervals.
- Tests and confidence sets are dual: the values θ_0 not rejected by a family of level-α tests form a $1-\alpha$ confidence set, and conversely.
- The Wald interval $\hat{\theta}_n \pm z_{1-\alpha/2}\,\mathrm{se}(\hat{\theta}_n)$ asymptotically covers $1-\alpha$ for all regular models, but the finite-sample coverage may be lower.
- One-sided intervals use $z_{1-\alpha}$ instead of $z_{1-\alpha/2}$. For two-sided intervals the equal-tailed choice is the norm.

Exercise 26.1. Let $X_1, \ldots, X_n$ be i.i.d. exponential with rate $\theta > 0$ and let $T = \sum_{i=1}^n X_i$. In exercise 13.1 we have seen that $2\theta T \sim \chi^2_{2n}$.

1. Explain why $Q = 2\theta T$ is a pivot for θ and derive the equal-tailed confidence interval for θ with level $1-\alpha$.

2. For $n = 10$ observations $\sum_i x_i = 25.0$, compute the 95% interval, using the values $\chi^2_{20,0.025} = 9.591$ and $\chi^2_{20,0.975} = 34.170$.

3. The MLE for the rate is $\hat{\theta} = 1/\bar{X}$ and the standard error is $\hat{\theta}/\sqrt{n}$, as shown in example 17.4. Compute the Wald interval from definition 26.7 for the given data and compare the two intervals.

Exercise 26.2. Let $X_1, \ldots, X_n$ be i.i.d. uniformly distributed on the set $[0, \theta]$ and let $M = \max_i X_i$.

1. Using the density of M from example 8.9, show that $\mathbb{P}_\theta(M/\theta \leq u) = u^n$ for $0 \leq u \leq 1$ and conclude that M/θ is a pivot.

2. Show that $[M,\ M/\alpha^{1/n}]$ is a confidence interval for θ with level $1-\alpha$ and compute the interval for $n = 5$, $\max_i x_i = 3.2$ and $\alpha = 0.05$.

3. Among all intervals $[M/b,\ M/a]$ with $0 < a < b \leq 1$ and $\mathbb{P}_\theta(a \leq M/\theta \leq b) = 1-\alpha$, show that the interval from the previous part is the shortest. (Hint: think about how the length of the interval changes when b is moved and a is adjusted to maintain the same coverage.)

Exercise 26.3. Let $X_1, \ldots, X_n$ be i.i.d. Poisson with mean $\theta > 0$. Then the MLE is $\hat{\theta}_n = \bar{X}$ and the Fisher information is $\mathcal{I}(\theta) = 1/\theta$ from example 11.6.

1. Find the Wald interval for θ with level $1-\alpha$ and determine the interval for level 95% when $n = 20$ is the number of counts with $\sum_i x_i = 30$.

2. In exercise 17.3 we have found the asymptotic distribution of the MLE $\hat{p}_0 = e^{-\bar{X}}$ for $p_0 = \mathbb{P}_\theta(X_1 = 0) = e^{-\theta}$. Use this distribution to find an approximate confidence interval for p_0 and work out the interval for the given data.

3. Another interval for p_0 can be found by applying the decreasing function $g(\theta) = e^{-\theta}$ to the boundaries of the interval from the first part. Show that this interval also has asymptotic coverage $1 - \alpha$, compute the interval for the given data and comment on the differences between the two intervals.

Exercise 26.4. Let $X_1, \ldots, X_n$ be i.i.d. $N(\mu, \sigma^2)$, where both parameters are unknown. For $\mu_0 \in \mathbb{R}$ consider the test which rejects $H_0 \colon \mu = \mu_0$ if $\sqrt{n}\,|\bar{X} - \mu_0|/S > t_{n-1,1-\alpha/2}$.

1. Show that the test has size α, independent of the value of σ^2.

2. Using lemma 26.6, determine the confidence set of values μ_0 which are not rejected. Check that your confidence set coincides with the t-interval from example 26.4.

3. For the data given in example 26.4, can you decide whether $H_0 \colon \mu = 9$ and $H_0 \colon \mu = 10$ should be rejected at level 5%, without having to compute the test statistic?

4. Now consider the one-sided test which rejects $H_0 \colon \mu = \mu_0$ if $\sqrt{n}\,(\bar{X} - \mu_0)/S > t_{n-1,1-\alpha}$. Invert this test and describe the resulting confidence set.

Lecture 27

Coverage and Level by Simulation

This is the R session for week 9 of the course. In lectures this week we have seen Wilks' theorem and confidence intervals. Both come with a stated probability: a 95% confidence interval covers the true parameter value with probability 0.95, and the likelihood-ratio test with a critical value from the chi-squared distribution has level 0.05, approximately. In this session we check both statements by simulation. The written reference for the R commands is in appendix B.

27.1 Coverage by Simulation

We first use the z-interval from example 26.3, with the numerical values $n = 16$ and $\sigma = 2$ (known) from that example, so that the 95% interval is $\bar{X} \pm 1.96 \cdot 0.5$. By definition 26.1 of the level 0.95, the random interval contains the fixed value θ in 95% of all samples. To verify this, we consider $\theta = 10$ and compute the interval for 100 samples. Before you execute the code, answer the following questions: (1) How many of the 100 intervals will not cover the value 10? (2) Will all intervals have the same width? (3) Will the misses be clustered or evenly distributed among the 100 samples?

```
theta <- 10
sigma <- 2
n     <- 16
z_interval <- function() {
  x <- rnorm(n, mean = theta, sd = sigma)
  mean(x) + c(-1, 1) * qnorm(0.975) * sigma / sqrt(n)
}
set.seed(2026)
ci <- replicate(100, z_interval())
covers <- ci[1, ] <= theta & theta <= ci[2, ]
sum(covers)
which(!covers)
```

```
## [1] 95
## [1]  1 29 34 63 83
```

As in lecture 12, `covers` is a vector of `TRUE` and `FALSE`, one for each interval, and the command `sum` can be used to count the `TRUE` values. In our example, 95

out of 100 intervals cover the truth, and the misses are for samples 1, 29, 34, 63 and 83. The exact number 95 is a coincidence of the seed: the number of intervals which cover the truth is binomially distributed with standard deviation 2.2, so 92 or 98 would have been just as unremarkable.

Figure 27.1 shows each interval as a horizontal line, with the intervals which cover the truth in grey and the misses in black. The script `R/S27-coverage.R` shows how the command `segments` can be used to create this plot.

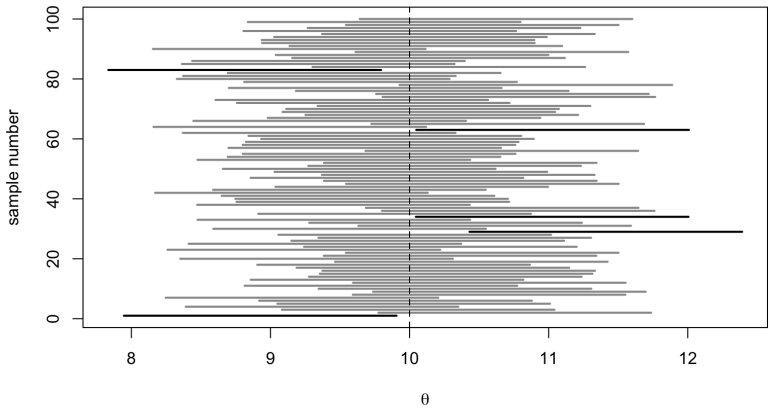


Figure 27.1: One hundred 95% confidence intervals for the mean of a $N(\theta, 4)$ sample of size $n = 16$, with true value $\theta = 10$ (dashed line). The grey intervals cover the truth, the black intervals do not.

The intervals in the figure all have the same width, since σ is known: the intervals move, while the dashed line stays fixed. The misses are scattered throughout the figure, since the samples are independent. If we had only observed sample 29, the interval $[10.43, 12.39]$ would have looked like any other, and there would have been no indication in the data that this was a miss. For this reason, the statement "θ is in the interval $[10.43, 12.39]$ with probability 0.95" is wrong: the value 10 is either in the interval or it is not, and in this instance it is not. The probability refers to the procedure, not to the computed interval. In the script `R/S27-coverage.R` the experiment is repeated with 10000 intervals, giving the coverage 0.947, which is within simulation error of 0.95. The interval is exact, and the simulation confirms this.

27.2 Wilks' Theorem in Action

We now turn from an exact statement to an approximate one. Assume that the variance of the normal sample is not known, so that the model has the two parameters θ and σ^2, and consider the likelihood-ratio test of $H_0 \colon \theta = \theta_0$, where H_0 leaves σ^2 free. By example 5.4, the log-likelihood is maximised over all parameters at $\hat{\sigma}^2 = \frac{1}{n}\sum_i (X_i - \bar{X})^2$, and the same computation with θ_0 in place of $\bar{X}$ gives the maximum $\hat{\sigma}_0^2 = \frac{1}{n}\sum_i (X_i - \theta_0)^2$ under H_0. In both cases the maximised log-likelihood equals $-\frac{n}{2}\log(2\pi\hat{\sigma}^2) - \frac{n}{2}$, with $\hat{\sigma}_0^2$ in place of $\hat{\sigma}^2$

under H_0, and thus by (25.1) we find

$$W = n \log \frac{\sum_{i=1}^{n}(X_i - \theta_0)^2}{\sum_{i=1}^{n}(X_i - \bar{X})^2}.$$

By Wilks' theorem (theorem 25.3), with $r = 1$ as in lecture 25, the distribution of W under H_0 converges to χ^2_1 as $n \to \infty$. In the case of known variance from lecture 25 the chi-squared distribution was exact, but here it is only an approximation, and we need to check how good it is for the n we have. We simulate W under H_0 for $n = 10$ and $n = 100$, using $\theta_0 = 10$ and the value $\sigma = 2$ from the first section, and we reject if W exceeds the critical value 3.841 of χ^2_1, as if the theorem were exact. Before you run the code, predict the two rejection rates. Will the rate for $n = 10$ be above or below 0.05?

```
theta0 <- 10
wilks <- function(n) {
  x <- rnorm(n, mean = theta0, sd = sigma)
  n * log(sum((x - theta0)^2) / sum((x - mean(x))^2))
}
set.seed(2026)
w10  <- replicate(10000, wilks(10))
w100 <- replicate(10000, wilks(100))
crit_w <- qchisq(0.95, df = 1)
c(n10 = mean(w10 > crit_w), n100 = mean(w100 > crit_w))
```

```
##    n10   n100
## 0.0745 0.0516
```

The rejection rate for $n = 100$ is 0.052, which is within simulation error of the nominal level. The rejection rate for $n = 10$ is 0.075: the test with nominal level 0.05 has type I errors half again as often as it should. Figure 27.2 shows why. In the quantile plots of the bottom row, made in the same way as the normal quantile plots of lecture 12, the points for $n = 10$ rise above the line from about 2 onwards, and thus the upper tail of W is heavier than that of the limit.

The script `R/S27-wilks.R` also computes the 0.95-quantile of the simulated values of W, which is the threshold that does give size 0.05: for $n = 10$ the threshold needs to be about 4.65 instead of 3.84, while for $n = 100$ the simulated quantile 3.90 is close to the chi-squared value. The error carries over to intervals: by lemma 26.6, the confidence interval obtained by inverting the test with critical value 3.84 covers the truth in only about 93% of all samples of size 10. For this model the exact distribution is known: exercise 25.1 uses the t distribution of appendix A to compare the two tests, and lecture 31 builds the exact test. In general, an asymptotic level is only guaranteed in the limit. An exact test should be used if one is available, and if not, a simulation under H_0 can be used to determine the actual size at a given n.

27.3 A Task to Take Home

Repeat the picture in figure 27.1 for the t-interval from example 26.4, with $n = 5$, `sigma` replaced by `sd(x)` and `qnorm(0.975)` replaced by `qt(0.975, df = 4)`.

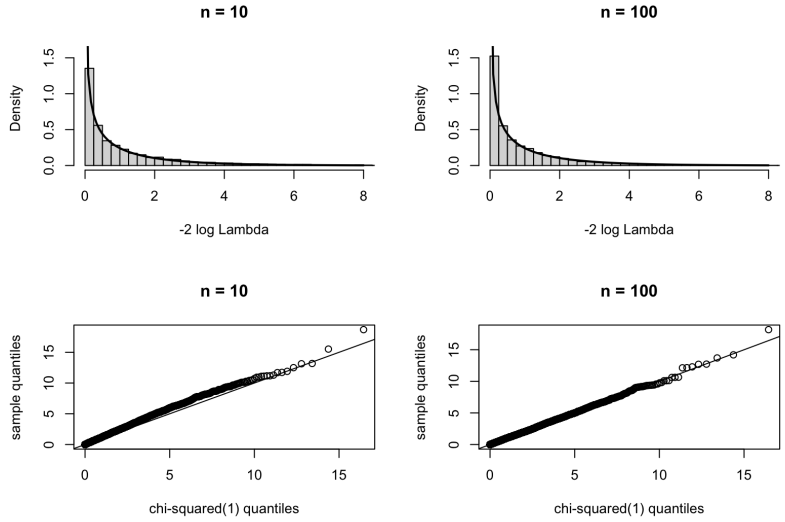


Figure 27.2: The likelihood-ratio statistic W for a normal mean with unknown variance, simulated 10000 times under H_0 for $n = 10$ (left) and $n = 100$ (right). Top: histograms with the density of the chi-squared distribution with one degree of freedom (truncated at 8). Bottom: the sorted values of the test statistic against the quantiles of the chi-squared distribution with one degree of freedom, with the identity line.

Estimate the long-run coverage from 10000 samples. Then repeat the experiment with `qnorm(0.975)` in place of the t quantile, and comment on your result.

Before the next session, you should type the commands used in this session, and should check that you get the same results as the text.

Summary.

- The written reference for the commands used in this session is appendix B.
- The confidence level is a property of the procedure: of 100 simulated 95% intervals, 95 contained the truth, and the misses looked like the other intervals. A computed confidence interval either contains the parameter or it does not.
- For the normal mean with unknown variance, the likelihood-ratio statistic is only approximately chi-squared: for $n = 10$ the chi-squared critical value 3.84 gives size 0.075 instead of 0.05, and the simulated 0.95-quantile is about 4.65. For $n = 100$ the approximation is good.
- If an exact test or interval is available, this should be used. If not, simulation can be used to determine the actual size or coverage at the given n.

Lecture 28

Bayesian Inference: Priors and Posteriors

So far the parameter θ has been a fixed but unknown number, and a sentence such as "θ is probably close to 0.3" had no meaning. In this lecture and in lecture 29 we will see that the parameter itself can be given a probability distribution: the *prior*, chosen before the data are seen, and the *posterior*, obtained after the data are seen. (We will see an update in pictures in lecture 30.)

28.1 Prior, Likelihood and Posterior

Since θ is now a random variable, the model density is a conditional density given the parameter, and in the Bayesian lectures we write it as $f(x|\theta)$, where $x = (x_1, \ldots, x_n)$ is the whole data set, so that $f(x|\theta) = \prod_i f(x_i|\theta)$ is the likelihood $L(\theta)$ of lecture 4.

Definition 28.1. Let θ be a random variable with values in Θ and density $\pi(\theta)$, and assume that given θ the data $X = (X_1, \ldots, X_n)$ have joint density $f(x|\theta)$. Then the density π is called the *prior density* (or prior) of θ and the conditional density $\pi(\theta|x)$ of θ given $X = x$ is called the *posterior density* (or posterior). The parameters in the prior are called *hyperparameters*.

From now on, unless otherwise specified, π will denote a prior or posterior density. In lecture 19, we used the same letter to denote the mixture weight for a two-component model.

The posterior density can be found using Bayes' theorem as given in proposition A.20 in appendix A, where the parameter plays the role of A and the data play the role of B.

Lemma 28.2. *In the situation of definition 28.1, the posterior density is given by*

$$\pi(\theta|x) = \frac{f(x|\theta)\,\pi(\theta)}{m(x)}, \qquad \text{where } m(x) = \int_\Theta f(x|\vartheta)\,\pi(\vartheta)\,\mathrm{d}\vartheta \tag{28.1}$$

is the marginal density of the data. The formula holds for all x with $m(x) > 0$.

Proof. The joint density of (θ, X) is given by $f(x|\theta)\,\pi(\theta)$. Integrating out θ gives the marginal density $m(x)$ of X. Dividing the joint density by $m(x)$ gives the conditional density of θ given $X = x$. This completes the proof. □

The denominator $m(x)$ makes the posterior integrate to one and does not involve θ; in practice we never compute it directly, but write

$$\pi(\theta|x) \propto f(x|\theta)\,\pi(\theta), \tag{28.2}$$

where $\propto$ means "equal up to a factor free of θ", and we try to recognise the right-hand side as the kernel of a known density, whose normalising constant then comes from table A.2. In words, the posterior is proportional to the likelihood times the prior.

The posterior is the complete Bayesian answer: a statement like "$\mathbb{P}(\theta > 0.3|x) = 0.95$" now has a meaning. Point estimates are summaries of the posterior (section 28.3), and intervals and tests will be considered in lecture 29.

28.2 Conjugate Priors

The rule (28.2) works best if the posterior distribution belongs to the same family of distributions as the prior distribution, so that the update amounts to no more than a change of the hyperparameters.

Definition 28.3. A family $\mathcal{P}$ of prior distributions is *conjugate* for a model $\big(f(x|\theta)\,\big|\,\theta \in \Theta\big)$, if for every prior in $\mathcal{P}$ and every data set x the posterior $\pi(\theta|x)$ also belongs to $\mathcal{P}$.

For exponential families, conjugate priors can be written down by inspection: the likelihood depends on the natural parameter η only through the exponent $\eta T_n - nK(\eta)$ (lemma 7.5), and a prior with an exponent of the same form reproduces itself.

Proposition 28.4. *Let $f(x|\eta) = h(x)\exp\big(\eta T(x) - K(\eta)\big)$ with $\eta \in \mathcal{N}$ be a one-parameter exponential family in its natural parametrisation, and for real numbers t_0 and n_0 let*

$$\pi_{t_0,n_0}(\eta) \propto \exp\big(t_0\eta - n_0 K(\eta)\big), \qquad \eta \in \mathcal{N},$$

whenever the right-hand side has a finite integral over $\mathcal{N}$. Then these priors are conjugate for the model: if the prior is π_{t_0,n_0}, the data $x_1, \dots, x_n$ have natural statistic $T_n = \sum_{i=1}^n T(x_i)$ and the marginal density satisfies $0 < m(x) < \infty$, then the posterior is $\pi_{t_0+T_n,\, n_0+n}$.

Proof. Using (28.2) and lemma 7.5 we find

$$\begin{aligned}\pi(\eta|x) &\propto \prod_{i=1}^n h(x_i)\,\exp\big(\eta T_n - nK(\eta)\big)\,\exp\big(t_0\eta - n_0 K(\eta)\big)\\ &\propto \exp\big((t_0+T_n)\eta - (n_0+n)K(\eta)\big),\end{aligned}$$

since $\prod_i h(x_i)$ does not depend on η. The right-hand side is the kernel of $\pi_{t_0+T_n,n_0+n}$. Since $m(x) < \infty$ by assumption, this kernel has a finite integral and thus is a density. This shows that the posterior is $\pi_{t_0+T_n,n_0+n}$. This completes the proof. □

The update adds the natural statistic to t_0 and the sample size to n_0: the prior acts like n_0 earlier observations with statistic total t_0. We now work the recipe for our simplest model in the original parameter.

Example 28.5. Let $X_1, \ldots, X_n$ be i.i.d. Bernoulli with success probability $\theta \in (0, 1)$ and let the prior be the Beta(a, b) distribution with $a, b > 0$. From table A.2 we know that this distribution has density

$$\pi(\theta) = \frac{\theta^{a-1}(1-\theta)^{b-1}}{B(a,b)}, \qquad 0 < \theta < 1,$$

and the mean of this distribution is $a/(a+b)$. If $T = \sum_i x_i$ is the number of successes in the sample, then the likelihood is $f(x|\theta) = \theta^T(1-\theta)^{n-T}$, as in example 4.2, and thus we have

$$\pi(\theta|x) \propto \theta^T(1-\theta)^{n-T} \cdot \theta^{a-1}(1-\theta)^{b-1} = \theta^{a+T-1}(1-\theta)^{b+n-T-1},$$

where we have omitted the constant $1/B(a,b)$ for simplicity. The right-hand side is the kernel of a Beta$(a+T, b+n-T)$ density and thus we have found the posterior distribution: the Beta distributions are conjugate for the Bernoulli model, and the update adds the successes to a and the failures to b. The posterior mean is given by

$$\mathbb{E}(\theta|x) = \frac{a+T}{a+b+n} = \frac{a+b}{a+b+n} \cdot \frac{a}{a+b} + \frac{n}{a+b+n} \cdot \bar{x}, \tag{28.3}$$

which is a weighted average of the prior mean $a/(a+b)$ and the sample proportion $\bar{x} = T/n$, *i.e.* the maximum likelihood estimate from example 5.3. The prior can be interpreted as having observed $a+b$ additional samples, of which a were successes. This *equivalent sample size* is a measure for how the hyperparameters should be chosen in practice (exercise 28.1). As $n \to \infty$, the weight of the prior decreases and with enough data the prior will be forgotten. For $a = b = 2$, $n = 10$ and $T = 7$, for example, the posterior distribution is Beta$(9, 5)$ with mean $9/14 \approx 0.643$, between the prior mean 0.5 and the sample proportion 0.7.

In terms of the log-odds η, the example is an instance of proposition 28.4: with the Jacobian $\theta(1-\theta)$ the Beta(a,b) prior becomes $\pi_{a,a+b}$. Using the same arguments, we can show that normal priors are conjugate for the normal mean with known variance. The cumulant function in this case is a quadratic function of η (exercise 7.5). The result is given here without proof, and the proof, based on completing the square, is given as exercise 28.3.

Proposition 28.6. *Let $X_1, \ldots, X_n$ be i.i.d. $N(\theta, \sigma^2)$ with known variance σ^2 and let the prior be $\theta \sim N(m, \tau^2)$. Then the posterior distribution is $N(m_n, \tau_n^2)$ where*

$$\frac{1}{\tau_n^2} = \frac{1}{\tau^2} + \frac{n}{\sigma^2} \qquad \text{and} \qquad m_n = \frac{1/\tau^2}{1/\tau^2 + n/\sigma^2}\, m + \frac{n/\sigma^2}{1/\tau^2 + n/\sigma^2}\, \bar{x}.$$

The reciprocal of a variance is called a *precision*, and in these terms the proposition is easy to remember: precisions add, and the posterior mean is the precision-weighted average of the prior mean and $\bar{x}$. Again, the weight of the prior tends to zero as $n \to \infty$.

28.3 Point Estimates from the Posterior

If a single number is required for θ, the posterior mean, median and mode can all be used. If the errors are measured in terms of squared error, as in lecture 2, the posterior mean is the optimal choice.

Lemma 28.7. *Assume that the posterior distribution has finite variance. Then the posterior mean $\hat{\theta}_B = \mathbb{E}(\theta|x)$ minimises the posterior expected squared error, i.e. for every number d we have*

$$\mathbb{E}((\theta - d)^2 \,|\, x) \geq \mathbb{E}((\theta - \hat{\theta}_B)^2 \,|\, x) = \mathrm{Var}(\theta|x),$$

and the equality holds only for $d = \hat{\theta}_B$.

Proof. Adding and subtracting $\hat{\theta}_B$ inside the square, as in the proof of lemma 2.7, we get

$$\mathbb{E}((\theta - d)^2 \,|\, x) = \mathbb{E}((\theta - \hat{\theta}_B)^2 \,|\, x) + 2(\hat{\theta}_B - d)\,\mathbb{E}(\theta - \hat{\theta}_B|x) + (\hat{\theta}_B - d)^2,$$

where all expectations are taken with respect to the posterior. The middle term vanishes, because $\mathbb{E}(\theta|x) = \hat{\theta}_B$, and the last term is non-negative and zero only for $d = \hat{\theta}_B$. This completes the proof. □

The estimator $\hat{\theta}_B$ is called the *Bayes estimator* (under squared error) for θ. The lemma is a statement about fixed data. For fixed θ, and varying data, exercise 28.4 shows that the Bernoulli $\hat{\theta}_B$ is biased but consistent, and that neither the Bernoulli estimator nor $\bar{X}$ has smaller mean squared error for all θ.

28.4 Flat, Improper and Jeffreys Priors

Without prior knowledge to encode, the obvious choice is the *flat prior* which is a constant density on Θ. This prior does not prefer any value of θ. For the Bernoulli model this prior is $\mathrm{Beta}(1, 1)$ and from example 28.5 we know that the posterior mean is $(T + 1)/(n + 2)$, which is close to $\bar{x}$ but is shifted towards $1/2$. On an unbounded parameter space a constant density has infinite integral; a prior with infinite integral is called *improper*. Nevertheless, this prior is still in use today: if $f(x|\theta)\,\pi(\theta)$ has finite integral, it is normalised and used as the posterior. The finiteness of the integral must be checked in each case (exercise 28.2). For the normal mean, the flat prior on $\mathbb{R}$ is the limit $\tau \to \infty$ in proposition 28.6 and in this case leads to the proper posterior $N(\bar{x}, \sigma^2/n)$.

The flat prior has a more basic defect: flatness depends on the parametrisation. If θ is uniform on $(0, 1)$, then $\psi = \theta^2$ has density $1/(2\sqrt{\psi})$ by proposition A.19, and a prior which is meant to express no knowledge about θ expresses some knowledge about θ^2. The Fisher information provides a parametrisation-free rule, since proposition 11.9 shows how it transforms.

Definition 28.8. In the situation of section 11.1, the *Jeffreys prior* for θ is given by

$$\pi_J(\theta) \propto \sqrt{\mathcal{I}(\theta)}, \qquad \theta \in \Theta,$$

where $\mathcal{I}(\theta)$ is the Fisher information of an observation.

This prior is justified by the following invariance property, which the flat prior lacks.

Proposition 28.9. *Let $\psi = g(\theta)$ where $g\colon \Theta \to g(\Theta)$ is a twice continuously differentiable bijection with $g'(\theta) \neq 0$ for all θ and let $h = g^{-1}$.*

1. *If θ has the flat prior on Θ, then ψ has the density $|h'(\psi)|$ on $g(\Theta)$. The density is flat only if g is affine.*

2. *If θ has the Jeffreys prior $\sqrt{\mathcal{I}_\theta(\theta)}$, then ψ has the density $\sqrt{\mathcal{I}_\psi(\psi)}$. This is the Jeffreys prior computed using the ψ-parametrisation.*

Proof. Using the change-of-variable formula from proposition A.19, if a density π_θ for θ is given, then a density of $\pi_\psi(\psi) = \pi_\theta(h(\psi))\,|h'(\psi)|$ for ψ can be derived. The same substitution can be applied to improper densities. In the case of the flat prior $\pi_\theta \equiv 1$, we have $\pi_\psi(\psi) = |h'(\psi)|$, which is constant only if h, *i.e.* if g, is affine. For the Jeffreys prior we have $\pi_\theta(\theta) = \sqrt{\mathcal{I}_\theta(\theta)}$ and using proposition 11.9 and $h'(\psi) = 1/g'(\theta)$ for $\theta = h(\psi)$ we get

$$\pi_\psi(\psi) = \sqrt{\mathcal{I}_\theta(\theta)}\,|h'(\psi)| = \frac{\sqrt{\mathcal{I}_\theta(\theta)}}{|g'(\theta)|} = \sqrt{\frac{\mathcal{I}_\theta(\theta)}{g'(\theta)^2}} = \sqrt{\mathcal{I}_\psi(\psi)}.$$

This is the Jeffreys prior in the ψ-parametrisation. This completes the proof. □

The Jeffreys prior is a property of the model, rather than of the way it is labelled, and as such it can be proper or improper.

Example 28.10. For the Bernoulli model, from example 11.5 we know $\mathcal{I}(\theta) = 1/(\theta(1-\theta))$ and thus the Jeffreys prior is given by

$$\pi_J(\theta) \propto \theta^{-1/2}(1-\theta)^{-1/2}, \qquad 0 < \theta < 1.$$

This is the density of the Beta$(1/2, 1/2)$ distribution. The Jeffreys prior is proper and has more weight near the boundaries of the interval than the flat prior does. From example 28.5 we know that the posterior distribution is Beta$(T+1/2,\, n-T+1/2)$ and the mean of the posterior is $(T+1/2)/(n+1)$. This value is between the maximum likelihood estimate T/n and the estimate $(T+1)/(n+2)$ obtained using the flat prior. For large n, all three values coincide.

For location and scale models the Jeffreys prior is the same for every base density f_0.

Proposition 28.11. *Let f_0 be a fixed density such that the following models satisfy the conditions of section 11.1.*

1. *Consider the location model $f(x|\theta) = f_0(x-\theta)$ with $\theta \in \mathbb{R}$. The Fisher information in this model is constant as a function of θ and thus the Jeffreys prior is flat, $\pi_J(\theta) \propto 1$ on $\mathbb{R}$.*

2. *Consider the scale model $f(x|\sigma) = f_0(x/\sigma)/\sigma$ with $\sigma > 0$. The Fisher information in this model is $\mathcal{I}(\sigma) = c/\sigma^2$ for a constant $c > 0$. Thus, the Jeffreys prior is $\pi_J(\sigma) \propto 1/\sigma$ on $(0, \infty)$.*

Both priors are improper.

Exercise 28.5 verifies both statements by writing the score as a function of the standardised observation. The normal mean with known variance is a location model, so its Jeffreys prior is flat. Similarly, the normal standard deviation with known mean is a scale model with $\mathcal{I}(\sigma) = 2/\sigma^2$.

Summary.

- The data can be used to update the prior density $\pi(\theta)$ to the posterior $\pi(\theta|x) \propto f(x|\theta)\,\pi(\theta)$. The normalising constant can be found by recognising the kernel of a known density.
- Priors are conjugate for a family if the posterior always belongs to the same family. For a one-parameter exponential family in its natural parametrisation, the priors $\exp(t_0\eta - n_0 K(\eta))$ are conjugate. Beta priors are conjugate for the Bernoulli model, and normal priors are conjugate for the normal mean with known variance.
- For the Beta-Bernoulli and normal-normal pairs, the posterior mean is a weighted average of the prior mean and the maximum likelihood estimate. As n increases, the prior weight decreases. The posterior mean is the Bayes estimator for θ under squared error.
- A flat prior is not invariant under reparametrisation, but the Jeffreys prior $\pi_J(\theta) \propto \sqrt{\mathcal{I}(\theta)}$ is. The Jeffreys prior is $\text{Beta}(1/2, 1/2)$ for the Bernoulli model, is flat for a location parameter, and is proportional to $1/\sigma$ for a scale parameter. An improper prior can only be used if the posterior is proper.

Exercise 28.1. A Bernoulli model is to be fitted with a $\text{Beta}(a, b)$ prior for the success probability θ.

1. Find a and b such that the prior mean is 0.3 and the equivalent sample size $a + b$ is 10. You can find the prior standard deviation in table A.2.

2. Find a and b such that the prior has mean 0.3 and standard deviation 0.1. What is the equivalent sample size for this prior?

3. For the prior in the first part of the question, a sample of size $n = 20$ has $T = 9$ successes. Determine the posterior distribution and the posterior mean. Write the posterior mean as a weighted average of the prior mean and $\bar{x}$.

Exercise 28.2. The *Haldane prior* for the Bernoulli success probability is the limit $a, b \to 0$ of the kernel $\theta^{a-1}(1-\theta)^{b-1}$, *i.e.* the function $\pi(\theta) \propto \theta^{-1}(1-\theta)^{-1}$ for $0 < \theta < 1$.

1. Show that this prior is improper.

2. Show that the formal posterior $\pi(\theta|x) \propto \theta^{T-1}(1-\theta)^{n-T-1}$ is a proper distribution if and only if $0 < T < n$. Identify the resulting distribution.

3. For $0 < T < n$, determine the posterior mean and compare it to the maximum likelihood estimate.

4. Describe what happens for $T = 0$.

Exercise 28.3. Using the following steps, prove proposition 28.6.

1. Show that the likelihood, as a function of θ, satisfies $f(x|\theta) \propto \exp\bigl(-n(\theta - \bar{x})^2/(2\sigma^2)\bigr)$.

2. Multiply the prior density with this function and show, by completing the square in θ, that
$$\pi(\theta|x) \propto \exp\Bigl(-\frac{(\theta - m_n)^2}{2\tau_n^2}\Bigr)$$
where m_n and τ_n^2 are as in the proposition.

3. What happens to m_n and τ_n^2 when $\tau \to \infty$ with n fixed? And when $n \to \infty$ with τ fixed?

Exercise 28.4. Let $X_1, \ldots, X_n$ be i.i.d. Bernoulli with success probability θ and let $\hat{\theta}_B = (a + T)/(a + b + n)$ with $T = \sum_i X_i$ be the Bayes estimator from example 28.5. Now fix θ and let $\hat{\theta}_B$ be an estimator as in lecture 2.

1. What are the bias and variance of $\hat{\theta}_B$? Show that the estimator is biased for every $\theta \neq a/(a + b)$, but consistent.

2. For $a = b = 1$ and $n = 10$, compare the mean squared errors $\mathrm{MSE}(\hat{\theta}_B)$ and $\mathrm{MSE}(\bar{X})$ for $\theta = 1/2$ and $\theta = 0.1$.

Exercise 28.5. Let f_0 be a fixed density as in proposition 28.11 and let $\ell_0(z) = \log f_0(z)$.

1. For the location model $f(x|\theta) = f_0(x - \theta)$, show that the score is $-\ell_0'(x - \theta)$ and thus that the Fisher information does not depend on θ.

2. For the scale model $f(x|\sigma) = f_0(x/\sigma)/\sigma$, show that the score is $-(1 + z\ell_0'(z))/\sigma$ where $z = x/\sigma$ and that $\mathcal{I}(\sigma) = c/\sigma^2$ where c does not depend on σ.

3. For the $N(0, \sigma^2)$ model with parameter σ, show that $\mathcal{I}(\sigma) = 2/\sigma^2$.

Lecture 29

Credible Intervals and Bayesian Testing

Lecture 28 ended with the posterior distribution as the complete Bayesian answer. In this lecture we answer the interval question of lecture 26 and the testing question of lectures 22 to 25 in posterior terms, and then set the two approaches side by side on the mean of a normal sample; lecture 30 shows a comparison of this kind in R.

We use the two conjugate pairs from lecture 28: the Beta posterior from example 28.5 for the Bernoulli distribution and, for a sample $X_1, \dots, X_n$ from the normal distribution $N(\theta, \sigma^2)$ with known variance and prior $\theta \sim N(m, \tau^2)$, the normal posterior $N(m_n, \tau_n^2)$ from proposition 28.6, with

$$\frac{1}{\tau_n^2} = \frac{1}{\tau^2} + \frac{n}{\sigma^2} \qquad \text{and} \qquad m_n = \tau_n^2\Big(\frac{m}{\tau^2} + \frac{n\bar{x}}{\sigma^2}\Big). \tag{29.1}$$

We consider the data set from example 26.3: $n = 16$ observations with $\sigma = 2$ and $\bar{x} = 10.3$. Assume that before the data were collected we expected the mean to be around 12 and we chose the prior $N(12, 1)$. Then the posterior precision is $1/1 + 16/4 = 5$ and the posterior mean is $m_n = 0.2 \cdot (12 + 4 \cdot 10.3) = 10.64$, *i.e.* the posterior distribution is $N(10.64, 0.2)$.

29.1 Credible Intervals

A confidence interval, as given in definition 26.1, is a random interval which covers the fixed parameter with a given probability. In contrast, in a Bayesian model the data are fixed after observation, the parameter is random and an interval estimate is a set which contains most of the posterior probability.

Definition 29.1. Let $\pi(\theta|x)$ be the posterior density of θ given the data x and let $\alpha \in (0, 1)$. An interval $[l, u]$ is a *credible interval* for θ with *credibility* $1 - \alpha$, if

$$\mathbb{P}(l \le \theta \le u|x) = \int_l^u \pi(\vartheta|x)\,\mathrm{d}\vartheta = 1 - \alpha.$$

The *equal-tailed* credible interval is the interval $[q_{\alpha/2}, q_{1-\alpha/2}]$, between the $\alpha/2$- and $(1 - \alpha/2)$-quantiles of the posterior distribution.

The probability in the definition is a probability about θ, for the given data set: the sentence "θ is in $[l, u]$ with probability 0.95" (which we had to forbid in

lecture 26) is the statement that a 95% credible interval makes. The trade-off is that this probability depends on the prior distribution.

Any interval which has posterior probability $1-\alpha$ works, and the equal-tailed interval is the usual choice, because the edges of this interval are quantiles. The alternative would be to find the shortest such set.

Definition 29.2. A *highest posterior density* (HPD) set with credibility $1-\alpha$ is the set

$$H_c = \{\, \theta \in \Theta \mid \pi(\theta|x) \geq c \,\},$$

where $c > 0$ is chosen such that $\mathbb{P}(\theta \in H_c|x) = 1-\alpha$.

The set H_c is the smallest set with posterior probability $1-\alpha$. Any time we want to exchange part of this set with a region where the density is below c, we need to include more length in the interval to make up for the lost probability. For a unimodal posterior distribution, H_c forms an interval, called the *HPD interval*, where the two boundaries have the same posterior density. For a symmetric, unimodal posterior distribution, this interval will be an equal-tailed interval (see exercise 29.1).

Example 29.3. If the posterior distribution is equal to the normal distribution $N(m_n, \tau_n^2)$, then the density is symmetric around m_n and decreases as it gets further away from this point. Thus, the HPD interval is the equal-tailed interval

$$\left[\, m_n - z_{1-\alpha/2}\,\tau_n,\ m_n + z_{1-\alpha/2}\,\tau_n \,\right]. \tag{29.2}$$

For the data set and prior used in the section, we have the prior $N(12,1)$ and $\tau_n = \sqrt{0.2} = 0.447$, and the 95% credible interval is $10.64 \pm 1.96 \cdot 0.447$, *i.e.* the interval $[9.76, 11.52]$. The credible interval is shifted towards the prior mean and is shorter than the 95% confidence interval $[9.32, 11.28]$ from example 26.3, because the prior provides additional information. In the case of the flat prior, $\tau \to \infty$ in section 28.4, the posterior distribution is $N(\bar{x}, \sigma^2/n)$ and the credible interval (29.2) coincides with the confidence interval $\bar{x} \pm z_{1-\alpha/2}\,\sigma/\sqrt{n}$. The reason for this coincidence is explained in section 29.3.

For a skewed posterior the HPD interval is shifted towards the mode, slightly shorter than the equal-tailed interval, and the numerical determination of this interval is the topic of exercise 29.2, which shows the difference between the two intervals for the skewed posterior $\mathrm{Beta}(9,5)$ in example 28.5. As n increases, the difference between the intervals decreases, because the posterior distribution approaches a symmetric normal distribution and the equal-tailed interval is then appropriate.

29.2 Bayesian Testing

In the Bayesian model a hypothesis $H_0 \colon \theta \in \Theta_0$, in the sense of definition 22.1, is an event concerning the random variable θ and thus has a probability before and after the data are seen. Testing does not require new theory.

Definition 29.4. Let $H_0 \colon \theta \in \Theta_0$ and $H_1 \colon \theta \in \Theta_1$ be hypotheses with positive prior probability. Then the *posterior probability* of H_0 is given by

$$\mathbb{P}(H_0|x) = \mathbb{P}(\theta \in \Theta_0|x) = \int_{\Theta_0} \pi(\vartheta|x)\,\mathrm{d}\vartheta.$$

Similarly, we can find the posterior probability of H_1. A *Bayesian test* rejects H_0 if $\mathbb{P}(H_0|x)$ falls below a threshold or, if both hypotheses are treated equally, chooses the hypothesis with the higher posterior probability.

The threshold plays the role of the level α of definition 22.5, but the two numbers are not comparable: α bounds the probability of a type I error over repeated samples, whereas the threshold bounds the posterior probability of H_0 for the data in hand. Any asymmetry between the hypotheses can be expressed by the prior.

Example 29.5. We consider the running data set, testing $H_0\colon \theta \leq 10$ against $H_1\colon \theta > 10$. Using the prior $N(12, 1)$, we find the posterior distribution to be $N(10.64, 0.2)$. Thus we have

$$\mathbb{P}(H_0|x) = \mathbb{P}(\theta \leq 10|x) = \Phi\Big(\frac{10 - 10.64}{0.447}\Big) = \Phi(-1.43) = 0.076,$$

and consequently $\mathbb{P}(H_1|x) = 0.924$. A symmetric decision between the hypotheses favours H_1. Using the rule with threshold 0.05, we would not have rejected H_0. For the flat prior case, the posterior is $N(10.3, 0.25)$ and we get $\mathbb{P}(H_0|x) = \Phi(-0.6) = 0.274$. This is a much less certain result than in the example above: the prior $N(12, 1)$, which had only 2.3% of its mass below 10, contributed substantially to the evidence against H_0. The same approach can be used to find the posterior probability of a hypothesis about a proportion: for the posterior Beta$(9, 5)$ in example 28.5, the posterior probability of $H_0\colon \theta \leq 1/2$ is the value of the Beta distribution function at $1/2$. This can be computed using any statistical package (in R, the function is `pbeta`), and the result is 0.133. Exercise 29.3 shows how to compute this integral manually.

A simple hypothesis $\theta = \theta_0$ has probability zero under every prior with a density, and thus for two simple hypotheses the prior must be a two-point distribution. The posterior in this case is then simple.

Definition 29.6. Let $H_0\colon \theta = \theta_0$ and $H_1\colon \theta = \theta_1$ be simple hypotheses and let the prior have probability p_0 for θ_0 and $p_1 = 1 - p_0$ for θ_1. Then the *prior odds* of H_0 against H_1 are p_0/p_1 and the *posterior odds* are $\mathbb{P}(H_0|x)/\mathbb{P}(H_1|x)$.

Using the result of Bayes' theorem, proposition A.20, we find $\mathbb{P}(H_j|x) = p_j\, f(x|\theta_j)/m(x)$ for $j = 0, 1$ and dividing these two expressions we find that the marginal density $m(x)$ cancels out. This gives

$$\frac{\mathbb{P}(H_0|x)}{\mathbb{P}(H_1|x)} = \frac{p_0}{p_1} \cdot \frac{f(x|\theta_0)}{f(x|\theta_1)}, \tag{29.3}$$

i.e. the posterior odds are the prior odds times the likelihood ratio. In this context, the likelihood ratio is also called the *Bayes factor* in favour of H_0. This is the inverse of the ratio considered in the Neyman–Pearson lemma, theorem 23.2. From the posterior odds r we can derive the posterior probabilities as $\mathbb{P}(H_0|x) = r/(1 + r)$.

Example 29.7. We consider the running data set again, this time to test the hypothesis $H_0\colon \theta = 10$ against the hypothesis $H_1\colon \theta = 11$. Since the likelihood of

observing a normal sample is $f(x|\theta) \propto \exp\bigl(-n(\bar{x}-\theta)^2/(2\sigma^2)\bigr)$ from exercise 28.3, we find the Bayes factor to be

$$\frac{f(x|\theta_0)}{f(x|\theta_1)} = \exp\Bigl(\frac{n}{2\sigma^2}\bigl((\bar{x}-\theta_1)^2 - (\bar{x}-\theta_0)^2\bigr)\Bigr) = \exp\Bigl(\frac{n(\theta_1-\theta_0)(\theta_0+\theta_1-2\bar{x})}{2\sigma^2}\Bigr).$$

Here we have $n = 16$, $\sigma = 2$ and $\bar{x} = 10.3$ and the exponent is $16 \cdot 1 \cdot 0.4/8 = 0.8$ and the Bayes factor is $e^{0.8} = 2.23$: the data are closer to the value 10 than to 11 and the data favour H_0 by a factor of approximately two. If the prior odds are 1, the posterior odds are 2.23 and $\mathbb{P}(H_0|x) = 2.23/3.23 = 0.69$. We see that the Bayes factor multiplies all prior opinions by the same factor, and in cases where the prior is very strong, it can be difficult to overturn the prior opinion. The situation in exercise 29.4 is more complicated, because the prior odds are not equal.

29.3 Comparing Frequentist and Bayesian Answers

Now we have two different sets of answers, for the same data, for the normal mean. We conclude by comparing the two sets.

The first difference is in the nature of the statements. A 95% confidence interval is a statement about a procedure: it says that for repeated samples, the interval covers the fixed truth in 95% of cases. A 95% credible interval is a statement about the parameter, given the data and the prior. Similarly, a p-value is computed assuming H_0 is true, whereas $\mathbb{P}(H_0|x)$ is the probability of the hypothesis given the data.

The second observation is that for the normal mean with known variance and a flat prior, both numbers coincide. Example 29.3 shows the two intervals coinciding, and the tests coincide as well: for $H_0\colon \theta \leq \theta_0$ the flat-prior posterior probability $\Phi\bigl(\sqrt{n}\,(\theta_0 - \bar{x})/\sigma\bigr)$ equals, by the symmetry of Φ, the p-value $1 - \Phi\bigl(\sqrt{n}\,(\bar{x}-\theta_0)/\sigma\bigr)$ of the one-sided z-test from example 22.6. That example tests the simple null $\theta = \theta_0$, but since $\mathbb{P}_\theta(\bar{X} \geq \bar{x})$ increases with θ, like the power function of the test, the supremum in definition 22.8 for the composite null $\theta \leq \theta_0$ is attained at θ_0 and gives the same p-value. For the running data set both values equal 0.274. This is the case because the flat-prior posterior density in θ has the same shape as the sampling density of $\bar{X}$ in $\bar{x}$. This is the reason why the misreadings in lectures 22 and 26 were so tempting: the forbidden sentence about the p-value is, for the simplest model, a true sentence about the flat-prior posterior, and only the interpretation needs to be changed.

The coincidence is only true for special models and priors. A genuine prior will move the Bayesian answer closer to the prior mean and will concentrate the answer around this value, as in the case of $N(12, 1)$ above. If the prior is correct, this is an advantage, but if the prior is wrong this effect will work against us. The frequentist guarantee from definition 26.1 applies for all θ and thus cannot be violated in this way. Exercise 29.5, however, shows that the frequentist coverage of a credible interval can be worse than 95% when θ is far from the prior mean.

As the sample size increases, the influence of the prior decreases. For the normal mean, equation (29.1) shows that the weight of the prior mean in m_n goes to zero and that $\tau_n^2 \leq \sigma^2/n$ so that both the posterior mean and the posterior standard deviation converge to the values they would have with flat prior. The same effect holds for all regular models. The general version of this

result is the Bayesian analogue of theorem 17.1: under the regularity conditions from definition 16.1, and for any continuous, positive prior density at the true value, the posterior distribution is approximately

$$N\Big(\hat{\theta}_n,\ \frac{1}{\mathcal{I}_n(\hat{\theta}_n)}\Big) \qquad \text{for large } n, \tag{29.4}$$

where $\hat{\theta}_n$ is the maximum likelihood estimator. We state this result without proof. The idea is that the log-posterior is $\ell(\theta) + \log \pi(\theta)$ up to a constant, and that the log-likelihood grows like n while $\log \pi(\theta)$ does not. From (29.4) we know that the equal-tailed credible interval is approximately the Wald interval $\hat{\theta}_n \pm z_{1-\alpha/2}\, \mathrm{se}(\hat{\theta}_n)$ of proposition 26.8, independently of the prior: for large sample size both approaches will determine the same interval with approximately frequentist coverage $1-\alpha$.

For a point null hypothesis against a two-sided alternative the two approaches can differ substantially. We do not pursue this here.

Summary.

- A credible interval with credibility $1-\alpha$ has posterior probability $1-\alpha$: this is a probability statement about θ for the given data, taking the prior into account. The equal-tailed interval is based on posterior quantiles, the HPD interval is the shortest one, and for symmetric, unimodal posteriors both intervals coincide.
- For the normal mean with known variance and normal prior, the $1-\alpha$ credible interval is $m_n \pm z_{1-\alpha/2}\, \tau_n$; for the flat prior this interval coincides with the confidence interval $\bar{x} \pm z_{1-\alpha/2}\, \sigma/\sqrt{n}$.
- A Bayesian test compares posterior probabilities of the hypotheses, or rejects H_0 if $\mathbb{P}(H_0|x)$ falls below a threshold. For two simple hypotheses, the posterior odds are the prior odds times the likelihood ratio, known as the Bayes factor.
- The level of a test and the confidence level of an interval are statements about the procedure applied to repeated samples, whereas the credibility of an interval is a statement about the parameter given one data set. For the normal mean with known variance and flat prior, the intervals, and the one-sided test and p-value, agree numerically while the statements do not have the same meaning.
- An informative prior will shift and shorten the Bayesian answer; as $n \to \infty$ the posterior is approximately $N(\hat{\theta}_n, 1/\mathcal{I}_n(\hat{\theta}_n))$ and then the credible and Wald intervals coincide.

Exercise 29.1. Let the posterior density $\pi(\theta|x)$ be continuous, strictly increasing on $(-\infty, \mu]$ and strictly decreasing on $[\mu, \infty)$.

1. Show that the HPD set H_c from definition 29.2 is an interval $[l, u]$ with $l < \mu < u$ and $\pi(l|x) = \pi(u|x)$.

2. Assume that the density is symmetric about μ, *i.e.* that $\pi(\mu + t|x) = \pi(\mu - t|x)$ for all t. Show that the HPD interval with credibility $1 - \alpha$ coincides with the equal-tailed interval $[q_{\alpha/2}, q_{1-\alpha/2}]$.

3. Now assume that the posterior density is continuous and strictly decreasing on $[0, \infty)$ and zero elsewhere. Show that the HPD set with credibility $1-\alpha$ is the one-sided interval $[0, q_{1-\alpha}]$ and that this interval is shorter than the equal-tailed interval.

Exercise 29.2. In example 28.5, where a Bernoulli sample with $n = 10$ and $T = 7$ successes was combined with the prior Beta$(2, 2)$ to obtain the posterior distribution Beta$(9, 5)$ with mean $9/14 = 0.643$, the quantiles of the Beta distribution had to be found numerically. Using any statistical package, and using the `qbeta` function in R, we find the 0.025- and 0.975-quantiles of Beta$(9, 5)$ to be 0.386 and 0.861.

1. Write down the equal-tailed 95% credible interval for θ and compare the mid-point of this interval to the posterior mean.

2. Show that the posterior density has its maximum at $\theta = 2/3$, *i.e.* that the posterior distribution is skewed with the mode above the mean.

3. The HPD interval for Beta$(9, 5)$, found numerically, is $[0.401, 0.874]$. Using definition 29.2, explain why this interval is shifted towards the mode, compared to the equal-tailed interval. Also explain why the difference between the two intervals decreases as n increases.

Exercise 29.3. Assume that we have observed two successes in three Bernoulli trials with success probability θ, and the prior distribution of θ is flat on the interval $(0, 1)$.

1. Determine the posterior distribution of θ.

2. Calculate the posterior probability of $H_0 \colon \theta \leq 1/2$ by integrating the posterior density. Also, calculate the posterior odds of H_0 against $H_1 \colon \theta > 1/2$.

3. The prior odds of H_0 against H_1 were 1. By what factor have the data changed these odds?

4. Write down the equations for the boundaries of the 95% equal-tailed credible interval and explain why these equations are best solved numerically.

Exercise 29.4. In the situation of example 22.6, a sample of size $n = 25$ from $N(\theta, 4)$ has mean $\bar{x} = 10.9$. We want to compare the simple hypotheses $H_0 \colon \theta = 10$ and $H_1 \colon \theta = 11$.

1. Compute the Bayes factor in favour of H_0.

2. Before the data were available, H_0 was three times as likely as H_1. Compute the posterior odds and the posterior probability of H_0.

3. In lecture 22 we have seen that the one-sided p-value 0.0122 can be found for the data. Compare your two numbers and explain why they can disagree, despite the fact that both values are against H_0.

4. What value must the prior odds in favour of H_0 have for the posterior probability of H_0 to be greater than one half?

Exercise 29.5. Let $X_1, \ldots, X_n$ be i.i.d. with $N(\theta, \sigma^2)$ where σ^2 is known, and let the prior be $N(m, \tau^2)$. Let $w = (1/\tau^2)/(1/\tau^2 + n/\sigma^2)$ be the weight of the prior mean in m_n. Consider the credible interval (29.2) to be a random interval, as introduced in lecture 26, where θ is fixed and the data are random.

1. Show that $\tau_n^2 = (1-w)\sigma^2/n$ and that under $\mathbb{P}_\theta$ the posterior mean satisfies $m_n - \theta \sim N\big(w(m-\theta),\, (1-w)^2\sigma^2/n\big)$.

2. Determine the coverage probability $\mathbb{P}_\theta\big(m_n - z_{1-\alpha/2}\tau_n \le \theta \le m_n + z_{1-\alpha/2}\tau_n\big)$ as a function of θ.

3. For $n = 16$, $\sigma = 2$, $\tau = 1$, $m = 12$ and $\alpha = 0.05$, determine the coverage probability for $\theta = 12$ and for $\theta = 10$ and discuss the result as $|\theta - m| \to \infty$.

4. Is the credible interval a confidence interval with level 0.95 in the sense of definition 26.1? What happens to the coverage probability as $n \to \infty$ with θ fixed?

Lecture 30

Confidence Versus Credibility

This is the R session for week 10. In lectures this week we have introduced priors, posteriors and credible intervals. In this session we compute a confidence interval and a credible interval for the same data, and we look at how a prior turns into a posterior as observations are added. As before, appendix B contains a written reference for the R commands used.

30.1 The Same Data, Two Intervals

We take the data of example 26.9, $T = 37$ successes in $n = 100$ Bernoulli trials, and compute two intervals for the success probability θ. The first is the Wald interval $\bar{x} \pm 1.96\sqrt{\bar{x}(1-\bar{x})/n}$ of that example. For the second we use the Bayesian model of lecture 28 with the flat prior: the posterior is Beta$(1+T,\, 1+n-T)$ by example 28.5, and the interval between its 2.5% and 97.5% quantiles is the equal-tailed credible interval with credibility 0.95 from definition 29.1. The command `qbeta` can be used to find these quantiles. Before you run the code, predict whether the credible interval will be wider or narrower than the Wald interval, and whether it will sit to the left or to the right of it.

```
n <- 100
total <- 37
xbar <- total / n
se   <- sqrt(xbar * (1 - xbar) / n)
xbar + c(-1, 1) * qnorm(0.975) * se
a <- 1 + total
b <- 1 + n - total
qbeta(c(0.025, 0.975), a, b)
```

```
## [1] 0.2753721 0.4646279
## [1] 0.2817646 0.4680688
```

The two intervals, $[0.275, 0.465]$ and $[0.282, 0.468]$, agree to within 0.01. The credible interval is slightly shifted towards 1/2, since the flat prior adds one success and one failure to the data. Although the numbers are close, the two statements are different. The Wald interval comes with the guarantee that the recipe used to compute it covers the truth in approximately 95% of samples, and it makes no statement about this particular interval. The credible interval means that, given the observations and the flat prior, the probability that θ lies

between 0.282 and 0.468 is 0.95. Thus the sentence which was wrong for the confidence intervals of section 27.1 is the correct interpretation here, since θ is a random variable in the Bayesian model. A different prior would give a different interval, and the numerical agreement is a property of the flat prior and of the sample size.

30.2 From Prior to Posterior

We now look at the update from example 28.5 one observation at a time. In the script `R/S30-posterior.R` we simulate 100 Bernoulli observations with $\theta = 0.3$. After k observations with s successes, the prior $\text{Beta}(a, b)$ has turned into the posterior $\text{Beta}(a+s,\ b+k-s)$, and we plot this posterior density for three priors: the flat prior $\text{Beta}(1, 1)$, the Jeffreys prior $\text{Beta}(1/2, 1/2)$ from example 28.10 and the misplaced prior $\text{Beta}(12, 3)$, with mean 0.8 and equivalent sample size 15. In the simulated sequence, the first five observations are all failures, there are 2 successes among the first 20 and 28 among all 100. Before you look at figure 30.1, sketch the panel for $k = 5$: which curves are steep, where are their peaks and how wide are the curves?

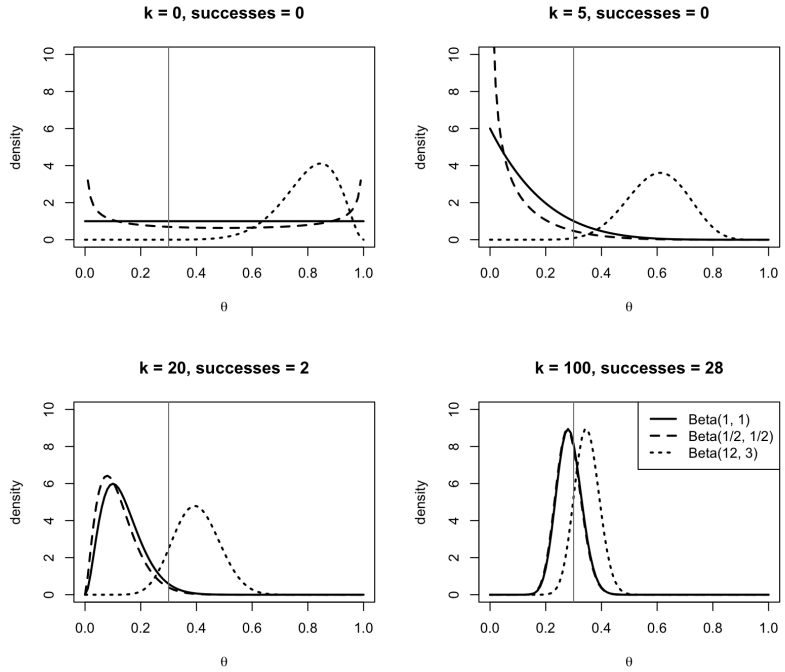


Figure 30.1: Posterior densities for the probability of success in a Bernoulli trial after $k = 0$, 5, 20 and 100 observations of one simulated sequence with $\theta = 0.3$ (grey vertical line), for the priors $\text{Beta}(1, 1)$ (solid), $\text{Beta}(1/2, 1/2)$ (dashed) and $\text{Beta}(12, 3)$ (dotted).

At $k = 0$ the three curves in figure 30.1 are the priors, and we can see that the Jeffreys prior is not flat. At $k = 5$ the two weak posteriors take their largest values at 0, since no successes have been observed so far, while the misplaced posterior has only moved from 0.8 to around 0.6. At $k = 20$ all three posteriors are humps, narrower than before. The two weak posteriors sit well below

the truth after an unlucky start, and the misplaced one is closest to 0.3; this shows how much the prior still matters after twenty observations, not that the misplaced prior is correct. At $k = 100$ the two weak posteriors coincide and the misplaced posterior overlaps them: as the weight of the prior decreases, as discussed in example 28.5, every posterior concentrates near the truth.

30.3 A Task to Take Home

For the data of section 30.1, compute the equal-tailed credible interval with credibility 0.95 under the prior Beta$(12, 3)$ from section 30.2, and compare it with the two intervals found there. Then compute all three intervals for $T = 4$ successes in $n = 10$ trials. Write a short paragraph about how the influence of the prior depends on the sample size.

After the session, you should type the commands used in this session, and should check that you get the same results as the text.

Summary.

- The written reference for the commands used in this session is appendix B.
- For the same Bernoulli data, the Wald interval and the credible interval with flat prior agree to within 0.01, but they are different statements: the credible interval is a probability statement about θ, conditioned on the data and on the prior, and thus depends on the prior.
- The Beta posterior becomes sharper as observations are added. After 20 observations the prior is still noticeable, but after 100 its influence is small.

Lecture 31

Standard Tests Revisited

The most commonly used statistical tests, the z-test, the t-test, the chi-squared test for a variance and the F-test, are often taught as recipes. In this lecture we will consider these tests again, from the perspective of the module: each of these tests is based on a pivot in the sense of lecture 26, and the test is performed by comparing a statistic to a quantile from a table. We will work through the one-sample t-test in detail, including how to read the tables, and will state the variance, paired, two-sample and F-tests in a similar way. We will conclude by giving a table which shows which test is appropriate for which question.

31.1 Tests from Pivots

Every test in this lecture is based on the following principle: the test statistic is known to follow a certain distribution under the null hypothesis, and the test rejects if the statistic exceeds a quantile of this distribution, taken from a table. Using the language of lecture 26, the test statistic is a pivot $Q(X, \theta)$ for the hypothesised value $\theta = \theta_0$ and the test is the one which, when inverted by lemma 26.6, gives the confidence interval based on this pivot.

We use the notation z_p, $t_{k,p}$ and $\chi^2_{k,p}$ for the quantiles from lecture 26, so that $\mathbb{P}(T \leq t_{k,p}) = p$ for $T \sim t_k$. A table shows these quantiles for a few values of p. For a one-sided test at level α, we need the $(1-\alpha)$-quantile, and for a two-sided test at the same level we need the $(1-\alpha/2)$-quantile. A p-value in the sense of definition 22.8 cannot be found exactly in a table, but can be bracketed between two tabulated levels; we will see an example of this approach in example 31.2, below.

The most basic example is the z-test from lecture 22: if $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ are i.i.d. with known σ^2, then $Z = \sqrt{n}\,(\bar{X} - \mu_0)/\sigma$ is standard normally distributed under $H_0 \colon \mu = \mu_0$ and in examples 22.6 and 22.7 we can reject for $Z > z_{1-\alpha}$ or $|Z| > z_{1-\alpha/2}$ for the one-sided and two-sided alternative, respectively. The drawback of this test is that the variance of the data must be known. The rest of the lecture is concerned with how to replace Z when σ^2 is unknown.

31.2 The One-Sample t-Test

Let $X_1, \dots, X_n$ be i.i.d. $N(\mu, \sigma^2)$, where both parameters are unknown, and let $\bar{X}$ and S^2 be the sample mean and sample variance, respectively. If we replace σ in the definition of Z by the estimate S, we get the test statistic

$$T = \frac{\sqrt{n}\,(\bar{X} - \mu_0)}{S}. \tag{31.1}$$

In example 26.4 we have seen that for $\mu = \mu_0$ this is the ratio of the standard normal variable $\sqrt{n}\,(\bar{X} - \mu_0)/\sigma$ to the square root of $V/(n-1)$, where $V = (n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$, and that the two values are independent of each other, both by proposition A.4. Consequently, $T \sim t_{n-1}$ by definition A.5. This leads to the following tests.

Proposition 31.1. *Let $n \geq 2$ and let $X_1, \dots, X_n$ be i.i.d. $N(\mu, \sigma^2)$, where the parameters μ and σ^2 are unknown, $\mu_0 \in \mathbb{R}$ and let T be given by (31.1). Then the three tests of $H_0 \colon \mu = \mu_0$, which reject if*

$$T > t_{n-1,1-\alpha}, \qquad T < -t_{n-1,1-\alpha}, \qquad \text{or} \qquad |T| > t_{n-1,1-\alpha/2},$$

for the alternatives $\mu > \mu_0$, $\mu < \mu_0$ and $\mu \neq \mu_0$, respectively, have size α, independent of the value of σ^2.

Proof. Under H_0 we have $T \sim t_{n-1}$ for every value of σ^2, as shown above. By the definition of the quantile we have $\mathbb{P}(T > t_{n-1,1-\alpha}) = 1 - (1-\alpha) = \alpha$. This is the size of the first test. The density of the t distribution is symmetric about zero and thus we have $\mathbb{P}(T < -t_{n-1,1-\alpha}) = \mathbb{P}(T > t_{n-1,1-\alpha}) = \alpha$ for the second test and for the third test we have $\mathbb{P}(|T| > t_{n-1,1-\alpha/2}) = 2 \cdot \alpha/2 = \alpha$. Since these probabilities do not depend on σ^2, the probability of rejection under H_0 is α for every value of σ^2, and thus the size of each test is α. This completes the proof. □

These are the one-sided and two-sided *one-sample t-tests*. Since $t_{n-1,p} > z_p$ for every $p > 1/2$, the t-test requires a larger standardised deviation than the z-test before it can reject the hypothesis, and this is the price we pay for estimating σ. As n increases, this price decreases as the t_{n-1} distribution gets closer to the standard normal distribution. As in lecture 22, the one-sided tests have level α for the composite null hypotheses $\mu \leq \mu_0$ and $\mu \geq \mu_0$, since the probability of rejection increases as μ moves into the alternative. Unlike in lecture 22, where the power function was explicit, we state this monotonicity without proof.

Example 31.2. A quantity has nominal value 10, and ten measurements give

$$10.2,\ 9.7,\ 11.1,\ 10.8,\ 10.4,\ 9.9,\ 11.3,\ 10.6,\ 10.1,\ 10.9.$$

Assume that the measurements are i.i.d. $N(\mu, \sigma^2)$. We want to test $H_0 \colon \mu = 10$ against $H_1 \colon \mu \neq 10$ at level $\alpha = 0.05$. The sum of the observations is 105, so that $\bar{x} = 10.5$. The sum of the squared deviations from $\bar{x}$ is 2.52, so that $s^2 = 2.52/9 = 0.28$ and $s = 0.529$. The observed value of the test statistic is

$$t = \frac{\sqrt{10}\,(10.5 - 10)}{0.529} = 2.99.$$

The row corresponding to $n-1=9$ degrees of freedom in the t-table shows that we have $t_{9,0.95}=1.833$, $t_{9,0.975}=2.262$, $t_{9,0.99}=2.821$ and $t_{9,0.995}=3.250$. For a two-sided test with level 0.05 we need $t_{9,0.975}=2.262$ and since $|t|=2.99>2.262$ we can reject H_0: the data are not compatible with a mean of 10. The same row also brackets the p-value. The observed value 2.99 falls between $t_{9,0.99}$ and $t_{9,0.995}$ and thus the two-sided p-value, $2\,\mathbb{P}(T\geq 2.99)$ for $T\sim t_9$, is between 0.01 and 0.02; the two-sided test also rejects at level 0.02 but not at level 0.01. If instead the question had been whether the measurements were high, the one-sided test for $H_0\colon \mu=10$ against $H_1\colon \mu>10$ at level 0.05 would have compared $t=2.99$ to $t_{9,0.95}=1.833$ and would have rejected H_0, with a p-value between 0.005 and 0.01.

By exercise 25.1, the two-sided t-test is the generalised likelihood-ratio test for this problem, with the exact t_{n-1} quantile in place of the χ^2_1 approximation of Wilks' theorem, theorem 25.3.

31.3 The Chi-Squared Test for a Variance

The second question we can ask a normal sample, about the spread of the sample, is whether the variance equals a given value σ_0^2. In proposition A.4 we have seen that the pivot $(n-1)S^2/\sigma^2\sim\chi^2_{n-1}$ does not depend on the mean, and the test will follow the structure of the test from the previous section.

Proposition 31.3. *Let $n\geq 2$ and let $X_1,\dots,X_n$ be i.i.d. $N(\mu,\sigma^2)$, where μ and σ^2 are unknown, $\sigma_0^2>0$ and let*

$$V=\frac{(n-1)S^2}{\sigma_0^2}. \tag{31.2}$$

Each of the three tests for $H_0\colon \sigma^2=\sigma_0^2$, which reject if

$$V>\chi^2_{n-1,1-\alpha},\qquad V<\chi^2_{n-1,\alpha},\qquad \text{or}\qquad V\notin[\chi^2_{n-1,\alpha/2},\ \chi^2_{n-1,1-\alpha/2}],$$

for the alternatives $\sigma^2>\sigma_0^2$, $\sigma^2<\sigma_0^2$ and $\sigma^2\neq\sigma_0^2$, respectively, has size α, whatever the value of μ.

Proof. Under H_0 we have $V=(n-1)S^2/\sigma^2\sim\chi^2_{n-1}$ by proposition A.4, for every value of μ. The first two rejection probabilities are $1-(1-\alpha)=\alpha$ and α by the definition of the quantiles. For the third, the events $V<\chi^2_{n-1,\alpha/2}$ and $V>\chi^2_{n-1,1-\alpha/2}$ are disjoint and each have probability $\alpha/2$, and thus the rejection probability is α. This completes the proof. □

Unlike the t distribution, the chi-squared distribution is not symmetric and so we need different values from the table for the two tails of a two-sided test. This is the inverse of the confidence interval from example 26.5. Similarly, the one-sided test against $\sigma^2>\sigma_0^2$ has level α for the composite null hypothesis $\sigma^2\leq\sigma_0^2$, since the probability of the test to reject the null hypothesis increases with σ^2. This is exercise 31.2.

The numerical carrying-out of the test is also part of that exercise, and it follows the pattern of example 31.2 with the chi-squared table in place of the t-table. Note that tests about a variance have low power for small samples, because S^2 is a much more variable estimator than $\bar{X}$.

31.4 Paired and Two-Sample Problems

The tests so far have considered a single sample. Here we will consider three more standard tests, which compare two sets of measurements each, and are again pivot tests.

The first of these three tests is the *paired t-test*. Assume that we have observed data in pairs (X_i, Y_i), $i = 1, \ldots, n$, *e.g.* before and after measurements for the same subject. Since the two measurements for one subject are not independent, we can reduce the data for each pair to the difference $D_i = X_i - Y_i$, model the differences as i.i.d. $N(\mu_D, \sigma_D^2)$ and then apply proposition 31.1 with $\mu_0 = 0$: the test statistic is $T = \sqrt{n}\,\bar{D}/S_D$, constructed from the sample mean and variance of the differences, and is t_{n-1} distributed under $H_0\colon \mu_D = 0$. The model is only for the differences, as the example in exercise 31.3 shows.

The second test is the *two-sample t-test*. Here we assume that $X_1, \ldots, X_m$ are i.i.d. $N(\mu_X, \sigma^2)$ and that $Y_1, \ldots, Y_n$ are i.i.d. $N(\mu_Y, \sigma^2)$, where the two samples are independent of each other, and where we assume that the unknown variances of the two populations are the same. We consider $H_0\colon \mu_X = \mu_Y$. The estimator for the common variance is now

$$S_p^2 = \frac{(m-1)S_X^2 + (n-1)S_Y^2}{m+n-2},$$

and the test statistic is

$$T = \frac{\bar{X} - \bar{Y}}{S_p\sqrt{1/m + 1/n}} \sim t_{m+n-2} \qquad \text{under } H_0. \tag{31.3}$$

The distribution of this test statistic is similar to the distribution of the test statistic for the one-sample case: under H_0 the numerator is $N\big(0, \sigma^2(1/m + 1/n)\big)$-distributed, and $(m+n-2)S_p^2/\sigma^2$ is the sum of two independent χ^2_{m-1} and χ^2_{n-1} variables and thus is χ^2_{m+n-2} by definition A.2, and numerator and denominator are independent of each other by proposition A.4, applied to each of the two samples, and the independence of the samples. Thus, the test can be used to test H_0 against $\mu_X \neq \mu_Y$, rejecting if $|T| > t_{m+n-2,1-\alpha/2}$, and the one-sided variants of the test use $t_{m+n-2,1-\alpha}$.

The third test is the *F-test for two variances*. In the same two-sample setting, but now with variances σ_X^2 and σ_Y^2 which are allowed to be different, the ratio $(S_X^2/\sigma_X^2)/(S_Y^2/\sigma_Y^2)$ is the ratio of two independent chi-squared variables, divided by the degrees of freedom. Thus, it is $F_{m-1,n-1}$ distributed by definition A.5. Under $H_0\colon \sigma_X^2 = \sigma_Y^2$ the unknown variances cancel and the statistic $F = S_X^2/S_Y^2$ is $F_{m-1,n-1}$ distributed. The two-sided test rejects if F does not fall into the interval $[F_{m-1,n-1,\alpha/2},\, F_{m-1,n-1,1-\alpha/2}]$, where $F_{k,l,p}$ is the p-quantile of $F_{k,l}$. Normally, tables only give the upper quantiles and the lower quantiles can be found by using $F_{k,l,p} = 1/F_{l,k,1-p}$. This test is also used to check the assumption of equal variances in (31.3).

31.5 Which Test?

Table 31.1 summarises the six tests. The choice between them is determined by the following three questions: Is the parameter in question a mean or a variance? Is there one sample or two? If two samples, are the observations paired or independent? All six tests assume that the data are normally distributed.

Question	H_0	Statistic	Null distribution
one mean, σ^2 known	$\mu = \mu_0$	$\sqrt{n}\,(\bar{X} - \mu_0)/\sigma$	$N(0,1)$
one mean, σ^2 unknown	$\mu = \mu_0$	$\sqrt{n}\,(\bar{X} - \mu_0)/S$	t_{n-1}
paired differences	$\mu_D = 0$	$\sqrt{n}\,\bar{D}/S_D$	t_{n-1}
one variance, μ unknown	$\sigma^2 = \sigma_0^2$	$(n-1)S^2/\sigma_0^2$	χ^2_{n-1}
two means, common σ^2	$\mu_X = \mu_Y$	$\frac{\bar{X}-\bar{Y}}{S_p\sqrt{1/m+1/n}}$	t_{m+n-2}
two variances	$\sigma_X^2 = \sigma_Y^2$	S_X^2/S_Y^2	$F_{m-1,n-1}$

Table 31.1: The standard tests for normal data. The pivot for each test is evaluated under H_0, and the final column gives the table from which the critical value can be found (using the $(1-\alpha)$-quantile for a one-sided and the $(1-\alpha/2)$-quantile (together with the $(\alpha/2)$-quantile for the chi-squared and F tests) for a two-sided alternative).

Summary.

- Each of the standard tests for normal data is based on the pivot from lecture 26, evaluated for the null value, and compared to a quantile of the known null distribution. The test is the inverse of the corresponding confidence interval.
- The one-sample t-test uses $T = \sqrt{n}\,(\bar{X} - \mu_0)/S \sim t_{n-1}$ under $H_0\colon \mu = \mu_0$ and rejects for $T > t_{n-1,1-\alpha}$ (one-sided) or $|T| > t_{n-1,1-\alpha/2}$ (two-sided). It is the likelihood-ratio test with an exact critical value.
- The chi-squared test for a variance uses $V = (n-1)S^2/\sigma_0^2 \sim \chi^2_{n-1}$ under $H_0\colon \sigma^2 = \sigma_0^2$. A two-sided test requires both lower and upper quantiles, since the distribution is not symmetric.
- The paired t-test is a one-sample t-test for the differences. Similarly, the two-sample t-test with pooled variance and the F-test for two variances follow this pattern, using the t_{m+n-2} and $F_{m-1,n-1}$ distributions.
- The table lists only a few quantiles and often the p-value is bracketed between two tabulated levels.

Exercise 31.1. Eight independent measurements of a quantity with nominal value 20 are given by

$$19.1,\ 18.6,\ 20.3,\ 19.4,\ 18.9,\ 19.8,\ 18.2,\ 19.7.$$

Assume that the measurements are i.i.d. $N(\mu, \sigma^2)$, where both parameters are unknown. From the t-table we find $t_{7,0.95} = 1.895$, $t_{7,0.975} = 2.365$, $t_{7,0.99} = 2.998$ and $t_{7,0.995} = 3.499$.

1. Determine $\bar{x}$, s^2 and the observed value of the test statistic T from (31.1) for $\mu_0 = 20$.
2. For $H_0\colon \mu = 20$, test $H_1\colon \mu < 20$ at levels 0.05 and 0.01. Bracket the p-value.
3. For $H_0\colon \mu = 20$, test $H_1\colon \mu \neq 20$ at the same two levels and bracket the p-value.

4. Calculate the 99% t-interval for μ from example 26.4 and discuss how it relates to your answer in the previous part.

Exercise 31.2. A sample of $n = 15$ i.i.d. $N(\mu, \sigma^2)$ observations has sample variance $s^2 = 2.9$, and the process generating the sample is specified to have variance at most 1.5. From the chi-squared table we find $\chi^2_{14,0.025} = 5.629$, $\chi^2_{14,0.05} = 6.571$, $\chi^2_{14,0.95} = 23.685$, $\chi^2_{14,0.975} = 26.119$ and $\chi^2_{14,0.99} = 29.141$.

1. Test $H_0 \colon \sigma^2 \leq 1.5$ against $H_1 \colon \sigma^2 > 1.5$ at levels 0.05 and 0.01. Bracket the p-value.

2. Test $H_0 \colon \sigma^2 = 1.5$ against $H_1 \colon \sigma^2 \neq 1.5$ at level 0.05.

3. Show that the power function of the one-sided test from proposition 31.3 is given by $\beta(\sigma^2) = \mathbb{P}(W > \chi^2_{n-1,1-\alpha}\, \sigma_0^2/\sigma^2)$ where $W \sim \chi^2_{n-1}$, and that this power function is increasing in σ^2. Deduce that the test has size α for the composite null hypothesis $\sigma^2 \leq \sigma_0^2$.

4. Using the power function, explain how it can happen that the one-sided test from part (a) and the two-sided test from part (b) can give different results at the same level.

Exercise 31.3. The following data give the reaction times of six subjects before and after a training period.

subject	1	2	3	4	5	6
before (x_i)	72	65	80	58	69	75
after (y_i)	68	66	74	55	63	70

We want to know whether the training reduced the mean reaction time. From the t-table we find $t_{5,0.95} = 2.015$, $t_{5,0.975} = 2.571$, $t_{5,0.99} = 3.365$ and $t_{10,0.95} = 1.812$.

1. Write down the model and hypotheses for the paired t-test and carry out the test at levels 0.05 and 0.01.

2. Determine a 95% confidence interval for the mean reduction in reaction time.

3. Your colleague, not knowing that the two rows in the data frame give paired data, performs a two-sample t-test (31.3). The sample variances of the two rows are $s_X^2 = 59.77$ and $s_Y^2 = 42.80$, respectively. What is the test statistic and the decision of your colleague at level 0.05? Justify your answer by showing that the paired analysis is appropriate, and that the two analyses are different.

Exercise 31.4. In example 26.4 we have observed a sample of $n = 16$ values, where $\bar{x} = 10.3$ and $s = 2.1$, and we have found the 95% t-interval $[9.18, 11.42]$ for μ. In example 26.5, for the same data, we have found the 95% interval $[2.41, 10.56]$ for σ^2. In exercise 26.4 we have seen that the hypothesis $H_0 \colon \mu = 9$ was rejected at level 0.05, using lemma 26.6 alone. Now use the values $t_{15,0.95} = 1.753$, $t_{15,0.975} = 2.131$, $\chi^2_{15,0.025} = 6.262$ and $\chi^2_{15,0.975} = 27.488$.

1. Perform a two-sided t-test for $H_0 \colon \mu = 9$ and state the decision. Repeat the test for $H_0 \colon \mu = 11.5$ and check that the result is consistent with the interval.

2. For $H_0 : \sigma^2 = \sigma_0^2$, perform a two-sided test at level 0.05 for $\sigma_0^2 = 2$ and $\sigma_0^2 = 9$, either from the interval alone or by computing the test statistic (31.2).

3. Perform a one-sided test for $H_0 : \mu = 9.5$ against $H_1 : \mu > 9.5$ at level 0.05. Why cannot the two-sided interval be used to answer this test? Which confidence set does this correspond to?

Lecture 32

The Exponential Model: From Start to Finish

In this final lecture we will consider the exponential distribution with unknown rate. We will follow the approach a statistician would take when considering this model: we will consider the model and the data, how to estimate the parameter, how well the estimate is performing, what intervals and tests we can use, and how things change when we consider Bayesian inference. The individual steps of this approach are based on results we have seen in previous lectures, and the aim of this lecture is to see how these results can be combined in a worked example. We choose the exponential distribution because this is the running example of the module, and because most of the individual steps in the analysis should already be familiar from lectures and exercises.

32.1 The Model and the Data

Assume that the lifetime of a component of a machine is exponentially distributed with rate $\theta > 0$. Then the lifetimes $X_1, \ldots, X_n$ of n components are i.i.d. with density $f(x;\theta) = \theta e^{-\theta x}$ for $x \geq 0$. The mean lifetime is $\mathbb{E}_\theta(X_i) = 1/\theta$, the variance is $1/\theta^2$ (table A.2). In example 7.4 we have seen that this is a one-parameter exponential family with natural statistic $T(x) = x$ and natural parameter $\eta(\theta) = -\theta$. Thus, by lemma 7.5, the joint density of the sample is

$$f(x_1, \ldots, x_n; \theta) = \theta^n \exp\Big(-\theta \sum_{i=1}^n x_i\Big)$$

for $x_1, \ldots, x_n \geq 0$. Since this density depends on the data only through $T = \sum_{i=1}^n X_i$, we can use the factorisation theorem (proposition 8.3) to conclude that T is sufficient for θ. Since the family has full rank in the sense of definition 7.8 (the natural parameter space is the open half-line), we can use proposition 10.6 to conclude that T is also complete. Thus, from lectures 7 to 10 we know that all our results can be based on T alone, and that any unbiased function of T will be the best unbiased estimator for the corresponding quantity.

As data we have observed $n = 10$ lifetimes (in years) of the components, and the sum of these values is $\sum_i x_i = 25.0$, *i.e.* we have $\bar{x} = 2.5$. The manufacturer claims that the mean lifetime is two years, *i.e.* $\theta_0 = 0.5$, and we want to test whether the components of the machine last longer than the manufacturer claims.

32.2 Point Estimation

The log-likelihood is $\ell(\theta) = n \log \theta - \theta T$, with derivative $\ell'(\theta) = n/\theta - T$ and second derivative $-n/\theta^2 < 0$, and thus the likelihood equation has the unique root $\hat{\theta} = n/T = 1/\bar{X}$, which is the maximum. Thus $\hat{\theta}$ is the MLE for θ (example 5.2). For our data we find $\hat{\theta} = 10/25 = 0.4$. By the invariance of the MLE, proposition 5.6, the MLE for the mean lifetime $1/\theta$ is $\bar{X} = 2.5$ years.

The MLE for the rate is biased: exercise 2.3 shows $\mathbb{E}_\theta(\hat{\theta}) = n\theta/(n-1)$, so that $\hat{\theta}$ overestimates θ by a factor of $n/(n-1)$. The corrected estimator $\tilde{\theta} = (n-1)/T$ is unbiased. Since $\tilde{\theta}$ is an unbiased function of the complete sufficient statistic T, the Lehmann–Scheffé theorem 10.5 shows that this is the UMVUE for θ. The variance of the UMVUE can be found as $\theta^2/(n-2)$ (exercise 5.3). For the data considered here, we find $\tilde{\theta} = 9/25 = 0.36$.

What is the optimal performance of an estimator? From example 11.8 we know that the Fisher information of one observation is $\mathcal{I}(\theta) = 1/\theta^2$ and thus we have $\mathcal{I}_n(\theta) = n/\theta^2$. Since the conditions of section 11.1 hold, the Cramér–Rao inequality (theorem 13.2) gives the bound θ^2/n for the variance of any unbiased estimator satisfying (13.1). The variance of the UMVUE is $\theta^2/(n-2)$, which is larger than the bound and by the discussion after corollary 13.6 no unbiased estimator can attain the bound, since the bound is only attained for an affine function of the natural statistic T and no such function is unbiased for the rate. The efficiency $(n-2)/n$ of $\tilde{\theta}$ tends to 1 as $n \to \infty$. The situation for the mean $1/\theta$ is different, since $\bar{X}$ is unbiased with variance $1/(n\theta^2)$. This variance equals the bound for the mean (exercise 11.6) and thus $\bar{X}$ is efficient.

For large samples, theorems 16.3 and 17.1 apply, because the regularity conditions are satisfied: $\hat{\theta}_n$ is consistent and $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{\mathrm{d}} N(0, \theta^2)$. From corollary 17.3 we know that the standard error is

$$\mathrm{se}(\hat{\theta}) = \frac{1}{\sqrt{\mathcal{I}_n(\hat{\theta})}} = \frac{\hat{\theta}}{\sqrt{n}}.$$

For the data at hand we get $\mathrm{se}(\hat{\theta}) = 0.4/\sqrt{10} = 0.126$. This is the number we would expect to see next to the estimate.

32.3 Interval Estimation

Proposition 26.8 turns the standard error into the approximate 95% confidence interval $\hat{\theta} \pm 1.96\ \mathrm{se}(\hat{\theta})$. For the data considered here, this confidence interval is 0.4 ± 0.248, *i.e.* $[0.152, 0.648]$. The interval is symmetric about the estimate, and the probability of the interval covering the correct value is only approximately 95%, since the interval is based on the normal limit at $n = 10$.

For the exponential distribution we can do better, because an exact pivot is available. In exercise 13.1 we have seen that $2\theta T \sim \chi^2_{2n}$ for every value of θ, so $Q = 2\theta T$ is a pivot in the sense of lecture 26. Solving $\chi^2_{2n,\alpha/2} \leq 2\theta T \leq \chi^2_{2n,1-\alpha/2}$ for θ gives the exact interval

$$\left[\frac{\chi^2_{2n,\alpha/2}}{2T},\ \frac{\chi^2_{2n,1-\alpha/2}}{2T}\right]$$

with coverage exactly $1-\alpha$ for every n. With $\chi^2_{20,0.025} = 9.591$ and $\chi^2_{20,0.975} = 34.170$ we find the interval $[9.591/50, 34.170/50] = [0.192, 0.683]$, as in exercise 26.1. The exact interval is shifted to the right relative to the Wald interval and is not symmetric about 0.4: it makes use of the fact that the sampling distribution of $\hat\theta$ is skewed, which the normal approximation does not. Both intervals are 95% confidence intervals in the sense of definition 26.1: the statement is about the procedure, and for the interval in front of us the parameter is either inside the interval or it is not.

32.4 Testing

We now test the manufacturer's claim. We take $H_0\colon \theta = \theta_0 = 0.5$ and start, as in lecture 23, with a simple alternative $H_1\colon \theta = \theta_1$ where $\theta_1 < \theta_0$ (a longer mean lifetime). The likelihood ratio is

$$\frac{f(x;\theta_1)}{f(x;\theta_0)} = \Big(\frac{\theta_1}{\theta_0}\Big)^n \exp((\theta_0-\theta_1)T),$$

which is an increasing function of T, since $\theta_0 - \theta_1 > 0$. By the Neyman–Pearson lemma, theorem 23.2, the most powerful test at level α rejects H_0 if $T > c$, where c is chosen such that $\mathbb{P}_{\theta_0}(T > c) = \alpha$. Under H_0 the pivot gives $2\theta_0 T \sim \chi^2_{2n}$, and thus $c = \chi^2_{2n,1-\alpha}/(2\theta_0)$. For $\alpha = 0.05$ we find $c = 31.410/1 = 31.41$. The critical region $\{T > 31.41\}$ does not depend on θ_1 and thus, by proposition 23.4 with the inequalities reversed, the same test is most powerful against all alternatives $\theta < \theta_0$. Since the observed value $T = 25.0$ is less than the critical value, we do not reject H_0 at the 5% level: the data do not show that the components have a longer lifetime than claimed. The p-value from definition 22.8 is $\mathbb{P}_{\theta_0}(T \geq 25.0) = \mathbb{P}(\chi^2_{20} \geq 25.0) = 0.20$.

The power function of definition 22.4 can be found by evaluating the same pivot at the alternative: $\beta(\theta) = \mathbb{P}_\theta(T > c) = \mathbb{P}(\chi^2_{2n} > 2\theta c)$. For example $\beta(0.25) = \mathbb{P}(\chi^2_{20} > 15.71) = 0.73$. If the true mean lifetime of the components is four years, then a sample of ten components has probability 0.73 of detecting this.

For the two-sided alternative $\theta \neq \theta_0$ there is no uniformly most powerful test, by the argument of proposition 23.5: the two one-sided tests reject on opposite tails of T. We use the generalised likelihood-ratio test of definition 25.1 instead. The statistic is

$$W = 2(\ell(\hat\theta) - \ell(\theta_0)) = 2n\Big(\log\frac{\hat\theta}{\theta_0} - 1 + \frac{\theta_0}{\hat\theta}\Big),$$

and by Wilks' theorem 25.3 it is approximately χ^2_1-distributed under H_0, since one parameter is fixed. For our data, $W = 20(\log 0.8 - 1 + 1.25) = 0.54$, well below $\chi^2_{1,0.95} = 3.84$, and thus again H_0 is not rejected. Exercise 32.1 verifies the formula for W.

32.5 The Bayesian Answer

Since we can assume θ to be random with prior density $\pi(\theta)$, and since the likelihood function is $\theta^n e^{-\theta T}$ as a function of θ, a prior of the same shape as the

likelihood, namely the Gamma(a, b) density $\pi(\theta) \propto \theta^{a-1}e^{-b\theta}$, is conjugate. This is a special case of proposition 28.4: the posterior distribution is given by

$$\pi(\theta|x) \propto \theta^{a-1}e^{-b\theta}\,\theta^n e^{-\theta T} = \theta^{a+n-1}e^{-(b+T)\theta},$$

which is again the Gamma$(a+n, b+T)$ distribution. The update increases a by n and b by T. For the natural parameter $\eta = -\theta$ the Jacobian is 1, and thus the Gamma(a, b) prior is the prior $\pi_{b,a-1}$ of proposition 28.4. In the example of the manufacturer's claim about the lifetime of the components, we centre the prior at the claimed rate by choosing $a = 2$ and $b = 4$, so that the prior mean is $a/b = 0.5 = \theta_0$. This prior is weak, since the update adds the sample size $n = 10$ to the small shape $a = 2$, and thus the data outweigh the prior. The posterior distribution is then Gamma$(12, 29)$ and, using lemma 28.7, the Bayes estimator for θ, using squared error loss, is given by

$$\mathbb{E}(\theta|x) = \frac{a+n}{b+T} = \frac{b}{b+T}\cdot\frac{a}{b} + \frac{T}{b+T}\cdot\frac{n}{T} = 0.138\cdot 0.5 + 0.862\cdot 0.4 = 0.414.$$

This is a weighted average of the prior mean and the MLE, where the weight of the data increases as T increases.

The credible interval of definition 29.1 comes from the same chi-squared fact as before: for integer $2a$, if $\theta \sim$ Gamma$(a+n, b+T)$ then $2(b+T)\theta \sim \chi^2_{2(a+n)}$, and thus the equal-tailed 95% credible interval is $[\chi^2_{24,0.025}, \chi^2_{24,0.975}]/(2\cdot 29) = [0.214, 0.679]$. For the manufacturer's claim, the posterior probability of long lifetimes is $\mathbb{P}(\theta < 0.5|x) = \mathbb{P}(\chi^2_{24} < 29) = 0.78$, up from the prior probability 0.59. Unlike the p-value, this is a probability about the parameter, computed after the data have been seen.

The Jeffreys prior from definition 28.8 is $\pi_J(\theta) \propto \sqrt{\mathcal{I}(\theta)} = 1/\theta$. This is the improper limit $a, b \to 0$ of the Gamma prior. Using this prior, the posterior distribution is Gamma(n, T). The credible interval is $[\chi^2_{2n,\alpha/2}, \chi^2_{2n,1-\alpha/2}]/(2T)$. This is the same as the exact confidence interval introduced in section 32.3. Also, $\mathbb{P}(\theta \geq \theta_0|x) = \mathbb{P}(\chi^2_{2n} \geq 2\theta_0 T)$. This is the p-value of the one-sided test. While the numbers coincide, the statements are different: the confidence interval covers the fixed θ in 95% of repetitions, whereas the credible interval covers the random θ with probability 0.95 given the data; and the p-value is a probability about the data under H_0, whereas $\mathbb{P}(\theta \geq \theta_0|x)$ is a probability about the hypothesis. This illustrates the comparison from section 29.3, for a concrete model, and it is special that the numbers coincide for the Jeffreys prior and one-sided hypotheses.

32.6 Looking Back and Looking Forward

Every question about the exponential model we have been able to answer so far has used a small number of different tools: the exponential-family structure of the model allowed us to find a complete sufficient statistic, the likelihood function allowed us to find the estimator, the Fisher information allowed us to find the standard error and the bound, a pivot or the normal limit allowed us to find the interval, the Neyman–Pearson lemma and Wilks' theorem allowed us to find tests, and Bayes' theorem allowed us to find the posterior distribution. The same list of tools applies to most regular models, and it is this list which we can use for a model not considered in these notes: write down the likelihood function,

find the sufficient statistic, solve the likelihood equation and check that the root is a maximum, compute the information, and use the results to compute the standard error, the interval, and the tests. The exact pivot is available because of the special form of the exponential distribution. In general we will only have the large-sample results from lectures 16, 17 and 25, and some care is needed when these results are used.

This is also the material we will study after this module. The standard errors computed by all regression packages are the inverse Fisher information from lecture 14, the tests which compare fitted models from two different analyses are the likelihood-ratio tests from lecture 25, mixture models and hidden Markov models are fitted using the EM algorithm from lecture 19, and Bayesian computation, from the simple conjugate updating to the simulation methods of later modules, is based on the posterior distribution from lecture 28. The models there will be much larger, but the questions we ask of these models will be similar to the ones we considered in this module.

Summary.

- For the exponential distribution with rate θ, the sum T is complete and sufficient, the MLE is $1/\bar{X}$, the UMVUE is $(n-1)/T$, and no unbiased estimator can achieve the Cramér–Rao bound θ^2/n, although $\bar{X}$ is efficient for the mean.
- The standard error $\hat{\theta}/\sqrt{n}$ can be used to compute the Wald interval. The pivot $2\theta T \sim \chi^2_{2n}$ can be used to compute an exact interval, an exact one-sided test with known power function, and the p-value.
- A two-sided test can be based on $W = 2n(\log(\hat{\theta}/\theta_0) - 1 + \theta_0/\hat{\theta})$ and Wilks' theorem.
- For the Gamma(a, b) prior, the posterior distribution is Gamma$(a + n, b + T)$, the posterior mean is a weighted average of the prior mean and the MLE, and the Jeffreys prior $1/\theta$ gives the exact interval and p-value numerically, but with a different meaning.
- The same short list of tools can be used to answer most questions about regular models.

Exercise 32.1. Let $X_1, \ldots, X_n$ be i.i.d. with exponential rate $\theta > 0$ and let $\hat{\theta} = 1/\bar{X}$ be the MLE.

1. Show that the generalised likelihood-ratio statistic for $H_0 \colon \theta = \theta_0$ against $H_1 \colon \theta \neq \theta_0$ is given by $W = 2n\big(\log(\hat{\theta}/\theta_0) - 1 + \theta_0/\hat{\theta}\big)$, and that W depends on the data only through $u = \theta_0 \bar{X}$, where $W = 2n(u - 1 - \log u)$.

2. Show that $u \mapsto u - 1 - \log u$ is zero at $u = 1$ and positive elsewhere, *i.e.* that $W \geq 0$ with equality if and only if $\hat{\theta} = \theta_0$.

3. For $n = 10$ and $\bar{x} = 2.5$, determine the value of W for $\theta_0 = 0.5$ and for $\theta_0 = 1$ and state the conclusion of the test at level 0.05 for both cases.

Exercise 32.2. Let $X_1, \dots, X_n$ be i.i.d. exponential with rate $\theta > 0$ and let $T = \sum_{i=1}^{n} X_i$.

1. Show that the Jeffreys prior for θ is $\pi_J(\theta) \propto 1/\theta$. Show that this prior is improper.

2. Show that the posterior distribution under the Jeffreys prior is proper for every sample with $T > 0$. Identify the posterior distribution as a Gamma distribution.

3. Using proposition 28.9 or otherwise, find the Jeffreys prior for the mean $\mu = 1/\theta$. Show that this prior is the image of π_J under the map $\theta \mapsto 1/\theta$.

4. Show that the equal-tailed 95% credible interval for θ, under the Jeffreys prior, coincides with the exact confidence interval from section 32.3, but in one sentence explain why the two intervals make different statements.

Exercise 32.3. Let $X_1, \dots, X_n$ with $n \geq 2$ be i.i.d. uniform on $(0, \theta)$, where $\theta > 0$ is unknown, and let $X_{(1)} = \min_i X_i$ and $X_{(n)} = \max_i X_i$. Without proof, we state that, given $X_{(n)} = m$, the remaining $n-1$ observations are i.i.d. uniform on $(0, m)$ and $\operatorname{Var}_\theta\big((n+1)X_{(n)}/n\big) = \theta^2/\big(n(n+2)\big)$.

(a) Find the density of $X_{(1)}$ and show that $\mathbb{E}_\theta(X_{(1)}) = \theta/(n+1)$.

(b) Show that $U = (n+1)X_{(1)}$ is an unbiased estimator for θ. Determine $\operatorname{Var}_\theta(U)$ and show that U is not a consistent estimator for θ.

(c) Show that $X_{(n)}$ is sufficient for θ.

(d) Determine $V = \mathbb{E}_\theta(U \mid X_{(n)})$ and show that $V = (n+1)X_{(n)}/n$. Which theorem guarantees, without computation, that V is a statistic, unbiased for θ and with a variance of at most that of U?

(e) Explain why V is the UMVUE for θ. The Cramér–Rao formula, for this model, gives the value θ^2/n, which is larger than the variance of V. Explain why there is no contradiction.

Exercise 32.4. Seed trays are bought from two suppliers. Each tray contains $m = 10$ seeds, and each seed has probability $p_A = 0.2$ of germinating if the tray comes from supplier A, and probability $p_B = 0.7$ if the tray comes from supplier B, independently of the other seeds. Both probabilities are known. It is unknown what proportion $\pi \in (0, 1)$ of trays come from supplier A (here π is the mixture weight, as in lecture 19), but the supplier of individual trays is not recorded. For tray i, the number X_i of germinated seeds is observed, and we write $Z_i = 1$ if the tray came from supplier A and $Z_i = 2$ otherwise. Let f_A and f_B be the probability weights of the Binomial$(10, 0.2)$ and Binomial$(10, 0.7)$ distributions.

(a) Write down the probability weights of the number X_i of germinated seeds and the observed-data log-likelihood $\ell(\pi)$ for n trays. Explain why the likelihood equation cannot be solved analytically and write down the complete-data log-likelihood $\ell_c(\pi)$.

(b) Derive the E step of the EM algorithm, showing that $Q(\pi|\pi_k)$ can be found from $\ell_c(\pi)$ by replacing the indicator $\mathbf{1}_{\{Z_i=1\}}$ by the responsibility

$$\gamma_{i,k} = \frac{\pi_k f_A(x_i)}{\pi_k f_A(x_i) + (1-\pi_k) f_B(x_i)}.$$

(c) Derive the M step, $\pi_{k+1} = \frac{1}{n}\sum_{i=1}^{n} \gamma_{i,k}$, and show that this is a maximum of $\pi \mapsto Q(\pi|\pi_k)$.

(d) For five trays, the number of germinated seeds was 2, 7, 1, 8 and 3. The probability weights of the two distributions are given by

x	1	2	3	7	8
$f_A(x)$	0.2684	0.3020	0.2013	0.000786	0.0000737
$f_B(x)$	0.000138	0.00145	0.00900	0.2668	0.2335

Given $\pi_0 = 1/2$, perform one iteration of the EM algorithm and comment on your responsibilities.

(e) Assume now that p_A is unknown, and p_B is still known. Derive the M step for p_A. State the ascent property of the EM algorithm, and explain what the property does and does not guarantee for the sequence $(\pi_k, p_{A,k})$.

Preparing for the Exam

This chapter summarises what you need to do to prepare for the examination: it includes a map of the module, a summary of important results, information about what you were required to prove and what you were only required to know, advice on how to write your answers, details of common mistakes to avoid, and information on how to use the practice paper. None of this is new material, but it is gathered together here from different parts of the module for revision purposes.

The Module in One Page

The module is divided into two parts. The first part, covering lectures 1 to 20, deals with estimation: the question is what an estimator is, and how one can measure quality of an estimator. In this part of the module we also learn how estimators can be constructed using the method of moments and the maximum likelihood method, how a sufficient statistic can be used to compress the data and how conditioning on the data can be used to improve an estimator. In this part of the module we also learn about the Fisher information and how it can be used to give a bound on the variance of all unbiased estimators. We learn how the maximum likelihood estimator behaves for large sample size and how to compute the maximum likelihood estimator when some of the data are missing. Throughout this part of the module we will consider the exponential families, introduced in lecture 7. These families of distributions provide the sufficient statistic, the completeness of the sufficient statistic, the Fisher information and, in lecture 28, the conjugate priors. The second part of the module, covering lectures 22 to 31, deals with testing and interval estimation. In this part of the module we learn about the language of tests, the best test for simple hypotheses, the likelihood ratio for composite hypotheses, the duality of tests and confidence intervals, the Bayesian approach to the same questions and the standard tests for normal data. Table 32.1 shows the lectures for each week of the module and the results of each lecture and the later lectures which depend on the results of these lectures.

The Results You Need to Know

For each result in this section, you must know the statement and conditions, and you must be able to reproduce the proof. A proof is reconstructed from the idea, rather than being memorised, so for each result we have written down the idea in one line. If this line is not enough for you to work out the argument, then you need to spend some time revising the proof.

Week	Lectures and main results	Used later in
1	1: model, statistic, estimator (definitions 1.1, 1.3 and 1.4). 2: bias and mean squared error, lemma 2.7; consistency, lemma 2.11.	every later lecture; the bias corrections of lecture 5; the consistency proof of lecture 16
2	4: likelihood, score and method of moments (definitions 4.1, 4.4 and 4.6). 5: the MLE (definition 5.1), the likelihood equation and its check, invariance, proposition 5.6; failures, section 5.3.	lectures 11, 16, 17, 19, 25 and 32
3	7: exponential families (definition 7.1), the cumulant function, lemma 7.10, full rank (definition 7.8). 8: sufficiency (definition 8.1), the factorisation theorem (proposition 8.3), minimal sufficiency (proposition 8.11), maximum and minimum (proposition 8.8).	lectures 10, 11, 13, 28 and 32
4	10: Rao–Blackwell, proposition 10.1; completeness (definition 10.3) and the completeness facts, proposition 10.6; Lehmann–Scheffé, theorem 10.5. 11: Fisher information (definition 11.2), the identities, lemma 11.3, additivity and reparametrisation (propositions 11.4 and 11.9).	lectures 13, 16, 17, 26, 28, 29 and 32
5	13: the Cramér–Rao inequality, theorem 13.2 and corollary 13.3; efficiency (definition 13.4) and attainment, proposition 13.5; the uniform counterexample, example 13.7. 14: the information matrix (definition 14.2), proposition 14.5, the normal matrix of example 14.6.	lectures 17, 25 and 32
6	16: the regularity conditions (definition 16.1), lemma 16.2 and theorem 16.3. 17: asymptotic normality, theorem 17.1; standard errors, corollary 17.3.	lectures 25, 26, 29 and 32
7	19: the EM algorithm (definition 19.2), the normal mixture of example 19.1. 20: the ascent property, proposition 20.2; fixed points, proposition 20.3.	lecture 32
8	22: tests, errors, power, size and level (definitions 22.2 to 22.5), the p-value (definition 22.8), the z-tests of examples 22.6 and 22.7. 23: the Neyman–Pearson lemma, theorem 23.2; one-sided optimality, proposition 23.4; no two-sided UMP test, proposition 23.5.	lectures 25, 26, 31 and 32
9	25: the generalised likelihood ratio (definition 25.1), Wilks' theorem 25.3, the exact case of example 25.2. 26: confidence intervals and pivots (definitions 26.1 and 26.2), duality, lemma 26.6, the Wald interval (definition 26.7 and proposition 26.8).	lectures 29, 31 and 32
10	28: posterior, lemma 28.2; conjugacy, proposition 28.4 and example 28.5; the normal–normal model, proposition 28.6; the posterior mean, lemma 28.7; Jeffreys priors, definition 28.8 and proposition 28.9. 29: credible and HPD intervals (definitions 29.1 and 29.2), Bayesian tests (definitions 29.4 and 29.6, example 29.5), the comparison of section 29.3.	lecture 32
11	31: the t-test and the variance test (propositions 31.1 and 31.3), table 31.1. 32: the exponential model worked from start to finish, a synthesis of the whole module.	

Table 32.1: The topics week by week.

We begin with estimation.

- Lemma 2.7, the decomposition MSE $=$ Var $+$ bias^2: add and subtract the mean inside the square, and the cross term vanishes. Lemma 2.11, a vanishing mean squared error gives consistency: Markov's inequality, lemma 2.10, applied to $(\hat{\theta}_n - \theta)^2$.
- Proposition 5.6, the MLE for $g(\theta)$ is $g(\hat{\theta})$: a relabelling of the parameter does not change which distribution has the largest likelihood.
- Lemma 7.10, the moments of the natural statistic are derivatives of K: differentiate the normalising integral under the integral sign, once for the mean and twice for the variance.
- Proposition 8.3, in the discrete case: write the joint probability as $\mathbb{P}_\theta(T = t)\,\mathbb{P}_\theta(X = x | T = t)$, and observe that sufficiency is the statement that the second factor is free of θ; for the converse, sum the factorisation over the data vectors with the same value of T.
- Proposition 10.1: the conditional expectation given a sufficient statistic is a statistic, the tower property keeps it unbiased, and the law of total variance, proposition A.11, shows that the variance can only decrease.
- Theorem 10.5: two unbiased functions of a complete statistic have a difference with mean zero, and thus are equal, so that the unbiased function of T is unique; Rao–Blackwell shows that no other unbiased estimator beats it.
- Lemma 11.3: differentiate $\int f(x;\theta)\,\mathrm{d}x = 1$ once to see that the score has mean zero, lemma 11.1, and a second time, using the quotient rule, to obtain $\mathcal{I}_n = -\mathbb{E}_\theta(\ell'')$.
- Theorem 13.2 and corollary 13.3, the Cramér–Rao inequality: the covariance of an unbiased estimator for $g(\theta)$ with the score equals $g'(\theta)$, by differentiating $\mathbb{E}_\theta(W) = g(\theta)$ under the integral sign, and the Cauchy–Schwarz inequality, lemma A.8, does the rest. Proposition 13.5 is the equality case of Cauchy–Schwarz: the estimator must be an affine function of the score.
- Lemma 16.2: by the law of large numbers the average log-likelihood ratio converges to
$$\mathbb{E}_{\theta_0}\left(\log\left(f(X;\theta)/f(X;\theta_0)\right)\right),$$
and Jensen's inequality, lemma A.7, makes this expectation negative.
- Theorem 16.3: on the event that $L(\theta_0)$ exceeds both $L(\theta_0 - \varepsilon)$ and $L(\theta_0 + \varepsilon)$, whose probability tends to one by the previous lemma, the log-likelihood has an interior maximum on $[\theta_0 - \varepsilon, \theta_0 + \varepsilon]$ and thus a root of the likelihood equation within ε of θ_0; uniqueness of the root makes it the MLE.
- Theorem 17.1: expand the score about θ_0, solve for $\sqrt{n}(\hat{\theta}_n - \theta_0)$, and apply the central limit theorem to the score, the law of large numbers to the second derivative, consistency and (R4) to the remainder and Slutsky's theorem, proposition A.16, to combine the three and to discard the events of vanishing probability on which the expansion fails. Corollary 17.3 follows by replacing θ_0 by $\hat{\theta}_n$ in the variance, again by Slutsky's theorem.

- Proposition 20.2: write $\ell = Q - H$ as in (20.1); the M step increases Q, and Jensen's inequality shows that $H(\theta|\theta_k)$ is largest at $\theta = \theta_k$, so that H cannot increase.

We now turn our attention to testing, intervals and the Bayesian approach.

- Theorem 23.2: the integrand $(\mathbf{1}_C - \mathbf{1}_{C'})(f(x;\theta_1) - k\, f(x;\theta_0))$ is non-negative everywhere, and the theorem follows by integrating this non-negative function to get $\beta_C(\theta_1) - \beta_{C'}(\theta_1) \geq k(\beta_C(\theta_0) - \beta_{C'}(\theta_0)) \geq 0$. From proposition 23.4 we know that a region which satisfies this condition for every $\theta_1 > \theta_0$ is most powerful against every one-sided alternative at once. From proposition 23.5 we know that there is no two-sided uniformly most powerful test, because such a test would have to agree with both one-sided tests, which reject on opposite tails.

- Lemma 26.6: the parameter θ_0 is in the interval $C(X)$ if and only if the test for θ_0 does not reject the value, and thus the probability of the interval covering θ_0 is at least one minus the level of the test.

- Proposition 26.8: the interval covers θ if and only if the standardised error lies between $\pm z_{1-\alpha/2}$, and by corollary 17.3 the probability of this event converges to $1 - \alpha$.

- Lemma 28.2: this is an application of Bayes' theorem, proposition A.20, where the parameter is the event and the data are the evidence. Proposition 28.4: $\exp(t_0\eta - n_0 K(\eta))$ can be multiplied by a likelihood of the form $\exp(\eta T_n - nK(\eta))$ to obtain an exponent of the same shape. Lemma 28.7: the same add-and-subtract argument as in lemma 2.7, but now under the posterior distribution.

- Proposition 28.9: the change of variables for densities, proposition A.19, provides the factor $|h'(\psi)|$ and the reparametrisation rule for the information, proposition 11.9, provides the same factor under the square root.

- Propositions 31.1 and 31.3: the statistics are pivots from lecture 26, evaluated at the null value, and the null distribution is given by proposition A.4 and definition A.5.

In addition to the theorems, the examination requires the recall of definitions and standard values, requiring only memory. The definitions are given in table 32.1 and should be given in the form in the notes, including the quantifier "for all $\theta \in \Theta$" wherever required. The standard values are the MLEs for the families in tables 1.1 and 1.2, as worked in lecture 5 and its exercises, the Fisher information for one observation for the same families, as summarised in lecture 11, the normal information matrix (14.1) and its inverse, the one-sided and two-sided z-tests and their power functions from lecture 22, and the Beta–Bernoulli and normal–normal updates from lecture 28.

What Is (Not) Proved

The following list summarises the results the module assumes to be known without proof. For each result, the examinable material is the statement, including any conditions, and how to use the result in a calculation.

- Wilks' theorem 25.3. Know the setting (25.3), the conclusion $W_n \xrightarrow{\mathrm{d}} \chi^2_r$ under H_0, and how to use the result: compute W from the two maximised log-likelihoods, count r as the number of constraints, and read off $\chi^2_{r,1-\alpha}$ from the table. The argument in lecture 25 for the case $k = r = 1$ is an explanation, not a proof. Also note that in example 25.2 the statement is exact for every n.

- The continuous case of the factorisation theorem (proposition 8.3). The proof of the discrete case is examinable, but for continuous data the statement of the proposition is used in both directions, with the indicator of a parameter-dependent support included in g or h as appropriate. The minimal-sufficiency criterion, proposition 8.11, is in a similar position: the proof was only sketched, but use of the result is examinable.

- The completeness results, proposition 10.6. Both results can be quoted as needed, and in the case of the exponential family the full-rank condition from definition 7.8 must be mentioned. These two are the only completeness results you need to know.

- The information matrix identity, proposition 14.3. Know the statement, that the information matrix is minus the expected Hessian of the log-likelihood; the proof mirrors the one-parameter case and was omitted.

- The vector Cramér–Rao bound, proposition 14.5. Know the statement, that the difference of the covariance matrix and the inverse information is positive semi-definite, and the consequence for the variances of single coordinates, namely $\mathrm{Var}_\theta(T_j) \geq (\mathcal{I}_n^{-1})_{jj}$, together with the example of the normal distribution. The proof is not examinable.

- The vector case of theorem 17.1, as given at the end of lecture 17: the MLE of a vector parameter is asymptotically normal with covariance matrix $\mathcal{I}_n(\theta_0)^{-1}$. Know the statement and what it means for the normal distribution, where the diagonal matrix makes $\bar{X}$ and $\hat{\sigma}^2$ asymptotically independent.

- The invariance of the MLE (proposition 5.6), the normal–normal update (proposition 28.6) and the Jeffreys priors for location and scale models (proposition 28.11). Know the statements and how to use them; the proofs are exercises.

- The minimality of the highest posterior density set of definition 29.2. Know the statement, that H_c is the smallest set with posterior probability $1 - \alpha$. The argument given in lecture 29 is an explanation, not a proof.

- The large-sample normal approximation (29.4) to the posterior distribution. Know the statement, that under the regularity conditions and for a prior with continuous positive density the posterior is approximately $N(\hat{\theta}_n, 1/\mathcal{I}_n(\hat{\theta}_n))$, and its consequence, that the equal-tailed credible interval is approximately the Wald interval whatever the prior. The argument given in lecture 29 is an explanation, not a proof.

- The level of the one-sided t-tests for the composite null hypotheses $\mu \leq \mu_0$ and $\mu \geq \mu_0$. Know the statement, that the probability of rejection increases

as μ moves into the alternative, so that the tests of proposition 31.1 have level α for these hypotheses. This monotonicity was not proved.

The material in appendix A is background knowledge which is used without proof, but the examination expects you to name the results when they are used. All the other material from the content lectures, including the MLE asymptotics from lectures 16 and 17, and the Lehmann–Scheffé theorem, was proved in full and is examinable as such.

Writing an Answer

The model answers in appendix C show the level of detail expected. Every statistical argument in this module has the same five steps, and an answer which makes each step visible is a complete answer.

1. State the model and the parameter. Write down the density or the probability weights, including the support and the parameter space, before doing anything with it. Many likelihood mistakes, in particular for models where the support depends on the parameter, are caused by leaving out the indicator of the support.

2. Name the result used: "by the factorisation theorem", "by Wilks' theorem". Show how each answer was obtained, not only the answer.

3. Check the conditions. If the result requires a sufficient statistic to be complete, as for the Lehmann–Scheffé theorem 10.5, you should mention where the completeness comes from. If the result requires regularity conditions, as for the Cramér–Rao inequality, theorem 13.2, you should say that the support does not depend on the parameter. For a Wald interval or for Wilks' theorem 25.3, you should say that the statement is an approximation for large samples. If a question asks why a result does not apply, this step is the whole answer.

4. Compute the result. Keep the computation on the same page as the argument that justifies it, so that your answer is easy to follow.

5. Give the conclusion in words, in the language of the original question: "the test rejects H_0 at the 5% level". Use the correct kind of statement: a confidence interval is a range of values which cover the parameter with a given probability, as repeated samples are taken. A credible interval is a range of values which have a given posterior probability, given the data. A p-value is not the probability that H_0 is true.

For a question like "show that" something is true, the answer is given in the question itself, and the argument is what counts. If you cannot answer one part of a question, attempt the later parts anyway, using the statement of the earlier part where they need it.

Three habits help in every question. Distinguish the information of one observation from that of the sample, $\mathcal{I}_n = n\mathcal{I}$, and say which one a formula uses. Verify that a root of the likelihood equation is a maximum. Count degrees of freedom by counting constraints, and remember that a two-sided chi-squared test needs two quantiles, because the distribution is not symmetric. The next section lists the mistakes made most often.

Common Mistakes

The following mistakes are common, and they are all easy to avoid once they are known.

- Treating the parameter as random in a frequentist statement. A confidence interval is a random interval which covers the fixed value θ with probability 0.95 before the data are seen; the sentence "θ lies in $[l, u]$ with probability 0.95" is a statement about a credible interval, definition 29.1, and is false for a confidence interval, definition 26.1.
- Using the Cramér–Rao inequality, or the asymptotic normality of the MLE, for a model whose support depends on the parameter. The uniform distribution on $(0, \theta)$ is a standard counterexample (examples 11.10 and 13.7); for such a model the estimator based on the maximum or the minimum has to be analysed directly.
- Claiming that an estimator is a UMVUE because it is unbiased and a function of a sufficient statistic. The Lehmann–Scheffé theorem requires the statistic to be complete, and this completeness needs to be shown, typically using proposition 10.6.
- Finding a root of the likelihood equation and calling it the MLE without checking that it is a maximum. The second derivative, or the concavity of the log-likelihood, is part of the answer, and for a model like the uniform distribution the maximum is not a root of the likelihood equation (section 5.3).
- Interpreting a non-rejection as a proof of H_0, or interpreting a p-value as the probability that H_0 is true. A test controls the probability of type I errors, and the p-value is a probability, computed under H_0, about the data, definition 22.8.
- Mixing up the one-observation and the sample information. The Cramér–Rao bound and the standard error use $\mathcal{I}_n(\theta) = n\mathcal{I}(\theta)$, and the limit in theorem 17.1 uses $\mathcal{I}(\theta)$ together with the factor $\sqrt{n}$; both forms are equivalent, but the n needs to be in one place or the other.
- Leaving out the indicator of the support when factorising a density. For the uniform distribution the factor $\mathbf{1}_{\{x_{(n)} \leq \theta\}}$ is the whole point of the factorisation, and without it the maximum would not be sufficient (example 8.4).
- Stopping at a number. An estimate of 0.111, a statistic of 3.61 or a posterior probability of 0.57 is not an answer until it has been compared to the relevant threshold and turned into a sentence about the question.

Using the Practice Paper

The exercises in appendix C are the best rehearsal for the examination you will get, and it is advisable to try them yourself before looking at the solutions.

A practice paper is most useful when it is sat before the answers are seen. If you have not yet done so, sit the paper under examination conditions, with the

answers written out in full. A paper which has been read through, with a mental note that each part looks manageable, tells you very little. A paper which has been sat tells you where the time went, which parts you could start but not finish, and which results you thought you knew but could not write down.

Two habits from doing the paper carry over to the examination. If you find yourself spending more than a quarter of an hour on one part of the paper, write down what you have done so far and then move on to the next question. It is better to have a partial answer for all parts of a question than to have a complete answer for only some parts.

Thank you for taking part in this module, and good luck in the examination.

Appendix A

Background from Probability

This appendix summarises some important results from probability which we use in the module. The prerequisite for understanding the contents of this appendix is the knowledge of the MATH2701 Statistical Methods module; all the results summarised here should be known from that module (or from an equivalent background). The purpose of this appendix is to fix notation and to provide a point of reference for the main text. None of the material in this appendix is examinable on its own. The results included here are mostly statements about fully specified distributions (known models). The opposite direction, where we try to infer an unknown parameter from data, is the one we are concerned with in the main text. We include proofs of the basic tools, because these are short and instructive. We only state the deeper limit theorems, because proofs of these results would take up valuable space and time in this text.

A.1 Expectation, Variance and Independence

We write $\mathbb{E}(X)$ for the expectation of a random variable X and $\mathrm{Var}(X)$ for its variance. We assume that the basic rules of algebra for expectations and variances are known. For reference, we state the following rules for expectations and variances.

Proposition A.1. *Let X and Y be random variables with finite variance, and let a and b be constants. Then*

$$\mathbb{E}(aX + bY) = a\,\mathbb{E}(X) + b\,\mathbb{E}(Y) \qquad \textit{and} \qquad \mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X).$$

The covariance is given by $\mathrm{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\,\mathbb{E}(Y)$ *and thus*

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X, Y).$$

If X and Y are independent, then $\mathbb{E}(XY) = \mathbb{E}(X)\,\mathbb{E}(Y)$ and thus the covariance is zero. Similarly, the variance of the sum can be computed as the sum of the variances.

More generally, for independent $X_1, \ldots, X_n$ the variance of the sum is the sum of the individual variances. We use this fact, for example, when we average a random sample. Since these identities are standard, we do not give proofs of these results here.

A.2 Standard Distributions

We will consider a small number of distributions in the module. Tables A.1 and A.2 summarise the probability weights or density, support, mean and variance of these distributions, for the discrete and for the continuous distributions respectively. The families of distributions considered in tables 1.1 and 1.2 are all included, together with their moments. The negative binomial, Pareto and Rayleigh distributions do not appear in tables 1.1 and 1.2; they are listed here for reference. The tables use the usual parameter names, but in the main text these families of distributions will be considered as parametric models where the unknown parameter is denoted by θ. For example, the rate parameter λ of the exponential distribution below is the θ of table 1.2. Throughout the module the exponential and gamma distributions are parametrised by their *rate*, as in the table and as in R's `dexp` and `dgamma`; some textbooks use the mean $1/\lambda$, or equivalently the scale, so a formula from elsewhere may need to be translated.

Family	$f(x)$	Support	Mean	Variance
Bernoulli(p)	$p^x(1-p)^{1-x}$	$\{0,1\}$	p	$p(1-p)$
Binomial(m,p)	$\binom{m}{x}p^x(1-p)^{m-x}$	$\{0,\ldots,m\}$	mp	$mp(1-p)$
Geometric(p)	$(1-p)^{x-1}p$	$\mathbb{N}$	$1/p$	$(1-p)/p^2$
Poisson(λ)	$\frac{\lambda^x}{x!}\,e^{-\lambda}$	$\mathbb{N}_0$	λ	λ
Negative binomial(r,p)	$\binom{x+r-1}{x}p^r(1-p)^x$	$\mathbb{N}_0$	$\frac{r(1-p)}{p}$	$\frac{r(1-p)}{p^2}$

Table A.1: The standard discrete distributions and their moments. Here $f(x)$ gives the probability weights. For the geometric distribution, the number of trials until the first success is considered, so the support starts at 1. In contrast, some texts and the statistical software R consider the number of failures before the first success, so the support starts at 0 (see appendix B).

Family	$f(x)$	Support	Mean	Variance
Uniform($0,\theta$)	$1/\theta$	$[0,\theta]$	$\theta/2$	$\theta^2/12$
Exponential(λ)	$\lambda\,e^{-\lambda x}$	$[0,\infty)$	$1/\lambda$	$1/\lambda^2$
Gamma(α,β)	$\frac{\beta^\alpha}{\Gamma(\alpha)}\,x^{\alpha-1}e^{-\beta x}$	$(0,\infty)$	α/β	α/β^2
Beta(α,β)	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$	$[0,1]$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
Normal(μ,σ^2)	$\frac{1}{\sqrt{2\pi\sigma^2}}\exp\!\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$	$\mathbb{R}$	μ	σ^2
Cauchy	$\frac{1}{\pi(1+x^2)}$	$\mathbb{R}$	none	none
Pareto(α)	$\alpha\,x^{-(\alpha+1)}$	$[1,\infty)$	$\frac{\alpha}{\alpha-1}$	$\frac{\alpha}{(\alpha-1)^2(\alpha-2)}$
Rayleigh(θ)	$\frac{2x}{\theta}\,e^{-x^2/\theta}$	$[0,\infty)$	$\sqrt{\pi\theta}/2$	$(1-\pi/4)\,\theta$

Table A.2: The standard continuous distributions and their moments. Here $f(x)$ is the density, Γ is the gamma function, which satisfies $\Gamma(x+1) = x\,\Gamma(x)$ for all $x > 0$ and $\Gamma(k) = (k-1)!$ for $k \in \mathbb{N}$, and $B(\alpha,\beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$ is the beta function. The exponential distribution is the special case Gamma($1,\lambda$). The mean of the Pareto distribution only exists for $\alpha > 1$ and the variance only for $\alpha > 2$.

In the module we often need the distribution of a sum of independent random

variables, for example when we compute the distribution of a total or of a sample mean. For several of the standard distributions the sum of independent copies again has one of the standard distributions, with only the parameters changed.

Remark. If $X_1, \ldots, X_n$ are independent, then the sum of Bernoulli(p) variables is Binomial(n, p), the sum of Poisson variables is Poisson with the sum of the means, the sum of Exponential(λ) variables (or, more generally, of Gamma variables with the same rate) is Gamma with the sum of the shapes, and the sum of independent normals is normal with the sum of the means and the sum of the variances.

Another family, introduced in exercise 25.2 of lecture 25, is the multinomial distribution. If each of n independent trials has one of k possible categories, with probabilities $p_1, \ldots, p_k$, where the probabilities must sum to one, then the vector of counts $(N_1, \ldots, N_k)$ is *multinomial* distributed with $\mathbb{E}(N_j) = np_j$ and $\mathrm{Var}(N_j) = np_j(1-p_j)$. The individual counts are Binomial(n, p_j) and the multinomial distribution is a generalisation of the binomial distribution to the case where there are more than two categories.

A.3 Sampling from a Normal Population

The following distributions are needed to describe the sampling distributions of statistics computed from a normal sample. They form the basis of the confidence intervals in lecture 26 and of the standard tests in lecture 31. The first of these distributions also arose in the comparison of estimators for the variance in lecture 2. We start our discussion by considering the chi-squared distribution.

Definition A.2. Let $Z_1, \ldots, Z_k$ be independent, standard normally distributed random variables. Then the distribution of $Z_1^2 + \cdots + Z_k^2$ is called the *chi-squared distribution with k degrees of freedom.* This distribution is denoted by χ_k^2.

The first two moments of the chi-squared distribution are needed frequently, for example to find the bias and the variance of an estimator built from a sum of squares.

Lemma A.3. *The χ_k^2 distribution has mean k and variance $2k$.*

The chi-squared distribution is a special case of the gamma distribution: the χ_k^2 distribution corresponds to the Gamma($k/2, 1/2$) distribution from table A.2, *i.e.* it has density $x^{k/2-1}e^{-x/2}/(2^{k/2}\Gamma(k/2))$ for $x > 0$. In statistics, the parametrisation in terms of degrees of freedom is more commonly used. The most important property of the chi-squared distribution for applications in statistics is that the sample mean and sample variance, being the two most common statistics for normally distributed samples, have known and independent distributions.

Proposition A.4. *Let $X_1, \ldots, X_n$ be i.i.d. $N(\mu, \sigma^2)$ variables. Then the sample mean $\bar{X}$ and the sample variance*

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

satisfy

$$\bar{X} \sim N\Big(\mu,\ \frac{\sigma^2}{n}\Big), \qquad \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1},$$

and the two statistics $\bar{X}$ *and* S^2 *are independent.*

The independence of $\bar{X}$ and S^2 is the least obvious of these statements; like the rest of the proposition, we use it without proof.

We now construct two more distributions, using the normal distribution and the chi-squared distribution as building blocks. These distributions describe what happens when trying to estimate an unknown variance.

Definition A.5. Let $Z \sim N(0,1)$, $V \sim \chi^2_k$ and $W \sim \chi^2_l$ be independent. Then the t *distribution with* k *degrees of freedom* is the distribution of

$$T = \frac{Z}{\sqrt{V/k}}$$

denoted by t_k, and the F *distribution with* k *and* l *degrees of freedom* is the distribution of

$$F = \frac{V/k}{W/l},$$

denoted by $F_{k,l}$.

Using definition A.5 and proposition A.4 one can understand the origin of the t statistic for a normal sample: by dividing the standardised mean by the square root of the scaled sample variance, the unknown σ is replaced with its estimate and a t_{n-1} variable is obtained. We will see this argument in lecture 26.

A.4 Some Inequalities

A small number of basic inequalities are used throughout the module, mostly to estimate the probability of an estimator being away from its target. The most basic of these, Markov's inequality, is proved in lecture 2 as lemma 2.10. The following application of Markov's inequality converts a bound on the variance into a bound on the probability of a large deviation from the mean.

Lemma A.6 (Chebyshev's inequality)**.** *Let* X *have finite variance and* $a > 0$*. Then*

$$\mathbb{P}(|X - \mathbb{E}(X)| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}.$$

Proof. The event $|X - \mathbb{E}(X)| \geq a$ coincides with the event $(X - \mathbb{E}(X))^2 \geq a^2$. By Markov's inequality, lemma 2.10, for the non-negative variable $(X - \mathbb{E}(X))^2$ with threshold a^2, we get

$$\mathbb{P}(|X - \mathbb{E}(X)| \geq a) = \mathbb{P}((X - \mathbb{E}(X))^2 \geq a^2) \leq \frac{\mathbb{E}((X - \mathbb{E}(X))^2)}{a^2} = \frac{\mathrm{Var}(X)}{a^2}.$$

This is the required inequality. □

The following inequality, which establishes a relation between the expectation of a convex function and the function of the expectation, is the basis of the information inequality for maximum likelihood and of the ascent property in the EM algorithm.

Lemma A.7 (Jensen's inequality)**.** *Let g be convex and X be a random variable with finite mean. Then*

$$g(\mathbb{E}(X)) \leq \mathbb{E}(g(X)).$$

If g is strictly convex, then the inequality is strict unless X is constant with probability one.

Proof. Since g is convex, at the point $m = \mathbb{E}(X)$ there is a supporting line with a constant slope c, *i.e.* a function $g(x) \geq g(m) + c\,(x - m)$ for all x. Substituting X for x and taking expectations, the linear term has expectation zero since $\mathbb{E}(X - m) = 0$, and thus

$$\mathbb{E}(g(X)) \geq g(m) + c\,\mathbb{E}(X - m) = g(m) = g(\mathbb{E}(X)).$$

If g is strictly convex, the supporting line at m is unique and thus $g(x) > g(m) + c\,(x - m)$ for all $x \neq m$. In this case, if we have equality $\mathbb{E}(g(X)) = g(m)$, the non-negative random variable $g(X) - g(m) - c\,(X - m)$ has expectation zero and thus is zero with probability one, *i.e.* $X = m$ with probability one. This proves the claimed inequality and completes the proof. □

Finally, the Cauchy–Schwarz inequality can be used to estimate a covariance in terms of variances. This result is, for example, used in the derivation of the Cramér–Rao lower bound in lecture 13.

Lemma A.8 (Cauchy–Schwarz inequality)**.** *Let X and Y be random variables with finite variances. Then we have*

$$\operatorname{Cov}(X, Y)^2 \leq \operatorname{Var}(X)\ \operatorname{Var}(Y).$$

Proof. We can assume that both variances are positive, since otherwise one of the random variables is constant and both sides of the inequality equal zero. Let $U = X - \mathbb{E}(X)$ and $V = Y - \mathbb{E}(Y)$, so that $\operatorname{Cov}(X, Y) = \mathbb{E}(UV)$, $\operatorname{Var}(X) = \mathbb{E}(U^2)$ and $\operatorname{Var}(Y) = \mathbb{E}(V^2)$, and we want to show $\mathbb{E}(UV)^2 \leq \mathbb{E}(U^2)\,\mathbb{E}(V^2)$. For any real number t, the quantity $\mathbb{E}\big((U - tV)^2\big)$ is an expectation of a square and thus is non-negative. This gives

$$\mathbb{E}(U^2) - 2t\,\mathbb{E}(UV) + t^2\,\mathbb{E}(V^2) \geq 0$$

for all t. A quadratic polynomial which never takes a negative value has non-positive discriminant and thus we have $4\,\mathbb{E}(UV)^2 - 4\,\mathbb{E}(U^2)\,\mathbb{E}(V^2) \leq 0$ and thus we get the required inequality. This completes the proof. □

A.5 Conditional Expectation

When one random variable carries information about another, we summarise it through conditional expectation. Given random variables X and Y, the conditional expectation $\mathbb{E}(X\,|\,Y)$ is the expectation of X computed as if the value of Y were known; because that value is itself random, $\mathbb{E}(X\,|\,Y)$ is a function of Y, and hence a random variable in its own right. It is computed from the conditional distribution of X given Y in the same way as an ordinary expectation is computed from a distribution, and in particular it is linear in X. Its own average recovers the ordinary expectation of X.

Proposition A.9 (Tower property)**.** *Let X and Y be random variables, where X has finite mean. Then*

$$\mathbb{E}\big(\mathbb{E}(X\,|\,Y)\big) = \mathbb{E}(X).$$

Since the value of $\mathbb{E}(X\,|\,Y)$ is computed from the value of Y, we can pull out all factors which are functions of Y from the conditional expectation.

Proposition A.10 (Taking out what is known)**.** *Let X and Y be random variables and let h be a function such that $h(Y)X$ has finite mean. Then*

$$\mathbb{E}\big(h(Y)\,X\,|\,Y\big) = h(Y)\,\mathbb{E}(X\,|\,Y).$$

In particular $\mathbb{E}\big(h(Y)\,|\,Y\big) = h(Y)$.

Conditioning on Y can only reduce the variance on average. The exact argument is given by the following decomposition, which we will use in lecture 10 when we improve an estimator by conditioning.

Proposition A.11 (Law of total variance)**.** *Let X be a random variable with finite variance and let Y be any random variable. Then*

$$\operatorname{Var}(X) = \mathbb{E}\big(\operatorname{Var}(X\,|\,Y)\big) + \operatorname{Var}\big(\mathbb{E}(X\,|\,Y)\big),$$

where $\operatorname{Var}(X\,|\,Y) = \mathbb{E}\big(X^2\,|\,Y\big) - \mathbb{E}(X\,|\,Y)^2$ *denotes the conditional variance.*

The conditional variance is the variance of X computed from the conditional distribution of X given Y, and in particular it is non-negative.

Proof. Let $m = \mathbb{E}(X\,|\,Y)$ for the conditional mean. Using the tower property, proposition A.9, for X^2 we find $\mathbb{E}(X^2) = \mathbb{E}\big(\mathbb{E}(X^2 \mid Y)\big)$ and using the definition of the conditional variance we have $\mathbb{E}\big(X^2 \mid Y\big) = \operatorname{Var}(X\,|\,Y) + m^2$. Taking expectations we get

$$\mathbb{E}(X^2) = \mathbb{E}\big(\operatorname{Var}(X\,|\,Y)\big) + \mathbb{E}(m^2).$$

Using the tower property again, we know $\mathbb{E}(m) = \mathbb{E}(X)$ and thus, after subtracting $\mathbb{E}(X)^2 = \mathbb{E}(m)^2$ on both sides, we can write $\mathbb{E}(X^2) - \mathbb{E}(X)^2$ as $\operatorname{Var}(X)$ on the left-hand side and $\mathbb{E}(m^2) - \mathbb{E}(m)^2$ as $\operatorname{Var}(m)$ on the right-hand side. This is the required identity and completes the proof. □

A.6 Convergence and Limit Theorems

The large-sample theory of the module is based on two different concepts of a sequence of random variables converging to a limit. The weaker of the two concepts requires that the random variables are unlikely to differ from the limit by at least a fixed amount.

Definition A.12. A sequence of random variables X_n *converges in probability* to a limit X, written as $X_n \xrightarrow{\text{p}} X$, if for every $\varepsilon > 0$

$$\mathbb{P}\big(|X_n - X| \geq \varepsilon\big) \longrightarrow 0 \qquad (n \to \infty).$$

The weaker (but still useful) convergence of distribution functions is used to approximate a whole distribution:

Definition A.13. A sequence of random variables X_n with distribution functions F_n converges *in distribution* to a limit X with distribution function F, written as $X_n \xrightarrow{\text{d}} X$, if $F_n(x) \to F(x)$ for all points x where F is continuous.

The two great theorems of large-sample statistics describe the behaviour of a sample mean of independent, identically distributed observations. The first theorem identifies the limit of the sample mean, the second describes the size and shape of the fluctuations around the mean.

Proposition A.14 (Weak law of large numbers)**.** *Let $X_1, X_2, \ldots$ be i.i.d. with finite mean μ. Then the sample mean $\bar{X}_n$ satisfies $\bar{X}_n \xrightarrow{\text{p}} \mu$ as $n \to \infty$.*

The assumption of finite mean is important. The Cauchy distribution from table A.2 has no finite mean and the distribution of the sample mean is the same as the distribution of a single observation for all n. Thus, in this case the sample mean does not converge as $n \to \infty$.

Theorem A.15 (Central limit theorem)**.** *Let $X_1, X_2, \ldots$ be i.i.d. with mean μ and finite variance σ^2. Then*

$$\sqrt{n}\,(\bar{X}_n - \mu) \xrightarrow{\text{d}} N(0, \sigma^2) \qquad (n \to \infty).$$

In order to combine these limits we will need rules which allow us to manipulate limits. Slutsky's theorem allows us to treat a term which converges to a constant as the constant itself, and the continuous mapping theorem allows us to apply a continuous function to a limit.

Proposition A.16 (Slutsky's theorem)**.** *Let $X_n \xrightarrow{\text{d}} X$ and $Y_n \xrightarrow{\text{p}} c$ for some constant c. Then $X_n + Y_n \xrightarrow{\text{d}} X + c$ and $X_n Y_n \xrightarrow{\text{d}} cX$. If $c \neq 0$ we have $X_n / Y_n \xrightarrow{\text{d}} X/c$.*

Since convergence in probability to a constant is the same as convergence in distribution to this constant, the proposition also applies to sums, products and ratios of two sequences which converge in probability to constants, and the limit then holds in probability.

Proposition A.17 (Continuous mapping theorem)**.** *Let g be continuous at the possible values of X. Then $X_n \xrightarrow{\text{d}} X$ implies $g(X_n) \xrightarrow{\text{d}} g(X)$ and $X_n \xrightarrow{\text{p}} X$ implies $g(X_n) \xrightarrow{\text{p}} g(X)$.*

An important application of this result is to apply the central limit theorem after applying a smooth transformation to the data, in order to determine the limiting distribution of a function of an asymptotically normal statistic. This method is known as the delta method. We will use the delta method in lecture 17 to compute standard errors for functions of the maximum likelihood estimator.

Proposition A.18 (Delta method)**.** *Let $\sqrt{n}\,(T_n - \theta) \xrightarrow{\text{d}} N(0, \tau^2)$ and let g be a differentiable function at θ with $g'(\theta) \neq 0$. Then*

$$\sqrt{n}\,(g(T_n) - g(\theta)) \xrightarrow{\text{d}} N(0,\ g'(\theta)^2\,\tau^2).$$

These five results are given without proof, and are from the material covered in a first course in probability. We use the results as a ready-made toolkit.

A.7 Transformation of a Random Variable

We sometimes know the distribution of a random variable and want to find the distribution of a smooth, one-to-one transformation of this random variable. The change-of-variable formula is a mathematical tool which allows us to find the density of the transformed random variable. This formula is, for example, used when we reparametrise a model, as discussed in the context of the Jeffreys prior in lecture 28.

Proposition A.19 (Change of variable)**.** *Let X be a continuous random variable with density f_X and let $Y = g(X)$ for a differentiable, strictly monotone function g with inverse h and with $g'(x) \neq 0$ for all x. Then Y has density*

$$f_Y(y) = f_X\big(h(y)\big)\,|h'(y)|$$

for all y in the range of g.

Proof. We only consider the case where g is increasing, the proof for the decreasing case is similar with changes in sign. The distribution function of Y is given by

$$F_Y(y) = \mathbb{P}(g(X) \leq y) = \mathbb{P}(X \leq h(y)) = F_X\big(h(y)\big),$$

where we used that g is increasing and has inverse h. Taking derivatives with respect to y, using the chain rule, we find $f_Y(y) = f_X(h(y))\,h'(y)$ and thus $h'(y) > 0$ equals its own absolute value as given in the formula. This completes the proof. □

The same argument applies to a vector of random variables: if X has a joint density f_X on $\mathbb{R}^k$ and $Y = g(X)$ for a differentiable, one-to-one map g with inverse h, then Y has joint density $f_Y(y) = f_X\big(h(y)\big)\,|\det h'(y)|$, where $h'(y)$ is the Jacobian matrix of h. For a linear map with determinant one, as in the solution to exercise 10.3 in appendix C, the factor is 1.

A.8 Bayes' Theorem

The final tool reverses the direction of conditioning. For random variables A and B with joint density $f(a, b)$, the conditional density of A given $B = b$ is $f(a|b) = f(a, b)/f(b)$, the joint density divided by the marginal density of B. Bayes' theorem expresses this conditional density in terms of the reverse conditional. It is the mechanism behind every posterior distribution in the Bayesian lectures.

Proposition A.20 (Bayes' theorem)**.** *Let A and B be random variables with joint density $f(a, b)$ and let B have strictly positive marginal density $f(b)$. Then the conditional density of A given $B = b$ is*

$$f(a|b) = \frac{f(b|a)\,f(a)}{f(b)},$$

where $f(b) = \int f(b|a)\,f(a)\,\mathrm{d}a$ is the marginal density of B.

If one of the two variables is discrete, the same formula holds with its probability weights in place of the density, and the integral in $f(b)$ becomes a sum over the values of A when A is discrete.

In the Bayesian lectures of the module, the unknown parameter plays the role of A with prior density $f(a)$ and the data are given by B. In this situation, proposition A.20 states that the posterior density is proportional to the product of the likelihood and the prior density. The denominator $f(B)$ does not depend on the parameter. This viewpoint is further developed in lectures 28 and 29.

Appendix B

Introduction to R

This appendix contains a short, self-contained reference to the R programming language as used in this module. We assume that you can execute commands in R and understand the output. We collect all R commands used in this module in one place, for your convenience. The aim is to provide enough information so that unfamiliarity with the R programming language does not hold you back.

Commands to be typed by you are enclosed in a framed box, and the R output is shown below. Each line of the R output is marked with `##` so that it can be distinguished from the input.

B.1 Getting Started

At Leeds the easiest way to run R is via the AppsAnywhere service, which provides access to RStudio on university machines, either in the cloud or using a remote desktop. No installation is required to use this method. Alternatively, to run R on your own computer, you can install the two free programs R from the Comprehensive R Archive Network at https://cran.r-project.org/ and RStudio Desktop from https://posit.co/download/rstudio-desktop/. RStudio provides a much more comfortable interface for using R. You need to install R first, before you can install RStudio.

Once RStudio is open, the large pane on the left is the *console*. This is where we type a command and see the result immediately. For simple calculations, the console is enough, but for anything we want to store or re-run later, we have to type the commands into a *script*, a plain text file, which we can open by clicking `File > New File > R Script`. We can then send individual lines of this file to the console, and we can save the whole file to disk.

Using the console is like using a calculator: we type an expression, and R prints the result.

```
1 + 1
```

```
## [1] 2
```

The `[1]` at the start of the output is a position marker, not part of the answer: it says that the first value printed on that line, here 2, is the first element of the result. For a single number this looks redundant, but it helps

once a result consists of many values spread out over several lines, because then each line starts with the index of the first value on that line.

To store a value for later use, we *assign* a name to it using the arrow `<-`, and we can later type the name to see the value. A list of numbers can be built using `c`, which combines values to form a vector.

```
x <- c(3, 1, 4, 1, 5)
x
mean(x)
```

```
## [1] 3 1 4 1 5
## [1] 2.8
```

Assignments to `x` do not print anything (this is intentional). If we type `x` on its own, R prints the five values. If we type `mean(x)`, R prints the average of the five values.

You will learn many commands, but it is impossible to remember them all. Every built-in function has a help page, which can be accessed by typing a question mark before the function name.

```
?mean
```

The page always has the same structure: the *Usage* block shows how to call the function, the *Arguments* block shows what each input argument does, and the *Examples* block at the bottom shows lines which can be copied into the console to see the function in action. The structure of a help page is the key to understanding an unfamiliar command. For example, if you are asked to use a function which you have never seen before, *e.g.* the exponential random-number generator, you can type `?rexp` to get the line `rexp(n, rate = 1)`. This line explains that the function is called `rexp`, takes a count `n` as its first argument, and has a 'rate' argument with a default value. Once you have learned how to read a help page, you will find that this approach is more useful than trying to memorise individual commands.

B.2 Vectors and Arithmetic

Most quantities we consider here are *vectors*, *i.e.* ordered sequences of numbers, constructed using the R command `c`. We can access elements of a vector by specifying either a single index or a range of indices. We can also select elements based on a condition. In the following code, all the text after the single `#` on each line is a *comment*, ignored by R. We use comments to explain what each line of code does.

```
x <- c(3, 1, 4, 1, 5, 9, 2, 6)
x[1]          # the first element
x[2:4]        # elements 2 to 4
x[x > 3]      # the elements bigger than 3
```

```
## [1] 3
## [1] 1 4 1
## [1] 4 5 9 6
```

Arithmetic on vectors is an operation which acts on all elements simultaneously. As a result, loops over the elements of a vector are rarely needed.

```
x + 10        # add 10 to every element
x^2           # square every element
```

```
## [1] 13 11 14 11 15 19 12 16
## [1]  9  1 16  1 25 81  4 36
```

Other functions operate on a vector to compute a single value. The most useful of these are `length` and `sum`, which return the count and the total, respectively, and the standard descriptive statistics.

```
length(x)
sum(x)
mean(x)
median(x)
var(x)
sd(x)
```

```
## [1] 8
## [1] 31
## [1] 3.875
## [1] 3.5
## [1] 7.553571
## [1] 2.748376
```

The functions `var` and `sd` compute the sample variance, using the divisor $n-1$, and the square root of the sample variance, known as the sample standard deviation. The command `summary` can be used to print several of these values.

```
summary(x)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   1.000   1.750   3.500   3.875   5.250   9.000
```

The command `quantile` can be used to compute sample quantiles. By default the result includes percentage labels, as shown in the output of `summary`, but if the argument `names = FALSE` is used, the labels are omitted. This is useful if the values are to be used in later computations.

```
quantile(x, c(0.25, 0.75))
quantile(x, c(0.25, 0.75), names = FALSE)
```

```
##  25%  75%
## 1.75 5.25
## [1] 1.75 5.25
```

The command `diff` can be used to compute the differences of consecutive elements of a vector. The result is one element shorter than the vector, and it shows by how much a sequence of values changes from one step to the next.

```
diff(x)
```

```
## [1] -2  3 -3  4  4 -7  4
```

Results for different settings are often best read in tabular form. The commands `cbind` and `rbind` can be used to construct tables from vectors, by specifying the vectors as columns and rows, respectively. Names given in the call to either of these commands are used as column or row labels, respectively.

```
ns <- c(10, 20, 40)
vars <- c(0.1, 0.05, 0.025)
cbind(n = ns, variance = vars)
rbind(flat = c(1, 1), jeffreys = c(0.5, 0.5))
```

```
##       n variance
## [1,] 10    0.100
## [2,] 20    0.050
## [3,] 40    0.025
##          [,1] [,2]
## flat      1.0  1.0
## jeffreys  0.5  0.5
```

The command `cor` can be used to compute the correlation of two vectors. A covariance matrix can be converted into the corresponding matrix of correlations by `cov2cor`, which is how we read the correlation of two estimates off a numerical fit.

```
covmat <- matrix(c(4, 2, 2, 9), nrow = 2)
cov2cor(covmat)
```

```
##           [,1]      [,2]
## [1,] 1.0000000 0.3333333
## [2,] 0.3333333 1.0000000
```

Some of the data sets which come with R are stored as a *data frame*, a table whose columns have names. The command `with` can be used to compute with those columns without naming the table each time.

```
with(sleep, mean(extra))
```

```
## [1] 1.54
```

B.3 Lists

A vector holds values of one kind, and arithmetic can be performed on all values at once. A *list* is a second basic container which can hold objects of different kinds and different lengths, *e.g.* a number, a vector, a matrix, or even a piece of text. Usually a list is provided with names for its parts. An entry of a list is accessed by name, using a dollar sign followed by the name.

```
fit <- list(estimate = 2.5, se = 0.4, data = c(2.1, 2.9, 2.5))
fit$estimate
fit$estimate + 1.96 * fit$se
```

```
## [1] 2.5
## [1] 3.284
```

The command `names` can be used to see which parts a list has, and `str` can be used to print a compact summary of the whole list, giving the kind and the size of each part.

```
names(fit)
str(fit)
```

```
## [1] "estimate" "se"       "data"
## List of 3
##  $ estimate: num 2.5
##  $ se      : num 0.4
##  $ data    : num [1:3] 2.1 2.9 2.5
```

A part of a list can be used like an ordinary value of its type: `fit$data` is an ordinary vector, so we can use `length(fit$data)` to get the length of the vector as 3, and all the rules from the previous section apply to this vector. Arithmetic is not possible on the list as a whole, *e.g.* `fit * 2` is an error, since a list can contain parts which do not make sense when multiplied.

Lists are useful because many R commands return a list. For example, a command which fits a model returns an estimate, an uncertainty measure, and some information about the computation, all in a list. We can then take the relevant part of the list. In the section about numerical maximum likelihood, below, `optim` returns the results in a list and the estimate can be found as `fit$par`.

B.4 Random Numbers and Distributions

The basis of simulation is the use of random numbers. The names of the distribution functions in R are constructed using the following scheme: a one-letter prefix, followed by the name of the distribution. For the normal distribution (`norm`) the corresponding prefixes are `r` for random samples, `d` for the density, `p` for the cumulative distribution function and `q` for the inverse of the cumulative distribution function, *i.e.* the quantile function.

Since the simulation should be repeatable, we first have to fix the state of the random-number generator. This can be done by calling `set.seed` with a fixed seed.

```
set.seed(2026)
rnorm(5)          # five draws from the standard normal
set.seed(2026)
rnorm(5)          # the same seed gives the same draws
```

```
## [1]  0.52058907 -1.07969076  0.13923812 -0.08474878 -0.66663962
## [1]  0.52058907 -1.07969076  0.13923812 -0.08474878 -0.66663962
```

The following three prefixes answer the usual kind of questions about the same distribution.

```
dnorm(0)          # the N(0,1) density at 0
pnorm(1.96)       # the probability of being at most 1.96
qnorm(0.975)      # the 0.975 quantile
```

```
## [1] 0.3989423
## [1] 0.9750021
## [1] 1.959964
```

The same four prefixes can be used for most standard distributions: instead of `norm` we can write `binom`, `pois`, `exp`, `gamma` and so on. For example, the command `rpois` can be used to generate Poisson draws, where `rpois(5, lambda = 3)` gives five draws from the Poisson distribution with mean 3. Thus, for a task which requires binomial or geometric draws, it is easy to write `rbinom` or `rgeom` and then just look at the parameters listed on the help page. The geometric distribution illustrates the need to read the help page: authors disagree on whether the geometric distribution counts the number of trials until the first success, or the number of failures before the first success. These notes count the number of trials, so that the support of the distribution in table A.1 starts at 1. In contrast, the help page for `rgeom` states that R counts the number of failures and thus starts the output at 0 and is one below our sequence. To compensate for this, we can use the command `rgeom` to simulate the geometric distribution from these notes by adding one to each of the generated values.

```
rbinom(5, size = 10, prob = 0.3)   # five Binomial(10, 0.3) draws
rgeom(5, prob = 0.3) + 1           # five Geometric(0.3) draws, starting
     at 1
```

```
## [1] 0 4 2 4 2
## [1]  4 14  3  3  2
```

B.5 Writing Your Own Code

A simulation repeats the same steps many times, so we can combine these steps into a *function* and then call this function as needed. The inputs of a function are given in brackets, and the output is the value of the last line of the function.

```
simulate_mean <- function(n, mu) {
  obs <- rnorm(n, mean = mu)
  mean(obs)
}
set.seed(1)
simulate_mean(10, 2)       # one estimate from a sample of size 10
```

```
## [1] 2.132203
```

To repeat a command several times, we can use a `for`-loop, where we store the results in a vector.

```
set.seed(1)
estimates <- numeric(5)
```

```
for (i in seq_len(5)) {
  estimates[i] <- simulate_mean(10, 2)
}
estimates
```

```
## [1] 2.132203 2.248845 1.866327 2.120730 2.134137
```

The function `replicate` can be used to do the same thing in one line. This is the most commonly used tool for this purpose.

```
set.seed(1)
replicate(5, simulate_mean(10, 2))
```

```
## [1] 2.132203 2.248845 1.866327 2.120730 2.134137
```

We used `seq_len(5)` instead of `1:5` above. While both forms work for positive numbers, the difference becomes apparent when the count may be zero: `1:0` counts down and returns two values, whereas `seq_len(0)` is empty and a loop over it does nothing (the correct behaviour). Using `seq_len` in this way avoids a subtle bug.

```
1:0
seq_len(0)
```

```
## [1] 1 0
## integer(0)
```

A function often has to be applied to each element of a vector, for example to compute an estimate for each of several parameter values. The command `sapply` can be used for this: it applies the function to every element and collects the results in a vector. The function can be written directly inside the call, without being given a name.

```
set.seed(1)
thetas <- c(0.5, 1, 2)
sapply(thetas, function(theta) mean(rexp(1000, rate = theta)))
```

```
## [1] 2.0626054 0.9835434 0.5409070
```

The three values estimate the means $1/\theta$, which are 2, 1 and 0.5.

A log-likelihood written with `sum` returns one number for the whole sample. If such a function is handed a vector of parameter values, it still returns a single number, which is wrong, and there is no error message. The command `sapply` can be used to evaluate the function at each value of a grid instead.

```
obs <- c(0.7, 1.4, 0.2)
loglik <- function(theta) sum(dexp(obs, rate = theta, log = TRUE))
loglik(c(1, 2))
sapply(c(1, 2), loglik)
```

```
## [1] -3.006853
## [1] -2.300000 -2.520558
```

Some computations depend on the input and may take either of two forms. We can use the construction `if (condition) a else b` to select one of two values depending on the condition. The result of this construction is a value, so we can assign it to a variable or return it from a function.

```
half_width <- function(n) {
  if (n < 2) NA else qt(0.975, df = n - 1) / sqrt(n)
}
half_width(1)
half_width(10)
```

```
## [1] NA
## [1] 0.7153569
```

The construction `if` tests a single condition. For a vector of conditions, the command `ifelse(condition, a, b)` can be used to select values element by element: where the condition holds, the result takes the value from `a`, and elsewhere the value from `b`. In a mixture model, for example, this can be used to give each observation the mean of its group.

```
z <- c(1, 0, 0, 1, 0)
ifelse(z == 1, 0, 3)
```

```
## [1] 0 3 3 0 3
```

The command `break` can be used to leave a `for`-loop before its last pass, usually inside an `if` which checks whether the work is done. After the loop has ended, the loop variable keeps its value from the last pass. In the following example we add up $1 + 2 + 3 + \ldots$ until the total exceeds 20. Assigning a value to the position just after the end of a vector makes the vector one entry longer, and thus the vector `total` grows by one entry in each pass.

```
total <- 0
for (k in 1:100) {
  total[k + 1] <- total[k] + k
  if (total[k + 1] > 20) break
}
k
total
```

```
## [1] 6
## [1]  0  1  3  6 10 15 21
```

The loop stopped in the sixth pass, when the total reached 21, and the remaining 94 passes were never run.

The value returned by a function is normally the value of the last line, but the command `return` can be used to exit the function earlier. A common use of this command is to put a 'guard' at the top of a function which takes care of special cases not handled by the rest of the function. For example, a negative log-likelihood which is to be minimised can refuse negative parameter values.

```
nll <- function(theta) {
```

```
  if (theta <= 0) return(Inf)
  -sum(dexp(obs, rate = theta, log = TRUE))
}
obs <- c(0.7, 1.4, 0.2)
nll(-1)
nll(1)
```

```
## [1] Inf
## [1] 2.3
```

B.6 Estimation by Simulation

We can study an estimator without any algebra, by simulating a large number of samples, computing the estimate for each sample, and then summarising the results. This is the Monte Carlo method. We will consider an example where we estimate the mean μ of a normal distribution using the sample mean, where $\mu = 2$ and samples of size $n = 10$.

```
mu <- 2
n  <- 10
set.seed(2026)
estimates <- replicate(2000, mean(rnorm(n, mean = mu)))
```

The vector `estimates` now contains 2000 sample means. We can take the average of these values to get an estimate of the truth, and the difference between the average and the truth is a measure of the bias of our estimator. Another measure of the quality of our estimator is the mean squared error, which we can estimate by taking the average of the squared differences between the estimates and the truth. For this estimator we know the exact value $\sigma^2/n = 1/10$, so we can use this value to check our estimate of the mean squared error.

```
mean(estimates) - mu           # empirical bias
mean((estimates - mu)^2)       # empirical mean squared error
1 / n                          # the theoretical value
```

```
## [1] 0.0105696
## [1] 0.101612
## [1] 0.1
```

An average of N simulated values is itself random. Its standard error, the *Monte Carlo standard error*, is estimated by $s/\sqrt{N}$, where s is the sample standard deviation of the values, and it should be reported together with the average.

```
sd(estimates) / sqrt(2000)
```

```
## [1] 0.007125697
```

The empirical bias 0.011 is less than two standard errors away from zero, so the simulation is consistent with an unbiased estimator.

A similar approach can be used to estimate a probability. A logical condition, *e.g.* `abs(estimates - mu) <= 0.5`, results in a vector of true and false values.

Since R interprets these values as ones and zeros, the average of these values is the proportion of true values.

```
mean(abs(estimates - mu) <= 0.5)
```

```
## [1] 0.8765
```

This idiom, the mean of a condition, can be used to estimate any probability by simulation, *e.g.* how often a confidence interval covers the truth.

For simulated values which take only a few different values, such as draws from a discrete distribution, the command `table` can be used to count how often each value occurs. Dividing the counts by the number of values turns them into proportions, which estimate the probability weights.

```
counts <- c(2, 0, 3, 2, 2)
table(counts)
table(counts) / length(counts)
```

```
## counts
## 0 2 3
## 1 3 1
## counts
##   0   2   3
## 0.2 0.6 0.2
```

Given two vectors of the same length, `table` counts every combination of their values instead, in the form of a two-way table.

To watch an estimate settle as data accumulate, `cumsum` can be used to compute the running totals, and the running mean is then the quotient of the running total and the running count.

```
set.seed(1)
obs  <- rnorm(200, mean = mu)
running <- cumsum(obs) / seq_len(200)
running[c(1, 10, 50, 200)]
```

```
## [1] 1.373546 2.132203 2.100448 2.035540
```

The running mean of the samples converges to 2 as the number of samples increases.

A simulated answer is only as good as the code behind it, and the usual failure is a plausible but wrong number instead of an error message. Two cheap habits can help to avoid this: try a small case first, and check the result against something known. For example, the mean of many standard normals should be close to zero.

```
mean(rnorm(1e5))
```

```
## [1] -0.002290606
```

There are two more idioms which come up repeatedly in the R sessions. First, if the expression to `replicate` returns a vector instead of a single number, *e.g.*

the two ends of an interval, the result is a matrix where there is one column for each repetition, and one can select a row of this matrix using `ci[1, ]`. Secondly, if a loop is iterating through different settings, *e.g.* a vector `ns` of sample sizes, one can use `seq_along(ns)` to get the counter values for the positions in the vector, and one can use `data.frame` to summarise the results in tabular form.

```
set.seed(2026)
ci <- replicate(4, mean(rnorm(10, mean = mu)) + c(-1, 1) * 1.96 / sqrt
    (10))
ci
ci[1, ]
ns    <- c(10, 40, 160)
width <- numeric(length(ns))
for (i in seq_along(ns)) width[i] <- 2 * 1.96 / sqrt(ns[i])
data.frame(n = ns, width = width)
```

```
##           [,1]     [,2]     [,3]     [,4]
## [1,] 0.7999089 1.173516 1.949872 1.431851
## [2,] 2.0395218 2.413128 3.189484 2.671464
## [1] 0.7999089 1.1735155 1.9498716 1.4318507
##     n     width
## 1  10 1.2396128
## 2  40 0.6198064
## 3 160 0.3099032
```

To summarise the contents of such a matrix row by row, one can use `apply`: the call `apply(ci, 1, mean)` applies the `mean` function to each row of the matrix (the `1` indicates that we want to operate on the rows, `2` would indicate the columns), *i.e.* this computes the average of the lower and upper ends of the four intervals. For means and sums specifically, R provides `rowMeans`, `colMeans`, `rowSums` and `colSums`, which do the same job more directly and are used by us when the summary is one of these four.

```
apply(ci, 1, mean)
```

```
## [1] 1.338787 2.578400
```

B.7 Plotting, and Getting Results into a Report

We can use plots to inspect data and results. The main functions used to create plots are `hist` (for histograms), `plot` (for points or lines) and `curve` (for plotting a function). The functions `abline` (to add a straight line), `lines` (to add a curve) and `points` (to add points) can be used to add to an existing plot. As an example, we consider the histogram of a sample, together with the density of the sample and the sample mean.

```
set.seed(2026)
obs <- rnorm(500, mean = 2, sd = 1)
hist(obs, freq = FALSE, breaks = 20, col = "grey85", main = "", xlab = "
    x")
curve(dnorm(x, mean = 2, sd = 1), add = TRUE, lwd = 2)
```

```
abline(v = mean(obs), lty = 2)
```

To compare the density to the histogram, we set `freq = FALSE` to put the histogram on a density scale and `add = TRUE` to plot the curve on top of the histogram. The result is shown in figure B.1.

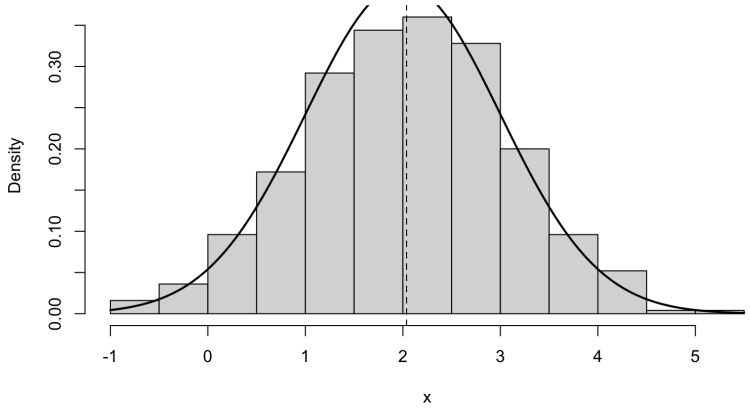


Figure B.1: A histogram on the density scale, with the true density overlaid and the sample mean marked by a dashed line.

When a plot shows several sets of points or lines, the command `legend` can be used to add a key. Its arguments are the position of the key, the labels and the plotting symbols (`pch`) or line types (`lty`) to show beside them, in the same order as the labels.

```
plot(1:10, (1:10)^2, pch = 16, xlab = "n", ylab = "value")
points(1:10, 10 * (1:10), pch = 1)
legend("topleft", legend = c("n^2", "10 n"), pch = c(16, 1))
```

To assess whether data are normally distributed, the function `qqnorm` can be used to plot the sorted values against normal quantiles. If the points lie close to a straight line, the data are approximately normally distributed.

```
qqnorm(obs)
```

To include the plot in a report, we have to save it to a file. We can open a file for graphics output using `pdf` (or `png`) and then use `dev.off` to close the file and indicate that all the output should go to the file instead of to the screen.

```
pdf("myplot.pdf", width = 6, height = 4)
hist(obs, freq = FALSE, breaks = 20)
curve(dnorm(x, mean = 2, sd = 1), add = TRUE)
dev.off()
```

In RStudio we can use the `Export` button above the plot pane to save the current figure. Numbers from the console can be copied and pasted into the report as they appear, and if we store the commands in a script, we can re-run the script to get the exact same result.

B.8 Numerical Maximum Likelihood

For many models the likelihood equation cannot be solved by hand, and we have to find the maximum likelihood estimate numerically. The tool we use here is the `optim` function from R, which *minimises* a function. This means that we have to give the negative of the log-likelihood function as an argument. As an example we consider here a gamma sample with known rate 1 and unknown shape parameter. For this example, the maximum likelihood estimate cannot be found analytically.

```
set.seed(2026)
obs <- rgamma(200, shape = 3, rate = 1)   # the true shape is 3
nll <- function(par, data) {
  -sum(dgamma(data, shape = par, rate = 1, log = TRUE))
}
fit <- optim(par = 1, fn = nll, data = obs, method = "BFGS", hessian =
    TRUE)
fit$par
```

```
## [1] 3.132089
```

A command like `optim` returns its results in a *list*. The individual results can be accessed by name, using a dollar sign as in `fit$par` above. The data reach `nll` through its second argument: any further named argument of `optim`, here `data = obs`, is passed on to the function, so that `nll` does not depend on a variable defined outside of it. In our experiment we use the starting value `par`, the method `"BFGS"` and `hessian = TRUE`, so that `optim` also returns the matrix of second derivatives of the negative log-likelihood at the optimum. This matrix is called the observed information. In the R command below, we use `solve` to invert this matrix and the square root of the diagonal elements to compute approximate standard errors. These can be used to compute a Wald confidence interval.

```
se <- sqrt(diag(solve(fit$hessian)))
se
fit$par + c(-1, 1) * 1.96 * se
```

```
## [1] 0.1153831
## [1] 2.905938 3.358240
```

The estimate 3.13 and its interval are close to the true shape 3, as expected. For a one-dimensional parameter, `optimise` is a simpler alternative to `optim`, which can be used to find the minimum in an interval, but does not return the information.

Appendix C

Solutions to the Exercises

This appendix contains worked solutions to the exercises found at the end of each lecture. The solutions are grouped by lecture and refer to the exercises by number.

C.1 Statistical Models and the Estimation Problem

Here we solve the exercises at the end of lecture 1.

Solution to exercise 1.1.

1. Since the tosses are independent and since each toss has probability θ of being heads, we can describe $X_1, \ldots, X_n$ as a sequence of independent and identically distributed Bernoulli trials with probability weights $f(x;\theta) = \theta^x(1-\theta)^{1-x}$ for $x \in \{0,1\}$ and parameter space $\Theta = (0,1)$.

2. We can use the proportion of heads in the sample, $\hat{\theta} = \bar{X} = (1/n)\sum_{i=1}^{n} X_i$, as an estimator.

3. The value $\hat{\theta}(X_1, \ldots, X_n)$ is a function of the random variables X_i before the coin tosses occur, so the value is still indeterminate. After the tosses have been made and the numbers $x_1, \ldots, x_n$ are known, $\hat{\theta}(x_1, \ldots, x_n)$ is a function of numbers, *i.e.* a single number.

□

Solution to exercise 1.2. A statistic is a function of the data which does not use the unknown parameter $\theta = (\mu, \sigma^2)$.

1. The range $\max_i X_i - \min_i X_i$ is computed from the data alone, so it is a statistic.

2. The value $(1/n)\sum_i (X_i - \mu)^2$ contains the unknown mean μ and thus cannot be computed from the data, so it is not a statistic. If μ was known, then it would be a statistic.

3. The sample variance $\big(1/(n-1)\big)\sum_i (X_i - \bar{X})^2$ uses the sample mean $\bar{X}$, which is computed from the data, in place of μ. Thus, it is a statistic.

□

Solution to exercise 1.3.

1. Since the observations are independent, the joint density is the product of the marginal densities:

$$f(x_1, \ldots, x_n; \theta) = \prod_{i=1}^{n} \frac{1}{\theta} = \frac{1}{\theta^n},$$

whenever all observations are in the support, *i.e.* when $0 \leq x_i \leq \theta$ for all i. Otherwise, the density is zero. This can also be written as the density being non-zero only when $\min_i x_i \geq 0$ and $\max_i x_i \leq \theta$.

2. Each observation satisfies $X_i \leq \theta$ with probability one, since the support is $[0, \theta]$. The maximum of numbers, none of which are larger than θ, cannot exceed θ. Thus, $\hat{\theta} = \max_i X_i \leq \theta$.

3. Since the distribution of each X_i is continuous, we have $\mathbb{P}_\theta(X_i = \theta) = 0$ for all i. The maximum equals θ only if at least one observation equals this value, and thus we get

$$\mathbb{P}_\theta(\hat{\theta} = \theta) \leq \sum_{i=1}^{n} \mathbb{P}_\theta(X_i = \theta) = 0.$$

Together with the first part of the solution this shows that $\hat{\theta} < \theta$ with probability one. Thus, the random variable $\theta - \hat{\theta}$ is non-negative and is zero with probability zero. A non-negative random variable with expectation zero equals zero with probability one, and thus we find $\mathbb{E}_\theta(\theta - \hat{\theta}) > 0$, which is the same as $\mathbb{E}_\theta(\hat{\theta}) < \theta$. Thus, the estimator never overestimates θ and underestimates θ on average.

□

C.2 Bias, Mean Squared Error and Consistency

Here we solve the exercises at the end of lecture 2.

Solution to exercise 2.1.

1. Since $\mathbb{E}_\theta(X_i) = \theta$, we can use lemma 2.2 to conclude $\mathbb{E}_\theta(\bar{X}) = \theta$. Thus we have $\text{bias}(\hat{\theta}) = \mathbb{E}_\theta(\bar{X}) - \theta = 0$ for all θ and the estimator $\hat{\theta}$ is unbiased.

2. Since the estimator is unbiased, the mean squared error equals the variance. Using lemma 2.2 and the fact that $\text{Var}_\theta(X_i) = \theta$ for the Poisson distribution, we find

$$\text{MSE}(\hat{\theta}) = \text{Var}_\theta(\bar{X}) = \frac{\theta}{n}.$$

Since this converges to zero as $n \to \infty$, we can use lemma 2.11 to conclude that the estimator is consistent.

□

Solution to exercise 2.2.

1. Using the given density

$$\mathbb{E}_\theta(\hat{\theta}) = \int_0^\theta t \cdot \frac{n\,t^{n-1}}{\theta^n}\,\mathrm{d}t = \frac{n}{\theta^n} \cdot \frac{\theta^{n+1}}{n+1} = \frac{n\theta}{n+1},$$

we find the bias to be $\operatorname{bias}(\hat{\theta}) = n\theta/(n+1) - \theta = -\theta/(n+1)$. Since this converges to zero as $n \to \infty$, the estimator is asymptotically unbiased. Since the bias is negative, the maximum systematically underestimates θ, as expected from exercise 1.3.

2. We have

$$\mathbb{E}_\theta(\hat{\theta}^2) = \int_0^\theta t^2 \cdot \frac{n\,t^{n-1}}{\theta^n}\,\mathrm{d}t = \frac{n\theta^2}{n+2},$$

and thus

$$\operatorname{Var}(\hat{\theta}) = \frac{n\theta^2}{n+2} - \Big(\frac{n\theta}{n+1}\Big)^2 = \frac{n\theta^2}{(n+2)(n+1)^2},$$

where we used $(n+1)^2 - n(n+2) = 1$ in the last step. By lemma 2.7 we find

$$\begin{aligned}\operatorname{MSE}(\hat{\theta}) &= \operatorname{Var}(\hat{\theta}) + \operatorname{bias}(\hat{\theta})^2 \\ &= \frac{n\theta^2}{(n+2)(n+1)^2} + \frac{\theta^2}{(n+1)^2} \\ &= \frac{2\theta^2}{(n+1)(n+2)}.\end{aligned}$$

Since this converges to zero as $n \to \infty$, we can conclude that $\hat{\theta}$ is consistent by lemma 2.11.

3. The mean squared error is of order $1/n^2$, while the mean squared error of the sample mean is of order $1/n$. Thus, the maximum converges to the target faster than the average does. The very fast convergence is caused by the support of the uniform model changing as a function of the parameter. This effect, mentioned in example 1.2, will be seen again in lecture 13.

□

Solution to exercise 2.3. Since $\hat{\theta} = n/T$, we have $\mathbb{E}_\theta(\hat{\theta}) = n\,\mathbb{E}_\theta(1/T)$ and using the gamma density of T we get

$$\mathbb{E}_\theta(1/T) = \int_0^\infty \frac{1}{x} \cdot \frac{\theta^n x^{n-1} e^{-\theta x}}{(n-1)!}\,\mathrm{d}x = \frac{\theta^n}{(n-1)!} \int_0^\infty x^{n-2} e^{-\theta x}\,\mathrm{d}x.$$

For $n \geq 2$, the remaining integral can be evaluated as the gamma integral $\int_0^\infty x^{n-2} e^{-\theta x}\,\mathrm{d}x = (n-2)!/\theta^{n-1}$. Thus we get

$$\mathbb{E}_\theta(\hat{\theta}) = \frac{n\,\theta^n}{(n-1)!} \cdot \frac{(n-2)!}{\theta^{n-1}} = \frac{n\theta}{n-1}.$$

The bias is given by

$$\operatorname{bias}(\hat{\theta}) = \frac{n\theta}{n-1} - \theta = \frac{\theta}{n-1}.$$

Since this is strictly positive, we see that $\hat{\theta}$ systematically overestimates θ on average and thus is biased. As $n \to \infty$, the bias converges to zero and thus $\hat{\theta}$ is asymptotically unbiased. This completes the solution. □

C.3 Likelihood and the Method of Moments

Here we solve the exercises at the end of lecture 4.

Solution to exercise 4.1.

1. Using the Poisson weights from table 1.1, we find the likelihood to be

$$L(\theta) = \prod_{i=1}^{n} \frac{\theta^{x_i}}{x_i!} e^{-\theta} = \frac{\theta^{\sum_i x_i}}{\prod_i x_i!} e^{-n\theta}.$$

   Taking logarithms we get

$$\ell(\theta) = \log\theta \sum_{i=1}^{n} x_i - n\theta - \sum_{i=1}^{n} \log(x_i!),$$

   as claimed.

2. Taking derivatives with respect to θ, and noting that the last term does not depend on θ, we find the score function to be

$$\ell'(\theta) = \frac{1}{\theta} \sum_{i=1}^{n} x_i - n.$$

   Setting this equal to zero and solving gives $\theta = (1/n)\sum_i x_i = \bar{x}$. Thus, the score function is zero at the sample mean.

3. For the Poisson distribution we have $\mu_1(\theta) = \mathbb{E}_\theta(X_1) = \theta$. Thus, the only moment equation we have is $\hat{\theta} = \bar{X}$: the method of moments estimator is the sample mean. This is the value at which the score function is zero, *i.e.* for the Poisson distribution the method of moments coincides with the maximum likelihood estimator from lecture 5.

□

Solution to exercise 4.2.

1. Since we have $\mu_1(p) = \mathbb{E}_p(X_1) = 1/p$ and the parameter is scalar, we get the moment equation $1/\hat{p} = \bar{X}$. Solving this equation we find

$$\hat{p} = \frac{1}{\bar{X}}.$$

2. Since each observation satisfies $X_i \geq 1$, we have $\bar{X} \geq 1$, with equality if and only if all observations equal one. Thus, the interval of $\hat{p} = 1/\bar{X}$ values where the estimate is possible is $(0, 1]$. Since for every $p \in (0, 1)$ the dataset with values in $\mathbb{N}$ has positive probability, no estimate in this range is incompatible with the data (this is in contrast to the example of the uniform distribution in the text).

□

Solution to exercise 4.3.

1. From table A.2 we know that $\mathbb{E}_\theta(X_1) = \mu$ and $\mathrm{Var}_\theta(X_1) = \sigma^2$. Thus, the first population moment is $\mu_1(\theta) = \mu$. Rewriting the relation $\mathrm{Var}(X_1) = \mathbb{E}(X_1^2) - \mathbb{E}(X_1)^2$ we find

$$\mu_2(\theta) = \mathbb{E}_\theta(X_1^2) = \mathrm{Var}_\theta(X_1) + \mathbb{E}_\theta(X_1)^2 = \sigma^2 + \mu^2,$$

as claimed.

2. Using the moments from the first part, the moment equations from definition 4.6 are

$$\hat{\mu} = m_1 = \bar{X} \qquad \text{and} \qquad \hat{\sigma}^2 + \hat{\mu}^2 = m_2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2.$$

From the first equation we find $\hat{\mu} = \bar{X}$ directly. Substituting this value into the second equation we find

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \bar{X}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2,$$

where the last equality sign comes from expanding the square $(X_i - \bar{X})^2 = X_i^2 - 2X_i\bar{X} + \bar{X}^2$ and using the relation $\sum_i X_i = n\bar{X}$. These are the estimators given in example 4.8 and thus the solution is complete.

□

Solution to exercise 4.4.

1. From table A.2 we know that the uniform model has $\mathbb{E}_\theta(X_1) = \theta/2$ and $\mathrm{Var}_\theta(X_1) = \theta^2/12$. Using lemma 2.2 we find $\mathbb{E}_\theta(\hat{\theta}) = 2\,\mathbb{E}_\theta(\bar{X}) = 2 \cdot \theta/2 = \theta$ and thus the estimator $\hat{\theta}$ is unbiased. Since the estimator is unbiased, the mean squared error equals the variance by lemma 2.7, and using lemma 2.2 again we get

$$\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(2\bar{X}) = 4 \cdot \frac{\theta^2/12}{n} = \frac{\theta^2}{3n}.$$

2. The maximum has smaller mean squared error, if and only if

$$\frac{2\theta^2}{(n+1)(n+2)} < \frac{\theta^2}{3n}.$$

This inequality is equivalent to $6n < (n+1)(n+2)$, *i.e.* to $0 < n^2 - 3n + 2 = (n-1)(n-2)$ for all $n \geq 3$. (The equality signs hold for $n = 1$ and $n = 2$.) Thus, for three or more observations the biased maximum is better than the unbiased method of moments estimator, and the $1/n^2$ rate of the maximum makes the difference increase quickly with n. This is another example of the lesson from lecture 2 that unbiasedness is not sufficient: the moment estimator squashes the data into the sample mean, whereas the maximum uses the fact that the support of the distribution moves with the parameter.

□

Solution to exercise 4.5.

1. The distribution $N(0, \theta^2)$ depends on θ only via θ^2. Thus, for different parameter values θ and $-\theta$, the distribution is the same, *e.g.* $\theta_1 = 1$ and $\theta_2 = -1$ both lead to $N(0, 1)$. Thus, the model is not identifiable.

2. To make the model identifiable, we can restrict ourselves to $\Theta' = (0, \infty)$: for $0 < \theta_1 < \theta_2$ the distributions $N(0, \theta_1^2)$ and $N(0, \theta_2^2)$ have different variances and thus are different. (The other half line $(-\infty, 0)$ would work equally well.)

3. Since the first moment $\mathbb{E}_\theta(X_1) = 0$ for all θ, we cannot use this moment to derive any information about the parameter. We can use the second moment instead. Since $\mu_2(\theta) = \mathbb{E}_\theta(X_1^2) = \theta^2$, the moment equation is $\hat{\theta}^2 = (1/n) \sum_i X_i^2$ and taking the positive root, as required for $\Theta' = (0, \infty)$, we get

$$\hat{\theta} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} X_i^2}\,.$$

   This completes the solution.

□

C.4 Maximum Likelihood Estimation

Here we solve the exercises at the end of lecture 5.

Solution to exercise 5.1.

1. Using the geometric weights, the likelihood function is

$$L(p) = \prod_{i=1}^{n} (1-p)^{x_i - 1} p = p^n \, (1-p)^{\sum_i x_i - n}.$$

   Taking logarithms we get $\ell(p) = n \log p + \left(\sum_i x_i - n\right) \log(1-p)$, as claimed.

2. Taking derivatives we find the score as

$$\ell'(p) = \frac{n}{p} - \frac{\sum_i x_i - n}{1-p}.$$

   Setting this equal to zero and clearing the denominators we find $n(1-p) = p(\sum_i x_i - n)$, *i.e.* $n = p \sum_i x_i$. The only solution to this equation is $p = n / \sum_i x_i = 1/\bar{x}$. Since all observations satisfy $x_i \geq 1$, we have $\sum_i x_i \geq n$ and thus

$$\ell''(p) = -\frac{n}{p^2} - \frac{\sum_i x_i - n}{(1-p)^2} < 0$$

   for all $p \in (0, 1)$. Thus, ℓ is strictly concave and the root is the global maximum. Thus, the MLE is $\hat{p} = 1/\bar{X}$. (If all observations equal one, *i.e.* if $\sum_i x_i = n$, then $\ell(p) = n \log p$ is strictly increasing and no maximiser exists in the open interval $(0, 1)$. The same boundary phenomenon as for the Bernoulli distribution in the text occurs. In this case, the formula $\hat{p} = 1/\bar{X}$ gives the boundary value 1.)

3. In exercise 4.2 we have found the method of moments estimator to be $\hat{p} = 1/\bar{X}$. Thus, for the geometric distribution, as for the Bernoulli, Poisson and exponential distributions, the two different construction methods coincide.

□

Solution to exercise 5.2.

1. The log-likelihood function is

$$\ell(\theta) = -2n \log \theta + \sum_{i=1}^{n} \log x_i - \frac{1}{\theta} \sum_{i=1}^{n} x_i,$$

and the score function is

$$\ell'(\theta) = -\frac{2n}{\theta} + \frac{1}{\theta^2} \sum_{i=1}^{n} x_i.$$

Setting the score equal to zero, we find $\sum_i x_i = 2n\theta$ and thus the only root is $\theta = \bar{x}/2$. Since the score is positive for $\theta < \bar{x}/2$ and negative for $\theta > \bar{x}/2$, this root is the global maximum and the MLE is $\hat{\theta} = \bar{X}/2$.

2. Since the model is described by $\text{Gamma}(2, 1/\theta)$ in the rate parametrisation, we can use table A.2 to find $\mathbb{E}_\theta(X_1) = 2\theta$. Using lemma 2.2 we find $\mathbb{E}_\theta(\bar{X}) = 2\theta$ and thus $\mathbb{E}_\theta(\hat{\theta}) = (1/2) \cdot 2\theta = \theta$ for all θ. Thus, $\hat{\theta}$ is unbiased.

3. From table A.2 we find the variance to be $\text{Var}_\theta(X_1) = 2\theta^2$. Since the estimator is unbiased, the mean squared error equals the variance and using lemma 2.2 again, we find

$$\text{MSE}(\hat{\theta}) = \text{Var}\Big(\frac{\bar{X}}{2}\Big) = \frac{1}{4} \cdot \frac{2\theta^2}{n} = \frac{\theta^2}{2n}.$$

Since this converges to zero as $n \to \infty$, we can conclude that the estimator is consistent by lemma 2.11.

□

Solution to exercise 5.3.

1. Since $\hat{\theta} = n/T$ and since T has gamma density $\theta^n x^{n-1} e^{-\theta x}/(n-1)!$ on $(0, \infty)$, we get the same integral as in exercise 2.3, but with one fewer power of x, *i.e.* for $n \geq 3$ we have

$$\begin{aligned}\mathbb{E}_\theta(\hat{\theta}^2) &= n^2 \int_0^\infty \frac{1}{x^2} \cdot \frac{\theta^n x^{n-1} e^{-\theta x}}{(n-1)!} \, \mathrm{d}x \\ &= \frac{n^2 \theta^n}{(n-1)!} \cdot \frac{(n-3)!}{\theta^{n-2}} = \frac{n^2}{(n-1)(n-2)} \theta^2,\end{aligned}$$

as claimed.

2. In the text we have seen that $\mathbb{E}_\theta(\hat\theta) = n\theta/(n-1)$. Using this value and the second moment from the previous part of the question, we get

$$\begin{aligned}\operatorname{Var}_\theta(\hat\theta) = \mathbb{E}_\theta(\hat\theta^2) - \mathbb{E}_\theta(\hat\theta)^2 &= \frac{n^2\theta^2}{(n-1)(n-2)} - \frac{n^2\theta^2}{(n-1)^2} \\ &= \frac{n^2\theta^2\big((n-1)-(n-2)\big)}{(n-1)^2(n-2)} = \frac{n^2\theta^2}{(n-1)^2(n-2)}.\end{aligned}$$

Since $\tilde\theta = (n-1)\hat\theta/n$, we find

$$\operatorname{Var}_\theta(\tilde\theta) = \frac{(n-1)^2}{n^2}\operatorname{Var}_\theta(\hat\theta) = \frac{\theta^2}{n-2}.$$

This completes the solution.

□

Solution to exercise 5.4.

1. Up to the factor $\nu+1$, which we can cancel, the score is $a/(\nu+a^2) + b/(\nu+b^2)$. Multiplying through by the positive quantity $(\nu+a^2)(\nu+b^2)$, the likelihood equation becomes

$$a(\nu+b^2) + b(\nu+a^2) = \nu(a+b) + ab(a+b) = (a+b)(\nu+ab) = 0,$$

as claimed.

2. For $\theta = m+t$ we have $a = d-t$ and $b = -d-t$, and up to an additive constant the log-likelihood is $-\frac{\nu+1}{2}\big(\log(\nu+a^2)+\log(\nu+b^2)\big)$. The product inside the logarithms is

$$\begin{aligned}(\nu+a^2)(\nu+b^2) &= (\nu+d^2+t^2-2dt)(\nu+d^2+t^2+2dt) \\ &= (\nu+d^2+t^2)^2 - 4d^2t^2 = (\nu+d^2)^2 + 2(\nu-d^2)\,t^2 + t^4,\end{aligned}$$

which is the polynomial $P(t)$ and thus $\ell(m+t) = C - \frac{\nu+1}{2}\log P(t)$.

3. Since the logarithm is increasing, $\ell(m+t)$ takes its largest value when $P(t)$ is smallest. Let $u = t^2 \geq 0$. Then we have $P = (\nu+d^2)^2 + 2(\nu-d^2)u + u^2$, *i.e.* we get a quadratic function of u with minimum at $u = d^2-\nu$. If $d^2 < \nu$, this minimum is at a negative value of u and thus P is increasing on $u \geq 0$ and takes its minimum at $t = 0$. The midpoint m is then the only maximiser of ℓ and, since the factor $\nu+ab = \nu-d^2+t^2$ from the first part of the solution has no real zero, m is the only root of the likelihood equation. If $d^2 > \nu$, then P takes its minimum at $u = d^2-\nu$, *i.e.* at $t = \pm\sqrt{d^2-\nu}$ which are the zeros of $\nu+ab$. These two values are the maximum likelihood estimates, and since P is even in t, they have equal likelihood. Between these two values, at $t = 0$, the polynomial P has a local maximum, because the coefficient $2(\nu-d^2)$ of t^2 is negative. Thus, ℓ has a local minimum at the midpoint. This completes the proof.

□

Solution to exercise 5.5. Since g is a bijection, there exists an inverse function $g^{-1}\colon \Psi \to \Theta$ and we can re-index the model, using ψ instead of θ, *i.e.* if we let $\tilde{f}(x;\psi) = f(x;g^{-1}(\psi))$ the family of distributions is unchanged, but the likelihood function is $\tilde{L}(\psi) = L(g^{-1}(\psi))$. For any $\psi \in \Psi$, the value $g^{-1}(\psi)$ falls into the set Θ and by the definition of $\hat{\theta}$ we have

$$\tilde{L}(\psi) = L(g^{-1}(\psi)) \leq L(\hat{\theta}) = \tilde{L}(g(\hat{\theta})).$$

Thus, $g(\hat{\theta})$ is a maximum of the function $\tilde{L}$ on the set Ψ and consequently is a maximum likelihood estimator for ψ. This completes the proof. □

Solution to exercise 5.6.

1. We have $s+t \geq s$ and thus the event $\{X > s+t\}$ is contained in the event $\{X > s\}$. We get

$$\mathbb{P}_\theta(X > s+t | X > s) = \frac{\mathbb{P}_\theta(X > s+t)}{\mathbb{P}_\theta(X > s)} = \frac{e^{-\theta(s+t)}}{e^{-\theta s}} = e^{-\theta t} = \mathbb{P}_\theta(X > t),$$

   i.e. the unconditional probability to survive beyond time t. We can conclude from this that, if a lifetime has already passed time s, the remaining lifetime has the same distribution as a fresh lifetime does. In other words, the time already elapsed does not change the distribution of the remaining lifetime.

2. The conditional density of X given the event $\{X > c\}$ is the density restricted to the set (c, ∞), normalised to be a density again. We get $f(x;\theta)/\mathbb{P}_\theta(X > c)$ for $x > c$. Thus we have

$$\mathbb{E}_\theta(X | X > c) = \frac{1}{e^{-\theta c}} \int_c^\infty x\, \theta e^{-\theta x}\, \mathrm{d}x.$$

   Using integration by parts, with $u = x$ and $v' = \theta e^{-\theta x}$, we find $v = -e^{-\theta x}$ and thus

$$\int_c^\infty x\, \theta e^{-\theta x}\, \mathrm{d}x = \left[-x\, e^{-\theta x}\right]_c^\infty + \int_c^\infty e^{-\theta x}\, \mathrm{d}x = c\, e^{-\theta c} + \frac{e^{-\theta c}}{\theta}.$$

   Dividing by $e^{-\theta c}$ gives $\mathbb{E}_\theta(X | X > c) = c + 1/\theta$ as required. This result is consistent with the first part of the solution: given survival until time c, the remaining lifetime $X - c$ is distributed like a fresh exponential lifetime. Thus, the conditional expectation is the unconditional mean $1/\theta$, plus the time c already elapsed, giving $c + 1/\theta$. This completes the solution.

□

Solution to exercise 5.7.

1. From the solution to exercise 4.1 we know that the score is given by $\ell'(\theta) = (1/\theta)\sum_i x_i - n$ and this score equals zero at $\theta = \bar{x}$. Let $S = \sum_i x_i \geq 1$. Then the second derivative is $\ell''(\theta) = -S/\theta^2 < 0$ for all $\theta > 0$ and thus ℓ is strictly concave. This shows that the root is the unique global maximum and thus the MLE is given by $\hat{\theta} = \bar{X}$.

2. The map $g(\theta) = e^{-\theta}$ is continuous and strictly decreasing and therefore maps $(0, \infty)$ onto $(0, 1)$ bijectively. Thus, by proposition 5.6, the MLE for $p_0 = g(\theta)$ is given by
$$\hat{p}_0 = e^{-\hat{\theta}} = e^{-\bar{X}}.$$

3. Using the independence of the observations and the given identity with $z = e^{-1/n}$ we get
$$\mathbb{E}_\theta(e^{-\bar{X}}) = \mathbb{E}_\theta\Big(\prod_{i=1}^n e^{-X_i/n}\Big) = \prod_{i=1}^n \mathbb{E}_\theta\Big((e^{-1/n})^{X_i}\Big) = \exp(n\theta(e^{-1/n} - 1)).$$
Since $e^u > 1 + u$ for all $u \neq 0$, we can choose $u = -1/n$ to get $e^{-1/n} - 1 > -1/n$ and thus $n(e^{-1/n} - 1) > -1$. This gives
$$\mathbb{E}_\theta(\hat{p}_0) = \exp(n\theta(e^{-1/n} - 1)) > e^{-\theta} = p_0,$$
and thus the estimator systematically overestimates p_0 on average and is biased for all n. On the other hand, expanding the exponential we find $n(e^{-1/n} - 1) = -1 + 1/(2n) - \cdots \to -1$ as $n \to \infty$ and thus $\mathbb{E}_\theta(\hat{p}_0) \to e^{-\theta}$ and the estimator is asymptotically unbiased. There is no contradiction to the unbiasedness of $\bar{X}$: unbiasedness is not preserved under the nonlinear map g, as discussed after proposition 5.6, whereas the maximum likelihood property is.

□

Solution to exercise 5.8. From lemma 2.2 we know $\mathbb{E}_\theta(\bar{X}) = \mathbb{E}_\theta(X_1) = 1/\theta$, and thus $\mathbb{E}_\theta(\hat{m}) = (\log 2)/\theta = m$ for all θ. This shows that $\hat{m}$ is unbiased. There is no contradiction: unbiasedness is a property of a parametrisation and is in general lost under nonlinear transformations of the parameter. Thus, the bias of the MLE $\hat{\theta} = 1/\bar{X}$ for the rate is not relevant for the bias of the estimator $\hat{m} = (\log 2)\,\bar{X}$ for the median, which is a linear function of $\bar{X}$. This completes the solution. □

C.5 Exponential Families

Solution to exercise 7.1.

1. We can write $x^{\alpha-1} = e^{(\alpha-1)\log x}$ and, collecting all factors into one exponential, for $x > 0$ we find
$$\begin{aligned} f(x; \alpha, \beta) &= \frac{\beta^\alpha}{\Gamma(\alpha)}\, e^{(\alpha-1)\log x}\, e^{-\beta x} \\ &= \exp\Big((\alpha - 1)\log x + (-\beta)\, x - \big(\log\Gamma(\alpha) - \alpha \log \beta\big)\Big). \end{aligned}$$
This is the form of definition 7.6 with $k = 2$, with $h(x) = 1$ for $x > 0$ and $h(x) = 0$ otherwise, natural statistic $(T_1(x), T_2(x)) = (\log x,\, x)$, natural parameter $(\eta_1(\alpha, \beta), \eta_2(\alpha, \beta)) = (\alpha - 1,\, -\beta)$ and $B(\alpha, \beta) = \log\Gamma(\alpha) - \alpha \log \beta$.

2. As α varies over $(0, \infty)$, the first coordinate $\eta_1 = \alpha - 1$ varies over $(-1, \infty)$ and as β varies over $(0, \infty)$, the second coordinate $\eta_2 = -\beta$ varies over $(-\infty, 0)$, independently of each other. Thus, the set of natural parameter values is the open quadrant $\{(\eta_1, \eta_2) \mid \eta_1 > -1,\ \eta_2 < 0\}$. This is a non-empty open subset of $\mathbb{R}^2$ and the family has full rank as in definition 7.8.

3. Knowing α, we can include the factor $x^{\alpha-1}$ into the function h: for $x > 0$ we have

$$f(x; \beta) = x^{\alpha-1} \exp\Big((-\beta)\, x - \big(\log \Gamma(\alpha) - \alpha \log \beta\big)\Big).$$

This is a one-parameter exponential family in the sense of definition 7.1, with $h(x) = x^{\alpha-1}$ for $x > 0$, natural statistic $T(x) = x$ and natural parameter $-\beta$.

□

Solution to exercise 7.2.

1. Using the notation $\mu = \theta$ and $\sigma^2 = \theta^2$ from example 7.7, we see that the density of $N(\theta, \theta^2)$ is of the form given in definition 7.6 with $k = 2$, where $h(x) = 1$, the natural statistic is (x, x^2) and the natural parameter is

$$\Big(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2}\Big) = \Big(\frac{1}{\theta}, -\frac{1}{2\theta^2}\Big).$$

Since the parameter θ is one-dimensional but $k = 2$, this is a curved exponential family.

2. Let $\eta_1 = 1/\theta$ and $\eta_2 = -1/(2\theta^2)$. As θ varies over the interval $(0, \infty)$, the first coordinate η_1 also varies over the interval $(0, \infty)$. Substituting $\theta = 1/\eta_1$ for the second coordinate, we find

$$\eta_2 = -\frac{1}{2\theta^2} = -\frac{\eta_1^2}{2}.$$

Thus, the set of natural parameter values is the curve $\big\{(\eta_1, -\eta_1^2/2) \mid \eta_1 > 0\big\}$. Any open subset of $\mathbb{R}^2$ which contains a point on the curve also contains a small disc around this point, and thus points off the curve, obtained by slightly changing η_2 while keeping η_1 fixed. Thus the curve does not contain any non-empty open subsets of $\mathbb{R}^2$. By definition 7.8, the family is not of full rank, as claimed.

□

Solution to exercise 7.3.

1. For $x \in \mathbb{N}$ we can write

$$\begin{aligned} f(x; p) = (1-p)^{x-1} p &= \exp\big((x-1)\log(1-p) + \log p\big) \\ &= \exp\Big(x \log(1-p) - \log \frac{1-p}{p}\Big). \end{aligned}$$

This is the form of definition 7.1 with $h(x) = 1$ for $x \in \mathbb{N}$ and $h(x) = 0$ otherwise, natural statistic $T(x) = x$, natural parameter $\eta(p) = \log(1-p)$ and $B(p) = \log\big((1-p)/p\big)$. By lemma 7.5 the natural statistic of a sample is $T_n = \sum_{i=1}^n X_i$.

2. For $\eta < 0$ the geometric series gives

$$\sum_{x=1}^{\infty} e^{\eta x} = \frac{e^\eta}{1 - e^\eta},$$

but for $\eta \geq 0$ the series diverges. Thus we have $\mathcal{N} = (-\infty, 0)$ and

$$K(\eta) = \eta - \log(1 - e^\eta), \qquad \eta < 0.$$

Substituting $\eta(p) = \log(1 - p)$, so that $e^\eta = 1 - p$, we get $K(\eta(p)) = \log(1 - p) - \log p = B(p)$ as required. As p ranges over $(0, 1)$, the natural parameter $\log(1-p)$ ranges over the entire open interval $(-\infty, 0)$, and thus the family has full rank.

3. We have

$$K'(\eta) = 1 + \frac{e^\eta}{1 - e^\eta} = \frac{1}{1 - e^\eta} \qquad \text{and} \qquad K''(\eta) = \frac{e^\eta}{(1 - e^\eta)^2}.$$

With $e^\eta = 1 - p$, by lemma 7.10, we find $\mathbb{E}_p(X) = 1/p$ and $\mathrm{Var}_p(X) = (1 - p)/p^2$, the values given in table A.1. The mean $1/p$ is the moment considered in exercise 4.2.

□

Solution to exercise 7.4.

1. For $x \in \{0, 1, \ldots, m\}$ we can write

$$f(x; \theta) = \binom{m}{x} \theta^x (1 - \theta)^{m-x} = \binom{m}{x} \exp\Big(x \log \frac{\theta}{1 - \theta} + m \log(1 - \theta)\Big).$$

This is the form of definition 7.1 with $h(x) = \binom{m}{x}$ for $x \in \{0, 1, \ldots, m\}$ and $h(x) = 0$ otherwise, natural statistic $T(x) = x$, natural parameter $\eta(\theta) = \log(\theta/(1 - \theta))$, the same log-odds as in example 7.2, and $B(\theta) = -m \log(1 - \theta)$.

2. Using the binomial theorem, we find

$$\sum_{x=0}^{m} \binom{m}{x} e^{\eta x} = (1 + e^\eta)^m$$

for all $\eta \in \mathbb{R}$. Thus, $\mathcal{N} = \mathbb{R}$ and $K(\eta) = m \log(1 + e^\eta)$. This is m times the cumulant function of the Bernoulli distribution.

3. Taking derivatives, we get

$$K'(\eta) = \frac{me^\eta}{1 + e^\eta} \qquad \text{and} \qquad K''(\eta) = \frac{me^\eta}{(1 + e^\eta)^2}.$$

Since $e^\eta/(1+e^\eta) = \theta$ and $1/(1+e^\eta) = 1-\theta$, lemma 7.10 gives $\mathbb{E}_\theta(X) = m\theta$ and $\mathrm{Var}_\theta(X) = m\theta(1 - \theta)$, in agreement with table A.1. For a sample of size n the natural statistic is $T_n = \sum_i X_i$ and, by the remark after lemma 7.10, we have $\mathbb{E}_\theta(T_n) = nm\theta$ and $\mathrm{Var}_\theta(T_n) = nm\theta(1 - \theta)$.

□

Solution to exercise 7.5.

1. Using the formula $K(\eta) = -\log(-\eta)$ on $\mathcal{N} = (-\infty, 0)$ we find $K'(\eta) = -1/\eta$ and $K''(\eta) = 1/\eta^2$. Substituting the natural parameter $\eta = -\theta$ from example 7.4, we can use lemma 7.10 to find $\mathbb{E}_\theta(X) = 1/\theta$ and $\mathrm{Var}_\theta(X) = 1/\theta^2$. This is the mean and variance of the exponential distribution with rate θ.

2. For $h(x) = (2\pi\sigma^2)^{-1/2} e^{-x^2/(2\sigma^2)}$ and $T(x) = x$ we have to evaluate

$$Z(\eta) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(-\frac{x^2 - 2\sigma^2\eta x}{2\sigma^2}\Big)\,\mathrm{d}x.$$

   We can complete the square to find $x^2 - 2\sigma^2\eta x = (x - \sigma^2\eta)^2 - \sigma^4\eta^2$ and thus

$$Z(\eta) = \exp\Big(\frac{\sigma^2\eta^2}{2}\Big) \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(-\frac{(x - \sigma^2\eta)^2}{2\sigma^2}\Big)\,\mathrm{d}x = \exp\Big(\frac{\sigma^2\eta^2}{2}\Big),$$

   since the remaining integrand is the density of $N(\sigma^2\eta, \sigma^2)$ and thus integrates to one. The integral is finite for all η and thus $\mathcal{N} = \mathbb{R}$ and $K(\eta) = \log Z(\eta) = \sigma^2\eta^2/2$. Taking derivatives we find $K'(\eta) = \sigma^2\eta$ and $K''(\eta) = \sigma^2$ and substituting $\eta = \mu/\sigma^2$ in lemma 7.10 we get $\mathbb{E}_\mu(X) = \mu$ and $\mathrm{Var}_\mu(X) = \sigma^2$, as expected. Finally, $K\big(\eta(\mu)\big) = \sigma^2(\mu/\sigma^2)^2/2 = \mu^2/(2\sigma^2)$, which is the function $B(\mu)$ from example 7.7.

□

Solution to exercise 7.6.

1. Writing the expectation as an integral (or a sum, for a discrete model), we find

$$\begin{aligned}\mathbb{E}_\eta(e^{sT(X)}) &= \int e^{sT(x)}\, h(x)\, e^{\eta T(x) - K(\eta)}\,\mathrm{d}x \\ &= e^{-K(\eta)} \int h(x)\, e^{(\eta+s)T(x)}\,\mathrm{d}x = e^{K(\eta+s)-K(\eta)},\end{aligned}$$

   where the last step uses the definition of K at the point $\eta + s \in \mathcal{N}$. Taking logarithms gives $\log \mathbb{E}_\eta(e^{sT(X)}) = K(\eta + s) - K(\eta) = C(s)$, as claimed.

2. By the first part we have $e^{C(s)} = \mathbb{E}_\eta(e^{sT(X)})$. Differentiating both sides with respect to s, and taking the derivative inside the expectation on the right, we get

$$\begin{aligned} C'(s)\, e^{C(s)} &= \mathbb{E}_\eta(T(X)\, e^{sT(X)}) \qquad \text{and} \\ (C''(s) + C'(s)^2) e^{C(s)} &= \mathbb{E}_\eta(T(X)^2\, e^{sT(X)}). \end{aligned}$$

   At $s = 0$ we have $C(0) = 0$, and thus $C'(0) = \mathbb{E}_\eta(T(X))$ and $C''(0) = \mathbb{E}_\eta(T(X)^2) - \mathbb{E}_\eta(T(X))^2 = \mathrm{Var}_\eta(T(X))$. On the other hand, from $C(s) = K(\eta + s) - K(\eta)$ we get directly $C'(0) = K'(\eta)$ and $C''(0) = K''(\eta)$. Comparing the two computations gives $\mathbb{E}_\eta(T(X)) = K'(\eta)$ and $\mathrm{Var}_\eta(T(X)) = K''(\eta)$, which is lemma 7.10.

3. For the Poisson distribution we have $T(x) = x$, $\eta = \log\theta$ and $K(\eta) = e^\eta$ by example 7.3 and the computation following definition 7.9, and $\mathcal{N} = \mathbb{R}$, so that $\eta + s \in \mathcal{N}$ for every s. Writing $z = e^s$, the first part gives

$$\mathbb{E}_\theta(z^X) = \mathbb{E}_\eta(e^{sX}) = \exp(e^{\eta+s} - e^\eta) = \exp(\theta z - \theta) = e^{\theta(z-1)},$$

which is the identity used in exercise 5.7.

□

Solution to exercise 7.7.

1. By lemma 7.5, applied in the natural parametrisation where $B = K$, the log-likelihood is

$$\ell(\eta) = \sum_{i=1}^n \log h(x_i) + \eta\, T_n - nK(\eta), \qquad T_n = \sum_{i=1}^n T(x_i),$$

and thus the score is $\ell'(\eta) = T_n - nK'(\eta)$. The likelihood equation $\ell'(\eta) = 0$ is therefore $K'(\eta) = T_n/n$, and by lemma 7.10 the left-hand side equals $\mathbb{E}_\eta(T(X_1))$. This is the stated equation.

2. Differentiating once more and using lemma 7.10 again, we find

$$\ell''(\eta) = -nK''(\eta) = -n\,\mathrm{Var}_\eta(T(X_1)) < 0$$

for all $\eta \in \mathcal{N}$. Thus ℓ is strictly concave on the interval $\mathcal{N}$, and a strictly concave function has at most one critical point, which is then its global maximum. Any root of the likelihood equation is therefore the unique maximiser of ℓ, *i.e.* the unique MLE for η.

3. For the Poisson distribution we have $T(x) = x$ and $K'(\eta) = e^\eta$, and thus the likelihood equation reads $e^\eta = \bar{x}$, with root $\hat{\eta} = \log\bar{x}$ whenever $\bar{x} > 0$. Since $\theta = e^\eta$ is a bijection between $\mathcal{N} = \mathbb{R}$ and $\Theta = (0, \infty)$, proposition 5.6 gives $\hat{\theta} = e^{\hat{\eta}} = \bar{X}$, the MLE of exercise 5.7. For the Bernoulli distribution we have $T(x) = x$ and $K'(\eta) = e^\eta/(1+e^\eta)$, and thus the likelihood equation reads $e^\eta/(1 + e^\eta) = \bar{x}$, with root $\hat{\eta} = \log(\bar{x}/(1 - \bar{x}))$ whenever $0 < \bar{x} < 1$. Since $\theta = e^\eta/(1 + e^\eta)$ is a bijection between $\mathbb{R}$ and $(0, 1)$, proposition 5.6 gives $\hat{\theta} = \bar{X}$, the MLE of example 5.3. The restriction $0 < \bar{x} < 1$ is the condition $0 < s < n$ discussed after that example: when it fails the likelihood equation has no root and the MLE does not exist in the open parameter space.

4. The equation of the first part says that the MLE is the parameter value under which the population mean of the natural statistic T agrees with its sample mean, *i.e.* the MLE is the method of moments estimator based on the moment $\mathbb{E}_\theta(T(X_1))$ rather than on $\mathbb{E}_\theta(X_1)$. Whenever $T(x) = x$, as for the Bernoulli, Poisson and exponential distributions and for the normal mean with known variance, the two moments are the same, and the MLE coincides with the method of moments estimator from definition 4.6; this is the coincidence observed in the remark at the end of the method of moments section of lecture 4. For the gamma distribution of exercise 7.1 the natural statistic is $(\log x,\, x)$, and thus maximum likelihood matches the

means of $\log X_1$ and X_1, whereas the method of moments in example 4.9 matched the means of X_1 and X_1^2. Different moments are matched, and thus the two estimators differ.

□

C.6 Sufficiency

Here we solve the exercises at the end of lecture 8.

Solution to exercise 8.1.

1. Since the observations are independent, the event $\{X = x\}$ is the intersection of the events $\{X_i = x_i\}$ and we have

$$\mathbb{P}_\theta(X = x | X_1 = x_1) = \frac{\mathbb{P}_\theta(X = x)}{\mathbb{P}_\theta(X_1 = x_1)} = \prod_{i=2}^{n} \theta^{x_i}(1-\theta)^{1-x_i}.$$

   For $n \geq 2$ this depends on θ: for $x = (x_1, 1, \ldots, 1)$ it equals θ^{n-1}. Thus the conditional distribution of the data given X_1 depends on θ and X_1 is not sufficient by definition 8.1. Knowing the first toss alone leaves the remaining $n-1$ tosses, which still carry information about θ.

2. The joint probability weights are given by

$$f(x;\theta) = \prod_{i=1}^{n} \theta^{x_i}(1-\theta)^{1-x_i} = \theta^{\sum_i x_i}(1-\theta)^{n-\sum_i x_i}$$

   for $x \in \{0,1\}^n$. This function is of the form (8.1) with $g(t;\theta) = \theta^t(1-\theta)^{n-t}$ and $h(x) = 1$. By proposition 8.3 the statistic T is sufficient, which confirms the direct computation in example 8.2.

3. Yes. The weights from the previous part depend on the data only through the total T, and the total is itself a function of the pair (X_1, T). Thus the weights can be written in the form

$$f(x;\theta) = h(x)\, \tilde{g}\big((x_1, t);\theta\big)$$

   with $\tilde{g}\big((x_1,t);\theta\big) = \theta^t(1-\theta)^{n-t}$ and with $h(x) = 1$, and the pair is sufficient by proposition 8.3. The pair retains more of the data than T does, so this is a sufficient statistic which is not minimal.

□

Solution to exercise 8.2.

1. Since the joint density is the product of the densities of the single observations, and since a product of indicators is the indicator of the intersection of the conditions, we find that all n conditions $-\theta \leq x_i \leq \theta$ are satisfied if and only if $|x_i| \leq \theta$ for all i, *i.e.* if and only if the largest absolute value is at most θ. Thus we have

$$f(x;\theta) = (2\theta)^{-n} \prod_{i=1}^{n} \mathbf{1}_{\{-\theta \leq x_i \leq \theta\}} = (2\theta)^{-n}\, \mathbf{1}_{\{\max_i |x_i| \leq \theta\}}.$$

2. Let $g(t;\theta) = (2\theta)^{-n}\mathbf{1}_{\{t\leq\theta\}}$ and $h(x) = 1$. Then g depends on the data only through $t = \max_i |x_i|$, the function h does not depend on θ and from the first part of the solution we know that $f(x;\theta) = h(x)\, g(\max_i |x_i|;\theta)$ for all x and θ. By proposition 8.3 the statistic $M = \max_i |X_i|$ is sufficient for θ. As in example 8.4, the indicator is where the data and the parameter interact; without it the density would appear not to depend on the data at all.

3. The condition $\max_i |x_i| \leq \theta$ holds if and only if $x_{(n)} \leq \theta$ and $x_{(1)} \geq -\theta$, and thus we can write

$$f(x;\theta) = (2\theta)^{-n}\, \mathbf{1}_{\{x_{(n)}\leq\theta\}}\, \mathbf{1}_{\{x_{(1)}\geq-\theta\}},$$

which depends on the data only through the pair $(x_{(1)}, x_{(n)})$. Taking this whole expression as g and $h(x) = 1$, proposition 8.3 shows that the pair is sufficient. The statistic M is the function $M = \max(X_{(n)}, -X_{(1)})$ of the pair: the pair is sufficient, but it retains more information than M does.

□

Solution to exercise 8.3.

1. For $x_1, \ldots, x_n \geq 0$ the joint density is

$$f(x;\theta) = \prod_{i=1}^{n} \theta e^{-\theta x_i} = \theta^n \exp\Big(-\theta \sum_{i=1}^{n} x_i\Big),$$

and it is zero if any x_i is negative. Thus the factorisation (8.1) holds with $g(t;\theta) = \theta^n e^{-\theta t}$ and $h(x) = \mathbf{1}_{\{\min_i x_i \geq 0\}}$, and $T = \sum_i X_i$ is sufficient by proposition 8.3.

2. For two data vectors x and y with non-negative entries, the first part gives

$$\frac{f(x;\theta)}{f(y;\theta)} = \exp\Big(-\theta\Big(\sum_i x_i - \sum_i y_i\Big)\Big).$$

If $\sum_i x_i = \sum_i y_i$, the ratio equals 1 for every θ. If the sums differ by $d \neq 0$, the ratio is $e^{-\theta d}$, which is a strictly monotone function of θ and thus not constant on $(0, \infty)$. The ratio is therefore free of θ if and only if $T(x) = T(y)$ and by proposition 8.11 we find that T is minimal sufficient.

3. Since $\bar{X} = T/n$, the MLE is given by $\hat{\theta} = 1/\bar{X} = n/T$, a function of T. For fixed data with all $x_i \geq 0$, the likelihood function is $L(\theta) = h(x)\, g(T;\theta) = \theta^n e^{-\theta T}$, since the indicator $h(x)$ equals 1. Thus the MLE, the unique maximiser of $\theta^n e^{-\theta T}$, depends on the data only through T. The same argument, using the factorisation in proposition 8.3, shows that a unique MLE is always a function of any sufficient statistic.

□

Solution to exercise 8.4.

1. With μ known, the joint density

$$f(x;\sigma^2) = (2\pi\sigma^2)^{-n/2}\exp\Big(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\Big)$$

depends on the data only through $t = \sum_i (x_i-\mu)^2$, so that (8.1) holds with $g(t;\sigma^2) = (2\pi\sigma^2)^{-n/2}\exp(-t/(2\sigma^2))$ and $h(x) = 1$. By proposition 8.3 the statistic $\sum_i (X_i-\mu)^2$ is sufficient for σ^2.

2. Expanding the squares in the exponent, the joint density is

$$\begin{aligned} f(x;\mu,\sigma^2) &= (2\pi\sigma^2)^{-n/2}\exp\Big(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\Big) \\ &= (2\pi\sigma^2)^{-n/2}\exp\Big(\frac{\mu}{\sigma^2}\sum_i x_i - \frac{1}{2\sigma^2}\sum_i x_i^2 - \frac{n\mu^2}{2\sigma^2}\Big), \end{aligned}$$

which depends on the data only through the pair $T = (\sum_i x_i, \sum_i x_i^2)$, and thus the factorisation (8.1) holds with $h(x) = 1$ and with g equal to the whole expression. By proposition 8.3 the pair T is sufficient for (μ,σ^2).

3. Using the expanded form of the joint density from the previous part, and writing $a = \sum_i x_i - \sum_i y_i$ and $b = \sum_i x_i^2 - \sum_i y_i^2$, the ratio of the densities of two data vectors is

$$\frac{f(x;\mu,\sigma^2)}{f(y;\mu,\sigma^2)} = \exp\Big(\frac{a\mu - b/2}{\sigma^2}\Big),$$

since the factors $(2\pi\sigma^2)^{-n/2}$ and $\exp(-n\mu^2/(2\sigma^2))$ cancel. If $a = b = 0$, which is the statement $T(x) = T(y)$, the ratio equals 1 for all parameter values. Conversely, suppose the ratio does not depend on (μ,σ^2), so that the exponent $(a\mu - b/2)/\sigma^2$ equals a constant c for all $\mu \in \mathbb{R}$ and all $\sigma^2 > 0$. Taking $\sigma^2 = 1$, the expression $a\mu - b/2$ is constant in μ, which forces $a = 0$ and then $b = -2c$. With $a = 0$ the exponent is $-b/(2\sigma^2)$, and this is constant in σ^2 only if $b = 0$. Thus the ratio is free of the parameters if and only if $T(x) = T(y)$, and T is minimal sufficient by proposition 8.11.

4. No. By the previous part T is minimal sufficient, and thus, by definition 8.10, T would have to be a function of $\bar{X}$ if $\bar{X}$ were sufficient. It is not: for $n = 2$ the data sets $x = (1,-1)$ and $y = (2,-2)$ have the same sample mean 0, but $\sum_i x_i^2 = 2$ and $\sum_i y_i^2 = 8$, so that $T(x) \neq T(y)$. Two data sets with the same value of $\bar{X}$ can therefore have non-proportional likelihood functions, and $\bar{X}$ alone loses information about σ^2.

□

Solution to exercise 8.5.

1. The exponential distribution with rate θ has distribution function $F(y) = 1 - e^{-\theta y}$ for $y \geq 0$, and thus $1 - F(y) = e^{-\theta y}$. Proposition 8.8 gives

$$f_{(1)}(y) = n\left(e^{-\theta y}\right)^{n-1}\theta e^{-\theta y} = n\theta\, e^{-n\theta y}, \qquad y \geq 0,$$

which is the exponential density with rate $n\theta$. Directly, the minimum exceeds y if and only if every observation does, and by independence

$$\mathbb{P}_\theta(X_{(1)} > y) = \prod_{i=1}^{n} \mathbb{P}_\theta(X_i > y) = \left(e^{-\theta y}\right)^n = e^{-n\theta y},$$

which is the probability that an exponential random variable with rate $n\theta$ exceeds y. The two computations agree.

2. From table A.2, an exponential random variable with rate $n\theta$ has mean $1/(n\theta)$ and variance $1/(n\theta)^2$. Thus $\mathbb{E}_\theta(nX_{(1)}) = n/(n\theta) = 1/\theta$, so $nX_{(1)}$ is unbiased for $1/\theta$, and $\operatorname{Var}_\theta(nX_{(1)}) = n^2/(n\theta)^2 = 1/\theta^2$.

3. By lemma 2.2 the sample mean has $\operatorname{Var}_\theta(\bar{X}) = 1/(n\theta^2)$, and thus we find that $\operatorname{Var}_\theta(nX_{(1)})$ equals $n \operatorname{Var}_\theta(\bar{X})$, so that the estimator based on the minimum has n times the variance of the sample mean. Worse, its variance does not decrease at all as n grows. Indeed, by the first part $nX_{(1)}$ has the exponential distribution with rate θ for every n, and thus its sampling distribution does not change with the sample size, and $nX_{(1)}$ is not a consistent estimator for $1/\theta$. It is not a function of $\sum_i X_i$ either: for $n = 2$ the data sets $(1, 3)$ and $(2, 2)$ have the same sum but minima 1 and 2. The estimator discards the information which the sufficient statistic retains, and lecture 10 shows how conditioning on a sufficient statistic repairs such an estimator.

□

C.7 Rao–Blackwell and Lehmann–Scheffé

Solution to exercise 10.1.

1. Since U takes only the values 0 and 1, we have $\mathbb{E}_\theta(U) = \mathbb{P}_\theta(X_1 = 0) = e^{-\theta}$ for all θ and thus U is an unbiased estimator for $e^{-\theta}$. The indicator function is a Bernoulli random variable with success probability $e^{-\theta}$, and thus we have $\operatorname{Var}_\theta(U) = e^{-\theta}(1 - e^{-\theta})$.

2. Let $T = X_1 + R$ where $R = \sum_{i=2}^{n} X_i$. By the given fact, R is Poisson with parameter $(n-1)\theta$ and T is Poisson with parameter $n\theta$, and R is independent of X_1. For $0 \le x \le t$, the event $\{X_1 = x,\ T = t\}$ is the event $\{X_1 = x,\ R = t - x\}$ and thus we get

$$\begin{aligned}
\mathbb{P}_\theta(X_1 = x | T = t) &= \frac{\mathbb{P}_\theta(X_1 = x)\,\mathbb{P}_\theta(R = t - x)}{\mathbb{P}_\theta(T = t)} \\
&= \frac{e^{-\theta}\theta^x}{x!} \cdot \frac{e^{-(n-1)\theta}\left((n-1)\theta\right)^{t-x}}{(t-x)!} \cdot \frac{t!}{e^{-n\theta}(n\theta)^t} \\
&= \frac{t!}{x!\,(t-x)!} \cdot \frac{(n-1)^{t-x}}{n^t} = \binom{t}{x}\left(\frac{1}{n}\right)^x\left(1 - \frac{1}{n}\right)^{t-x}.
\end{aligned}$$

As expected, the parameter θ cancels since T is sufficient. Given $T = t$, the first observation can be seen as the number of successes in t independent trials with success probability $1/n$, and in the model each of the t counted events is equally likely to have been any of the n observations.

3. The conditional expectation of an indicator function is a conditional probability. Thus, using the second part of the question, we find
$$V = \mathbb{E}_\theta(U|T) = \mathbb{P}_\theta(X_1 = 0|T) = \binom{T}{0}\Big(1 - \frac{1}{n}\Big)^T = \Big(1 - \frac{1}{n}\Big)^T.$$

4. Using the probability generating function with $z = 1 - 1/n$, we find
$$\mathbb{E}_\theta(V) = \mathbb{E}_\theta\Big(\Big(1 - \frac{1}{n}\Big)^T\Big) = \exp\Big(n\theta\Big(1 - \frac{1}{n} - 1\Big)\Big) = e^{-\theta}.$$
This agrees with the second statement of the Rao–Blackwell theorem, proposition 10.1.

5. From example 10.7 we know that the sum T is a complete and sufficient statistic, and V is a function of T which is unbiased for $e^{-\theta}$. Thus, by the Lehmann–Scheffé theorem, theorem 10.5, V is the UMVUE for $e^{-\theta}$. The MLE $e^{-\bar{X}} = e^{-T/n}$ is also a function of T, but in exercise 5.7 we have seen that $\mathbb{E}_\theta(e^{-\bar{X}}) = \exp\big(n\theta(e^{-1/n} - 1)\big) > e^{-\theta}$, and thus it is biased and cannot be the UMVUE. Nevertheless, for large n, the two estimators are close since $(1 - 1/n)^T = \exp\big(T\log(1 - 1/n)\big)$ and $\log(1 - 1/n) = -1/n - 1/(2n^2) - \cdots$ and thus $V \approx e^{-\bar{X}}e^{-\bar{X}/(2n)}$: the UMVUE is obtained by applying a small downward correction to the MLE in order to compensate for the upward bias.

□

Solution to exercise 10.2.

1. Since X_1 and X_2 are both Bernoulli with success probability θ, we have $\mathbb{E}_\theta(X_1 - X_2) = \theta - \theta = 0$ for all $\theta \in (0, 1)$ and thus $w(T) = X_1 - X_2$ is an unbiased estimator for zero. The variable $w(T)$ equals 1 on the event $\{X_1 = 1, X_2 = 0\}$ and, since the two events are independent, this happens with probability $\theta(1-\theta) > 0$. Thus we have $\mathbb{P}_\theta\big(w(T) = 0\big) \leq 1 - \theta(1-\theta) < 1$.

2. The function w satisfies $\mathbb{E}_\theta\big(w(T)\big) = 0$ for all θ, but $w(T)$ does not equal zero with probability one, and thus the condition from definition 10.3 is violated: the whole sample is not complete. Nevertheless, the statistic T is sufficient, because once the whole sample is given, there is no variability left. This completes the proof.

□

Solution to exercise 10.3.

1. The density $\theta e^{-\theta x}$ is that of the exponential distribution with rate θ, whose mean is $1/\theta$ by table A.2, and thus X_1 is unbiased for $1/\theta$. The joint density of the sample is
$$f(x_1, \ldots, x_n; \theta) = \prod_{i=1}^n \theta e^{-\theta x_i} = \theta^n \exp\Big(-\theta \sum_{i=1}^n x_i\Big)$$
for all $x_i \geq 0$, and zero otherwise. This is of the form (8.1) with $h(x) = \mathbf{1}_{\{\min_i x_i \geq 0\}}$, which carries the support, and $g(t; \theta) = \theta^n e^{-\theta t}$, and thus T is sufficient by the factorisation theorem, proposition 8.3.

2. The random variables X_1 and R are independent, X_1 has density $f(x;\theta) = \theta e^{-\theta x}$ and R has the quoted gamma density with $k = n-1$, and thus the pair (X_1, R) has joint density $f(x;\theta) f_{n-1}(r)$ for $x, r \geq 0$. The map $(x, r) \mapsto (x, s) = (x, x+r)$ is linear with determinant one, and thus by the change of variable formula, proposition A.19, the pair (X_1, T) has joint density

$$f_{X_1,T}(x,s) = f(x;\theta)\, f_{n-1}(s-x) = \theta e^{-\theta x} \cdot \frac{\theta^{n-1}(s-x)^{n-2} e^{-\theta(s-x)}}{(n-2)!}$$
$$= \frac{\theta^n (s-x)^{n-2} e^{-\theta s}}{(n-2)!}$$

for $0 < x < s$, and zero otherwise. Dividing by the density $f_n(s) = \theta^n s^{n-1} e^{-\theta s}/(n-1)!$ of T, we obtain the conditional density

$$f(x|s) = \frac{f_{X_1,T}(x,s)}{f_n(s)} = \frac{(n-1)!}{(n-2)!} \cdot \frac{(s-x)^{n-2}}{s^{n-1}} = \frac{(n-1)(s-x)^{n-2}}{s^{n-1}}$$

for $0 < x < s$. Both θ^n and $e^{-\theta s}$ have cancelled, and thus the conditional density does not depend on θ, which is the sufficiency of T seen from definition 8.1. For $n = 2$ the density is $1/s$ on $(0, s)$: given the total of two observations, the first is uniformly distributed over the possible values.

3. Substituting $x = su$, we find

$$\mathbb{E}_\theta(X_1|T=s) = \int_0^s x\, \frac{(n-1)(s-x)^{n-2}}{s^{n-1}}\, \mathrm{d}x = (n-1)\, s \int_0^1 u(1-u)^{n-2}\, \mathrm{d}u.$$

Integrating by parts, with u differentiated and $(1-u)^{n-2}$ integrated, gives

$$\int_0^1 u(1-u)^{n-2}\, \mathrm{d}u = \Big[-\frac{u(1-u)^{n-1}}{n-1}\Big]_0^1 + \frac{1}{n-1}\int_0^1 (1-u)^{n-1}\, \mathrm{d}u = \frac{1}{(n-1)n},$$

and thus $\mathbb{E}_\theta(X_1|T=s) = s/n$. Consequently $\mathbb{E}_\theta(X_1|T) = T/n = \bar{X}$: Rao–Blackwellising the single observation X_1, which has variance $1/\theta^2$, gives the sample mean, which has variance $1/(n\theta^2)$, and the reduction promised by proposition 10.1 is by a factor of n.

4. Since the X_i are i.i.d. and T is a symmetric function of them, the pair (X_i, T) has the same joint distribution for every i, and thus the conditional expectations $\mathbb{E}_\theta(X_i|T)$ are the same random variable for all i. Summing over i and using the linearity of conditional expectation, we find

$$n\, \mathbb{E}_\theta(X_1|T) = \sum_{i=1}^n \mathbb{E}_\theta(X_i|T) = \mathbb{E}_\theta(T|T) = T,$$

and thus $\mathbb{E}_\theta(X_1|T) = T/n$. This argument uses only the symmetry of an i.i.d. sample and none of the exponential form, and so it applies to every model in which $\sum_i X_i$ is sufficient.

□

Solution to exercise 10.4.

1. In exercise 2.2 we have found $\mathbb{E}_\theta(M) = n\theta/(n+1)$ and thus

$$\mathbb{E}_\theta(\hat{\theta}) = \frac{n+1}{n} \cdot \frac{n\theta}{n+1} = \theta$$

for all θ. Thus, $\hat{\theta}$ is unbiased.

2. From example 8.4 we know that the maximum M is sufficient for θ and by the second statement of proposition 10.6 it is complete. Since $\hat{\theta}$ is a function of M and is unbiased for θ, we can use the Lehmann–Scheffé theorem, theorem 10.5, to conclude that this is the UMVUE for θ.

3. From the solution to exercise 2.2 we know $\operatorname{Var}_\theta(M) = n\theta^2/\big((n+2)(n+1)^2\big)$ and thus

$$\operatorname{Var}_\theta(\hat{\theta}) = \Big(\frac{n+1}{n}\Big)^2 \cdot \frac{n\theta^2}{(n+2)(n+1)^2} = \frac{\theta^2}{n(n+2)}.$$

The variance of a single observation is $\theta^2/12$, as shown in table A.2, and thus, using lemma 2.2, we find

$$\operatorname{Var}_\theta(\tilde{\theta}) = 4 \operatorname{Var}_\theta(\bar{X}) = \frac{4\theta^2}{12n} = \frac{\theta^2}{3n}.$$

The ratio $\operatorname{Var}_\theta(\hat{\theta})/\operatorname{Var}_\theta(\tilde{\theta}) = 3/(n+2)$, which is always less than or equal to one for all $n \geq 1$, equals one only for $n = 1$, where both estimators equal $2X_1$. For large n, the variance of the UMVUE is of order $1/n^2$, compared to the order $1/n$ for $2\bar{X}$, similar to the result in exercise 2.2.

4. From proposition 10.1, the Rao–Blackwell theorem, we know that the conditional expectation $\mathbb{E}_\theta(\tilde{\theta}|M)$ is a random variable, which is a function of M and is unbiased for θ. Since M is complete, the first step of the proof of theorem 10.5 shows that there can only be one such function and thus we can conclude $\mathbb{E}_\theta(\tilde{\theta}|M) = (n+1)M/n$ with probability one. For $n = 1$ we have $M = X_1$ and $\tilde{\theta} = 2X_1 = 2M$, which coincides with the formula $(n+1)M/n = 2M$.

□

Solution to exercise 10.5.

1. Using the definition of the exponential family, we can write the density as

$$\begin{aligned} f(x; \mu, \sigma^2) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(-\frac{(x-\mu)^2}{2\sigma^2}\Big) \\ &= \exp\Big(\frac{\mu}{\sigma^2}\, x - \frac{1}{2\sigma^2}\, x^2 - \frac{\mu^2}{2\sigma^2} - \frac{1}{2}\log(2\pi\sigma^2)\Big), \end{aligned}$$

which is of the form given in definition 7.6 with $h(x) = 1$, natural statistic $T(x) = (x, x^2)$, natural parameter

$$\eta(\mu, \sigma^2) = \Big(\frac{\mu}{\sigma^2},\ -\frac{1}{2\sigma^2}\Big)$$

and $B(\mu, \sigma^2) = \mu^2/(2\sigma^2) + \frac{1}{2}\log(2\pi\sigma^2)$.

2. As (μ, σ^2) varies over $\mathbb{R} \times (0, \infty)$, the second component $\eta_2 = -1/(2\sigma^2)$ can take any value in $(-\infty, 0)$. For fixed σ^2, the first component $\eta_1 = \mu/\sigma^2$ can take any value in $\mathbb{R}$ as μ varies. Thus, the range of η is the open half-plane $\mathbb{R} \times (-\infty, 0)$, which is an open, non-empty subset of $\mathbb{R}^2$ and thus the family has full rank in the sense of definition 7.8.

3. From corollary 8.5 and the first statement of proposition 10.6 we know that the natural statistic $\left(\sum_i X_i, \sum_i X_i^2\right)$ of the sample is complete and sufficient. From the pair $\left(\sum_i X_i, \sum_i X_i^2\right)$ we can compute $\bar{X} = \sum_i X_i/n$ and $S^2 = \left(\sum_i X_i^2 - n\bar{X}^2\right)/(n-1)$, and conversely $\sum_i X_i = n\bar{X}$ and $\sum_i X_i^2 = (n-1)S^2 + n\bar{X}^2$, and thus the map between the two pairs is one-to-one. Since a one-to-one function of a complete sufficient statistic is itself complete and sufficient, the pair $(\bar{X}, S^2)$ is complete and sufficient.

□

C.8 Fisher Information

Solution to exercise 11.1.

1. The score for the observation is given by $\ell_1'(\theta) = X_1/\theta - (1 - X_1)/(1 - \theta)$. This takes values $1/\theta$ if $X_1 = 1$ (with probability θ) and $-1/(1-\theta)$ if $X_1 = 0$ (with probability $1 - \theta$). Thus we have

$$\mathbb{E}_\theta(\ell_1'(\theta)) = \theta \cdot \frac{1}{\theta} + (1-\theta) \cdot \left(-\frac{1}{1-\theta}\right) = 1 - 1 = 0,$$

   as given in lemma 11.1.

2. Combining the two fractions we get

$$\ell_1'(\theta) = \frac{X_1(1-\theta) - (1-X_1)\theta}{\theta(1-\theta)} = \frac{X_1 - \theta}{\theta(1-\theta)},$$

   which is a constant multiple of $X_1 - \theta$. Since $\mathrm{Var}_\theta(X_1) = \theta(1-\theta)$ we find

$$\mathrm{Var}_\theta(\ell_1'(\theta)) = \frac{\theta(1-\theta)}{\theta^2(1-\theta)^2} = \frac{1}{\theta(1-\theta)}.$$

   This is the value of $\mathcal{I}(\theta)$ in example 11.5 and thus, by lemma 11.3, the correct value.

3. From proposition 11.4 we know $\mathcal{I}_n(\theta) = n/(\theta(1-\theta))$ and thus $1/\mathcal{I}_n(\theta) = \theta(1-\theta)/n$. By lemma 2.2 the sample mean has $\mathrm{Var}_\theta(\bar{X}) = \mathrm{Var}_\theta(X_1)/n = \theta(1-\theta)/n$ and thus the variance of $\bar{X}$ equals the reciprocal of the information, as it did for the normal mean in example 11.7.

4. The probability weights of Y are $f(y;\theta) = \binom{m}{y}\theta^y(1-\theta)^{m-y}$ for $y \in \{0, 1, \ldots, m\}$, and thus the log-likelihood is $\ell_1(\theta) = \log\binom{m}{Y} + Y\log\theta + (m-Y)\log(1-\theta)$ with

$$\ell_1''(\theta) = -\frac{Y}{\theta^2} - \frac{m-Y}{(1-\theta)^2}.$$

Using $\mathbb{E}_\theta(Y) = m\theta$ we find

$$-\mathbb{E}_\theta(\ell_1''(\theta)) = \frac{m\theta}{\theta^2} + \frac{m - m\theta}{(1-\theta)^2} = \frac{m}{\theta} + \frac{m}{1-\theta} = \frac{m}{\theta(1-\theta)}.$$

This is m times the information of a single Bernoulli observation. Since Y has the same distribution as the sum of m independent Bernoulli observations with success probability θ, and since the likelihood of Y differs from the likelihood of these m observations only by a factor which does not depend on θ, the two have the same score and thus the same information, and proposition 11.4 gives the factor m.

□

Solution to exercise 11.2.

1. In the natural parametrisation we have $\ell_1(\eta) = \log h(X) + \eta T(X) - K(\eta)$ and thus $\ell_1'(\eta) = T(X) - K'(\eta)$ and $\ell_1''(\eta) = -K''(\eta)$. By lemma 7.10 we have $\mathbb{E}_\eta(T(X)) = K'(\eta)$ and $\operatorname{Var}_\eta(T(X)) = K''(\eta)$, and since the second derivative is non-random, lemma 11.3 gives $\mathcal{I}_\eta(\eta) = K''(\eta) = \operatorname{Var}_\eta(T(X))$.

2. Since the density integrates to one for every θ, we have

$$\int h(x) \exp(\eta(\theta) T(x)) \, \mathrm{d}x = e^{B(\theta)},$$

and the left-hand side equals $e^{K(\eta(\theta))}$ by the definition of the cumulant function. Thus we have $B(\theta) = K(\eta(\theta))$ and, using the chain rule, we find

$$\ell_1'(\theta) = \eta'(\theta) \Big(T(X) - K'(\eta(\theta)) \Big).$$

The factor $\eta'(\theta)$ is a constant and thus the variance of the score is

$$\mathcal{I}_\theta(\theta) = \eta'(\theta)^2 \operatorname{Var}_\theta(T(X)) = \eta'(\theta)^2 K''(\eta(\theta)),$$

as claimed.

3. The Bernoulli distribution has $T(x) = x$ and $\eta(\theta) = \log(\theta/(1-\theta))$ by example 7.2, so that $\eta'(\theta) = 1/(\theta(1-\theta))$. Using $\operatorname{Var}_\theta(X) = \theta(1-\theta)$ we find

$$\mathcal{I}_\theta(\theta) = \frac{\theta(1-\theta)}{(\theta(1-\theta))^2} = \frac{1}{\theta(1-\theta)},$$

which recovers example 11.5.

□

Solution to exercise 11.3.

1. The log-likelihood of a single observation is $\ell_1(\theta) = -2\log\theta + \log X_1 - X_1/\theta$ and thus the score is given by

$$\ell_1'(\theta) = -\frac{2}{\theta} + \frac{X_1}{\theta^2}.$$

From table A.2 or from the solution to exercise 5.2 we know $\mathbb{E}_\theta(X_1) = 2\theta$ and thus $\mathbb{E}_\theta(\ell_1'(\theta)) = -2/\theta + 2\theta/\theta^2 = 0$.

2. Taking derivatives again, we find $\ell_1''(\theta) = 2/\theta^2 - 2X_1/\theta^3$ and thus

$$\mathcal{I}(\theta) = -\mathbb{E}_\theta(\ell_1''(\theta)) = -\frac{2}{\theta^2} + \frac{2 \cdot 2\theta}{\theta^3} = \frac{2}{\theta^2}.$$

On the other hand, since the score is X_1/θ^2 plus a constant, we can find the variance as

$$\operatorname{Var}_\theta(\ell_1'(\theta)) = \frac{\operatorname{Var}_\theta(X_1)}{\theta^4} = \frac{2\theta^2}{\theta^4} = \frac{2}{\theta^2}.$$

Both approaches give the same result.

3. Since the density can be written as

$$f(x;\theta) = x \exp\Big(-\frac{x}{\theta} - 2\log\theta\Big),$$

we have $h(x) = x$, $T(x) = x$, $\eta(\theta) = -1/\theta$ and $B(\theta) = 2\log\theta$. Thus, the model is an exponential family as in definition 7.1. The natural parameter takes values $\eta < 0$ and the cumulant function is

$$K(\eta) = \log \int_0^\infty x\, e^{\eta x}\, \mathrm{d}x = \log \frac{1}{\eta^2} = -2\log(-\eta),$$

where $K'(\eta) = -2/\eta = 2\theta$ is the mean and $K''(\eta) = 2/\eta^2 = 2\theta^2$ is the variance, as predicted by lemma 7.10. Since $\eta'(\theta) = 1/\theta^2$, we find by exercise 11.2 $\mathcal{I}(\theta) = \eta'(\theta)^2 K''(\eta(\theta)) = 2\theta^2/\theta^4 = 2/\theta^2$. This is the same value as we obtained above.

4. From proposition 11.4 we know that the information of the sample is $\mathcal{I}_n(\theta) = 2n/\theta^2$ and thus $1/\mathcal{I}_n(\theta) = \theta^2/(2n) = \operatorname{MSE}(\hat\theta)$. Since $\hat\theta$ is unbiased, its mean squared error equals its variance and thus the variance of the MLE is the reciprocal of the Fisher information for all sample sizes, as it was for the sample mean in exercise 11.1 and example 11.7. In lecture 13 we will see that this is the smallest variance an unbiased estimator can have.

□

Solution to exercise 11.4.

1. The log-likelihood of a single observation is $\ell_1(p) = (X_1 - 1)\log(1-p) + \log p$ and thus the score is given by

$$\ell_1'(p) = \frac{1}{p} - \frac{X_1 - 1}{1-p}.$$

Using $\mathbb{E}_p(X_1 - 1) = 1/p - 1 = (1-p)/p$ we find

$$\mathbb{E}_p(\ell_1'(p)) = \frac{1}{p} - \frac{(1-p)/p}{1-p} = \frac{1}{p} - \frac{1}{p} = 0.$$

2. Taking derivatives again, we find $\ell_1''(p) = -1/p^2 - (X_1 - 1)/(1-p)^2$ and taking expectations we get

$$\begin{aligned}\mathcal{I}(p) = -\mathbb{E}_p(\ell_1''(p)) &= \frac{1}{p^2} + \frac{(1-p)/p}{(1-p)^2} \\ &= \frac{1}{p^2} + \frac{1}{p(1-p)} = \frac{(1-p)+p}{p^2(1-p)} = \frac{1}{p^2(1-p)}.\end{aligned}$$

From proposition 11.4 we know that the sample has information $\mathcal{I}_n(p) = n/(p^2(1-p))$.

3. Since the score is $-X_1/(1-p)$ plus a constant, we find

$$\operatorname{Var}_p(\ell_1'(p)) = \frac{\operatorname{Var}_p(X_1)}{(1-p)^2} = \frac{(1-p)/p^2}{(1-p)^2} = \frac{1}{p^2(1-p)}.$$

This agrees with the result from the previous part.

□

Solution to exercise 11.5.

1. Since $K(\eta) = e^\eta$ we have $K''(\eta) = e^\eta$ and using exercise 11.2 we find $\mathcal{I}_\eta(\eta) = e^\eta = \theta$. This is the variance of the natural statistic $T(X) = X$, as expected.

2. Using proposition 11.9 with η (the original parameter) and $\theta = g(\eta) = e^\eta$ (the new parameter), we find $g'(\eta) = e^\eta = \theta$. Thus we have

$$\mathcal{I}_\theta(\theta) = \frac{\mathcal{I}_\eta(\eta)}{g'(\eta)^2} = \frac{\theta}{\theta^2} = \frac{1}{\theta},$$

the value from example 11.6.

3. We have $p_0 = g(\theta) = e^{-\theta}$ with $g'(\theta) = -e^{-\theta}$ and using proposition 11.9 we find

$$\mathcal{I}_{p_0}(p_0) = \frac{1/\theta}{e^{-2\theta}} = \frac{e^{2\theta}}{\theta},$$

or using the notation p_0 we find $\mathcal{I}_{p_0}(p_0) = 1/(p_0^2 \log(1/p_0))$. As $\theta \to \infty$, the information goes to infinity. This makes sense, since $p_0 = e^{-\theta}$ converges to zero, and the information measures accuracy on the absolute scale of the parameter: a probability which is close to zero is estimated with small absolute error by any reasonable estimator, and the approximate variance $1/\mathcal{I}_n(p_0) = \theta e^{-2\theta}/n$ from lecture 17 is tiny, even if the error relative to the size of p_0 is not small.

□

Solution to exercise 11.6.

1. The map $g(\theta) = 1/\theta$ maps $(0, \infty)$ to $(0, \infty)$ and $g'(\theta) = -1/\theta^2 \neq 0$. From example 11.8 we know that $\mathcal{I}_\theta(\theta) = 1/\theta^2$. Thus, by proposition 11.9, we have

$$\mathcal{I}_\mu(\mu) = \frac{\mathcal{I}_\theta(\theta)}{g'(\theta)^2} = \frac{1/\theta^2}{1/\theta^4} = \theta^2 = \frac{1}{\mu^2}.$$

This is the value given in section 11.5.

2. Using proposition 11.4, in the μ-parametrisation, the information of the sample about μ is given by $n\mathcal{I}_\mu(\mu) = n/\mu^2$. From table A.2 we know that $\mathrm{Var}_\mu(X_1) = 1/\theta^2 = \mu^2$ and thus we have $\mathrm{Var}_\mu(\bar{X}) = \mu^2/n$, by lemma 2.2. The variance of the unbiased estimator $\bar{X}$ for μ equals the reciprocal μ^2/n of the information of the sample, and from lecture 13 we know that no unbiased estimator can be better.

□

Solution to exercise 11.7. The right-hand side of (11.1) with $k = 1$ is an integral over the whole real line, but $f(x;\theta) = 0$ and thus $\partial f/\partial\theta = 0$ for $x < 0$ and for $x > \theta$; since $x = \theta$ is a single point, it does not contribute to the integral; only $0 \le x \le \theta$, where $f(x;\theta) = 1/\theta$, is left. We have

$$\int_0^\theta \frac{\partial}{\partial\theta} f(x;\theta)\,\mathrm{d}x = \int_0^\theta \Big(-\frac{1}{\theta^2}\Big)\mathrm{d}x = -\frac{1}{\theta},$$

while the left-hand side can be found as

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\int_0^\theta f(x;\theta)\,\mathrm{d}x = \frac{\mathrm{d}}{\mathrm{d}\theta}\,1 = 0.$$

The difference between the two sides is $0 - (-1/\theta) = 1/\theta$, which is the value $f(\theta;\theta) = 1/\theta$ of the density at the upper limit of the integral. The derivative of the integral picks up this boundary term (from the moving upper limit) in addition to the derivative of the integrand, but the differentiation under the integral sign ignores this boundary term. This completes the solution. □

C.9 The Cramér–Rao Inequality

Here we solve the exercises at the end of lecture 13.

Solution to exercise 13.1.

1. For $y \ge 0$ we have

$$\mathbb{P}_\theta(Y_1 \le y) = \mathbb{P}_\theta\Big(X_1 \le \frac{y}{2\theta}\Big) = 1 - e^{-y/2},$$

and taking derivatives we find the density $\frac{1}{2}e^{-y/2}$ for $y \ge 0$. This is the exponential density with rate $1/2$, *i.e.* the Gamma$(1, 1/2)$ density from table A.2 and Gamma$(1, 1/2)$ is χ^2_2 by the statement quoted in the question. Thus we have $Y_1 \sim \chi^2_2$.

2. From the first part and definition A.2 we know that each $Y_i = 2\theta X_i$ has the distribution of $Z^2_{2i-1} + Z^2_{2i}$ for independent, standard normally distributed Z_{2i-1} and Z_{2i}. Since the X_i are independent, we can assume that all $2n$ are independent normal variables. Thus, $2\theta T = \sum_{i=1}^n Y_i$ has the distribution of $Z^2_1 + \cdots + Z^2_{2n}$, *i.e.* χ^2_{2n} by definition A.2.

3. The density of $T \sim \mathrm{Gamma}(n, \theta)$ is given by $f_T(t) = \theta^n t^{n-1} e^{-\theta t}/\Gamma(n)$ for $t \ge 0$. For $V = 2\theta T$, using proposition A.19 with inverse map $t = v/(2\theta)$ and derivative $1/(2\theta)$, we find

$$f_V(v) = f_T\Big(\frac{v}{2\theta}\Big)\,\frac{1}{2\theta} = \frac{\theta^n\big(v/(2\theta)\big)^{n-1}e^{-v/2}}{\Gamma(n)}\cdot\frac{1}{2\theta} = \frac{(1/2)^n\, v^{n-1}e^{-v/2}}{\Gamma(n)}$$

for $v \geq 0$. This is the Gamma$(n, 1/2)$ density again, by table A.2. Since $n = (2n)/2$, this is the χ^2_{2n} density and it agrees with the result from the previous part.

4. From lemma A.3 we know $\mathbb{E}_\theta(2\theta T) = 2n$ and $\mathrm{Var}_\theta(2\theta T) = 4n$, and thus $\mathbb{E}_\theta(T) = n/\theta$ and $\mathrm{Var}_\theta(T) = 4n/(4\theta^2) = n/\theta^2$. For a single observation, the mean is $1/\theta$ and the variance is $1/\theta^2$, as found in table A.2, and thus the sum of n independent copies has mean n/θ and variance n/θ^2, as shown above.

□

Solution to exercise 13.2.

1. From example 11.5 we know that $\mathcal{I}_n(\theta) = n/(\theta(1-\theta))$. Thus, the bound from corollary 13.3 is $1/\mathcal{I}_n(\theta) = \theta(1-\theta)/n$.

2. The sample mean is an unbiased estimator for θ and using lemma 2.2 we find $\mathrm{Var}_\theta(\bar{X}) = \theta(1-\theta)/n$, since the variance of a Bernoulli distribution is $\theta(1-\theta)$. This bound coincides with the bound from the definition, and thus $\bar{X}$ is efficient by definition 13.4, since the exchange identity (13.1) holds because the Bernoulli distribution is an exponential family with $\eta(\theta)$ in the interior of the natural parameter space and $\bar{X}$ has finite variance, as noted after lemma 13.1.

3. The score of the sample in example 11.5 is given by

$$\ell'(\theta) = \frac{T_n}{\theta} - \frac{n - T_n}{1-\theta} = \frac{T_n - n\theta}{\theta(1-\theta)} = \frac{n}{\theta(1-\theta)}\,(\bar{X} - \theta),$$

and thus we have $\bar{X} - \theta = c(\theta)\,\ell'(\theta)$ with $c(\theta) = \theta(1-\theta)/n$. Since $g(\theta) = \theta$ has $g'(\theta) = 1$, this is $g'(\theta)/\mathcal{I}_n(\theta)$ as predicted by proposition 13.5.

4. For $g(\theta) = \theta/(1-\theta)$ we have $g'(\theta) = 1/(1-\theta)^2$ and thus using theorem 13.2 we get the bound

$$\frac{g'(\theta)^2}{\mathcal{I}_n(\theta)} = \frac{1}{(1-\theta)^4} \cdot \frac{\theta(1-\theta)}{n} = \frac{\theta}{n(1-\theta)^3}.$$

The Bernoulli distribution is an exponential family with natural statistic $T(x) = x$ and $\mu(\theta) = \theta$. From the discussion after corollary 13.6 we know that only affine functions of the form $a + b\theta$ can be used to construct an efficient estimator. Since the odds are not an affine function of θ, there can be no efficient estimator for the odds.

5. Any estimator $W = W(X_1, \ldots, X_n)$ can take at most 2^n different values, one for each possible value of the sample $x \in \{0,1\}^n$, and the expectation of the estimator is

$$\mathbb{E}_\theta(W) = \sum_{x \in \{0,1\}^n} W(x)\,\theta^{\sum_i x_i}(1-\theta)^{n - \sum_i x_i},$$

which is a polynomial in θ of degree at most n. A polynomial function is bounded on $(0,1)$, whereas $\theta/(1-\theta) \to \infty$ as $\theta \to 1$ and thus no choice of W makes the estimator $\mathbb{E}_\theta(W)$ equal to the odds for all θ. Thus there is no unbiased estimator for the odds, efficient or otherwise.

□

Solution to exercise 13.3.

1. From example 11.6 we know that $\mathcal{I}_n(\theta) = n/\theta$. Thus, the Cramér–Rao bound for unbiased estimators for θ is θ/n. Since the Poisson distribution has mean and variance θ, lemma 2.2 shows that $\mathbb{E}_\theta(\bar{X}) = \theta$ and $\mathrm{Var}_\theta(\bar{X}) = \theta/n$. Thus, $\bar{X}$ is an unbiased estimator and satisfies the bound: it is an efficient estimator for θ, since the exchange identity (13.1) holds because the Poisson distribution is an exponential family with $\eta(\theta)$ in the interior of the natural parameter space and $\bar{X}$ has finite variance, as noted after lemma 13.1.

2. The log-likelihood function is $\ell(\theta) = T_n \log\theta - n\theta - \sum_i \log(X_i!)$ and thus the score function is

$$\ell'(\theta) = \frac{T_n}{\theta} - n = \frac{n}{\theta}\,(\bar{X} - \theta).$$

   Re-arranging this we find $\bar{X} - \theta = (\theta/n)\,\ell'(\theta)$, which is condition (13.2) with $W = \bar{X}$, $g(\theta) = \theta$ and $c(\theta) = \theta/n$; indeed $g'(\theta)/\mathcal{I}_n(\theta) = 1/(n/\theta) = \theta/n$.

3. From the discussion after corollary 13.6 we know that in a one-parameter exponential family, only affine functions of the mean $\mu(\theta)$ of the natural statistic allow for efficient estimators. In this case we have $\mu(\theta) = \theta$ and $g(\theta) = e^{-\theta}$ is not an affine function of θ. Thus, no efficient estimator for $e^{-\theta}$ exists. There is no contradiction in this result: example 10.7 shows that $(1-1/n)^{T_n}$ has the smallest variance amongst all unbiased estimators for $e^{-\theta}$, but this smallest variance is strictly larger than the Cramér–Rao bound. Efficiency is a stronger property than being UMVUE, and in this case the bound cannot be achieved.

□

Solution to exercise 13.4.

1. Example 11.8 gives $\mathcal{I}(\theta) = 1/\theta^2$, and thus $\mathcal{I}_n(\theta) = n/\theta^2$ and the bound is $1/\mathcal{I}_n(\theta) = \theta^2/n$.

2. Since $\tilde{\theta}$ is unbiased, its mean squared error is its variance, $\mathrm{Var}_\theta(\tilde{\theta}) = \theta^2/(n-2)$. The efficiency is

$$\mathrm{eff}_\theta(\tilde{\theta}) = \frac{\theta^2/n}{\theta^2/(n-2)} = \frac{n-2}{n},$$

   which is less than one for every $n \geq 3$ and tends to one as $n \to \infty$: the estimator falls short of the bound for every finite sample size, but only by a factor which disappears in the limit.

3. The exponential distribution with rate θ is an exponential family with $h(x) = 1$, $\eta(\theta) = -\theta$, $T(x) = x$ and $B(\theta) = -\log\theta$, as in example 7.4, and thus $\mu(\theta) = \mathbb{E}_\theta(X_1) = 1/\theta$. By the discussion after corollary 13.6, an efficient estimator exists only for affine functions $a + b/\theta$ of $\mu(\theta)$. The rate $g(\theta) = \theta$ is not of this form, and thus no unbiased estimator for θ is

efficient. The same conclusion can be reached directly: condition (13.2) with $\ell'(\theta) = n/\theta - T$ would read $W = \theta + c(\theta)n/\theta - c(\theta)T$, and since W does not depend on θ, the coefficients $-c(\theta)$ and $\theta + c(\theta)n/\theta$ would have to be constants b and a, giving $\mathbb{E}_\theta(W) = a + bn/\theta$, which cannot equal θ for all θ.

4. The natural parameter $\eta = -\theta$ ranges over the open interval $(-\infty, 0)$, and thus the family is of full rank and T is complete and sufficient by corollary 8.5 and proposition 10.6. Since $\tilde{\theta}$ is a function of T and unbiased for θ, the Lehmann–Scheffé theorem, theorem 10.5, shows that it is the UMVUE for θ. Together with the previous part this is an example of a UMVUE which is not efficient: the bound θ^2/n is not attained by any unbiased estimator, and the best unbiased estimator has variance $\theta^2/(n-2)$.

5. For $g(\theta) = 1/\theta$ we have $g'(\theta) = -1/\theta^2$, and theorem 13.2 gives the bound

$$\frac{g'(\theta)^2}{\mathcal{I}_n(\theta)} = \frac{1/\theta^4}{n/\theta^2} = \frac{1}{n\theta^2} = \frac{\mu^2}{n}.$$

The sample mean is unbiased for μ and has variance $\operatorname{Var}_\theta(X_1)/n = \mu^2/n$ by lemma 2.2, and thus $\bar{X}$ is efficient for μ, since the exchange identity (13.1) holds because the exponential distribution is an exponential family with $\eta(\theta)$ in the interior of the natural parameter space and $\bar{X}$ has finite variance, as noted after lemma 13.1. This is consistent with the previous parts: $\mu(\theta) = 1/\theta$ is the mean of the natural statistic, for which corollary 13.6 guarantees an efficient estimator, whereas the rate is a non-affine function of it. Whether an efficient estimator exists depends on which function of the parameter we set out to estimate.

□

Solution to exercise 13.5.

1. Since μ is given and $\tau = \sigma^2$ is the parameter, the log-likelihood function is $\ell(\tau) = -\frac{n}{2}\log(2\pi\tau) - \frac{1}{2\tau}\sum_i (X_i - \mu)^2$ and thus

$$\ell'(\tau) = -\frac{n}{2\tau} + \frac{1}{2\tau^2}\sum_{i=1}^n (X_i - \mu)^2, \qquad \ell''(\tau) = \frac{n}{2\tau^2} - \frac{1}{\tau^3}\sum_{i=1}^n (X_i - \mu)^2.$$

Since $\mathbb{E}((X_i - \mu)^2) = \tau$, we can use lemma 11.3 to find

$$\mathcal{I}_n(\tau) = -\mathbb{E}(\ell''(\tau)) = -\frac{n}{2\tau^2} + \frac{n\tau}{\tau^3} = \frac{n}{2\tau^2} = \frac{n}{2\sigma^4}.$$

Thus, the Cramér–Rao bound is $2\sigma^4/n$.

2. The $(X_i - \mu)/\sigma$ are i.i.d. standard normally distributed, *i.e.* $\sum_i (X_i - \mu)^2/\sigma^2 \sim \chi^2_n$ by definition A.2, with mean n and variance $2n$ by lemma A.3. Thus, $\hat{\sigma}^2 = \frac{\sigma^2}{n}\sum_i (X_i - \mu)^2/\sigma^2$ has mean σ^2 and variance $\sigma^4 \cdot 2n/n^2 = 2\sigma^4/n$. This is the bound and thus $\hat{\sigma}^2$ is efficient, since the exchange identity (13.1) holds because the normal distribution with known mean is an exponential family with $\eta(\theta)$ in the interior of the natural parameter space

and $\hat{\sigma}^2$ has finite variance, as noted after lemma 13.1. Alternatively, the score for the first part can be written as $\ell'(\sigma^2) = \frac{n}{2\sigma^4}(\hat{\sigma}^2 - \sigma^2)$, which satisfies condition (13.2) with $c(\sigma^2) = 2\sigma^4/n$.

3. From proposition A.4 we know that $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$, with variance $2(n-1)$. Thus we have

$$\operatorname{Var}(S^2) = \frac{\sigma^4}{(n-1)^2} \cdot 2(n-1) = \frac{2\sigma^4}{n-1} > \frac{2\sigma^4}{n}.$$

Thus, the sample variance does not achieve the bound, and the efficiency of the sample variance is $(n-1)/n$. Since the bound $2\sigma^4/n$ still holds for unknown μ, and since S^2 is the UMVUE for σ^2 by example 10.8, this is a second example of a best unbiased estimator which is not efficient: no unbiased estimator for σ^2 can achieve the bound if μ also needs to be estimated, and the loss of one degree of freedom is the price to be paid for not knowing μ.

□

Solution to exercise 13.6.

1. By proposition 11.4 the information of the sample is n times the information of a single observation, in either parametrisation. Proposition 11.9 gives the relation for a single observation, and multiplying both sides by n we find

$$\mathcal{I}_{n,\psi}(\psi) = \frac{\mathcal{I}_{n,\theta}(\theta)}{g'(\theta)^2} \qquad \text{for } \psi = g(\theta).$$

2. Since $\psi = g(\theta)$, the estimator W is an unbiased estimator for the parameter ψ, and corollary 13.3, applied in the ψ-parametrisation, bounds its variance by $1/\mathcal{I}_{n,\psi}(\psi)$. Using the first part we get

$$\frac{1}{\mathcal{I}_{n,\psi}(\psi)} = \frac{g'(\theta)^2}{\mathcal{I}_{n,\theta}(\theta)},$$

which is the bound of theorem 13.2 for $g(\theta)$. The two results thus give the same bound, and the bound does not depend on how the model is parametrised. This completes the proof.

□

C.10 Multiparameter Fisher Information

Here we solve the exercises at the end of lecture 14.

Solution to exercise 14.1.

1. The log-likelihood function in terms of (μ, σ) is

$$\ell(\mu, \sigma) = -\frac{n}{2}\log(2\pi) - n\log\sigma - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2,$$

and its first partial derivatives are $\partial\ell/\partial\mu = \sum_i (X_i - \mu)/\sigma^2$ and $\partial\ell/\partial\sigma = -n/\sigma + \sum_i (X_i - \mu)^2/\sigma^3$.

2. Taking derivatives again, we find

$$\frac{\partial^2 \ell}{\partial \mu^2} = -\frac{n}{\sigma^2}, \qquad \frac{\partial^2 \ell}{\partial \mu \, \partial \sigma} = -\frac{2}{\sigma^3} \sum_{i=1}^{n} (X_i - \mu),$$
$$\frac{\partial^2 \ell}{\partial \sigma^2} = \frac{n}{\sigma^2} - \frac{3}{\sigma^4} \sum_{i=1}^{n} (X_i - \mu)^2.$$

   Using $\mathbb{E}(X_i - \mu) = 0$ and $\mathbb{E}\big((X_i - \mu)^2\big) = \sigma^2$ we find that the mixed derivative has expectation zero, and the last term has expectation $n/\sigma^2 - 3n/\sigma^2 = -2n/\sigma^2$. Thus, by proposition 14.3, the information matrix is given by $\operatorname{diag}(n/\sigma^2, 2n/\sigma^2)$.

3. The inverse of the matrix from part (b) is given by $\operatorname{diag}\big(\sigma^2/n, \sigma^2/(2n)\big)$. Thus, by proposition 14.5, any unbiased estimator for σ has variance of at least $\sigma^2/(2n)$. The sample variance S^2 is an unbiased estimator for σ^2 by lemma 2.3, but the square root of the sample variance is not an unbiased estimator for σ. The function $g(x) = -\sqrt{x}$ is strictly convex on $(0, \infty)$ and by Jensen's inequality, lemma A.7, we have $-\sqrt{\mathbb{E}(S^2)} \leq \mathbb{E}(-\sqrt{S^2})$, *i.e.* $\mathbb{E}(S) \leq \sqrt{\mathbb{E}(S^2)} = \sigma$. The inequality is strict, since S^2 is not constant with probability one (by proposition A.4 it is a multiple of a chi-squared random variable). Thus, $\mathbb{E}(S) < \sigma$, the estimator S is biased for σ and the Cramér–Rao bound, which only concerns unbiased estimators, cannot be compared to $\operatorname{Var}(S)$ directly.

□

Solution to exercise 14.2.

1. We have $\ell_1(\alpha, \beta) = \alpha \log \beta - \log \Gamma(\alpha) + (\alpha - 1) \log X - \beta X$ and

$$\frac{\partial \ell_1}{\partial \alpha} = \log \beta - \psi(\alpha) + \log X \qquad \text{and} \qquad \frac{\partial \ell_1}{\partial \beta} = \frac{\alpha}{\beta} - X,$$

   where $\psi = (\log \Gamma)'$ is the derivative of $\log \Gamma$, which is smooth and thus differentiable on $(0, \infty)$. Taking derivatives again, we get $\partial^2 \ell_1 / \partial \alpha^2 = -\psi'(\alpha)$, $\partial^2 \ell_1 / \partial \alpha \, \partial \beta = 1/\beta$ and $\partial^2 \ell_1 / \partial \beta^2 = -\alpha/\beta^2$, as given in section 14.4. To verify the first identity of proposition 14.3, we note that the second component of the score has mean $\alpha/\beta - \mathbb{E}(X) = 0$, since $\mathbb{E}(X) = \alpha/\beta$ by table A.2.

2. If $\alpha = 2$ is known, the model consists of one parameter β and by lemma 11.3 the Fisher information is $\mathcal{I}(\beta) = -\mathbb{E}\big(\partial^2 \ell_1 / \partial \beta^2\big) = \alpha/\beta^2 = 2/\beta^2$. The lower right corner of $\mathcal{I}(\alpha, \beta)$ is the diagonal element $\mathcal{I}(\alpha, \beta)_{22}$, which by definition is the variance of $\partial \ell_1 / \partial \beta$ with α fixed to the true value, *i.e.* for known shape. The diagonal element of the inverse answers a different question, asking about the estimation of β when α is unknown. Part (d) shows that this value is larger.

3. The scale is $\theta = g(\beta) = 1/\beta$ where $g'(\beta) = -1/\beta^2$ and by proposition 11.9 we have

$$\mathcal{I}_\theta(\theta) = \frac{\mathcal{I}_\beta(\beta)}{g'(\beta)^2} = \frac{2/\beta^2}{1/\beta^4} = 2\beta^2 = \frac{2}{\theta^2}.$$

   This is the same value as in exercise 11.3.

4. We have $a = \psi'(\alpha)$, $b = -1/\beta$ and $c = \alpha/\beta^2$ and thus the determinant of $\mathcal{I}(\alpha, \beta)$ is $ac - b^2 = (\alpha\psi'(\alpha) - 1)/\beta^2$. The lower right element of the inverse is given by

$$(\mathcal{I}(\alpha, \beta)^{-1})_{22} = \frac{a}{ac - b^2} = \frac{\psi'(\alpha)\,\beta^2}{\alpha\psi'(\alpha) - 1}.$$

For a sample of size n, the information matrix is $n\mathcal{I}(\alpha, \beta)$, the inverse is $\mathcal{I}(\alpha, \beta)^{-1}/n$ and thus, by proposition 14.5, we find the lower bound $\beta^2\psi'(\alpha)/(n(\alpha\psi'(\alpha) - 1))$ for an unbiased estimator for β as claimed. The bound for known shape is $1/\mathcal{I}_n(\beta) = \beta^2/(n\alpha)$ and the ratio of the two is

$$\frac{\beta^2\psi'(\alpha)/(n(\alpha\psi'(\alpha) - 1))}{\beta^2/(n\alpha)} = \frac{\alpha\psi'(\alpha)}{\alpha\psi'(\alpha) - 1} = \frac{1}{1 - \rho^2},$$

where $\rho^2 = 1/(\alpha\psi'(\alpha))$ is as in example 14.10. For $\alpha = 2$ we have $\psi'(2) = \pi^2/6 - 1 \approx 0.6449$ and thus $\alpha\psi'(\alpha) \approx 1.2899$ and the ratio is approximately $1.2899/0.2899 \approx 4.45$: not knowing the shape increases the bound for the rate by a factor of approximately 4.45, as in example 14.10.

□

Solution to exercise 14.3.

1. Since the two samples are independent, the log-likelihood is the sum of the two sample log-likelihoods:

$$\ell(\mu_1, \mu_2) = -\frac{m + n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{m}(X_i - \mu_1)^2 - \frac{1}{2\sigma^2}\sum_{j=1}^{n}(Y_j - \mu_2)^2,$$

with partial derivatives $\partial\ell/\partial\mu_1 = \sum_i(X_i - \mu_1)/\sigma^2$ and $\partial\ell/\partial\mu_2 = \sum_j(Y_j - \mu_2)/\sigma^2$. The second partial derivatives are $\partial^2\ell/\partial\mu_1^2 = -m/\sigma^2$, $\partial^2\ell/\partial\mu_2^2 = -n/\sigma^2$ and $\partial^2\ell/\partial\mu_1\,\partial\mu_2 = 0$, since each first partial derivative involves only one of the two parameters. None of these second derivatives are random, and thus proposition 14.3 gives $\mathcal{I}_{m+n}(\mu_1, \mu_2) = \operatorname{diag}(m/\sigma^2, n/\sigma^2)$, without any expectations.

2. The difference is $\mu_1 - \mu_2 = a^\top\theta$ with $a = (1, -1)$ and the inverse information matrix is $\operatorname{diag}(\sigma^2/m, \sigma^2/n)$. By proposition 14.5 any unbiased estimator for $\mu_1 - \mu_2$ has variance at least

$$a^\top\mathcal{I}_{m+n}(\mu_1, \mu_2)^{-1}a = \frac{\sigma^2}{m} + \frac{\sigma^2}{n}.$$

The estimator $\bar{X} - \bar{Y}$ is unbiased for $\mu_1 - \mu_2$ by lemma 2.2 and, since the two samples are independent, the variance of this estimator is $\operatorname{Var}(\bar{X}) + \operatorname{Var}(\bar{Y}) = \sigma^2/m + \sigma^2/n$. This is the bound. Thus, $\bar{X} - \bar{Y}$ is an efficient estimator for the difference of the means.

3. If we include σ^2 as an additional parameter, the first derivative is

$$\frac{\partial\ell}{\partial\sigma^2} = -\frac{m + n}{2\sigma^2} + \frac{1}{2\sigma^4}\Big(\sum_{i=1}^{m}(X_i - \mu_1)^2 + \sum_{j=1}^{n}(Y_j - \mu_2)^2\Big).$$

The mixed derivatives with respect to μ_1 and σ^2 and with respect to μ_2 and σ^2 are $-\sum_i(X_i-\mu_1)/\sigma^4$ and $-\sum_j(Y_j-\mu_2)/\sigma^4$, both with expectation zero, and the second derivative with respect to σ^2 has expectation $(m+n)/(2\sigma^4) - (m+n)\sigma^2/\sigma^6 = -(m+n)/(2\sigma^4)$, as shown in example 14.6. Thus, the information matrix is

$$\mathcal{I}_{m+n}(\mu_1, \mu_2, \sigma^2) = \operatorname{diag}\Big(\frac{m}{\sigma^2}, \frac{n}{\sigma^2}, \frac{m+n}{2\sigma^4}\Big),$$

still diagonal, and thus σ^2 is orthogonal to both means. The upper left 2×2 block of the inverse is the inverse information matrix from part (b), and the Cramér–Rao bound for $\mu_1 - \mu_2$ is still $\sigma^2/m + \sigma^2/n$, and this bound is achieved by $\bar{X} - \bar{Y}$, whose definition does not depend on σ^2. For point estimation of the difference, not knowing the variance does not change anything. However, once we want to determine the accuracy of an estimate, the unknown variance starts to matter, since the bound depends on σ^2 and thus needs to be estimated. This is the reason why we go from the normal distribution to the t distribution in lectures 26 and 31.

□

C.11 Consistency of the MLE

Here we solve the exercises at the end of lecture 16.

Solution to exercise 16.1.

1. For (R1) we have two different rates and thus two different distributions: for example, the means $1/\theta$ are different. For (R2) the support is $[0, \infty)$ for all θ. For (R3) we have $\log f(x;\theta) = \log\theta - \theta x$ and the derivatives with respect to θ are $1/\theta - x$, $-1/\theta^2$ and $2/\theta^3$, all of which are continuous in θ. The exponential distribution is an exponential family, and thus by lecture 11 the conditions of section 11.1 hold, so that we can interchange the order of differentiation and integration twice. The Fisher information $\mathcal{I}(\theta) = 1/\theta^2$ is continuous and lies strictly between zero and infinity, so that (R3) holds. Finally, for (R4), the third derivative $2/\theta^3$ does not depend on x and for $|\theta - \theta_0| < \delta$ with $\delta = \theta_0/2$ we have $\theta > \theta_0/2$ and thus $2/\theta^3 \leq 16/\theta_0^3$. The constant $M(x) = 16/\theta_0^3$ bounds the third derivative and has finite expectation.

2. The likelihood equation $n/\theta - \sum_i x_i = 0$ has only one solution $\theta = 1/\bar{x}$, since $\sum_i x_i > 0$ with probability one. Since $\ell''(\theta) = -n/\theta^2 < 0$, the log-likelihood function is strictly concave and thus the root is the global maximum of the likelihood function, *i.e.* the MLE as in example 5.2. The remaining assumption of the theorem also holds: for all $\theta, \theta' > 0$ the log-likelihood ratio $\log\big(f(X_1;\theta)/f(X_1;\theta')\big) = \log(\theta/\theta') - (\theta - \theta')X_1$ has the finite mean $\log(\theta/\theta') - (\theta - \theta')/\theta'$ under $\mathbb{P}_{\theta'}$. Since all conditions of theorem 16.3 are satisfied, with the remark after its proof covering the sample $x_1 = \cdots = x_n = 0$, we find that $1/\bar{X}$ is consistent for θ.

3. By the law of large numbers, proposition A.14, we have $\bar{X} \xrightarrow{\text{p}} \mathbb{E}_\theta(X_1) = 1/\theta$. Since the function $g(y) = 1/y$ is continuous at $y = 1/\theta > 0$,

the continuous mapping theorem, proposition A.17, applies and we find $1/\bar{X} \xrightarrow{\text{p}} g(1/\theta) = \theta$. This proves the consistency of the MLE.

□

Solution to exercise 16.2.

1. Since $f(x;\theta) = \theta^x(1-\theta)^{1-x}$ for $x \in \{0,1\}$, we have

$$\log \frac{f(X_1;\theta)}{f(X_1;\theta_0)} = X_1 \log \frac{\theta}{\theta_0} + (1 - X_1) \log \frac{1-\theta}{1-\theta_0},$$

and taking expectations with $\mathbb{E}_{\theta_0}(X_1) = \theta_0$ gives

$$D(\theta) = \theta_0 \log \frac{\theta}{\theta_0} + (1-\theta_0) \log \frac{1-\theta}{1-\theta_0}.$$

2. Using the rule $\log y \leq y - 1$ for logarithms, we find

$$D(\theta) \leq \theta_0\Big(\frac{\theta}{\theta_0} - 1\Big) + (1-\theta_0)\Big(\frac{1-\theta}{1-\theta_0} - 1\Big) = (\theta - \theta_0) + (\theta_0 - \theta) = 0.$$

For $D(\theta) = 0$ to hold, we need to have equality in both applications of the inequality, *i.e.* we need $\theta/\theta_0 = 1$ and $(1-\theta)/(1-\theta_0) = 1$. Both of these conditions state $\theta = \theta_0$. Thus we have $D(\theta) < 0$ for all $\theta \neq \theta_0$, as stated in lemma 16.2.

□

Solution to exercise 16.3.

1. Since all observations are at most θ, we have $\hat{\theta}_n \leq \theta$ and thus $|\hat{\theta}_n - \theta| \geq \varepsilon$ is the same as $\hat{\theta}_n \leq \theta - \varepsilon$. This is the case if and only if all n observations are at most $\theta - \varepsilon$. Using independence we find

$$\mathbb{P}_\theta(|\hat{\theta}_n - \theta| \geq \varepsilon) = \Big(\frac{\theta - \varepsilon}{\theta}\Big)^n = \Big(1 - \frac{\varepsilon}{\theta}\Big)^n \longrightarrow 0 \qquad (n \to \infty),$$

since $0 < 1 - \varepsilon/\theta < 1$. For $\varepsilon \geq \theta$, the probability of this event is zero. Thus $\hat{\theta}_n \xrightarrow{\text{p}} \theta$ and the MLE is consistent.

2. Condition (R2) is violated, since the support $(0,\theta)$ depends on θ and (R3) is also violated: for fixed x, the function $\theta \mapsto f(x;\theta)$ has a jump from 0 to $1/\theta$ at $\theta = x$ and thus is not differentiable there. The MLE $\max_i x_i$ is unique, but it is not a root of the likelihood equation, since the equation has no roots: on the region $\theta \geq \max_i x_i$ where L is positive, the derivative $-n\theta^{-n-1}$ never equals zero. The proof of theorem 16.3 identifies the MLE with a root of the derivative of a differentiable likelihood, and neither of these is available here. Nevertheless, the statement is true, as can be seen by the direct computation in the first part of the question, or using the mean squared error from exercise 2.2.

□

C.12 Asymptotic Normality of the MLE

Here we solve the exercises at the end of lecture 17.

Solution to exercise 17.1.

1. From $\ell(\theta) = n \log \theta - \theta \sum_i X_i$ we find $\ell'(\theta) = n/\theta - n\bar{X}$, $\ell''(\theta) = -n/\theta^2$ and $\ell'''(\theta) = 2n/\theta^3$. Thus we have
$$A_n = \sqrt{n}\Big(\frac{1}{\theta_0} - \bar{X}\Big), \qquad B_n = \frac{1}{\theta_0^2}, \qquad C_n = \frac{\hat{\theta}_n - \theta_0}{\xi_n^3}.$$

2. The observations have mean $1/\theta_0$ and variance $1/\theta_0^2$. Thus, by the central limit theorem, $\sqrt{n}\,(\bar{X} - 1/\theta_0) \xrightarrow{\text{d}} N(0, 1/\theta_0^2)$, and since the normal distribution with mean zero is symmetric, the same holds for $A_n = -\sqrt{n}\,(\bar{X} - 1/\theta_0)$. This agrees with the limit $N\big(0, \mathcal{I}(\theta_0)\big)$ from the proof. The second derivative ℓ'' does not depend on the data, and thus $B_n = \mathcal{I}(\theta_0)$ holds exactly, without the law of large numbers.

3. On the event $|\hat{\theta}_n - \theta_0| < \theta_0/2$ the point ξ_n, which lies between θ_0 and $\hat{\theta}_n$, satisfies $\xi_n > \theta_0/2$, and thus $|C_n| \leq 8\,|\hat{\theta}_n - \theta_0|/\theta_0^3$. Let $\eta > 0$. Then the event $|C_n| \geq \eta$ is contained in the union of the events $|\hat{\theta}_n - \theta_0| \geq \theta_0/2$ and $|\hat{\theta}_n - \theta_0| \geq \eta\theta_0^3/8$. Since $\hat{\theta}_n$ is consistent by exercise 16.1, both events have probability tending to zero, and thus $C_n \xrightarrow{\text{p}} 0$.

□

Solution to exercise 17.2.

1. Since we have $\mathcal{I}(\theta) = 1/\big(\theta(1-\theta)\big)$, theorem 17.1 gives that $\sqrt{n}\,(\bar{X} - \theta) \xrightarrow{\text{d}} N\big(0, \theta(1-\theta)\big)$. Since the mean and variance of the Bernoulli distribution equal θ and $\theta(1-\theta)$, respectively, this is the central limit theorem, theorem A.15, for $\bar{X}$ and thus the two results coincide.

2. From proposition 5.6 we know that on the event $0 < \bar{X} < 1$ the MLE for ψ is given by $\hat{\psi}_n = \log\big(\bar{X}/(1-\bar{X})\big)$. Taking derivatives we find the derivative of $g(\theta) = \log\theta - \log(1-\theta)$ to be $g'(\theta) = 1/\theta + 1/(1-\theta) = 1/\big(\theta(1-\theta)\big)$. Let $T_n = \bar{X}$ on the event $0 < \bar{X} < 1$ and $T_n = 1/2$ otherwise. The difference $\sqrt{n}\,(T_n - \theta) - \sqrt{n}\,(\bar{X} - \theta)$ is zero on this event, whose complement has probability $\theta^n + (1-\theta)^n \to 0$. Thus the difference converges in probability to zero, and the central limit theorem and Slutsky's theorem, proposition A.16, give $\sqrt{n}\,(T_n - \theta) \xrightarrow{\text{d}} N\big(0, \theta(1-\theta)\big)$. The value $g(T_n)$ is defined for every sample and equals $\hat{\psi}_n$ on the event $0 < \bar{X} < 1$. Using the delta method, proposition A.18, for $g(T_n)$, and then the same argument with Slutsky's theorem once more, we find
$$\sqrt{n}\,(\hat{\psi}_n - \psi) \xrightarrow{\text{d}} N\Big(0,\ \frac{\theta(1-\theta)}{\theta^2(1-\theta)^2}\Big) = N\Big(0, \frac{1}{\theta(1-\theta)}\Big).$$

3. Substituting θ by $\hat{\theta}_n$ in the asymptotic variance, we find the standard error $\text{se}(\hat{\psi}_n) = 1/\sqrt{n\hat{\theta}_n(1-\hat{\theta}_n)}$ and thus the approximate 95% Wald interval $\hat{\psi}_n \pm 1.96/\sqrt{n\hat{\theta}_n(1-\hat{\theta}_n)}$. Both of these quantities are undefined if all observations are equal, an event of vanishing probability.

□

Solution to exercise 17.3.

1. Using $\mathcal{I}(\theta) = 1/\theta$ from example 11.6, we find $\sqrt{n}\,(\bar{X} - \theta) \xrightarrow{\text{d}} N(0, \theta)$. Since the Poisson distribution has mean and variance equal to θ, this is the central limit theorem for $\bar{X}$.

2. For $g(\theta) = e^{-\theta}$ we have $g'(\theta) = -e^{-\theta}$ and using the delta method we find

$$\sqrt{n}\,(\hat{p}_0 - p_0) \xrightarrow{\text{d}} N(0,\ e^{-2\theta}\theta).$$

   In exercise 5.7 we found $\mathbb{E}_\theta(\hat{p}_0) = \exp(n\theta(e^{-1/n} - 1))$ and expanding $e^{-1/n} = 1 - 1/n + 1/(2n^2) - \cdots$ we find that the bias is approximately $\theta e^{-\theta}/(2n)$, which is of order $1/n$. Since it is multiplied by $\sqrt{n}$, the bias goes to zero and thus is of a smaller order than the spread $1/\sqrt{n}$ and does not contribute in the limit: the estimator is biased for finite n, but is asymptotically normal with mean p_0.

3. By theorem 13.2 with $g(\theta) = e^{-\theta}$ and $\mathcal{I}_n(\theta) = n/\theta$, the Cramér–Rao bound for unbiased estimators for $e^{-\theta}$ is

$$\frac{g'(\theta)^2}{\mathcal{I}_n(\theta)} = \frac{\theta e^{-2\theta}}{n},$$

   and n times this bound is the asymptotic variance $\theta e^{-2\theta}$ from the previous part. By exercise 13.3 no unbiased estimator attains the bound for any fixed n, but the MLE $\hat{p}_0$ attains it in the limit. Asymptotic efficiency is thus a much weaker requirement than exact efficiency, and it is met by functions of the parameter which have no efficient estimator.

□

Solution to exercise 17.4.

1. Since $1/\mathcal{I}(p) = p^2(1-p)$, theorem 17.1 gives

$$\sqrt{n}\left(\frac{1}{\bar{X}} - p\right) \xrightarrow{\text{d}} N(0,\ p^2(1-p)).$$

2. By the central limit theorem, theorem A.15, we have $\sqrt{n}\,(\bar{X} - 1/p) \xrightarrow{\text{d}} N(0, (1-p)/p^2)$. The function $g(y) = 1/y$ has $g'(y) = -1/y^2$, and thus $g'(1/p) = -p^2 \neq 0$. The delta method, proposition A.18, gives

$$\sqrt{n}\left(\frac{1}{\bar{X}} - p\right) \xrightarrow{\text{d}} N\left(0,\ p^4 \cdot \frac{1-p}{p^2}\right) = N(0,\ p^2(1-p)),$$

   which is the same result.

3. The estimate is $\hat{p}_n = 1/2.5 = 0.4$. The standard error is

$$\text{se}(\hat{p}_n) = \frac{1}{\sqrt{\mathcal{I}_n(\hat{p}_n)}} = \hat{p}_n\sqrt{\frac{1-\hat{p}_n}{n}} = 0.4 \cdot \sqrt{\frac{0.6}{40}} \approx 0.049,$$

   and the approximate 95% Wald interval is $0.4 \pm 1.96 \cdot 0.049$, *i.e.* approximately [0.304, 0.496].

□

Solution to exercise 17.5.

1. For $x \geq 0$ and $n > x/\theta$, the event $n(\theta - \hat{\theta}_n) > x$ is the event $\hat{\theta}_n < \theta - x/n$, *i.e.* that all observations are less than $\theta - x/n$. Thus we have

$$\mathbb{P}_\theta(n(\theta - \hat{\theta}_n) > x) = \left(1 - \frac{x}{n\theta}\right)^n \longrightarrow e^{-x/\theta} \qquad (n \to \infty),$$

where we used the standard limit $(1 - a/n)^n \to e^{-a}$. The right-hand side is the probability that an exponential random variable with mean θ is greater than x. Thus, $n(\theta - \hat{\theta}_n)$ converges in distribution to the corresponding exponential distribution.

2. Since $\sqrt{n}\,(\hat{\theta}_n - \theta) = -n^{-1/2} \cdot n(\theta - \hat{\theta}_n)$, the second factor converges in distribution to the result from the first part, and the first factor is a constant which converges to zero. Thus, by Slutsky's theorem (proposition A.16), the product converges in distribution to $0 \cdot Y = 0$. This is equivalent to convergence in probability to zero, as noted after proposition A.16. Thus we cannot have a limit $N(0, \tau^2)$ where $\tau^2 > 0$, since the error of the MLE is of order $1/n$ and not of order $1/\sqrt{n}$. This is a manifestation of the super-efficiency from lecture 13, seen in the limiting distribution: the variance of the bias-corrected maximum is of order $1/n^2$, smaller than allowed by the Cramér–Rao bound, and thus neither the bound nor theorem 17.1 apply, since the model violates (R2).

□

C.13 The EM Algorithm

Here we solve the exercises at the end of lecture 19.

Solution to exercise 19.1.

1. Using the formula for the sum of a geometric series, we find

$$\begin{aligned}\mathbb{P}_p(X \geq m) &= \sum_{x=m}^{\infty} (1-p)^{x-1} p = p\,(1-p)^{m-1} \sum_{j=0}^{\infty} (1-p)^j \\ &= p\,(1-p)^{m-1} \cdot \frac{1}{p} = (1-p)^{m-1}.\end{aligned}$$

Since the event $\{X \geq m + k\}$ is contained in the event $\{X \geq m\}$, we can use the definition of conditional probability to get

$$\mathbb{P}_p(X \geq m + k | X \geq m) = \frac{(1-p)^{m+k-1}}{(1-p)^{m-1}} = (1-p)^k = \mathbb{P}_p(X \geq k+1).$$

This shows that, given that the first $m-1$ trials were failures, the number $X - (m-1)$ of additional trials required until the first success is still geometrically distributed. The information about the number of failures so far does not change the distribution of the number of additional failures required. This is the discrete analogue of the statement $\mathbb{P}_\theta(X > s+t | X > s) = \mathbb{P}_\theta(X > t)$ for the exponential distribution in exercise 5.6. The shift by one is due to the fact that the support of the geometric distribution starts at 1 instead of at 0.

2. Given $X \geq m$, the conditional probability weights of X are

$$\frac{f(x;p)}{\mathbb{P}_p(X \geq m)} = (1-p)^{x-m}p$$

for $x \geq m$. Substituting $x = m - 1 + j$ with $j \in \mathbb{N}$ we get

$$\begin{aligned}\mathbb{E}_p(X \mid X \geq m) &= \sum_{j=1}^{\infty}(m-1+j)(1-p)^{j-1}p \\ &= (m-1)\sum_{j=1}^{\infty}(1-p)^{j-1}p + \sum_{j=1}^{\infty} j(1-p)^{j-1}p \\ &= (m-1) + \mathbb{E}_p(X),\end{aligned}$$

where the first sum is the total probability 1 and the second sum is the mean of the geometric distribution. Using $\mathbb{E}_p(X) = 1/p$ we get the claim. Alternatively we can get the same result from the first part of the solution: given $X \geq m$, the value $X - (m-1)$ is geometrically distributed with parameter p and thus has mean $1/p$.

□

Solution to exercise 19.2.

1. Since we know π, the complete-data log-likelihood is the same as in example 19.1 and the terms $\log \pi$ and $\log(1-\pi)$ are now constants. The E step determines the responsibilities $\gamma_{i,k} = \pi\varphi_\sigma(x_i - \mu_{1,k})/(\pi\varphi_\sigma(x_i - \mu_{1,k}) + (1-\pi)\varphi_\sigma(x_i - \mu_{2,k}))$. Note that $Q(\theta|\theta_k)$ depends on $\theta = (\mu_1, \mu_2)$ only via the terms $-\frac{1}{2\sigma^2}\sum_i[\gamma_{i,k}(x_i-\mu_1)^2 + (1-\gamma_{i,k})(x_i-\mu_2)^2]$. The partial derivatives are as shown in the example, and setting them equal to zero gives the two mean updates of (19.1). The update for the weight is no longer needed.

2. Since $\varphi'_\sigma(u) = -(u/\sigma^2)\varphi_\sigma(u)$, we have $\frac{\partial}{\partial\mu_1}\varphi_\sigma(x-\mu_1) = \frac{x-\mu_1}{\sigma^2}\varphi_\sigma(x-\mu_1)$ and thus we can take derivatives of $\ell(\theta) = \sum_i \log(\pi\varphi_\sigma(x_i-\mu_1) + (1-\pi)\varphi_\sigma(x_i-\mu_2))$ to get

$$\begin{aligned}\frac{\partial\ell}{\partial\mu_1}(\theta) &= \sum_{i=1}^{n}\frac{\pi\varphi_\sigma(x_i-\mu_1)}{\pi\varphi_\sigma(x_i-\mu_1) + (1-\pi)\varphi_\sigma(x_i-\mu_2)}\cdot\frac{x_i-\mu_1}{\sigma^2} \\ &= \frac{1}{\sigma^2}\sum_{i=1}^{n}\gamma_i(\theta)\,(x_i-\mu_1),\end{aligned}$$

and similarly we find $\frac{\partial\ell}{\partial\mu_2}(\theta) = \frac{1}{\sigma^2}\sum_i(1-\gamma_i(\theta))(x_i-\mu_2)$. Setting both of these equal to zero gives $\mu_1 = \sum_i \gamma_i(\theta)x_i / \sum_i \gamma_i(\theta)$ and the corresponding equation for μ_2, stating that θ is mapped to itself by the updates of the first part. Thus, the likelihood equations and the fixed-point equations of the EM algorithm coincide. The computation also shows that the derivative of ℓ equals the derivative of $\theta' \mapsto Q(\theta'|\theta)$ at $\theta' = \theta$. In lecture 20 we will see that this identity holds in general (proposition 20.3).

3. As $\pi \to 1$, the responsibilities converge to $\gamma_{i,k} = 1$ for all observations since $\varphi_\sigma(x_i - \mu_{1,k}) > 0$. Thus, the update for μ_1 is now $\mu_{1,k+1} = \bar{x}$

independent of the initial value, and this is the MLE for the mean of a single normal sample in example 5.4, reached in one step. Since $1-\gamma_{i,k} \sim (1-\pi)\varphi_\sigma(x_i-\mu_{2,k})/\varphi_\sigma(x_i-\mu_{1,k})$, the update for μ_2 converges to the weighted mean of the x_i with weights $\varphi_\sigma(x_i-\mu_{2,k})/\varphi_\sigma(x_i-\mu_{1,k})$. At $\pi=1$ itself the mixture density does not involve μ_2, and the data carry no information about this mean.

□

C.14 Properties of the EM Algorithm

Here we solve the exercises at the end of lecture 20.

Solution to exercise 20.1.

1. The log-likelihood function for the given data is

$$\ell(\pi) = \sum_{i=1}^{n} \log\bigl(\pi f_1(x_i) + (1-\pi) f_2(x_i)\bigr).$$

   Taking derivatives we get the likelihood equation

$$\ell'(\pi) = \sum_{i=1}^{n} \frac{f_1(x_i) - f_2(x_i)}{\pi f_1(x_i) + (1-\pi) f_2(x_i)} = 0.$$

   This equation cannot be solved analytically for general densities.

2. The density of the complete data is $f_c(x,1;\pi) = \pi f_1(x)$ and $f_c(x,2;\pi) = (1-\pi)f_2(x)$. Thus we have

$$\ell_c(\pi) = \sum_{i=1}^{n} \Bigl[\mathbf{1}_{\{Z_i=1\}} \bigl(\log\pi + \log f_1(x_i)\bigr) + \mathbf{1}_{\{Z_i=2\}} \bigl(\log(1-\pi) + \log f_2(x_i)\bigr) \Bigr].$$

   Using Bayes' rule we find the conditional probability of $Z_i = 1$ given $X_i = x_i$ under π_k as $\gamma_i(\pi_k) = \pi_k f_1(x_i)/(\pi_k f_1(x_i) + (1-\pi_k) f_2(x_i))$. Substituting the indicators using these responsibilities we get

$$Q(\pi \mid \pi_k) = \log\pi \sum_{i=1}^{n} \gamma_i(\pi_k) + \log(1-\pi) \sum_{i=1}^{n} \bigl(1-\gamma_i(\pi_k)\bigr) + \text{const},$$

   where the constant term combines the terms $\log f_1(x_i)$ and $\log f_2(x_i)$, which do not depend on π. Setting the derivative $\sum_i \gamma_i/\pi - \sum_i (1-\gamma_i)/(1-\pi)$ equal to zero we find $\pi_{k+1} = \frac{1}{n}\sum_i \gamma_i(\pi_k)$ and the second derivative $-\sum_i \gamma_i/\pi^2 - \sum_i (1-\gamma_i)/(1-\pi)^2$ is negative for every $\pi \in (0,1)$, so $Q(\,\cdot\,\mid\pi_k)$ is concave on $(0,1)$ and the stationary point is the global maximum.

3. Using the notation $f_i = f(x_i;\pi) = \pi f_1(x_i) + (1-\pi) f_2(x_i)$ for the denominator, we have

$$\begin{aligned} \gamma_i(\pi) - \pi &= \frac{\pi f_1(x_i) - \pi\bigl(\pi f_1(x_i) + (1-\pi) f_2(x_i)\bigr)}{f_i} \\ &= \frac{\pi(1-\pi)\bigl(f_1(x_i) - f_2(x_i)\bigr)}{f_i}. \end{aligned}$$

Summing over i and comparing to the first part of the solution we find $\frac{1}{n}\sum_i \gamma_i(\pi) - \pi = \pi(1-\pi)\,\ell'(\pi)/n$. A value $\pi^* \in (0,1)$ is a fixed point of the iteration if and only if the left-hand side is zero at π^* and since $\pi^*(1-\pi^*) > 0$ this is the case if and only if $\ell'(\pi^*) = 0$. Thus the fixed points of EM are the roots of the likelihood equation. Proposition 20.3 gives one direction of this equivalence, and for this model the converse holds as well.

□

Solution to exercise 20.2.

1. Since $\pi_0 = 1/2$, the weights cancel, and dividing the numerator and denominator by $\varphi_1(x_i - 2)$ gives

$$\gamma_{i,0} = \frac{1}{1 + \exp\bigl(\frac{1}{2}(x_i-2)^2 - \frac{1}{2}(x_i-4)^2\bigr)} = \frac{1}{1+e^{2x_i-6}}.$$

   Thus, to four decimal digits we have

$$(\gamma_{1,0},\dots,\gamma_{5,0}) = (0.9975,\ 0.9820,\ 0.0180,\ 0.0025,\ 0.0003),$$

   and the two small observations are assigned nearly completely to group 1, the three large ones to group 2. With $\sum_i \gamma_{i,0} = 2.0003$, $\sum_i \gamma_{i,0} x_i = 1.0891$ and $\sum_i (1-\gamma_{i,0}) x_i = 17.9109$, the updates (19.1) give

$$\theta_1 = \Bigl(\frac{2.0003}{5},\ \frac{1.0891}{2.0003},\ \frac{17.9109}{2.9997}\Bigr) = (0.4001,\ 0.5445,\ 5.9709).$$

2. At $\theta_0 = (1/2, 2, 4)$, the mixture density of observation i is $\frac{1}{2}\varphi_1(x_i-2) + \frac{1}{2}\varphi_1(x_i-4)$. Using the standard normal density values $\varphi_1(1) = 0.2420$, $\varphi_1(2) = 0.05399$, $\varphi_1(3) = 0.004432$, $\varphi_1(4) = 0.0001338$ and $\varphi_1(5) = 0.0000015$, we find the five densities to be 0.027062, 0.12320, 0.12320, 0.027062 and 0.0022167. The corresponding logarithms are

$$-3.6096,\quad -2.0939,\quad -2.0939,\quad -3.6096,\quad -6.1118$$

   and thus $\ell(\theta_0) = -17.5188$. At $\theta_1 = (0.4001, 0.5445, 5.9709)$ the densities are 0.13763, 0.14389, 0.14939, 0.23922 and 0.14094. The corresponding logarithms are

$$-1.9832,\quad -1.9387,\quad -1.9012,\quad -1.4304,\quad -1.9595$$

   and thus $\ell(\theta_1) = -9.2129$. The log-likelihood has increased by 8.31 as required by proposition 20.2. The largest contribution, 4.15, comes from observation $x_5 = 7$: under θ_0 this observation is three standard deviations away from the nearer mean 4 and has tiny density, whereas under θ_1 it is about one standard deviation away from the mean 5.97. The observations $x_1 = 0$ and $x_4 = 6$ contribute 1.63 and 2.18, respectively, for the same reason. The observations $x_2 = 1$ and $x_3 = 5$, which were always within one standard deviation of a mean, contribute little.

This completes the solution. □

C.15 Hypothesis Testing

Here we solve the exercises at the end of lecture 22.

Solution to exercise 22.1.

1. By equation (22.1) the power at θ_1 is $\beta(\theta_1) = \Phi(\delta - z_{1-\alpha})$, where $\delta = \sqrt{n}\,(\theta_1 - \theta_0)/\sigma$. Since Φ is strictly increasing with $\Phi(z_{1-\gamma}) = 1 - \gamma$, we have $\beta(\theta_1) \geq 1 - \gamma$ if and only if $\delta - z_{1-\alpha} \geq z_{1-\gamma}$, *i.e.* if and only if

$$\sqrt{n} \geq \frac{(z_{1-\alpha} + z_{1-\gamma})\,\sigma}{\theta_1 - \theta_0}.$$

   Since $1 - \gamma > \alpha$ we have $z_{1-\gamma} > z_\alpha = -z_{1-\alpha}$, and thus the right-hand side is positive; squaring both sides gives the stated condition.

2. With $z_{0.95} = 1.645$ and $z_{0.9} = 1.282$ the bound is

$$n \geq \Big(\frac{(1.645 + 1.282) \cdot 2}{0.5}\Big)^2 = 11.708^2 = 137.1,$$

   and thus $n = 138$ observations are needed.

3. The bound is proportional to $\sigma^2/(\theta_1 - \theta_0)^2$. Halving the difference $\theta_1 - \theta_0$ multiplies the required sample size by four, to $n = 549$ in the numerical example (the bound becomes 548.3), and doubling σ has the same effect. Detecting a difference half as large costs four times as many observations.

□

Solution to exercise 22.2.

1. Replacing δ by $-\delta$ swaps the two terms, and thus we find

$$g(-\delta) = \Phi(-z + \delta) + \Phi(-\delta - z) = g(\delta).$$

   The value $\theta_0 + h$ corresponds to $\delta = \sqrt{n}\,h/\sigma$ and the value $\theta_0 - h$ to $-\delta$, and thus we have $\beta(\theta_0 + h) = \beta(\theta_0 - h)$ for all h.

2. Let $\Phi'(x) = e^{-x^2/2}/\sqrt{2\pi}$ be the standard normal density. Then

$$g'(\delta) = -\Phi'(-z - \delta) + \Phi'(\delta - z) = \Phi'(z - \delta) - \Phi'(z + \delta),$$

   since Φ' is symmetric. For $\delta > 0$ we have $|z - \delta| < z + \delta$, since $z > 0$, and since $\Phi'(x)$ is strictly decreasing in $|x|$ we find $g'(\delta) > 0$. Thus g is strictly increasing on $[0, \infty)$, and by the first part it is strictly decreasing on $(-\infty, 0]$. The minimum of g is therefore $g(0) = 2\Phi(-z) = 2(1 - \Phi(z_{1-\alpha/2})) = \alpha$. Since δ is an increasing function of θ with $\delta = 0$ for $\theta = \theta_0$, the power function β has its minimum value α at θ_0.

□

Solution to exercise 22.3.

1. Under H_0 the total T is binomially distributed with parameters 10 and $1/2$. Thus, the test size is

$$\beta(1/2) = \mathbb{P}_{1/2}(T \geq c) = \frac{1}{2^{10}} \sum_{k=c}^{10} \binom{10}{k}.$$

2. We can compute the binomial coefficients $\binom{10}{10} = 1$, $\binom{10}{9} = 10$ and $\binom{10}{8} = 45$ to find the sizes $1/1024 = 0.0010$ for $c = 10$, $11/1024 = 0.0107$ for $c = 9$ and $56/1024 = 0.0547$ for $c = 8$. The smallest value c with size less than or equal to 0.05 is $c = 9$ with size 0.0107. Since T can only take the values $0, \ldots, 10$, every $c \leq 0$ gives size 1 and every $c \geq 11$ gives size 0. Thus the size can only take the twelve values given by the formula from the first part for $c = 0, \ldots, 11$. The value 0.05 is not one of these (the size jumps from 0.0107 to 0.0547 between $c = 9$ and $c = 8$).

3. The power at θ is $\mathbb{P}_\theta(T \geq 9) = 10\theta^9(1-\theta) + \theta^{10} = \theta^9(10 - 9\theta)$. This gives the values 0.149 for $\theta = 0.7$, 0.376 for $\theta = 0.8$ and 0.736 for $\theta = 0.9$. A coin which shows heads in four out of five tosses on average is detected in fewer than 40% of all samples of ten tosses. With ten observations the test can only detect large deviations from fairness with reasonable probability, and a failure to reject the hypothesis is not very meaningful.

4. The p-value is the probability of the event that the total under H_0 is larger than or equal to the observed value, *i.e.* the probability $\mathbb{P}_{1/2}(T \geq 9) = 11/1024 = 0.0107$. Since this probability is less than or equal to 0.05, we can reject H_0 at a significance level of 0.05. This agrees with the critical region $T \geq 9$ from the second part of the question.

□

Solution to exercise 22.4.

1. The maximum of the sample is at most m, if and only if all observations are at most m. Since the observations are independent, for $0 \leq m \leq \theta$ we have

$$\mathbb{P}_\theta(M \leq m) = \prod_{i=1}^{n} \mathbb{P}_\theta(X_i \leq m) = \Big(\frac{m}{\theta}\Big)^n.$$

   This is also the integral of the density from example 8.9.

2. For $c \leq \theta_0$ the size is $\mathbb{P}_{\theta_0}(M > c) = 1 - (c/\theta_0)^n$, as found in the first part. Setting this equal to α gives $c = \theta_0(1-\alpha)^{1/n}$.

3. For $\theta > \theta_0 \geq c$ the power function is

$$\beta(\theta) = \mathbb{P}_\theta(M > c) = 1 - \Big(\frac{c}{\theta}\Big)^n = 1 - (1-\alpha)\Big(\frac{\theta_0}{\theta}\Big)^n.$$

   This function increases from α for $\theta = \theta_0$ to 1 as $\theta \to \infty$. For $n = 5$, $\theta_0 = 1$ and $\alpha = 0.05$ we have $c = 0.95^{1/5} = 0.990$, and for $\theta = 1.2$ the power is $1 - 0.95/1.2^5 = 1 - 0.95 \cdot 0.402 = 0.618$.

4. Under H_0 all observations are in the interval $[0, \theta_0]$ and thus we have $\mathbb{P}_{\theta_0}(M > \theta_0) = 0$. The second test has size 0 and is a level α test for every α. The power function of this test is $\mathbb{P}_\theta(M > \theta_0) = 1 - (\theta_0/\theta)^n$ for $\theta > \theta_0$. This function equals 0.598 for $\theta = 1.2$ in the numerical example. The difference between the two power functions is $\alpha(\theta_0/\theta)^n$, and this is positive for every alternative, *i.e.* the first test is more powerful everywhere on H_1, but the first test has type I error probability α which the second test avoids. In the context of section 22.3, the first test is to be preferred, since we have agreed on a level α, and a test which does not use the level would be a waste of power.

□

Solution to exercise 22.5.

1. The observed value of the standardised statistic is $z = \sqrt{16}\,(52.3 - 50)/4 = 2.3$ and the one-sided p-value is $p = 1 - \Phi(2.3) = 0.0107$. Since $0.0107 \leq 0.05$ we can reject H_0 at level 0.05, but since $0.0107 > 0.01$ we cannot reject at level 0.01.

2. The two-sided p-value is $2\big(1 - \Phi(2.3)\big) = 0.0214$. Again we reject at level 0.05 and we cannot reject at level 0.01.

3. The mean θ is a fixed, unknown number and not a random variable, so we cannot determine a probability for the statement $\theta = 50$. The p-value is a probability about the data, assuming that $\theta = 50$. A correct interpretation is the following. If the true mean were 50, then a sample mean of at least 52.3 would occur with probability 0.011, and thus the data are significant at level 0.05 but not at level 0.01.

4. Under H_0 the statistic Z is standard normally distributed. Since Φ is continuous and strictly increasing, for $u \in (0, 1)$ we have

$$\begin{aligned}\mathbb{P}_{\theta_0}\big(1 - \Phi(Z) \leq u\big) &= \mathbb{P}_{\theta_0}\big(\Phi(Z) \geq 1 - u\big) \\ &= \mathbb{P}_{\theta_0}(Z \geq z_{1-u}) = 1 - \Phi(z_{1-u}) = u,\end{aligned}$$

and thus, as lemma 22.9 states, the p-value is uniformly distributed on the interval $(0, 1)$ under H_0. In particular we have the equation $\mathbb{P}_{\theta_0}\big(p(X) \leq \alpha\big) = \alpha$, which is the statement that the test has size α.

□

C.16 The Neyman–Pearson Lemma

Here we solve the exercises at the end of lecture 23.

Solution to exercise 23.1.

1. The joint probability weights are $f(x; \theta) = \theta^s e^{-n\theta} / \prod_i x_i!$ with $s = \sum_i x_i$. Thus the likelihood ratio is

$$R(x) = \Big(\frac{\theta_1}{\theta_0}\Big)^s e^{-n(\theta_1 - \theta_0)}.$$

Since $\theta_1/\theta_0 > 1$, this function is strictly increasing as a function of s and the inequality $R(x) > k$ is equivalent to $s > c$ with $c = \big(\log k + n(\theta_1 - \theta_0)\big)/\log(\theta_1/\theta_0)$. By theorem 23.2 the test which rejects when $S > c$ is most powerful among all tests whose level is its size $\mathbb{P}_{\theta_0}(S > c)$. This size can be found using the Poisson distribution with mean $n\theta_0$, which contains the parameters n and θ_0 only. For a given size the threshold c is the same for all $\theta_1 > \theta_0$. By proposition 23.4 the test is uniformly most powerful against $H_1 \colon \theta > \theta_0$.

2. Under H_0, the sum S is Poisson-distributed with mean 5 and thus we have $\mathbb{P}_{\theta_0}(S > 8) = 1 - 0.9319 = 0.0681 > 0.05$, whereas $\mathbb{P}_{\theta_0}(S > 9) = 1 - 0.9682 = 0.0318 \leq 0.05$. The smallest acceptable threshold for the test is $c = 9$: we can reject $S \geq 10$ and the size of this test is 0.0318.

3. Under $\theta_1 = 2$, the sum is Poisson-distributed with mean 10 and thus the power of the test is $\mathbb{P}_{\theta_1}(S > 9) = 1 - 0.4579 = 0.5421$.

4. The size of the test $\{S > c\}$ is $1 - \mathbb{P}_{\theta_0}(S \leq c)$. This value can only take the values listed for $c = 0, 1, 2, \ldots$ and the jumps between these values are from $c = 8$ to $c = 9$, *i.e.* 0.0681 to 0.0318. The value 0.05 is not one of the possible values. The test from part (b) has size 0.0318 and the lemma shows that this is the most powerful test at level 0.0318. The test is also a test of level 0.05, but the lemma does not show that this is the most powerful test among all tests with level 0.05. Indeed, at level 0.05 another critical region can have more power: the region $\{S \geq 10\} \cup \{S = 0\}$ has size $0.0318 + e^{-5} = 0.0385$, and its power exceeds that of the test from part (b) by $\mathbb{P}_{\theta_1}(S = 0) = e^{-10} > 0$.

□

Solution to exercise 23.2.

1. The joint probability weights are $f(x; p) = p^s(1-p)^{n-s}$ with $s = \sum_i x_i$. Under H_0, these weights equal 2^{-n} for all x. Thus we have

$$R(x) = \frac{p_1^s(1-p_1)^{n-s}}{2^{-n}} = \Big(\frac{p_1}{1-p_1}\Big)^s \big(2(1-p_1)\big)^n,$$

as claimed. For $p_1 > 1/2$, we have $p_1/(1-p_1) > 1$ and thus $R(x)$ is strictly increasing in s. The inequality $R(x) > k$ is equivalent to $s > c$ for some c. By theorem 23.2, the most powerful test is to reject when $S > c$, where c is determined by the distribution of S under H_0, *i.e.* the binomial distribution with parameters n and $1/2$. Since this distribution does not depend on p_1, the same region is optimal for every alternative $p_1 > 1/2$, and by proposition 23.4 the test from exercise 22.3 is uniformly most powerful against $H_1 \colon p > 1/2$.

2. Since $\mathbb{P}_{1/2}(S \geq 14) = 0.0577 > 0.05$ and $\mathbb{P}_{1/2}(S \geq 15) = 0.0207 \leq 0.05$, the largest critical region which does not exceed size 0.05 is $\{S \geq 15\}$. This region has size 0.0207. The power of this test at $p_1 = 0.7$ is $\mathbb{P}_{0.7}(S \geq 15) = 1 - 0.5836 = 0.4164$. For $n = 10$, the corresponding region $\{S \geq 9\}$ from exercise 22.3 has smaller size $11/1024 = 0.0107$ but the power at $p_1 = 0.7$ is $10 \cdot 0.7^9 \cdot 0.3 + 0.7^{10} = 0.1493$. By doubling the sample size, we can use a

test with size closer to the nominal value 0.05 and nearly triple the power at this alternative.

3. For $p_1 < 1/2$, we have $p_1/(1-p_1) < 1$ and thus $R(x)$ is strictly decreasing in s. The inequality $R(x) > k$ is now equivalent to $s < c$ and the most powerful test will be a test which rejects for small values of S. For $n = 20$, with the same size 0.0207, the test will reject for $S \leq 5$, by the symmetry of the binomial distribution with parameter 1/2. Thus, the two one-sided problems will result in tests which reject on the opposite tails of the distribution of S and there can be no single critical region which satisfies both tests.

□

Solution to exercise 23.3.

1. For $x_1, \ldots, x_n \geq 0$ we have

$$f(x;\theta_0) = \theta_0^{-n}\mathbf{1}_{\{m\leq\theta_0\}} \qquad \text{and} \qquad f(x;\theta_1) = \theta_1^{-n}\mathbf{1}_{\{m\leq\theta_1\}},$$

and since $\theta_0 < \theta_1$ there are three cases. If $m \leq \theta_0$, both densities are positive and

$$f(x;\theta_1) = \theta_1^{-n} = (\theta_0/\theta_1)^n\theta_0^{-n} = k\, f(x;\theta_0).$$

If $\theta_0 < m \leq \theta_1$, we have $f(x;\theta_0) = 0 < f(x;\theta_1)$, and thus $f(x;\theta_1) > k\, f(x;\theta_0)$. If $m > \theta_1$, both densities vanish and equality holds. Consequently

$$\{x \mid f(x;\theta_1) > k\, f(x;\theta_0)\} = \{\theta_0 < m \leq \theta_1\},$$
$$\{x \mid f(x;\theta_1) = k\, f(x;\theta_0)\} = \{m \leq \theta_0\} \cup \{m > \theta_1\}.$$

2. Since $0 < c < \theta_0$, the size of C is $\mathbb{P}_{\theta_0}(M > c) = 1 - (c/\theta_0)^n = 1 - (1-\alpha) = \alpha$. The region $C = \{m > c\}$ contains the set $\{\theta_0 < m \leq \theta_1\}$ of part (a), because $c < \theta_0$, and it is contained in the union of the two sets of part (a), which is the whole sample space. Thus C satisfies the condition of theorem 23.2 with $k = (\theta_0/\theta_1)^n$ and by the first statement of the theorem it is most powerful at level α. The power of this test is

$$\mathbb{P}_{\theta_1}(M > c) = 1 - \Big(\frac{c}{\theta_1}\Big)^n = 1 - (1-\alpha)\Big(\frac{\theta_0}{\theta_1}\Big)^n.$$

The threshold $c = \theta_0(1-\alpha)^{1/n}$ does not depend on θ_1 and the argument from above applies to all $\theta_1 > \theta_0$ with the corresponding k and thus, by proposition 23.4, the test is uniformly most powerful against $H_1 \colon \theta > \theta_0$.

3. The two parts of C'' are disjoint and thus the size of this region is $\mathbb{P}_{\theta_0}(M > \theta_0) + \mathbb{P}_{\theta_0}(M \leq \theta_0\alpha^{1/n}) = 0 + \alpha = \alpha$. The power of this test is

$$\mathbb{P}_{\theta_1}(M > \theta_0) + \mathbb{P}_{\theta_1}(M \leq \theta_0\alpha^{1/n}) = 1 - \Big(\frac{\theta_0}{\theta_1}\Big)^n + \alpha\Big(\frac{\theta_0}{\theta_1}\Big)^n = 1 - (1-\alpha)\Big(\frac{\theta_0}{\theta_1}\Big)^n.$$

This is the same power as for C. Thus both regions are most powerful at level α and, by the second statement of theorem 23.2, they can only differ

on the boundary set. And indeed they do: the two regions differ on the sets $C \setminus C'' = \{c < m \le \theta_0\}$ and $C'' \setminus C = \{m \le \theta_0 \alpha^{1/n}\}$, both of which are contained in $\{m \le \theta_0\}$ where $f(x;\theta_1) = k\, f(x;\theta_0)$. The argument in the uniqueness comment in section 23.2 used the fact that the likelihood ratio has a continuous distribution, but in this case this is not true: on the set $\{M \le \theta_0\}$, which has probability one under H_0 and probability $(\theta_0/\theta_1)^n$ under θ_1, the ratio $R(X)$ is constant and thus the boundary set has positive probability, despite the model having densities.

□

Solution to exercise 23.4. Since T is sufficient, we can use the factorisation theorem (proposition 8.3) to find functions g and h such that

$$f(x;\theta) = h(x)\, g\big(T(x);\theta\big)$$

for all x and all parameter values θ. Now let us choose an x with $f(x;\theta_0) > 0$. Then neither of the two factors in the expression for $f(x;\theta_0) = h(x)\, g\big(T(x);\theta_0\big)$ can be zero and we get

$$R(x) = \frac{f(x;\theta_1)}{f(x;\theta_0)} = \frac{h(x)\, g\big(T(x);\theta_1\big)}{h(x)\, g\big(T(x);\theta_0\big)} = \frac{g\big(T(x);\theta_1\big)}{g\big(T(x);\theta_0\big)}.$$

Since we can cancel the factor $h(x)$ on both sides, the right-hand side only depends on x via $T(x)$. This completes the proof. □

C.17 Likelihood-Ratio Tests

Here we solve the exercises at the end of lecture 25.

Solution to exercise 25.1.

1. By example 5.4 the log-likelihood

$$\ell(\mu,\sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i-\mu)^2$$

is maximised over Θ at $\hat\mu = \bar X$ and $\hat\sigma^2 = Q/n$. Substituting these values, the sum of squares becomes Q and the second term becomes $Q/(2Q/n) = n/2$, and thus the maximised log-likelihood is

$$\ell(\hat\mu,\hat\sigma^2) = -\frac{n}{2}\log\Big(\frac{2\pi Q}{n}\Big) - \frac{n}{2}.$$

2. Under H_0 the mean is fixed at μ_0 and only $v = \sigma^2$ is free. Writing $Q_0 = \sum_i (X_i - \mu_0)^2$, the function to maximise is $-\frac{n}{2}\log(2\pi v) - Q_0/(2v)$, which is the function of example 5.4 with Q_0 in place of Q, and thus by the same derivative computation it is maximised at $\hat\sigma_0^2 = Q_0/n$, with maximum $-\frac{n}{2}\log(2\pi Q_0/n) - \frac{n}{2}$. The identity (2.1) with μ_0 in place of μ gives $Q_0 = Q + n(\bar X - \mu_0)^2$, and thus

$$\hat\sigma_0^2 = \frac{Q + n(\bar X - \mu_0)^2}{n}.$$

3. By (25.1) and the two previous parts, the terms $-n/2$ cancel and we get

$$W = 2\Big(-\frac{n}{2}\log\frac{2\pi Q}{n} + \frac{n}{2}\log\frac{2\pi Q_0}{n}\Big) = n\log\frac{Q_0}{Q} = n\log\Big(1+\frac{n(\bar X-\mu_0)^2}{Q}\Big).$$

Since $Q=(n-1)S^2$, we have $n(\bar X-\mu_0)^2/Q = n(\bar X-\mu_0)^2/((n-1)S^2) = t^2/(n-1)$, which gives the stated formula. The function $t^2 \mapsto n\log(1+t^2/(n-1))$ is strictly increasing, and thus the event $W \geq c$ is the event $|t| \geq c'$ for a suitable c': the generalised likelihood-ratio test rejects for large values of $|t|$.

4. The full parameter $\theta=(\mu,\sigma^2)$ has $k=2$ coordinates, and H_0 fixes one of them, $\mu=\mu_0$, leaving σ^2 free, so that $r=1$. The normal model satisfies the regularity conditions, and thus theorem 25.3 gives $W \xrightarrow{\mathrm{d}} \chi^2_1$ under H_0. The approximate test rejects when $W \geq 3.841$, *i.e.* when $t^2 \geq (n-1)(e^{3.841/n}-1)$. For $n=10$ this is $t^2 \geq 9\,(e^{0.3841}-1) = 9\cdot 0.4683 = 4.215$, or $|t| \geq 2.053$, whereas the exact test rejects when $|t| \geq 2.262$. The approximate test rejects more often than the exact one, and its true size is $\mathbb{P}(|t_9| \geq 2.053) = 0.070$ instead of the nominal 0.05: the chi-squared approximation overstates the evidence against H_0 for small n, because the t_9 distribution has heavier tails than the normal limit which the approximation assumes. For large n the two tests merge, since $n\log(1+t^2/(n-1)) \to t^2$ by $\log(1+x)\approx x$ and the t_{n-1} distribution tends to the standard normal one; for $n=100$ the two thresholds are 1.969 and 1.984. Whenever the exact distribution is available, as here, it is the one to use, and lecture 31 does so.

□

Solution to exercise 25.2.

1. Taking logarithms of the probability weights gives

$$\ell(p) = \log\frac{n!}{n_1!\cdots n_m!} + \sum_{j=1}^m n_j \log p_j,$$

where the first term does not depend on p. To maximise $\sum_j n_j\log p_j$ subject to $\sum_j p_j = 1$ we consider the Lagrangian $\sum_j n_j\log p_j - \lambda(\sum_j p_j - 1)$. Setting its partial derivative with respect to p_j to zero gives $n_j/p_j = \lambda$, *i.e.* $p_j = n_j/\lambda$ for every j, and summing over j and using the constraint we find $1 = n/\lambda$, so that $\lambda = n$ and $\hat p_j = n_j/n$. This stationary point is the maximum, because $\sum_j n_j\log p_j$ is a concave function of p and the constraint set is convex. If some n_j is zero, the corresponding $\hat p_j = 0$ lies on the boundary of the parameter space; the formula still gives the maximum, since any probability assigned to an unobserved outcome only lowers the likelihood.

2. Since H_0 is simple, the restricted maximiser is p^0, and by (25.1) we get

$$W = 2(\ell(\hat p) - \ell(p^0)) = 2\Big(\sum_{j=1}^m n_j\log\frac{n_j}{n} - \sum_{j=1}^m n_j\log p^0_j\Big) = 2\sum_{j=1}^m n_j\log\frac{n_j}{n\,p^0_j},$$

where the combinatorial constant cancels. A term with $n_j = 0$ contributes nothing to either sum, which is the convention $0\log 0 = 0$.

3. With $(p_1, \dots, p_{m-1})$ as the parameter, as in section 25.4, the null hypothesis fixes all $k = m - 1$ coordinates, so that $r = m - 1$. The counts are the sums of n i.i.d. indicator vectors, one per repetition, whose distribution is a full-rank exponential family with support not depending on p, so that the regularity conditions hold at every interior point p^0. Theorem 25.3 therefore gives $W \xrightarrow{\mathrm{d}} \chi^2_{m-1}$ under H_0.

4. Under $H_0 \colon p_j^0 = 1/6$ the expected counts are $np_j^0 = 10$ for every face. The terms $n_j \log(n_j/10)$ are

$$13 \log 1.3 = 3.411, \quad 8 \log 0.8 = -1.785, \quad 11 \log 1.1 = 1.048,$$
$$6 \log 0.6 = -3.065, \quad 12 \log 1.2 = 2.188, \quad 10 \log 1 = 0,$$

with sum 1.797, and thus $w = 2 \cdot 1.797 = 3.59$. Since $3.59 < 11.07 = \chi^2_{5,0.95}$, we do not reject H_0 at level 0.05; the approximate p-value is $\mathbb{P}(\chi^2_5 \geq 3.59) = 0.61$. The observed counts are consistent with a fair die.

5. Write $x_j = d_j/(np_j^0)$, so that $n_j = np_j^0(1 + x_j)$ and $n_j/(np_j^0) = 1 + x_j$. Using $\log(1 + x) = x - x^2/2 + O(x^3)$ we get

$$\begin{aligned} n_j \log \frac{n_j}{np_j^0} &= np_j^0 \, (1 + x_j)\Big(x_j - \frac{x_j^2}{2} + O(x_j^3)\Big) \\ &= np_j^0\Big(x_j + \frac{x_j^2}{2} + O(x_j^3)\Big) = d_j + \frac{d_j^2}{2np_j^0} + O(x_j^3), \end{aligned}$$

since $np_j^0 x_j = d_j$ and $np_j^0 x_j^2 = d_j^2/(np_j^0)$. Summing over j, the linear terms add up to $\sum_j d_j = 0$, and doubling the result gives

$$W \approx \sum_{j=1}^{m} \frac{d_j^2}{np_j^0} = \sum_{j=1}^{m} \frac{(n_j - np_j^0)^2}{np_j^0},$$

up to terms of third order in the relative deviations x_j. For the die the deviations from 10 are $3, -2, 1, -4, 2, 0$, and Pearson's statistic is $(9 + 4 + 1 + 16 + 4 + 0)/10 = 3.4$, close to the value 3.59 of W; the two statistics agree to the extent that the relative deviations are small, which for these data they are.

□

Solution to exercise 25.3.

1. By example 4.2 the likelihood is $L(p) = p^S(1 - p)^{n-S}$, so that $\ell(p) = S \log p + (n - S) \log(1 - p)$, and by example 5.3 it is maximised over $(0, 1)$ at $\hat{p} = S/n$ when $0 < S < n$. Since H_0 is simple, the restricted maximiser is p_0, and (25.1) gives

$$W = 2(\ell(\hat{p}) - \ell(p_0)) = 2\Big(S \log \frac{\hat{p}}{p_0} + (n - S) \log \frac{1 - \hat{p}}{1 - p_0}\Big).$$

For $S = 0$ or $S = n$ the supremum of L over $(0, 1)$ is approached at the boundary, where one of the two terms is absent, and the convention $0 \log 0 = 0$ gives the correct value.

2. The parameter is one-dimensional and H_0 fixes it, so that $k = r = 1$; the Bernoulli model satisfies the regularity conditions for $p_0 \in (0,1)$, as noted in lecture 16. Theorem 25.3 gives $W \xrightarrow{\mathrm{d}} \chi^2_1$ under H_0, and the approximate level α test rejects when $W \geq \chi^2_{1,1-\alpha}$, which for $\alpha = 0.05$ means $W \geq 3.841$.

3. We have $\hat{p} = 32/50 = 0.64$ and $p_0 = 0.5$, and thus

$$w = 2\big(32 \log 1.28 + 18 \log 0.72\big) = 2\,(7.8995 - 5.9131) = 3.973.$$

   Since $3.973 \geq 3.841$, we reject $H_0 \colon p = 1/2$ at level 0.05. The approximate p-value is $\mathbb{P}(\chi^2_1 \geq 3.973) = 2(1 - \Phi(\sqrt{3.973})) = 2(1 - \Phi(1.993)) = 0.046$, just below 0.05, so that the evidence against a fair coin is real but not strong.

4. The map $g(p) = \log(p/(1-p))$ is a strictly increasing bijection from $(0,1)$ onto $\mathbb{R}$ with inverse $p = e^\psi/(1+e^\psi)$, and the likelihood in the new parametrisation is $L_\psi(\psi) = L(g^{-1}(\psi))$. The null hypothesis $\psi = \psi_0$ is the same set of distributions as $p = p_0$ with $p_0 = g^{-1}(\psi_0)$, and thus the numerator of the ratio is $L_\psi(\psi_0) = L(p_0)$ in both cases. Since the supremum $\sup_{p\in(0,1)} L(p)$ is attained at the MLE $\hat{p}$ by the first part, and g is a bijection from $(0,1)$ onto $\mathbb{R}$, the two suprema $\sup_{\psi\in\mathbb{R}} L_\psi(\psi)$ and $\sup_{p\in(0,1)} L(p)$ run over the same set of values, and thus the denominator is $\sup_{\psi\in\mathbb{R}} L_\psi(\psi) = \sup_{p\in(0,1)} L(p) = L(\hat{p})$. Both ratios are therefore equal, and so are the two statistics W. This is the invariance noted in section 25.1; by contrast the standard errors of exercise 17.2 do depend on the parametrisation.

□

C.18 Confidence Intervals

Here we solve the exercises at the end of lecture 26.

Solution to exercise 26.1.

1. The function $Q = 2\theta T$ is a function of the data and the parameter, and from exercise 13.1 we know that under $\mathbb{P}_\theta$ the distribution of Q is χ^2_{2n} for all θ, independent of θ. Thus, Q is a pivot. Using $a = \chi^2_{2n,\alpha/2}$ and $b = \chi^2_{2n,1-\alpha/2}$ we find $\mathbb{P}_\theta(a \leq 2\theta T \leq b) = 1 - \alpha$ and since the function Q is increasing in θ, the two inequalities can be solved for θ directly:

$$a \leq 2\theta T \leq b \quad \Longleftrightarrow \quad \frac{\chi^2_{2n,\alpha/2}}{2T} \leq \theta \leq \frac{\chi^2_{2n,1-\alpha/2}}{2T}.$$

   Thus, the interval

$$\big[\chi^2_{2n,\alpha/2}/(2T),\ \chi^2_{2n,1-\alpha/2}/(2T)\big]$$

   is a confidence interval for θ with level $1 - \alpha$ and the probability of this interval covering the correct value is exactly $1 - \alpha$.

2. We have $2T = 50$ and $2n = 20$ and thus the 95% interval is

$$[9.591/50,\ 34.170/50] = [0.192,\ 0.683].$$

3. The MLE for the rate is $\hat{\theta} = 1/\bar{x} = 10/25 = 0.4$ and the standard error of the MLE is $0.4/\sqrt{10} = 0.126$. Thus the Wald interval is $0.4 \pm 1.96 \cdot 0.126$, which is the interval $[0.152, 0.648]$. The exact interval is asymmetric about the estimate, extending further to the right than to the left. This is caused by the right skew of the distribution of $\hat{\theta}$ for small exponential samples; the Wald interval, being symmetric, extends to lower values than the exact interval does. At $n = 10$ the normal approximation used in the Wald interval is rough, as noted in section 17.5, and the exact interval should be used instead.

□

Solution to exercise 26.2.

1. From example 8.9 we know that the sample maximum M has the density $f_M(t) = nt^{n-1}/\theta^n$ for $0 \leq t \leq \theta$. Thus, for $0 \leq u \leq 1$ we have

$$\mathbb{P}_\theta\Big(\frac{M}{\theta} \leq u\Big) = \mathbb{P}_\theta(M \leq u\theta) = \int_0^{u\theta} \frac{nt^{n-1}}{\theta^n}\,\mathrm{d}t = \frac{(u\theta)^n}{\theta^n} = u^n.$$

   The distribution function of M/θ does not depend on θ and thus M/θ is a pivot.

2. Since $M \leq \theta$ always holds, we have

$$\mathbb{P}_\theta\Big(M \leq \theta \leq \frac{M}{\alpha^{1/n}}\Big) = \mathbb{P}_\theta\Big(\alpha^{1/n} \leq \frac{M}{\theta} \leq 1\Big) = 1 - \big(\alpha^{1/n}\big)^n = 1 - \alpha,$$

   where we used the first part of the solution. Thus, $[M, M/\alpha^{1/n}]$ is a confidence interval with level $1 - \alpha$. For $n = 5$ and $\alpha = 0.05$ we have $\alpha^{1/n} = 0.05^{0.2} = 0.549$ and thus the interval is $[3.2,\ 3.2/0.549] = [3.2, 5.83]$.

3. The condition for coverage is $b^n - a^n = 1 - \alpha$. This condition can be used to find $a = (b^n - 1 + \alpha)^{1/n}$ as a function of b and thus, by choosing the interval $[M/b, M/a]$, we can find the length $M(1/a - 1/b)$ of the interval. Taking derivatives we find $b^{n-1} = a^{n-1}\,\mathrm{d}a/\mathrm{d}b$ and thus

$$\frac{\mathrm{d}}{\mathrm{d}b}\Big(\frac{1}{a} - \frac{1}{b}\Big) = -\frac{1}{a^2}\frac{\mathrm{d}a}{\mathrm{d}b} + \frac{1}{b^2} = -\frac{b^{n-1}}{a^{n+1}} + \frac{1}{b^2} = \frac{1}{b^2}\Big(1 - \Big(\frac{b}{a}\Big)^{n+1}\Big) < 0,$$

   since $b > a$. The length of the interval decreases as b increases and thus is smallest for the largest possible value $b = 1$, where $a = \alpha^{1/n}$. This is the interval from the previous part of the solution. The reason for this result is that the density nu^{n-1} of the pivot is increasing on $[0, 1]$ and thus the probability $1 - \alpha$ is concentrated on the shortest possible interval which is placed at the upper end of the density.

□

Solution to exercise 26.3.

1. The standard error is $\operatorname{se}(\hat\theta_n) = 1/\sqrt{n\mathcal{I}(\bar X)} = \sqrt{\bar X/n}$, and the Wald interval is $\bar X \pm z_{1-\alpha/2}\sqrt{\bar X/n}$. For the data we have $\bar x = 1.5$ and $\sqrt{1.5/20} = 0.274$, and the 95% interval is $1.5 \pm 1.96 \cdot 0.274$, which is $[0.96, 2.04]$.

2. In exercise 17.3 we found $\sqrt{n}\,(\hat p_0 - p_0) \xrightarrow{\text{d}} N(0, e^{-2\theta}\theta)$, and replacing θ by $\bar X$ in the variance gives the standard error $\operatorname{se}(\hat p_0) = e^{-\bar X}\sqrt{\bar X/n}$, in agreement with the delta-method formula $|g'(\hat\theta_n)|\operatorname{se}(\hat\theta_n)$ for $g(\theta) = e^{-\theta}$. The approximate interval is $e^{-\bar X} \pm z_{1-\alpha/2}\, e^{-\bar X}\sqrt{\bar X/n}$. For the data we have $\hat p_0 = e^{-1.5} = 0.223$ and $\operatorname{se}(\hat p_0) = 0.223 \cdot 0.274 = 0.061$, and the 95% interval is $0.223 \pm 1.96 \cdot 0.061$, which is $[0.103, 0.343]$.

3. Write $[L, U]$ for the interval of the first part. Since g is decreasing, the event $L \le \theta \le U$ is the same as the event $g(U) \le g(\theta) \le g(L)$, and thus the interval $[e^{-U}, e^{-L}]$ covers p_0 exactly when $[L, U]$ covers θ. Its coverage probability is that of the Wald interval, which converges to $1-\alpha$ by proposition 26.8. For the data we get $[e^{-2.04}, e^{-0.96}] = [0.130, 0.382]$. The two intervals differ because g is not linear: the delta-method interval is symmetric about $\hat p_0$, while the transformed interval is shifted towards larger values and is asymmetric about $\hat p_0 = 0.223$. Both have asymptotic coverage $1-\alpha$, and for large n the two nearly coincide, since g is close to linear over a short interval. A practical advantage of the transformed interval is that it always lies inside $(0, \infty)$, whereas the delta-method interval can extend below zero when $\hat p_0$ is small. It stays below 1 only when $L > 0$.

□

Solution to exercise 26.4.

1. Under H_0, the test statistic $\sqrt{n}\,(\bar X - \mu_0)/S$ is the pivot T from example 26.4 and has t_{n-1} distribution, independent of the value of σ^2. Since the t distribution is symmetric, we have

$$\begin{aligned}\mathbb{P}_{\mu_0,\sigma^2}(|T| > t_{n-1,1-\alpha/2}) &= \mathbb{P}(T > t_{n-1,1-\alpha/2}) + \mathbb{P}(T < -t_{n-1,1-\alpha/2}) \\ &= \frac{\alpha}{2} + \frac{\alpha}{2} = \alpha,\end{aligned}$$

   and the test has size α.

2. The acceptance region of the test for μ_0 is $A(\mu_0) = \{x \mid |\bar x - \mu_0| \le t_{n-1,1-\alpha/2}\, s/\sqrt{n}\}$. By lemma 26.6, the set of values not rejected,

$$\begin{aligned}C(X) &= \Big\{\mu_0 \mid |\bar X - \mu_0| \le t_{n-1,1-\alpha/2}\,\frac{S}{\sqrt{n}}\Big\} \\ &= \Big[\,\bar X - t_{n-1,1-\alpha/2}\,\frac{S}{\sqrt{n}},\ \bar X + t_{n-1,1-\alpha/2}\,\frac{S}{\sqrt{n}}\,\Big],\end{aligned}$$

   is a confidence set with level $1-\alpha$ and it is the t-interval from example 26.4.

3. The 95% t-interval for the given data is $[9.18, 11.42]$. By duality, $H_0\colon \mu = \mu_0$ is rejected at level 5% if and only if μ_0 is outside this interval. Thus, $H_0\colon \mu = 9$ is rejected since $9 < 9.18$, and $H_0\colon \mu = 10$ is not rejected since 10 is inside the interval.

4. The one-sided test does not reject μ_0 if and only if $\sqrt{n}\,(\bar{X} - \mu_0)/S \leq t_{n-1,1-\alpha}$, *i.e.* if $\mu_0 \geq \bar{X} - t_{n-1,1-\alpha}\,S/\sqrt{n}$. Thus, the set of values not rejected is the one-sided interval $\left[\bar{X} - t_{n-1,1-\alpha}\,S/\sqrt{n},\ \infty\right)$, whose left-hand boundary is a lower confidence bound for μ with level $1-\alpha$. Under H_0, the test statistic is t_{n-1} distributed, and thus the test has size α. Since the bound misses μ exactly when the test of the true value rejects, its coverage probability is exactly $1-\alpha$.

□

C.19 Bayesian Inference: Priors and Posteriors

Here we solve the exercises at the end of lecture 28.

Solution to exercise 28.1.

1. The prior mean is $a/(a+b) = 0.3$ and since $a + b = 10$ we have $a = 3$ and $b = 7$. From table A.2 we find that the variance of the Beta$(3, 7)$ distribution is

$$\frac{ab}{(a+b)^2(a+b+1)} = \frac{21}{100 \cdot 11} \approx 0.0191,$$

and thus the prior standard deviation is approximately 0.138.

2. Let $\mu = a/(a+b)$ denote the prior mean and $s = a + b$ the equivalent sample size. Then we have $a = \mu s$ and $b = (1-\mu)s$. Using the formula for the variance from the table, we get

$$\frac{ab}{(a+b)^2(a+b+1)} = \frac{\mu(1-\mu)}{s+1}.$$

Given $\mu = 0.3$ and a variance of $0.1^2 = 0.01$ we find that we need $s + 1 = 0.21/0.01 = 21$, *i.e.* $s = 20$ and thus we have $a = 6$ and $b = 14$. The equivalent sample size is 20: the smaller spread corresponds to a prior which gives twice the number of observations.

3. From example 28.5 we know that the posterior distribution is Beta$(3 + 9, 7 + 11)$ = Beta$(12, 18)$. The mean of this distribution is $12/30 = 0.4$. In the form (28.3) we can write this as

$$\frac{10}{30} \cdot 0.3 + \frac{20}{30} \cdot 0.45 = 0.1 + 0.3 = 0.4,$$

where $\bar{x} = 9/20 = 0.45$. The data have twice the weight of the prior, and the posterior mean is two thirds of the way between the prior mean and the sample proportion.

□

Solution to exercise 28.2.

1. On the interval $(0, 1/2]$ we have $1 - \theta \geq 1/2$ and thus $\theta^{-1}(1-\theta)^{-1} \geq \theta^{-1}$ there. Since $\int_0^{1/2} \theta^{-1}\,\mathrm{d}\theta = \infty$, the integral of the prior over $(0, 1)$ is infinite and the prior is improper.

2. Consider the kernel $\theta^{\alpha-1}(1-\theta)^{\beta-1}$ for real α and β. Near $\theta = 0$ the factor $(1-\theta)^{\beta-1}$ is bounded between two positive constants and thus the kernel is integrable near 0 if and only if $\theta^{\alpha-1}$ is, which is equivalent to $\alpha > 0$. By symmetry the kernel is integrable near $\theta = 1$ if and only if $\beta > 0$ and thus, since it is bounded away from the endpoints, the integral over $(0, 1)$ is finite if and only if $\alpha > 0$ and $\beta > 0$, in which case it equals $B(\alpha, \beta)$. For the formal posterior we have $\alpha = T$ and $\beta = n - T$ and thus the posterior is proper if and only if $T > 0$ and $n - T > 0$, *i.e.* if and only if $0 < T < n$. In this case it is the Beta$(T, n - T)$ distribution.

3. From table A.2 we find that the posterior mean is $T/(T+n-T) = T/n = \bar{x}$. This is the maximum likelihood estimate from example 5.3. The Haldane prior is the choice of prior where the Bayes estimator and the maximum likelihood estimator coincide. This happens for an equivalent sample size of zero in (28.3).

4. For $T = 0$ the likelihood $(1-\theta)^n$ has its maximum at $\theta = 0$ and does not vanish there, while the prior has a non-integrable singularity at $\theta = 0$. Since the data does not contain any successes, the infinite mass of the prior near zero cannot be compensated and thus the integral of $\theta^{-1}(1-\theta)^{n-1}$ is infinite: there is no posterior distribution in this case. In informal language, the improper prior suggests that θ is very close to either 0 or 1 and a data set with no successes cannot exclude the first of these two possibilities.

□

Solution to exercise 28.3.

1. We have

$$\begin{aligned} f(x|\theta) &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(-\frac{(x_i-\theta)^2}{2\sigma^2}\Big) \\ &= (2\pi\sigma^2)^{-n/2} \exp\Big(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\theta)^2\Big). \end{aligned}$$

   Using the identity (2.1) from lecture 2, which holds for all numbers in place of the mean, and using θ, we find $\sum_i (x_i-\theta)^2 = \sum_i (x_i-\bar{x})^2 + n(\bar{x}-\theta)^2$. Since the first term does not depend on θ and since the factor $(2\pi\sigma^2)^{-n/2}$ does not depend on θ, we have

$$f(x|\theta) \propto \exp\big(-n(\theta-\bar{x})^2/(2\sigma^2)\big).$$

2. Since the prior density, as a function of θ, is proportional to $\exp\big(-(\theta - m)^2/(2\tau^2)\big)$ and since we can group the terms in the exponent of the posterior density as powers of θ, we get

$$\begin{aligned} \pi(\theta|x) &\propto \exp\Big(-\frac{n(\theta-\bar{x})^2}{2\sigma^2} - \frac{(\theta-m)^2}{2\tau^2}\Big) \\ &\propto \exp\Big(-\frac{1}{2}\Big(\frac{n}{\sigma^2}+\frac{1}{\tau^2}\Big)\theta^2 + \Big(\frac{n\bar{x}}{\sigma^2}+\frac{m}{\tau^2}\Big)\theta\Big), \end{aligned}$$

where we have omitted the terms which do not contain θ. Using the abbreviations $1/\tau_n^2 = 1/\tau^2 + n/\sigma^2$ and $m_n = \tau_n^2(m/\tau^2 + n\bar{x}/\sigma^2)$ we can write the exponent as $-\theta^2/(2\tau_n^2) + \theta m_n/\tau_n^2$ and completing the square we find

$$-\frac{\theta^2}{2\tau_n^2} + \frac{\theta m_n}{\tau_n^2} = -\frac{(\theta - m_n)^2}{2\tau_n^2} + \frac{m_n^2}{2\tau_n^2},$$

where we have omitted the last term which does not contain θ. Thus we have $\pi(\theta|x) \propto \exp(-(\theta - m_n)^2/(2\tau_n^2))$ and this is the kernel of the $N(m_n, \tau_n^2)$ density and, since the posterior is a density, this is the corresponding normal distribution. If we multiply out m_n we get the weighted form of the posterior given in proposition 28.6.

3. As $\tau \to \infty$ with n fixed, the prior precision $1/\tau^2$ goes to zero and thus $\tau_n^2 \to \sigma^2/n$ and $m_n \to \bar{x}$: the posterior converges to $N(\bar{x}, \sigma^2/n)$, *i.e.* to the posterior under the flat prior discussed in section 28.4. As $n \to \infty$ with τ fixed, the weight $(1/\tau^2)/(1/\tau^2 + n/\sigma^2)$ of the prior mean goes to zero and thus $m_n - \bar{x} \to 0$ and $\tau_n^2 \leq \sigma^2/n$ goes to zero: the posterior is now concentrated around $\bar{x}$ and the influence of the prior vanishes.

□

Solution to exercise 28.4.

1. Under $\mathbb{P}_\theta$, the total number T of successes is binomially distributed with $\mathbb{E}_\theta(T) = n\theta$ and $\mathrm{Var}_\theta(T) = n\theta(1-\theta)$. Since $\hat{\theta}_B$ is an affine function of T, we have

$$\mathbb{E}_\theta(\hat{\theta}_B) = \frac{a + n\theta}{a + b + n} \qquad \text{and}$$
$$\mathrm{bias}(\hat{\theta}_B) = \frac{a + n\theta - (a+b+n)\theta}{a+b+n} = \frac{a - (a+b)\theta}{a+b+n},$$

which equals zero if and only if $\theta = a/(a+b)$, the prior mean, and

$$\mathrm{Var}_\theta(\hat{\theta}_B) = \frac{n\theta(1-\theta)}{(a+b+n)^2}.$$

The bias is of order $1/n$ and the variance is of order $1/n$, both tending to zero as $n \to \infty$. Thus, by corollary 2.13, the estimator is consistent, although for fixed n it is biased for all $\theta \neq a/(a+b)$.

2. For $a = b = 1$ and $n = 10$ we have $\hat{\theta}_B = (T+1)/12$ and thus, by lemma 2.7,

$$\mathrm{MSE}(\hat{\theta}_B) = \frac{10\theta(1-\theta) + (1-2\theta)^2}{144}, \qquad \mathrm{MSE}(\bar{X}) = \frac{\theta(1-\theta)}{10}.$$

At $\theta = 1/2$ the bias is zero and we have $\mathrm{MSE}(\hat{\theta}_B) = 2.5/144 \approx 0.0174$ compared to $\mathrm{MSE}(\bar{X}) = 0.025$. Thus, the Bayes estimator is better. At $\theta = 0.1$ we have $\mathrm{MSE}(\hat{\theta}_B) = (0.9 + 0.64)/144 \approx 0.0107$ compared to $\mathrm{MSE}(\bar{X}) = 0.009$ and thus $\bar{X}$ is better. Neither estimator has smaller mean squared error for all θ: the Bayes estimator is better in the middle of the parameter range, where the prior mean is close to the truth, and the reduced variance does not compensate for the bias introduced by the shrinkage towards $1/2$ at the ends of the range.

□

Solution to exercise 28.5. For the first part of the question we can take derivatives. The log-likelihood function for a single observation is $\ell_0(x-\theta)$ and thus the score function is

$$\frac{\partial}{\partial\theta}\log f(x|\theta) = \frac{\partial}{\partial\theta}\,\ell_0(x-\theta) = -\ell_0'(x-\theta).$$

Since $Z = X - \theta$, with density f_0, for all values of θ, the Fisher information is

$$\mathcal{I}(\theta) = \mathbb{E}(\ell_0'(X-\theta)^2) = \int \ell_0'(z)^2\, f_0(z)\,\mathrm{d}z.$$

This does not depend on θ.

For the second part of the question we have $\log f(x|\sigma) = \ell_0(x/\sigma) - \log\sigma$ and thus the score function is

$$\frac{\partial}{\partial\sigma}(\ell_0(x/\sigma) - \log\sigma) = -\ell_0'(x/\sigma)\,\frac{x}{\sigma^2} - \frac{1}{\sigma} = -\frac{1+z\ell_0'(z)}{\sigma}$$

where $z = x/\sigma$. Since $Z = X/\sigma$ has density f_0 for all σ, we find

$$\mathcal{I}(\sigma) = \frac{1}{\sigma^2}\,\mathbb{E}\Big((1+Z\ell_0'(Z))^2\Big) = \frac{c}{\sigma^2}, \qquad \text{where } c = \int (1+z\ell_0'(z))^2 f_0(z)\,\mathrm{d}z$$

does not depend on σ.

Finally, the $N(0,\sigma^2)$ model is the scale model with base density the standard normal density and with $\ell_0(z) = -z^2/2 - \log\sqrt{2\pi}$ and thus $\ell_0'(z) = -z$. In this case we have $1 + z\ell_0'(z) = 1 - z^2$ and using the substitutions $\mathbb{E}(Z^2) = 1$ and $\mathbb{E}(Z^4) = 3$ for the standard normal density Z we find

$$c = \mathbb{E}((1-Z^2)^2) = 1 - 2\mathbb{E}(Z^2) + \mathbb{E}(Z^4) = 2.$$

Thus $\mathcal{I}(\sigma) = 2/\sigma^2$, as claimed. □

C.20 Credible Intervals and Bayesian Testing

Here we solve the exercises at the end of lecture 29.

Solution to exercise 29.1.

1. Since the posterior density is monotone on each half-line and integrates to one, it converges to zero as $\theta \to \pm\infty$. The condition $\mathbb{P}(\theta \in H_c|x) = 1-\alpha \in (0,1)$ excludes $c \geq \pi(\mu|x)$, where H_c contains at most the point μ and has probability zero, and thus $0 < c < \pi(\mu|x)$. On $(-\infty,\mu]$ the density is continuous and strictly increasing from 0 to $\pi(\mu|x)$ and thus there is exactly one $l < \mu$ with $\pi(l|x) = c$ and $\pi(\theta|x) \geq c$ for $\theta \leq \mu$ if and only if $\theta \geq l$. Similarly, there is exactly one $u > \mu$ with $\pi(u|x) = c$ and the density is at least c on $[\mu,\infty)$ exactly on $[\mu,u]$. Thus we have $H_c = [l,u]$ with $\pi(l|x) = c = \pi(u|x)$.

2. Let $l = \mu - s$ and $u = \mu + t$ with $s,t > 0$. By symmetry we have $\pi(\mu+s|x) = \pi(\mu-s|x) = c = \pi(\mu+t|x)$ and since the density is strictly decreasing on $[\mu,\infty)$ we find $s = t$. The two tails $(-\infty,\mu-t)$ and $(\mu+t,\infty)$ have equal posterior probability by symmetry and together they have probability α, *i.e.* each tail has probability $\alpha/2$. Thus we have $l = q_{\alpha/2}$ and $u = q_{1-\alpha/2}$, *i.e.* we have found the equal-tailed interval.

3. The set $\{\theta \mid \pi(\theta|x) \geq c\}$ is for $0 < c < \pi(0|x)$ the interval $[0, u]$ with $\pi(u|x) = c$, obtained by using the same argument as in the first part, but only for the decreasing branch of the density, and using the condition $\mathbb{P}(\theta \leq u|x) = 1-\alpha$ to get $u = q_{1-\alpha}$. The equal-tailed interval $[q_{\alpha/2}, q_{1-\alpha/2}]$ is then obtained from $[0, q_{1-\alpha}]$ by removing the piece $[0, q_{\alpha/2})$ and adding the piece $(q_{1-\alpha}, q_{1-\alpha/2}]$, both of which have posterior probability $\alpha/2$. The density is at least $d_1 = \pi(q_{\alpha/2}|x)$ on the first piece and at most $d_2 = \pi(q_{1-\alpha}|x)$ on the second piece, where $d_1 > d_2$ since the density is strictly decreasing. Writing λ_1 and λ_2 for the lengths of the two pieces we find $\alpha/2 \geq d_1\lambda_1$ and $\alpha/2 \leq d_2\lambda_2$ and thus $\lambda_1 \leq \alpha/(2d_1) < \alpha/(2d_2) \leq \lambda_2$: the added piece is longer than the removed piece and the equal-tailed interval is longer than the HPD interval.

□

Solution to exercise 29.2.

1. The equal-tailed 95% credible interval is $[0.386, 0.861]$, with posterior probability 2.5% cut off in each tail. The midpoint of this interval is $(0.386 + 0.861)/2 = 0.624$, which is slightly below the posterior mean $9/14 = 0.643$.

2. The posterior density is proportional to $\theta^8(1-\theta)^4$ on $(0, 1)$. Setting the derivative of the logarithm of the posterior density, *i.e.* $8\log\theta + 4\log(1-\theta)$, equal to zero, gives the maximum of the density at $8/\theta = 4/(1-\theta)$ and thus the mode is at $\theta = 8/12 = 2/3 \approx 0.667$. Since the mode is above the mean 0.643, the posterior distribution is skewed.

3. By definition 29.2 of the HPD interval, the HPD interval contains the values with the highest posterior density and thus is shifted towards the mode, so that it lies to the right of the equal-tailed interval. The HPD interval is the shortest interval which has posterior probability 0.95, in this case of length 0.473 against 0.475. As n increases, the posterior distribution approaches a symmetric, normal distribution and the mode and mean move together. In the limit, for a symmetric, unimodal posterior, both intervals coincide; this is a result from exercise 29.1.

□

Solution to exercise 29.3.

1. Since the prior is flat, we have $\text{Beta}(1, 1)$ and using $n = 3$ and $T = 2$ from example 28.5 we find the posterior $\text{Beta}(3, 2)$. Since $B(3, 2) = \Gamma(3)\Gamma(2)/\Gamma(5) = 2/24 = 1/12$, the density of this posterior is $\pi(\theta|x) = 12\theta^2(1-\theta)$ for $0 < \theta < 1$.

2. We can find the posterior probability of H_0 by integrating the density:

$$\mathbb{P}(\theta \leq 1/2|x) = 12\int_0^{1/2}(\theta^2 - \theta^3)\,\mathrm{d}\theta = 12\Big(\frac{1}{24} - \frac{1}{64}\Big) = \frac{5}{16} = 0.3125.$$

Thus we have $\mathbb{P}(H_1|x) = 11/16$ and the posterior odds for H_0 against H_1 are $5/11 \approx 0.45$.

3. Since the prior odds are 1, the data have multiplied the odds by the posterior odds $5/11 \approx 0.45$.

4. The bounds l and u satisfy $12\int_0^l(\theta^2-\theta^3)\,\mathrm{d}\theta = 0.025$ and $12\int_0^u(\theta^2-\theta^3)\,\mathrm{d}\theta = 0.975$, *i.e.* we have

$$4l^3 - 3l^4 = 0.025 \qquad \text{and} \qquad 4u^3 - 3u^4 = 0.975.$$

These are polynomial equations of degree four and while they can be solved explicitly, the solutions are cumbersome and not very instructive; for a general Beta posterior, the bounds will not be algebraic. The quantiles are best found numerically, for example using the R function `qbeta` to get $l = 0.194$ and $u = 0.932$ and thus the 95% equal-tailed credible interval is $[0.194, 0.932]$.

□

Solution to exercise 29.4.

1. Using the formula from example 29.7, with $n = 25$, $\sigma^2 = 4$, $\theta_0 = 10$, $\theta_1 = 11$ and $\bar{x} = 10.9$, we find that the Bayes factor in favour of H_0 is

$$\frac{f(x|\theta_0)}{f(x|\theta_1)} = \exp\Big(\frac{25 \cdot 1 \cdot (21 - 21.8)}{8}\Big) = e^{-2.5} = 0.082.$$

The data give evidence in favour of H_1 over H_0 by a factor of $e^{2.5} \approx 12.2$.

2. The prior odds are 3 and thus the posterior odds are $3 \cdot 0.082 = 0.246$ and $\mathbb{P}(H_0|x) = 0.246/1.246 = 0.198$. The data have changed the prior opinion, which was three to one in favour of H_0, to a posterior opinion of approximately four to one against it.

3. The p-value 0.0122 is the probability, under $\theta = 10$, of obtaining a sample mean of at least 10.9. This value refers to the data, includes outcomes more extreme than the one observed and makes no reference to any alternative. In contrast, the posterior probability 0.198 is a probability for the hypothesis; it uses only the observed data, compares the likelihoods at exactly 10 and 11 and takes into account the prior odds, which were chosen in favour of H_0. Even if the prior odds had been 1, the posterior probability would have been $0.082/1.082 = 0.076$, or six times the p-value. Both numbers show that the data are evidence against H_0, but in different ways. Neither number is a rewording of the other.

4. The posterior odds are the prior odds times $e^{-2.5}$, and the posterior odds are greater than 1 if and only if the prior odds are greater than $e^{2.5} \approx 12.2$. Only a prior which assumed H_0 to be more likely than H_1, by a factor of more than twelve, would result in H_0 being the more likely hypothesis after these data are observed.

□

Solution to exercise 29.5.

1. By (29.1) we have $1-w = (n/\sigma^2)/(1/\tau^2 + n/\sigma^2) = (n/\sigma^2)\,\tau_n^2$, and thus $\tau_n^2 = (1-w)\sigma^2/n$. The posterior mean is $m_n = wm + (1-w)\bar{X}$, and thus $m_n - \theta = w(m-\theta) + (1-w)(\bar{X}-\theta)$. Under $\mathbb{P}_\theta$ we have $\bar{X} - \theta \sim N(0, \sigma^2/n)$ by proposition A.4, and an affine function of a normal variable is normal, so that $m_n - \theta \sim N\big(w(m-\theta), (1-w)^2\sigma^2/n\big)$.

2. Write $z = z_{1-\alpha/2}$ and $\delta = \sqrt{n}\,(m-\theta)/\sigma$ for the standardised distance between the prior mean and the true value. The interval covers θ if and only if $|m_n - \theta| \le z\tau_n = z\sqrt{1-w}\,\sigma/\sqrt{n}$. Subtracting the mean $w(m-\theta)$ and dividing by the standard deviation $(1-w)\sigma/\sqrt{n}$ turns $m_n - \theta$ into a standard normal variable, and the bounds into $(\pm z\sqrt{1-w} - w\delta)/(1-w)$. Thus the coverage probability is

$$\mathbb{P}_\theta(\theta \in \text{interval}) = \Phi\Big(\frac{z\sqrt{1-w} - w\delta}{1-w}\Big) - \Phi\Big(\frac{-z\sqrt{1-w} - w\delta}{1-w}\Big).$$

   It depends on θ only through δ, and it is symmetric in δ.

3. Here $1/\tau^2 = 1$ and $n/\sigma^2 = 4$, and thus $w = 0.2$, $1-w = 0.8$, $\sqrt{1-w} = 0.894$ and $z\sqrt{1-w} = 1.753$. At $\theta = m = 12$ we have $\delta = 0$ and the coverage probability is $2\Phi(1.753/0.8) - 1 = 2\Phi(2.19) - 1 = 0.971$, above the nominal 0.95. At $\theta = 10$ we have $\delta = 4 \cdot 2/2 = 4$ and $w\delta = 0.8$, and the coverage probability is $\Phi(0.953/0.8) - \Phi(-2.553/0.8) = \Phi(1.19) - \Phi(-3.19) = 0.883 - 0.001 = 0.882$, below the nominal value. As $|\theta - m| \to \infty$ we have $|\delta| \to \infty$, both arguments of Φ tend to the same infinity and the coverage probability tends to zero: the interval is shifted towards the prior mean by a fixed fraction w of the distance, and once that shift exceeds the half-width the interval almost never contains the truth.

4. It is not a confidence interval with level 0.95, because definition 26.1 requires coverage probability at least 0.95 for every θ, and the previous part exhibits values of θ where it is smaller. The credible interval trades the uniform guarantee for better behaviour near the prior mean, which is where a well-chosen prior expects the truth to be. As $n \to \infty$ with θ fixed we have $w \approx \sigma^2/(n\tau^2) \to 0$, and thus $w\delta \approx \sigma(m-\theta)/(\tau^2\sqrt{n}) \to 0$ and $\sqrt{1-w} \to 1$, so that the coverage probability tends to $2\Phi(z) - 1 = 1 - \alpha$ for every θ. This is the large-sample agreement of section 29.3 seen from the frequentist side.

□

C.21 Standard Tests Revisited

Here we solve the exercises at the end of lecture 31.

Solution to exercise 31.1.

1. We have the sum of observations 154 and thus $\bar{x} = 154/8 = 19.25$. The sum of squared deviations from $\bar{x}$ equals 3.30 and thus we have $s^2 = 3.30/7 = 0.471$ and $s = 0.687$. The observed value of the test statistic is

$$t = \frac{\sqrt{8}\,(19.25 - 20)}{0.687} = \frac{2.828 \cdot (-0.75)}{0.687} = -3.09.$$

2. From proposition 31.1 we know that the test against $\mu < 20$ rejects if $t < -t_{7,1-\alpha}$. For level 0.05 the critical value is $-t_{7,0.95} = -1.895$ and for level 0.01 the critical value is $-t_{7,0.99} = -2.998$; since -3.09 lies below both of these critical values, the hypothesis H_0 is rejected at both levels. The p-value is given by $\mathbb{P}(T \leq -3.09) = \mathbb{P}(T \geq 3.09)$ for $T \sim t_7$, by symmetry, and since 3.09 is between $t_{7,0.99} = 2.998$ and $t_{7,0.995} = 3.499$, the value is between 0.005 and 0.01.

3. The two-sided test rejects if $|t| > t_{7,1-\alpha/2}$. For level 0.05 we have $|t| = 3.09 > t_{7,0.975} = 2.365$ and thus H_0 is rejected. For level 0.01 we have $3.09 < t_{7,0.995} = 3.499$ and thus H_0 is not rejected. The two-sided p-value is $2\,\mathbb{P}(T \geq 3.09)$, which is twice the value of the one-sided test, so it is between 0.01 and 0.02.

4. The 99% interval is $\bar{x} \pm t_{7,0.995}\, s/\sqrt{8} = 19.25 \pm 3.499 \cdot 0.687/2.828 = 19.25 \pm 0.849$. This interval is $[18.40, 20.10]$. The interval contains the value 20 and by lemma 26.6 this is equivalent to the non-rejection of $H_0\colon \mu = 20$ by the two-sided test with level 0.01 in the previous part. The 95% interval $19.25 \pm 2.365 \cdot 0.243 = [18.68, 19.82]$ does not contain 20 and thus the rejection at level 0.05 is consistent with the results of the other parts.

□

Solution to exercise 31.2.

1. The value of the test statistic (31.2) is $v = 14 \cdot 2.9/1.5 = 27.07$. We can compare this value to quantiles of the χ^2_{14} distribution: at level 0.05 the critical value is $\chi^2_{14,0.95} = 23.685$ and since $27.07 > 23.685$ we can reject H_0, whereas at level 0.01 the critical value is $\chi^2_{14,0.99} = 29.141$ and since $27.07 < 29.141$ we cannot reject H_0. The p-value, $\mathbb{P}(W \geq 27.07)$ for $W \sim \chi^2_{14}$, is between 0.01 and 0.025, since v is between $\chi^2_{14,0.975} = 26.119$ and $\chi^2_{14,0.99} = 29.141$.

2. The two-sided test with level 0.05 does not reject, if

$$v \in [\chi^2_{14,0.025}, \chi^2_{14,0.975}] = [5.629, 26.119].$$

   Since $27.07 > 26.119$, we reject H_0.

3. Let $c = \chi^2_{n-1,1-\alpha}$. Assuming the true variance is σ^2, the value $W = (n-1)S^2/\sigma^2$ is χ^2_{n-1} distributed by proposition A.4 and thus $V = (n-1)S^2/\sigma_0^2 = W\sigma^2/\sigma_0^2$. We have

$$\beta(\sigma^2) = \mathbb{P}_{\sigma^2}(V > c) = \mathbb{P}\Big(W > \frac{c\,\sigma_0^2}{\sigma^2}\Big).$$

   As σ^2 increases, the threshold $c\sigma_0^2/\sigma^2$ decreases and the probability of W exceeding a smaller number increases, *i.e.* β increases as a function of σ^2. By definition 22.5, the size of the test for the composite null hypothesis $\sigma^2 \leq \sigma_0^2$ is $\sup_{\sigma^2 \leq \sigma_0^2} \beta(\sigma^2) = \beta(\sigma_0^2) = \mathbb{P}(W > c) = \alpha$, where the bound is achieved at the boundary.

4. The two-sided test with level α rejects large values of V only if $V > \chi^2_{n-1,1-\alpha/2}$. Since this is a larger threshold than the $\chi^2_{n-1,1-\alpha}$ of the one-sided test, the power of the two-sided test against the alternative $\sigma^2 > \sigma_0^2$ is smaller than the power of the one-sided test. An observed value between the two thresholds is rejected by the one-sided test but not by the two-sided test. The two-sided test has power above α against the alternative $\sigma^2 < \sigma_0^2$, where the one-sided test has probability of rejection below α. For the given data, both tests reject the hypothesis at level 0.05, but at level 0.025 the one-sided test, with threshold $\chi^2_{14,0.975} = 26.119$, still rejects, whereas the two-sided test compares 27.07 to $\chi^2_{14,0.9875} = 28.42$ and does not reject.

□

Solution to exercise 31.3.

1. We can describe the differences $D_i = X_i - Y_i$ using i.i.d. $N(\mu_D, \sigma_D^2)$, where both parameters are unknown. We test $H_0\colon \mu_D = 0$ against $H_1\colon \mu_D > 0$, since we expect the reaction time to be smaller after training. The differences are $4, -1, 6, 3, 6, 5$ and the sum of these differences is 23, *i.e.* we have $\bar{d} = 3.833$. The sum of squared deviations from $\bar{d}$ is 34.83, and thus we have $s_D^2 = 34.83/5 = 6.967$ and $s_D = 2.639$. The test statistic from proposition 31.1, for $\mu_0 = 0$, is

$$t = \frac{\sqrt{6} \cdot 3.833}{2.639} = 3.56,$$

where 5 is the number of degrees of freedom. Since $3.56 > t_{5,0.95} = 2.015$ we can reject the hypothesis at level 0.05 and since $3.56 > t_{5,0.99} = 3.365$ we can also reject the hypothesis at level 0.01: the data show a reduction in reaction time.

2. The 95% t-interval from example 26.4 for μ_D is $\bar{d} \pm t_{5,0.975}\, s_D/\sqrt{6} = 3.833 \pm 2.571 \cdot 2.639/2.449 = 3.833 \pm 2.770$, which is $[1.06, 6.60]$.

3. The row means are $\bar{x} = 419/6 = 69.83$ and $\bar{y} = 396/6 = 66.00$, the pooled variance is $s_p^2 = (5 \cdot 59.77 + 5 \cdot 42.80)/10 = 51.28$ and thus we have $s_p = 7.161$. The test statistic (31.3) is

$$t = \frac{69.83 - 66.00}{7.161\sqrt{1/6 + 1/6}} = \frac{3.833}{4.135} = 0.93,$$

where 10 is the number of degrees of freedom. Since $0.93 < t_{10,0.95} = 1.812$ the colleague does not reject the hypothesis at level 0.05. The two-sample test is wrong, because the two measurements for one subject are not independent (a slow person is likely to be slow after training, too), whereas the two-sample test assumes that the twelve observations are independent. The difference in opinion is caused by the variability of the data. Between subjects the standard deviation of the reaction times is about 7. The two-sample test compares the mean difference 3.83 to this spread. Inside subjects the differences have standard deviation only 2.64. The paired test compares 3.83 to this smaller spread. By taking differences we remove the effect of the subjects from the comparison, and this is the idea of the paired test.

□

Solution to exercise 31.4.

1. For $\mu_0 = 9$ the value of the test statistic (31.1) is $t = 4\,(10.3 - 9)/2.1 = 2.48$ and, since $|t| > t_{15,0.975} = 2.131$, the two-sided test rejects at level 0.05, as found using duality in exercise 26.4. For $\mu_0 = 11.5$ we have $t = 4\,(10.3 - 11.5)/2.1 = -2.29$ and $|t| = 2.29 > 2.131$, so H_0 is rejected again. This agrees with the interval, since 11.5 is outside the interval $[9.18, 11.42]$.

2. From lemma 26.6 we know that the two-sided test from proposition 31.3 rejects σ_0^2 at level 0.05 if and only if σ_0^2 is outside the 95% confidence interval $[2.41, 10.56]$ from example 26.5. This is the inverse of the test, so $\sigma_0^2 = 2$ is rejected and $\sigma_0^2 = 9$ is not. Looking at the values directly, for $(n-1)s^2 = 15 \cdot 4.41 = 66.15$ the test statistic corresponding to $\sigma_0^2 = 2$ is $v = 66.15/2 = 33.08$, which is greater than $\chi^2_{15,0.975} = 27.488$, so H_0 is rejected. For $\sigma_0^2 = 9$ the test statistic $v = 66.15/9 = 7.35$ falls into the interval $[6.262, 27.488]$ and thus H_0 is not rejected.

3. For $\mu_0 = 9.5$ the test statistic is $t = 4\,(10.3 - 9.5)/2.1 = 1.52$ and, since $1.52 \leq t_{15,0.95} = 1.753$, the one-sided test does not reject at level 0.05. The two-sided interval is the inverse of the two-sided test with level α, *i.e.* the upper tail of the two-sided test has probability $\alpha/2$. If we carried out a one-sided test by checking whether μ_0 is less than the lower bound of the interval, we would obtain the one-sided test at level $\alpha/2 = 0.025$, not at level 0.05. Thus, the one-sided test with 0.05 is not the same as the two-sided test. The confidence set corresponding to the one-sided test with 0.05 is, as in exercise 26.4, the one-sided interval $[\bar{x} - t_{15,0.95}\, s/\sqrt{n},\ \infty) = [10.3 - 1.753 \cdot 0.525,\ \infty) = [9.38, \infty)$, and 9.5 is contained in this interval, as expected.

□

C.22 The Exponential Model: From Start to Finish

Here we solve the exercises at the end of lecture 32.

Solution to exercise 32.1.

1. The log-likelihood function is $\ell(\theta) = n \log \theta - \theta T$, with $T = \sum_i x_i = n\bar{x}$, and the MLE is $\hat{\theta} = n/T$. Using $\hat{\theta} T = n$ and $\theta_0 T = n\theta_0/\hat{\theta}$ we find

$$W = 2(\ell(\hat{\theta}) - \ell(\theta_0)) = 2\big(n \log \hat{\theta} - n - n \log \theta_0 + n\theta_0/\hat{\theta}\big) = 2n\Big(\log \frac{\hat{\theta}}{\theta_0} - 1 + \frac{\theta_0}{\hat{\theta}}\Big).$$

   Using $u = \theta_0 \bar{x} = \theta_0/\hat{\theta}$ we have $\log(\hat{\theta}/\theta_0) = -\log u$ and $\theta_0/\hat{\theta} = u$, and thus $W = 2n(u - 1 - \log u)$.

2. Let $g(u) = u - 1 - \log u$ for $u > 0$. Then $g(1) = 0$ and $g'(u) = 1 - 1/u$. Since this value is negative for $u < 1$ and positive for $u > 1$, $u = 1$ must be the unique minimum of g and $g(u) > 0$ for $u \neq 1$. Since $W = 2ng(u)$, we have $W \geq 0$, with equality if and only if $u = 1$, *i.e.* if and only if $\hat{\theta} = \theta_0$.

3. For $\theta_0 = 0.5$ we have $u = 1.25$ and $W = 20\,(1.25 - 1 - \log 1.25) = 20 \cdot 0.0269 = 0.54$. This value is less than $\chi^2_{1,0.95} = 3.84$, so $H_0\colon \theta = 0.5$ is not rejected at level 0.05. For $\theta_0 = 1$ we have $u = 2.5$ and $W = 20\,(2.5 - 1 - \log 2.5) = 20 \cdot 0.584 = 11.67$. This value is greater than 3.84, so $H_0\colon \theta = 1$ is rejected: the data are not compatible with a mean lifetime of one year.

This completes the solution. □

Solution to exercise 32.2.

1. From example 11.8 we know that $\mathcal{I}(\theta) = 1/\theta^2$. Thus, the Jeffreys prior is $\pi_J(\theta) \propto \sqrt{\mathcal{I}(\theta)} = 1/\theta$. Since $\int_0^1 \theta^{-1}\,\mathrm{d}\theta = \infty$, the prior has infinite integral and thus is improper.

2. The posterior density is proportional to $\theta^{-1} \cdot \theta^n e^{-\theta T} = \theta^{n-1} e^{-\theta T}$. This is the Gamma$(n, T)$ density up to the constant $T^n/\Gamma(n)$. Since $T > 0$, this constant is finite, so the posterior is proper and corresponds to the Gamma(n, T) distribution.

3. From exercise 11.6 we know that the information about the mean is $\mathcal{I}_\mu(\mu) = 1/\mu^2$. Thus, the Jeffreys prior for μ is $\pi_J(\mu) \propto 1/\mu$. Using the transformation $\pi_J(\theta)$ with $\theta = 1/\mu$ and $|\mathrm{d}\theta/\mathrm{d}\mu| = 1/\mu^2$ we find the transformed density $\pi_J(1/\mu)/\mu^2 \propto \mu/\mu^2 = 1/\mu$. This is the same prior as $\pi_J(\mu)$ above, as expected from proposition 28.9.

4. For $\theta \sim \text{Gamma}(n, T)$ we have $2T\theta \sim \text{Gamma}(n, 1/2) = \chi^2_{2n}$ and thus the equal-tailed 95% credible interval is $[\chi^2_{2n,0.025}/(2T),\ \chi^2_{2n,0.975}/(2T)]$. These are the same numbers as for the exact confidence interval. The confidence interval covers the fixed value θ in 95% of repeated samples. The credible interval, on the other hand, covers the random variable θ with posterior probability 0.95 given the observed data.

This completes the solution. □

Solution to exercise 32.3.

(a) The distribution function of one observation is $F(y) = y/\theta$ for $0 \le y \le \theta$, with density $f(y) = 1/\theta$ there. By proposition 8.8 the minimum has density

$$f_{(1)}(y) = n(1 - F(y))^{n-1} f(y) = \frac{n}{\theta}\Big(1 - \frac{y}{\theta}\Big)^{n-1}, \qquad 0 \le y \le \theta,$$

and zero otherwise. Substituting $u = y/\theta$ we find

$$\mathbb{E}_\theta(X_{(1)}) = \theta \int_0^1 n\,u(1-u)^{n-1}\,\mathrm{d}u = \theta\, n\, B(2, n) = \theta\, n\, \frac{1!\,(n-1)!}{(n+1)!} = \frac{\theta}{n+1},$$

using the beta function of table A.2, or equivalently the mean $1/(n+1)$ of the Beta$(1, n)$ distribution of $X_{(1)}/\theta$.

(b) By part (a) and the linearity of expectation, $\mathbb{E}_\theta(U) = (n+1)\,\theta/(n+1) = \theta$ for all θ, and thus U is unbiased. The same substitution gives

$$\begin{aligned}\mathbb{E}_\theta(X_{(1)}^2) &= \theta^2 \int_0^1 n\,u^2(1-u)^{n-1}\,\mathrm{d}u = \theta^2\, n\, B(3,n) \\ &= \theta^2\,\frac{2n\,(n-1)!}{(n+2)!} = \frac{2\theta^2}{(n+1)(n+2)},\end{aligned}$$

and thus

$$\begin{aligned}\mathrm{Var}_\theta(X_{(1)}) &= \frac{2\theta^2}{(n+1)(n+2)} - \frac{\theta^2}{(n+1)^2} \\ &= \frac{\theta^2\big(2(n+1)-(n+2)\big)}{(n+1)^2(n+2)} = \frac{n\,\theta^2}{(n+1)^2(n+2)}\end{aligned}$$

and $\mathrm{Var}_\theta(U) = (n+1)^2\,\mathrm{Var}_\theta(X_{(1)}) = n\theta^2/(n+2)$. Since U is unbiased, $\mathrm{MSE}(U) = \mathrm{Var}_\theta(U) \to \theta^2$ as $n \to \infty$, and thus the mean squared error does not tend to zero. This alone does not disprove consistency, since lemma 2.11 is only a sufficient condition; but directly, for $0 < \varepsilon < \theta$ we have

$$\begin{aligned}\mathbb{P}_\theta(|U-\theta| \geq \varepsilon) &\geq \mathbb{P}_\theta(U < \theta - \varepsilon) = \mathbb{P}_\theta\Big(X_{(1)} < \frac{\theta-\varepsilon}{n+1}\Big) \\ &= 1 - \Big(1 - \frac{\theta-\varepsilon}{(n+1)\theta}\Big)^n \to 1 - e^{-(\theta-\varepsilon)/\theta} > 0,\end{aligned}$$

and thus U does not converge to θ in probability: U is not consistent in the sense of definition 2.9. The reason is that U uses only the smallest observation, which carries little information about the upper endpoint θ.

(c) Writing the density of one observation as $\theta^{-1}\mathbf{1}_{\{0\leq x\leq\theta\}}$, the joint density is

$$f(x;\theta) = \theta^{-n}\prod_{i=1}^n \mathbf{1}_{\{0\leq x_i\leq\theta\}} = \theta^{-n}\,\mathbf{1}_{\{x_{(n)}\leq\theta\}}\cdot\mathbf{1}_{\{x_{(1)}\geq 0\}},$$

since all x_i are at most θ if and only if the largest is, and all are non-negative if and only if the smallest is. This is of the form (8.1) with $g(x_{(n)};\theta) = \theta^{-n}\mathbf{1}_{\{x_{(n)}\leq\theta\}}$ and $h(x) = \mathbf{1}_{\{x_{(1)}\geq 0\}}$, and thus $X_{(n)}$ is sufficient for θ by proposition 8.3. The indicator involving θ has to stay in g.

(d) Conditionally on $X_{(n)} = m$, the other $n-1$ observations are i.i.d. uniform on $(0,m)$, and $X_{(1)}$ is their minimum. By part (a) with $n-1$ in place of n and m in place of θ, the minimum of $n-1$ i.i.d. uniform observations on $(0,m)$ has mean m/n, and thus

$$\mathbb{E}_\theta(X_{(1)}\,|\,X_{(n)} = m) = \frac{m}{n}, \qquad V = \mathbb{E}_\theta(U\,|\,X_{(n)}) = (n+1)\,\frac{X_{(n)}}{n}.$$

The Rao–Blackwell theorem, proposition 10.1, applies because $X_{(n)}$ is sufficient by part (c) and U is unbiased with finite variance by part (b). It guarantees that V does not depend on θ, which the formula confirms, that V is unbiased for θ, and that $\mathrm{Var}_\theta(V) \leq \mathrm{Var}_\theta(U)$.

(e) By proposition 10.6 the maximum $X_{(n)}$ is complete, and it is sufficient by part (c). Since V is a function of $X_{(n)}$ and unbiased for θ, the Lehmann–Scheffé theorem, theorem 10.5, shows that V is the UMVUE for θ. Its variance $\theta^2/(n(n+2))$ is smaller than $\operatorname{Var}_\theta(U) = n\theta^2/(n+2)$ by a factor n^2, and it lies below the Cramér–Rao value θ^2/n. There is no contradiction, because theorem 13.2 does not apply to this model: its conditions, section 11.1, require the support of the density not to depend on θ, and here the support is $(0,\theta)$. The identity $\operatorname{Cov}_\theta(W, \ell'(\theta)) = 1$ on which the proof rests fails, because differentiating $\int W f \,\mathrm{d}x$ with respect to θ picks up a term from the moving boundary of the integral. The value θ^2/n is therefore not a bound at all for this model, and estimators based on the maximum are super-efficient in the sense of example 13.7.

□

Solution to exercise 32.4.

(a) The tray comes from supplier A with probability π, and given the supplier the count is binomial, and thus by the law of total probability

$$f(x;\pi) = \pi f_A(x) + (1-\pi) f_B(x), \qquad x \in \{0, 1, \dots, 10\},$$

a two-component mixture with known components. For n independent trays the observed-data log-likelihood is

$$\ell(\pi) = \sum_{i=1}^{n} \log\bigl(\pi f_A(x_i) + (1-\pi) f_B(x_i)\bigr),$$
$$\ell'(\pi) = \sum_{i=1}^{n} \frac{f_A(x_i) - f_B(x_i)}{\pi f_A(x_i) + (1-\pi) f_B(x_i)}.$$

The likelihood equation $\ell'(\pi) = 0$ is, after clearing denominators, a polynomial equation of degree $n-1$ in π, and the sum inside the logarithm prevents the simplification which made the examples of lecture 5 solvable. The complete data are the pairs (X_i, Z_i) with $f_c(x, 1; \pi) = \pi f_A(x)$ and $f_c(x, 2; \pi) = (1-\pi) f_B(x)$, and thus

$$\ell_c(\pi) = \sum_{i=1}^{n} \Bigl[\mathbf{1}_{\{Z_i = 1\}} \bigl(\log \pi + \log f_A(x_i)\bigr) + \mathbf{1}_{\{Z_i = 2\}} \bigl(\log(1-\pi) + \log f_B(x_i)\bigr) \Bigr],$$

which is linear in the indicators.

(b) By definition 19.2 the E step computes $Q(\pi \mid \pi_k) = \mathbb{E}_{\pi_k}\bigl(\ell_c(\pi) \mid X\bigr)$. Since ℓ_c is linear in the indicators, and since the pairs are independent, this replaces $\mathbf{1}_{\{Z_i=1\}}$ by its conditional expectation $\mathbb{P}_{\pi_k}(Z_i = 1 \mid X_i = x_i)$, which by Bayes' rule, proposition A.20, is

$$\gamma_{i,k} = \frac{\mathbb{P}_{\pi_k}(Z_i = 1)\, \mathbb{P}(X_i = x_i \mid Z_i = 1)}{\mathbb{P}_{\pi_k}(X_i = x_i)} = \frac{\pi_k f_A(x_i)}{\pi_k f_A(x_i) + (1-\pi_k) f_B(x_i)},$$

and $\mathbf{1}_{\{Z_i=2\}}$ by $1 - \gamma_{i,k}$. Thus

$$Q(\pi \mid \pi_k) = \sum_{i=1}^{n} \Bigl[\gamma_{i,k} \log \pi + (1 - \gamma_{i,k}) \log(1-\pi) \Bigr] + \text{terms not involving } \pi.$$

(c) Differentiating with the $\gamma_{i,k}$ held fixed gives

$$\frac{\partial Q}{\partial \pi} = \frac{\sum_i \gamma_{i,k}}{\pi} - \frac{\sum_i (1-\gamma_{i,k})}{1-\pi} = \frac{\sum_i \gamma_{i,k} - n\pi}{\pi(1-\pi)},$$

which vanishes at $\pi_{k+1} = \frac{1}{n}\sum_i \gamma_{i,k}$, is positive to the left of this value and negative to the right. Thus π_{k+1} is the maximiser; equivalently, the second derivative $-\sum_i \gamma_{i,k}/\pi^2 - \sum_i (1-\gamma_{i,k})/(1-\pi)^2$ is negative for every $\pi \in (0,1)$, so $Q(\,\cdot\,|\pi_k)$ is concave on $(0,1)$ and π_{k+1} is its global maximiser. The update is the complete-data MLE of example 5.3 with the unobserved indicators replaced by responsibilities.

(d) With $\pi_0 = 1/2$ the weights cancel and $\gamma_{i,0} = f_A(x_i)/(f_A(x_i) + f_B(x_i))$. From the table we get for $x = (2, 7, 1, 8, 3)$, to four decimal places,

$$(\gamma_{1,0}, \ldots, \gamma_{5,0}) = (0.9952,\ 0.0029,\ 0.9995,\ 0.0003,\ 0.9572),$$

and thus $\sum_i \gamma_{i,0} = 2.9551$ and $\pi_1 = 2.9551/5 = 0.5910$. The responsibilities are all close to 0 or 1. Since the two component distributions barely overlap, each count identifies its supplier almost with certainty, and the trays with 1, 2 and 3 germinated seeds are assigned to supplier A, the trays with 7 and 8 to supplier B. The tray with 3 seeds is the least certain, at 0.957. Further iterations move π_k only slightly, to the fixed point 0.594; the observed-data log-likelihood rises from -10.30 at π_0 to -10.22 at π_1, as proposition 20.2 requires.

(e) With p_A unknown, $\log f_A(x_i) = x_i \log p_A + (m - x_i)\log(1-p_A) + \text{const}$ can no longer be dropped from Q, which now contains the term

$$\sum_i \gamma_{i,k}\big(x_i \log p_A + (m - x_i)\log(1-p_A)\big).$$

Setting the derivative of this term with respect to p_A to zero gives

$$\frac{\sum_i \gamma_{i,k} x_i}{p_A} - \frac{\sum_i \gamma_{i,k}(m - x_i)}{1-p_A} = 0, \qquad \textit{i.e.} \qquad p_{A,k+1} = \frac{\sum_i \gamma_{i,k} x_i}{m \sum_i \gamma_{i,k}},$$

the responsibility-weighted proportion of germinated seeds among the trays attributed to supplier A, with the responsibilities now computed from $(\pi_k, p_{A,k})$. For the data of part (d) and the responsibilities computed there this gives $p_{A,1} = 5.884/29.55 = 0.199$. The ascent property, proposition 20.2, states that the observed-data log-likelihood never decreases along the EM sequence, $\ell(\pi_{k+1}, p_{A,k+1}) \geq \ell(\pi_k, p_{A,k})$ for all k. It guarantees that the sequence of log-likelihood values converges, and by proposition 20.3 a point at which the iteration comes to rest is a stationary point of ℓ; it does not guarantee that this point is the global maximum, since a poor starting value can lead to a local maximum, and it says nothing about the speed of convergence.

□

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